Contents

EM Algorithm for the Student-t Dynamic Factor Model

Lorenzo Giordani

What is a DGP?

Before writing down the model, it is worth pausing on what a *Data Generating Process* (DGP) is and why we bother specifying one at all.

A DGP is a mathematical description of *how we assume the data were produced by the world*. It is an assumption — and a strong one. When we postulate a DGP, we are committing to a specific stochastic mechanism that we believe (or at least find useful to pretend) has generated the observations we see in the dataset.

There is a subtle but important point lurking beneath this definition. At a fundamental level, the numbers we observe — GDP growth in 2008Q4, unemployment in March 2020, the federal funds rate today — exist as isolated quantities, each living in its own "cloud" of possibility. Nothing in the raw data forces them to be related. A priori, every observation could be the realisation of an entirely independent random draw, with no structural connection to any other observation.

To assume a DGP is to make precisely the opposite claim: it is to *disturb* this state of mutual independence and to force the observations into a specific combinatorial relationship — a fixed mathematical structure that ties them together. The freedom of each observation to be "anything" is replaced by the constraint of being "the value the DGP would produce". The DGP is therefore both a tool of inference and a hypothesis about the world.

Why, then, do we accept this imposition? Because once a DGP is in place we can ask a very precise question:

> *Given that the data are generated by this mechanism, what is the probability of observing the specific dataset* $\mathbf{Y}$ *that we actually have?*

This probability, viewed as a function of the unknown parameters, is the *likelihood*. Maximising it over the parameters gives the maximum likelihood estimator: the parameter value that makes our observed data the most plausible outcome under the assumed mechanism. This is the intuition behind MLE and, as we shall see, the object we ultimately want to maximise in this notebook. Every step that follows is motivated by this single question.

Of course, reality is not literally generated by our DGP. The entire exercise is conditional on the modelling assumptions being a reasonable approximation. Making those assumptions explicit — and understanding which of them matter most — is part of the scientific contribution of the model. In this work the key departure from the standard Gaussian DGP is the use of Student-t innovations, which allows heavy tails and extreme realisations to be part of the model rather than being treated as anomalies — a property that will prove essential once we turn to Growth-at-Risk forecasting, where the left tail of the GDP distribution is the object of primary interest.

Observed Data

We observe M time series from $t = 1$ to $t = T$. At each time t the observation is a vector $y_t \in \mathbb{R}^M$:

$$y_t = \begin{bmatrix} y_{1,t} \\ y_{2,t} \\ \vdots \\ y_{M,t} \end{bmatrix}, \qquad t = 1, \ldots, T.$$

Stacking all observations column by column — each column is one time period and each row is one series — gives the $M \times T$ data matrix

$$\mathbf{Y} \;=\; \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{M,1} & y_{M,2} & \cdots & y_{M,T} \end{bmatrix} \;=\; \begin{bmatrix} y_1 \mid y_2 \mid \cdots \mid y_T \end{bmatrix} \;\in\; \mathbb{R}^{M \times T}.$$

Throughout the text we use **bold upper-case letters** for matrices (so $\mathbf{Y}$, $\mathbf{F}$, $\mathbf{\Lambda}$, $\mathbf{E}$, ...) and *lower-case* italic letters for column vectors indexed by time (y_t, f_t, ε_t, ...). Notice the convention: in $\mathbf{Y}$, each *column* is a time period and each *row* is one series. This is the standard Bańbura–Modugno (2014) orientation; it may feel backwards if you are used to the "long" data format where each row is an observation (as in pandas or R data frames), but it is the natural convention for state-space models because it makes matrix products like $\mathbf{\Lambda F}$ directly interpretable as stacked observation equations.

Preprocessing: Stationarity, Centring and Standardisation

Before the data matrix $\mathbf{Y}$ enters the model, each time series undergoes a two-step preprocessing: first it is transformed to achieve *stationarity*, then it is *centred and standardised*.

Stationarity. The factor model (3) assumes a stable VAR(1) law of motion for the factors, which requires the underlying data to have a time-invariant mean and variance. Most raw macroeconomic series are *not* stationary in levels: GDP, employment, industrial production typically exhibit strong upward trends, while price indices grow roughly exponentially. Applying a factor model directly to such series would confuse the common trend with the common factor, producing misleading loadings and unstable estimates.

The standard remedy is to apply a series-specific transformation that makes each variable stationary.

Centring and standardisation. Once the data are stationary, each series i is centred at its sample mean $\bar{y}_i$ and rescaled by its sample standard deviation s_i:

$$\tilde{y}_{i,t} \;=\; \frac{y_{i,t} - \bar{y}_i}{s_i}, \qquad i = 1, \ldots, M, \;\; t = 1, \ldots, T.$$

The transformed $\tilde{y}_{i,t}$ has sample mean zero and sample standard deviation one by construction. In what follows, when we write $\mathbf{Y}$ we mean the matrix of *pre-processed* series — stationary, centred, and standardised — and the DGP we specify in the next subsections is the DGP of these transformed observations.

Why centring and standardising. Two purposes.

First, it puts all series on a comparable scale. A macroeconomic variable expressed in percentage points (like GDP growth) and one expressed in basis points (like a credit spread) otherwise have enormously different sample variances, and the larger-variance series would dominate the factor extraction purely by accident of units.

Second, centring ensures that the location parameter of the idiosyncratic errors and of the factors in the DGP can be fixed at zero without loss of generality:

$$\mathbb{E}[y_t] \;=\; 0, \qquad \mathbb{E}[f_t] \;=\; 0, \qquad \mathbb{E}[\varepsilon_t] \;=\; 0.$$

A non-zero mean in any of these quantities would be absorbed by an intercept in the observation equation; pre-centring the data removes this ambiguity from the outset. The mean-zero assumptions for the latent factors f_t and the idiosyncratic errors ε_t in the DGP specified below are therefore coherent with the scaled data fed into the model.

Standardisation does not Gaussianise the data. A point worth emphasising — and one easy to overlook — is that standardisation does *not* alter the shape of the distribution of each series, only its location and scale. The transformation $\tilde{y}_{i,t} = (y_{i,t} - \bar{y}_i)/s_i$ is an affine map, and affine maps preserve all higher-order standardised moments. If Y is a random variable with mean μ, variance σ^2, skewness γ_3, and excess kurtosis γ_4, then the standardised variable $\tilde{Y} = (Y - \mu)/\sigma$ satisfies

$$\mathbb{E}[\tilde{Y}] \;=\; 0, \qquad \text{Var}(\tilde{Y}) \;=\; 1, \qquad \text{Skew}(\tilde{Y}) \;=\; \gamma_3, \qquad \text{Kurt}(\tilde{Y}) \;=\; \gamma_4 + 3.$$

Only the first two moments are normalised; the third and fourth moments — and therefore the entire shape of the distribution beyond location and scale — pass through the transformation unchanged.

The implication is conceptually important for our framework: if a macroeconomic growth rate exhibits fat tails or asymmetry in its raw form, those features are still present in $\tilde{y}_{i,t}$. The common practice of describing standardised data as "unit-variance" should not be misread as "Gaussian". In particular, the empirical evidence on the heavy-tailed nature of macroeconomic growth rates — the basis for our Student-t specification of the DGP — remains fully relevant after standardisation. Pre-processing places the data on a common, comparable scale; it does not erase the distributional features that motivate the modelling choices in the next subsections.

Back to the original scale. After estimation, any quantity of economic interest — point forecasts, quantile forecasts, conditional densities — can be translated back to the original scale of each series by reversing the standardisation: multiply by s_i and add $\bar{y}_i$. If the raw series has been log-differenced, the model's forecast is a forecast of the growth rate, which can be aggregated into a forecast of the level. The zero-mean, unit-variance convention is therefore a matter of model parametrisation, not a loss of information — all quantities of interest are recoverable in the original units whenever needed.

Factor Structure: The DGP of the Observations

We now specify the first half of the data-generating process: how the observations y_t are produced from a smaller set of latent variables. This is the *observation equation*.

The generative model. We assume that each observation vector y_t is generated as a linear combination of r latent common factors plus an idiosyncratic disturbance:

$$y_t \;=\; \mathbf{\Lambda} f_t \;+\; \varepsilon_t, \qquad t = 1, \dots, T. \tag{1}$$

The interpretation is generative: at each time t, nature first draws the latent factors f_t (whose dynamics we specify in the next subsection), then loads them linearly onto the M series via the matrix $\mathbf{\Lambda}$, and finally adds a series-specific noise term ε_t. The observed y_t is what we see — the sum of a *common signal* $\mathbf{\Lambda} f_t$ and an *idiosyncratic component* ε_t.

The dimensions of the objects involved are:

- $y_t \in \mathbb{R}^M$ — vector of observations at time t,
- $f_t \in \mathbb{R}^r$ — vector of latent common factors, with $r \ll M$,
- $\mathbf{\Lambda} \in \mathbb{R}^{M \times r}$ — matrix of *factor loadings*, describing how each series reacts to each factor,
- $\varepsilon_t \in \mathbb{R}^M$ — vector of idiosyncratic errors specific to each series.

The core economic content of (1) is that the co-movement across the M series is summarised by the small number r of unobserved common factors, while series-specific fluctuations are absorbed into ε_t. The loadings $\mathbf{\Lambda}$ encode *who responds to what*: the element $\mathbf{\Lambda}_{ij}$ is the sensitivity of series i to factor j.

Matrix form across time. It is often convenient to work with the entire sample at once. Stacking the latent factors column by column yields the $r \times T$ matrix

$$\mathbf{F} \;=\; \begin{bmatrix} f_{1,1} & f_{1,2} & \cdots & f_{1,T} \\ f_{2,1} & f_{2,2} & \cdots & f_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ f_{r,1} & f_{r,2} & \cdots & f_{r,T} \end{bmatrix} \;=\; \big[\, f_1 \mid f_2 \mid \cdots \mid f_T \,\big] \;\in\; \mathbb{R}^{r \times T},$$

and the idiosyncratic errors analogously stack into

$$\mathbf{E} \;=\; \big[\, \varepsilon_1 \mid \varepsilon_2 \mid \cdots \mid \varepsilon_T \,\big] \;\in\; \mathbb{R}^{M \times T}.$$

With these conventions, (1) admits the compact matrix form

$$\underbrace{\mathbf{Y}}_{M\times T} \;=\; \underbrace{\mathbf{\Lambda}}_{M\times r}\, \underbrace{\mathbf{F}}_{r\times T} \;+\; \underbrace{\mathbf{E}}_{M\times T}. \tag{2}$$

Written out fully entry by entry, (2) reads

$$\underbrace{\begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{M,1} & y_{M,2} & \cdots & y_{M,T} \end{bmatrix}}_{\mathbf{Y} \in \mathbb{R}^{M\times T}} = \underbrace{\begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,r} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{M,1} & \lambda_{M,2} & \cdots & \lambda_{M,r} \end{bmatrix}}_{\mathbf{\Lambda} \in \mathbb{R}^{M\times r}} \underbrace{\begin{bmatrix} f_{1,1} & f_{1,2} & \cdots & f_{1,T} \\ f_{2,1} & f_{2,2} & \cdots & f_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ f_{r,1} & f_{r,2} & \cdots & f_{r,T} \end{bmatrix}}_{\mathbf{F} \in \mathbb{R}^{r\times T}} + \underbrace{\begin{bmatrix} \varepsilon_{1,1} & \varepsilon_{1,2} & \cdots & \varepsilon_{1,T} \\ \varepsilon_{2,1} & \varepsilon_{2,2} & \cdots & \varepsilon_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{M,1} & \varepsilon_{M,2} & \cdots & \varepsilon_{M,T} \end{bmatrix}}_{\mathbf{E} \in \mathbb{R}^{M\times T}}$$

Each column of this identity reproduces exactly the period-t relation $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$ from (1), while the matrix form makes visible the joint structure across the entire sample. The product $\mathbf{\Lambda}\,\mathbf{F}$ is an $M \times T$ matrix whose columns are $\mathbf{\Lambda} f_1, \mathbf{\Lambda} f_2, \ldots, \mathbf{\Lambda} f_T$. Equations (1) and (2) are therefore the same model, read either one time period at a time or all at once.

Equation (1) describes how observations are generated from factors and errors, but it says nothing about how the factors $\{f_t\}$ evolve over time, nor about the distribution of the idiosyncratic errors ε_t. The next two subsections specify these two ingredients, completing the DGP.

Factor Dynamics: VAR(1) with Student-t Shocks

We now specify the second half of the DGP: how the latent factors $\{f_t\}$ evolve over time. We assume a VAR(1) law of motion, but we depart from the Gaussian baseline and allow for *heavy-tailed* innovations. Instead of assuming $u_t \sim \mathcal{N}(0, \mathbf{Q})$, we posit

$$f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \qquad u_t \overset{iid}{\sim} t_{\nu_u}(0,\, \mathbf{Q}), \tag{3}$$

where $t_{\nu_u}(0, \mathbf{Q})$ denotes a multivariate Student-t distribution with location 0, scale matrix $\mathbf{Q} \in \mathbb{R}^{r \times r}$ (symmetric positive definite), and $\nu_u > 2$ degrees of freedom. The constraint $\nu_u > 2$ ensures that the innovation has a finite second moment, which is required for standard covariance-based estimation.

A point worth fixing immediately is that $\mathbf{Q}$ is the *scale matrix* of the multivariate Student-t distribution, not the covariance matrix of u_t. The two coincide only in the Gaussian limit $\nu_u \to \infty$. For finite $\nu_u > 2$, the relation between the two is

$$\mathrm{Cov}(u_t) \;=\; \frac{\nu_u}{\nu_u - 2}\, \mathbf{Q},$$

which exceeds $\mathbf{Q}$ by a factor strictly greater than one and which diverges as $\nu_u \to 2^+$. In what follows, when we estimate $\mathbf{Q}$ in the M-step we are estimating the scale matrix, not the covariance matrix; the covariance is recovered from $\mathbf{Q}$ and ν_u via the formula above whenever it is needed for interpretation.

The transition matrix $\mathbf{A} \in \mathbb{R}^{r \times r}$ governs the persistence of the factors over time. Crucially, we allow $\mathbf{A}$ to be *full* rather than diagonal. A diagonal $\mathbf{A}$ would force each factor to evolve independently of the others — the real-activity factor would react only to its own past, the financial factor only to its own past, and so on. A full $\mathbf{A}$ instead allows *cross-factor dynamics*: a shock to the financial factor at time $t-1$ can propagate to the real factor at time t through the off-diagonal element $\mathbf{A}_{RF}$.

This matters economically. Financial stress does not affect real activity instantly, but with a lag measurable in months — the cross-factor dynamics in $\mathbf{A}$ are precisely what capture this transmission. Keeping $\mathbf{A}$ unrestricted is what makes the model suitable for Growth-at-Risk analysis, where the propagation from financial stress to real tail risk is the central economic mechanism.

The scale matrix $\mathbf{Q}$ is allowed to be non-diagonal, so that shocks to different factors can be *contemporaneously correlated*. Off-diagonal entries of $\mathbf{Q}$ capture shocks that hit multiple blocks of the economy simultaneously (for example, a macroeconomic announcement that moves both real and financial factors on impact). Together with the cross-dynamic structure of $\mathbf{A}$, this gives the model considerable flexibility in describing how shocks originate and spread.

Matrix form across time. Stacking across time, the state equation can be written compactly. Define the shock matrix $\mathbf{U} = [\, u_1 \mid u_2 \mid \cdots \mid u_T \,] \in \mathbb{R}^{r \times T}$ and the lagged-factor matrix $\mathbf{F}_{-1} = [\, f_0 \mid f_1 \mid \cdots \mid f_{T-1} \,] \in \mathbb{R}^{r \times T}$. Then (3) reads

$$\underbrace{\mathbf{F}}_{r \times T} = \underbrace{\mathbf{A}}_{r \times r} \underbrace{\mathbf{F}_{-1}}_{r \times T} + \underbrace{\mathbf{U}}_{r \times T}. \tag{4}$$

Written out fully entry by entry, (4) reads

$$\underbrace{\begin{bmatrix} f_{1,1} & f_{1,2} & \cdots & f_{1,T} \\ f_{2,1} & f_{2,2} & \cdots & f_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ f_{r,1} & f_{r,2} & \cdots & f_{r,T} \end{bmatrix}}_{\mathbf{F} \in \mathbb{R}^{r \times T}} = \underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,r} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r,1} & a_{r,2} & \cdots & a_{r,r} \end{bmatrix}}_{\mathbf{A} \in \mathbb{R}^{r \times r}} \underbrace{\begin{bmatrix} f_{1,0} & f_{1,1} & \cdots & f_{1,T-1} \\ f_{2,0} & f_{2,1} & \cdots & f_{2,T-1} \\ \vdots & \vdots & \ddots & \vdots \\ f_{r,0} & f_{r,1} & \cdots & f_{r,T-1} \end{bmatrix}}_{\mathbf{F}_{-1} \in \mathbb{R}^{r \times T}} + \underbrace{\begin{bmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,T} \\ u_{2,1} & u_{2,2} & \cdots & u_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ u_{r,1} & u_{r,2} & \cdots & u_{r,T} \end{bmatrix}}_{\mathbf{U} \in \mathbb{R}^{r \times T}}$$

The dimensions check out: the product $\mathbf{A}\,\mathbf{F}_{-1}$ is $r \times T$ (summing over the shared dimension r), matching the shapes of $\mathbf{F}$ and $\mathbf{U}$. Each column of this identity reproduces the period-t equation $f_t = \mathbf{A} f_{t-1} + u_t$ from (3).

Stability. For the VAR(1) to be stable — and hence for the factors to have a well-defined long-run distribution — we require all eigenvalues of $\mathbf{A}$ to lie strictly inside the unit circle. We do not impose this as a hard constraint during estimation, but in all empirical applications the estimated $\widehat{\mathbf{A}}$ satisfies this condition.

Idiosyncratic Errors

For the idiosyncratic component we mirror the modelling choice made on the factor side and allow heavy-tailed behaviour, while keeping the specification otherwise standard. Specifically:

$$\varepsilon_t \overset{iid}{\sim} t_{\nu_\varepsilon}(0, \mathbf{R}), \qquad \mathbf{R} \in \mathbb{R}^{M \times M} \text{ diagonal}, \tag{5}$$

with $\nu_\varepsilon > 2$ the degrees-of-freedom parameter governing the heaviness of the tails on the idiosyncratic side.

As said before, a point worth fixing immediately is that $\mathbf{R}$ is the *scale matrix* of the multivariate Student-t distribution, not the covariance matrix of ε_t. The two coincide only in the Gaussian limit $\nu_\varepsilon \to \infty$. For finite $\nu_\varepsilon > 2$, the relation between the two is

$$\operatorname{Cov}(\varepsilon_t) = \frac{\nu_\varepsilon}{\nu_\varepsilon - 2}\,\mathbf{R},$$

which exceeds $\mathbf{R}$ by a factor strictly greater than one and which diverges as $\nu_\varepsilon \to 2^+$. In what follows, when we estimate $\mathbf{R}$ in the M-step we are estimating the scale matrix, not the covariance matrix; the covariance is recovered from $\mathbf{R}$ and ν_ε via the formula above whenever it is needed for interpretation.

Diagonal scale matrix. The scale matrix $\mathbf{R}$ is taken to be diagonal. This means that the idiosyncratic shocks at time t are *uncorrelated* across series: $\operatorname{Cov}(\varepsilon_{i,t}, \varepsilon_{j,t}) = 0$ for $i \neq j$. All cross-sectional co-movement across the M series is therefore attributed to the common factors — which is precisely what makes the factor representation identifiable. A non-diagonal $\mathbf{R}$ would introduce cross-correlations observationally indistinguishable, in practice, from the contribution of an additional latent factor, and we would lose the clean separation between common and series-specific variation.

A subtle but important point: in the Gaussian case, diagonal scale matrix and diagonal covariance matrix imply *statistical independence* of the components of ε_t. In the multivariate Student-t case, this is no longer true. Even with $\mathbf{R}$ diagonal, the components of ε_t are *uncorrelated but not independent*: they share the common mixing weight w_t^ε in the scale-mixture representation introduced shortly. Two different idiosyncratic shocks at the same time t are therefore linked through the common tail-event signal — a small w_t^ε inflates the variance of all components simultaneously. This is a feature of the multivariate Student-t, not a bug, and reflects the natural notion that "unusual times" for the economy tend to coincide across series even after the common factors have been accounted for.

Serial uncorrelation. The errors ε_t are assumed to be serially uncorrelated (*iid* over time). Any persistence observed in the data is thereby attributed to the common factors through the VAR dynamics on f_t, rather than to the idiosyncratic component. One could extend the model by giving each idiosyncratic series its own AR(1) dynamics, as in Bańbura and Modugno (2014) or Cascaldi-Garcia (2023); doing so adds M autoregressive parameters and requires augmenting the state vector from dimension r to $r + M$. We do not pursue this extension in the baseline specification: the heavy-tailed distribution of ε_t is already the main source of enrichment relative to the Gaussian DFM, and adding idiosyncratic persistence on top would dilute the message and reduce the parsimony of the model. We view this as a natural direction for robustness checks rather than a core element of the model.

Two degrees-of-freedom parameters. Importantly, we allow $\nu_\varepsilon \neq \nu_u$. There is no economic reason to impose a common tail behaviour on *aggregate* shocks (hitting the common factors and driving systemic co-movements) and *series-specific* shocks (hitting individual variables independently). Aggregate shocks are the natural carriers of crises, pandemics, and policy-induced regime shifts, and are plausibly the main source of fat tails in macroeconomic data. Idiosyncratic shocks, by contrast, may behave more regularly or more erratically depending on the data quality and sectoral composition of the dataset. Whether ν_u is smaller or larger than ν_ε in a given application is an empirical question that the estimation procedure will answer.

Stacking across time, the matrix form of (5) is

$$\mathbf{E} \;=\; \left[\, \varepsilon_1 \mid \varepsilon_2 \mid \cdots \mid \varepsilon_T \,\right] \;\in\; \mathbb{R}^{M \times T},$$

with each column drawn independently from $t_{\nu_\varepsilon}(0, \mathbf{R})$.

The Student-t Distribution: From Univariate to Multivariate

The Student-t distribution is the core building block of the entire model. Because it plays a role analogous to the Gaussian in the classical DFM — but with the crucial difference of heavy tails — it is worth spending some time to build a complete and concrete understanding of where it comes from and, above all, why it can be rewritten as a mixture of Gaussians. This last property is what makes the EM algorithm work: it restores, conditional on a latent weight, the same Gaussian algebra we know from the standard case.

We build up gradually. We start in one dimension with the univariate Student-t, contrast it with the univariate Gaussian, then introduce the Gamma distribution (which we will need shortly), derive the scale-mixture representation in one dimension, and only then generalise to the multivariate case.

The Univariate Student-t Distribution

Let X be a scalar random variable. We say that X follows a univariate Student-t distribution with location $\mu \in \mathbb{R}$, scale $\sigma^2 > 0$ and degrees of freedom $\nu > 0$, and we write

$$X \sim t(\mu, \sigma^2, \nu),$$

if its density is

$$p(x \mid \mu, \sigma^2, \nu) \;=\; \frac{\Gamma\big(\frac{\nu+1}{2}\big)}{\Gamma\big(\frac{\nu}{2}\big)\,\sqrt{\nu\pi}\,\sigma} \left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{-\frac{\nu+1}{2}}. \tag{6}$$

Side-by-side comparison with the Gaussian. Recall the univariate Gaussian density:

$$\mathcal{N}\text{-density:} \qquad p(x \mid \mu, \sigma^2) \;=\; \underbrace{\frac{1}{\sqrt{2\pi}\,\sigma}}_{\text{normalising constant}} \cdot \underbrace{\exp\left(-\frac{1}{2}\,\frac{(x-\mu)^2}{\sigma^2}\right)}_{\text{bell-shaped kernel}}.$$

Every Gaussian density is made of two pieces: a *normalising constant* that does not depend on x (only on the parameters, and its job is to make the density integrate to one) and a *kernel* that depends on x and describes the shape.

The Student-t density (6) has *exactly the same two-piece structure*, but both pieces are modified:

$$t\text{-density:} \qquad p(x \mid \mu, \sigma^2, \nu) \;=\; \underbrace{\frac{\Gamma\big(\frac{\nu+1}{2}\big)}{\Gamma\big(\frac{\nu}{2}\big)\,\sqrt{\nu\pi}\,\sigma}}_{\text{normalising constant}} \cdot \underbrace{\left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{-\frac{\nu+1}{2}}}_{\text{polynomial-decay kernel}}.$$

Let us look at the two modifications one at a time.

The kernel The Gaussian kernel and the Student-t kernel differ in one crucial respect:

$$\text{Gaussian:} \quad \exp\Big(-\tfrac{1}{2}\,z^2\Big), \qquad\qquad \text{Student-}t: \quad \big[1 + \tfrac{1}{\nu}\,z^2\big]^{-\frac{\nu+1}{2}},$$

where $z = (x-\mu)/\sigma$ is the standardised deviation. The Gaussian kernel decays *exponentially* in z^2: for large $|z|$, the density vanishes extremely fast. The Student-t kernel decays only *polynomially* in z^2 — as $|z| \to \infty$, the kernel behaves like $|z|^{-(\nu+1)}$, so the probability of extreme observations decays with a power law rather than exponentially. This is the same power-law tail behaviour documented empirically by Fagiolo, Napoletano, and Roventini (2008) for GDP growth rates across OECD countries.This is the precise mathematical manifestation of the expression "heavy tails": the Student-t assigns non-negligible probability to realisations far from μ, whereas the Gaussian effectively rules them out. For macroeconomic data, where recessions, financial crises and policy regime shifts do produce such extreme observations with empirically relevant frequency, polynomial decay is a much more realistic assumption than exponential decay.

The normalising constant: the Gamma functions. The expression $\Gamma(\frac{\nu+1}{2})/\Gamma(\frac{\nu}{2})$ looks exotic but serves exactly the same purpose as $1/\sqrt{2\pi}$ in the Gaussian: it is the constant that makes the density integrate to one. It is worth spending a moment on where the Gamma functions come from, since the same algebra will resurface several times in the derivations that follow.

The logic is the standard one for any probability density. We have a kernel $k(x)$ — the part of the density that depends on x — and we need to find the constant C such that the full density $C \cdot k(x)$ integrates to one. Compute the integral of the kernel alone:

$$I \;=\; \int_{-\infty}^{+\infty} k(x)\,dx \;=\; \int_{-\infty}^{+\infty} \left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{-\frac{\nu+1}{2}} dx. \tag{7}$$

Then setting $C = 1/I$ ensures that the density integrates to one:

$$\int_{-\infty}^{+\infty} \frac{1}{I}\,k(x)\,dx \;=\; \frac{1}{I}\int_{-\infty}^{+\infty} k(x)\,dx \;=\; \frac{1}{I}\cdot I \;=\; 1.$$

The normalising constant is therefore the reciprocal of the kernel integral. To find it we compute I explicitly; we shall show that

$$I \;=\; \sqrt{\nu\pi}\,\sigma\,\frac{\Gamma(\nu/2)}{\Gamma((\nu+1)/2)},$$

so that

$$C \;=\; \frac{1}{I} \;=\; \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\,\sqrt{\nu\pi}\,\sigma},$$

which is exactly the constant appearing in front of the kernel in (6).

We begin by reducing the integrand to a standard form. Substituting $z = (x-\mu)/\sigma$ gives $dx = \sigma\,dz$, and the integral becomes

$$I \;=\; \sigma \int_{-\infty}^{+\infty} \left[1 + \frac{z^2}{\nu}\right]^{-\frac{\nu+1}{2}} dz.$$

A second substitution $u = z^2/\nu$, with $z = \sqrt{\nu u}$ and $dz = \frac{1}{2}\sqrt{\nu/u}\,du$, together with the symmetry of the integrand in z (which allows us to restrict to $z > 0$ and double), transforms this into

$$I \;=\; \sqrt{\nu}\,\sigma \int_0^{+\infty} u^{-1/2}\,(1+u)^{-\frac{\nu+1}{2}}\,du.$$

At this point the integral is in a form that matches a classical special function. The Beta function, for $\alpha, \beta > 0$, is defined by

$$B(\alpha,\beta) \;=\; \int_0^{+\infty} \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}}\,du \;=\; \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \tag{8}$$

The first equality is one of the two standard integral representations of $B(\alpha, \beta)$; the second is its expression in terms of Gamma functions. Both results are classical; a self-contained proof of the Beta–Gamma identity is given in the remark below.

Comparing (8) with the integral we need to evaluate, we identify $\alpha = 1/2$ (from $\alpha - 1 = -1/2$) and $\beta = \nu/2$ (from $\alpha + \beta = (\nu+1)/2$), giving

$$\int_0^{+\infty} u^{-1/2}\,(1+u)^{-\frac{\nu+1}{2}}\,du \;=\; B\big(\tfrac{1}{2},\,\tfrac{\nu}{2}\big) \;=\; \frac{\Gamma(1/2)\,\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}.$$

Combining this with the classical value $\Gamma(1/2) = \sqrt{\pi}$ yields

$$I \;=\; \sqrt{\nu}\,\sigma \cdot \frac{\sqrt{\pi}\,\Gamma(\nu/2)}{\Gamma((\nu+1)/2)} \;=\; \sqrt{\nu\pi}\,\sigma \cdot \frac{\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}.$$

The normalising constant of the Student-t density is therefore $1/I = \Gamma((\nu+1)/2)/[\Gamma(\nu/2)\sqrt{\nu\pi}\,\sigma]$, exactly the constant appearing in front of the kernel in (6).

The underlying parallel with the Gaussian. This calculation is instructive because it makes explicit why the normalising constants of Gaussian and Student-t look so different: the two distributions have kernels that integrate to very different objects.

- The Gaussian kernel $\exp(-z^2/2)$ decays exponentially. Its integral reduces — after completing the square — to the celebrated Gauss integral $\int_{-\infty}^{+\infty} e^{-z^2/2}\,dz = \sqrt{2\pi}$, which explains the factor $1/\sqrt{2\pi}$ in the Gaussian normalising constant.
- The Student-t kernel $[1 + z^2/\nu]^{-(\nu+1)/2}$ decays only polynomially. Its integral reduces to a Beta function, which is expressible as a ratio of Gamma functions — whence the $\Gamma(\cdot)/\Gamma(\cdot)$ appearing in the Student-t normalising constant.

In other words, $\sqrt{2\pi}$ is to exponential decay what $\Gamma(\nu/2)/\Gamma((\nu+1)/2)$ is to polynomial decay: the specific transcendental constant that each kernel happens to integrate to.

Remark: the Beta–Gamma identity. For completeness we recall why

$$B(\alpha, \beta) \;=\; \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Starting from the definition of the Gamma function, $\Gamma(\alpha) = \int_0^\infty s^{\alpha-1}e^{-s}\,ds$, the product $\Gamma(\alpha)\,\Gamma(\beta)$ is a double integral over $(s,t) \in \mathbb{R}^2_{>0}$. Changing variables to (u, v) with $s = uv$, $t = u(1-v)$, $u > 0$, $v \in (0,1)$ — a classical substitution with Jacobian u — factorises the double integral as the product of an integral in u (which yields $\Gamma(\alpha+\beta)$) and an integral in v (which yields $B(\alpha, \beta)$ in its other classical representation, $\int_0^1 v^{\alpha-1}(1-v)^{\beta-1}\,dv$). Rearranging gives the identity.

The Three Parameters of the Student-t

The Student-t has *three parameters*, whereas the Gaussian has two. This is its key distinguishing feature. Before listing them, we stress an important conceptual point: the parameters μ and σ^2 of the Student-t should *not* be read as the mean and variance of the distribution. They are *shape parameters* — location and scale — which always exist, for every $\nu > 0$, and which describe the centre of symmetry and the horizontal stretching of the density. The mean and the variance are, by contrast, *moments*: integrals $\mathbb{E}[X] = \int x\, p(x)\, dx$ and $\mathrm{Var}(X) = \int (x - \mathbb{E}[X])^2 p(x)\, dx$ that exist only when they converge. For the Student-t, convergence fails for small enough ν, and when the moments *do* exist, they stand in a precise but non-trivial relation to the shape parameters:

$$\mathbb{E}[X] = \mu \quad (\text{only if } \nu > 1), \qquad \mathrm{Var}(X) = \frac{\nu}{\nu - 2}\,\sigma^2 \quad (\text{only if } \nu > 2).$$

With this distinction firmly in mind, we turn to the three parameters one at a time.

Location μ. The centre of symmetry of the density. It is *not*, in general, the mean: the mean is an integral that may or may not converge, while the location is a shape parameter defined irrespective of integrability. When the mean exists — that is, for $\nu > 1$ — symmetry forces $\mathbb{E}[X] = \mu$. For $\nu \leq 1$ (including the Cauchy case $\nu = 1$), the distribution is still perfectly well-defined and perfectly symmetric around μ, but the mean simply does not exist.

Scale σ^2. Controls the horizontal stretching of the density. It is *not*, in general, the variance. The variance is a moment that exists only for $\nu > 2$, and even when it exists it is strictly larger than σ^2: $\mathrm{Var}(X) = \frac{\nu}{\nu-2}\,\sigma^2$. The gap is a direct consequence of the heavier tails: a larger variance reflects the non-negligible probability that X lands far from μ. The identity $\mathrm{Var}(X) = \sigma^2$ holds only in the Gaussian limit $\nu \to \infty$.

Degrees of freedom $\nu > 0$. The genuinely new parameter, absent in the Gaussian. It controls the *tail heaviness* and, at the same time, determines which moments exist: the k-th moment of X is finite if and only if $\nu > k$. This is the parameter that makes the Student-t a richer family than the Gaussian.

What ν does: a smooth interpolation. The degrees-of-freedom parameter ν is not a single number with a single meaning — it parametrises a whole family of distributions that smoothly bridges two very different regimes.

- **$\nu \to \infty$: Gaussian limit.** As ν grows, the Student-t density converges pointwise to the Gaussian density:
$$t(\mu, \sigma^2, \nu) \xrightarrow{\nu \to \infty} \mathcal{N}(\mu, \sigma^2).$$
The tails become exponentially light, the variance converges to σ^2, and all moments become finite.

- **$\nu = 1$: Cauchy distribution.** At the opposite extreme, setting $\nu = 1$ gives the notorious Cauchy distribution. This is a pathological case: the mean does not exist, the variance does not exist, and in fact no moment of any order exists. Yet the density is perfectly well-defined and integrates to one. It is the "heaviest-tailed" member of the Student-t family.

- **Moderate ν (say 3 to 10): the empirically relevant range.** In this regime the Student-t has well-defined mean ($\nu > 1$) and variance ($\nu > 2$), but its tails are visibly heavier than the Gaussian's. A useful summary is the *kurtosis*, which for the Student-t with $\nu > 4$ equals

$$\mathrm{Kurt}(X) \;=\; 3 + \frac{6}{\nu - 4},$$

  diverging as $\nu \to 4^+$ and approaching the Gaussian value 3 as $\nu \to \infty$. For $\nu = 5$, kurtosis is 9 — three times the Gaussian value — and the probability of three-standard-deviation realisations is roughly four to five times larger than under a matched-variance Gaussian. The reason is mechanical: the Student-t kernel decays only polynomially in $|z|$, so the density at $|z| = 3$ remains non-negligible, while the Gaussian kernel has already collapsed to near-zero by exponential decay. Polynomial tails "hold up" where exponential tails "crash". This is the regime of interest for macroeconomic time series.

One useful rule of thumb is that the k-th moment of the Student-t exists if and only if $\nu > k$. This is why we will routinely assume $\nu > 2$ when we need a well-defined variance, as is the case throughout the DFM.

Interlude: The Gamma Distribution

Before proceeding to the scale-mixture representation we need to introduce one more distribution: the Gamma. It will appear as the distribution of the latent weights that make the Student-t a mixture of Gaussians, so familiarity with it is indispensable. A brief note on terminology: in what follows we use the term *Gamma distribution* to refer to a probability distribution on $(0, \infty)$ parametrised by a shape and a rate. This is distinct from the *Gamma function* $\Gamma(\alpha)$ used in the previous derivation to normalise the Student-t density. The Gamma distribution *uses* the Gamma function in its density, but the two are conceptually different objects — the first is a random-variable distribution, the second is a transcendental function on $\mathbb{R}_{>0}$.

A positive scalar random variable $W > 0$ follows a Gamma distribution with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$, written

$$W \sim \mathrm{Gamma}(\alpha,\, \beta),$$

if its density is

$$p(w \mid \alpha, \beta) \;=\; \frac{\beta^\alpha}{\Gamma(\alpha)}\, w^{\alpha-1}\, e^{-\beta w}, \qquad w > 0. \tag{9}$$

Its moments are elementary:

$$\mathbb{E}[W] = \frac{\alpha}{\beta}, \qquad \mathrm{Var}(W) = \frac{\alpha}{\beta^2}.$$

For our purposes, the only Gamma distribution that matters is the one with $\alpha = \beta = \nu/2$:

$$W \sim \mathrm{Gamma}\big(\tfrac{\nu}{2},\, \tfrac{\nu}{2}\big).$$

This choice has two remarkable properties that will be essential:

1. $\mathbb{E}[W] = \nu/2 \,/\, \nu/2 = 1$. The expected weight is exactly one — so multiplying or dividing by W does not change the scale of the distribution on average.

2. $\mathrm{Var}(W) = (\nu/2)/(\nu/2)^2 = 2/\nu$. The variance of W is controlled by ν: small ν produces highly dispersed weights, large ν produces weights tightly concentrated around one.

These two facts together mean that, in the $\nu \to \infty$ limit, the distribution of W degenerates to a point mass at 1: with no variance, W is almost surely equal to its mean. This will be precisely how the Student-t collapses to the Gaussian.

The Scale-Mixture Representation: Univariate Case

We are now ready to state and prove the key result.

Proposition 1 (Scale-mixture representation — univariate)**.** *Let $\nu > 0$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$. The following two statements are equivalent:*

(i) $X \sim t(\mu, \sigma^2, \nu)$*;*

(ii) There exists a latent positive random variable W such that

$$X \mid W \sim \mathcal{N}\Big(\mu, \frac{\sigma^2}{W}\Big), \qquad W \sim \mathit{Gamma}\big(\tfrac{\nu}{2}, \tfrac{\nu}{2}\big).$$

In other words, a Student-t is a Gaussian whose variance is itself a random variable, with a specific Gamma distribution.

Proof. The marginal density of X in the hierarchical model (ii) is obtained by integrating the weight W out of the joint density:

$$p(x) \;=\; \int_0^\infty p(x \mid w)\, p(w)\, dw. \tag{10}$$

The integrand is the product of a Gaussian density in x, conditional on w, and the Gamma prior on w:

$$\begin{aligned} p(x \mid w) \;&=\; \frac{1}{\sqrt{2\pi\sigma^2/w}} \exp\Big(-\frac{w(x-\mu)^2}{2\sigma^2}\Big) \;=\; \frac{\sqrt{w}}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{w(x-\mu)^2}{2\sigma^2}\Big), \\ p(w) \;&=\; \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)}\, w^{\nu/2-1} \exp\Big(-\frac{\nu w}{2}\Big). \end{aligned}$$

Their product is

$$p(x \mid w)\, p(w) \;=\; \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)\,\sqrt{2\pi\sigma^2}}\, w^{(\nu+1)/2-1} \exp\Big\{-\frac{w}{2}\Big[\nu + \frac{(x-\mu)^2}{\sigma^2}\Big]\Big\}. \tag{11}$$

Notice that we have combined the two powers of w (the $\sqrt{w}$ from the Gaussian and the $w^{\nu/2-1}$ from the Gamma) into a single factor $w^{(\nu+1)/2-1}$, and we have collected the two exponentials into a single one whose argument is linear in w.

Recognise a Gamma kernel in w. The expression in (11) has the *form* $w^{a-1}e^{-bw}$ with

$$a \;=\; \frac{\nu+1}{2}, \qquad b \;=\; \frac{1}{2}\Big[\nu + \frac{(x-\mu)^2}{\sigma^2}\Big].$$

This is exactly the kernel of a Gamma density in w with shape a and rate b — except, of course, for the wrong normalising constant (the density (11) is the joint $p(x, w)$, not the conditional $p(w \mid x)$).

Integrate w out using the Gamma identity. Recall the fundamental identity defining the Gamma function:

$$\int_0^\infty w^{a-1}\, e^{-bw}\, dw \;=\; \frac{\Gamma(a)}{b^{\,a}}, \qquad a>0,\; b>0. \tag{12}$$

This is just a change of variable ($u = bw$) applied to the definition $\Gamma(a) = \int_0^\infty u^{a-1} e^{-u}\, du$.

Plugging (11) into (10) and applying (12) with our specific a and b:

$$\begin{aligned} p(x) \;&=\; \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)\,\sqrt{2\pi\sigma^2}} \int_0^\infty w^{(\nu+1)/2-1}\, \exp\Big\{-\frac{w}{2}\Big[\nu + \tfrac{(x-\mu)^2}{\sigma^2}\Big]\Big\}\, dw \\ &=\; \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)\,\sqrt{2\pi\sigma^2}} \cdot \frac{\Gamma\big(\frac{\nu+1}{2}\big)}{\Big\{\frac{1}{2}\Big[\nu + \frac{(x-\mu)^2}{\sigma^2}\Big]\Big\}^{(\nu+1)/2}}. \end{aligned}$$

Simplify. We now manipulate the right-hand side to bring it into the canonical form (6). Write the denominator as

$$\left\{\frac{1}{2}\left[\nu + \frac{(x-\mu)^2}{\sigma^2}\right]\right\}^{(\nu+1)/2} \;=\; \frac{1}{2^{(\nu+1)/2}} \cdot \nu^{(\nu+1)/2} \cdot \left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{(\nu+1)/2},$$

where we factored ν out of the bracket in the last step. Substituting back:

$$p(x) \;=\; \frac{(\nu/2)^{\nu/2}\, 2^{(\nu+1)/2}}{\Gamma(\nu/2)\,\sqrt{2\pi\sigma^2}\;\nu^{(\nu+1)/2}} \cdot \Gamma\big(\tfrac{\nu+1}{2}\big) \cdot \left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{-(\nu+1)/2}.$$

The kernel is already the Student-t kernel we wanted. What remains is to show that the constant in front equals the Student-t normalising constant.

Group the powers of ν and 2:

$$\frac{(\nu/2)^{\nu/2}\, 2^{(\nu+1)/2}}{\nu^{(\nu+1)/2}} \;=\; \frac{\nu^{\nu/2}}{2^{\nu/2}} \cdot \frac{2^{(\nu+1)/2}}{\nu^{(\nu+1)/2}} \;=\; \frac{2^{(\nu+1)/2-\nu/2}}{\nu^{(\nu+1)/2-\nu/2}} \;=\; \frac{2^{1/2}}{\nu^{1/2}} \;=\; \sqrt{\tfrac{2}{\nu}}.$$

So the constant simplifies to

$$\frac{\sqrt{2/\nu}\;\Gamma\big(\frac{\nu+1}{2}\big)}{\Gamma(\nu/2)\,\sqrt{2\pi\sigma^2}} \;=\; \frac{\Gamma\big(\frac{\nu+1}{2}\big)}{\Gamma(\nu/2)\,\sqrt{\nu\pi}\,\sigma},$$

where we used $\sqrt{2/\nu}\cdot 1/\sqrt{2\pi\sigma^2} = 1/\sqrt{\nu\pi\sigma^2} = 1/(\sqrt{\nu\pi}\,\sigma)$.

Conclusion. Putting the constant and the kernel back together,

$$p(x) \;=\; \frac{\Gamma\big(\frac{\nu+1}{2}\big)}{\Gamma(\nu/2)\,\sqrt{\nu\pi}\,\sigma} \left[1 + \frac{1}{\nu}\,\frac{(x-\mu)^2}{\sigma^2}\right]^{-(\nu+1)/2},$$

which is exactly the univariate Student-t density (6). The hierarchical model (ii) therefore induces the marginal distribution $X \sim t(\mu, \sigma^2, \nu)$ stated in (i), and the two representations are equivalent. $\square$

How to read this result. The hierarchical description in (ii) tells a story:

1. First, nature draws a *weight* W from a Gamma distribution. The weight is a random positive number, centred at one, whose dispersion is controlled by ν.

2. Conditional on W, nature then draws the observation X from a Gaussian whose variance is σ^2/W.

The Student-t distribution is therefore a Gaussian *with a random variance*. Each observation is drawn from its own "personal" Gaussian, whose spread depends on the weight realised for that observation.

Why this produces heavy tails. The intuition behind the heavy tails is now completely transparent.

If W happens to realise a small value (a "bad draw"), the conditional variance σ^2/W is large — the Gaussian from which X is drawn is very spread out, and the realisation X can fall far from μ without being particularly surprising. Conversely, if W is large, the conditional Gaussian is tight and X stays close to μ.

The marginal distribution of X therefore combines the typical Gaussian "bell" corresponding to typical weights $W \approx 1$ with a mixture of increasingly spread-out Gaussians, corresponding to the occasional small-W realisations. The second ingredient is what generates the heavy tails: from time to time a small W is drawn, the Gaussian becomes diffuse, and an extreme X is produced. How often this happens is governed precisely by how dispersed the Gamma distribution of W is — which, as we saw, is controlled by ν.

The Gaussian limit revisited. We can now reread the $\nu \to \infty$ limit through the scale-mixture lens. As ν grows, the Gamma distribution of W concentrates around its mean of one: $\mathrm{Var}(W) = 2/\nu \to 0$. In the limit, $W = 1$ with probability one, and the hierarchy collapses to

$$X \mid W = 1 \;\sim\; \mathcal{N}(\mu, \sigma^2) \qquad \Longrightarrow \qquad X \sim \mathcal{N}(\mu, \sigma^2).$$

The random-variance mechanism is switched off and we recover an ordinary Gaussian. Small ν, by contrast, produces a Gamma distribution with substantial mass near zero — many small W's, many diffuse Gaussians, and hence many large realisations: a heavy-tailed distribution.

The Multivariate Student-t Distribution

With the univariate story in place, the step up to the multivariate case is conceptually straightforward: we replace the scalar $x \in \mathbb{R}$ with a vector $x \in \mathbb{R}^m$, the scalar location μ with a vector $\mu \in \mathbb{R}^m$, and the scalar scale σ^2 with a *scale matrix* $\Sigma \in \mathbb{R}^{m\times m}$ (symmetric, positive definite). The scalar squared standardised deviation $(x-\mu)^2/\sigma^2$ becomes the *Mahalanobis-squared form* $(x-\mu)'\Sigma^{-1}(x-\mu)$, and the scalar σ in the normalising constant becomes $|\Sigma|^{1/2}$ — the square root of the determinant of Σ. Both objects require a proper geometric interpretation, to which we shall devote a subsection below.

A random vector $x \in \mathbb{R}^m$ is said to follow a multivariate Student-t distribution with location μ, scale matrix Σ and $\nu > 0$ degrees of freedom, written $x \sim t(\mu, \Sigma, \nu)$, if its density is

$$p(x \mid \mu, \Sigma, \nu) \;=\; \frac{\Gamma\big(\frac{\nu+m}{2}\big)}{\Gamma\big(\frac{\nu}{2}\big)\,(\nu\pi)^{m/2}\,|\Sigma|^{1/2}}\left[1+\frac{1}{\nu}\,(x-\mu)'\Sigma^{-1}(x-\mu)\right]^{-\frac{\nu+m}{2}}. \tag{13}$$

Parallel with the univariate case. Comparing (13) with the univariate density (6), one can read off the multivariate generalisation piece by piece. The normalising constant now contains $\Gamma((\nu+m)/2)$ instead of $\Gamma((\nu+1)/2)$, because the integral normalising the density is now m-dimensional rather than one-dimensional (the same algebra — change of variables to the standardised quadratic form, reduction to a Beta integral, application of the Beta–Gamma identity — extends to m dimensions essentially unchanged, with $\Gamma(1/2)\sqrt{\pi}$ from the univariate case replaced by the corresponding m-dimensional volume factor $(\nu\pi)^{m/2}$ in the denominator, and the exponent in the kernel updated accordingly). For the same reason, the kernel exponent becomes $-(\nu+m)/2$ rather than $-(\nu+1)/2$. Finally, the scalar σ in the denominator is replaced by $|\Sigma|^{1/2}$, the square root of the determinant of the scale matrix.

This last point is the only genuinely new ingredient. The quantity $|\Sigma|^{1/2}$ plays exactly the role that σ played in the univariate case: it is the multivariate measure of overall spread, the scalar summary of "how large" the scale matrix is. We make this precise in the geometric interlude below.

Parameters and their interpretation. The roles of the three parameters carry over from the univariate case with no conceptual change — only lifted from scalars to vectors/matrices. The same careful distinction between *shape parameters* (which always exist) and *moments* (which exist only under integrability conditions) applies.

Location $\mu \in \mathbb{R}^m$ — the centre of symmetry of the density. When the mean exists, that is for $\nu > 1$, symmetry forces $\mathbb{E}[x] = \mu$. For $\nu \le 1$ the density is still perfectly well-defined and symmetric around μ, but the mean fails to exist component-wise.

Scale matrix $\Sigma \in \mathbb{R}^{m\times m}$ — symmetric, positive definite, controlling the shape and the stretching of the density's level sets. It is *not*, in general, the covariance matrix. When the covariance exists (that is for $\nu > 2$), the two are related by

$$\mathrm{Cov}(x) \;=\; \frac{\nu}{\nu-2}\,\Sigma, \qquad \text{valid for } \nu > 2,$$

so $\mathrm{Cov}(x)$ is always strictly larger than Σ (in the positive-definite ordering), and the gap vanishes only in the Gaussian limit $\nu \to \infty$. We emphasise this distinction because in the DGP that follows we will assume $u_t \sim t(0, \mathbf{Q}, \nu_u)$ and $\varepsilon_t \sim t(0, \mathbf{R}, \nu_\varepsilon)$: the matrices $\mathbf{Q}$ and $\mathbf{R}$ are scale matrices, not covariance matrices. When in the empirical part we report estimated "variances" of the shocks, what we have estimated directly is $\mathbf{Q}$ and $\mathbf{R}$, and the actual covariances are recovered by the $\frac{\nu}{\nu-2}$ rescaling above.

Degrees of freedom $\nu > 0$ — as in the univariate case, a single scalar governing tail heaviness. The moment-existence rule "the k-th moment exists iff $\nu > k$" continues to hold component-wise in the multivariate case. All univariate intuition transfers unchanged.

Symmetry: no skewness. Both the univariate and multivariate Student-t are *symmetric around* μ. There is no skewness parameter. This is a deliberate modelling choice in the present work: all departure from Gaussianity is encoded in ν, which thickens the tails symmetrically on both sides of μ. Asymmetry — the central object of the Growth-at-Risk literature — would require a different distributional family (such as the skewed-t of Azzalini and Capitanio, 2003), which however complicates the mixture representation considerably. In this model the asymmetry will not be built into the distributional assumption itself, but will emerge endogenously at the second stage of the framework, through the quantile regression step.

Geometric Interlude: Mahalanobis Distance, Ellipsoids, Determinants

The Student-t density (13) and the Gaussian density share the same underlying geometry. Both are fully determined, once μ is fixed, by the scale/covariance matrix Σ — specifically by two objects derived from Σ: the Mahalanobis distance $(x-\mu)'\Sigma^{-1}(x-\mu)$ in the kernel, and the determinant $|\Sigma|$ in the normalising constant. We now unpack these two ingredients carefully. This material is not new — it is the same geometry one encounters in the Gaussian case — but since the rest of the derivation will repeatedly manipulate these objects, it pays off to fix the geometric intuition firmly. A particularly important point, which we will develop in depth, is the role played by the eigenvalues of Σ: they provide the unifying language in which Mahalanobis distance, ellipsoid volume, determinant, and degeneracy can all be understood as expressions of the same underlying object.

A first remark on notation: $|\cdot|$ for matrices means determinant. When applied to a matrix, the symbol $|\cdot|$ denotes the *determinant*, not the absolute value of a number. So $|\Sigma|$, $|\mathbf{Q}|$, $|\mathbf{R}|$ are determinants of positive-definite matrices; they are positive scalars summarising how "large" a matrix is in a specific, volumetric sense. We will derive this volumetric interpretation rigorously in a moment, after introducing the spectral decomposition. As a preview, two simple cases give the right intuition: if $\Sigma = \sigma^2 I_m$ (isotropic case), then $|\Sigma| = \sigma^{2m}$ — every dimension contributes equally; and if Σ is diagonal with entries $s_1, \ldots, s_m$, then $|\Sigma| = s_1 \cdot s_2 \cdots s_m$ — each dimension contributes independently. These are the two "simple cases" that the spectral decomposition will let us extend to general Σ, by viewing it as diagonal in the basis of its eigenvectors.

From the univariate to the multivariate density. Before entering the geometric analysis we recall how the multivariate Gaussian density is built from its univariate cousin, and by direct analogy how the multivariate Student-t density is built from its univariate cousin.

Start from $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$p(x) \;=\; \underbrace{\frac{1}{\sqrt{2\pi}\,\sigma}}_{\text{normalising constant}} \cdot \underbrace{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}_{\text{bell-shaped kernel}}.$$

To move to m dimensions we perform two generalisations. First, we replace the scalar squared distance $(x-\mu)^2/\sigma^2$ with a *multivariate distance* that accounts for both direction-dependent stretching and correlation among coordinates. The natural generalisation is the *Mahalanobis distance*:

$$(x-\mu)'\,\Sigma^{-1}\,(x-\mu),$$

a quadratic form that measures how far x is from μ, stretching and rotating distances according to the correlation structure encoded in Σ. When $\Sigma = I_m$, the Mahalanobis distance reduces to the ordinary squared Euclidean distance $\|x-\mu\|^2$.

Second, we replace the normalising constant $1/(\sqrt{2\pi}\,\sigma)$ with its multivariate analogue. In m dimensions we need a factor $(2\pi)^{m/2}$ — one $\sqrt{2\pi}$ per dimension — and a factor $|\Sigma|^{1/2}$, the square root of the determinant, which plays the role of σ in the univariate case (why this is precisely the right replacement will be derived below).

Putting these pieces together yields the familiar multivariate Gaussian density,

$$p(x) \;=\; \frac{1}{(2\pi)^{m/2}\,|\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}\,(x-\mu)'\Sigma^{-1}(x-\mu)\Big)\,.$$

The multivariate Student-t density (13) is obtained from this Gaussian by exactly the same two replacements that turn a univariate Gaussian into a univariate Student-t: replace the exponential kernel with a polynomial one, adjust the normalising constant accordingly. The geometric content — the Mahalanobis distance and the determinant of Σ — stays identical.

The spectral decomposition of Σ. The central tool for understanding the geometry of Σ is its *spectral decomposition*. Because Σ is symmetric and positive definite, the spectral theorem guarantees that Σ can be written as

$$\Sigma \;=\; V \Lambda V', \tag{14}$$

where $V \in \mathbb{R}^{m\times m}$ is an *orthogonal* matrix (that is, $V'V = VV' = I_m$) and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_m)$ is a diagonal matrix whose entries are the *eigenvalues* of Σ, all strictly positive. The columns $v_1, \ldots, v_m$ of V are the corresponding *eigenvectors*, which form an orthonormal basis of $\mathbb{R}^m$.

Proof of the spectral theorem (sketch). The proof has three essential steps. First, since Σ is symmetric over the reals, all its eigenvalues are real. To see this, let v be a possibly complex eigenvector with eigenvalue λ, so that $\Sigma v = \lambda v$. Taking conjugate-transpose and using symmetry $\Sigma' = \Sigma$ gives $v^*\Sigma = \bar{\lambda}\, v^*$. Now $v^*\Sigma v = \lambda\, v^* v$ from the first equation and $v^*\Sigma v = \bar{\lambda}\, v^* v$ from the second. Since $v^* v$ is a positive real (a norm squared), we conclude $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$.

Second, eigenvectors corresponding to distinct eigenvalues are orthogonal. Let $\Sigma v_i = \lambda_i v_i$ and $\Sigma v_j = \lambda_j v_j$ with $\lambda_i \neq \lambda_j$. Then $v_j' \Sigma v_i = \lambda_i\, v_j' v_i$, and by symmetry also equal to $(\Sigma v_j)' v_i = \lambda_j\, v_j' v_i$. Therefore $(\lambda_i - \lambda_j) v_j' v_i = 0$, and since $\lambda_i \neq \lambda_j$, we have $v_j' v_i = 0$.

Third, for eigenvalues with multiplicity greater than one, an orthonormal basis of the corresponding eigenspace can always be chosen (for example via Gram–Schmidt). Collecting all eigenvectors across all eigenspaces gives an orthonormal basis of $\mathbb{R}^m$, which as columns of V satisfies $VV' = I$. Writing $\Sigma V = V\Lambda$ and right-multiplying by $V' = V^{-1}$ yields $\Sigma = V\Lambda V'$ as claimed.

Finally, positive definiteness of Σ means $v'\Sigma v > 0$ for all $v \neq 0$; taking v to be an eigenvector with eigenvalue λ gives $v'\Sigma v = \lambda\, v' v > 0$, so all eigenvalues must be strictly positive. □

Geometric reading of the spectral decomposition. The decomposition $\Sigma = V\Lambda V'$ tells us that Σ acts on vectors in a very specific way. Given a vector x, the operation Σx can be read from right to left as three successive steps. First, $V'x$ is a rotation of x: it expresses x in the basis of the eigenvectors $v_1, \ldots, v_m$. Second, $\Lambda V' x$ is a component-wise stretching by the eigenvalues λ_i: along the i-th eigenvector direction, the vector is stretched by a factor λ_i. Third, $V\Lambda V' x$ rotates the result back into the original basis. Geometrically: Σ rotates into a basis that diagonalises it, stretches by the eigenvalues in that basis, then rotates back. The eigenvectors point in the directions along which Σ's action is simple (pure scaling), and the eigenvalues measure how much stretching happens in each direction.

This is the unifying picture: Σ encodes a collection of *principal directions* (the eigenvectors), each with its own *principal scale* (the square root of the corresponding eigenvalue, $\sqrt{\lambda_i}$, for reasons we see next). All the geometric quantities derived from Σ — ellipsoid, volume, determinant, Mahalanobis distance — reduce in this basis to elementary expressions in the eigenvalues.

The ellipsoid of a covariance/scale matrix. We have defined the Mahalanobis distance but not yet given it a clear geometric meaning. Its level sets provide the key. In one dimension, the "typical region" of a Gaussian $\mathcal{N}(\mu, \sigma^2)$ is simply an *interval* around μ: for instance, $[\mu - \sigma, \mu + \sigma]$

contains about 68% of the probability mass. In m dimensions, the analogous object is an *ellipsoid* — the set of all points at a fixed Mahalanobis distance from the centre:

$$E_c \;\equiv\; \{\, x \in \mathbb{R}^m : (x-\mu)'\Sigma^{-1}(x-\mu) \leq c \,\},$$

for some constant $c > 0$. As c increases, the ellipsoid grows and contains more probability mass; different values of c trace out the level sets of the density. The shape, orientation and size of E_c are entirely determined by Σ, and with the spectral decomposition (14) in hand we can read this off cleanly.

Using $\Sigma = V\Lambda V'$, the inverse is $\Sigma^{-1} = V\Lambda^{-1}V'$, where Λ^{-1} is the diagonal matrix with entries $1/\lambda_i$. Substituting the change of coordinates $y = V'(x-\mu)$ — a rotation that expresses $x - \mu$ in the basis of the eigenvectors of Σ — the Mahalanobis distance becomes

$$(x-\mu)'\Sigma^{-1}(x-\mu) \;=\; y'\Lambda^{-1}y \;=\; \sum_{i=1}^{m} \frac{y_i^2}{\lambda_i}.$$

So in the eigenvector basis, the ellipsoid E_c is simply

$$\sum_{i=1}^{m} \frac{y_i^2}{\lambda_i} \;\leq\; c,$$

which is the standard equation of an ellipsoid with semi-axes of length $\sqrt{c\,\lambda_i}$ along each axis. The picture is now complete. In the original coordinates the ellipsoid has its *axes* aligned with the eigenvectors $v_1, \dots, v_m$ of Σ (the columns of V), and its *semi-axis lengths* equal to $\sqrt{c\,\lambda_i}$ — so proportional to the square roots of the eigenvalues. Large eigenvalues correspond to long axes (directions of high variability); small eigenvalues to short axes (directions of low variability).

Three canonical examples in two dimensions fix the intuition. When $\Sigma = \sigma^2 I$ (equal variance, no correlation), both eigenvalues are equal to σ^2 and the ellipsoid reduces to a *circle* of radius $\sigma\sqrt{c}$ — the distribution is equally spread in all directions. When $\Sigma = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$ with $\sigma_1 > \sigma_2$ (no correlation but unequal variances), the eigenvalues are σ_1^2 and σ_2^2, the eigenvectors are the coordinate axes, and the ellipsoid is stretched more along the first axis than the second. When Σ has non-zero off-diagonal entries (correlation between components), the eigenvectors are no longer aligned with the coordinate axes, and the ellipsoid is *rotated* by an angle determined by the eigenvector directions.

Three useful ways to compute a determinant. Three classical results, all standard in linear algebra, will be used throughout the derivations:

- *Multiplicative property:* for any two square matrices A, B of the same size, $\det(AB) = \det(A)\,\det(B)$.
- *Diagonal and triangular matrices:* the determinant of a diagonal (or, more generally, triangular) matrix is the product of its diagonal entries. In particular, if $D = \mathrm{diag}(d_1, \dots, d_m)$, then $|D| = d_1\, d_2 \cdots d_m$.
- *Spectral expression:* for any square matrix A with eigenvalues $\lambda_1, \dots, \lambda_m$ (counted with multiplicity), $\det(A) = \prod_{i=1}^{m} \lambda_i$. For symmetric matrices this follows directly from the spectral decomposition, as we now derive for Σ.

The determinant as the volume of the ellipsoid. The determinant $|\Sigma|$ measures the *volume* of the Mahalanobis ellipsoid. This follows directly from the spectral decomposition.

By a classical result, the determinant of any square matrix equals the product of its eigenvalues (counted with multiplicity). For symmetric matrices this fact can be read off immediately from the spectral decomposition. Using the multiplicative property of the determinant, $\det(AB) = \det(A)\det(B)$,

$$|\Sigma| \;=\; |V\Lambda V'| \;=\; |V|\,|\Lambda|\,|V'| \;=\; |\Lambda| \;=\; \prod_{i=1}^{m} \lambda_i,$$

where we used $|V|\,|V'| = |VV'| = |I| = 1$ (so $|V|^2 = 1$, giving $|V| = \pm 1$, and the two factors of $|V|$ cancel), and the fact that the determinant of a diagonal matrix is the product of its diagonal entries.

The volume of an m-dimensional ellipsoid with semi-axes $a_1, \ldots, a_m$ is a classical result of geometry:

$$\text{Volume} \;=\; \frac{\pi^{m/2}}{\Gamma(m/2+1)}\, a_1\, a_2 \cdots a_m.$$

For our Mahalanobis ellipsoid E_c the semi-axes are $\sqrt{c\,\lambda_i}$, so

$$\text{Volume}(E_c) \;=\; \frac{\pi^{m/2}}{\Gamma(m/2+1)}\, c^{m/2}\, \sqrt{\lambda_1 \lambda_2 \cdots \lambda_m} \;=\; \frac{\pi^{m/2}\, c^{m/2}}{\Gamma(m/2+1)}\, |\Sigma|^{1/2}.$$

For the unit Mahalanobis ball ($c = 1$), this collapses to a constant times $|\Sigma|^{1/2}$ — confirming that the determinant is, up to an m-dependent constant, the volume of the ellipsoid in the original coordinates. The consequence is intuitive: large $|\Sigma|$ means a large ellipsoid, which means the distribution is very spread out and the density has a low peak; small $|\Sigma|$ means a small ellipsoid, which means the distribution is concentrated and the density has a high peak.

Degeneracy: when an eigenvalue tends to zero. The picture above assumed all eigenvalues $\lambda_i > 0$. What happens when one or more eigenvalues approach zero?

The ellipsoid collapses. If $\lambda_k \to 0$ for some k, the k-th semi-axis $\sqrt{c\lambda_k} \to 0$, and the ellipsoid degenerates to a lower-dimensional object — a "flat" ellipsoid of dimension $m-1$ embedded in $\mathbb{R}^m$. In terms of the determinant, $|\Sigma| = \prod_i \lambda_i \to 0$: the "volume" of E_c vanishes. In terms of the distribution, the probability mass concentrates on the $(m-1)$-dimensional subspace orthogonal to the degenerate eigenvector v_k. The random variable x no longer has a proper m-dimensional density: the distribution is *degenerate*.

Equivalently, Σ becomes *singular* (Σ^{-1} does not exist), and the Mahalanobis distance $(x - \mu)'\Sigma^{-1}(x-\mu)$ is undefined. In the log-likelihood, the term $-\frac{1}{2}\log|\Sigma|$ diverges to $+\infty$ — the log-likelihood "explodes".

This is not a mere theoretical concern. In the EM algorithm derived in later sections, we estimate the scale matrices $\mathbf{Q}$ and $\mathbf{R}$ from data. In finite samples, an unconstrained estimator can easily produce a near-singular $\mathbf{Q}$ or $\mathbf{R}$, especially if the number of observations is small relative to the dimension or if the model is overparameterised. Practically this requires two safeguards: (i) an initialisation that yields positive-definite starting values; (ii) monitoring of the smallest eigenvalues of $\mathbf{Q}$ and $\mathbf{R}$ across EM iterations, with a numerical floor imposed when they approach zero. We return to these practical issues when discussing the M-step and its numerical implementation.

Covariance versus scale: the determinant factor. We noted earlier that in the multivariate Student-t, the scale matrix Σ and the covariance matrix $\mathrm{Cov}(x)$ (when it exists, for $\nu > 2$) are related by $\mathrm{Cov}(x) = \frac{\nu}{\nu-2}\,\Sigma$. The implication for the determinant follows immediately. Since the determinant is multilinear and $\mathrm{Cov}(x) = \frac{\nu}{\nu-2}\,\Sigma$ differs from Σ by a scalar multiple in each dimension:

$$|\mathrm{Cov}(x)| \;=\; \left(\frac{\nu}{\nu-2}\right)^m \,|\Sigma|.$$

The inflation factor $[\nu/(\nu-2)]^m$ applies *per dimension*, so in the multivariate case the gap between the "scale volume" $|\Sigma|$ and the "covariance volume" $|\mathrm{Cov}(x)|$ grows exponentially with m. For the Gaussian limit $\nu \to \infty$, the factor collapses to 1 and the two coincide, as expected.

Linear change of variables and the Jacobian. A final property that will come up repeatedly in the derivations below: under a linear transformation $Y = AX + b$, with $A \in \mathbb{R}^{m\times m}$ invertible, the covariance (or scale) matrix transforms as $\mathrm{Cov}(Y) = A\,\mathrm{Cov}(X)\,A'$, and its determinant transforms as

$$|\mathrm{Cov}(Y)| \;=\; |A|^2\,|\mathrm{Cov}(X)|.$$

Intuitively, the transformation stretches volumes by a factor $|A|$ in each of m directions, so the volume of the covariance ellipsoid — and hence its determinant — is multiplied by $|A|^2$. (The square arises because the determinant scales like a squared length, just as an area scales with the square of a linear measure.) This is the multivariate analogue of the familiar fact that $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$ in one dimension.

Why $|\Sigma|^{1/2}$ plays the role of σ. We can now return to the question posed at the start of this interlude: why is $|\Sigma|^{1/2}$ the right replacement for σ in the multivariate normalising constant? With the spectral decomposition in hand, the answer is transparent.

Consider the whitening change of variables $z = \Sigma^{-1/2}(x-\mu)$, where $\Sigma^{-1/2} = V\Lambda^{-1/2}V'$ is the symmetric inverse square root of Σ. This is the transformation that "sphericises" the distribution: if $X \sim \mathcal{N}(\mu, \Sigma)$ then $Z \sim \mathcal{N}(0, I_m)$, a standard multivariate Gaussian. The Jacobian of the change of variables is

$$\left|\frac{\partial x}{\partial z}\right| \;=\; |\Sigma^{1/2}| \;=\; \prod_{i=1}^m \sqrt{\lambda_i} \;=\; \sqrt{\prod_{i=1}^m \lambda_i} \;=\; |\Sigma|^{1/2}.$$

So $|\Sigma|^{1/2}$ is precisely the volumetric "stretching" that $\Sigma^{1/2}$ applies when mapping the standard sphere in z-space to the Mahalanobis ellipsoid in x-space. The factor of $|\Sigma|^{1/2}$ in the denominator of the normalising constant compensates for this stretching — one $\sqrt{\lambda_i}$ per dimension — just as dividing by σ in the univariate case compensates for the stretch of the density by a factor σ.

A concrete check in two dimensions: suppose $m = 2$ and $\Sigma = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$ (diagonal, independent variables). Then $\lambda_1 = \sigma_1^2$, $\lambda_2 = \sigma_2^2$, so $|\Sigma|^{1/2} = \sqrt{\sigma_1^2\sigma_2^2} = \sigma_1\sigma_2$, and the density factorises as

$$p(x) \;=\; \frac{1}{2\pi\sigma_1\sigma_2}\exp\!\Big(-\tfrac{(x_1-\mu_1)^2}{2\sigma_1^2} - \tfrac{(x_2-\mu_2)^2}{2\sigma_2^2}\Big) \;=\; p_1(x_1)\cdot p_2(x_2),$$

the product of two independent univariate normals — as it should. The factor $|\Sigma|^{1/2} = \sigma_1\sigma_2$ in the denominator is doing exactly the same job as σ in the univariate case, one factor per dimension. The general case simply replaces $\sigma_1\cdots\sigma_m$ with $\sqrt{\lambda_1\cdots\lambda_m} = |\Sigma|^{1/2}$, which allows for rotated ellipsoids as well as axis-aligned ones.

In the log-likelihood, the term $-\frac{1}{2}\log|\Sigma|$ that arises from the log of the normalising constant acts as a *penalty on large determinants* — a distribution that is too spread out (large $|\Sigma|$) assigns low density to every individual point, and the log-likelihood penalises this. Conversely, a near-singular Σ (small $|\Sigma|$) would artificially inflate the log-likelihood, which is one of the reasons why degeneracy must be prevented during estimation.

Summary: the three geometric objects. The multivariate Gaussian and the multivariate Student-t share the same geometric skeleton. Three objects are worth fixing firmly in memory.

The *Mahalanobis distance* $(x-\mu)'\Sigma^{-1}(x-\mu)$ is a scalar measure of how far x is from μ, weighted by the inverse of the scale. In the eigenvector basis it becomes $\sum_i y_i^2/\lambda_i$: a weighted sum of squared coordinates where each direction is divided by its own principal scale.

The *Mahalanobis ellipsoid* is the level set of that distance — a rotated and stretched ball centred at μ, whose axes point in the eigenvector directions $v_1, \ldots, v_m$ and whose semi-axis lengths are $\sqrt{c\,\lambda_i}$. Eigenvalues are the scales along principal axes; eigenvectors are the principal directions.

The *determinant* $|\Sigma| = \prod_i \lambda_i$ is the volumetric summary of the ellipsoid, entering the normalising constant as $|\Sigma|^{1/2}$ and the log-likelihood as $\frac{1}{2}\log|\Sigma|$. It vanishes when any eigenvalue approaches zero, signalling that the distribution is becoming degenerate.

Everything that will follow — the closed-form M-step updates for $\mathbf{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$, $\mathbf{R}$, the trace trick, the quadratic forms in the E-step, and the numerical update for ν_u, ν_ε — manipulates the same three objects: Mahalanobis distance, determinant, and (implicitly) spectral decomposition of the scale matrices. The log-determinant penalty $-\frac{T}{2}\log|\Sigma|$ governs the updates of the scale parameters $\mathbf{Q}$ and $\mathbf{R}$; the Mahalanobis quadratic form $(\cdot)'\Sigma^{-1}(\cdot)$ governs the updates of $\mathbf{\Lambda}$, $\mathbf{A}$, and ν. All these expressions ultimately reduce to the eigenvalues of the underlying scale matrix. The Student-t changes the kernel shape from exponential to polynomial decay but leaves this geometric skeleton entirely intact. This is why the scale-mixture representation works: conditional on the weight, the density is Gaussian again, and the same Mahalanobis/determinant machinery applies — with each observation simply scaled by its own weight.

The Scale-Mixture Representation: Multivariate Case

The scale-mixture representation extends from the univariate case to the multivariate one essentially unchanged:

Proposition 2 (Scale-mixture representation — multivariate)**.** *Let $\nu > 0$, $\mu \in \mathbb{R}^m$, $\Sigma \in \mathbb{R}^{m\times m}$ symmetric positive definite. The following are equivalent:*

(i) $x \sim t(\mu, \Sigma, \nu)$;

(ii) There exists a latent positive scalar W such that

$$x \mid W \sim \mathcal{N}\Big(\mu, \frac{\Sigma}{W}\Big), \qquad W \sim \mathit{Gamma}\big(\tfrac{\nu}{2}, \tfrac{\nu}{2}\big).$$

Several features of this multivariate version deserve emphasis.

A single scalar weight governs the whole vector. The random weight W is a scalar, not a vector or a matrix. *The same W scales the entire scale matrix Σ uniformly.* This means that a "bad draw" of W produces a Gaussian that is diffuse *in every direction simultaneously* — not in some components more than others. In our DGP, this will correspond to entire time periods being "anomalous" (all components of u_t or ε_t more variable than normal), as opposed to specific components being anomalous in isolation. This is a coherent modelling choice for macroeconomic aggregate shocks, where crises typically hit all variables simultaneously.

The marginal distribution is obtained as in the univariate case. Marginalising W out by the same Gamma integral used in the univariate proof yields exactly the multivariate Student-t density (13). Concretely:

$$p(x \mid \mu, \Sigma, \nu) \;=\; \int_0^\infty \mathcal{N}(x \mid \mu, \Sigma/W)\ \mathrm{Gamma}(W \mid \nu/2, \nu/2)\ dW.$$

Substituting the two densities, collecting the factors of W, and using the Gamma integral $\int_0^\infty W^{\alpha-1}e^{-\beta W}\,dW = \Gamma(\alpha)/\beta^\alpha$ with $\alpha = (\nu+m)/2$ and $\beta = [\nu + (x-\mu)'\Sigma^{-1}(x-\mu)]/2$, the integral evaluates to the multivariate Student-t density (13). The only algebraic difference relative to the univariate calculation is the appearance of the dimension m in the Gamma-function arguments and in the exponent of the kernel.

Why the Scale-Mixture Matters for the EM Algorithm

We close this section by explaining why the scale-mixture representation is more than a curiosity — it is the technical device that makes the EM algorithm feasible in the heavy-tailed case. The argument is entirely conceptual at this stage: we will see the concrete mechanics once we actually derive the E-step and M-step later in the chapter.

The difficulty with the Student-t density directly. Look back at the multivariate Student-t density (13). The Mahalanobis quadratic form $(x-\mu)'\Sigma^{-1}(x-\mu)$ — which in the Gaussian case sits alone in the exponent — now sits *inside* a power function $[\,1+\cdot\,]^{-(\nu+m)/2}$. Taking the logarithm does not linearise the expression: one obtains

$$\log p(x \mid \mu, \Sigma, \nu) \;=\; \text{const} \;-\; \tfrac{1}{2}\log|\Sigma| \;-\; \tfrac{\nu+m}{2}\log\Big[1 + \tfrac{1}{\nu}(x-\mu)'\Sigma^{-1}(x-\mu)\Big].$$

The appearance of $\log[\,1+(\cdot)\,]$ means that the log-density is *not* a quadratic function of x — it is only *logarithmically*-quadratic, a much more awkward object.

Why does this matter? Because the entire EM machinery for the Gaussian DFM is built on quadratic log-likelihoods. As we saw already in the Gaussian derivation, the trace trick, the separation between observed and latent quantities, the closed-form first-order conditions for the parameters — all of this relies on the log-density being a quadratic form in the latent variables. The logarithm of the Student-t density breaks this structure, and with it the possibility of solving the M-step in closed form.

The fix: condition on the weight. The scale-mixture representation resolves the difficulty in one stroke. If we enlarge the latent space to include the weight W alongside the usual latent objects (in our DFM, the factors F), then *conditional on* W each observation is Gaussian:

$$x \mid W \sim \mathcal{N}\big(\mu, \Sigma/W\big).$$

The log-density conditional on W is therefore

$$\log p(x \mid W, \mu, \Sigma) \;=\; \text{const} \;+\; \tfrac{m}{2}\log W \;-\; \tfrac{1}{2}\log|\Sigma| \;-\; \tfrac{W}{2}\,(x-\mu)'\Sigma^{-1}(x-\mu),$$

which *is* quadratic in x — with the weight W simply appearing as a multiplicative factor in front of the Mahalanobis form. All the Gaussian algebra is restored, at the cost of carrying one additional latent scalar for each observation.

Why this cost is cheap. Enlarging the latent space from F to (F, W) sounds expensive, but in fact it is not, and this is the genuine pay-off of the representation. Two reasons:

1. The *posterior distribution* of W given (x, μ, Σ, ν) turns out to be a Gamma distribution again — a consequence of Gaussian/Gamma conjugacy. Hence the posterior mean of W, which is what the E-step will end up needing, is available in closed form. We will derive this explicitly when we reach the E-step.

2. The M-step, in turn, will be expressible in the same form as the Gaussian M-step, with the only modification being that sums over time are *weighted* by the posterior expectations of the W's. No closed form is lost.

The upshot, which we shall see materialise concretely in the following sections, is that the Student-t DFM is solvable by the same EM loop as the Gaussian DFM — only with weights entering the sums. The intuition for what these weights do is already clear from the univariate scale-mixture picture: observations from "anomalous" time periods (those for which a small W was drawn, generating an extreme realisation) receive a small posterior weight and contribute less to the parameter updates. Heavy-tailed robustness emerges naturally, from conjugacy alone, without any ad-hoc down-weighting scheme.

With this conceptual roadmap in place, we now have all the probabilistic ingredients needed to write down the DGP in hierarchical form — which is what we do in the next section — and then turn to the derivation of the E-step and M-step.

Complete DGP: Summing Up

Before moving on to estimation, let us gather all the modelling assumptions we have made and write down the complete Data Generating Process in a single place. This subsection fixes the final form of the model and sets the stage for the likelihood-based approach of the next chapter.

What we have assumed. We have committed to the following set of assumptions:

- the M observed series are driven by $r \ll M$ common latent factors through a linear observation equation, plus an idiosyncratic error specific to each series;
- the factors follow a VAR(1) with heavy-tailed Student-t innovations, with degrees of freedom ν_u;
- the idiosyncratic errors are *iid* over time and drawn from a Student-t with diagonal scale matrix $\mathbf{R}$ and degrees of freedom ν_ε, with ν_ε possibly different from ν_u;
- both Student-t distributions are represented in *scale-mixture form*, which introduces two sequences of latent positive weights $\{w_t^u\}$ and $\{w_t^\varepsilon\}$, Gamma-distributed and independent of everything else.

The scale-mixture representation is not an additional modelling assumption — it is a re-expression of the Student-t assumption that we adopt for computational convenience. It will allow us to work with conditionally Gaussian densities in everything that follows.

The Complete DGP. Collecting everything, the full system is a *state-space model with heavy-tailed innovations*, represented as a hierarchical Gaussian scale-mixture. Point-wise (for a generic time t):

State equation:

$$f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \qquad u_t \mid w_t^u \sim \mathcal{N}(0,\, \mathbf{Q}/w_t^u), \qquad w_t^u \overset{iid}{\sim} \text{Gamma}\big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\big). \tag{15}$$

Observation equation:

$$y_t \;=\; \boldsymbol{\Lambda}\, f_t \;+\; \varepsilon_t, \qquad \varepsilon_t \mid w_t^\varepsilon \sim \mathcal{N}(0,\, \mathbf{R}/w_t^\varepsilon), \qquad w_t^\varepsilon \overset{iid}{\sim} \text{Gamma}\big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\big), \tag{16}$$

with initial condition $f_0 \sim \mathcal{N}(0, \boldsymbol{\Sigma}_0)$. All weights $\{w_t^u, w_t^\varepsilon\}_{t=1}^T$ are mutually independent and independent of the innovations.

Equivalently, in *matrix form*, with the notation introduced earlier — $\mathbf{Y} = [y_1 \mid \ldots \mid y_T] \in \mathbb{R}^{M\times T}$, $\mathbf{F} = [f_1 \mid \ldots \mid f_T] \in \mathbb{R}^{r\times T}$, $\mathbf{F}_{-1} = [f_0 \mid \ldots \mid f_{T-1}] \in \mathbb{R}^{r\times T}$, $\mathbf{E} = [\varepsilon_1 \mid \ldots \mid \varepsilon_T] \in \mathbb{R}^{M\times T}$, and $\mathbf{U} = [u_1 \mid \ldots \mid u_T] \in \mathbb{R}^{r\times T}$ — the same system reads

$$\mathbf{F} \;=\; \mathbf{A}\,\mathbf{F}_{-1} \;+\; \mathbf{U}, \qquad\qquad \mathbf{Y} \;=\; \boldsymbol{\Lambda}\,\mathbf{F} \;+\; \mathbf{E}. \tag{17}$$

The two representations (15)–(16) and (17) are the same model, written one period at a time or all periods at once.

Parameters and latent quantities. Before proceeding we make explicit the distinction between what we want to *estimate* and what we treat as *latent* (unobserved).

The **parameter vector** collects the objects we shall estimate from the data:

$$\theta = \{ \mathbf{\Lambda}, \ \mathbf{A}, \ \mathbf{R}, \ \mathbf{Q}, \ \nu_\varepsilon, \ \nu_u \}.$$

It contains the factor loadings ($\mathbf{\Lambda}$), the VAR transition matrix ($\mathbf{A}$), the scale matrix of the idiosyncratic errors ($\mathbf{R}$, diagonal), the scale matrix of the factor innovations ($\mathbf{Q}$, not necessarily diagonal), and the two degrees-of-freedom parameters (ν_ε, ν_u) that govern tail heaviness.

The **latent quantities** — present in the model but not observed — are three:

$$\mathbf{F} \in \mathbb{R}^{r\times T}, \qquad W^u \in \mathbb{R}^T_{>0}, \qquad W^\varepsilon \in \mathbb{R}^T_{>0},$$

namely the matrix of common factors, and the two row vectors collecting the Gamma weights, $W^u = [w_1^u, \ldots, w_T^u]$ and $W^\varepsilon = [w_1^\varepsilon, \ldots, w_T^\varepsilon]$. We shall see in the next chapter that the estimation strategy (the EM algorithm) must integrate out *all three* of these latent objects simultaneously — this is precisely the additional complication introduced by the heavy-tailed assumption relative to the Gaussian baseline, which has only $\mathbf{F}$ as its latent variable.

Finally, the **observed quantity** is the matrix

$$\mathbf{Y} \in \mathbb{R}^{M\times T}$$

which collects all the data. Everything else in the model is either an unknown parameter or a latent variable.

Time-invariance of the degrees of freedom. A point worth emphasising: both ν_u and ν_ε are assumed *constant over the sample period*. The tail heaviness of the factor shocks (governed by ν_u) and of the idiosyncratic shocks (governed by ν_ε) is a structural feature of the DGP, not a time-varying quantity. What *does* vary across time are the latent weights w_t^u and w_t^ε: for each period t, a fresh weight is drawn independently from the common Gamma distribution $\text{Gamma}(\nu_u/2, \nu_u/2)$ or $\text{Gamma}(\nu_\varepsilon/2, \nu_\varepsilon/2)$. The weights are the stochastic element that captures how "anomalous" each specific period is — small weights correspond to periods when an extreme realisation was drawn, large weights to ordinary periods — while the ν's parameterise the *distribution* from which these weights are drawn, constant throughout the sample.

A useful analogy: one can think of ν as the shape of a shared probability distribution, and the w_t's as independent draws from that distribution. The shape is fixed; the draws vary. If one wanted to model time-varying tail heaviness — allowing ν_u to change over time in response to regime shifts or other state variables — this would require substantially more elaborate dynamics, for example a score-driven specification as in the generalised autoregressive score (GAS) framework of Creal, Koopman, and Lucas (2013), or a stochastic-volatility model with a dynamic tail index. We do not pursue this direction here. The Growth-at-Risk framework we build in Stage 2 captures time-varying *asymmetry* and *tail magnitude* in a different and arguably more economically natural way: through the quantile regression of GDP growth on the estimated factors, where the tails of the conditional distribution of future GDP growth respond endogenously to the current values of the factors. Time variation in tail risk is therefore encoded in the quantile-regression profile — specifically in how $\beta_F(\tau)$ and $\beta_R(\tau)$ vary across τ — rather than in a time-varying ν.

How are the degrees of freedom estimated? The two parameters ν_u and ν_ε are estimated jointly with $\mathbf{A}, \mathbf{\Lambda}, \mathbf{Q}$ and $\mathbf{R}$ by the EM algorithm developed in the next chapter. At the initialisation step, we set provisional values $\nu_u^{(0)} = \nu_\varepsilon^{(0)} = 10$ — a moderately heavy-tailed default that imposes no strong prior but is well within the range typically observed in macroeconomic data. These are *starting* values, not estimates: the EM algorithm then iterates, and at each M-step the degrees of freedom are updated by maximising the expected complete-data log-likelihood over (ν_u, ν_ε). The resulting first-order condition takes the form of a non-linear equation involving the digamma function $\psi(\cdot)$, which does not admit a closed-form solution; we solve it numerically using Brent's method on each EM iteration. As the algorithm converges, so do the estimates of the degrees of freedom: the final values $\hat{\nu}_u$ and $\hat{\nu}_\varepsilon$ are the maximum-likelihood estimates of the tail-heaviness parameters given the observed data.

Reading the model: a generative interpretation. The DGP (15)–(16) can be read as a recipe for generating data. At each time period t:

1. Nature draws two positive scalar weights: one for the factor shock, $w_t^u \sim \text{Gamma}(\nu_u/2, \nu_u/2)$, and one for the idiosyncratic shock, $w_t^\varepsilon \sim \text{Gamma}(\nu_\varepsilon/2, \nu_\varepsilon/2)$.

2. Conditional on w_t^u, nature draws a factor innovation u_t from a Gaussian with scale $\mathbf{Q}/w_t^u$, and updates the factor by $f_t = \mathbf{A} f_{t-1} + u_t$.

3. Conditional on w_t^ε, nature draws an idiosyncratic shock ε_t from a Gaussian with scale $\mathbf{R}/w_t^\varepsilon$.

4. The observation is finally composed as $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$.

Marginalising the weights out, steps (1)–(3) are equivalent to drawing $u_t \sim t(0, \mathbf{Q}, \nu_u)$ and $\varepsilon_t \sim t(0, \mathbf{R}, \nu_\varepsilon)$ directly from the Student-t. But the hierarchical, step-by-step reading is the one we shall exploit throughout the rest of the chapter, because it turns Student-t problems into Gaussian problems conditional on the weights.

With the DGP fully specified, we now have everything in place to pose the estimation question. We adopt a maximum likelihood viewpoint: *given the observed data* $\mathbf{Y}$*, and under the assumption that those data were generated by the DGP above, what values of* θ *make the observations the most plausible?*

This question is harder than it looks, and for a reason that has nothing to do with the Student-t: the likelihood $p(\mathbf{Y} \mid \theta)$ is itself a marginalisation over *three* blocks of latent quantities — the factors $\mathbf{F}$, the factor-shock weights W^u, and the idiosyncratic weights W^ε — each of which has dimension growing with the sample size:

$$p(\mathbf{Y} \mid \theta) \; = \; \iiint p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta) \, d\mathbf{F} \, dW^u \, dW^\varepsilon.$$

Direct maximisation of this integrated likelihood is computationally infeasible: the dimension of the integration domain is $(r+2)T$, which is prohibitively large for any realistic sample size. The next section takes up this problem and explains why — and then introduces the EM algorithm as the standard way around it.

The Likelihood Problem

Now that we have fixed the DGP, we can pose the estimation problem. The approach is classical maximum likelihood: find the parameter vector θ that makes the observed data most probable under the assumed generating process. The road from this intuitive statement to an actual algorithm runs into one serious obstacle — the presence of latent variables — and the bulk of this section is dedicated to explaining precisely what that obstacle is, why it cannot be circumvented by direct computation, and why it is the natural starting point for the EM construction that follows.

What We Observe and What We Do Not

We recall the distinction between observed and latent quantities.

Observed: the data matrix

$$\mathbf{Y} \;=\; [\, y_1 \mid y_2 \mid \cdots \mid y_T \,] \;\in\; \mathbb{R}^{M \times T},$$

that is, $M \times T$ numbers sitting in front of us.

Latent: three objects generated internally by the DGP but never directly observed:

- the factor matrix $\mathbf{F} = [\, f_1 \mid \cdots \mid f_T \,] \in \mathbb{R}^{r \times T}$;
- the weight vector on the factor side, $W^u = [\, w_1^u, \ldots, w_T^u \,] \in \mathbb{R}^T_{>0}$;
- the weight vector on the idiosyncratic side, $W^\varepsilon = [\, w_1^\varepsilon, \ldots, w_T^\varepsilon \,] \in \mathbb{R}^T_{>0}$.

Every one of these objects is part of our DGP, but none of them is ever measured. The estimation procedure must treat them as random variables to be integrated out of the likelihood — not as auxiliary numbers that we can plug in.

To be estimated: the parameter vector

$$\theta \;=\; \{\, \mathbf{\Lambda},\; \mathbf{A},\; \mathbf{R},\; \mathbf{Q},\; \nu_\varepsilon,\; \nu_u \,\}.$$

What is the Likelihood?

The *likelihood* is the probability (or, in the continuous case, the probability density) of the observed data as a function of the parameters:

$$L(\theta) \;=\; p(\text{observed data} \mid \theta).$$

The interpretation is: "If the world is truly governed by the parameter θ, how plausible would it be to observe the data we actually have?" Crucially, the data are *fixed* — they are what nature has handed us — and θ is the variable. We seek the value of θ that maximises L: the parameter value under which our observations look the least surprising.

A classical coin example fixes the intuition. If you flip a coin 100 times and observe 42 heads, the likelihood under a parameter p (the true probability of heads) is $L(p) \;=\; p^{42}(1-p)^{58}$. Maximising over p gives the MLE $\hat{p} = 42/100 = 0.42$ — the probability value that makes the observed sequence most plausible. In our case the data are $\mathbf{Y}$ and the parameter is θ, but the logic is identical.

The Marginal Likelihood

In our setup, however, there is a subtlety. The DGP does not directly give us $p(\mathbf{Y} \mid \theta)$. What the DGP specifies is the *joint* distribution of all the quantities in the model:

$$p\big(\mathbf{Y},\, \mathbf{F},\, W^u,\, W^\varepsilon \mid \theta\big),$$

which is the density we would obtain if the observation mechanism were transparent — if we could see not only $\mathbf{Y}$ but also the latent factors and the latent Gamma weights. This is the natural object the DGP defines, because the DGP generates the data *sequentially*:

1. draw the weights w_t^u and w_t^ε from their Gamma priors;
2. draw the factor innovations $u_t \mid w_t^u$ conditionally Gaussian, and update $f_t = \mathbf{A} f_{t-1} + u_t$;
3. draw the idiosyncratic shocks $\varepsilon_t \mid w_t^\varepsilon$ conditionally Gaussian;
4. compose the observation $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$.

Every step is an explicit conditional density, and their product factorises into $p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta)$.

But the object that maximum likelihood asks us to work with is not the joint distribution: it is the *marginal distribution* of the observed data alone,

$$L(\theta) \;=\; p(\mathbf{Y} \mid \theta).$$

The two are linked by the law of total probability: the marginal is obtained from the joint by integrating out the latent variables,

$$\boxed{p(\mathbf{Y} \mid \theta) \;=\; \iiint p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big)\, d\mathbf{F}\, dW^u\, dW^\varepsilon.} \tag{18}$$

This expression says something simple conceptually: the probability of observing $\mathbf{Y}$ is the sum (integral), over *all possible realisations of the latent variables*, of the joint probability of observing $\mathbf{Y}$ together with those realisations. We do not know which realisation of $(\mathbf{F}, W^u, W^\varepsilon)$ actually occurred, so we must consider every possibility, weighted by its probability, and add everything up.

Equivalently, using the product rule to factor the joint density,

$$p(\mathbf{Y} \mid \theta) \;=\; \iiint p(\mathbf{Y} \mid \mathbf{F}, W^\varepsilon, \theta)\, p(\mathbf{F} \mid W^u, \theta)\, p(W^u \mid \theta)\, p(W^\varepsilon \mid \theta)\, d\mathbf{F}\, dW^u\, dW^\varepsilon,$$

which makes clearer how each latent quantity enters: the weights W^u, W^ε have Gamma priors, the factors $\mathbf{F}$ have a conditionally Gaussian distribution given W^u, and the observations $\mathbf{Y}$ have a conditionally Gaussian distribution given $\mathbf{F}$ and W^ε.

The log-likelihood. For reasons that will become clear below (sums are easier to work with than products, and the exponential family densities simplify dramatically in logs), in what follows we will work with the log-likelihood rather than the likelihood itself:

$$\ell(\theta) \;=\; \log p(\mathbf{Y} \mid \theta) \;=\; \log \iiint p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big)\, d\mathbf{F}\, dW^u\, dW^\varepsilon.$$

The maximum likelihood estimator is

$$\hat{\theta} \;=\; \arg\max_{\theta}\; \ell(\theta).$$

Why Direct Optimisation Is Infeasible

The maximum likelihood estimator is easy to *write down* but hard to *compute*. The source of the difficulty is the integral in (18). We now unpack, with some care, exactly why this is so.

(a) The dimension of the integration domain. Look at what we are integrating over. The integration variable is the triple $(\mathbf{F}, W^u, W^\varepsilon)$, which lives in

$$\mathbb{R}^{r\times T} \times \mathbb{R}^T_{>0} \times \mathbb{R}^T_{>0},$$

a space of dimension $rT + T + T = (r+2)T$. For a typical macroeconomic application with, say, $r = 3$ factors and $T = 240$ monthly observations (twenty years of data), this is an integration in $(3+2)\cdot 240 = 1200$ dimensions. Integration in such a high-dimensional space cannot be performed by numerical quadrature: quadrature methods require a grid whose cost grows exponentially in dimension, which is hopeless here.

(b) The shape of the integrand. Even ignoring dimensionality, the integrand $p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta)$ is a complicated function. In the Gaussian DFM, conditional on $\mathbf{F}$, each y_t is Gaussian in $\mathbf{F}$, and one can integrate the factors out analytically — the result is another Gaussian in $\mathbf{Y}$, from which a "prediction-error form" of the likelihood can be obtained via the Kalman filter. Our setup is more awkward. Conditional on (W^u, W^ε) the model is Gaussian and the same Kalman-filter decomposition applies; but then one still has to integrate out W^u and W^ε, which enter the Gaussian covariances as scaling factors — making the marginal, after integration over W, no longer Gaussian but Student-t. The integral over the weights is the integral that *defines* the Student-t from the Gaussian scale-mixture; doing it once is straightforward, doing it *inside* a maximisation loop with θ still unspecified is not. This last integral is precisely what makes direct optimisation intractable; the EM algorithm developed below sidesteps the integral by treating the weights as latent variables to be estimated alongside the factors, rather than integrated out.

(c) The maximisation is the real bottleneck. Even if we could *evaluate* the integral at a fixed θ — say by Monte Carlo simulation, at a considerable computational cost — we would still have to *maximise* this stochastic, high-dimensional, non-convex function of θ numerically. Each gradient step would require re-computing the integral. For a parameter vector with dozens of components (all entries of $\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$, plus ν_u, ν_ε), the cumulative cost is prohibitive.

(d) The logarithm and the integral do not commute. There is one final, more subtle point that deserves emphasis. Even if we could evaluate $p(\mathbf{Y} \mid \theta)$ exactly, we would still have to compute

$$\log \int p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta)\, d\mathbf{F}\, dW^u\, dW^\varepsilon,$$

that is, *the logarithm of an integral*. This is structurally awkward for optimisation, because we cannot push the logarithm inside the integral and reduce the problem to maximising an integral of log-densities (which would be much easier, because $\log p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta)$ is, in our DGP, a sum of simple terms). The inequality

$$\log \mathbb{E}[X] \neq \mathbb{E}[\log X]$$

— Jensen's inequality, to which we will return — is precisely what stands between us and an easy optimisation problem. In fact, as we will see, the E-step of the EM algorithm sidesteps this difficulty by replacing the intractable 'log of an integral' with a tractable lower bound — a quantity of the form 'integral of a log' that can be maximised in closed form.

The Key Observation: the Complete-Data Problem Would Be Easy

The structure of the problem suggests a way out. Observe that the *joint* density inside the integral,

$$p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big),$$

factorises by construction into a product of simple conditional Gaussians and Gammas — the four ingredients of the DGP:

$$p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta) \;=\; \underbrace{p(\mathbf{Y} \mid \mathbf{F}, W^\varepsilon, \theta)}_{\text{Gaussian}} \; \underbrace{p(\mathbf{F} \mid W^u, \theta)}_{\text{Gaussian}} \; \underbrace{p(W^u \mid \theta)}_{\text{Gamma}} \; \underbrace{p(W^\varepsilon \mid \theta)}_{\text{Gamma}}.$$

Its logarithm is a sum of quadratic forms (from the Gaussian blocks) plus some power and exponential terms (from the Gamma blocks), all of them of a standard, tractable shape. This is the *complete-data log-likelihood*:

$$\ell_c(\theta) \;=\; \log p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big),$$

the log-likelihood we would have if the latent variables were observed.

The EM algorithm is built on the following trade-off:

- maximising the marginal $\log p(\mathbf{Y} \mid \theta)$ is *what we want* but *hard*;
- maximising the complete $\log p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta)$ is *easy* but requires information we do not have.

The EM algorithm resolves the trade-off by *pretending* we have the latent variables, replacing them with their best posterior guess, and iterating. The precise construction — which makes the idea rigorous via the ELBO bound and Jensen's inequality — is the subject of the next section.

The EM Intuition: Evidence Lower Bound (ELBO)

We left the previous section with the following state of affairs. The log-likelihood we want to maximise is

$$\ell(\theta) \;=\; \log p(\mathbf{Y} \mid \theta) \;=\; \log \int\!\!\int\!\!\int p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big)\, d\mathbf{F}\, dW^u\, dW^\varepsilon,$$

which is a logarithm of an integral in $(r+2)T$ dimensions — intractable to compute and, intractable to maximise over θ. The underlying joint density, by contrast, is a product of simple conditional Gaussians and Gammas; its logarithm is a sum of elementary terms that we would have no trouble maximising if only we could get rid of the integral and of the outer logarithm.

The EM algorithm is the answer to this tension. In this section we construct it step by step, starting from a simple algebraic trick and ending with a principled alternating optimisation scheme. The construction is independent of our particular DGP — it would apply to any latent variable model — but we carry it out with our specific latent triple $(\mathbf{F}, W^u, W^\varepsilon)$ so that every integral we write down is the one we will actually need later.

The Key Trick

To solve this computational problem we resort to an *escamotage* (a clever trick). We will *pretend* to have observed not only the data $\mathbf{Y}$ but also the latent triple $(\mathbf{F}, W^u, W^\varepsilon)$. This is, of course, *not true* in reality — the whole point of the problem is that we do not observe them. But the pretence helps with the calculations.

Instead of working directly with the marginal log-likelihood $\ell(\theta) = \log p(\mathbf{Y} \mid \theta)$, we shall work with the *complete-data log-likelihood*:

$$\ell_c(\theta) \;=\; \log p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big), \tag{19}$$

as if we also observed the latent quantities. As noted before, this object is a sum of standard log-densities (Gaussians and Gammas) — a much friendlier mathematical object than the marginal log-likelihood.

Of course we cannot literally use (19) — we still do not know the latent variables. What we will do is replace them, in a controlled way, with a *probability distribution* over the latent space. This is the next ingredient.

Introducing the Auxiliary Distribution q

We introduce $q(\mathbf{F}, W^u, W^\varepsilon)$ — a probability distribution over the latent triple. This distribution is chosen by us; it is *not necessarily the true one*, it is simply a tool that will let us bound the marginal likelihood from below.

Since q is a valid probability distribution, it satisfies the two usual properties:

$$q(\mathbf{F}, W^u, W^\varepsilon) \;\geq\; 0, \qquad \int q(\mathbf{F}, W^u, W^\varepsilon)\, d\mathbf{F}\, dW^u\, dW^\varepsilon \;=\; 1.$$

These are the only constraints we impose on q at this stage; its shape is otherwise free. The machinery of EM will determine a specific, optimal choice for q — but for now q is simply a generic distribution over the latent space.

For the sake of notational brevity, we will write $\mathcal{Z}$ for the full latent triple:

$$\mathcal{Z} \;\equiv\; (\mathbf{F},\, W^u,\, W^\varepsilon), \qquad d\mathcal{Z} \;\equiv\; d\mathbf{F}\, dW^u\, dW^\varepsilon.$$

So $q(\mathcal{Z})$ is a distribution over the latent space, and the marginal likelihood can be written compactly as $p(\mathbf{Y} \mid \theta) = \int p(\mathbf{Y}, \mathcal{Z} \mid \theta)\, d\mathcal{Z}$. This is purely a shorthand: every statement we make about $\mathcal{Z}$ translates directly into statements about the original triple.

Manipulating the Marginal Likelihood

Starting from

$$p(\mathbf{Y} \mid \theta) \;=\; \int p(\mathbf{Y}, \mathcal{Z} \mid \theta)\, d\mathcal{Z},$$

we multiply and divide the integrand by $q(\mathcal{Z})$. Since we are multiplying and dividing by the same quantity, we are not changing the value of the integral:

$$p(\mathbf{Y} \mid \theta) \;=\; \int q(\mathcal{Z}) \cdot \frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\, d\mathcal{Z}. \tag{20}$$

This looks like a pointless manipulation, but it will pay off in a moment: the right-hand side has the structure of an *expected value* taken under the distribution q.

Why does this integral equal an expectation? Recall the definition of the expected value of a function of a random variable. If $\mathcal{Z}$ is continuous with density $q(\mathcal{Z})$, the expectation of any function $g(\mathcal{Z})$ is *by definition* the integral

$$\mathbb{E}_q[g(\mathcal{Z})] \;=\; \int g(\mathcal{Z}) \cdot q(\mathcal{Z})\, d\mathcal{Z}.$$

This is just a weighted average of g over all possible values of $\mathcal{Z}$, where the weights are given by the density $q(\mathcal{Z})$.

It may help to recall the discrete analogy. If $\mathcal{Z}$ could only take values $\{\mathcal{Z}_1, \mathcal{Z}_2, \mathcal{Z}_3, \ldots\}$ with probabilities $\{q_1, q_2, q_3, \ldots\}$, the expectation would be

$$\mathbb{E}_q[g(\mathcal{Z})] \;=\; \sum_i g(\mathcal{Z}_i) \cdot q_i.$$

The continuous case is the same idea — an "infinite weighted sum" — with the integral replacing the sum and the density $q(\mathcal{Z})$ replacing the point probabilities. Either notation, $\mathbb{E}_q[g(\mathcal{Z})]$ or $\int g(\mathcal{Z})q(\mathcal{Z})\, d\mathcal{Z}$, denotes the same object: the average value of g when $\mathcal{Z}$ has density q.

Now setting $g(\mathcal{Z}) = p(\mathbf{Y}, \mathcal{Z} \mid \theta)/q(\mathcal{Z})$, equation (20) becomes precisely an expectation:

$$p(\mathbf{Y} \mid \theta) \;=\; \int q(\mathcal{Z}) \cdot \underbrace{\frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}}_{g(\mathcal{Z})}\, d\mathcal{Z} \;=\; \mathbb{E}_q\left[\frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right].$$

Taking logarithms on both sides:

$$\ell(\theta) \;=\; \log p(\mathbf{Y} \mid \theta) \;=\; \log \mathbb{E}_q\left[\frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right]. \tag{21}$$

We have rewritten the log-likelihood as the logarithm of an expectation under q. This is progress — because there is a classical inequality, due to Jensen, that will let us replace the right-hand side with a lower bound that is much easier to work with.

Applying Jensen's Inequality

Jensen's inequality states that for a concave function ϕ and any random variable X,

$$\phi(\mathbb{E}[X]) \;\geq\; \mathbb{E}\big[\phi(X)\big].$$

The logarithm is a concave function, so applying Jensen with $\phi = \log$:

$$\log\big(\mathbb{E}[X]\big) \;\geq\; \mathbb{E}\big[\log(X)\big].$$

Setting $X = p(\mathbf{Y}, \mathcal{Z} \mid \theta)/q(\mathcal{Z})$ and taking expectations under q:

$$\log \mathbb{E}_q\left[\frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right] \;\geq\; \mathbb{E}_q\left[\log \frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right].$$

Combining this with (21):

$$\ell(\theta) \;\geq\; \mathbb{E}_q\left[\log \frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right]. \tag{22}$$

We have found a *lower bound* on the log-likelihood. The bound holds for every choice of q. Crucially, the logarithm and the expectation have swapped places: instead of "log of an integral", we now have "integral of a log", which is a much more tractable object.

The Evidence Lower Bound (ELBO)

Using $\log(a/b) = \log a - \log b$, we can split the right-hand side of (22) into two pieces:

$$\ell(\theta) \;\geq\; \mathbb{E}_q[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})].$$

This lower bound has a name — the *Evidence Lower Bound* (ELBO) — and a standard symbol:

$$\boxed{\mathcal{L}(q, \theta) \;=\; \mathbb{E}_q[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})].} \tag{23}$$

Equivalently, before splitting,

$$\mathcal{L}(q, \theta) \;=\; \mathbb{E}_q\left[\log \frac{p(\mathbf{Y}, \mathcal{Z} \mid \theta)}{q(\mathcal{Z})}\right].$$

The ELBO depends on two inputs: the auxiliary distribution $q(\mathcal{Z})$ and the parameter vector θ. By construction, for every q,

$$\mathcal{L}(q, \theta) \;\leq\; \ell(\theta).$$

Reading the two terms. The first term of (23), $\mathbb{E}_q[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)]$, is the expected *complete-data log-likelihood* — the object we introduced in (19), now averaged under q rather than evaluated at a specific $\mathcal{Z}$. Since $\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)$ is a sum of simple Gaussian and Gamma log-densities, its expectation under q is a straightforward object to work with — in sharp contrast with the marginal log-likelihood $\ell(\theta)$.

The second term, $-\mathbb{E}_q[\log q(\mathcal{Z})]$, depends only on q and not on θ. It is the *entropy* of q:

$$H(q) \;=\; -\mathbb{E}_q[\log q(\mathcal{Z})] \;=\; -\int q(\mathcal{Z}) \log q(\mathcal{Z})\, d\mathcal{Z},$$

a scalar measuring how spread out the distribution q is. A distribution concentrated around a single point has low entropy (close to zero or even $-\infty$); a very flat distribution has high entropy. We shall see in a moment that this term plays the role of a *regulariser* on the choice of q.

A Background Note: Entropy and KL Divergence

Before closing the ELBO argument we collect two notions from information theory that we will need repeatedly. Readers already familiar with them can skip this subsection.

Entropy. The entropy of a probability distribution q on a space $\mathcal{Z}$ is

$$H(q) \;=\; -\mathbb{E}_q[\log q(\mathcal{Z})] \;=\; -\int q(\mathcal{Z}) \log q(\mathcal{Z})\, d\mathcal{Z}.$$

It measures how *spread out* or *uncertain* the distribution q is. A degenerate distribution (point mass) has entropy $-\infty$; a uniform distribution on a large set has high entropy. Entropy appears naturally whenever one computes $\mathbb{E}_q[\log q]$ — for example, inside the ELBO.

Kullback–Leibler divergence. The Kullback–Leibler (KL) divergence between two distributions q and p on the same space is

$$D_{KL}(q \| p) \;=\; \mathbb{E}_q\left[\log \frac{q(\mathcal{Z})}{p(\mathcal{Z})}\right] \;=\; \int q(\mathcal{Z}) \log \frac{q(\mathcal{Z})}{p(\mathcal{Z})}\, d\mathcal{Z}.$$

It measures how *different* q is from p. Two fundamental properties hold:

- $D_{KL}(q\|p) \geq 0$ always (Gibbs' inequality);
- $D_{KL}(q\|p) = 0$ if and only if $q = p$ almost everywhere.

The KL divergence is *not* a symmetric distance — in general $D_{KL}(q\|p) \neq D_{KL}(p\|q)$ — but it is a natural measure of dissimilarity on the space of probability distributions. We will use it to measure how far our chosen q is from the "ideal" choice, which will turn out to be the posterior $p(\mathcal{Z} \mid \mathbf{Y}, \theta)$.

The Gap Between ELBO and Log-Likelihood

We are ready for the central result of this section — the exact expression of the gap between $\mathcal{L}(q,\theta)$ and $\ell(\theta)$. This result is what will tell us, precisely, which q is the "right" one to choose.

A rewriting of the ELBO. Starting from the definition (23) and applying the product rule to the joint density inside the logarithm:

$$\log p(\mathbf{Y}, \mathcal{Z} \mid \theta) \;=\; \log p(\mathcal{Z} \mid \mathbf{Y}, \theta) \;+\; \log p(\mathbf{Y} \mid \theta).$$

Substituting into the ELBO:

$$\begin{aligned} \mathcal{L}(q,\theta) \;&=\; \mathbb{E}_q[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})] \\ &=\; \mathbb{E}_q[\log p(\mathcal{Z} \mid \mathbf{Y}, \theta)] \;+\; \mathbb{E}_q[\log p(\mathbf{Y} \mid \theta)] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})]. \end{aligned}$$

The middle term simplifies: $\log p(\mathbf{Y} \mid \theta)$ does not depend on $\mathcal{Z}$, so taking the expectation under q is trivial — $\mathbb{E}_q[\log p(\mathbf{Y} \mid \theta)] = \log p(\mathbf{Y} \mid \theta) = \ell(\theta)$. Rearranging:

$$\mathcal{L}(q,\theta) \;=\; \ell(\theta) \;-\; \Big\{ \mathbb{E}_q[\log q(\mathcal{Z})] \;-\; \mathbb{E}_q[\log p(\mathcal{Z} \mid \mathbf{Y}, \theta)] \Big\},$$

and the quantity in braces is exactly $D_{KL}(q\|p(\mathcal{Z} \mid \mathbf{Y}, \theta))$. Therefore:

$$\boxed{\mathcal{L}(q,\theta) \;=\; \underbrace{\ell(\theta)}_{\text{what we want}} \;-\; \underbrace{D_{KL}\big(q \,\|\, p(\mathcal{Z} \mid \mathbf{Y}, \theta)\big)}_{\geq 0}.} \tag{24}$$

What this decomposition tells us. Equation (24) is the keystone of the whole EM construction. It states three things simultaneously:

- **The ELBO is always a lower bound.** Since $D_{KL} \geq 0$, we have $\mathcal{L}(q,\theta) \leq \ell(\theta)$ for every q — confirming what Jensen's inequality already told us, but now with the *exact value* of the gap.
- **The bound is tight when q is the posterior.** The gap $D_{KL}(q\|p(\mathcal{Z} \mid \mathbf{Y}, \theta))$ equals zero *if and only if* $q(\mathcal{Z}) = p(\mathcal{Z} \mid \mathbf{Y}, \theta)$, i.e. q equals the posterior distribution of the latent variables given the data and the current parameters. For this specific choice of q, the ELBO coincides exactly with the log-likelihood: $\mathcal{L}(q,\theta) = \ell(\theta)$.
- **We have a principled criterion for choosing q.** The "best possible q", in the sense of closing the gap with the log-likelihood, is not an arbitrary choice — it is the posterior $p(\mathcal{Z} \mid \mathbf{Y}, \theta)$. Computing this posterior (or summaries of it sufficient to evaluate the ELBO) is precisely what the E-step of the EM algorithm will do.

A geometric reading. The decomposition (24) admits a useful visual interpretation. For a fixed θ, $\ell(\theta)$ is a single number (we cannot easily compute it, but it exists). The ELBO $\mathcal{L}(q,\theta)$ is a lower bound that depends on the choice of q: changing q moves the bound up or down, while $\ell(\theta)$ stays fixed. The vertical gap between the two — the height by which $\mathcal{L}$ falls short of ℓ — is exactly $D_{KL}(q \,\|\, p(\mathcal{Z} \mid \mathbf{Y},\theta))$. By choosing q cleverly we can shrink this gap; the optimal choice $q = p(\mathcal{Z} \mid \mathbf{Y},\theta)$ closes it entirely, lifting the ELBO up to coincide with the log-likelihood.

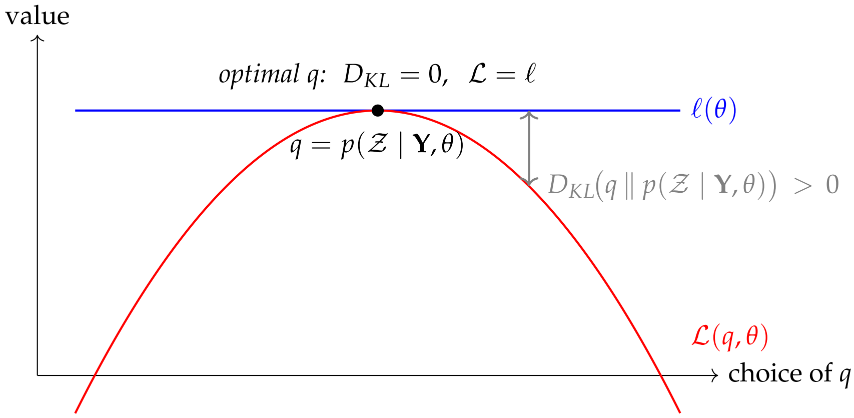


Figure 1: The ELBO as a function of q, at fixed θ. The log-likelihood $\ell(\theta)$ is a horizontal line (independent of q); the ELBO $\mathcal{L}(q,\theta)$ is bounded above by ℓ, with vertical gap equal to $D_{KL}(q \,\|\, p(\mathcal{Z} \mid \mathbf{Y},\theta))$. The bound is tight at the optimal choice $q = p(\mathcal{Z} \mid \mathbf{Y},\theta)$, where the KL gap is zero and the ELBO coincides with the log-likelihood.

This is the geometry of the E-step: *at fixed θ, choose q to push the ELBO up against the log-likelihood as much as possible.* The M-step then exploits this by maximising the (now tight) ELBO over θ, which is equivalent to maximising the log-likelihood itself at the current q. Iterating between these two moves is the EM algorithm.

The upshot is that we are no longer fumbling in the dark when we choose q. The ELBO itself, through its decomposition (24), tells us what the optimal q is.

The Optimisation Problem and Alternating Maximisation

We have set up a surrogate objective function $\mathcal{L}(q,\theta)$ that bounds the true log-likelihood from below and equals it at the "right" q. The natural idea is to maximise $\mathcal{L}(q,\theta)$ in place of the intractable $\ell(\theta)$:

$$\max_{q,\theta} \mathcal{L}(q,\theta).$$

This is a joint maximisation in two "variables" — a distribution $q(\mathcal{Z})$ and a parameter vector θ. A simultaneous optimisation over both is not feasible: the space of admissible distributions q is infinite-dimensional, and $\mathcal{L}$ is a complicated non-convex function of θ. We therefore resort to *alternating maximisation*, also known as *coordinate ascent*. In each iteration we hold one of the two arguments fixed and maximise over the other:

1. **E-step**: fix θ, maximise $\mathcal{L}(q,\theta)$ over q;

2. **M-step**: fix q, maximise $\mathcal{L}(q,\theta)$ over θ;

3. Iterate until convergence.

This yields a sequence of iterates $\theta^{(0)},\theta^{(1)},\theta^{(2)},\ldots$, which — as we discuss at the end of this section — is guaranteed to produce a monotonically non-decreasing sequence of log-likelihood values.

Before moving on to the concrete derivations, it is worth spelling out *what each of these two steps does* in some detail, because the roles of q and θ and the meaning of "holding one fixed" become very concrete once we look closely.

The E-Step (Expectation Step)

Fix $\theta = \theta^{(j)}$, the current parameter estimate, and maximise the ELBO with respect to the auxiliary distribution q:

$$\max_{q} \mathcal{L}(q,\theta^{(j)}) \;=\; \mathbb{E}_q[\log p(\mathbf{Y},\mathcal{Z}\mid\theta^{(j)})] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})].$$

The decomposition (24) we derived in the previous subsection gives us the answer immediately. Recall that

$$\mathcal{L}(q,\theta^{(j)}) \;=\; \ell(\theta^{(j)}) \;-\; D_{KL}\big(q(\mathcal{Z})\,\|\,p(\mathcal{Z}\mid\mathbf{Y},\theta^{(j)})\big).$$

The first term is fixed (we have fixed $\theta^{(j)}$, so $\ell(\theta^{(j)})$ is just a number). The second term is a non-negative quantity that we subtract. Maximising $\mathcal{L}$ over q therefore amounts to *minimising the KL divergence*, and the KL divergence is zero *if and only if* q equals the posterior. Thus:

$$\boxed{q^{(j+1)}(\mathcal{Z}) \;=\; p\big(\mathcal{Z}\mid\mathbf{Y},\theta^{(j)}\big).} \tag{25}$$

The optimal q is the *posterior distribution of the latent variables given the observed data and the current parameters*. Not an approximation, not a surrogate — the exact posterior.

What the E-step actually computes. Knowing that $q^{(j+1)} = p(\mathcal{Z}\mid\mathbf{Y},\theta^{(j)})$ identifies the optimal q but does not by itself give the M-step what it needs. The M-step, which will follow, needs to evaluate the ELBO at this specific choice of q in order to maximise it over θ. What does this require in practice?

The term in the ELBO that depends on θ is the expected complete-data log-likelihood, which in our notation reads

$$L(\theta,\theta^{(j)}) \;\equiv\; \mathbb{E}_{q^{(j+1)}}\big[\log p(\mathbf{Y},\mathcal{Z}\mid\theta)\big] \;=\; \mathbb{E}_{\theta^{(j)}}\big[\log p(\mathbf{Y},\mathcal{Z}\mid\theta)\,\big|\,\mathbf{Y}\big]. \tag{26}$$

We have used two equivalent notations: the first emphasises that the expectation is taken under the distribution computed in the E-step; the second — standard in the EM literature — emphasises that this is a conditional expectation under $\theta^{(j)}$, given the observed data $\mathbf{Y}$. We will use both interchangeably.

The complete-data log-likelihood $\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)$ is a sum of log-Gaussian and log-Gamma densities — in full detail, it will contain terms such as:

- products involving the factors: f_t, $f_t f_t'$, $f_t f_{t-1}'$;
- scalar multiples of these factor products by the weights: $w_t^u \cdot f_t f_{t-1}'$, $w_t^\varepsilon \cdot f_t f_t'$, $w_t^\varepsilon \cdot y_t f_t'$;
- terms in the weights alone: $\log w_t^u$, w_t^u, $\log w_t^\varepsilon$, w_t^ε.

Since the latent quantities $\mathcal{Z}$ are unobserved, we cannot evaluate these terms directly. But now we *do* have a distribution for $\mathcal{Z}$ — the posterior $q^{(j+1)}$ — and we can replace every latent quantity by its *expectation under* $q^{(j+1)}$. Taking these expectations is precisely what the E-step does. Concretely, the E-step produces the following summaries:

Smoothed factor moments:

- $\hat{f}_{t|T}^{(j)} \equiv \mathbb{E}_{\theta^{(j)}}[f_t \mid \mathbf{Y}]$ — smoothed factor means;
- $\mathbb{E}_{\theta^{(j)}}[f_t f_t' \mid \mathbf{Y}]$ — smoothed second moments;
- $\mathbb{E}_{\theta^{(j)}}[f_t f_{t-1}' \mid \mathbf{Y}]$ — smoothed cross-period second moments;
- $\mathbb{E}_{\theta^{(j)}}[f_{t-1} f_{t-1}' \mid \mathbf{Y}]$ — lagged second moments.

Posterior weight expectations:

- $\hat{w}_t^{u,(j)} \equiv \mathbb{E}_{\theta^{(j)}}[w_t^u \mid \mathbf{Y}]$ — posterior mean of the factor-side weight at time t;
- $\hat{w}_t^{\varepsilon,(j)} \equiv \mathbb{E}_{\theta^{(j)}}[w_t^\varepsilon \mid \mathbf{Y}]$ — posterior mean of the idiosyncratic-side weight at time t;
- $\mathbb{E}_{\theta^{(j)}}[\log w_t^u \mid \mathbf{Y}]$, $\mathbb{E}_{\theta^{(j)}}[\log w_t^\varepsilon \mid \mathbf{Y}]$ — posterior expected log-weights (needed for updating ν_u and ν_ε).

Once all of these quantities are computed, $L(\theta, \theta^{(j)})$ is a fully explicit function of θ alone — every latent variable has been replaced by a number — and is ready for the M-step.

How are these quantities actually computed? The computation naturally splits along the latent structure of the model:

- The *smoothed factor moments* are obtained via the **Kalman filter** (a forward recursion giving $p(f_t \mid y_1, \ldots, y_t, \theta^{(j)})$) followed by the **Kalman smoother** (a backward recursion giving $p(f_t \mid \mathbf{Y}, \theta^{(j)})$). The smoother is what lets us use future observations to sharpen our estimate of past factors. These recursions require, as inputs, the current $\theta^{(j)}$ and — crucially for our Student-t setup — the posterior weight expectations as well.
- The *posterior weight expectations* are obtained in closed form through the Gaussian–Gamma conjugacy previewed earlier: conditional on the data and the factors, the posterior of each w is a Gamma distribution, whose mean can be written explicitly.

The two computations are coupled — the Kalman recursions need the weights, the weight posteriors need the factors — and we resolve this coupling by iterating within the E-step (an ECM inner loop), as we shall see.

The M-Step (Maximisation Step)

Fix $q = q^{(j+1)}$, the posterior just computed in the E-step, and maximise the ELBO with respect to the parameter vector θ:

$$\theta^{(j+1)} \;=\; \arg\max_{\theta}\; \mathcal{L}(q^{(j+1)}, \theta).$$

With q held fixed, the ELBO simplifies considerably:

$$\mathcal{L}(q^{(j+1)}, \theta) \;=\; \underbrace{\mathbb{E}_{q^{(j+1)}}[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)]}_{\text{depends on } \theta} \;-\; \underbrace{\mathbb{E}_{q^{(j+1)}}[\log q^{(j+1)}(\mathcal{Z})]}_{\text{does not depend on } \theta}.$$

The second term is the (negative) entropy of $q^{(j+1)}$. Since $q^{(j+1)}$ has been fixed in the E-step and does not depend on θ, the entire entropy term is a constant from the M-step's point of view. Maximising $\mathcal{L}(q^{(j+1)}, \theta)$ over θ is therefore equivalent to maximising just the first term, which is precisely the expected complete-data log-likelihood $L(\theta, \theta^{(j)})$ from (26):

$$\boxed{\theta^{(j+1)} \;=\; \arg\max_{\theta}\; L(\theta, \theta^{(j)}) \;=\; \arg\max_{\theta}\; \mathbb{E}_{\theta^{(j)}}\big[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta) \,\big|\, \mathbf{Y}\big].} \tag{27}$$

What are we actually maximising? The object $\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)$ is the complete-data log-likelihood — the log-likelihood we *would* maximise if we had observed $\mathcal{Z}$ alongside $\mathbf{Y}$. Under our DGP it is a sum of simple log-Gaussian and log-Gamma densities, and in its raw form it is a quadratic function of the factors $\mathbf{F}$ (conditional on the weights W^u, W^ε), plus linear-and-log terms in the weights. If we could see $\mathcal{Z}$, maximising it over θ would be a textbook exercise in closed-form OLS-type formulas.

But we cannot see $\mathcal{Z}$. What the M-step does instead is maximise the *expectation* of this complete-data log-likelihood under the posterior $q^{(j+1)}$: it maximises the "average log-likelihood", averaged over all possible paths of $\mathcal{Z}$, weighted by how likely each path is given the data and $\theta^{(j)}$.

Why is this easier than the original problem? The crucial simplification is that *the expectation has turned every latent quantity into a fixed number*. Wherever the complete-data log-likelihood contained a term like $f_t f_{t-1}'$ or $w_t^\varepsilon \cdot y_t f_t'$, that term has been replaced by its posterior expectation — a concrete scalar or matrix computed in the E-step. Once this substitution is made, the object we maximise in the M-step is a function of θ *alone*, with no latent variables left inside. It is a clean, finite-dimensional optimisation problem.

Better still: the dependence of this function on θ is exactly the same as the dependence of a Gaussian log-likelihood on its parameters, modulo the presence of the posterior weights as multiplicative scaling factors. As a consequence, the M-step will yield *closed-form* updates for $\mathbf{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$, $\mathbf{R}$ — the same OLS-type formulas as in the Gaussian DFM, but with sums over t weighted by the posterior expectations $\hat{w}_t^u$ and $\hat{w}_t^\varepsilon$. Only the updates for the degrees of freedom ν_u and ν_ε will require a one-dimensional numerical root-finding step (they have no closed-form maximiser due to the digamma function arising in the Gamma log-density). We defer all of this later.

Intuition behind the M-step. The M-step answers a transparent question:

> *"Given that the latent variables $\mathcal{Z}$ were distributed as $q^{(j+1)}$, which θ makes the observed data $\mathbf{Y}$ most likely?"*

It is as if we had *observed* the latent variables — but with their specific unknown values replaced by their expected values under our best current guess of their distribution. This is what makes EM tractable: we are alternating between (i) "filling in" the latent variables with their best current guess (E-step), and (ii) estimating θ as if we had seen the filled-in values (M-step).

Summary: E-Step and M-Step Side by Side

The two steps are summarised in Table 1.

Step	Fixed	Optimised	Output
E-step	$\theta = \theta^{(j)}$	$q(\mathcal{Z})$	$q^{(j+1)} = p(\mathcal{Z} \mid \mathbf{Y}, \theta^{(j)})$ *Via:* Kalman filter/smoother (factor moments); Gaussian–Gamma conjugacy (weight expectations).
M-step	$q = q^{(j+1)}$	θ	$\theta^{(j+1)} = \arg\max_\theta \mathbb{E}_{q^{(j+1)}}[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)]$ *Via:* closed-form weighted-OLS for $\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$; numerical root-finding for ν_u, ν_ε.

Table 1: Structure of the EM algorithm for the Student-t DFM. E-step and M-step alternate, each holding one argument of the ELBO fixed and maximising over the other.

Monotonicity and convergence. Each EM iteration increases the ELBO (or leaves it unchanged). This is easy to see: the E-step maximises $\mathcal{L}$ over q with θ fixed, so $\mathcal{L}(q^{(j+1)}, \theta^{(j)}) \geq \mathcal{L}(q^{(j)}, \theta^{(j)})$; the M-step maximises $\mathcal{L}$ over θ with q fixed, so $\mathcal{L}(q^{(j+1)}, \theta^{(j+1)}) \geq \mathcal{L}(q^{(j+1)}, \theta^{(j)})$. Chaining the two,

$$\mathcal{L}(q^{(j+1)}, \theta^{(j+1)}) \;\geq\; \mathcal{L}(q^{(j)}, \theta^{(j)}).$$

Moreover, by the ELBO decomposition (24), *after the E-step* we have $\mathcal{L}(q^{(j+1)}, \theta^{(j)}) = \ell(\theta^{(j)})$ (the KL gap is zero). Combining these two facts, one can show that the log-likelihood itself is monotonically non-decreasing:

$$\ell(\theta^{(j+1)}) \;\geq\; \ell(\theta^{(j)}).$$

This is the crucial property that makes EM a valid algorithm: although we never compute $\ell(\theta)$ directly and never maximise it directly, the values $\ell(\theta^{(0)}), \ell(\theta^{(1)}), \ldots$ form a monotone sequence, and under mild regularity conditions they converge to a (local) maximum of the likelihood.

A geometric reading of the iteration. The argument we have just given admits a useful visual interpretation, illustrated in Figure 2. The blue curve is the true log-likelihood $\ell(\theta)$ — the object we want to maximise but cannot evaluate directly. At the start of iteration j, the E-step constructs a lower bound (the red curve), tangent to $\ell(\theta)$ at the current parameter value $\theta^{(j)}$: the tangency reflects the fact that, after the E-step, the KL gap is zero. The M-step then maximises this tangent lower bound, producing $\theta^{(j+1)}$. At the new parameter value, ℓ is necessarily at least as high as the maximum of the tangent bound, which in turn is at least as high as $\ell(\theta^{(j)})$. Iterating, the EM algorithm climbs the log-likelihood through a sequence of *tangent surrogate maximisations*.

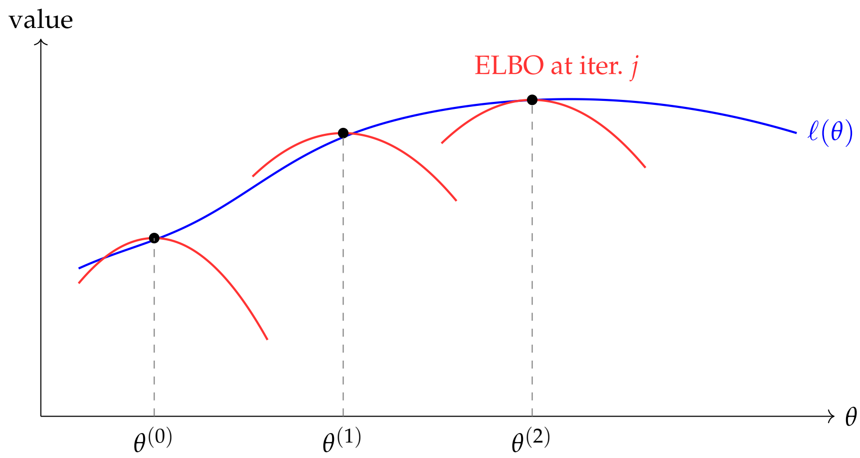


Figure 2: Geometric reading of the EM algorithm. The blue curve is the log-likelihood $\ell(\theta)$. At each iteration, the E-step constructs a tangent lower bound (red), maximising which yields the next parameter estimate. The procedure climbs the log-likelihood through a sequence of tangent surrogate maximisations.

The Bańbura–Modugno perspective. It is worth noting that the logic we have just followed — ELBO, Jensen's inequality, alternating maximisation — is the "variational" view of EM, popular in the machine-learning literature. The econometric tradition following Bańbura and Modugno (2014) presents EM more directly: one simply *defines* the objective $L(\theta, \theta^{(j)}) = \mathbb{E}_{\theta^{(j)}}[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta) \mid \mathbf{Y}]$ and iterates: compute the expectation under $\theta^{(j)}$ (E-step), maximise over θ (M-step). The two perspectives are mathematically equivalent — as we have shown, the ELBO equals $L(\theta, \theta^{(j)})$ up to a constant once q is chosen as the posterior — but the variational derivation has the pedagogical advantage of making the role of q, the meaning of the bound, and the reason for monotonic convergence fully transparent.

Deriving the Complete-Data Log-Likelihood

Setup and Notation

We now assemble the *complete-data log-likelihood* — the object the M-step will maximise once the expectation over the latent variables has been taken. Because the model has more latent structure than the Gaussian DFM, this section is longer than its Gaussian counterpart. Every intermediate step is made explicit.

Recall the full hierarchical system introduced before:

$$\begin{aligned}
&\textbf{State:} && f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \quad u_t \mid w_t^u \sim \mathcal{N}\big(0,\; \mathbf{Q}/w_t^u\big), \quad w_t^u \overset{iid}{\sim} \mathrm{Gamma}\big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\big),\\
&\textbf{Observation:} && y_t \;=\; \mathbf{\Lambda}\, f_t \;+\; \varepsilon_t, \quad \varepsilon_t \mid w_t^\varepsilon \sim \mathcal{N}\big(0,\; \mathbf{R}/w_t^\varepsilon\big), \quad w_t^\varepsilon \overset{iid}{\sim} \mathrm{Gamma}\big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\big),
\end{aligned}$$

with initial state $f_0 \sim \mathcal{N}(0, \mathbf{\Sigma}_0)$.

We can also collect the system in compact matrix form. Stacking columns across time:

$$\underbrace{\mathbf{Y}}_{M\times T} \;=\; \mathbf{\Lambda}\; \underbrace{\mathbf{F}}_{r\times T} \;+\; \underbrace{\mathbf{E}}_{M\times T}, \qquad \underbrace{\mathbf{F}}_{r\times T} \;=\; \mathbf{A}\; \underbrace{\mathbf{F}_{-1}}_{r\times T} \;+\; \underbrace{\mathbf{U}}_{r\times T},$$

We recall the notation fixed so far:

- $\mathbf{Y} = [y_1 \mid \cdots \mid y_T] \in \mathbb{R}^{M\times T}$ — the observed data matrix;
- $\mathbf{F} = [f_1 \mid \cdots \mid f_T] \in \mathbb{R}^{r\times(T+1)}$ — the full latent factor path;
- $\mathbf{F}_{-1} = [f_0 \mid f_1 \mid \cdots \mid f_{T-1}] \in \mathbb{R}^{r\times T}$ — the lagged factor matrix (one-period shift of $\mathbf{F}$);
- $W^u = [w_1^u, \ldots, w_T^u] \in \mathbb{R}^T_{>0}$ — latent weights on the factor side;
- $W^\varepsilon = [w_1^\varepsilon, \ldots, w_T^\varepsilon] \in \mathbb{R}^T_{>0}$ — latent weights on the idiosyncratic side;
- $\theta = \{\mathbf{\Lambda},\ \mathbf{A},\ \mathbf{Q},\ \mathbf{R},\ \nu_u,\ \nu_\varepsilon\}$ — parameter vector;
- $\mathcal{Z} = (\mathbf{F}, W^u, W^\varepsilon)$ — the full latent triple.

The object we want to compute is

$$\ell_c(\theta) \;=\; \log p\big(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta\big).$$

Working assumption: complete data. Throughout this section we assume that the data matrix $\mathbf{Y}$ is fully observed (no missing entries). This assumption simplifies the exposition and lets us focus on the structural ingredients of the complete-data log-likelihood. With missing data, only Block A is formally affected — the observation equation is restricted to the observed entries via a selection matrix $\mathbf{W}_t$ — but this modification propagates through the Kalman filter, the weight update, and the row-by-row M-step formulas for $\mathbf{\Lambda}$ and $\mathbf{R}$. The full treatment deferred in another section.

Factoring the Joint Distribution

Using the product rule of probability, the joint density factorises into five conditional blocks, one per DGP equation:

$$
\begin{aligned}
p\big(\mathbf{Y},\mathbf{F},W^u,W^\varepsilon \mid \theta\big) \;=\; & \underbrace{p(\mathbf{Y}\mid \mathbf{F},W^\varepsilon,\theta)}_{\text{Block A: observation}} \cdot \underbrace{p(\mathbf{F}\mid W^u,\theta)}_{\text{Blocks B and C: transitions + initial state}} \\
& \cdot \underbrace{p(W^u\mid\theta)}_{\text{Block D: Gamma prior on } W^u} \cdot \underbrace{p(W^\varepsilon\mid\theta)}_{\text{Block E: Gamma prior on } W^\varepsilon} .
\end{aligned}
$$

The factorisation reflects the causal structure of the DGP: nature first draws the weights (which depend only on θ, via the Gamma priors), then draws the factors (whose conditional distribution given W^u and θ is Gaussian), then finally draws the observations (Gaussian given the factors and W^ε). Below we work out each block separately.

Block A — Observation density. Conditional on the factor f_t and the idiosyncratic weight w_t^ε, the observation y_t is Gaussian:

$$y_t \mid f_t, w_t^\varepsilon \;\sim\; \mathcal{N}\big(\mathbf{\Lambda} f_t,\ \mathbf{R}/w_t^\varepsilon\big).$$

The mean $\mathbf{\Lambda} f_t$ is the systematic component; the matrix $\mathbf{R}/w_t^\varepsilon$ is the *conditional covariance matrix* of y_t given f_t and w_t^ε. Note carefully: *inside the scale-mixture representation*, conditionally on the weight, this matrix is a genuine covariance — the Gaussian $\mathcal{N}(\mathbf{\Lambda} f_t, \mathbf{R}/w_t^\varepsilon)$ has mean $\mathbf{\Lambda} f_t$ and variance $\mathbf{R}/w_t^\varepsilon$ in the usual sense. The distinction between scale matrix and covariance matrix emphasised before applies to the *marginal* distribution of ε_t after integrating the weight out: marginally, $\varepsilon_t \sim t(0,\mathbf{R},\nu_\varepsilon)$ and the marginal covariance is $\frac{\nu_\varepsilon}{\nu_\varepsilon-2}\mathbf{R}$, not $\mathbf{R}$. Here, working within the hierarchical representation, we are always in the conditional world and $\mathbf{R}/w_t^\varepsilon$ is the covariance matrix to be used.

A small realisation of w_t^ε inflates the conditional covariance and produces a diffuse Gaussian — the mechanism through which heavy tails emerge when w_t^ε is integrated out. Conditional on $(\mathbf{F}, W^\varepsilon)$, observations at different times are independent, so

$$p(\mathbf{Y}\mid \mathbf{F},W^\varepsilon,\theta) \;=\; \prod_{t=1}^{T} p(y_t \mid f_t, w_t^\varepsilon, \theta).$$

Block B — Factor transition density. Conditional on the previous factor f_{t-1} and the factor weight w_t^u, the current factor is Gaussian:

$$f_t \mid f_{t-1}, w_t^u \;\sim\; \mathcal{N}\big(\mathbf{A} f_{t-1},\ \mathbf{Q}/w_t^u\big).$$

As in Block A, $\mathbf{Q}/w_t^u$ is the *conditional covariance matrix* of f_t within the hierarchical representation — not the scale matrix. The scale matrix $\mathbf{Q}$ appears instead in the marginal distribution $u_t \sim t(0,\mathbf{Q},\nu_u)$, with marginal covariance $\frac{\nu_u}{\nu_u-2}\mathbf{Q}$.

By the Markov property of the VAR(1) (conditional on the weights, f_t depends on the past only through f_{t-1}) and the chain rule,

$$p(\mathbf{F}\mid W^u,\theta) \;=\; p(f_0\mid\theta)\prod_{t=1}^{T} p(f_t \mid f_{t-1}, w_t^u, \theta).$$

Block C — Initial state density. The recursion requires a distribution for f_0; we adopt

$$f_0 \sim \mathcal{N}(0, \mathbf{\Sigma}_0),$$

with $\mathbf{\Sigma}_0$ the initial-state covariance. This block is not weighted: f_0 pre-dates the dynamic system and is not touched by the heavy-tailed innovations. As a consequence, the contribution of Block C to the complete-data log-likelihood is identical to its Gaussian counterpart, and its contribution to the M-step updates is independent of ν_u and ν_ε.

Block D — Gamma prior on W^u. The factor-side weights are *iid* Gamma across time:

$$p(W^u \mid \theta) \;=\; \prod_{t=1}^{T} p(w_t^u \mid \theta), \qquad w_t^u \sim \text{Gamma}(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}).$$

Note the dependence on θ: this block contains ν_u, and is the only block — together with Block E — in which ν_u and ν_ε appear. This fact will be crucial for the update of the degrees of freedom in the M-step.

Block E — Gamma prior on W^ε. Analogously for the idiosyncratic side:

$$p(W^\varepsilon \mid \theta) \;=\; \prod_{t=1}^{T} p(w_t^\varepsilon \mid \theta), \qquad w_t^\varepsilon \sim \text{Gamma}(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}).$$

Computing Each Block

We now take the logarithm of each block, applying the multivariate Gaussian log-density for Blocks A, B, C and the Gamma log-density for Blocks D and E. For reference, the generic multivariate Gaussian $X \sim \mathcal{N}(\mu, \Sigma)$ in m dimensions has density

$$p(x) \;=\; \frac{1}{(2\pi)^{m/2}\,|\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\Big),$$

and log-density

$$\log p(x) \;=\; -\tfrac{m}{2}\log(2\pi) - \tfrac{1}{2}\log|\Sigma| - \tfrac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu).$$

The three terms have a clear interpretation:

- $-\frac{m}{2}\log(2\pi)$: a normalising constant, depending only on the dimension m — never on the parameters;
- $-\frac{1}{2}\log|\Sigma|$: penalises large determinants — a very spread-out distribution is penalised;
- $-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)$: penalises observations far from the mean μ, in the Mahalanobis sense.

The generic Gamma(α, β) with shape $\alpha > 0$ and rate $\beta > 0$ has density

$$p(w) \;=\; \frac{\beta^\alpha}{\Gamma(\alpha)}\, w^{\alpha-1}\, e^{-\beta w}, \qquad w > 0,$$

and log-density

$$\log p(w) \;=\; \alpha\log\beta - \log\Gamma(\alpha) + (\alpha-1)\log w - \beta w.$$

The complete-data log-likelihood as a sum. Taking the logarithm of the joint factorisation turns the product of the five blocks into a sum of five log-densities:

$$\begin{aligned}\ell_c(\theta) &= \log p(\mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon \mid \theta) \\ &= \underbrace{\log p(\mathbf{Y} \mid \mathbf{F}, W^\varepsilon, \theta)}_{\text{Block A}} + \underbrace{\log p(\mathbf{F} \mid W^u, \theta)}_{\text{Blocks B and C}} \\ &\quad + \underbrace{\log p(W^u \mid \theta)}_{\text{Block D}} + \underbrace{\log p(W^\varepsilon \mid \theta)}_{\text{Block E}}.\end{aligned}$$

We now apply the log-density formulae above to each of the five blocks.

Block A — Observation Equation, M Variables

We have $y_t \mid f_t, w_t^\varepsilon \sim \mathcal{N}(\mathbf{\Lambda} f_t,\, \mathbf{R}/w_t^\varepsilon)$ where:

- y_t is $M \times 1$, so $m = M$;
- the mean is $\mu = \mathbf{\Lambda} f_t$ ($M \times 1$);
- the conditional covariance is $\Sigma = \mathbf{R}/w_t^\varepsilon$ ($M \times M$, diagonal, scaled by the weight. Note that the diagonality of $\mathbf{R}$ is preserved by the weighting: $\mathbf{R}/w_t^\varepsilon$ is also diagonal for every t, with diagonal entries r_i/w_t^ε for $i = 1, \ldots, M$. The inverse $(\mathbf{R}/w_t^\varepsilon)^{-1}$ is therefore obtained simply as $\operatorname{diag}(w_t^\varepsilon/r_1, \ldots, w_t^\varepsilon/r_M)$. This sparsity pattern propagates through the entire algorithm: each M-step update preserves $\mathbf{R}$ diagonal by construction, with no constraint being imposed during optimisation).

Plug into the multivariate normal formula.

$$p(y_t \mid f_t, w_t^\varepsilon, \theta) \;=\; \frac{1}{(2\pi)^{M/2} \left|\mathbf{R}/w_t^\varepsilon\right|^{1/2}} \exp\Big(-\tfrac{1}{2}(y_t - \mathbf{\Lambda} f_t)' \,(\mathbf{R}/w_t^\varepsilon)^{-1}\, (y_t - \mathbf{\Lambda} f_t)\Big).$$

Taking logs.

$$\log p(y_t \mid f_t, w_t^\varepsilon, \theta) \;=\; -\tfrac{M}{2}\log(2\pi) \;-\; \tfrac{1}{2}\log\left|\mathbf{R}/w_t^\varepsilon\right| \;-\; \tfrac{1}{2}(y_t - \mathbf{\Lambda} f_t)' \,(\mathbf{R}/w_t^\varepsilon)^{-1}\, (y_t - \mathbf{\Lambda} f_t).$$

Simplify the weight-dependent pieces. Two identities from linear algebra allow us to isolate the weight. For the log-determinant,

$$\log\left|\mathbf{R}/w_t^\varepsilon\right| \;=\; \log\left|(w_t^\varepsilon)^{-1}\,\mathbf{R}\right| \;=\; -M \log w_t^\varepsilon \;+\; \log|\mathbf{R}|,$$

where we used $|c \cdot A| = c^m\, |A|$ for a scalar c and an $m \times m$ matrix A. For the inverse,

$$(\mathbf{R}/w_t^\varepsilon)^{-1} \;=\; w_t^\varepsilon \cdot \mathbf{R}^{-1}.$$

Substituting both into the log-density:

$$\log p(y_t \mid f_t, w_t^\varepsilon, \theta) \;=\; -\tfrac{M}{2}\log(2\pi) \;+\; \tfrac{M}{2}\log w_t^\varepsilon \;-\; \tfrac{1}{2}\log|\mathbf{R}| \;-\; \tfrac{w_t^\varepsilon}{2}\,(y_t - \mathbf{\Lambda} f_t)'\mathbf{R}^{-1}(y_t - \mathbf{\Lambda} f_t).$$

The weight has come out in two places: as an additive term $+\frac{M}{2}\log w_t^\varepsilon$ (from the log-determinant) and as a multiplicative factor on the Mahalanobis quadratic form.

Sum over $t = 1, \ldots, T$**.** By conditional independence over time given $(\mathbf{F}, W^{\varepsilon})$,

$$\log p(\mathbf{Y} \mid \mathbf{F}, W^{\varepsilon}, \theta) \;=\; \sum_{t=1}^{T} \log p(y_t \mid f_t, w_t^{\varepsilon}, \theta).$$

Expanding the sum:

$$= \sum_{t=1}^{T} \Big[-\tfrac{M}{2} \log(2\pi) \;+\; \tfrac{M}{2} \log w_t^{\varepsilon} \;-\; \tfrac{1}{2} \log |\mathbf{R}| \;-\; \tfrac{w_t^{\varepsilon}}{2} (y_t - \boldsymbol{\Lambda} f_t)' \mathbf{R}^{-1} (y_t - \boldsymbol{\Lambda} f_t) \Big]$$

$$= -\tfrac{TM}{2} \log(2\pi) \;+\; \tfrac{M}{2} \sum_{t=1}^{T} \log w_t^{\varepsilon} \;-\; \tfrac{T}{2} \log |\mathbf{R}| \;-\; \tfrac{1}{2} \sum_{t=1}^{T} w_t^{\varepsilon} \, (y_t - \boldsymbol{\Lambda} f_t)' \mathbf{R}^{-1} (y_t - \boldsymbol{\Lambda} f_t).$$

Drop the constant and box the result. The term $-\frac{TM}{2} \log(2\pi)$ does not depend on any parameter in θ nor on any latent variable, so we drop it:

$$\boxed{\text{Block A} \;=\; -\tfrac{T}{2} \log |\mathbf{R}| \;+\; \tfrac{M}{2} \sum_{t=1}^{T} \log w_t^{\varepsilon} \;-\; \tfrac{1}{2} \sum_{t=1}^{T} w_t^{\varepsilon} \, (y_t - \boldsymbol{\Lambda} f_t)' \mathbf{R}^{-1} (y_t - \boldsymbol{\Lambda} f_t) \;+\; \text{const.}} \tag{28}$$

Three observations are worth making on (28):

- The determinant term $-\frac{T}{2} \log |\mathbf{R}|$ is identical to the Gaussian case — it penalises a large idiosyncratic scale matrix.
- The new term $+\frac{M}{2} \sum_t \log w_t^{\varepsilon}$ contains only the weights, not the structural parameters, and will contribute to the update of ν_{ε} together with the corresponding term in Block E.
- The Mahalanobis quadratic is now *weighted* by w_t^{ε}. This is the technical origin of the "weighted OLS" structure of the M-step: observations at time periods with small w_t^{ε} (extreme y_t's, in the sense of the model) contribute less to the parameter updates. Robustness emerges automatically.

Block B — Factor Transition Equation, r Factors

We have $f_t \mid f_{t-1}, w_t^u \sim \mathcal{N}(\mathbf{A} f_{t-1}, \, \mathbf{Q}/w_t^u)$ where:

- f_t is $r \times 1$, so $m = r$;
- the mean is $\mu = \mathbf{A} f_{t-1}$ $(r \times 1)$;
- the conditional covariance is $\Sigma = \mathbf{Q}/w_t^u$ ($r \times r$, not necessarily diagonal, scaled by the weight).

Plug into the multivariate normal formula.

$$p(f_t \mid f_{t-1}, w_t^u, \theta) \;=\; \frac{1}{(2\pi)^{r/2} \left| \mathbf{Q}/w_t^u \right|^{1/2}} \exp\Big(-\tfrac{1}{2} (f_t - \mathbf{A} f_{t-1})' \, (\mathbf{Q}/w_t^u)^{-1} \, (f_t - \mathbf{A} f_{t-1}) \Big).$$

Taking logs.

$$\log p(f_t \mid f_{t-1}, w_t^u, \theta) \;=\; -\tfrac{r}{2} \log(2\pi) \;-\; \tfrac{1}{2} \log \left| \mathbf{Q}/w_t^u \right| \;-\; \tfrac{1}{2} (f_t - \mathbf{A} f_{t-1})' \, (\mathbf{Q}/w_t^u)^{-1} \, (f_t - \mathbf{A} f_{t-1}).$$

Simplify the weight-dependent pieces. Exactly as in Block A, $|\mathbf{Q}/w_t^u| = (w_t^u)^{-r}\,|\mathbf{Q}|$, so $\log|\mathbf{Q}/w_t^u| = -r\log w_t^u + \log|\mathbf{Q}|$, and $(\mathbf{Q}/w_t^u)^{-1} = w_t^u\,\mathbf{Q}^{-1}$. Substituting:

$$\log p(f_t \mid f_{t-1}, w_t^u, \theta) \;=\; -\tfrac{r}{2}\log(2\pi) \;+\; \tfrac{r}{2}\log w_t^u \;-\; \tfrac{1}{2}\log|\mathbf{Q}| \;-\; \tfrac{w_t^u}{2}(f_t-\mathbf{A}f_{t-1})'\mathbf{Q}^{-1}(f_t-\mathbf{A}f_{t-1}).$$

Sum over $t = 1, \ldots, T$. By the Markov property (conditional on $(\mathbf{F}_{-1}, W^u)$, each transition is independent):

$$\log p(\mathbf{F} \mid \mathbf{F}_{-1}, W^u, \theta) \;=\; \sum_{t=1}^{T} \log p(f_t \mid f_{t-1}, w_t^u, \theta).$$

Expanding the sum:

$$= \sum_{t=1}^{T}\Big[-\tfrac{r}{2}\log(2\pi) \;+\; \tfrac{r}{2}\log w_t^u \;-\; \tfrac{1}{2}\log|\mathbf{Q}| \;-\; \tfrac{w_t^u}{2}(f_t-\mathbf{A}f_{t-1})'\mathbf{Q}^{-1}(f_t-\mathbf{A}f_{t-1})\Big]$$

$$= -\tfrac{Tr}{2}\log(2\pi) \;+\; \tfrac{r}{2}\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{T}{2}\log|\mathbf{Q}| \;-\; \tfrac{1}{2}\sum_{t=1}^{T} w_t^u\,(f_t-\mathbf{A}f_{t-1})'\mathbf{Q}^{-1}(f_t-\mathbf{A}f_{t-1}).$$

Drop the constant and box the result.

$$\boxed{\text{Block B} \;=\; -\tfrac{T}{2}\log|\mathbf{Q}| \;+\; \tfrac{r}{2}\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{1}{2}\sum_{t=1}^{T} w_t^u\,(f_t-\mathbf{A}f_{t-1})'\mathbf{Q}^{-1}(f_t-\mathbf{A}f_{t-1}) \;+\; \text{const.}} \tag{29}$$

The structural parallel with Block A is perfect: a determinant penalty on $\mathbf{Q}$, a linear-in-log w_t^u term contributing to the ν_u update, and a quadratic form weighted by w_t^u.

Block C — Initial State, r Factors

We specify $f_0 \sim \mathcal{N}(0, \mathbf{\Sigma}_0)$ where:

- f_0 is $r \times 1$, so $m = r$;
- the mean is $\mu = 0$;
- the covariance is $\mathbf{\Sigma}_0$ ($r \times r$).

This block is not weighted: the initial state pre-dates the dynamic system and does not interact with the heavy-tailed innovations. It is therefore structurally identical to the Gaussian case.

Plug into the formula.

$$p(f_0 \mid \theta) \;=\; \frac{1}{(2\pi)^{r/2}\,|\mathbf{\Sigma}_0|^{1/2}}\exp\Big(-\tfrac{1}{2}\,f_0'\,\mathbf{\Sigma}_0^{-1}\,f_0\Big).$$

Taking logs.

$$\log p(f_0 \mid \theta) \;=\; -\tfrac{r}{2}\log(2\pi) \;-\; \tfrac{1}{2}\log|\mathbf{\Sigma}_0| \;-\; \tfrac{1}{2}\,f_0'\,\mathbf{\Sigma}_0^{-1}\,f_0.$$

Drop the constant and box.

$$\boxed{\text{Block C} \;=\; -\tfrac{1}{2}\log|\mathbf{\Sigma}_0| \;-\; \tfrac{1}{2}\,f_0'\,\mathbf{\Sigma}_0^{-1}\,f_0 \;+\; \text{const.}} \tag{30}$$

Being a single density (one time period only, not a sum over t), this block contributes a negligible amount to the log-likelihood when T is large. In practice, the role of Block C is mainly to initialise the Kalman filter recursions; for $T \to \infty$ the parameter estimates become independent of the initial condition.

Block D — Gamma Prior on W^u, T Weights

We specify $w_t^u \overset{iid}{\sim} \text{Gamma}(\nu_u/2,\ \nu_u/2)$ where $\alpha = \beta = \nu_u/2$ for each t.

Plug into the Gamma formula.

$$p(w_t^u \mid \theta) \;=\; \frac{(\nu_u/2)^{\nu_u/2}}{\Gamma(\nu_u/2)}\,(w_t^u)^{\nu_u/2-1}\,e^{-(\nu_u/2)\,w_t^u}, \qquad w_t^u > 0.$$

Taking logs.

$$\log p(w_t^u \mid \theta) \;=\; \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\big(\tfrac{\nu_u}{2}\big) \;+\; \big(\tfrac{\nu_u}{2}-1\big)\log w_t^u \;-\; \tfrac{\nu_u}{2}\,w_t^u.$$

Sum over $t = 1, \dots, T$. By independence of the weights:

$$\begin{aligned}
\log p(W^u \mid \theta) \;&=\; \sum_{t=1}^{T} \log p(w_t^u \mid \theta) \\
&=\; \sum_{t=1}^{T} \Big[\, \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\big(\tfrac{\nu_u}{2}\big) \;+\; \big(\tfrac{\nu_u}{2}-1\big)\log w_t^u \;-\; \tfrac{\nu_u}{2}\,w_t^u \,\Big] \\
&=\; T\Big[\, \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\big(\tfrac{\nu_u}{2}\big)\,\Big] \;+\; \big(\tfrac{\nu_u}{2}-1\big)\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{\nu_u}{2}\sum_{t=1}^{T} w_t^u.
\end{aligned}$$

Box the result. There is no additive constant to drop here: all terms depend either on ν_u (the parameter) or on the latent weights.

$$\boxed{\begin{aligned}
\text{Block D} \;=\; & T\Big[\, \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\big(\tfrac{\nu_u}{2}\big)\,\Big] \\
& +\; \big(\tfrac{\nu_u}{2}-1\big)\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{\nu_u}{2}\sum_{t=1}^{T} w_t^u.
\end{aligned}} \tag{31}$$

The structure of Block D is important and worth anatomising in detail. Three types of terms appear:

- A term depending on ν_u but *not* on the weights: $T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma(\tfrac{\nu_u}{2})\big]$. This term will be entirely responsible for the (lack of) closed-form update of ν_u in the M-step: the $\log\Gamma$ means that its derivative involves the *digamma function* $\psi(\cdot)$, and setting the derivative to zero yields an implicit equation in ν_u that must be solved numerically.

- A term linear in $\sum_t \log w_t^u$, multiplied by $(\nu_u/2 - 1)$. Taking the E-step expectation replaces $\log w_t^u$ by its posterior mean $\mathbb{E}[\log w_t^u \mid \mathbf{Y}, \theta^{(j)}]$, a quantity we will compute in closed form (Gaussian–Gamma conjugacy).

- A term linear in $\sum_t w_t^u$, multiplied by $-\nu_u/2$. Taking the E-step expectation replaces w_t^u by $\hat{w}_t^{u,(j)} = \mathbb{E}[w_t^u \mid \mathbf{Y}, \theta^{(j)}]$.

Notice that Block D contains *only* ν_u and the weights W^u. It does not contain $\boldsymbol{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$, nor any of the idiosyncratic-side objects. This cleanly decouples the update of ν_u from the updates of the other parameters in the M-step: when maximising over $\boldsymbol{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$, the entire Block D is an irrelevant additive constant; when maximising over ν_u, Blocks A, B, C, and E are irrelevant constants (the weights in Blocks A and B, once taken in expectation, do not depend on ν_u themselves).

Block E — Gamma Prior on W^{ε}, T Weights

By direct analogy with Block D, substituting $\nu_u \to \nu_\varepsilon$ and $w_t^u \to w_t^\varepsilon$.

Plug into the Gamma formula.

$$p(w_t^\varepsilon \mid \theta) \;=\; \frac{(\nu_\varepsilon/2)^{\nu_\varepsilon/2}}{\Gamma(\nu_\varepsilon/2)}\,(w_t^\varepsilon)^{\nu_\varepsilon/2-1}\,e^{-(\nu_\varepsilon/2)\,w_t^\varepsilon}, \qquad w_t^\varepsilon > 0.$$

Take logs and sum. Going through the same algebra as Block D:

$$\boxed{\begin{aligned}\text{Block E} \;&=\; T\left[\tfrac{\nu_\varepsilon}{2}\log\tfrac{\nu_\varepsilon}{2} \;-\; \log\Gamma\!\left(\tfrac{\nu_\varepsilon}{2}\right)\right] \\ &\quad+\;\left(\tfrac{\nu_\varepsilon}{2}-1\right)\sum_{t=1}^{T}\log w_t^\varepsilon \;-\; \tfrac{\nu_\varepsilon}{2}\sum_{t=1}^{T} w_t^\varepsilon.\end{aligned}} \tag{32}$$

All observations made about Block D transfer directly, with the subscript ε replacing u: the decoupling of ν_ε from the structural parameters, the $\log\Gamma$ term implying numerical root-finding, and the linear dependence on $\sum_t w_t^\varepsilon$ and $\sum_t \log w_t^\varepsilon$.

The Trace Trick with Weights

Before assembling the complete log-likelihood we recall a useful identity. The *trace* of a matrix A is the sum of its diagonal entries, $\mathrm{tr}(A) = \sum_i a_{ii}$. For any vector x and square matrix M,

$$x'\,M\,x \;=\; \mathrm{tr}(Mxx').$$

This "trace trick" is useful for two reasons. First, the trace is a linear operator, so it commutes with expectations:

$$\mathbb{E}[x'Mx] \;=\; \mathbb{E}[\mathrm{tr}(Mxx')] \;=\; \mathrm{tr}(M\,\mathbb{E}[xx']).$$

This will let us replace the latent outer product xx' inside the trace with its posterior expectation — exactly what the E-step needs. Second, the trace is additive over sums, so $\sum_t \mathrm{tr}(A_t) = \mathrm{tr}(\sum_t A_t)$.

Weighted case. In our setting, each summand in Blocks A and B is multiplied by a scalar weight (w_t^ε or w_t^u). Since scalars commute with everything, the weighted version of the trace trick is:

$$\sum_{t=1}^{T} w_t^\varepsilon\,(y_t - \mathbf{\Lambda} f_t)'\mathbf{R}^{-1}(y_t - \mathbf{\Lambda} f_t) \;=\; \mathrm{tr}\!\left[\mathbf{R}^{-1}\sum_{t=1}^{T} w_t^\varepsilon\,(y_t - \mathbf{\Lambda} f_t)(y_t - \mathbf{\Lambda} f_t)'\right],$$

$$\sum_{t=1}^{T} w_t^u\,(f_t - \mathbf{A} f_{t-1})'\mathbf{Q}^{-1}(f_t - \mathbf{A} f_{t-1}) \;=\; \mathrm{tr}\!\left[\mathbf{Q}^{-1}\sum_{t=1}^{T} w_t^u\,(f_t - \mathbf{A} f_{t-1})(f_t - \mathbf{A} f_{t-1})'\right].$$

The weights enter the sum of outer products and stay there. This is the form of the quadratic terms we will use throughout the M-step.

Complete-Data Log-Likelihood: Final Form

Combining Blocks A through E, we obtain the full complete-data log-likelihood. Using the weighted trace trick on the quadratic terms in Blocks A and B, and dropping all additive constants independent of θ and of the latent variables:

$$\boxed{\begin{aligned}
\ell_c(\theta) \;=\; & -\tfrac{1}{2}\log|\mathbf{\Sigma}_0| \;-\; \tfrac{1}{2}\, f_0'\,\mathbf{\Sigma}_0^{-1}\, f_0 \\
& -\tfrac{T}{2}\log|\mathbf{Q}| \;+\; \tfrac{r}{2}\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{1}{2}\,\mathrm{tr}\!\left[\mathbf{Q}^{-1}\sum_{t=1}^{T} w_t^u\,(f_t-\mathbf{A}f_{t-1})(f_t-\mathbf{A}f_{t-1})'\right] \\
& -\tfrac{T}{2}\log|\mathbf{R}| \;+\; \tfrac{M}{2}\sum_{t=1}^{T}\log w_t^\varepsilon \;-\; \tfrac{1}{2}\,\mathrm{tr}\!\left[\mathbf{R}^{-1}\sum_{t=1}^{T} w_t^\varepsilon\,(y_t-\mathbf{\Lambda}f_t)(y_t-\mathbf{\Lambda}f_t)'\right] \\
& +\,T\left[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\!\left(\tfrac{\nu_u}{2}\right)\right] \;+\; \left(\tfrac{\nu_u}{2}-1\right)\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{\nu_u}{2}\sum_{t=1}^{T} w_t^u \\
& +\,T\left[\tfrac{\nu_\varepsilon}{2}\log\tfrac{\nu_\varepsilon}{2} \;-\; \log\Gamma\!\left(\tfrac{\nu_\varepsilon}{2}\right)\right] \;+\; \left(\tfrac{\nu_\varepsilon}{2}-1\right)\sum_{t=1}^{T}\log w_t^\varepsilon \;-\; \tfrac{\nu_\varepsilon}{2}\sum_{t=1}^{T} w_t^\varepsilon.
\end{aligned}} \tag{33}$$

This is the complete-data log-likelihood of the Student-t DFM. It contains five groups of terms: one from each of the five blocks A–E. The Gaussian DFM is recovered as a limiting case: when $\nu_u, \nu_\varepsilon \to \infty$, all weights collapse to $w_t^u = w_t^\varepsilon = 1$, the $\log w$ terms vanish, and the ν-dependent terms in Blocks D and E become constants that drop out — leaving exactly the Gaussian expression familiar from the standard DFM.

Identifying the Blocks of Parameters

Before moving to the E-step, it is helpful to group the terms of (33) by the parameter they depend on. This decomposition will be directly reflected in the M-step, where we will maximise each group separately.

Parameter	Relevant terms in $\ell_c(\theta)$
$\mathbf{\Sigma}_0$	$-\frac{1}{2}\log|\mathbf{\Sigma}_0| - \frac{1}{2}f_0'\mathbf{\Sigma}_0^{-1}f_0$ (Block C)
$\mathbf{A}$	Quadratic in Block B: $-\frac{1}{2}\mathrm{tr}\big[\mathbf{Q}^{-1}\sum_t w_t^u(f_t-\mathbf{A}f_{t-1})(f_t-\mathbf{A}f_{t-1})'\big]$
$\mathbf{Q}$	Block B log-det and quadratic
$\mathbf{\Lambda}$	Quadratic in Block A: $-\frac{1}{2}\mathrm{tr}\big[\mathbf{R}^{-1}\sum_t w_t^\varepsilon(y_t-\mathbf{\Lambda}f_t)(y_t-\mathbf{\Lambda}f_t)'\big]$
$\mathbf{R}$	Block A log-det and quadratic
ν_u	Block D: $T\big[\frac{\nu_u}{2}\log\frac{\nu_u}{2} - \log\Gamma\big(\frac{\nu_u}{2}\big)\big] + \frac{\nu_u}{2}\sum_t(\log w_t^u - w_t^u) - \sum_t \log w_t^u$
ν_ε	Block E: $T\big[\frac{\nu_\varepsilon}{2}\log\frac{\nu_\varepsilon}{2} - \log\Gamma\big(\frac{\nu_\varepsilon}{2}\big)\big] + \frac{\nu_\varepsilon}{2}\sum_t(\log w_t^\varepsilon - w_t^\varepsilon) - \sum_t \log w_t^\varepsilon$

Two features of this decomposition are worth stressing:

- The updates of $\mathbf{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$, $\mathbf{R}$ will be *weighted OLS*: identical in structure to the Gaussian case, except that the sum over t carries multiplicative factors $\hat{w}_t^\varepsilon$ or $\hat{w}_t^u$ — the posterior weight expectations from the E-step.

- The updates of ν_u, ν_ε are entirely *decoupled* from the other parameters. Blocks D and E contain only the degrees of freedom and the weights. As we shall see, the first-order conditions for ν involve the digamma function and therefore do not admit a closed form, requiring a numerical root-finding step. This is the ECME feature of the algorithm — expectation-conditional-maximisation extended.

The Expected Complete-Data Log-Likelihood

As established in the discussion of alternating maximisation, the M-step cannot directly maximise the complete-data log-likelihood $\ell_c(\theta)$ because it depends on the latent variables $\mathcal{Z} = \{\mathbf{F}, W^u, W^\varepsilon\}$, which are not observed. To move forward, we must resolve this dependency by taking a conditional expectation.

We define our surrogate objective function, $\Xi(\theta \mid \theta^{(j)})$, as the expectation of the complete-data log-likelihood under the posterior distribution computed at the current parameter estimate $\theta^{(j)}$:

$$\Xi(\theta \mid \theta^{(j)}) \;\equiv\; \mathbb{E}_{\theta^{(j)}}\big[\ell_c(\theta) \mid \mathbf{Y}\big]. \tag{34}$$

By the linearity of the expectation operator, $\Xi(\theta \mid \theta^{(j)})$ is simply the sum of the expectations of the five blocks (A–E) identified in the previous section. Crucially, since $\ell_c(\theta)$ is a linear-in-parameters function of the latent quantities, the expectation operator can be pushed inside the summations.

This transformation effectively "linearises" the latent structure: every unobserved term (whether a factor, a weight, or their product) is replaced by its best estimate given the available data. Maximising Ξ is thus equivalent to performing a series of weighted regressions where the "missing variables" has been filled in by its posterior moments.

$$\boxed{\begin{aligned}
\Xi(\theta \mid \theta^{(j)}) \;=\; & -\tfrac{1}{2}\log|\mathbf{\Sigma}_0| \;-\; \tfrac{1}{2}\operatorname{tr}\big(\mathbf{\Sigma}_0^{-1}\mathbb{E}_{\theta^{(j)}}[f_0 f_0' \mid \mathbf{Y}]\big) \\
& -\tfrac{T}{2}\log|\mathbf{Q}| \;+\; \tfrac{r}{2}\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[\log w_t^u \mid \mathbf{Y}] \\
& -\tfrac{1}{2}\operatorname{tr}\Bigg[\mathbf{Q}^{-1}\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[w_t^u\,(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})' \mid \mathbf{Y}]\Bigg] \\
& -\tfrac{T}{2}\log|\mathbf{R}| \;+\; \tfrac{M}{2}\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[\log w_t^\varepsilon \mid \mathbf{Y}] \\
& -\tfrac{1}{2}\operatorname{tr}\Bigg[\mathbf{R}^{-1}\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[w_t^\varepsilon\,(y_t - \mathbf{\Lambda}f_t)(y_t - \mathbf{\Lambda}f_t)' \mid \mathbf{Y}]\Bigg] \\
& + T\left[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} \;-\; \log\Gamma\big(\tfrac{\nu_u}{2}\big)\right] \;+\; \big(\tfrac{\nu_u}{2}-1\big)\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[\log w_t^u \mid \mathbf{Y}] \;-\; \tfrac{\nu_u}{2}\sum_{t=1}^{T}\hat{w}_t^{u,(j)} \\
& + T\left[\tfrac{\nu_\varepsilon}{2}\log\tfrac{\nu_\varepsilon}{2} \;-\; \log\Gamma\big(\tfrac{\nu_\varepsilon}{2}\big)\right] \;+\; \big(\tfrac{\nu_\varepsilon}{2}-1\big)\sum_{t=1}^{T}\mathbb{E}_{\theta^{(j)}}[\log w_t^\varepsilon \mid \mathbf{Y}] \;-\; \tfrac{\nu_\varepsilon}{2}\sum_{t=1}^{T}\hat{w}_t^{\varepsilon,(j)}.
\end{aligned}} \tag{35}$$

This expression clarifies the "step forward" mentioned earlier: every unobserved quantity has been replaced by a numerical summary produced by the E-step. Maximising Ξ is now a deterministic optimisation problem where the parameters $\theta = \{\mathbf{A}, \mathbf{\Lambda}, \mathbf{Q}, \mathbf{R}, \nu_u, \nu_\varepsilon\}$ are the only variables.

Preview of the E-Step

We have just established that $\Xi(\theta \mid \theta^{(j)})$ requires posterior expectations of every latent quantity appearing in $\ell_c(\theta)$. Anticipating what the next section derives in detail, we record here the list of objects the E-step must produce:

- **Factor moments**, for the Blocks A, B, C terms: $\mathbb{E}[f_t \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[f_t f_t' \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[f_0 f_0' \mid \mathbf{Y}, \theta^{(j)}]$.
- **Posterior weight means**, for the w-multiplied terms in Blocks A, B and the linear w terms in Blocks D, E: $\hat{w}_t^{u,(j)} = \mathbb{E}[w_t^u \mid \mathbf{Y}, \theta^{(j)}]$ and $\hat{w}_t^{\varepsilon,(j)} = \mathbb{E}[w_t^\varepsilon \mid \mathbf{Y}, \theta^{(j)}]$.
- **Posterior log-weight means**, for the $\log w$ terms in Blocks A, B, D, E (needed essentially only for the ν updates): $\mathbb{E}[\log w_t^u \mid \mathbf{Y}, \theta^{(j)}]$ and $\mathbb{E}[\log w_t^\varepsilon \mid \mathbf{Y}, \theta^{(j)}]$.
- **Weighted cross-moments**, for the quadratic terms: $\mathbb{E}[w_t^\varepsilon f_t f_t' \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[w_t^u f_t f_t' \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[w_t^u f_t f_{t-1}' \mid \mathbf{Y}, \theta^{(j)}]$, $\mathbb{E}[w_t^u f_{t-1} f_{t-1}' \mid \mathbf{Y}, \theta^{(j)}]$.

The factor moments are obtained via the Kalman filter/smoother applied to the conditionally Gaussian model (conditional on the weights). The posterior weight quantities are obtained in closed form through the Gaussian–Gamma conjugacy. The weighted cross-moments require care because they mix factors and weights—a property known as *non-separability of expectations*. These are not simple products of marginal expectations because, in the posterior, the latent factors and the weights are conditional on the same data $\mathbf{Y}$ and thus become dependent. The treatment of these cross-moments is one of the subtler points of the derivation, and it is precisely what distinguishes the robust DFM from a standard Kalman filter with pre-determined weights. These computations are the subject of the next section.

Initialisation of the EM Algorithm

The EM algorithm is a local optimisation method: at every iteration it increases the log-likelihood (weakly), but what it ultimately converges to depends on the starting point $\theta^{(0)}$. A badly initialised EM can converge to a sub-optimal local maximum, get stuck on the boundary of the parameter space (for instance producing a near-singular scale matrix), or converge extremely slowly. A good initialisation, by contrast, often brings the algorithm to the vicinity of the global maximum in just a handful of iterations.

For the standard Gaussian DFM, there is a well-established initialisation strategy due to Bańbura and Modugno (2014): initialise the factors by Principal Component Analysis (PCA), then estimate all other parameters via ordinary least squares on the PCA factors. We extend this strategy to the Student-t DFM by adding two new ingredients: initial values for the degrees-of-freedom parameters ν_u, ν_ε, and initial values for the latent weights W^u, W^ε. A further practical complication, addressed in this section, is that the observation matrix $\mathbf{Y}$ typically contains missing entries — both at the end of the sample (ragged edge from non-synchronous releases) and within the sample (mixed-frequency variables observed only every third period). Since PCA requires a complete data matrix, we describe how $\mathbf{Y}$ is preliminarily completed before the PCA step, with the caveat that this filled-in version is used *only* for initialisation: once the EM iterations begin, the model relies exclusively on the selection-matrix framework that we will see later and the imputed entries are discarded.

Why Initialisation Matters

The complete-data log-likelihood (33) is not globally concave in θ. It has multiple local maxima, corresponding to different factor orderings, factor sign choices (a factor is identified only up to a sign), and — more substantively — to different allocations of variance between the common and idiosyncratic components. The EM algorithm, being a monotone ascent procedure, necessarily converges to *some* local maximum, but which one is determined entirely by $\theta^{(0)}$.

Moreover, the Student-t extension enlarges the parameter space with two additional parameters (ν_u, ν_ε) that are notoriously difficult to estimate well. A poor initial value of ν can lead the algorithm either to collapse onto the Gaussian limit ($\nu \to \infty$) — wasting the heavy-tailed flexibility of the model — or to a degenerate heavy-tailed regime ($\nu \to 0$) where a handful of outliers dominate the likelihood. A sensible initialisation of ν_u and ν_ε is therefore at least as important as the initialisation of the structural parameters.

Initialising the Factors by PCA

The strategy begins with a fact that is specific to factor models: *if* the factors were observed, estimating all other parameters would reduce to standard regressions. So, the natural starting point is to *construct* an initial guess for the factors directly from the data.

Principal Component Analysis provides exactly such a construction. Given the standardised data matrix $\tilde{\mathbf{Y}} \in \mathbb{R}^{M \times T}$ — where each row has been centred and rescaled to unit variance — form the sample covariance matrix

$$\widehat{\mathbf{\Sigma}}_Y = \frac{1}{T} \tilde{\mathbf{Y}} \tilde{\mathbf{Y}}' \in \mathbb{R}^{M \times M}.$$

Let $v_1, \ldots, v_r \in \mathbb{R}^M$ be the eigenvectors of $\widehat{\mathbf{\Sigma}}_Y$ corresponding to the r largest eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$, and collect them in the $M \times r$ matrix $V = [v_1 \mid \cdots \mid v_r]$.

The initial factor estimates are obtained by projecting the standardised data onto the PCA directions:

$$\mathbf{F}^{(0)} \;=\; V'\,\tilde{\mathbf{Y}} \;\in\; \mathbb{R}^{r\times T}. \tag{36}$$

That is, at time t, the k-th initial factor is $f_{k,t}^{(0)} = v_k'\,\tilde{y}_t$, the inner product of the k-th PCA direction with the standardised observation. The columns of $\mathbf{F}^{(0)}$ are the *principal component scores* of the data.

Geometrically, the PCA factors are the directions of maximum variance in the data — by construction they capture the largest share of the co-movement across series, which is exactly what latent factors are meant to represent. This is why PCA is such a natural starting point: under mild regularity conditions, as $M, T \to \infty$, the PCA estimator consistently recovers the true factor space (up to a rotation). In finite samples it is a consistent but inefficient estimator; the EM iterations will refine it.

Handling mixed frequencies and ragged edges. In the presence of mixed-frequency data and ragged edges, the sample covariance matrix $\widehat{\mathbf{\Sigma}}_Y$ cannot be computed directly because the panel is unbalanced. To address this we follow the broad logic of Giannone, Reichlin, and Small (2008) and Bańbura and Modugno (2014), performing PCA on a balanced version of the panel obtained by filling the missing entries in two distinct ways.

For quarterly series — which are structurally missing in non-quarter-end months — we disaggregate the quarterly observation into monthly values that satisfy the Mariano–Murasawa aggregation (106). Assuming locally constant monthly values within each quarter, we solve the Mariano–Murasawa restriction recursively across quarters, obtaining an implied monthly series that is consistent with the aggregation scheme adopted in the model. This is more principled than a simple linear interpolation, as it preserves the economic structure of the quarterly observation.

For generic missings arising from ragged-edge publication delays or sample gaps, we fill with draws from a standard Gaussian $\mathcal{N}(0,1)$. This choice is consistent with the centred and standardised data convention adopted at the beginning: each series has empirical mean zero and empirical variance one by construction, so unit-variance Gaussian draws are of the same scale as the observed data. We are aware that this is a deliberate simplification: the empirical distribution of macroeconomic growth rates exhibits heavier tails than the Gaussian, and Gaussian fills therefore underweight the extreme realisations the model is meant to capture. This bias is acceptable here because the imputed values affect only the initialisation $\theta^{(0)}$; once the EM iterations begin, the Kalman filter handles missing entries natively and the fill values are discarded.

These initialisation choices are deliberately simple and conservative: they are used only to produce a starting point $\theta^{(0)}$ for the EM iteration. Once the EM algorithm begins, missing entries are handled rigorously by the Kalman filter, which operates natively on incomplete observation vectors; the initialisation-phase fills are replaced by posterior conditional expectations within a few iterations, so the choice of filling scheme affects convergence speed but not the final parameter estimates.

Block-by-block PCA for economic identification. The procedure described in equation (36) extracts the initial factors via global PCA on the full standardised covariance matrix $\widehat{\mathbf{\Sigma}}_Y \in \mathbb{R}^{M\times M}$. This is the textbook approach, but in our application it requires an additional step: each of the r eigenvectors $v_1, \ldots, v_r$ returned by the global eigendecomposition has nonzero entries on *all*

M series, and a post-hoc reordering is needed to associate each principal component with an economic block. Since the loading matrix $\mathbf{\Lambda}$ is structurally restricted to be block-diagonal across the three blocks (real, financial, other), as discussed in later sections, the global approach would yield an initial $\mathbf{\Lambda}^{(0)}$ that does not respect this restriction, requiring a non-trivial re-projection onto the block-diagonal subspace.

We therefore adopt a structurally consistent variant of the initialisation: *block-by-block PCA*. For each economic block $b \in \{R, F, X\}$, we extract the sub-panel $\tilde{\mathbf{Y}}^b \in \mathbb{R}^{M_b \times T}$ corresponding to the M_b series of block b, compute its own sample covariance

$$\widehat{\mathbf{\Sigma}}_Y^b \;=\; \frac{1}{T}\,\tilde{\mathbf{Y}}^b\,(\tilde{\mathbf{Y}}^b)' \;\in\; \mathbb{R}^{M_b \times M_b},$$

and eigendecompose:

$$\widehat{\mathbf{\Sigma}}_Y^b \;=\; \mathbf{V}^b\,\mathbf{\Lambda}^b\,(\mathbf{V}^b)', \qquad \mathbf{V}^b \in \mathbb{R}^{M_b \times M_b},\ \mathbf{\Lambda}^b = \mathrm{diag}(\lambda_1^b, \ldots, \lambda_{M_b}^b) \text{ with } \lambda_1^b \geq \cdots \geq \lambda_{M_b}^b \geq 0.$$

The matrix $\widehat{\mathbf{\Sigma}}_Y^b$ is exactly the b-th diagonal block of the global covariance $\widehat{\mathbf{\Sigma}}_Y$; the off-diagonal blocks (cross-block sample covariances) are simply not used at this stage. The r_b leading eigenvectors of $\mathbf{V}^b$ are the principal directions of variation *within* the block:

$$\mathbf{V}_{(r_b)}^b \;=\; [\,v_1^b,\ v_2^b,\ \ldots,\ v_{r_b}^b\,] \;\in\; \mathbb{R}^{M_b \times r_b}.$$

The corresponding factor scores for the block are obtained by projecting the sub-panel onto these eigenvectors:

$$\mathbf{F}^{b,(0)} \;=\; \big(\mathbf{V}_{(r_b)}^b\big)'\,\tilde{\mathbf{Y}}^b \;\in\; \mathbb{R}^{r_b \times T}.$$

Stacking the block factors yields the full initial factor matrix:

$$\mathbf{F}^{(0)} \;=\; \begin{bmatrix} \mathbf{F}^{R,(0)} \\ \mathbf{F}^{F,(0)} \\ \mathbf{F}^{X,(0)} \end{bmatrix} \;\in\; \mathbb{R}^{r \times T}, \qquad r \;=\; r_R + r_F + r_X.$$

Why this works (population intuition). Under the structural assumption of the model, each series i in block b loads only on the factor f^b of its own block, so within-block sample variability is generated entirely by f^b:

$$\mathrm{Cov}(y_t^b, y_t^b) \;=\; \mathbf{\Lambda}^b\,\mathrm{Var}(f_t^b)\,\mathbf{\Lambda}^{b\prime} \;+\; \mathbf{R}^b.$$

The dominant principal component of $\widehat{\mathbf{\Sigma}}_Y^b$ is therefore a consistent estimator of $\mathbf{\Lambda}^b$ (up to a scalar rotation), without needing to invoke cross-block covariances. The information contained in the cross-block sample covariances is not lost: it re-enters the model through the dynamic correlations of the factors captured by $\mathbf{A}$, and is recovered automatically within a few EM iterations.

When does the global $\widehat{\mathbf{\Sigma}}_Y$ collapse onto the diagonal blocks? The global sample covariance $\widehat{\mathbf{\Sigma}}_Y$ would be exactly equal to the block-diagonal arrangement of $\widehat{\mathbf{\Sigma}}_Y^R, \widehat{\mathbf{\Sigma}}_Y^F, \widehat{\mathbf{\Sigma}}_Y^X$ if and only if all cross-block sample covariances were zero:

$$\frac{1}{T}\sum_{t=1}^{T} \tilde{y}_{i,t}^b\,\tilde{y}_{j,t}^{b'} \;=\; 0 \qquad \forall\, i \in b,\ j \in b',\ b \neq b'.$$

In finite samples this never holds exactly — some cross-block sample correlation is always present, both because of finite-sample noise and because the factors themselves are dynamically correlated through $\mathbf{A}$ in the population. At the population level, $\mathbf{\Sigma}_Y$ is block-diagonal only when the factors are also cross-block uncorrelated, which is generally not the case in our specification (the model allows for full $\mathbf{A}$). Our block-by-block approach is therefore an *approximation* that ignores the (typically small) cross-block sample correlations at initialisation, in exchange for an exact block-diagonal structure of $\mathbf{\Lambda}^{(0)}$ from the outset.

When would the global PCA be needed instead? Two natural cases require the global eigendecomposition of $\widehat{\mathbf{\Sigma}}_Y$ rather than its block-by-block variant:

- *Single-factor specifications* ($r = 1$, no block partitioning). There is only one latent dimension, and all M series may load on it. The global covariance must be used to find this single common direction of variation. This corresponds to the standard one-factor specifications of Stock and Watson 2002a and others.
- *Multi-factor specifications with full* $\mathbf{\Lambda}$. If the loading matrix is not restricted to be block-diagonal (every series may load on every factor), the natural initialisation extracts the top r principal components of the global matrix $\widehat{\mathbf{\Sigma}}_Y$. This is the setting of Doz, Giannone, Reichlin 2012 and the broad PCA-based DFM literature.

In our setting, neither condition applies: we have multiple factors ($r = r_R + r_F + r_X = 3$) and the loadings are block-diagonal by construction. The block-by-block PCA is the natural counterpart to this structural restriction, and the global $\widehat{\mathbf{\Sigma}}_Y$ plays no role in our initialisation procedure.

An illustrative example. In our 20-series specification with $r_R = r_F = r_X = 1$, the block-by-block PCA reduces to three separate eigendecompositions:

- *Real block* ($M_R = 9$ series: INDPRO, IPFINAL, IPMANSICS, PAYEMS, MANEMP, RPI, CMRMTSPLx, RETAILx, GDPC1). Eigendecompose $\widehat{\mathbf{\Sigma}}_Y^R \in \mathbb{R}^{9\times 9}$, retain the first eigenvector $v_1^R \in \mathbb{R}^9$, and compute $f_t^R = (v_1^R)' \tilde{y}_t^R$ for $t = 1, \ldots, T$.
- *Financial block* ($M_F = 7$ series: S&P 500, S&P PE ratio, T10YFFM, T1YFFM, BAAFFM, TB3SMFFM, NFCI). Eigendecompose $\widehat{\mathbf{\Sigma}}_Y^F \in \mathbb{R}^{7\times 7}$, retain $v_1^F \in \mathbb{R}^7$, and compute $f_t^F = (v_1^F)' \tilde{y}_t^F$.
- *Other block* ($M_X = 4$ series: CPIAUCSL, PCEPI, UMCSENTx, TWEXAFEGSMTHx). Eigendecompose $\widehat{\mathbf{\Sigma}}_Y^X \in \mathbb{R}^{4\times 4}$, retain $v_1^X \in \mathbb{R}^4$, and compute $f_t^X = (v_1^X)' \tilde{y}_t^X$.

The resulting factor matrix is $\mathbf{F}^{(0)} \in \mathbb{R}^{3\times T}$, with the real, financial, and other factors stacked in this order. The initial loading matrix $\mathbf{\Lambda}^{(0)} \in \mathbb{R}^{20\times 3}$ is then obtained by series-by-series scalar OLS: each row of $\tilde{\mathbf{Y}}$ is regressed only on the factor of its own block, with off-block entries set to zero by construction. This ensures that $\mathbf{\Lambda}^{(0)}$ is block-diagonal from the first EM iteration, in line with the structural restriction of the model.

The choice of one factor per block: a modelling assumption. Retaining only the *first* eigenvector of each block covariance matrix is a deliberate modelling assumption, not a technical necessity. It encodes the hypothesis that a *single* latent force drives the common comovement within each economic block — one "real activity" factor, one "financial conditions" factor, one "price/other" factor. Formally, we set $r_R = r_F = r_X = 1$, so that each block contributes one column to $\mathbf{F}^{(0)}$.

Nothing in the procedure forces this choice. For any block b we could retain the leading $r_b > 1$ eigenvectors of $\widehat{\mathbf{\Sigma}}_Y^b$, extracting, say, three factors from the nine real series. Each additional eigenvector captures the next orthogonal direction of maximal residual variance within the block, so adding factors *monotonically increases the share of block variance explained and lowers the reconstruction error* $\|\tilde{\mathbf{Y}}^b - \mathbf{\Lambda}^b \mathbf{F}^b\|$. In the limit $r_b = M_b$ — retaining *all* eigenvectors — the reconstruction is exact and the block variance is explained in full.

That limit is also useless: with $r_b = M_b$ the "factors" are merely a rotation of the original series, no dimension reduction has occurred, and there is nothing latent left to interpret. This makes explicit the trade-off that governs the choice of r_b. More factors mean a better statistical fit (more variance explained, smaller residuals) but weaker *economic interpretability*: a single real factor admits a clean reading as "the state of real activity", whereas a second or third real factor typically captures finer, harder-to-name subcomponents (sectoral rotations, lead-lag patterns) whose economic meaning is ambiguous and whose role in a Growth-at-Risk narrative is unclear.

We resolve this trade-off in favour of interpretability. The one-factor-per-block specification keeps the model parsimonious, gives each factor an unambiguous economic label, and aligns with the three-block structure that motivates the whole exercise. The cost — some within-block variance left in the idiosyncratic component — is acceptable precisely because our goal is interpretable monitoring and density nowcasting, not maximal in-sample variance explained. Should a diagnostic later reveal that one block carries a strong, economically meaningful second factor, the framework extends immediately by raising the corresponding r_b, at the cost of the interpretability just discussed.

PCA as a standalone estimator vs. the state-space approach. PCA is a legitimate and widely-used estimator for static factor models, and even for some dynamic factor models in large cross-sections (see, e.g., Stock and Watson, 2002a). For our setting — monitoring, nowcasting, and Growth-at-Risk applications on macroeconomic data — the iterative EM-based state-space approach is preferred, for three reasons highlighted in the literature (see, e.g., Doz, Giannone, and Reichlin, 2011; Cascaldi-Garcia et al., 2023).

- **Dynamic efficiency.** PCA is a purely static procedure and ignores the VAR law of motion of the factors. The state-space formulation exploits this structure through the Kalman filter, producing more efficient factor estimates, especially in shorter samples.
- **Native handling of missing data.** The classical PCA estimator requires a balanced panel. Although extensions exist (Stock and Watson, 2002b), they recover missing entries by iterative imputation rather than by exploiting dynamic relationships across periods. The Kalman filter, by contrast, handles missing observations natively, and mixed-frequency variables are accommodated through a time-varying selection matrix without the need for ad-hoc completion of the panel.
- **Robustness to outliers.** Standard PCA is equivalent to a Gaussian OLS-type estimator and is sensitive to extreme idiosyncratic shocks, which can distort the estimated factor space. Our Student-t EM framework down-weights such observations endogenously through the posterior weights $\hat{w}_t^\varepsilon$ — a robustness property that is absent in the classical PCA-only approach.

PCA remains a consistent and computationally inexpensive tool for initialisation $\theta^{(0)}$, and we use it in this role. The full estimation, however, proceeds via the iterative EM algorithm with Kalman filter/smoother recursions described in the following sections.

Initialising Λ, A, Q, R from the PCA Factors

Once $\mathbf{F}^{(0)}$ is fixed, each of the structural parameters has a natural initial value obtained by treating $\mathbf{F}^{(0)}$ *as if it were observed* and running standard regressions.

Factor loadings $\mathbf{\Lambda}^{(0)}$. Regress each row of $\tilde{\mathbf{Y}}$ on the initial factors $\mathbf{F}^{(0)}$:

$$\mathbf{\Lambda}^{(0)} \;=\; \tilde{\mathbf{Y}}\,(\mathbf{F}^{(0)})'\,\left[\mathbf{F}^{(0)}\,(\mathbf{F}^{(0)})'\right]^{-1} \;\in\; \mathbb{R}^{M\times r}. \tag{37}$$

This is the OLS estimator of the factor loadings under the pretence that $\mathbf{F}^{(0)}$ is observed. When the PCA factors are orthonormalised (so that $\mathbf{F}^{(0)}\,(\mathbf{F}^{(0)})' = T\cdot I_r$), this simplifies to $\mathbf{\Lambda}^{(0)} = T^{-1}\,\tilde{\mathbf{Y}}\,(\mathbf{F}^{(0)})'$.

Treatment of quarterly series at initialisation. Equation (37) is applied uniformly to all series, including the quarterly GDP series, using the *fully balanced panel* $\tilde{\mathbf{Y}}$ — that is, the panel after the Mariano–Murasawa fill for the quarterly entries and the $\mathcal{N}(0,1)$ fill for the ragged edge. Each (filled) series is regressed directly on its block factor f^b, without the composite Mariano–Murasawa regressor ϕ^b that will be introduced in the M-step. This choice deserves a brief comment.

A natural alternative would be to initialise the quarterly loading using only the genuinely observed quarter-end values — roughly one third of the sample — rather than the MM-filled monthly values. We deliberately do not do this, for two reasons. First, consistency: the initial factors $\mathbf{F}^{(0)}$ are themselves extracted from the balanced panel, so the loadings should be estimated on the same balanced panel; treating the quarterly series differently from the monthly ones would break this uniformity. Second, empirical performance: in our dataset, regressing GDP growth on the monthly real factor using only the quarter-end observations yields a small *negative* loading, which is economically implausible (GDP growth should comove positively with the real-activity factor), whereas the balanced-panel regression yields a small but correctly signed positive loading and a substantially lower idiosyncratic variance. The isolated quarter-end observations are too weakly aligned with the monthly factor at the raw-correlation level to provide a reliable starting point.

We stress that this is purely an initialisation choice. The rigorous treatment of the mixed-frequency aggregation — in which the quarterly observation is matched against the composite regressor $\phi_t^b = \frac{1}{3}f_t^b + \frac{2}{3}f_{t-1}^b + f_{t-2}^b + \frac{2}{3}f_{t-3}^b + \frac{1}{3}f_{t-4}^b$ and evaluated only on the quarter-end observation set $\mathcal{T}_j^Q$ — enters in the M-step and corrects any imprecision in $\mathbf{\Lambda}^{Q,(0)}$ within the first few EM iterations.

VAR transition matrix $\mathbf{A}^{(0)}$. Estimate a VAR(1) on the initial factors by OLS:

$$\mathbf{A}^{(0)} \;=\; \Big(\sum_{t=2}^{T} f_t^{(0)}\,f_{t-1}^{(0)\prime}\Big)\Big(\sum_{t=2}^{T} f_{t-1}^{(0)}\,f_{t-1}^{(0)\prime}\Big)^{-1}. \tag{38}$$

Factor innovation scale $\mathbf{Q}^{(0)}$. Compute the VAR residuals $\hat{u}_t^{(0)} = f_t^{(0)} - \mathbf{A}^{(0)}\,f_{t-1}^{(0)}$ and take their sample covariance:

$$\mathbf{Q}^{(0)} \;=\; \frac{1}{T}\sum_{t=2}^{T}\hat{u}_t^{(0)}\,\hat{u}_t^{(0)\prime}. \tag{39}$$

Idiosyncratic scale $\mathbf{R}^{(0)}$. Compute the observation-equation residuals $\hat{\varepsilon}_t^{(0)} = \tilde{y}_t - \mathbf{\Lambda}^{(0)} f_t^{(0)}$ and take their sample variances, keeping only the diagonal (since $\mathbf{R}$ is assumed diagonal):

$$\mathbf{R}^{(0)} \;=\; \operatorname{diag}\left(\frac{1}{T} \sum_{t=1}^{T} \hat{\varepsilon}_t^{(0)} \, \hat{\varepsilon}_t^{(0)\prime} \right). \tag{40}$$

Initial factor covariance $\mathbf{\Sigma}_0^{(0)}$. A standard choice is the unconditional covariance implied by the initial VAR, obtained as the solution to the discrete Lyapunov equation $\mathbf{\Sigma}_0 = \mathbf{A}^{(0)} \mathbf{\Sigma}_0 \mathbf{A}^{(0)\prime} + \mathbf{Q}^{(0)}$, which exists uniquely whenever $\mathbf{A}^{(0)}$ is stable (all eigenvalues strictly inside the unit circle). A simpler and often adequate alternative is to set $\mathbf{\Sigma}_0^{(0)} = I_r$, expressing a flat initial uncertainty.

Initialising the Student-t Parameters

The two new ingredients specific to the Student-t DFM are the degrees-of-freedom parameters ν_u, ν_ε and the latent weights W^u, W^ε. We initialise them as follows.

Degrees of freedom. A conventional and robust starting point is

$$\nu_u^{(0)} \;=\; \nu_\varepsilon^{(0)} \;=\; 10. \tag{41}$$

The value $\nu = 10$ is a compromise between two concerns:

- it is small enough to allow visibly heavier tails than the Gaussian (at $\nu = 10$, the Student-t kurtosis is $3 + 6/(\nu - 4) = 4$, against the Gaussian value of 3);
- it is large enough that the Student-t is still numerically close to the Gaussian, so that the first few EM iterations behave approximately as in the Gaussian DFM — a regime in which we know the initialisation of the structural parameters via PCA works well.

Lower initial values of ν (for example $\nu^{(0)} = 4$) give the algorithm a stronger incentive to detect heavy tails early, but risk instability in the first few iterations when the PCA factors are still poor estimates. Higher values ($\nu^{(0)} = 30$) effectively tell the algorithm "start Gaussian", but risk leaving the algorithm in the Gaussian basin of attraction if heavy tails are mild. The choice $\nu^{(0)} = 10$ is empirically well-behaved in both cases. Robustness checks with alternative initial values are a standard diagnostic.

Latent weights. With ν_u, ν_ε fixed and the structural parameters initialised, the weights W^u, W^ε can be initialised at their *prior means*:

$$w_t^{u,(0)} \;=\; 1, \qquad w_t^{\varepsilon,(0)} \;=\; 1, \qquad t = 1, \ldots, T. \tag{42}$$

Recall that under $\text{Gamma}(\nu/2, \nu/2)$ the prior mean of the weights is exactly 1. Setting $w_t^{(0)} = 1$ therefore says: "initially, treat every time period as Gaussian with weight 1". The first iteration of the E-step will update these weights based on the conjugate posterior — time periods where the initial residuals are large (in Mahalanobis sense) will receive a smaller updated weight, down-weighting them in the subsequent M-step.

This choice has a concrete consequence. *With $w_t^{(0)} = 1$ for every t, the very first iteration of the algorithm reduces exactly to a Gaussian DFM iteration*, with the PCA initialisation (36)–(40). Only from the second iteration onwards do the posterior weight updates start differentiating time periods, and the heavy-tailed features of the model begin to operate. This "warm start" is deliberate: we want the algorithm to latch on to the Gaussian solution first, and only subsequently discover which time periods deserve to be treated as outliers.

Summary of the Initialisation Procedure

Putting all the pieces together, the initialisation proceeds as follows:

1. **Stationarise and standardise the data**: each row of $\mathbf{Y}$ is first transformed to stationarity (log-differencing, differencing, or other series-specific transformation) following the conventions described earlier, then centred at the sample mean and rescaled to unit sample standard deviation. The result is $\tilde{\mathbf{Y}}$.

2. **Handle missing observations**: To obtain a balanced panel for PCA, we apply a two-fold strategy:

   - *Mixed frequencies*: quarterly series are converted to a monthly frequency via MM to provide a continuous baseline.
   - *Ragged edges*: any remaining missing values at the beginning or end of the sample are filled by drawing from a Normal distribution with mean zero and unit variance.

   This hybrid approach ensures a complete dataset for the initial eigendecomposition while preserving the possibility of extreme values in the unsampled periods, consistent with the heavy-tailed nature of the model.

3. **Compute the PCA factors block-by-block**: for each block $b \in \{R, F, X\}$, form the sub-panel $\tilde{\mathbf{Y}}^b$ corresponding to the series of block b, eigendecompose its sample covariance $\widehat{\mathbf{\Sigma}}_Y^b = T^{-1}\,\tilde{\mathbf{Y}}^b\,(\tilde{\mathbf{Y}}^b)'$, and take its leading r_b eigenvectors V^b. The initial factors of block b are $\mathbf{F}^{b,(0)} = (V^b)'\,\tilde{\mathbf{Y}}^b$. Stack the block factors into the full factor matrix $\mathbf{F}^{(0)} = [\mathbf{F}^{R,(0)}; \mathbf{F}^{F,(0)}; \mathbf{F}^{X,(0)}] \in \mathbb{R}^{r\times T}$. See the discussion in the previous subsection for the rationale.

4. **Initialise the loadings block-by-block**: for each series i in block b, run a scalar OLS regression of $\tilde{y}_i$ on $\mathbf{F}^{b,(0)}$ alone (no off-block regressors). The off-block entries of $\mathbf{\Lambda}^{(0)}$ are set to zero by construction, so that $\mathbf{\Lambda}^{(0)}$ is block-diagonal from the start.

5. **Initialise the VAR** on $\mathbf{F}^{(0)}$: equations (38) and (39).

6. **Initialise the idiosyncratic scale** from the residuals: equation (40).

7. **Initialise the initial-state covariance** by Lyapunov or I_r.

8. **Initialise the degrees of freedom** at $\nu_u^{(0)} = \nu_\varepsilon^{(0)} = 10$.

9. **Initialise the weights** at $w_t^{u,(0)} = w_t^{\varepsilon,(0)} = 1$ for all t.

The resulting parameter vector

$$\theta^{(0)} \;=\; \left\{\,\mathbf{\Lambda}^{(0)},\; \mathbf{A}^{(0)},\; \mathbf{Q}^{(0)},\; \mathbf{R}^{(0)},\; \nu_u^{(0)},\; \nu_\varepsilon^{(0)}\,\right\}$$

together with the initial latent values $\mathbf{F}^{(0)}, W^{u,(0)}, W^{\varepsilon,(0)}$ provides a complete starting point. The EM algorithm will now iterate: at each step, the E-step computes the posterior distribution of the latent variables given $\theta^{(j)}$, and the M-step updates $\theta^{(j+1)}$ accordingly. We turn to the E-step in the next section.

A practical remark on robustness checks. In empirical applications, it is standard to verify that the final estimates $\hat{\theta}$ do not depend sensitively on the initialisation. Two checks are particularly informative: (i) run the algorithm from multiple random perturbations of $\theta^{(0)}$ and verify that the final estimates cluster around the same point; (ii) run the algorithm from $\nu^{(0)} \in \{4, 10, 30\}$

and verify that $\hat{v}_u$ and $\hat{v}_\varepsilon$ converge to similar values regardless of the starting point. If the final estimates vary substantially across initialisations, the likelihood surface is poorly identified and the model specification should be reconsidered.

The E-Step

We are now in a position to derive the E-step in detail. Recall that the E-step holds the parameter vector $\theta^{(j)}$ fixed and computes the posterior distribution of the latent variables given the observed data and the current parameters:

$$q^{(j+1)}(\mathcal{Z}) \;=\; p\big(\mathcal{Z} \mid \mathbf{Y}, \theta^{(j)}\big), \qquad \mathcal{Z} \;=\; (\mathbf{F},\, W^u,\, W^\varepsilon).$$

This posterior is the input that the M-step will use in the next section. Concretely, the M-step does not need the full posterior — it only needs specific expectations of functions of $\mathcal{Z}$. The job of the E-step is to compute those expectations.

What the E-Step Must Compute

Before diving into the derivations, let us restate precisely which quantities the E-step is expected to produce. Looking back at the complete-data log-likelihood (33), every latent quantity that appears inside it will need to be replaced by its posterior expectation under $\theta^{(j)}$. These fall into four categories:

(i) First-order factor moments. The smoothed factor means

$$\hat{f}^{(j)}_{t|T} \;=\; \mathbb{E}_{\theta^{(j)}}\big[\, f_t \;\big|\; \mathbf{Y} \,\big], \qquad t = 0, 1, \ldots, T,$$

needed for the linear terms in the observation equation's residual and in the state-transition residual.

(ii) Second-order factor moments. The M-step requires the smoothed second moments for all time steps, including the initial state $t = 0$:

$$\begin{aligned}
\mathbb{E}_{\theta^{(j)}}[\, f_t f_t' \mid \mathbf{Y} \,] \;&=\; P^{(j)}_{t|T} \;+\; \hat{f}^{(j)}_{t|T} \hat{f}^{(j)\prime}_{t|T}, \qquad t = 0, 1, \ldots, T \\
\mathbb{E}_{\theta^{(j)}}[\, f_t f_{t-1}' \mid \mathbf{Y} \,] \;&=\; P^{(j)}_{t,t-1|T} \;+\; \hat{f}^{(j)}_{t|T} \hat{f}^{(j)\prime}_{t-1|T}, \qquad t = 1, \ldots, T
\end{aligned}$$

where $P^{(j)}_{t|T}$ is the smoothed factor covariance and $P^{(j)}_{t,t-1|T}$ is the smoothed lag-one cross-covariance. The moment for $t = 0$ is necessary for the update of the initial state distribution.

(iii) Posterior weight expectations.

$$\hat{w}^{u,(j)}_t \;=\; \mathbb{E}_{\theta^{(j)}}\big[\, w^u_t \,\big|\, \mathbf{Y} \,\big], \qquad \hat{w}^{\varepsilon,(j)}_t \;=\; \mathbb{E}_{\theta^{(j)}}\big[\, w^\varepsilon_t \,\big|\, \mathbf{Y} \,\big],$$

and the corresponding *log-weight expectations*

$$\widehat{\log w}^{u,(j)}_t \;=\; \mathbb{E}_{\theta^{(j)}}\big[\, \log w^u_t \,\big|\, \mathbf{Y} \,\big], \qquad \widehat{\log w}^{\varepsilon,(j)}_t \;=\; \mathbb{E}_{\theta^{(j)}}\big[\, \log w^\varepsilon_t \,\big|\, \mathbf{Y} \,\big].$$

The plain expectations enter the M-step updates for $\{\boldsymbol{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}\}$ as weights on the sums over t. The log-weight expectations enter only the M-step updates for $\{\nu_u, \nu_\varepsilon\}$.

(iv) Weighted first- and second-order moments. The expansion of the quadratic forms in Blocks A and B requires weighted moments of both first and second order. Specifically:

$$\mathbb{E}_{\theta^{(j)}}[w_t^\varepsilon f_t \mid \mathbf{Y}], \qquad \mathbb{E}_{\theta^{(j)}}[w_t^u f_t \mid \mathbf{Y}], \qquad \mathbb{E}_{\theta^{(j)}}[w_t^u f_{t-1} \mid \mathbf{Y}],$$

and the corresponding second-order weighted moments

$$\mathbb{E}_{\theta^{(j)}}[w_t^\varepsilon f_t f_t' \mid \mathbf{Y}], \qquad \mathbb{E}_{\theta^{(j)}}[w_t^u f_t f_t' \mid \mathbf{Y}], \qquad \mathbb{E}_{\theta^{(j)}}[w_t^u f_t f_{t-1}' \mid \mathbf{Y}], \qquad \mathbb{E}_{\theta^{(j)}}[w_t^u f_{t-1} f_{t-1}' \mid \mathbf{Y}].$$

These are *not* simply products of marginal expectations (e.g., $\hat{w}_t \hat{f}_t$ or $\hat{w}_t \hat{f}_t \hat{f}_t'$), because under the posterior the weights and the factors are not independent. We compute these moments by using the law of iterated expectations, conditioning on the weights to exploit the Gaussian-conditional structure of the state-space model.

Summary of Required Expectations

The following table summarises the latent quantities computed during the E-step and identifies which structural parameters rely on them during the subsequent M-step.

Latent Moment	Expectation	M-step Parameters
First-order Factor	$\hat{f}_{t\|T}$	Initial state
Second-order Factor	$\mathbb{E}[f_t f_{t-k}' \| \mathbf{Y}]$	Initial state
Posterior Weights	$\hat{w}_t^u, \hat{w}_t^\varepsilon$	ν_u, ν_ε
Log-weight Expectations	$\widehat{\log w_t^u}, \widehat{\log w_t^\varepsilon}$	ν_u, ν_ε
Weighted Factor Moments	$\mathbb{E}[w_t f_t \mid \mathbf{Y}], \mathbb{E}[w_t f_t f_{t-k}' \mid \mathbf{Y}]$	$\mathbf{A}, \mathbf{\Lambda}, \mathbf{Q}, \mathbf{R}$

Table 2: Mapping of E-step outputs to M-step parameter updates.

As shown, while the log-weights are exclusively used to refine the degrees-of-freedom parameters, the weighted second-order moments are the engine of the robust estimation, allowing the algorithm to down-weight outliers when updating the transition and observation matrices.

The strategy for computing all of these objects exploits the *conditional structure* of the model. Even though the full joint posterior $p(\mathbf{F}, W^u, W^\varepsilon \mid \mathbf{Y})$ is complicated, its conditional decomposition is tractable:

- *Conditional on the weights*, the factors follow a linear Gaussian state-space model, and the Kalman filter/smoother delivers their moments exactly.

- *Conditional on the factors*, the weights are a posteriori Gamma-distributed, thanks to Gaussian–Gamma conjugacy, and their moments are available in closed form.

The two computations are coupled — each needs the other's output — and the resolution of this coupling is the subject of following sections. We begin, however, with the simpler of the two: the conditional posterior of the weights.

The Computational Loop: Coupling Weights and Factors

The interdependence between weights and factors suggests a natural iterative strategy for the E-step. Although the joint posterior of $\mathcal{Z} = (\mathbf{F}, W^u, W^\varepsilon)$ is complicated, conditioning on the right argument breaks it into two tractable subproblems:

1. **Weights given factors (conjugacy).** Conditional on the current smoothed factors $\hat{f}_{t|T}$, the posterior of each weight w_t^u and w_t^ε is a Gamma distribution, by Gaussian–Gamma conjugacy. From this posterior we extract the two summaries that the M-step needs: the posterior mean $\hat{w}_t$ (entering the weighted-OLS updates for $\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$) and the posterior log-mean $\widehat{\log w_t}$ (entering the updates for ν_u, ν_ε).

2. **Factors given weights (weighted Kalman).** Conditional on the current weights, the model reduces to a time-heteroscedastic linear Gaussian state-space system, in which the factor noise covariance is $\mathbf{Q}/w_t^u$ and the observation noise covariance is $\mathbf{R}/w_t^\varepsilon$ — time-varying, but at each t a known matrix. The Kalman filter and smoother then deliver, in closed form, the smoothed factor means $\hat{f}_{t|T}$ and the smoothed factor covariances $P_{t|T}$.

These two subproblems are the building blocks of the E-step. The weighted moments required by Block A and Block B ($\mathbb{E}[w_t f_t \mid \mathbf{Y}]$, $\mathbb{E}[w_t f_t f_t' \mid \mathbf{Y}]$, $\mathbb{E}[w_t f_t f_{t-1}' \mid \mathbf{Y}]$) are then assembled from the two block outputs via the law of iterated expectations.

The two computations are coupled — each depends on the output of the other — so the E-step cannot be a single one-shot pass. We resolve this coupling through an inner loop that alternates between the two updates until both stabilise, a standard ECM-style construction discussed in detail below. The mapping of E-step outputs to the M-step updates is summarised in Table 2.

The next sections develop the two ingredients in order: first the conjugate posterior of the weights, then the Kalman recursions for the factors, and finally the weighted moments and their assembly.

Conjugacy: The Posterior of the Latent Weights

We now derive in full the conditional posterior distributions

$$p\big(w_t^\varepsilon \mid \mathbf{Y}, \mathbf{F}, \theta\big), \qquad p\big(w_t^u \mid \mathbf{Y}, \mathbf{F}, \theta\big).$$

The calculation is an elegant application of Gaussian–Gamma conjugacy and yields the posteriors *in closed form* — a Gamma distribution whose hyperparameters depend on the residuals of the conditionally Gaussian model.

Posterior of the Idiosyncratic Weight w_t^ε

The prior on w_t^ε is

$$p(w_t^\varepsilon \mid \theta) \;=\; \frac{(\nu_\varepsilon/2)^{\nu_\varepsilon/2}}{\Gamma(\nu_\varepsilon/2)}\,(w_t^\varepsilon)^{\nu_\varepsilon/2-1}\,e^{-(\nu_\varepsilon/2)\,w_t^\varepsilon}, \qquad w_t^\varepsilon > 0,$$

and, conditional on f_t and w_t^ε, the observation density is

$$p(y_t \mid f_t, w_t^\varepsilon, \theta) \;=\; \frac{1}{(2\pi)^{M/2}\,|\mathbf{R}/w_t^\varepsilon|^{1/2}}\exp\Big(-\tfrac{1}{2}\,(y_t - \mathbf{\Lambda} f_t)'\,(\mathbf{R}/w_t^\varepsilon)^{-1}\,(y_t - \mathbf{\Lambda} f_t)\Big)\,.$$

Apply Bayes' theorem. The conditional posterior of w^ε_t given y_t, f_t and θ is proportional, as a function of w^ε_t, to the product of the prior and the likelihood contribution of y_t:

$$p(w^\varepsilon_t \mid y_t, f_t, \theta) \;\propto\; p(y_t \mid f_t, w^\varepsilon_t, \theta)\, p(w^\varepsilon_t \mid \theta).$$

We are not conditioning on the other y_s with $s \neq t$ because, by the conditional independence structure of the model, the weight at time t depends only on the observation at time t (given the factor at time t). We will return to the full conditioning set shortly.

Collect the terms depending on w^ε_t. Retain only the factors that depend on w^ε_t. Using the identities $|\mathbf{R}/w^\varepsilon_t|^{1/2} = (w^\varepsilon_t)^{-M/2}\,|\mathbf{R}|^{1/2}$ and $(\mathbf{R}/w^\varepsilon_t)^{-1} = w^\varepsilon_t\,\mathbf{R}^{-1}$ (as derived before for Block A):

$$p(y_t \mid f_t, w^\varepsilon_t, \theta) \;\propto\; (w^\varepsilon_t)^{M/2}\, \exp\!\Big(-\tfrac{w^\varepsilon_t}{2}\, d^\varepsilon_t\Big)\,,$$

where we have defined the *squared Mahalanobis residual* on the idiosyncratic side:

$$d^\varepsilon_t \;\equiv\; (y_t - \mathbf{\Lambda} f_t)'\, \mathbf{R}^{-1}\, (y_t - \mathbf{\Lambda} f_t). \tag{43}$$

This scalar is a non-negative quantity measuring, in Mahalanobis sense, how far the observation y_t is from the factor-implied mean $\mathbf{\Lambda} f_t$. Large d^ε_t means that, conditional on the current factors and parameters, the observation at time t is an outlier in the idiosyncratic direction.

Similarly, the prior contributes

$$p(w^\varepsilon_t \mid \theta) \;\propto\; (w^\varepsilon_t)^{\nu_\varepsilon/2-1}\, \exp\!\big(-\tfrac{\nu_\varepsilon}{2}\, w^\varepsilon_t\big)\,.$$

Multiply and recognise a Gamma kernel. Multiplying the two:

$$p(w^\varepsilon_t \mid y_t, f_t, \theta) \;\propto\; (w^\varepsilon_t)^{M/2}\, (w^\varepsilon_t)^{\nu_\varepsilon/2-1} \exp\!\Big(-\tfrac{w^\varepsilon_t}{2} d^\varepsilon_t\Big)\, \exp\!\big(-\tfrac{\nu_\varepsilon}{2} w^\varepsilon_t\big)\,.$$

Collecting powers and exponentials:

$$p(w^\varepsilon_t \mid y_t, f_t, \theta) \;\propto\; (w^\varepsilon_t)^{\frac{\nu_\varepsilon+M}{2}-1}\, \exp\!\Big(-\tfrac{\nu_\varepsilon+d^\varepsilon_t}{2}\, w^\varepsilon_t\Big)\,.$$

This is, up to the normalising constant, precisely the kernel of a Gamma density with shape $(\nu_\varepsilon + M)/2$ and rate $(\nu_\varepsilon + d^\varepsilon_t)/2$. Normalisation being unique, we conclude:

$$\boxed{\,w^\varepsilon_t \mid y_t,\, f_t,\, \theta \;\sim\; \mathrm{Gamma}\!\left(\frac{\nu_\varepsilon + M}{2},\; \frac{\nu_\varepsilon + d^\varepsilon_t}{2}\right).\,} \tag{44}$$

Remark on the conditioning set. We derived (44) conditioning on y_t and f_t only. What about the other observations $y_s, s \neq t$, and the other factors $f_s, s \neq t$, and indeed the other weights? The model's conditional independence structure guarantees that they do not enter. Given f_t and w^ε_t, the observation y_t is conditionally independent of everything else. Hence

$$p(w^\varepsilon_t \mid \mathbf{Y}, \mathbf{F}, W^u, W^\varepsilon_{-t}, \theta) \;=\; p(w^\varepsilon_t \mid y_t, f_t, \theta),$$

where W^ε_{-t} denotes W^ε with the t-th entry removed. The idiosyncratic weight at time t depends only on the observation and factor at the same time — an extremely useful property because it means that weights at different times can be updated independently.

Posterior of the Factor-Side Weight w_t^u

The derivation for w_t^u is exactly analogous, with the state equation replacing the observation equation. Skipping the identical intermediate steps:

$$p(w_t^u \mid f_t, f_{t-1}, \theta) \;\propto\; (w_t^u)^{r/2} \exp\Big(-\tfrac{w_t^u}{2}\, d_t^u\Big) \cdot (w_t^u)^{\nu_u/2-1} \exp\big(-\tfrac{\nu_u}{2} w_t^u\big)\,,$$

where

$$d_t^u \;\equiv\; (f_t - \mathbf{A} f_{t-1})'\, \mathbf{Q}^{-1}\, (f_t - \mathbf{A} f_{t-1}) \tag{45}$$

is the squared Mahalanobis residual on the factor side. Collecting and recognising the Gamma kernel:

$$\boxed{w_t^u \mid f_t,\, f_{t-1},\, \theta \;\sim\; \mathrm{Gamma}\left(\frac{\nu_u + r}{2},\; \frac{\nu_u + d_t^u}{2}\right).} \tag{46}$$

Note the dimension change: on the observation side the "added" dimension in the shape parameter was M (the dimension of y_t); on the factor side it is r (the dimension of f_t). This makes intuitive sense — each observation dimension adds one degree of freedom to the posterior.

Posterior Moments of the Weights

Having identified the posterior distributions as Gamma, we can write down the specific moments required for the next steps of the EM algorithm. It is important to distinguish between two types of expectations:

- **The plain expectation** $\mathbb{E}[w_t]$, which acts as a scaling factor (precision) for the variances. This is the "weight" that the Kalman filter will use to handle outliers.
- **The log-expectation** $\mathbb{E}[\log w_t]$, which is required exclusively for the M-step updates of the degrees-of-freedom parameters ν_u and ν_ε.

For a generic Gamma(α, β) distribution, these moments are available in closed form:

$$\mathbb{E}[W] \;=\; \frac{\alpha}{\beta}, \qquad \mathbb{E}[\log W] \;=\; \psi(\alpha) \;-\; \log \beta,$$

where $\psi(x)$ is the **digamma function**. Mathematically, the digamma function arises as the derivative of the log-gamma function, $\psi(x) = \frac{d}{dx} \log \Gamma(x)$. In the context of the EM algorithm, its appearance is a standard result of the exponential family property of the Gamma distribution: when we take the expectation of the log-likelihood with respect to a Gamma posterior, the derivative of the normalizing constant $\Gamma(\alpha)$ naturally generates the ψ function.Applying these formulae to our posteriors (44) and (46):

Posterior means of the weights.

$$\boxed{\hat{w}_t^\varepsilon \;\equiv\; \mathbb{E}[w_t^\varepsilon \mid y_t, f_t, \theta] \;=\; \frac{\nu_\varepsilon + M}{\nu_\varepsilon + d_t^\varepsilon}, \qquad \hat{w}_t^u \;\equiv\; \mathbb{E}[w_t^u \mid f_t, f_{t-1}, \theta] \;=\; \frac{\nu_u + r}{\nu_u + d_t^u}.} \tag{47}$$

Posterior log-means of the weights.

$$\boxed{\widehat{\log w}_t^{\,\varepsilon} \;\equiv\; \mathbb{E}[\log w_t^\varepsilon \mid y_t, f_t, \theta] \;=\; \psi\left(\frac{\nu_\varepsilon + M}{2}\right) \;-\; \log\left(\frac{\nu_\varepsilon + d_t^\varepsilon}{2}\right),} \tag{48}$$

and analogously

$$\boxed{\widehat{\log w}_t^{\,u} \;=\; \psi\left(\frac{\nu_u + r}{2}\right) \;-\; \log\left(\frac{\nu_u + d_t^u}{2}\right).} \tag{49}$$

Anatomy of the weight formula. The expression for $\hat{w}_t^\varepsilon$ is worth pausing on, because it is the technical and statistical heart of the heavy-tailed robustness of the model. Read the formula as follows:

$$\hat{w}_t^\varepsilon \;=\; \frac{\nu_\varepsilon + M}{\nu_\varepsilon + d_t^\varepsilon}.$$

The numerator $\nu_\varepsilon + M$ is a constant that depends only on the degrees of freedom and the dimension of the observations — it does not vary over time. The denominator $\nu_\varepsilon + d_t^\varepsilon$ is, apart from the constant ν_ε, equal to the Mahalanobis distance of y_t from $\mathbf{\Lambda} f_t$. The formula therefore says:

> *Times when the observation y_t is close to its factor-implied mean (d_t^ε small) receive a posterior weight close to* 1*; times when the observation is far from its factor-implied mean (d_t^ε large) receive a posterior weight much smaller than* 1*.*

This is exactly the mechanism of *automatic outlier down-weighting*: anomalous observations, in Mahalanobis sense, automatically receive less weight in the subsequent M-step updates, without any user-specified threshold or tuning parameter. The down-weighting is smooth — the weight decays as $1/d_t^\varepsilon$ in the extreme tail — and the decay rate is governed entirely by the estimated ν_ε.

In the Gaussian limit $\nu_\varepsilon \to \infty$, the formula becomes $\hat{w}_t^\varepsilon \to 1$ for every t, regardless of d_t^ε. No down-weighting is performed, and the algorithm reduces to the Gaussian DFM. This is consistent with the fact that the Gaussian has no heavy tails: there is no notion of "outlier" to down-weight.

Trigamma and Fisher information (for later use). For completeness, we also record the *trigamma function* $\psi'(x) = \frac{d}{dx}\psi(x)$, which is the second derivative of $\log \Gamma(x)$. The variance of $\log W$ under a $\mathrm{Gamma}(\alpha, \beta)$ distribution is $\psi'(\alpha)$. This quantity will appear when we set up the Newton iteration for the ν_u and ν_ε updates.

Why the Conjugacy Works: A Pedagogical Remark

The conjugacy that gave us (44) and (46) is not a happy accident — it is a consequence of how the Gaussian and Gamma densities interact. To see the structure, consider a generic Gaussian scale-mixture $x \mid w \sim \mathcal{N}(0, \Sigma/w)$ with prior $w \sim \mathrm{Gamma}(\alpha, \beta)$. The likelihood contribution of x to the posterior of w is

$$p(x \mid w) \;\propto\; w^{m/2} \exp(-\tfrac{w}{2}\, x'\Sigma^{-1}x),$$

which, *as a function of w*, has the form of a Gamma kernel with shape $m/2 + 1$ and rate $x'\Sigma^{-1}x/2$. Multiplying by the $\mathrm{Gamma}(\alpha, \beta)$ prior simply adds the shape and rate parameters:

$$p(w \mid x) \;\sim\; \mathrm{Gamma}\big(\alpha + m/2,\; \beta + \tfrac{1}{2}x'\Sigma^{-1}x\big).$$

The Gamma family is "closed" under multiplication by Gaussian likelihoods in this way — a property known as *conjugacy*. Our specific posterior formulas (44) and (46) are simply this generic result instantiated with $\alpha = \beta = \nu/2$ and the appropriate Mahalanobis distance.

This conjugacy is what makes the Student-t DFM tractable via EM. Without it, the posterior of the weights would not be available in closed form, and one would have to resort to Monte Carlo methods (MCMC, variational approximations) to evaluate the E-step. With it, the weights are no more expensive to compute than the Mahalanobis distances themselves.

The Kalman Filter Conditional on the Weights

We now turn to the other half of the E-step: computing the posterior moments of the factors. The key observation is the one we made at the beginning: *conditional on the weights W^u and W^ε*, the model is a linear Gaussian state-space model, and the posterior moments of the factors can be computed exactly by the Kalman filter and smoother.

To see this conditional Gaussian structure clearly, recall the hierarchical DGP:

$$\begin{aligned} \textbf{State:} \quad & f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \qquad & u_t \mid w_t^u \;\sim\; \mathcal{N}\big(0,\; \mathbf{Q}/w_t^u\big), \\ \textbf{Observation:} \quad & y_t \;=\; \boldsymbol{\Lambda}\, f_t \;+\; \varepsilon_t, \qquad & \varepsilon_t \mid w_t^\varepsilon \;\sim\; \mathcal{N}\big(0,\; \mathbf{R}/w_t^\varepsilon\big). \end{aligned}$$

Once we condition on specific values of the weights $\{w_t^u, w_t^\varepsilon\}$, the time-varying Gaussian covariances $\mathbf{Q}/w_t^u$ and $\mathbf{R}/w_t^\varepsilon$ are just deterministic functions of t. The system becomes a *linear Gaussian state-space model with time-varying covariances*:

$$\begin{aligned} f_t \;&=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \quad u_t \;\sim\; \mathcal{N}\big(0,\; \tilde{\mathbf{Q}}_t\big), \quad \tilde{\mathbf{Q}}_t \;\equiv\; \mathbf{Q}/w_t^u, \\ y_t \;&=\; \boldsymbol{\Lambda}\, f_t \;+\; \varepsilon_t, \qquad \varepsilon_t \;\sim\; \mathcal{N}\big(0,\; \tilde{\mathbf{R}}_t\big), \quad \tilde{\mathbf{R}}_t \;\equiv\; \mathbf{R}/w_t^\varepsilon. \end{aligned} \tag{50}$$

To this system the standard Kalman filter and smoother apply directly, with the only modification that the covariances $\tilde{\mathbf{Q}}_t$ and $\tilde{\mathbf{R}}_t$ are now time-varying. This is a genuine advantage of the scale-mixture representation: no new machinery is needed — we simply feed the weighted covariances into the ordinary Kalman recursions.

For completeness, we derive the filter equations below, emphasising where the weights enter. The reader familiar with the Gaussian Kalman filter can skip the derivations and note only the places where $\tilde{\mathbf{Q}}_t$ and $\tilde{\mathbf{R}}_t$ appear in place of $\mathbf{Q}$ and $\mathbf{R}$.

Notation for the Conditional Moments

Following standard notation, for each t we track two sets of moments:

- *Predicted moments* — the mean and covariance of f_t given information up to $t-1$:

$$\hat{f}_{t|t-1} \;=\; \mathbb{E}[f_t \mid y_1, \ldots, y_{t-1}, W^u, W^\varepsilon], \qquad P_{t|t-1} \;=\; \mathrm{Cov}(f_t \mid y_1, \ldots, y_{t-1}, W^u, W^\varepsilon).$$

- *Filtered moments* — the mean and covariance of f_t given information up to and including t:

$$\hat{f}_{t|t} \;=\; \mathbb{E}[f_t \mid y_1, \ldots, y_t, W^u, W^\varepsilon], \qquad P_{t|t} \;=\; \mathrm{Cov}(f_t \mid y_1, \ldots, y_t, W^u, W^\varepsilon).$$

All conditional distributions are Gaussian thanks to the conditional Gaussianity of the state-space model given the weights. To keep the notation light we suppress the conditioning on W^u, W^ε in what follows — it is implicit throughout.

The filter initialises at $t = 0$ with

$$\hat{f}_{0|0} \;=\; 0, \qquad P_{0|0} \;=\; \boldsymbol{\Sigma}_0,$$

which are the mean and covariance of the initial state.

Forward Recursion: the Filter

The forward Kalman filter is a *recursion* that runs sequentially from $t = 1$ to $t = T$. At each step it takes as input the filtered moments from the previous period $(\hat{f}_{t-1|t-1}, P_{t-1|t-1})$ and the new observation y_t, and produces as output the filtered moments at the current period $(\hat{f}_{t|t}, P_{t|t})$. The recursion is initialised at $t = 0$ with the prior moments

$$(\hat{f}_{0|0},\ P_{0|0}) \ = \ (0,\ \mathbf{\Sigma}_0),$$

introduced earlier, which encode the initial-state distribution of f_0 *before any observation has been seen*. Strictly speaking, $(\hat{f}_{0|0}, P_{0|0})$ are not "filtered" in the usual sense (they are not conditioned on any y_t); they simply provide the starting point that lets the recursion begin.

Each iteration of the filter consists of two sub-steps — a *prediction* step (from $t-1$ to t, using the state equation) and an *update* step (incorporating y_t, using the observation equation):

$$\underbrace{\left(\hat{f}_{t-1|t-1},\ P_{t-1|t-1}\right)}_{\text{filtered at } t-1} \xrightarrow{\text{predict}} \underbrace{\left(\hat{f}_{t|t-1},\ P_{t|t-1}\right)}_{\text{predicted at } t} \xrightarrow{\text{update with } y_t} \underbrace{\left(\hat{f}_{t|t},\ P_{t|t}\right)}_{\text{filtered at } t}.$$

The filtered moments at t become the input of the next iteration at $t+1$, and the recursion proceeds. Concretely, the first iteration ($t = 1$) starts from $(\hat{f}_{0|0}, P_{0|0})$, predicts $(\hat{f}_{1|0}, P_{1|0})$, then updates using y_1 to produce $(\hat{f}_{1|1}, P_{1|1})$. The second iteration takes $(\hat{f}_{1|1}, P_{1|1})$ and produces $(\hat{f}_{2|2}, P_{2|2})$, and so on until $t = T$.

Prediction step. Starting from the filtered moments $(\hat{f}_{t-1|t-1}, P_{t-1|t-1})$, we use the state equation $f_t = \mathbf{A} f_{t-1} + u_t$ with $u_t \sim \mathcal{N}(0, \tilde{\mathbf{Q}}_t)$ to compute the predicted moments:

$$\boxed{\hat{f}_{t|t-1} \ = \ \mathbf{A}\,\hat{f}_{t-1|t-1}, \qquad P_{t|t-1} \ = \ \mathbf{A}\,P_{t-1|t-1}\,\mathbf{A}' \ + \ \tilde{\mathbf{Q}}_t.} \tag{51}$$

The formula for the mean is obtained by taking conditional expectation on both sides of the state equation. The formula for the covariance uses that (i) f_{t-1} and u_t are conditionally independent given the weights, and (ii) u_t has conditional covariance $\tilde{\mathbf{Q}}_t$. The $\tilde{\mathbf{Q}}_t = \mathbf{Q}/w_t^u$ is where the factor-side weight enters the filter: a small w_t^u inflates $\tilde{\mathbf{Q}}_t$, which in turn inflates the predicted covariance $P_{t|t-1}$.

Update step. We now incorporate the new observation y_t. The observation equation is $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \tilde{\mathbf{R}}_t)$.

The *innovation* is the difference between the actual observation and what the filter predicted:

$$\eta_t \ \equiv \ y_t \ - \ \mathbf{\Lambda}\,\hat{f}_{t|t-1}.$$

Its conditional covariance — the *innovation covariance* — is:

$$S_t \ = \ \mathbf{\Lambda}\,P_{t|t-1}\,\mathbf{\Lambda}' \ + \ \tilde{\mathbf{R}}_t. \tag{52}$$

This covariance has two sources of uncertainty: the uncertainty about the factor $(\mathbf{\Lambda} P_{t|t-1} \mathbf{\Lambda}')$ and the idiosyncratic noise $(\tilde{\mathbf{R}}_t)$. The $\tilde{\mathbf{R}}_t = \mathbf{R}/w_t^\varepsilon$ is where the idiosyncratic-side weight enters the filter: a small w_t^ε inflates S_t, which — via the Kalman gain defined below — reduces the weight placed on y_t in the update.

The *Kalman gain*:

$$K_t \ = \ P_{t|t-1}\,\mathbf{\Lambda}'\,S_t^{-1}. \tag{53}$$

Intuitively, K_t is the matrix of regression coefficients of the state update on the innovation — how much we move the factor estimate per unit of innovation. When $\tilde{\mathbf{R}}_t$ is large relative to $\mathbf{\Lambda} P_{t|t-1}\mathbf{\Lambda}'$ (i.e. the observation is noisy compared with our prior uncertainty about the factor), S_t^{-1} is small and the gain is small: we do not move much. Conversely, a precise observation (small $\tilde{\mathbf{R}}_t$) produces a large gain and a large update.

Finally, the filtered moments:

$$\boxed{\hat{f}_{t|t} \;=\; \hat{f}_{t|t-1} \;+\; K_t\,\eta_t, \qquad P_{t|t} \;=\; P_{t|t-1} \;-\; K_t\,\mathbf{\Lambda}\,P_{t|t-1}.} \tag{54}$$

The mean update says: take the predicted mean and correct it by the innovation, weighted by the gain. The covariance update says: the posterior uncertainty is the prior uncertainty minus the information contributed by the new observation.

Summary of weight entry points. The weights enter the Kalman filter in exactly two places:

- $\tilde{\mathbf{Q}}_t = \mathbf{Q}/w_t^u$ appears in the prediction-step covariance (51);
- $\tilde{\mathbf{R}}_t = \mathbf{R}/w_t^\varepsilon$ appears in the innovation covariance (52), and hence indirectly in the Kalman gain and the update.

Otherwise, the filter is identical to the Gaussian Kalman filter. If all weights equal 1, we recover exactly the standard Gaussian Kalman filter of Bańbura and Modugno (2014).

Interpretation: a time-varying signal-to-noise ratio. A useful way to read the weighted Kalman filter is as a filter with *time-varying signal-to-noise ratios*. At each time t:

- If w_t^ε is large (the observation is informative about the factor), the update step trusts y_t more — the Kalman gain is large, and the filter moves its estimate significantly towards what y_t suggests.
- If w_t^ε is small (the observation is likely an outlier), the effective noise covariance is inflated, the Kalman gain is small, and the filter largely ignores y_t.
- If w_t^u is small (the innovation itself is likely a heavy-tailed realisation), the predicted covariance is inflated and the filter becomes more willing to accept a large deviation in the factor path.

This behaviour is exactly the kind of *adaptive robustness* that heavy-tailed state-space models are designed to provide. Extreme observations and extreme innovations are accommodated without distorting the estimates in nearby time periods — a property that the Gaussian Kalman filter, with its fixed covariances, does not share.

The Kalman Smoother Conditional on the Weights

The filter produces $\{\hat{f}_{t|t}, P_{t|t}\}$ for $t = 1, \ldots, T$ — the posterior moments of f_t given data *up to time t*. But for the E-step we need the *smoothed* moments, which condition on the entire sample:

$$\hat{f}_{t|T} \;=\; \mathbb{E}[f_t \mid \mathbf{Y}, W^u, W^\varepsilon, \theta], \qquad P_{t|T} \;=\; \mathrm{Cov}(f_t \mid \mathbf{Y}, W^u, W^\varepsilon, \theta).$$

The smoother refines the filter's estimates by exploiting information from future observations, providing a *minimum mean-square-error* (MMSE) estimate of the factor path over the entire sample period $[1, T]$. Intuitively: observations at times $s > t$ carry information about f_t (through the VAR dynamics), and the smoother propagates this backward in time.

Recursion structure. The smoother is a *backward recursion* that runs sequentially from $t = T - 1$ down to $t = 0$, processing one time period at a time in reverse order. At each step, it takes as input the smoothed moments at $t+1$ and the filter outputs at t, and produces as output the smoothed moments at t:

$$\underbrace{\left(\hat{f}_{t+1|T},\ P_{t+1|T}\right)}_{\text{smoothed at } t+1} + \underbrace{\left(\hat{f}_{t|t},\ P_{t|t}\right)}_{\text{filtered at } t} \xrightarrow{\text{smooth}} \underbrace{\left(\hat{f}_{t|T},\ P_{t|T}\right)}_{\text{smoothed at } t}.$$

The recursion is initialised at the end of the sample ($t = T$), where the smoothed moments coincide with the filter's final output: $\hat{f}_{T|T}$ and $P_{T|T}$ are simply taken from the filter, since there are no future observations to incorporate beyond time T.

Smoother depends on filter outputs. A crucial point: the smoother does *not* reprocess the observations y_t directly. It works entirely with the filter outputs — the filtered moments $\{\hat{f}_{t|t}, P_{t|t}\}$ and the predicted moments $\{\hat{f}_{t+1|t}, P_{t+1|t}\}$ already produced by the forward pass. The role of the smoother is to combine these forward-pass quantities with the smoothed moments already computed for later periods, propagating "future information" backward through the dynamics encoded in $\mathbf{A}$.

Concrete example: a sample of length $T = 3$. To make the recursion concrete, consider a sample of length $T = 3$. The forward filter has produced $\{\hat{f}_{t|t}, P_{t|t}\}$ for $t = 0, 1, 2, 3$ and $\{\hat{f}_{t+1|t}, P_{t+1|t}\}$ for $t = 0, 1, 2$. The smoother proceeds as follows:

- **Initialise at $t = 3$:** $\hat{f}^{\text{smooth}}_{3|3} = \hat{f}^{\text{filter}}_{3|3}$ and $P^{\text{smooth}}_{3|3} = P^{\text{filter}}_{3|3}$.
- **Step at $t = 2$:** use $(\hat{f}_{3|3}, P_{3|3})$ (just initialised) and the filter outputs at $t = 2$ to compute $(\hat{f}_{2|3}, P_{2|3})$.
- **Step at $t = 1$:** use $(\hat{f}_{2|3}, P_{2|3})$ (computed in the previous step) and the filter outputs at $t = 1$ to compute $(\hat{f}_{1|3}, P_{1|3})$.
- **Step at $t = 0$:** use $(\hat{f}_{1|3}, P_{1|3})$ and the filter outputs at $t = 0$ to compute $(\hat{f}_{0|3}, P_{0|3})$ — the smoothed moments of the initial state.

At the end of the backward pass, the smoothed moments are available for every $t \in \{0, 1, \ldots, T\}$.

Backward Recursion: the Rauch–Tung–Striebel Smoother

The standard Rauch–Tung–Striebel (RTS) smoother produces the smoothed moments by a backward recursion, initialised at $t = T$ with the filter's final output:

$$\hat{f}^{\text{smooth}}_{T|T} = \hat{f}^{\text{filter}}_{T|T}, \qquad P^{\text{smooth}}_{T|T} = P^{\text{filter}}_{T|T},$$

For $t = T-1, T-2, \ldots, 0$, the smoother computes:

$$J_t = P_{t|t}\, \mathbf{A}'\, P^{-1}_{t+1|t}, \tag{55}$$

the *smoother gain*, analogous to the Kalman gain but for the backward pass. The smoothed mean and covariance are:

$$\boxed{\hat{f}_{t|T} = \hat{f}_{t|t} + J_t \left(\hat{f}_{t+1|T} - \hat{f}_{t+1|t}\right),} \tag{56}$$

$$\boxed{P_{t|T} = P_{t|t} + J_t \left(P_{t+1|T} - P_{t+1|t}\right) J_t'.} \tag{57}$$

The mean update (56) says: take the filtered mean and correct it by the discrepancy between the smoothed and predicted means at $t+1$, propagated backwards via J_t. The covariance update (57) is analogous: the smoother reduces the covariance by the amount of information that future observations contributed.

How the weights enter the smoother. The smoother itself, as written in (55)–(57), does *not* reference $\tilde{\mathbf{Q}}_t$ or $\tilde{\mathbf{R}}_t$ explicitly — it works entirely with the filter outputs $\{P_{t|t}, P_{t+1|t}\}$ and the transition matrix $\mathbf{A}$. The weights' influence on the smoother is therefore *indirect*: they shape the filter outputs $P_{t|t}$ and $P_{t+1|t}$ (via $\tilde{\mathbf{Q}}_t, \tilde{\mathbf{R}}_t$), and these in turn determine J_t and all the smoothed quantities.

This is not a coincidence — the smoother formula derives directly from the linear Gaussian structure of the state-space model, conditional on the weights. The weights enter only through the conditional covariances, and once those are fixed, the smoother proceeds deterministically.

The Lag-One Smoothed Covariance

One additional quantity is needed for the M-step: the smoothed lag-one cross-covariance $P_{t,t-1|T} = \mathrm{Cov}(f_t, f_{t-1} \mid \mathbf{Y})$. This enters the second-order moment $\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}] = P_{t,t-1|T} + \hat{f}_{t|T}\,\hat{f}_{t-1|T}'$, which is required for the update of $\mathbf{A}$ and $\mathbf{Q}$ in the M-step.

The RTS smoother yields this quantity via a separate backward recursion. Initialise at

$$P_{T,T-1|T} \;=\; (I - K_T \mathbf{\Lambda})\,\mathbf{A}\,P_{T-1|T-1},$$

and for $t = T-1, T-2, \ldots, 1$:

$$P_{t,t-1|T} \;=\; P_{t|t}\,J_{t-1}' \;+\; J_t\left(P_{t+1,t|T} - \mathbf{A}\,P_{t|t}\right)J_{t-1}'. \tag{58}$$

The recursion for $P_{t,t-1|T}$ essentially tracks how the uncertainty about the transition from $t-1$ to t is reduced as we move backwards from the end of the sample. While the filtered covariance $P_{t|t}$ only accounts for information up to time t, the term $(P_{t+1,t|T} - \mathbf{A}P_{t|t})$ represents the 'smoothing correction' that flows back from the future estimates.

We omit the full derivation (see de Jong, 1988, or Shumway and Stoffer, 1982) — it is algebraically lengthy but conceptually a simple extension of the main smoother.

Summary of the E-Step So Far

Collecting the results:

- **Given the weights**, the Kalman filter and smoother produce the smoothed factor means $\{\hat{f}_{t|T}\}$, covariances $\{P_{t|T}\}$, and lag-one cross-covariances $\{P_{t,t-1|T}\}$. All required factor moments follow by the identities $\mathbb{E}[f_t f_t' \mid \mathbf{Y}] = P_{t|T} + \hat{f}_{t|T}\hat{f}_{t|T}'$ and $\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}] = P_{t,t-1|T} + \hat{f}_{t|T}\hat{f}_{t-1|T}'$.
- **Given the factors**, the Gaussian–Gamma conjugacy produces the posterior Gamma distributions (44) and (46) in closed form, from which the weight expectations $\hat{w}_t^{\varepsilon}, \hat{w}_t^{u}$ and log-weight expectations $\widehat{\log w}_t^{\varepsilon}, \widehat{\log w}_t^{u}$ follow immediately via (47)–(49).

Neither computation can be performed in isolation: the filter needs the weights to fill in $\tilde{\mathbf{Q}}_t, \tilde{\mathbf{R}}_t$, and the weight formulas need the factor residuals d_t^u, d_t^{ε} to be evaluated. How to resolve this coupling is the subject of the next section.

Resolving the Coupling: ECM with an Inner Iteration

We have reached the technical core of the E-step. The two conditional computations — factor moments given weights, weight moments given factors — are each tractable in isolation, but neither is the unconditional object we actually want. The M-step needs the *unconditional* posterior expectations $\mathbb{E}[\cdot \mid \mathbf{Y}]$, not conditional ones.

Two strategies are possible. The first, is a *single-pass* approximation: evaluate the weights using the factor estimates from the previous outer EM iteration, then run the filter/smoother once. This is computationally cheap and empirically robust, but does not maximise the ELBO exactly within each outer iteration. The second strategy — which we adopt here, following Liu and Rubin (1994) and the ECME literature — is to iterate the filter/smoother and the weight computations *within* each E-step until internal convergence. This resolves the coupling exactly at the cost of an inner loop.

We call this variant the *ECM E-step*, where ECM stands for Expectation-Conditional-Maximisation (Meng and Rubin, 1993). The inner iteration can be viewed as a coordinate ascent on the ELBO with θ held fixed: at each step we maximise over one "coordinate" of the posterior (the weights, or the factors) while holding the other fixed at its current value.

What our "ECM E-step" actually is. A precise positioning of our scheme in the EM family is in order. The classical EM algorithm (Dempster, Laird, and Rubin, 1977) requires the E-step to compute the *exact* posterior $p(\mathcal{Z} \mid \mathbf{Y}, \theta)$ — but in our model this posterior does not factorise across factors and weights, and is not available in closed form.

Two natural sub-problems *are* tractable in isolation. The Kalman filter and smoother, conditional on known weights, deliver in closed form the smoothed factor moments $\hat{f}_{t|T}, P_{t|T}$. Symmetrically, the Gaussian–Gamma conjugacy, conditional on known factor residuals, delivers in closed form the posterior weights $\hat{w}_t$ and log-weights $\widehat{\log w_t}$. Each piece by itself is therefore solvable; what is not solvable in one shot is the joint problem, because the Kalman recursions presuppose the weights as known and the conjugacy presupposes the factors as known — a circular dependency.

The standard remedy is to break the circle by *alternating* between the two sub-problems within the E-step: hold one fixed, solve for the other, swap, repeat. The inner loop converges to the best factorised approximation

$$q(\mathcal{Z}) \;\approx\; q_F(\mathbf{F})\, q_{W^u}(W^u)\, q_{W^\varepsilon}(W^\varepsilon),$$

in which factors and weights are treated as posteriorly independent. This is the standard mean-field variational approximation, and our algorithm is therefore a *variational EM* in the sense of Beal (2003).

The M-step itself follows the expectation-conditional-maximisation logic of Meng and Rubin (1993) and Liu and Rubin (1994): we maximise over disjoint blocks of θ — closed-form weighted-OLS for $\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}$ and a one-dimensional numerical step for ν_u, ν_ε. The combination of a variational E-step (with inner alternation) and a conditional-maximisation M-step is what we mean here by the ECM E-step.

The Inner Loop

The inner loop of the ECM E-step, for a given outer parameter $\theta^{(j)}$, proceeds as follows. Let k index the inner iteration, and denote by $\{\hat{f}_{t|T}^{[k]}, P_{t|T}^{[k]}\}$ and $\{\hat{w}_t^{u,[k]}, \hat{w}_t^{\varepsilon,[k]}\}$ the factor and weight estimates at iteration k of the inner loop.

Inner initialisation ($k=0$). Initialise the weights at their prior mean,

$$\hat{w}_t^{u,[0]} \;=\; 1, \qquad \hat{w}_t^{\varepsilon,[0]} \;=\; 1, \qquad \text{for } t = 1,\dots,T.$$

This is the same convention as the outer initialisation at $j=0$, but applied fresh at each outer iteration: we do not carry forward the weights from the previous outer iteration, since $\theta^{(j)}$ has changed.

Inner iteration ($k \geq 1$). At each inner iteration perform two sub-steps:

1. **(F-update)** Given the current weights $\{\hat{w}_t^{u,[k-1]}, \hat{w}_t^{\varepsilon,[k-1]}\}$, run the Kalman filter (51)–(54) and smoother (56)–(57) with time-varying covariances $\tilde{\mathbf{Q}}_t = \mathbf{Q}^{(j)}/\hat{w}_t^{u,[k-1]}$ and $\tilde{\mathbf{R}}_t = \mathbf{R}^{(j)}/\hat{w}_t^{\varepsilon,[k-1]}$. Obtain the updated smoothed factor moments $\{\hat{f}_{t|T}^{[k]}, P_{t|T}^{[k]}, P_{t,t-1|T}^{[k]}\}$.

2. **(W-update)** Given the current smoothed factor moments, compute the Mahalanobis residuals

$$d_t^{\varepsilon,[k]} \;=\; (y_t - \mathbf{\Lambda}^{(j)}\hat{f}_{t|T}^{[k]})'\,(\mathbf{R}^{(j)})^{-1}\,(y_t - \mathbf{\Lambda}^{(j)}\hat{f}_{t|T}^{[k]}) \;+\; \mathrm{tr}\big((\mathbf{R}^{(j)})^{-1}\,\mathbf{\Lambda}^{(j)}\,P_{t|T}^{[k]}\,(\mathbf{\Lambda}^{(j)})'\big),$$

$$d_t^{u,[k]} \;=\; (\hat{f}_{t|T}^{[k]} - \mathbf{A}^{(j)}\hat{f}_{t-1|T}^{[k]})'\,(\mathbf{Q}^{(j)})^{-1}\,(\hat{f}_{t|T}^{[k]} - \mathbf{A}^{(j)}\hat{f}_{t-1|T}^{[k]}) \;+\; \mathrm{tr}\big((\mathbf{Q}^{(j)})^{-1}\,V_t^{[k]}\big),$$

where $V_t^{[k]} = P_{t|T}^{[k]} + \mathbf{A}^{(j)}P_{t-1|T}^{[k]}\mathbf{A}^{(j)\prime} - \mathbf{A}^{(j)}P_{t,t-1|T}^{[k]\prime} - P_{t,t-1|T}^{[k]}\mathbf{A}^{(j)\prime}$ is the smoothed covariance of the innovation $f_t - \mathbf{A}^{(j)}f_{t-1}$. These residuals account for both the point estimate of the factor and its posterior uncertainty. Then compute the updated weight expectations via (47):

$$\hat{w}_t^{\varepsilon,[k]} \;=\; \frac{\nu_\varepsilon^{(j)} + M}{\nu_\varepsilon^{(j)} + d_t^{\varepsilon,[k]}}, \qquad \hat{w}_t^{u,[k]} \;=\; \frac{\nu_u^{(j)} + r}{\nu_u^{(j)} + d_t^{u,[k]}}.$$

Computing the Mahalanobis residuals in the augmented state. The residuals $d_t^{\varepsilon,[k]}$ and $d_t^{u,[k]}$ defined above are written, for clarity, in terms of the monthly factor $f_t \in \mathbb{R}^r$ and the monthly transition matrix $\mathbf{A}$. In a single-frequency model this is exactly what the Kalman recursions return. In our mixed-frequency setting, however, the temporal aggregation of the quarterly series forces the state to be augmented with four lags of the factor,

$$\tilde{f}_t = \big(f_t', f_{t-1}', f_{t-2}', f_{t-3}', f_{t-4}'\big)' \in \mathbb{R}^{5r},$$

and it is on this augmented state that the filter and smoother operate (we will see this in later sections). Rather than re-deriving the residual formulas directly in the $5r$-dimensional space, we keep the compact monthly expressions and simply read the required monthly quantities off the appropriate blocks of the augmented smoothed moments. This keeps the theory transparent while remaining faithful to the quantities the implementation actually computes.

Factor-side residual. The residual d_t^u involves only the latent factors, through the one-step innovation $f_t - \mathbf{A}f_{t-1}$, and therefore does not interact with the observation equation or the quarterly aggregation at all. The key observation is that the current and lagged factors are *both* contained in the augmented state at the single time index t, by construction of the companion form:

$$\hat{f}_{t|T} = \big(\hat{\tilde{f}}_{t|T}\big)_{1:r}, \qquad \hat{f}_{t-1|T} = \big(\hat{\tilde{f}}_{t|T}\big)_{r+1:2r}.$$

Likewise, the three covariance blocks needed to form the smoothed innovation covariance V_t are sub-blocks of the augmented smoothed covariance $P_{t|T} \in \mathbb{R}^{5r\times 5r}$:

$$\mathrm{Var}(f_t) = \big(P_{t|T}\big)_{1:r,\,1:r}, \quad \mathrm{Var}(f_{t-1}) = \big(P_{t|T}\big)_{r+1:2r,\,r+1:2r}, \quad \mathrm{Cov}(f_t, f_{t-1}) = \big(P_{t|T}\big)_{1:r,\,r+1:2r}.$$

Substituting these blocks into the definition of V_t gives

$$V_t \;=\; \left(P_{t|T}\right)_{1:r,\,1:r} \;+\; \mathbf{A}\left(P_{t|T}\right)_{r+1:2r,\,r+1:2r}\mathbf{A}' \;-\; \mathbf{A}\left(P_{t|T}\right)'_{1:r,\,r+1:2r} \;-\; \left(P_{t|T}\right)_{1:r,\,r+1:2r}\mathbf{A}',$$

so that the entire factor-side residual is assembled from a single augmented covariance matrix at time t, with no need to access the smoothed state at $t-1$ or a separately stored lag-one covariance.

This internal-block construction is not merely a notational convenience: it is numerically equivalent to the alternative route that reads f_{t-1} and its covariance from the smoothed state at time $t-1$ and uses the lag-one smoothed covariance $P_{t,t-1|T}$. We verified that the two routes agree to machine precision (the maximum discrepancy across the sample is of order 10^{-15}). The equivalence is a direct consequence of the companion structure of $\tilde{\mathbf{A}}$, which places a copy of f_{t-1} as the second block of $\tilde{f}_t$: the smoother is internally consistent, so the cross-covariance between the first two blocks of $\tilde{f}_{t|T}$ *is* the lag-one covariance of the monthly factor. We rely on this check rather than assuming it, because an off-by-one error in the block indexing would otherwise propagate silently into every weight update.

Idiosyncratic residual. The residual d_t^{ε} does involve the observation equation, and here the mixed-frequency structure enters through the augmented loading matrix $\tilde{\mathbf{\Lambda}}$. The model-implied observation $\mathbf{\Lambda}\hat{f}_{t|T}$ is computed as $\tilde{\mathbf{\Lambda}}\,\hat{\tilde{f}}_{t|T}$, and the posterior-uncertainty trace term uses the full augmented covariance $P_{t|T}$ together with $\tilde{\mathbf{\Lambda}}$. For the monthly series this reproduces $\mathbf{\Lambda}^M\hat{f}_{t|T}$ exactly, since the lag-blocks of $\tilde{\mathbf{\Lambda}}$ are zero in those rows; for the quarterly series it produces the Mariano–Murasawa weighted aggregate $\frac{1}{3}f_t + \frac{2}{3}f_{t-1} + f_{t-2} + \frac{2}{3}f_{t-3} + \frac{1}{3}f_{t-4}$ of the five monthly factors, which is exactly the quantity the quarterly observation measures. The residual is thus correct for both frequencies under a single expression.

Finally, missing data are handled by the selection matrix $\mathbf{W}_t$, which restricts the residual to the m_t series actually observed at time t and replaces M by m_t in the weight formula (47), as detailed later. This is the same mechanism that unifies the ragged edge and the quarterly mask in the filter: a series is simply absent from the residual whenever it is unobserved, whether because of a publication lag or because it is a quarterly series in a non-quarter-end month. Apart from these substitutions, the residual expressions are identical to the single-frequency, fully-observed case.

Convergence. The inner iteration terminates when the weights stop changing significantly. A standard criterion is

$$\max_t \left|\hat{w}_t^{\varepsilon,[k]} - \hat{w}_t^{\varepsilon,[k-1]}\right| \;+\; \max_t \left|\hat{w}_t^{u,[k]} - \hat{w}_t^{u,[k-1]}\right| \;<\; \varepsilon_{\text{inner}}$$

for some tolerance $\varepsilon_{\text{inner}}$, typically 10^{-4} or 10^{-5}. In practice the inner loop converges quickly — usually in 3 to 10 iterations — because each sub-step is a coordinate-wise optimisation and the landscape is well-behaved.

Monotonicity of the inner loop. An important property is that the inner iteration is *monotone in the ELBO with θ held fixed*. To see this, recall that the ELBO decomposes as

$$\mathcal{L}(q,\theta) \;=\; \ell(\theta) \;-\; D_{\text{KL}}\big(q(\mathcal{Z}) \,\big\|\, p(\mathcal{Z} \mid \mathbf{Y},\theta)\big).$$

Within an inner iteration, $\theta = \theta^{(j)}$ is fixed, so $\ell(\theta^{(j)})$ is a constant. Maximising $\mathcal{L}$ over q therefore means minimising the KL divergence. The inner loop approximates the true posterior by the product of two conditional Gaussians (one for the factors given the weights, one for the weights given the factors), and each sub-step minimises the KL divergence with respect to one conditional while holding the other fixed — a coordinate descent. Coordinate descent on a KL divergence is monotone, so the inner loop indeed increases $\mathcal{L}(q, \theta^{(j)})$ at each step (or leaves it unchanged), until a stationary point is reached.

Instability of the inner fixed point at small ν, and a best-iterate safeguard. The monotonicity just established holds for the ELBO as a function of q, but it does not guarantee that the simple iterate $\left(\hat{f}^{[k]}, \hat{w}^{[k]}\right)$ of the inner loop converges to the fixed point along a contracting path. It should be stressed that the phenomenon discussed here lives entirely *inside the E-step*: it concerns the inner coordinate-ascent loop that, for a *fixed* outer parameter $\theta^{(j)}$, alternates between the factor update (given the weights) and the weight update (given the factors). The outer M-step plays no role; what is at stake is solely whether the inner loop reaches its own stationary point cleanly.

When the degrees of freedom are small, the coordinate-ascent map may have a *locally repelling* fixed point. The diagnostic symptom is characteristic: the inner-loop increment first decreases towards a near-minimum (the iterate approaches the fixed point) and then *grows* again as the iteration overshoots and drifts away, until the iteration cap is reached. The mechanism is the strong non-linearity of the Student-t weights in the Mahalanobis residuals at small ν: a small change in the weights shifts the smoothed factor moments, which in turn shifts the weights by *more* than the previous step, so the local curvature amplifies rather than damps the correction. Heuristically, the regime is most fragile when the relevant residual is computed from few effective dimensions, since the weights are then more reactive to small movements of the factors; aggregating over many series tends to stabilise the corresponding weights. Crucially, this instability is confined to the handful of outer iterations in which a degrees-of-freedom parameter transits the small-ν band: the outer EM loop proceeds normally before and after, and the algorithm converges to the same fixed point regardless.

The iteration cap does not, and cannot, cure the instability. A natural first reaction is to enlarge the inner-loop iteration cap, on the assumption that the loop merely needs more iterations to converge. This intuition is mistaken, and seeing why clarifies the nature of the problem. Because the fixed point is locally repelling, the increment does not oscillate its way down to the tolerance: after touching its minimum it *increases* again. Granting more iterations therefore does not bring the iterate closer to the fixed point — it lets it drift *further away*. Raising the cap leaves the smallest attainable increment essentially unchanged — it is a property of the trajectory's closest approach to the repelling fixed point, not of how long one is willing to iterate — while the final increment at the cap only grows and the location of the transient merely shifts by a few outer iterations. In this band the inner loop simply cannot reach the inner tolerance under plain coordinate ascent, for *any* cap.

Given this, returning the *last* inner iterate would be the worst choice, since it is the one that has drifted farthest from the fixed point, delivering weights inconsistent with $\theta^{(j)}$. Feeding such weights to the M-step yields a $\theta^{(j+1)}$ optimised against an off-fixed-point posterior. We therefore adopt a simple and robust safeguard: rather than the last iterate, the inner loop returns the *best* iterate visited — the one with the smallest increment, i.e. the closest to the fixed point — whenever it exits by reaching the cap without meeting the tolerance. This best-iterate rule hands the M-step the most internally consistent weights available, contains the overshoot, and is inert in the overwhelming majority of outer iterations, where the inner loop converges normally and the best iterate is simply the last one. It does not, however, force the closest approach to satisfy the tolerance exactly: a small residual inconsistency can survive in the transit band, and this is the proximate source of the occasional monotonicity violation discussed next.

On occasional, transient violations of monotonicity. In exact EM the monitored objective is non-decreasing at every iteration by construction. The present algorithm is a *variational* (mean-field) ECM, and the monitored quantity is the full ELBO, which should likewise be non-decreasing in the ideal case. In practice, however, isolated and transient decreases can occur, and it is more honest to acknowledge this than to engineer it away. Two distinct sources are worth separating. The first is the mean-field gap itself: the ELBO is monotone only up to the Kullback–Leibler divergence between the factorised posterior and the true joint posterior, which produces fluctuations of a relative magnitude comparable to floating-point precision — numerically invisible and far below the convergence tolerance. The second, more substantive source is the inner-loop instability described above: in the small-ν transit band the weights handed to the M-step are not exactly at the inner fixed point, so the following outer step can register a small, *genuine* (non-numerical) decrease in the full ELBO. Such an episode is localised to the iteration in which the unstable band is crossed, after which the trajectory resumes its monotone ascent to convergence.

We regard this as an acceptable feature rather than a defect, for three reasons. First, the decrease is tied to a fully understood mechanism — the repelling inner fixed point at small ν — and not to an unidentified error: a coding mistake in a gradient, a sign error, or a mis-normalisation would produce *persistent* or *large* decreases, whereas what is observed is a transient dip confined to a known parameter band. Second, what matters for the validity of the estimate is not bitwise monotonicity at every single step but that the algorithm *converges* and that any violations are *few and small*: a small number of episodes, each of a relative magnitude far below the scale of the optimisation, leaves the converged θ unaffected, as confirmed by the satisfaction of the convergence criterion and by the validity checks on the final parameters. Third, forcing exact monotonicity here would require either a fundamentally different inner solver — for instance a damped or accelerated coordinate ascent designed to tame the repelling region — or an ad hoc suppression of the diagnostic; the former is a disproportionate intervention for a transient of this size, the latter a loss of transparency. The robust and honest choice is therefore to retain the best-iterate safeguard, to monitor the number and magnitude of monotonicity violations as a routine diagnostic, and to treat a fit as sound when those violations are few and small and the algorithm converges to a valid stationary point.

A practical illustration of the phenomenon's locality is provided by the recovery study: on the synthetic panels examined there the EM converges with no monotonicity violations at all, whereas the fit on the real data exhibits a single transient dip. This is not because synthetic data are violation-free by construction, but because the convergence path from a given initialisation may cross the unstable small-ν band more or less abruptly — or, as in those runs, follow a trajectory that does not trigger an off-fixed-point step at the critical iteration. The episode is thus genuinely a property of the path through parameter space, not of the data being real or simulated; its rarity and small magnitude are what matter, and both are borne out in practice.

A warning on approximation. As anticipated, the inner loop converges to the factorised approximation $q(\mathcal{Z}) = q_F(\mathbf{F})q_{W^u}(W^u)q_{W^\varepsilon}(W^\varepsilon)$, not to the exact posterior. In practice the factorisation is sufficiently accurate that the convergence properties of the algorithm mirror those of standard EM, especially for moderately large T. The implications of this approximation for the formal guarantees of the procedure are briefly discussed at the end of the chapter.

Weighted Cross-Moments

We now address the final set of quantities the M-step will need: the *weighted cross-moments*. These are expectations where the latent weights multiply the factor products, and they appear directly in the first-order conditions for the parameter updates. Specifically, we require:

$$\mathbb{E}\big[w_t^\varepsilon y_t f_t' \mid \mathbf{Y}\big], \quad \mathbb{E}\big[w_t^\varepsilon f_t f_t' \mid \mathbf{Y}\big], \quad \mathbb{E}\big[w_t^u f_t f_{t-1}' \mid \mathbf{Y}\big], \quad \mathbb{E}\big[w_t^u f_{t-1} f_{t-1}' \mid \mathbf{Y}\big], \quad \mathbb{E}[w_t^u f_t f_t' \mid \mathbf{Y}].$$

They appear in the M-step updates for $\mathbf{A}$, $\mathbf{\Lambda}$, $\mathbf{Q}$, and $\mathbf{R}$, where the weights enter multiplicatively in front of factor products inside sums over t.

A subtlety: weights and factors are not posterior-independent. One might be tempted to factorise

$$\mathbb{E}\big[\, w_t^u\, f_t f_{t-1}' \,\big|\, \mathbf{Y} \big] \;\stackrel{?}{=}\; \mathbb{E}\big[\, w_t^u \,\big|\, \mathbf{Y} \big] \cdot \mathbb{E}\big[\, f_t f_{t-1}' \,\big|\, \mathbf{Y} \big] \;=\; \hat{w}_t^u \cdot \mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}].$$

This factorisation is *not exact in general*, because w_t^u and (f_t, f_{t-1}) are not independent under the true posterior $p(\cdot \mid \mathbf{Y}, \theta)$. The weight w_t^u depends on the innovation $f_t - \mathbf{A} f_{t-1}$ through the Mahalanobis residual d_t^u, and this residual itself depends on the factors — so conditioning on $\mathbf{Y}$ does not break the dependence between w_t^u and (f_t, f_{t-1}).

However, the mean-field approximation we introduced in the ECM E-step *does* make this factorisation exact. Recall that the mean-field q assumes

$$q(\mathbf{F}, W^u, W^\varepsilon) \;=\; q_F(\mathbf{F})\, q_{W^u}(W^u)\, q_{W^\varepsilon}(W^\varepsilon).$$

Under this factorised q, the factors and the weights are independent by construction, so under this variational distribution q, we have:

$$\mathbb{E}_q\big[\, w_t^u\, f_t f_{t-1}' \,\big] \;=\; \mathbb{E}_q\big[\, w_t^u \,\big] \cdot \mathbb{E}_q\big[\, f_t f_{t-1}' \,\big] \;=\; \hat{w}_t^u \cdot \big(P_{t,t-1|T} + \hat{f}_{t|T} \hat{f}_{t-1|T}' \big).$$

This is the formula we will use in the M-step. As elsewhere in this chapter, the formula is written under the working assumption that $\mathbf{Y}$ contains no missing entries; the modifications introduced by missing data are treated in later sections.

Summary of the M-step substitutions. Putting everything together, the substitutions the E-step provides to the M-step are as follows. Wherever the complete-data log-likelihood (33) contains a latent term, the M-step replaces it by the corresponding object listed below:

Latent term	**E-step substitution**
f_t	$\hat{f}_{t\|T}$
$f_t f_t'$	$P_{t\|T} + \hat{f}_{t\|T} \hat{f}_{t\|T}'$
$f_t f_{t-1}'$	$P_{t,t-1\|T} + \hat{f}_{t\|T} \hat{f}_{t-1\|T}'$
w_t^u	$\hat{w}_t^u$
w_t^ε	$\hat{w}_t^\varepsilon$
$\log w_t^u$	$\widehat{\log w}_t^{\,u}$
$\log w_t^\varepsilon$	$\widehat{\log w}_t^{\,\varepsilon}$
$w_t^\varepsilon f_t f_t'$	$\hat{w}_t^\varepsilon \cdot (P_{t\|T} + \hat{f}_{t\|T} \hat{f}_{t\|T}')$
$w_t^u f_t f_t'$	$\hat{w}_t^u \cdot (P_{t\|T} + \hat{f}_{t\|T} \hat{f}_{t\|T}')$
$w_t^u f_t f_{t-1}'$	$\hat{w}_t^u \cdot (P_{t,t-1\|T} + \hat{f}_{t\|T} \hat{f}_{t-1\|T}')$
$w_t^u f_{t-1} f_{t-1}'$	$\hat{w}_t^u \cdot (P_{t-1\|T} + \hat{f}_{t-1\|T} \hat{f}_{t-1\|T}')$
$w_t^\varepsilon y_t f_t'$	$\hat{w}_t^\varepsilon \cdot y_t \hat{f}_{t\|T}'$

These substitutions, combined with the closed-form factor moments from the Kalman smoother and the closed-form weight moments from the Gaussian–Gamma conjugacy, fully equip the M-step to proceed. In the next section we carry out the M-step derivation in detail, deriving each parameter update one by one.

Summary of the E-Step

We close this section by summarising the E-step algorithmically. For each outer iteration indexed by j:

1. **Initialise the inner loop**: $k \leftarrow 0$, set $\hat{w}_t^{u,[0]} = \hat{w}_t^{\varepsilon,[0]} = 1$ for all t.
2. **Inner iteration**: for $k = 1, 2, \ldots$ until convergence:
   (a) Run the Kalman filter (51)–(54) and smoother (56)–(57) with weighted covariances $\tilde{\mathbf{Q}}_t = \mathbf{Q}^{(j)} / \hat{w}_t^{u,[k-1]}$ and $\tilde{\mathbf{R}}_t = \mathbf{R}^{(j)} / \hat{w}_t^{\varepsilon,[k-1]}$.
   (b) Compute smoothed factor moments $\{\hat{f}_{t|T}^{[k]}, P_{t|T}^{[k]}, P_{t,t-1|T}^{[k]}\}$.
   (c) Compute Mahalanobis residuals $d_t^{\varepsilon,[k]}, d_t^{u,[k]}$ and updated weights $\hat{w}_t^{\varepsilon,[k]}, \hat{w}_t^{u,[k]}$ (and their log-expectations) via (47)–(49).
   (d) Check convergence; if not converged, set $k \leftarrow k + 1$ and go to (a).
3. **Output** the converged factor and weight moments: $\{\hat{f}_{t|T}, P_{t|T}, P_{t,t-1|T}\}$, $\{\hat{w}_t^\varepsilon, \hat{w}_t^u\}$, and $\{\widehat{\log w}_t^{\,\varepsilon}, \widehat{\log w}_t^{\,u}\}$.

These are the inputs to the M-step, which we develop next.

A Note on the Degrees of Freedom. It is important to underscore that during the entirety of the E-step—including the Kalman recursions—the degrees-of-freedom parameters ν_u and ν_ε are held fixed at their current estimates $\theta^{(j)}$. These parameters define the "shape" of the heavy tails used to compute the posterior weights. Only once the E-step has reached convergence and provided the required expectations (such as $\widehat{\log w_t}$) will the subsequent M-step update the values of ν to better fit the observed distribution of residuals. This iterative refinement—using ν to find outliers, then using outliers to update ν—is the mechanism by which the EM algorithm learns the optimal level of robustness for the model.

The M-Step: Deriving the Parameter Update Equations

We have now all the ingredients needed to derive the M-step. The E-step produced, for the current parameter vector $\theta^{(j)}$, the posterior expectations of all latent quantities appearing in the complete-data log-likelihood. The M-step uses these expectations to compute the next parameter vector $\theta^{(j+1)}$ by maximising the *expected* complete-data log-likelihood over θ.

In this section we carry out this maximisation for each component of $\theta = \{\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}, \nu_u, \nu_\varepsilon\}$, one parameter at a time. For the four structural parameters $\{\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}\}$ the update equations will admit *closed-form solutions* — weighted versions of the OLS-type estimators that arise in the Gaussian DFM. For the two degrees-of-freedom parameters $\{\nu_u, \nu_\varepsilon\}$, no closed form exists (the digamma function appears in the first-order condition), and we will resort to a one-dimensional numerical root-finding step, following the ECME construction of Liu and Rubin (1994).

Setup: The Expected Complete-Data Log-Likelihood

Recall from (33) the complete-data log-likelihood:

$$
\begin{aligned}
\ell_c(\theta) \;=\; & -\tfrac{1}{2}\log|\mathbf{\Sigma}_0| \;-\; \tfrac{1}{2} f_0'\mathbf{\Sigma}_0^{-1} f_0 \\
& -\tfrac{T}{2}\log|\mathbf{Q}| \;+\; \tfrac{r}{2}\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{1}{2}\operatorname{tr}\Big[\mathbf{Q}^{-1}\sum_{t=1}^{T} w_t^u\,(f_t-\mathbf{A}f_{t-1})(f_t-\mathbf{A}f_{t-1})'\Big] \\
& -\tfrac{T}{2}\log|\mathbf{R}| \;+\; \tfrac{M}{2}\sum_{t=1}^{T}\log w_t^\varepsilon \;-\; \tfrac{1}{2}\operatorname{tr}\Big[\mathbf{R}^{-1}\sum_{t=1}^{T} w_t^\varepsilon\,(y_t-\mathbf{\Lambda}f_t)(y_t-\mathbf{\Lambda}f_t)'\Big] \\
& +\; T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma\big(\tfrac{\nu_u}{2}\big)\big] \;+\; \big(\tfrac{\nu_u}{2}-1\big)\sum_{t=1}^{T}\log w_t^u \;-\; \tfrac{\nu_u}{2}\sum_{t=1}^{T} w_t^u \\
& +\; T\big[\tfrac{\nu_\varepsilon}{2}\log\tfrac{\nu_\varepsilon}{2} - \log\Gamma\big(\tfrac{\nu_\varepsilon}{2}\big)\big] \;+\; \big(\tfrac{\nu_\varepsilon}{2}-1\big)\sum_{t=1}^{T}\log w_t^\varepsilon \;-\; \tfrac{\nu_\varepsilon}{2}\sum_{t=1}^{T} w_t^\varepsilon.
\end{aligned}
$$

In the M-step we cannot maximise $\ell_c(\theta)$ directly because it contains the latent variables $\mathbf{F}, W^u, W^\varepsilon$. Following the variational logic, we instead maximise the *expected* complete-data log-likelihood

$$
L(\theta, \theta^{(j)}) \;\equiv\; \mathbb{E}_{\theta^{(j)}}\big[\,\ell_c(\theta)\,\big|\,\mathbf{Y}\,\big], \tag{59}
$$

where the expectation is taken under the posterior computed in the E-step at iteration j. Substituting each latent quantity by its posterior expectation, we get:

$$
\begin{aligned}
L(\theta, \theta^{(j)}) \;=\;& \underbrace{-\tfrac{1}{2}\log|\boldsymbol{\Sigma}_0| \;-\; \tfrac{1}{2}\,\mathbb{E}[f_0'\boldsymbol{\Sigma}_0^{-1}f_0 \mid \mathbf{Y}]}_{\text{initial-state block}} \\
& \underbrace{-\tfrac{T}{2}\log|\mathbf{Q}| \;-\; \tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{Q}^{-1}\,\mathcal{S}^{u}_{ff}(\mathbf{A})\Big]}_{\text{state-transition block}} \\
& \underbrace{-\tfrac{T}{2}\log|\mathbf{R}| \;-\; \tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{R}^{-1}\,\mathcal{S}^{\varepsilon}_{yy}(\boldsymbol{\Lambda})\Big]}_{\text{observation block}} \\
& + \underbrace{\text{terms involving } \nu_u, \widehat{\log w}^{\,u}_t, \hat{w}^u_t}_{\text{Block D}} \\
& + \underbrace{\text{terms involving } \nu_\varepsilon, \widehat{\log w}^{\,\varepsilon}_t, \hat{w}^\varepsilon_t}_{\text{Block E}} + \text{const},
\end{aligned} \tag{60}
$$

where we introduced shorthand notation for the two *weighted smoothed sums of squared residuals*:

$$
\mathcal{S}^{u}_{ff}(\mathbf{A}) \;\equiv\; \sum_{t=1}^{T} \mathbb{E}\big[\, w^u_t\,(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})' \,\big|\, \mathbf{Y}\big], \tag{61}
$$

$$
\mathcal{S}^{\varepsilon}_{yy}(\boldsymbol{\Lambda}) \;\equiv\; \sum_{t=1}^{T} \mathbb{E}\big[\, w^\varepsilon_t\,(y_t - \boldsymbol{\Lambda}f_t)(y_t - \boldsymbol{\Lambda}f_t)' \,\big|\, \mathbf{Y}\big]. \tag{62}
$$

These objects are the analogues, for the Student-t DFM, of the unweighted sums that appear in the Gaussian case. The *weighted* adjective is central: every contribution at time t is multiplied by the corresponding posterior weight expectation $\hat{w}^u_t$ or $\hat{w}^\varepsilon_t$ (see (35)).

The decoupling of parameters in the M-step. Looking at (60) or at (35) one sees that the parameters decouple cleanly across the blocks:

- $\boldsymbol{\Sigma}_0$ appears only in the initial-state block;
- $\mathbf{A}$ and $\mathbf{Q}$ appear only in the state-transition block;
- $\boldsymbol{\Lambda}$ and $\mathbf{R}$ appear only in the observation block;
- ν_u appears only in Block D;
- ν_ε appears only in Block E.

This means we can maximise $L(\theta, \theta^{(j)})$ over each parameter *independently* of the others — the maximisation splits into several independent sub-problems. Below we tackle them one by one.

Sequential conditional maximisation. A further simplification, exploited by Bańbura and Modugno (2014), is that within the state-transition block the parameters $\mathbf{A}$ and $\mathbf{Q}$ can be updated sequentially: first maximise over $\mathbf{A}$ with $\mathbf{Q}$ held at its current value, then update $\mathbf{Q}$ using the new $\mathbf{A}$. The analogous sequential update holds for $(\boldsymbol{\Lambda}, \mathbf{R})$. The reason for this sequential scheme is purely practical: a fully simultaneous maximisation over $(\mathbf{A}, \mathbf{Q})$ — or $(\boldsymbol{\Lambda}, \mathbf{R})$ — would require solving a coupled non-linear system in which the optimal $\mathbf{Q}$ depends on $\mathbf{A}$ through the residuals, and the optimal $\mathbf{A}$ depends on $\mathbf{Q}$ through the weighting. Updating sequentially decouples the two problems: each becomes a standard weighted-OLS update with closed-form solution. This is formally a *conditional-maximisation* step rather than a full simultaneous maximisation, but the two yield the same fixed point and the monotone convergence of EM is preserved (Meng and Rubin, 1993). We adopt this sequential scheme throughout.

Update of the VAR Transition Matrix A

We derive the update for $\mathbf{A}$ in full detail. This derivation is *the prototype* for the other three structural updates $(\mathbf{\Lambda}, \mathbf{Q}, \mathbf{R})$: once the logic is clear here, the others follow by direct analogy.

Isolate the terms in $L(\theta, \theta^{(j)})$ that depend on A. Looking at (60), only the state-transition block contains $\mathbf{A}$. More precisely, $\mathbf{A}$ appears inside $\mathcal{S}^u_{ff}(\mathbf{A})$; all other parts of $L(\theta, \theta^{(j)})$ are constants with respect to $\mathbf{A}$. So the relevant objective is:

$$L_{\mathbf{A}}(\mathbf{A}) \;=\; -\tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{Q}^{-1}\,\mathcal{S}^u_{ff}(\mathbf{A})\Big] \;+\; \text{const.} \tag{63}$$

(We have dropped the $-\frac{T}{2}\log|\mathbf{Q}|$ term as well, since $\mathbf{Q}$ is held fixed during the $\mathbf{A}$-update.)

Expand the quadratic inside the expectation. Write out the product $(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})'$:

$$(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})' \;=\; f_t f_t' \;-\; f_t f_{t-1}'\mathbf{A}' \;-\; \mathbf{A}f_{t-1}f_t' \;+\; \mathbf{A}f_{t-1}f_{t-1}'\mathbf{A}'.$$

Multiplying by w^u_t and taking the expectation under the posterior (using the mean-field factorisation established before, so that w and ff' separate):

$$\begin{aligned}\mathbb{E}\big[\,w^u_t\,(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})' \,\big|\, \mathbf{Y}\,\big] \;=\; &\hat{w}^u_t\,\mathbb{E}[f_t f_t' \mid \mathbf{Y}] \;-\; \hat{w}^u_t\,\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}]\,\mathbf{A}' \\ &-\; \mathbf{A}\,\hat{w}^u_t\,\mathbb{E}[f_{t-1}f_t' \mid \mathbf{Y}] \;+\; \mathbf{A}\,\hat{w}^u_t\,\mathbb{E}[f_{t-1}f_{t-1}' \mid \mathbf{Y}]\,\mathbf{A}'.\end{aligned}$$

The matrix $\mathbf{A}$ is a fixed (non-random) parameter in the M-step, so it passes through the expectation unchanged.

We introduce compact notation for the sums. Define the following *weighted* posterior moments, each a fixed $r \times r$ matrix computed in the E-step:

$$\boxed{\begin{aligned}\mathcal{P}^u_{11} \;&\equiv\; \sum_{t=1}^{T} \hat{w}^u_t\,\mathbb{E}[f_t f_t' \mid \mathbf{Y}],\\ \mathcal{P}^u_{10} \;&\equiv\; \sum_{t=1}^{T} \hat{w}^u_t\,\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}],\\ \mathcal{P}^u_{00} \;&\equiv\; \sum_{t=1}^{T} \hat{w}^u_t\,\mathbb{E}[f_{t-1} f_{t-1}' \mid \mathbf{Y}].\end{aligned}} \tag{64}$$

Note the superscript u: these sums are weighted by $\hat{w}^u_t$, the factor-side weights, because they come from Block B of the complete log-likelihood. By transposing, $(\mathcal{P}^u_{10})' = \sum_t \hat{w}^u_t\,\mathbb{E}[f_{t-1}f_t' \mid \mathbf{Y}]$.

Substituting, the sum inside the trace becomes:

$$\mathcal{S}^u_{ff}(\mathbf{A}) \;=\; \mathcal{P}^u_{11} \;-\; \mathcal{P}^u_{10}\,\mathbf{A}' \;-\; \mathbf{A}\,(\mathcal{P}^u_{10})' \;+\; \mathbf{A}\,\mathcal{P}^u_{00}\,\mathbf{A}'.$$

The objective (63) becomes:

$$L_{\mathbf{A}}(\mathbf{A}) \;=\; -\tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{Q}^{-1}\big(\mathcal{P}^u_{11} - \mathcal{P}^u_{10}\mathbf{A}' - \mathbf{A}(\mathcal{P}^u_{10})' + \mathbf{A}\mathcal{P}^u_{00}\mathbf{A}'\big)\Big] \;+\; \text{const.} \tag{65}$$

Differentiate with respect to A. We now apply matrix calculus to differentiate the trace expression. The three relevant matrix derivative identities are:

- $\dfrac{\partial}{\partial X}\,\mathrm{tr}(AX'B) \;=\; BA\,,$
- $\dfrac{\partial}{\partial X}\,\mathrm{tr}(AXB) \;=\; A'B'\,,$
- $\dfrac{\partial}{\partial X}\,\mathrm{tr}(AXBX') \;=\; AXB' \;+\; A'XB\,,$

with the special case (for B symmetric) $\dfrac{\partial}{\partial X}\,\mathrm{tr}(AXBX') = 2AXB$ when A is also symmetric. We apply these identities term by term.

Term 1: $\mathrm{tr}(\mathbf{Q}^{-1}\mathcal{P}^u_{11})$ does not depend on $\mathbf{A}$; its derivative is zero.

Term 2: $\mathrm{tr}(\mathbf{Q}^{-1}\mathcal{P}^u_{10}\mathbf{A}')$ is of the form $\mathrm{tr}(AX'B)$ with $A = \mathbf{Q}^{-1}\mathcal{P}^u_{10}$, $X = \mathbf{A}$, $B = I$. But the trace of a matrix equals the trace of its transpose, so $\mathrm{tr}(\mathbf{Q}^{-1}\mathcal{P}^u_{10}\mathbf{A}') = \mathrm{tr}(\mathbf{A}(\mathcal{P}^u_{10})'\mathbf{Q}^{-1})$. This is of the form $\mathrm{tr}(XA)$ with $A = (\mathcal{P}^u_{10})'\mathbf{Q}^{-1}$, whose derivative with respect to X is $A' = \mathbf{Q}^{-1}\mathcal{P}^u_{10}$ (using the symmetry of $\mathbf{Q}$).

Term 3: $\mathrm{tr}(\mathbf{Q}^{-1}\mathbf{A}(\mathcal{P}^u_{10})')$ is of the form $\mathrm{tr}(AXB)$ with $A = \mathbf{Q}^{-1}$, $X = \mathbf{A}$, $B = (\mathcal{P}^u_{10})'$. Its derivative is $A'B' = \mathbf{Q}^{-1}\,\mathcal{P}^u_{10}$ (again using symmetry of $\mathbf{Q}$).

So Terms 2 and 3 contribute the same derivative, $\mathbf{Q}^{-1}\,\mathcal{P}^u_{10}$ — which makes sense because the two traces are equal by the cyclic property of trace. Their combined contribution is $2\,\mathbf{Q}^{-1}\,\mathcal{P}^u_{10}$.

Term 4: $\mathrm{tr}(\mathbf{Q}^{-1}\mathbf{A}\mathcal{P}^u_{00}\mathbf{A}')$ is of the form $\mathrm{tr}(AXBX')$ with $A = \mathbf{Q}^{-1}$, $X = \mathbf{A}$, $B = \mathcal{P}^u_{00}$. Both A and B are symmetric, so its derivative is $2\,\mathbf{Q}^{-1}\,\mathbf{A}\,\mathcal{P}^u_{00}$.

Assembling:

$$\frac{\partial}{\partial \mathbf{A}}\,\mathrm{tr}\Big[\mathbf{Q}^{-1}\big(\mathcal{P}^u_{11} - \mathcal{P}^u_{10}\mathbf{A}' - \mathbf{A}(\mathcal{P}^u_{10})' + \mathbf{A}\mathcal{P}^u_{00}\mathbf{A}'\big)\Big] \;=\; 0 \;-\; \mathbf{Q}^{-1}\mathcal{P}^u_{10} \;-\; \mathbf{Q}^{-1}\mathcal{P}^u_{10} \;+\; 2\mathbf{Q}^{-1}\mathbf{A}\mathcal{P}^u_{00}.$$

Simplifying, and including the $-\frac{1}{2}$ prefactor from $L_{\mathbf{A}}$:

$$\frac{\partial L_{\mathbf{A}}}{\partial \mathbf{A}} \;=\; -\tfrac{1}{2}\,\big(-2\,\mathbf{Q}^{-1}\mathcal{P}^u_{10} + 2\,\mathbf{Q}^{-1}\mathbf{A}\mathcal{P}^u_{00}\big) \;=\; \mathbf{Q}^{-1}\,\mathcal{P}^u_{10} \;-\; \mathbf{Q}^{-1}\,\mathbf{A}\,\mathcal{P}^u_{00}. \tag{66}$$

Set the gradient to zero. The first-order condition is

$$\mathbf{Q}^{-1}\,\mathcal{P}^u_{10} \;-\; \mathbf{Q}^{-1}\,\mathbf{A}\,\mathcal{P}^u_{00} \;=\; 0.$$

Left-multiply by $\mathbf{Q}$ (positive definite, hence invertible):

$$\mathcal{P}^u_{10} \;-\; \mathbf{A}\,\mathcal{P}^u_{00} \;=\; 0 \qquad \Longrightarrow \qquad \mathbf{A}\,\mathcal{P}^u_{00} \;=\; \mathcal{P}^u_{10}.$$

Right-multiply by $(\mathcal{P}^u_{00})^{-1}$ (invertibility of which is generically guaranteed when $T \gg r$ and the factors are not collinear):

$$\boxed{\mathbf{A}^{(j+1)} \;=\; \mathcal{P}^u_{10}\,(\mathcal{P}^u_{00})^{-1} \;=\; \left(\sum_{t=1}^{T} \hat{w}^u_t\,\mathbb{E}[f_t f'_{t-1} \mid \mathbf{Y}]\right)\left(\sum_{t=1}^{T} \hat{w}^u_t\,\mathbb{E}[f_{t-1} f'_{t-1} \mid \mathbf{Y}]\right)^{-1}.} \tag{67}$$

Interpretation. Compare (67) with the corresponding Gaussian update from the standard DFM of Bańbura and Modugno (2014):

$$\mathbf{A}^{(j+1)}_{\text{Gaussian}} \;=\; \left(\sum_{t=1}^{T} \mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}]\right)\left(\sum_{t=1}^{T} \mathbb{E}[f_{t-1} f_{t-1}' \mid \mathbf{Y}]\right)^{-1}.$$

The two differ by exactly one ingredient: the presence of $\hat{w}_t^u$ as a multiplicative weight inside every sum. The structure of the estimator is otherwise *identical*.

Statistically, (67) is the estimator you would obtain by running a *weighted least squares regression* of f_t on f_{t-1}, where observation t receives weight $\hat{w}_t^u$. Time periods with a small $\hat{w}_t^u$ — that is, periods in which the factor innovation $f_t - \mathbf{A} f_{t-1}$ is large in Mahalanobis sense — contribute less to the estimate. This is the exact mechanism by which the Student-t assumption down-weights outlier innovations without any explicit outlier-detection step: the weights are computed automatically in the E-step, and the M-step simply uses them.

The Gaussian limit $\nu_u \to \infty$ recovers $\hat{w}_t^u \to 1$ for all t, and (67) collapses to the unweighted Gaussian formula. The heavy-tailed extension is therefore a strict generalisation: the Gaussian case is embedded as a special case of the Student-t case, obtained by setting $\nu_u = \infty$.

A check on the dimensions. As a sanity check: $\mathbf{A}$ is $r \times r$. Both $\mathcal{P}_{10}^u$ and $\mathcal{P}_{00}^u$ are $r \times r$. The product $\mathcal{P}_{10}^u (\mathcal{P}_{00}^u)^{-1}$ is therefore $r \times r$, as required.

Update of the Factor Loading Matrix $\boldsymbol{\Lambda}$

The derivation of the update for $\boldsymbol{\Lambda}$ mirrors the one for $\mathbf{A}$ almost exactly, with the observation equation replacing the state equation. The analogy is perfect: wherever $\mathbf{A}$ multiplied f_{t-1}, now $\boldsymbol{\Lambda}$ multiplies f_t; wherever $\hat{w}_t^u$ weighted the state-side quantities, now $\hat{w}_t^\varepsilon$ weights the observation-side quantities; wherever $\mathbf{Q}$ appeared, now $\mathbf{R}$ appears.

Isolate the terms in $L(\theta, \theta^{(j)})$ that depend on $\boldsymbol{\Lambda}$. Looking at (60), only the observation block contains $\boldsymbol{\Lambda}$:

$$L_{\boldsymbol{\Lambda}}(\boldsymbol{\Lambda}) \;=\; -\tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{R}^{-1}\,\mathcal{S}_{yy}^{\varepsilon}(\boldsymbol{\Lambda})\Big] \;+\; \text{const.} \tag{68}$$

As in the $\mathbf{A}$-update, the $-\frac{T}{2}\log|\mathbf{R}|$ term is dropped since $\mathbf{R}$ is held fixed during the $\boldsymbol{\Lambda}$-update.

Expand the quadratic inside the expectation.

$$(y_t - \boldsymbol{\Lambda} f_t)(y_t - \boldsymbol{\Lambda} f_t)' \;=\; y_t y_t' \;-\; y_t f_t' \boldsymbol{\Lambda}' \;-\; \boldsymbol{\Lambda} f_t y_t' \;+\; \boldsymbol{\Lambda} f_t f_t' \boldsymbol{\Lambda}'.$$

Multiplying by w_t^ε and taking the posterior expectation (using the same mean-field factorisation exploited for $\mathbf{A}$, so that weights and factor products separate):

$$\begin{aligned}
\mathbb{E}\big[\, w_t^\varepsilon\,(y_t - \boldsymbol{\Lambda} f_t)(y_t - \boldsymbol{\Lambda} f_t)' \,\big|\, \mathbf{Y} \,\big] \;=\;& \hat{w}_t^\varepsilon\, y_t y_t' \;-\; \hat{w}_t^\varepsilon\, y_t\, \mathbb{E}[f_t' \mid \mathbf{Y}]\,\boldsymbol{\Lambda}' \\
&-\; \boldsymbol{\Lambda}\,\hat{w}_t^\varepsilon\,\mathbb{E}[f_t \mid \mathbf{Y}]\, y_t' \;+\; \boldsymbol{\Lambda}\,\hat{w}_t^\varepsilon\,\mathbb{E}[f_t f_t' \mid \mathbf{Y}]\,\boldsymbol{\Lambda}'.
\end{aligned}$$

Two simplifications are automatic: y_t is observed and hence passes through the expectation as a constant; $\boldsymbol{\Lambda}$ is a fixed parameter and passes through too.

Introduce compact notation for the sums. Define, in full analogy with (64):

$$\boxed{\begin{aligned} \mathcal{S}^{\varepsilon}_{yf} &\equiv \sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\, y_t\, \mathbb{E}[f_t' \mid \mathbf{Y}], \\ \mathcal{P}^{\varepsilon}_{11} &\equiv \sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\, \mathbb{E}[f_t f_t' \mid \mathbf{Y}], \\ \mathcal{S}^{\varepsilon}_{yy} &\equiv \sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\, y_t y_t'. \end{aligned}} \tag{69}$$

Note the superscript ε: these sums are weighted by $\hat{w}^{\varepsilon}_t$ (idiosyncratic-side weights), because they come from Block A. The cross-moment matrix $\mathcal{S}^{\varepsilon}_{yf}$ is $M \times r$, while $\mathcal{P}^{\varepsilon}_{11}$ is $r \times r$.

Note also that $\mathcal{P}^{\varepsilon}_{11} \neq \mathcal{P}^{u}_{11}$ in general — same second-order factor moments, but weighted by different sequences of weights. This is a critical distinction that arises only in the Student-t DFM (in the Gaussian case all weights are 1 and the two sums coincide).

Substituting into (68):

$$L_{\mathbf{\Lambda}}(\mathbf{\Lambda}) = -\tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{R}^{-1}\Big(\mathcal{S}^{\varepsilon}_{yy} - \mathcal{S}^{\varepsilon}_{yf}\mathbf{\Lambda}' - \mathbf{\Lambda}(\mathcal{S}^{\varepsilon}_{yf})' + \mathbf{\Lambda}\,\mathcal{P}^{\varepsilon}_{11}\,\mathbf{\Lambda}'\Big)\Big] + \text{const.} \tag{70}$$

Differentiate with respect to $\mathbf{\Lambda}$. The structure of (70) is identical to that of (65), with the replacements $\mathbf{A} \to \mathbf{\Lambda}$, $\mathbf{Q} \to \mathbf{R}$, $\mathcal{P}^{u}_{11} \to \mathcal{S}^{\varepsilon}_{yy}$, $\mathcal{P}^{u}_{10} \to \mathcal{S}^{\varepsilon}_{yf}$, $\mathcal{P}^{u}_{00} \to \mathcal{P}^{\varepsilon}_{11}$. Applying the same matrix calculus identities term by term:

$$\frac{\partial L_{\mathbf{\Lambda}}}{\partial \mathbf{\Lambda}} = \mathbf{R}^{-1}\,\mathcal{S}^{\varepsilon}_{yf} - \mathbf{R}^{-1}\,\mathbf{\Lambda}\,\mathcal{P}^{\varepsilon}_{11}. \tag{71}$$

Set the gradient to zero and solve. Imposing (71) equal to zero, left-multiplying by $\mathbf{R}$, and right-multiplying by $(\mathcal{P}^{\varepsilon}_{11})^{-1}$:

$$\boxed{\mathbf{\Lambda}^{(j+1)} = \mathcal{S}^{\varepsilon}_{yf}\,(\mathcal{P}^{\varepsilon}_{11})^{-1} = \left(\sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\, y_t\, \mathbb{E}[f_t' \mid \mathbf{Y}]\right)\left(\sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\, \mathbb{E}[f_t f_t' \mid \mathbf{Y}]\right)^{-1}.} \tag{72}$$

Interpretation. Equation (72) is the *weighted OLS estimator of a multivariate regression of y_t on f_t*, with observation weights $\hat{w}^{\varepsilon}_t$. Time periods in which y_t is far (in Mahalanobis sense) from its factor-implied value $\mathbf{\Lambda} f_t$ — that is, periods classified as idiosyncratic outliers — receive a small $\hat{w}^{\varepsilon}_t$ and contribute less to the estimate.

The contrast with the Gaussian DFM is clean: the Bańbura–Modugno (2014) formula

$$\mathbf{\Lambda}^{(j+1)}_{\text{Gaussian}} = \left(\sum_{t=1}^{T} y_t\, \mathbb{E}[f_t' \mid \mathbf{Y}]\right)\left(\sum_{t=1}^{T} \mathbb{E}[f_t f_t' \mid \mathbf{Y}]\right)^{-1}$$

is recovered in the Gaussian limit $\nu_{\varepsilon} \to \infty$, for which $\hat{w}^{\varepsilon}_t \to 1$.

A note on missing data. The formula (72) is written under the complete-data assumption maintained throughout this chapter; the modifications introduced by missing entries in $\mathbf{Y}$ are treated later, where the row-by-row update structure of $\mathbf{\Lambda}$ takes a different form.

A check on the dimensions. $\mathbf{\Lambda}$ is $M \times r$. The numerator $\mathcal{S}_{yf}^{\varepsilon}$ is $M \times r$; the denominator $(\mathcal{P}_{11}^{\varepsilon})^{-1}$ is $r \times r$. Their product is $M \times r$, as required.

Update of the Factor Innovation Scale Matrix Q

The update for $\mathbf{Q}$ is structurally different from those for $\mathbf{A}$ and $\mathbf{\Lambda}$: rather than a linear parameter inside a quadratic form, $\mathbf{Q}$ is a *covariance-like matrix* whose update arises from balancing a determinant penalty against a weighted sum of outer products. The derivation uses a standard matrix calculus result due to Magnus and Neudecker.

Isolate the terms in $L(\theta, \theta^{(j)})$ that depend on Q. From the state-transition block of (60), the $\mathbf{Q}$-dependent terms are:

$$L_{\mathbf{Q}}(\mathbf{Q}) \;=\; -\tfrac{T}{2}\,\log|\mathbf{Q}| \;-\; \tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{Q}^{-1}\,\mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)})\Big] \;+\; \text{const.} \tag{73}$$

Two remarks are in order.

First, the sum $\mathcal{S}_{ff}^{u}$ is now evaluated at the *updated* $\mathbf{A}^{(j+1)}$ from (67) rather than at $\mathbf{A}^{(j)}$. This is the sequential conditional maximisation scheme we discussed before: we first update $\mathbf{A}$, then use the updated value when updating $\mathbf{Q}$.

Second, both the log-determinant term and the trace term now depend on $\mathbf{Q}$. This is a genuine optimisation in $\mathbf{Q}$, not a simple regression.

Simplify the sum using the new $\mathbf{A}^{(j+1)}$. Expand $\mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)})$ using the same passages of the $\mathbf{A}$ update

$$\mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)}) \;=\; \mathcal{P}_{11}^{u} \;-\; \mathcal{P}_{10}^{u}\,\mathbf{A}^{(j+1)\prime} \;-\; \mathbf{A}^{(j+1)}\,(\mathcal{P}_{10}^{u})' \;+\; \mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime}.$$

Substituting the first-order condition (67), namely $\mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u} = \mathcal{P}_{10}^{u}$ and hence $(\mathcal{P}_{10}^{u})' = \mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime}$:

$$\begin{aligned}
\mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)}) \;&=\; \mathcal{P}_{11}^{u} \;-\; \mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime} \;-\; \mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime} \;+\; \mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime} \\
&=\; \mathcal{P}_{11}^{u} \;-\; \mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime}.
\end{aligned}$$

Equivalently, using $\mathbf{A}^{(j+1)}\,\mathcal{P}_{00}^{u}\,\mathbf{A}^{(j+1)\prime} = \mathbf{A}^{(j+1)}\,(\mathcal{P}_{10}^{u})'$:

$$\mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)}) \;=\; \mathcal{P}_{11}^{u} \;-\; \mathbf{A}^{(j+1)}\,(\mathcal{P}_{10}^{u})'. \tag{74}$$

This simplified form — valid *only at the updated* $\mathbf{A}$ — is the one we will use. It reflects the fact that at the optimum the "cross-term" is absorbed into the diagonal-like structure of the residuals.

The objective (73) becomes:

$$L_{\mathbf{Q}}(\mathbf{Q}) \;=\; -\tfrac{T}{2}\,\log|\mathbf{Q}| \;-\; \tfrac{1}{2}\,\mathrm{tr}\big[\mathbf{Q}^{-1}\,C_{\mathbf{Q}}\big] \;+\; \text{const},$$

where we have defined the fixed $r \times r$ matrix

$$C_{\mathbf{Q}} \;\equiv\; \mathcal{S}_{ff}^{u}(\mathbf{A}^{(j+1)}) \;=\; \mathcal{P}_{11}^{u} \;-\; \mathbf{A}^{(j+1)}\,(\mathcal{P}_{10}^{u})'. \tag{75}$$

Differentiate using the standard covariance-matrix identity. The following two matrix calculus identities, due to Magnus and Neudecker, are the workhorses of this kind of derivation. For a symmetric positive-definite matrix $\mathbf{S}$:

- $\dfrac{\partial}{\partial \mathbf{S}} \log|\mathbf{S}| \;=\; \mathbf{S}^{-1},$
- $\dfrac{\partial}{\partial \mathbf{S}} \operatorname{tr}(\mathbf{S}^{-1}C) \;=\; -\,\mathbf{S}^{-1}\,C\,\mathbf{S}^{-1} \quad$ (for C symmetric).

Applying these to $L_{\mathbf{Q}}(\mathbf{Q})$:

$$\frac{\partial L_{\mathbf{Q}}}{\partial \mathbf{Q}} \;=\; -\tfrac{T}{2}\,\mathbf{Q}^{-1} \;-\; \tfrac{1}{2}\big(-\,\mathbf{Q}^{-1}\,C_{\mathbf{Q}}\,\mathbf{Q}^{-1}\big) \;=\; -\tfrac{T}{2}\,\mathbf{Q}^{-1} \;+\; \tfrac{1}{2}\,\mathbf{Q}^{-1}\,C_{\mathbf{Q}}\,\mathbf{Q}^{-1}.$$

Set the gradient to zero and solve. The first-order condition is

$$-\tfrac{T}{2}\,\mathbf{Q}^{-1} \;+\; \tfrac{1}{2}\,\mathbf{Q}^{-1}\,C_{\mathbf{Q}}\,\mathbf{Q}^{-1} \;=\; 0.$$

Multiplying both sides by 2 on the left:

$$-\,T\,\mathbf{Q}^{-1} \;+\; \mathbf{Q}^{-1}\,C_{\mathbf{Q}}\,\mathbf{Q}^{-1} \;=\; 0.$$

Left-multiplying by $\mathbf{Q}$:

$$-\,T\,I_r \;+\; C_{\mathbf{Q}}\,\mathbf{Q}^{-1} \;=\; 0 \qquad\Longrightarrow\qquad C_{\mathbf{Q}}\,\mathbf{Q}^{-1} \;=\; T\,I_r.$$

Finally right-multiplying by $\mathbf{Q}$:

$$C_{\mathbf{Q}} \;=\; T\,\mathbf{Q} \qquad\Longrightarrow\qquad \mathbf{Q} \;=\; \tfrac{1}{T}\,C_{\mathbf{Q}}.$$

Substituting back the definition of $C_{\mathbf{Q}}$:

$$\boxed{\mathbf{Q}^{(j+1)} \;=\; \frac{1}{T}\Big(\mathcal{P}^{u}_{11} \;-\; \mathbf{A}^{(j+1)}\,(\mathcal{P}^{u}_{10})'\Big) \;=\; \frac{1}{T}\sum_{t=1}^{T} \hat{w}^{u}_{t}\,\mathbb{E}\big[\,(f_t - \mathbf{A}^{(j+1)} f_{t-1})(f_t - \mathbf{A}^{(j+1)} f_{t-1})' \,\big|\, \mathbf{Y}\big].} \tag{76}$$

The two expressions on the right are equal by (74); we give both because the first is the form most convenient for computation, while the second is the form that makes the statistical interpretation transparent.

Interpretation. Equation (76) is the *weighted sample covariance of the smoothed factor innovations*. Each period contributes its squared innovation $(f_t - \mathbf{A}^{(j+1)} f_{t-1})(f_t - \mathbf{A}^{(j+1)} f_{t-1})'$ evaluated under the posterior, weighted by $\hat{w}^{u}_{t}$. Periods with small $\hat{w}^{u}_{t}$ (large innovations, the heavy-tailed events) contribute less to the estimated scale.

A more detailed discussion of the normalisation $1/T$ — and why it does not coincide with the alternative $1/\sum_t \hat{w}^{u}_{t}$ — is deferred to the dedicated paragraph below.

The link between the Gaussian model and our model is governed by the degrees of freedom ν_u. In the Gaussian limit, where $\nu_u \to \infty$, the tails of the state-innovation distribution thin out and the probability of "shocks" being treated as outliers vanishes. Mathematically, this implies that the posterior expectations of the state-side weights collapse to unity:

$$\hat{w}^{u}_{t} \to 1 \quad \text{for all } t = 1, \ldots, T. \tag{77}$$

When these weights are substituted into our updates, the weighted sums $\mathcal{P}_{11}^u$ and $\mathcal{P}_{10}^u$ revert to the standard unweighted moments used in the Gaussian EM algorithm. A point of potential confusion arises when comparing our update for $\mathbf{Q}$ in (76) with the expression provided by Bańbura and Modugno (2014) for the Gaussian DFM:

$$\mathbf{Q}^{(j+1)} = \frac{1}{T}\left(\sum_{t=1}^{T} \mathbb{E}[f_t f_t' \mid \mathbf{Y}] - \mathbf{A}^{(j+1)} \sum_{t=1}^{T} \mathbb{E}[f_{t-1} f_t' \mid \mathbf{Y}]\right). \tag{78}$$

At first glance, this might look different from our simplified form $\frac{1}{T}(\mathcal{P}_{11}^u - \mathbf{A}^{(j+1)}(\mathcal{P}_{10}^u)')$. However, the two are algebraically identical once the first-order condition for $\mathbf{A}^{(j+1)}$ is satisfied. Our formulation is simply the "compact" version of the same estimator, generalized to include the weight $\hat{w}_t^u$.

On the normalisation $1/T$: a closer look. The $1/T$ factor in (76) merits a careful explanation, because at first sight it might seem inconsistent with the weighted-average logic that one would expect from a robust estimator. Three points clarify the situation.

The $1/T$ comes from the first-order condition, not from counting weights. The expected log-likelihood $L_{\mathbf{Q}}(\mathbf{Q}) = -\frac{T}{2}\log|\mathbf{Q}| - \frac{1}{2}\mathrm{tr}(\mathbf{Q}^{-1}C_{\mathbf{Q}})$ contains the term $-\frac{T}{2}\log|\mathbf{Q}|$ as a determinant penalty, and the factor T here is precisely the number of time periods — each period contributes one observation equation in the underlying complete-data log-likelihood, and there are T of them. The MLE balances this T against the trace term, yielding $\mathbf{Q}^{(j+1)} = C_{\mathbf{Q}}/T$. The denominator T is therefore *structural*, not statistical: it reflects the dimension of the observation set, not the sum of posterior weights.

Weights act differentially, not globally. The posterior weights $\hat{w}_t^u$ enter *inside* the sum that defines $C_{\mathbf{Q}}$, multiplying each squared innovation $(f_t - \mathbf{A}f_{t-1})(f_t - \mathbf{A}f_{t-1})'$ at its own time period. Their effect is to give different weight to different periods within the average — outlier periods (large d_t^u, hence small $\hat{w}_t^u$) contribute less, while typical periods ($\hat{w}_t^u \approx 1$) contribute essentially as in the Gaussian case. The robustness of the Student-t estimator comes from this *differential weighting* across t, not from a uniform global rescaling. Outlier down-weighting and the $1/T$ normalisation are therefore two independent ingredients: the first reshapes the sum, the second sets its scale.

Large-sample consistency: $\sum_t \hat{w}_t^u \approx T$. There is, however, a useful asymptotic relation between the two ingredients that explains why the Student-t and Gaussian estimators have comparable global magnitudes. Recall that $\hat{w}_t^u = (\nu_u + r)/(\nu_u + d_t^u)$, where d_t^u is the smoothed Mahalanobis residual. Under the true Student-t data-generating process, d_t^u/r follows a scaled F-type distribution with expected value $\nu_u/(\nu_u - 2)$, so that

$$\mathbb{E}[d_t^u] \;=\; r \cdot \frac{\nu_u}{\nu_u - 2}.$$

A first-order Taylor expansion of $\hat{w}_t^u$ around this expected value (equivalently, a Jensen-type approximation) gives

$$\mathbb{E}[\hat{w}_t^u] \;\approx\; \frac{\nu_u + r}{\nu_u + \mathbb{E}[d_t^u]} \;=\; \frac{\nu_u + r}{\nu_u + r\,\nu_u/(\nu_u - 2)} \;\xrightarrow{\nu_u \to \infty}\; 1.$$

For finite ν_u the relation is approximate but tight: at $\nu_u = 10$ and $r = 3$, for instance, $\mathbb{E}[d_t^u] \approx 3.75$ and $\mathbb{E}[\hat{w}_t^u] \approx 0.95$ — close to one but slightly below. By the law of large numbers, the sample average of the weights converges to its expected value, so

$$\frac{1}{T}\sum_{t=1}^{T} \hat{w}_t^u \;\xrightarrow{T \to \infty}\; \mathbb{E}[\hat{w}_t^u] \;\approx\; 1, \qquad \text{equivalently} \qquad \sum_{t=1}^{T} \hat{w}_t^u \;\approx\; T \quad \text{for large } T.$$

This is a property of the Gamma posterior under the conjugacy structure: the distribution of d_t^u is calibrated such that the sum of posterior weights matches the sample size up to a small finite-ν_u correction. Practically, this means that the *global* magnitude of $\mathbf{Q}^{(j+1)}$ remains well-calibrated regardless of the particular configuration of weights — outlier down-weighting redistributes mass across periods but does not bias the overall scale of the estimate.

Equivalence with a weighted-mean formula. Combining the two facts above, in large samples the formula (76) can be rewritten as

$$\mathbf{Q}^{(j+1)} \;\approx\; \frac{1}{\sum_t \hat{w}_t^u} C_{\mathbf{Q}} \;\approx\; \mathbb{E}_w[\,(f_t - \mathbf{A} f_{t-1})(f_t - \mathbf{A} f_{t-1})'\,]_{\text{weighted}},$$

where the right-hand side is the weighted sample average of the innovations, with weights $\hat{w}_t^u / \sum_s \hat{w}_s^u$. The estimator therefore admits both interpretations: a maximum-likelihood formula with $1/T$ normalisation, and (asymptotically) a weighted sample mean of the innovations. The two coincide as $T \to \infty$ because $\sum_t \hat{w}_t^u / T \to 1$.

This equivalence is also why $\mathbf{Q}^{(j+1)}$ collapses smoothly to the Gaussian formula (78) in the limit $\nu_u \to \infty$: in that limit, every $\hat{w}_t^u \to 1$ exactly, so $\sum_t \hat{w}_t^u = T$ exactly, and the two interpretations coincide for any sample size. For finite ν_u, the two interpretations differ in a small finite-sample correction that is negligible for moderately large T.

A check on the dimensions. $C_{\mathbf{Q}}$ is $r \times r$, divided by the scalar T: the result is $r \times r$, consistent with $\mathbf{Q}$.

Update of the Idiosyncratic Scale Matrix R

The update for $\mathbf{R}$ parallels that for $\mathbf{Q}$ but with one important wrinkle: $\mathbf{R}$ is *diagonal* by assumption. This constraint must be imposed at the maximisation step; if we simply applied the formulas used for $\mathbf{Q}$, we would obtain a full symmetric matrix. The projection onto the diagonal class is straightforward, but let us be explicit about how it arises.

Isolate the terms in $L(\theta, \theta^{(j)})$ that depend on R. From the observation block of (60):

$$L_{\mathbf{R}}(\mathbf{R}) \;=\; -\tfrac{T}{2} \log |\mathbf{R}| \;-\; \tfrac{1}{2} \operatorname{tr}\!\Big[\mathbf{R}^{-1}\, \mathcal{S}_{yy}^{\varepsilon}(\mathbf{\Lambda}^{(j+1)})\Big] \;+\; \text{const}, \tag{79}$$

where $\mathcal{S}_{yy}^{\varepsilon}(\mathbf{\Lambda}^{(j+1)})$ is evaluated at the updated $\mathbf{\Lambda}^{(j+1)}$ from the previous step.

Simplify the sum using the updated $\mathbf{\Lambda}^{(j+1)}$. Expanding the quadratic and using the first-order condition (72), namely $\mathbf{\Lambda}^{(j+1)}\, \mathcal{P}_{11}^{\varepsilon} = \mathcal{S}_{yf}^{\varepsilon}$ and hence $(\mathcal{S}_{yf}^{\varepsilon})' = \mathcal{P}_{11}^{\varepsilon}\, \mathbf{\Lambda}^{(j+1)\prime}$, we obtain (by the same algebra as in the $\mathbf{Q}$-update):

$$\mathcal{S}_{yy}^{\varepsilon}(\mathbf{\Lambda}^{(j+1)}) \;=\; \mathcal{S}_{yy}^{\varepsilon} \;-\; \mathbf{\Lambda}^{(j+1)}\,(\mathcal{S}_{yf}^{\varepsilon})'. \tag{80}$$

This is the $M \times M$ matrix of weighted sample residual outer products, at the updated loadings. Define $C_{\mathbf{R}} \equiv \mathcal{S}_{yy}^{\varepsilon}(\mathbf{\Lambda}^{(j+1)})$.

Unconstrained maximisation (if R were free). If we treated $\mathbf{R}$ as a free symmetric matrix (no diagonality constraint), the same argument as for $\mathbf{Q}$ would give

$$\mathbf{R}^{\text{unconstr.},(j+1)} \;=\; \tfrac{1}{T}\, C_{\mathbf{R}} \;=\; \tfrac{1}{T}\,\Big(\mathcal{S}_{yy}^{\varepsilon} \;-\; \mathbf{\Lambda}^{(j+1)}\,(\mathcal{S}_{yf}^{\varepsilon})'\Big),$$

a full $M \times M$ symmetric matrix.

Impose the diagonality constraint. The DGP assumption requires $\mathbf{R}$ diagonal. We therefore impose the constraint explicitly. The restricted maximiser is obtained by retaining only the diagonal entries of the unconstrained solution:

$$\boxed{\mathbf{R}^{(j+1)} \;=\; \operatorname{diag}\big(\tfrac{1}{T}\,C_{\mathbf{R}}\big) \;=\; \operatorname{diag}\Big(\tfrac{1}{T}\,\big(\mathcal{S}^{\varepsilon}_{yy} \;-\; \mathbf{\Lambda}^{(j+1)}\,(\mathcal{S}^{\varepsilon}_{yf})'\big)\Big)\,,} \tag{81}$$

where the $\operatorname{diag}(\cdot)$ operator extracts the diagonal of its argument and embeds it back as a diagonal matrix, zeroing off-diagonal entries.

Why the diagonal projection is correct. That the restricted maximiser is indeed the diagonal of the unrestricted one can be verified directly. Write $\mathbf{R} = \operatorname{diag}(r_1, \ldots, r_M)$ and differentiate $L_{\mathbf{R}}$ with respect to each r_i separately. The log-determinant becomes $-\frac{T}{2}\sum_i \log r_i$, and the trace becomes $-\frac{1}{2}\sum_i (C_{\mathbf{R}})_{ii}/r_i$. Setting $\partial L_{\mathbf{R}}/\partial r_i = 0$ yields $r_i = (C_{\mathbf{R}})_{ii}/T$, i.e. the i-th diagonal entry of the unrestricted solution — confirming (81).

Equivalent residual-based form. Equation (81) can be written more transparently in terms of *smoothed idiosyncratic residuals*. Define, for each series i and each time t, the smoothed residual

$$\hat{\varepsilon}^{(j+1)}_{i,t} \;=\; y_{i,t} \;-\; \mathbf{\Lambda}^{(j+1)}_{i\cdot}\,\hat{f}_{t|T},$$

where $\mathbf{\Lambda}^{(j+1)}_{i\cdot}$ is the i-th row of $\mathbf{\Lambda}^{(j+1)}$. Then the i-th diagonal entry of $\mathbf{R}^{(j+1)}$ is approximately (exactly, up to the posterior variance of f_t, which is absorbed into $\mathcal{S}^{\varepsilon}_{yy}$)

$$r^{(j+1)}_i \;\approx\; \frac{1}{T}\sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\,\big(\hat{\varepsilon}^{(j+1)}_{i,t}\big)^2.$$

Each diagonal entry is the *weighted mean squared smoothed residual* for series i: observations with small $\hat{w}^{\varepsilon}_t$ (outliers) contribute less to the estimated idiosyncratic variance, exactly as in a robust regression.

Two equivalent forms and comparison with Bańbura and Modugno (2014). The update (81) admits two equivalent but pedagogically distinct expressions. The first highlights the residual-based interpretation:

$$\mathbf{R}^{(j+1)} \;=\; \operatorname{diag}\left(\frac{1}{T}\sum_{t=1}^{T} \hat{w}^{\varepsilon}_t\,\mathbb{E}\big[\,(y_t - \mathbf{\Lambda}^{(j+1)} f_t)(y_t - \mathbf{\Lambda}^{(j+1)} f_t)'\,\big|\,\mathbf{Y}\,\big]\right). \tag{82}$$

By expanding the quadratic form and substituting the first-order condition $\mathbf{\Lambda}^{(j+1)}\,\mathcal{P}^{\varepsilon}_{11} = \mathcal{S}^{\varepsilon}_{yf}$, this becomes the equivalent "compact" form most convenient for computation, which is the form already given in (81).

Both expressions are the robust generalisation of the standard Gaussian update for the idiosyncratic scale. The algebraic structure is identical to that of Bańbura and Modugno (2014), with one critical difference: every term inside the sums is multiplied by the posterior weight $\hat{w}^{\varepsilon}_t$. In the Gaussian limit $\nu_\varepsilon \to \infty$, all weights collapse to one and the original BM estimator is recovered exactly. For finite ν_ε, periods identified as idiosyncratic outliers receive $\hat{w}^{\varepsilon}_t < 1$ and contribute less to the estimate, shielding the diagonal entries of $\mathbf{R}$ from the variance inflation typically caused by extreme shocks.

It is worth reiterating that $\mathbf{R}^{(j+1)}$ is the *scale matrix* of the idiosyncratic distribution, not its covariance. In the Gaussian case the two coincide; here, the conditional covariance of ε_t at time t is $\mathbf{R}/w^{\varepsilon}_t$, and the unconditional covariance is $\frac{\nu_\varepsilon}{\nu_\varepsilon - 2}\,\mathbf{R}$. The discussion of the normalisation $1/T$ developed for $\mathbf{Q}$ applies verbatim to $\mathbf{R}$ and is not repeated here.

A check on the dimensions. $C_{\mathbf{R}}$ is $M \times M$; its diagonal projection is $M \times M$ diagonal, consistent with the assumed structure of $\mathbf{R}$.

Interim Summary: The Updates for the Structural Parameters

We have now derived in full the M-step updates for the four structural parameters. They are, in the order in which they should be computed:

$$\begin{aligned}
\mathbf{A}^{(j+1)} &= \mathcal{P}_{10}^{u}\,(\mathcal{P}_{00}^{u})^{-1}, \\
\mathbf{\Lambda}^{(j+1)} &= \mathcal{S}_{yf}^{\varepsilon}\,(\mathcal{P}_{11}^{\varepsilon})^{-1}, \\
\mathbf{Q}^{(j+1)} &= \tfrac{1}{T}\,\big(\mathcal{P}_{11}^{u} - \mathbf{A}^{(j+1)}\,(\mathcal{P}_{10}^{u})'\big), \\
\mathbf{R}^{(j+1)} &= \operatorname{diag}\Big(\tfrac{1}{T}\,\big(\mathcal{S}_{yy}^{\varepsilon} - \mathbf{\Lambda}^{(j+1)}\,(\mathcal{S}_{yf}^{\varepsilon})'\big)\Big).
\end{aligned}$$

All four are closed-form expressions. Every sum is weighted by the posterior weight expectations $\hat{w}_t^u$ or $\hat{w}_t^\varepsilon$ from the E-step. In the Gaussian limit $\nu_u, \nu_\varepsilon \to \infty$ the weights collapse to 1 and the four updates reduce to the Gaussian DFM updates of Bańbura and Modugno (2014).

What remains, and what cannot be done in closed form, is the update of the two degrees-of-freedom parameters (ν_u, ν_ε). We develop that derivation in the next section.

Update of the Degrees of Freedom ν_u and ν_ε

We now turn to the two parameters that cannot be updated in closed form: the degrees of freedom ν_u and ν_ε. The reason is structural: these parameters enter the complete-data log-likelihood through terms involving the function $\log\Gamma(\nu/2)$, whose derivative is the *digamma function* ψ — a transcendental function that does not admit algebraic manipulation in the same way as polynomials or rationals. As a consequence, setting the first-order condition to zero produces an *implicit equation* in ν, solvable only numerically.

This is the technical reason why the algorithm we construct is not an ordinary EM but rather an *ECME* (Expectation-Conditional-Maximisation Either) algorithm in the sense of Liu and Rubin (1994): "conditional maximisation" because different parameters are maximised one at a time; "either" because the maximisation for ν is not performed by closed-form analytical maximisation but by numerical methods on the first-order condition. The convergence guarantees of EM carry through to ECME without modification, provided each conditional maximisation is performed exactly.

Isolating the ν_u-Dependent Terms

We focus on ν_u; the derivation for ν_ε is identical with the subscript substitution $u \to \varepsilon$ and the dimension substitution $r \to M$.

Looking at the expected complete-data log-likelihood (60), ν_u appears only in Block D — all other blocks are constants with respect to ν_u. The ν_u-relevant terms are:

$$L_{\nu_u}(\nu_u) \;=\; T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma\big(\tfrac{\nu_u}{2}\big)\big] \;+\; \big(\tfrac{\nu_u}{2} - 1\big)\,\mathbb{E}_{\theta^{(j)}}\big[\sum_t \log w_t^u \,\big|\, \mathbf{Y}\big] \;-\; \tfrac{\nu_u}{2}\,\mathbb{E}_{\theta^{(j)}}\big[\sum_t w_t^u \,\big|\, \mathbf{Y}\big]. \quad (83)$$

Using the shorthand (the posterior weight moments computed in the E-step):

$$\bar{w}^u \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\hat{w}_t^u, \qquad \overline{\log w}^u \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\widehat{\log w}_t^{\,u},$$

the objective becomes, after dividing by T (which does not affect the argmax):

$$\frac{L_{\nu_u}(\nu_u)}{T} = \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma\big(\tfrac{\nu_u}{2}\big) + \big(\tfrac{\nu_u}{2} - 1\big)\,\overline{\log w}^{u} - \tfrac{\nu_u}{2}\,\varpi^{u}. \tag{84}$$

The two quantities ϖ^u and $\overline{\log w}^{u}$ are fixed *numerical constants* at the time of the ν_u-update: they have been computed in the E-step from the current parameter values and do not depend on ν_u itself. The objective is therefore a function of a *single scalar variable* ν_u, on which we now perform a one-dimensional maximisation.

The First-Order Condition

Differentiate $L_{\nu_u}(\nu_u)$ with respect to ν_u. Recall the definition of the digamma function:

$$\psi(x) \equiv \frac{d}{dx}\log\Gamma(x),$$

and note the chain rule: $\frac{d}{d\nu_u}\log\Gamma(\nu_u/2) = \frac{1}{2}\,\psi(\nu_u/2)$. We differentiate the four terms in (84) one at a time.

Term 1: $\frac{d}{d\nu_u}\big[\frac{\nu_u}{2}\log\frac{\nu_u}{2}\big]$. Apply the product rule with $u = \nu_u/2$, so that $u\log u$ and $\frac{du}{d\nu_u} = 1/2$:

$$\frac{d}{d\nu_u}\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2}\big] = \tfrac{1}{2}\log\tfrac{\nu_u}{2} + \tfrac{\nu_u}{2}\cdot\tfrac{1}{\nu_u/2}\cdot\tfrac{1}{2} = \tfrac{1}{2}\log\tfrac{\nu_u}{2} + \tfrac{1}{2}.$$

Term 2: $\frac{d}{d\nu_u}[-\log\Gamma(\nu_u/2)] = -\frac{1}{2}\,\psi(\nu_u/2)$.

Term 3: $\frac{d}{d\nu_u}\Big[\big(\frac{\nu_u}{2} - 1\big)\,\overline{\log w}^{u}\Big] = \frac{1}{2}\,\overline{\log w}^{u}$.

Term 4: $\frac{d}{d\nu_u}\big[-\frac{\nu_u}{2}\,\varpi^u\big] = -\frac{1}{2}\,\varpi^u$.

Summing the four contributions:

$$\frac{d}{d\nu_u}\left[\frac{L_{\nu_u}(\nu_u)}{T}\right] = \tfrac{1}{2}\log\tfrac{\nu_u}{2} + \tfrac{1}{2} - \tfrac{1}{2}\,\psi\big(\tfrac{\nu_u}{2}\big) + \tfrac{1}{2}\,\overline{\log w}^{u} - \tfrac{1}{2}\,\varpi^u. \tag{85}$$

Set the derivative equal to zero. Setting (85) to zero and multiplying by 2:

$$\boxed{\log\tfrac{\nu_u}{2} - \psi\big(\tfrac{\nu_u}{2}\big) + 1 + \overline{\log w}^{u} - \varpi^u = 0.} \tag{86}$$

This is the first-order condition for ν_u. It is a *scalar implicit equation*: the unknown ν_u appears in two places, one logarithmic and one digamma, and no algebraic manipulation reduces this to a closed form. Identical reasoning for ν_ε gives:

$$\log\tfrac{\nu_\varepsilon}{2} - \psi\big(\tfrac{\nu_\varepsilon}{2}\big) + 1 + \overline{\log w}^{\varepsilon} - \varpi^\varepsilon = 0, \tag{87}$$

where ϖ^ε and $\overline{\log w}^{\varepsilon}$ are the corresponding E-step summaries on the idiosyncratic side.

The function to root-find. To solve (86) we define the function

$$g_u(\nu_u) \equiv \log\tfrac{\nu_u}{2} - \psi\big(\tfrac{\nu_u}{2}\big) + 1 + \overline{\log w}^{u} - \varpi^u, \tag{88}$$

and look for the root $g_u(\nu_u) = 0$ on $\nu_u > 0$. We now establish some structural properties of g_u that guarantee the root exists, is unique, and can be found efficiently by numerical root-finding.

Properties of the First-Order Condition

Before attacking the numerical root-finding, it is worth pausing on two properties of g_u: monotonicity (ensures the root is unique) and the sign of the constant term (ensures the root exists).

Monotonicity. Differentiating g_u with respect to ν_u:

$$\frac{dg_u}{d\nu_u} = \frac{1}{2} \cdot \frac{1}{\nu_u/2} - \frac{1}{2}\psi'(\tfrac{\nu_u}{2}) = \frac{1}{\nu_u} - \frac{1}{2}\psi'(\tfrac{\nu_u}{2}),$$

where $\psi'(x) = \frac{d}{dx}\psi(x)$ is the *trigamma function* — the second derivative of $\log \Gamma$.

A classical inequality (see for example Abramowitz and Stegun, 1964) states that for all $x > 0$,

$$\psi'(x) > \frac{1}{x},$$

so $\frac{1}{2}\psi'(\nu_u/2) > 1/\nu_u$, and therefore

$$\frac{dg_u}{d\nu_u} < 0 \quad \text{for all } \nu_u > 0.$$

The function g_u is *strictly monotonically decreasing* on $(0, \infty)$. This implies that it has *at most one root*.

Existence of the root: signs at the boundaries. We check the limits.

As $\nu_u \to 0^+$: $\log(\nu_u/2) \to -\infty$ and $\psi(\nu_u/2) \to -\infty$, but the rate at which ψ diverges dominates — specifically, $\psi(x) \sim -1/x$ as $x \to 0^+$, whereas $\log(x) \to -\infty$ more slowly. So $\log(\nu_u/2) - \psi(\nu_u/2) \to +\infty$, and $g_u(\nu_u) \to +\infty$.

As $\nu_u \to \infty$: A standard asymptotic expansion gives $\psi(x) = \log(x) - 1/(2x) + O(1/x^2)$ as $x \to \infty$. So $\log(\nu_u/2) - \psi(\nu_u/2) = 1/\nu_u + O(1/\nu_u^2) \to 0^+$, and

$$g_u(\nu_u) \longrightarrow 0 + 1 + \overline{\log w}^u - \bar{w}^u = 1 + \left(\overline{\log w}^u - \bar{w}^u\right).$$

Jensen's inequality: for any random positive w, we have $\mathbb{E}[\log w] \leq \log \mathbb{E}[w]$, so in posterior expectation $\overline{\log w}^u \leq \log \bar{w}^u$, and since $\log x \leq x - 1$ for $x > 0$ we have $\log \bar{w}^u \leq \bar{w}^u - 1$. Therefore

$$\overline{\log w}^u - \bar{w}^u \leq -1,$$

with equality only in the degenerate case where all $\hat{w}_t^u = 1$. In all non-trivial cases we have $\overline{\log w}^u - \bar{w}^u < -1$, which gives

$$\lim_{\nu_u \to \infty} g_u(\nu_u) = 1 + \left(\overline{\log w}^u - \bar{w}^u\right) < 0.$$

Conclusion. The function g_u is continuous on $(0, \infty)$, strictly monotonically decreasing, with $g_u(\nu_u) \to +\infty$ at $\nu_u \to 0^+$ and $g_u(\nu_u) \to$ negative value as $\nu_u \to \infty$. By the intermediate value theorem, the equation $g_u(\nu_u) = 0$ has *exactly one root* on $(0, \infty)$. This root is the update $\nu_u^{(j+1)}$.

The only degenerate case is when $\overline{\log w}^u = \bar{w}^u - 1 + \delta$ with $\delta \to 0$ (that is, when the weights are essentially all 1). In this regime, $g_u(\nu_u) \to 0$ at $\nu_u \to \infty$, and the root escapes to infinity — which correctly corresponds to the Gaussian limit $\nu_u \to \infty$. We will discuss below how to handle this case in practice.

The Numerical Root-Finding Step: Brent's Method

With the root $\nu_u^{(j+1)}$ guaranteed to exist and be unique, the remaining task is computational: find it numerically. There are two standard options.

Option 1: Newton's method. Given the monotonicity, one could apply Newton's method to g_u:

$$\nu_u^{[k+1]} = \nu_u^{[k]} - \frac{g_u(\nu_u^{[k]})}{g_u'(\nu_u^{[k]})} = \nu_u^{[k]} - \frac{g_u(\nu_u^{[k]})}{1/\nu_u^{[k]} - (1/2)\,\psi'(\nu_u^{[k]}/2)}.$$

Newton's method has quadratic convergence near the root and is extremely fast. However, it has two drawbacks in our context:

- it requires evaluating the trigamma function ψ', which adds computational cost (and numerical care near $\nu_u \to 0$ where $\psi'(\nu_u/2) \sim 4/\nu_u^2$ explodes);
- without safeguards, Newton can overshoot into negative values of ν_u or fail to converge from a bad starting point. For example, when $\hat{\nu}_u^{(j)}$ happens to be very small and the first Newton step is large, the iterate $\nu_u^{[1]}$ may exit the positive real line.

Option 2: Brent's method (our choice). *Brent's method* (Brent, 1973) is a robust univariate root-finder that combines three ideas:

1. *Bisection*, which always halves a bracketing interval and is guaranteed to converge linearly;
2. *Secant method*, which uses a linear interpolation between two iterates to accelerate convergence;
3. *Inverse quadratic interpolation*, which uses three iterates and a quadratic fit to achieve near-Newton convergence rates without requiring derivative evaluations.

Brent's method guarantees convergence to a root *provided an initial bracketing interval* $[a, b]$ *is given* in which $g_u(a)$ and $g_u(b)$ have opposite signs. The algorithm adaptively chooses among the three methods at each iteration: if the interpolation step is well-behaved (strictly inside the bracket, with sufficient improvement) it is accepted; otherwise the algorithm falls back to bisection to maintain the bracket.

In our setting, a safe bracket is provided by the monotonicity and limit analysis above. For instance, one can use

$$[a, b] = [\,\nu_u^{\min}, \nu_u^{\max}\,] = [\,2.001, 1000\,],$$

where $\nu_u^{\min} > 2$ ensures the second moment of the Student-t exists (a natural constraint), and $\nu_u^{\max} = 1000$ is large enough to be effectively "Gaussian". The user should verify at the start of each iteration that $g_u(\nu_u^{\min}) > 0$ and $g_u(\nu_u^{\max}) < 0$; if the signs do not bracket, it means the root lies outside $[a, b]$ and one should extend the interval accordingly. In practice, if $g_u(\nu^{\max}) > 0$ (the derivative is still positive at the upper bound), the optimum lies above $\nu^{\max}$, and we set $\nu_u^{(j+1)} = \nu^{\max}$, effectively treating the process as Gaussian.

Why Brent over Newton. The trade-off between the two methods is clear. Newton converges in fewer iterations (often 5–10) but can fail catastrophically without safeguards. Brent requires more iterations (typically 20–40) but is *guaranteed* to converge as long as the bracket is valid. Importantly, Brent does *not* require evaluating ψ' — only ψ and log — so each function evaluation is cheaper. For an EM algorithm where the ν-update is performed many times, the additional Brent iterations are a small overhead, and the robustness is worth much more.

For these reasons we adopt Brent's method. In Python, the call `scipy.optimize.brentq(g_u, a, b)` implements this algorithm and returns the root within machine precision.

Initialisation of the bracket. A practical detail: the bracket $[2.001, 1000]$ is conservative but sometimes wasteful. A more efficient choice uses the previous iteration's value as a starting point and constructs a narrow bracket around it:

$$[a, b] \;=\; \big[\,\max(2.001, \nu_u^{(j)}/2),\, \min(1000, 2\,\nu_u^{(j)})\,\big],$$

checking the sign condition and expanding the interval if it fails. This reduces the expected number of Brent iterations, especially once the algorithm is near convergence.

The Idiosyncratic-Side Update: ν_ε

The derivation for ν_ε is, in its structure, identical to the one we have just carried out for ν_u: the relevant terms of the expected complete-data log-likelihood come from Block E (the Gamma prior on W^ε), and the resulting first-order condition has the same functional form as (86). All the properties established above — monotonicity, uniqueness of the root, and the applicability of Brent's method — carry over without modification. Nevertheless, a few quantitative differences between the two cases deserve a brief explicit treatment.

The first-order condition. By the same algebra applied to Block E of the expected complete-data log-likelihood, the first-order condition for ν_ε is

$$\log\tfrac{\nu_\varepsilon}{2} \;-\; \psi\big(\tfrac{\nu_\varepsilon}{2}\big) \;+\; 1 \;+\; \overline{\log w}^{\varepsilon} \;-\; \bar{w}^{\varepsilon} \;=\; 0, \tag{89}$$

where $\bar{w}^{\varepsilon}$ and $\overline{\log w}^{\varepsilon}$ are the idiosyncratic-side counterparts of the factor-side posterior summaries:

$$\bar{w}^{\varepsilon} \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\hat{w}_t^{\varepsilon}, \qquad \overline{\log w}^{\varepsilon} \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\widehat{\log w}_t^{\varepsilon}.$$

The root-finding function becomes

$$g_\varepsilon(\nu_\varepsilon) \;\equiv\; \log\tfrac{\nu_\varepsilon}{2} \;-\; \psi\big(\tfrac{\nu_\varepsilon}{2}\big) \;+\; 1 \;+\; \overline{\log w}^{\varepsilon} \;-\; \bar{w}^{\varepsilon},$$

and the update is $\nu_\varepsilon^{(j+1)} = \mathrm{Brent}(g_\varepsilon;\, [a, b])$ on the same bracket $[2.001, 1000]$.

Summary of the ν-Update

Putting it all together, the M-step updates for (ν_u, ν_ε) proceed as follows. At each outer iteration j:

1. From the E-step, retrieve the posterior weight summaries: $\bar{w}^u$, $\overline{\log w}^{u}$, $\bar{w}^{\varepsilon}$, $\overline{\log w}^{\varepsilon}$.

2. Define the two functions:

$$\begin{aligned} g_u(\nu_u) \;&=\; \log\tfrac{\nu_u}{2} - \psi\big(\tfrac{\nu_u}{2}\big) + 1 + \overline{\log w}^{u} - \bar{w}^u, \\ g_\varepsilon(\nu_\varepsilon) \;&=\; \log\tfrac{\nu_\varepsilon}{2} - \psi\big(\tfrac{\nu_\varepsilon}{2}\big) + 1 + \overline{\log w}^{\varepsilon} - \bar{w}^{\varepsilon}. \end{aligned}$$

3. Find the roots numerically:

$$\nu_u^{(j+1)} \;=\; \mathrm{Brent}\big(g_u;\, [a, b]\big), \qquad \nu_\varepsilon^{(j+1)} \;=\; \mathrm{Brent}\big(g_\varepsilon;\, [a, b]\big).$$

What is different in practice. Although the equations are identical in form, there are two quantitative differences between the two updates worth noting.

First, the dimension of the underlying distribution differs: in the E-step derivation, the conjugacy argument yielded posterior shape parameters of $\frac{\nu_u+r}{2}$ for the factor-side weights and $\frac{\nu_\varepsilon+M}{2}$ for the idiosyncratic-side weights. Since in typical macroeconomic applications M (number of observed series) is much larger than r (number of factors), the idiosyncratic weights $\hat{w}_t^\varepsilon$ are estimated from more information per period and tend to be more tightly concentrated than the factor weights $\hat{w}_t^u$.

Second, in the presence of missing data, the summation $\sum_t w_t^\varepsilon$ that defines $\bar{w}^\varepsilon$ must be adjusted: for each observation t, only the *observed* components of y_t contribute to the likelihood, and the effective dimension M in the conjugacy argument becomes m_t, the number of observed components of y_t at time t. By construction, m_t equals the number of rows of the selection matrix $\mathbf{W}_t$, or equivalently $m_t = \operatorname{rank}(\mathbf{W}_t)$. The posterior mean and log-moments of w_t^ε are computed using this time-varying dimension m_t rather than the full M. This is the standard refinement handled by the missing-data formalism that we will see later and does not alter the functional form of (89).

Interpretation of the First-Order Conditions

The two implicit equations for ν_u and ν_ε — (86) and its idiosyncratic-side counterpart (89) — can be given a clean and unifying statistical interpretation as the *matching of concavity gaps* of the logarithm, one under the Gamma prior and one under the posterior. The treatment below covers both cases simultaneously.

Let W denote either of the two weight sequences (w^u or w^ε) and let ν denote the corresponding degrees-of-freedom parameter. Let $\bar{w}$ and $\overline{\log w}$ denote the E-step summaries of the posterior mean and mean-log of the weights — that is, either $(\bar{w}^u, \overline{\log w}^u)$ or $(\bar{w}^\varepsilon, \overline{\log w}^\varepsilon)$. The unified first-order condition is

$$\underbrace{\left[\ \log\tfrac{\nu}{2} - \psi\left(\tfrac{\nu}{2}\right)\ \right]}_{\Delta_{\text{prior}}(\nu)} + 1 \ = \ \underbrace{\bar{w} - \overline{\log w}}_{\Delta_{\text{post}}}. \tag{90}$$

Both sides have the structure of a concavity gap of the logarithm, one computed under the Gamma prior and one computed under the posterior distribution of the weights given the data.

The prior gap. Consider first the left-hand side. If $W \sim \text{Gamma}(\nu/2, \nu/2)$, then $\mathbb{E}[W] = 1$ (so $\log \mathbb{E}[W] = 0$) and $\mathbb{E}[\log W] = \psi(\nu/2) - \log(\nu/2)$. Therefore

$$\log \mathbb{E}[W] \ - \ \mathbb{E}[\log W] \ = \ \log\tfrac{\nu}{2} \ - \ \psi\left(\tfrac{\nu}{2}\right) \ = \ \Delta_{\text{prior}}(\nu),$$

which is precisely the concavity gap of log under the Gamma prior with shape and rate both equal to $\nu/2$. This is a deterministic function of ν alone, decreasing monotonically from $+\infty$ at $\nu \to 0^+$ (where the Gamma is highly dispersed, with W taking extreme values on both sides of its mean) to 0 at $\nu \to \infty$ (where the Gamma concentrates at $W = 1$ and no gap remains).

The posterior gap. The right-hand side, $\bar{w} - \overline{\log w}$, is the empirical counterpart: $\mathbb{E}_{\text{post}}[W] - \mathbb{E}_{\text{post}}[\log W]$ computed using the posterior summaries from the E-step. Applying Jensen's inequality twice — first $\mathbb{E}[\log W] \le \log \mathbb{E}[W]$ (concavity of log), then $\log \mathbb{E}[W] \le \mathbb{E}[W] - 1$ (the elementary bound $\log x \le x - 1$) — we have

$$\bar{w} - \overline{\log w} \ \ge \ 1,$$

with equality only in the degenerate case $W \equiv 1$ almost surely.

The matching condition. The first-order condition (90) therefore reads: *choose ν so that the prior concavity gap plus one equals the posterior concavity gap*. Equivalently, the algorithm picks the value of ν that produces a prior Gamma distribution whose inherent "Jensen gap for log" matches what the data imply.

This has an intuitive consequence: a large posterior gap (strongly dispersed weights — a sign that the data contain observations far from the typical region, drawn under small posterior W's) forces the algorithm to choose a *small* ν, i.e. a heavy-tailed DGP. A posterior gap close to 1 (weights tightly concentrated around unity) yields a *large* ν, pushing the model toward the Gaussian limit. The concavity-gap matching is the precise statistical mechanism by which the EM algorithm "reads" the tail heaviness from the posterior weights and translates it into a degrees-of-freedom estimate.

Application to ν_u and ν_ε. Equation (90) applies identically to both first-order conditions — one substitutes the factor-side summaries $(\bar{w}^u, \overline{\log w}^u)$ to get the update for ν_u, or the idiosyncratic-side summaries $(\bar{w}^\varepsilon, \overline{\log w}^\varepsilon)$ to get the update for ν_ε. The functional form of the matching is identical in the two cases, but the numerical values of the posterior gaps are generally different, reflecting the distinct information content of factor-side and idiosyncratic-side weights. Two quantitative differences between the two cases are worth noting:

- The dimension of the underlying Student-t distribution differs: the factor-side conjugacy argument yields posterior shape parameters of $\frac{\nu_u+r}{2}$, whereas the idiosyncratic-side yields $\frac{\nu_\varepsilon+M}{2}$. Since in typical macroeconomic applications M (number of series) is much larger than r (number of factors), the idiosyncratic weights $\hat{w}_t^\varepsilon$ are estimated from more information per period and tend to be more tightly concentrated than the factor weights $\hat{w}_t^u$. Accordingly, the idiosyncratic posterior gap $\bar{w}^\varepsilon - \overline{\log w}^\varepsilon$ tends to be smaller, and the estimated $\hat{\nu}_\varepsilon$ to be larger, than their factor-side counterparts.

- In the presence of missing data, the effective dimension in the idiosyncratic conjugacy argument becomes time-varying: $m_t = \text{rank}(\mathbf{W}_t)$, the number of observed components at time t. This refinement is handled by the missing-data formalism and does not alter the functional form of the matching condition (90), only the computation of the posterior summaries feeding into it.

This gives a clean intuition for what "estimating ν" means: the algorithm is asking *how heavy-tailed must the Gamma prior be to produce the observed dispersion of the posterior weights?* If the posterior weights are close to 1 for all t (low dispersion), the answer is: very concentrated prior, meaning ν_u large, meaning little departure from Gaussianity. If the posterior weights show a long tail of small values (high dispersion), the answer is: dispersed prior, meaning ν_u small, meaning heavy tails.

Missing Data and Ragged Edge

Up to this point we have assumed that the observation matrix $\mathbf{Y}$ is fully observed — every series i measured at every time t. This is not realistic for nowcasting applications. Macroeconomic data typically exhibit a *ragged edge* at the right end of the sample: different series are released with different publication lags, so at any given vintage date some series extend further in time than others. A typical dataset ends with a "staircase" pattern of NaN values in its most recent rows.

In this section we extend the model and the EM algorithm to handle this missing-data structure. The extension is elegant: introducing a *selection matrix* $\mathbf{W}_t$ per time period, the entire machinery of the E-step and M-step carries through with only one non-trivial modification — the update for $\boldsymbol{\Lambda}$ and $\mathbf{R}$ must now be derived series by series rather than as a single matrix equation.

A remark The initialization of the parameters $\theta^{(0)}$ requires a balanced dataset, which we construct through a specialized two-step procedure designed to handle the multi-frequency nature of macroeconomic indicators. First, we address the mixed-frequency problem—specifically for quarterly variables like GDP—by implementing the Mariano and Murasawa (2003) framework. Unlike simple linear interpolation, this approach treats quarterly observations as the temporal aggregation of a latent monthly process, ensuring that the disaggregated monthly series honors the accounting constraints of the national accounts. Second, for unsystematic missing data and the ragged edge, we perform a preliminary data imputation where missing values are replaced using sampling from a Normal. Since our framework is built on the Student-t distribution, this initialization ensures that the starting factor estimates are not contaminated by outliers, providing a robust 'warm start' for the subsequent ECME iterations.It is important to distinguish between this initialization phase and the iterative EM estimation. While initial parameter values are obtained via PCA on this balanced version of the dataset, this 'plug-in' approach is discarded once the EM algorithm begins. During the subsequent E-step and M-step, the model relies exclusively on the selection matrix framework, ensuring that only genuine observations contribute to the likelihood and that the idiosyncratic variances are not biased by artificial data points.

Setup: The Selection Matrix $\mathbf{W}_t$

At each time t, let $m_t \leq M$ denote the number of series observed (non-missing) at that time. Define the *selection matrix* $\mathbf{W}_t \in \{0,1\}^{m_t \times M}$ as a 0/1 matrix with exactly one non-zero entry per row, placed in the column corresponding to an observed series. Formally,

$$(\mathbf{W}_t)_{k,i} \;=\; \begin{cases} 1 & \text{if } k\text{-th observed series at time } t \text{ is series } i, \\ 0 & \text{otherwise.} \end{cases}$$

The matrix $\mathbf{W}_t$ therefore *selects* from the full vector $y_t \in \mathbb{R}^M$ only the components that are actually observed:

$$y_t^{\text{obs}} \;\equiv\; \mathbf{W}_t \, y_t \;\in\; \mathbb{R}^{m_t}.$$

Equivalently, $\mathbf{W}_t$ is built from the identity I_M by removing the rows corresponding to missing series. The matrix is *known and fixed*: it is determined entirely by the release schedule of the dataset, not estimated.

Real-time implementation. In a real-time nowcasting setting, the selection matrices $\{\mathbf{W}_t\}_{t=1}^{T}$ depend on the vintage date: as new releases arrive, the most recent $\mathbf{W}_t$ matrices acquire additional non-zero rows (previously missing series become observed). The algorithm we describe here is invariant to this: given any configuration of $\{\mathbf{W}_t\}$, the EM iteration produces the corresponding parameter estimates and factor moments. The treatment of forecasts and quantile updates as new information arrives is developed in the second stage of the framework.

Example. If $M = 4$ and at time t the second series is missing while all others are observed, then $m_t = 3$ and

$$\mathbf{W}_t \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad y_t^{\text{obs}} \;=\; \mathbf{W}_t\, y_t \;=\; \begin{pmatrix} y_{1,t} \\ y_{3,t} \\ y_{4,t} \end{pmatrix}.$$

The missing $y_{2,t}$ is removed from the observed vector.

Observation equation under missing data. The observation equation $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$ transforms into its "observed-components" version by left-multiplying with $\mathbf{W}_t$:

$$y_t^{\text{obs}} \;=\; (\mathbf{W}_t \mathbf{\Lambda})\, f_t \;+\; \mathbf{W}_t\, \varepsilon_t, \tag{91}$$

with the two components of the right-hand side interpretable as:

- $\mathbf{W}_t \mathbf{\Lambda} \in \mathbb{R}^{m_t \times r}$: the *effective loading matrix* at time t — only the rows of $\mathbf{\Lambda}$ corresponding to observed series survive.
- $\mathbf{W}_t \varepsilon_t \in \mathbb{R}^{m_t}$: the *observed idiosyncratic errors*, with covariance $\mathbf{W}_t(\mathbf{R}/w_t^{\varepsilon})\mathbf{W}_t' = \mathbf{W}_t \mathbf{R}\, \mathbf{W}_t'/w_t^{\varepsilon}$. When $\mathbf{R}$ is diagonal, $\mathbf{W}_t \mathbf{R} \mathbf{W}_t'$ is also diagonal, with diagonal entries equal to the variances of the observed series.

The model thus becomes a linear state-space model with *time-varying observation dimension m_t*. Everything we did in the fully-observed case generalises by this simple substitution.

The state equation is unaffected. The state equation $f_t = \mathbf{A} f_{t-1} + u_t$ makes no reference to $\mathbf{Y}$ at all. It only involves the latent factors, which are always "present" at every t (even if we don't have any observations to constrain them). Hence the state dynamics — and consequently the M-step updates for $\mathbf{A}, \mathbf{Q}, \nu_u$ — are **unchanged** by the missing-data issue.

The only parts of the algorithm that need modification are those that directly involve y_t:

- The Kalman filter update step (which "absorbs" y_t);
- The weight update $\hat{w}_t^{\varepsilon}$ (which depends on the idiosyncratic residual d_t^{ε} calculated only over the m_t observed dimensions);
- The M-step updates for $\mathbf{\Lambda}$ and $\mathbf{R}$.

We treat each of these in turn.

The Complete-Data Log-Likelihood with Missing Observations

Before deriving the modified E-step and M-step, we make explicit how the complete-data log-likelihood is modified by the presence of the selection matrices $\mathbf{W}_t$. The modification is local — it touches only the observation-equation block (Block A in our earlier decomposition) — and leaves the state-equation block, the initial-state block, and the two Gamma-prior blocks entirely unchanged.

Block A revisited. In the fully-observed case, conditional on the latent quantities $(\mathbf{F}, W^u, W^\varepsilon)$, the observation block contributes

$$\ell_c^{\text{A, full}} = -\frac{1}{2}\sum_{t=1}^{T}\Big[M\log(2\pi) - M\log w_t^\varepsilon + \log|\mathbf{R}| + w_t^\varepsilon\,(y_t - \mathbf{\Lambda} f_t)^\top \mathbf{R}^{-1}(y_t - \mathbf{\Lambda} f_t)\Big].$$

With missing data, y_t is no longer fully observed: the observed sub-vector is $y_t^{\text{obs}} = \mathbf{W}_t y_t \in \mathbb{R}^{m_t}$, and the corresponding observation equation becomes $y_t^{\text{obs}} = \mathbf{W}_t \mathbf{\Lambda} f_t + \mathbf{W}_t \varepsilon_t$. Since $\varepsilon_t \mid w_t^\varepsilon \sim \mathcal{N}(0, \mathbf{R}/w_t^\varepsilon)$ is multivariate Gaussian and $\mathbf{W}_t$ is deterministic, the observed innovations satisfy

$$\mathbf{W}_t \varepsilon_t \mid w_t^\varepsilon \;\sim\; \mathcal{N}\big(0,\; \mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top / w_t^\varepsilon\big),$$

with effective scale matrix $\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top \in \mathbb{R}^{m_t \times m_t}$. The contribution of period t to the observation block is therefore the log-density of an m_t-dimensional Gaussian:

$$\log p(y_t^{\text{obs}} \mid f_t, w_t^\varepsilon, \theta) \;=\; -\frac{1}{2}\Bigg[m_t \log(2\pi) \;-\; m_t \log w_t^\varepsilon \;+\; \log|\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top| \\ +\; w_t^\varepsilon\,(y_t^{\text{obs}} - \mathbf{W}_t \mathbf{\Lambda} f_t)^\top (\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top)^{-1}(y_t^{\text{obs}} - \mathbf{W}_t \mathbf{\Lambda} f_t)\Bigg].$$

Summing over t:

$$\ell_c^{\text{A, miss}} \;=\; -\frac{1}{2}\sum_{t=1}^{T}\Bigg[m_t \log(2\pi) \;-\; m_t \log w_t^\varepsilon \;+\; \log|\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top| \\ +\; w_t^\varepsilon\,(y_t^{\text{obs}} - \mathbf{W}_t \mathbf{\Lambda} f_t)^\top (\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top)^{-1}(y_t^{\text{obs}} - \mathbf{W}_t \mathbf{\Lambda} f_t)\Bigg]. \tag{92}$$

The two changes relative to the fully-observed expression are: the fixed dimension M is replaced by the time-varying effective dimension m_t, and the matrices $\mathbf{\Lambda}$ and $\mathbf{R}$ are pre- and post-multiplied by $\mathbf{W}_t$ wherever they appear.

Simplification under diagonal R. The expression (92) simplifies considerably under the working assumption that $\mathbf{R}$ is diagonal: $\mathbf{R} = \operatorname{diag}(r_1, r_2, \ldots, r_M)$. The selection matrix $\mathbf{W}_t$ extracts the rows and columns of $\mathbf{R}$ corresponding to the observed series, leaving the diagonal sub-matrix

$$\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top \;=\; \operatorname{diag}\big(\, r_i \;:\; i \in \operatorname{obs}(t)\,\big),$$

where $\operatorname{obs}(t) \subseteq \{1, 2, \ldots, M\}$ denotes the indices of the observed series at time t. Its determinant and inverse are elementary,

$$|\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top| \;=\; \prod_{i \in \operatorname{obs}(t)} r_i, \qquad (\mathbf{W}_t \mathbf{R} \mathbf{W}_t^\top)^{-1} \;=\; \operatorname{diag}\big(\, 1/r_i \;:\; i \in \operatorname{obs}(t)\,\big),$$

and the quadratic form decomposes additively over the observed series. Substituting into (92):

$$\ell_c^{\text{A, miss}} \;=\; -\frac{1}{2}\sum_{t=1}^{T}\sum_{i\in\text{obs}(t)}\Big[\,\log(2\pi)-\log w_t^{\varepsilon}+\log r_i+\frac{w_t^{\varepsilon}}{r_i}\,(y_{i,t}-\boldsymbol{\Lambda}_{i\cdot}f_t)^2\,\Big]. \tag{93}$$

Equivalently, swapping the order of summation,

$$\ell_c^{\text{A, miss}} \;=\; -\frac{1}{2}\sum_{i=1}^{M}\sum_{t\in\mathcal{T}_i}\Big[\,\log(2\pi)-\log w_t^{\varepsilon}+\log r_i+\frac{w_t^{\varepsilon}}{r_i}\,(y_{i,t}-\boldsymbol{\Lambda}_{i\cdot}f_t)^2\,\Big], \tag{94}$$

where $\mathcal{T}_i = \{\, t : y_{i,t} \text{ is observed} \,\}$ is the observation set of series i. The sum has been re-organised by *series* rather than by period, with each series contributing only over its own observation set. This is the form that makes the M-step updates row-by-row.

Where the M-step formulas come from. Once (94) is written down, the M-step updates derived earlier follow by exactly the same first-order calculations as in the fully-observed case, restricted to the observation sets $\mathcal{T}_i$. Differentiating with respect to the i-th row of $\boldsymbol{\Lambda}$ gives a weighted OLS update that depends only on $t \in \mathcal{T}_i$; differentiating with respect to r_i gives an estimator normalised by $|\mathcal{T}_i|$ — the count of t's effectively entering the sum — rather than by T. This normalisation is the natural maximum-likelihood result and is the source of the difference, discussed previously, with their $1/T$ convention.

The remaining blocks of the complete-data log-likelihood — the state equation, the initial-state distribution, and the two Gamma priors on the weights — make no reference to y_t and are therefore identical to the fully-observed case. As a consequence, the M-step updates of $\mathbf{A}$, $\mathbf{Q}$, ν_u are unchanged; only the updates of $\boldsymbol{\Lambda}$ and $\mathbf{R}$ acquire missing-data modifications, and only the update of ν_ε acquires the time-varying effective dimension m_t in place of the fixed M.

Summary. The introduction of the selection matrix $\mathbf{W}_t$ produces a single, local modification to the complete-data log-likelihood: the M-dimensional observation block is replaced by a time-varying m_t-dimensional one. Under the diagonal-$\mathbf{R}$ working assumption, this becomes a re-organisation of the observation sum by series, with each series summing only over its own observation set. The M-step updates derived in the next subsections follow by the same algebraic manipulations as in the fully-observed case, applied to this modified expression.

The Kalman Filter with Missing Data

The Kalman filter derived before adapts directly to the missing-data case by simply replacing y_t, $\mathbf{\Lambda}$, and $\mathbf{R}/w_t^\varepsilon$ with their observed counterparts. Specifically, at the update step of iteration t:

Prediction step (unchanged).

$$\hat{f}_{t|t-1} \;=\; \mathbf{A}\,\hat{f}_{t-1|t-1}, \qquad P_{t|t-1} \;=\; \mathbf{A}\,P_{t-1|t-1}\,\mathbf{A}' \;+\; \tilde{\mathbf{Q}}_t.$$

The prediction step does not involve y_t at all — it is therefore identical to the fully-observed case.

Update step (modified). The innovation, its covariance, the Kalman gain, and the posterior moments now use only the observed components:

$$\eta_t^{\text{obs}} \;=\; y_t^{\text{obs}} \;-\; (\mathbf{W}_t\mathbf{\Lambda})\,\hat{f}_{t|t-1} \;\in\; \mathbb{R}^{m_t}, \tag{95}$$

$$S_t^{\text{obs}} \;=\; (\mathbf{W}_t\mathbf{\Lambda})\,P_{t|t-1}\,(\mathbf{W}_t\mathbf{\Lambda})' \;+\; \mathbf{W}_t\,\tilde{\mathbf{R}}_t\,\mathbf{W}_t' \;\in\; \mathbb{R}^{m_t\times m_t}, \tag{96}$$

$$K_t^{\text{obs}} \;=\; P_{t|t-1}\,(\mathbf{W}_t\mathbf{\Lambda})'\,(S_t^{\text{obs}})^{-1} \;\in\; \mathbb{R}^{r\times m_t}, \tag{97}$$

$$\hat{f}_{t|t} \;=\; \hat{f}_{t|t-1} \;+\; K_t^{\text{obs}}\,\eta_t^{\text{obs}}, \tag{98}$$

$$P_{t|t} \;=\; P_{t|t-1} \;-\; K_t^{\text{obs}}\,(\mathbf{W}_t\mathbf{\Lambda})\,P_{t|t-1}. \tag{99}$$

Every object that used to be M-dimensional (or $M \times M$) is now m_t-dimensional (or $m_t \times m_t$). The structural form is identical to the full-data filter — only the dimensions adapt. A practical consequence is computational efficiency: the inversion $(S_t^{\text{obs}})^{-1}$ is performed in dimension $m_t \le M$, which can be substantially smaller than M in the presence of extensive missing data — for instance, at vintage dates near the release calendar boundary.

The limiting case $m_t = 0$. If at some time t *no* observations are available (all series missing — a situation that arises, for example, in early out-of-sample forecasts where no data have yet been released for the target period), we skip the update step entirely: $\hat{f}_{t|t} = \hat{f}_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$. The filter simply propagates the prediction forward through the state dynamics. This behaviour is automatic from the formulas above: when $m_t = 0$, the selection matrix $\mathbf{W}_t$ has no rows and all observed quantities vanish, leaving only the prediction.

Smoother. The Kalman smoother (55)–(57) depends only on the *filtered* outputs $\hat{f}_{t|t}$, $P_{t|t}$ and on $\mathbf{A}$, $\tilde{\mathbf{Q}}_t$. Since the filter outputs already incorporate the missing-data adjustments, the smoother **does not require any further modification**. It runs as before.

The Weight Update with Missing Data

The posterior of w_t^ε derived before depended on the squared Mahalanobis residual $d_t^\varepsilon = (y_t - \mathbf{\Lambda} f_t)'\mathbf{R}^{-1}(y_t - \mathbf{\Lambda} f_t)$. With missing data, this quantity cannot be computed (we don't have all components of y_t). The natural fix is to compute the Mahalanobis residual using only the *observed* components:

$$d_t^{\varepsilon,\text{obs}} \;=\; (y_t^{\text{obs}} - \mathbf{W}_t\mathbf{\Lambda}\,\hat{f}_{t|T})'\,(\mathbf{W}_t\mathbf{R}\mathbf{W}_t')^{-1}\,(y_t^{\text{obs}} - \mathbf{W}_t\mathbf{\Lambda}\,\hat{f}_{t|T}) \;+\; \operatorname{tr}\Big[(\mathbf{W}_t\mathbf{R}\mathbf{W}_t')^{-1}\,(\mathbf{W}_t\mathbf{\Lambda})\,P_{t|T}\,(\mathbf{W}_t\mathbf{\Lambda})'\Big], \tag{100}$$

using only the m_t observed series at time t. The corresponding posterior Gamma distribution now has shape parameter $(\nu_\varepsilon + m_t)/2$ (not $(\nu_\varepsilon + M)/2$), because only m_t observation dimensions contribute:

$$\boxed{w_t^\varepsilon \mid y_t^{\text{obs}}, f_t, \theta \;\sim\; \text{Gamma}\left(\frac{\nu_\varepsilon + m_t}{2}, \frac{\nu_\varepsilon + d_t^{\varepsilon,\text{obs}}}{2}\right).} \tag{101}$$

Consequently the posterior mean of the weight becomes

$$\hat{w}_t^\varepsilon \;=\; \frac{\nu_\varepsilon + m_t}{\nu_\varepsilon + d_t^{\varepsilon,\text{obs}}}. \tag{102}$$

The structure is identical to the full-data case (47), but with m_t replacing M. A time period with fewer observations receives a shape parameter closer to $\nu_\varepsilon/2$ — that is, closer to the prior — which is natural: less information available, more weight on the prior. This time-varying adjustment of the shape parameter ensures that the robustness of the filter scales dynamically with the amount of information available. In periods with sparse data, the model relies more heavily on the prior degrees of freedom, whereas in periods with a full data release, the weight becomes more sensitive to the Mahalanobis distance of the innovations.

The weight update for w_t^u is unaffected. The factor-side weight w_t^u depends only on the state residual d_t^u, which involves the factors (always fully present). Hence $\hat{w}_t^u$ is computed exactly as in the full-data case.

The M-Step with Missing Data

We now turn to the M-step. The four structural parameters respond differently to the missing-data adjustment:

- $\mathbf{A}$ and $\mathbf{Q}$ involve only factors — **updates unchanged**.
- ν_u and ν_ε involve only weights — **updates unchanged in form**, only the shape parameter of the weight posterior changes as noted above.
- $\boldsymbol{\Lambda}$ and $\mathbf{R}$ involve y_t — **updates must be rederived**, because the missing components of y_t must be handled appropriately.

The update for $\boldsymbol{\Lambda}$ in the fully-observed case was (72):

$$\boldsymbol{\Lambda}^{(j+1)} \;=\; \left(\sum_t \hat{w}_t^\varepsilon\, y_t\, \mathbb{E}[f_t' \mid \mathbf{Y}]\right)\left(\sum_t \hat{w}_t^\varepsilon\, \mathbb{E}[f_t f_t' \mid \mathbf{Y}]\right)^{-1}.$$

This formula **does not generalise** straightforwardly to missing data: the sum $\sum_t y_t \mathbb{E}[f_t' \mid \mathbf{Y}]$ is ill-defined for missing $y_{i,t}$ entries, and the matrix $\sum_t \hat{w}_t^\varepsilon \mathbb{E}[f_t f_t' \mid \mathbf{Y}]$ treats every time period on equal footing — but a time t where series i is missing contains no information about the i-th row of $\boldsymbol{\Lambda}$.

The correct generalisation is to derive the update *series by series*: each row $\boldsymbol{\Lambda}_{i\cdot}$ of the loading matrix is estimated using only the time periods where series i is observed.

Update for $\boldsymbol{\Lambda}$, Row by Row

For each series $i = 1, \ldots, M$, define the *observation set* $\mathcal{T}_i \subseteq \{1, \ldots, T\}$ of time periods at which series i is observed. The i-th row of $\boldsymbol{\Lambda}$, denoted $\boldsymbol{\Lambda}_{i\cdot} \in \mathbb{R}^{1\times r}$, contributes to the expected complete-data log-likelihood only through the terms where series i is observed.

Extracting from (60) only the contribution of series i, and following the same derivation but restricted to $t \in \mathcal{T}_i$:

$$\boxed{\Lambda_{i\cdot}^{(j+1)} \;=\; \left(\sum_{t\in\mathcal{T}_i} \hat{w}_t^{\varepsilon}\, y_{i,t}\, \mathbb{E}[f_t' \mid \mathbf{Y}]\right) \left(\sum_{t\in\mathcal{T}_i} \hat{w}_t^{\varepsilon}\, \mathbb{E}[f_t f_t' \mid \mathbf{Y}]\right)^{-1}.} \tag{103}$$

The numerator is $1 \times r$, the denominator is $r \times r$, the product is $1 \times r$ — consistent with one row of $\mathbf{\Lambda}$.

This row-by-row update is algebraically equivalent to the vectorized global update found in Bańbura and Modugno (2014), thanks to the diagonal structure of $\mathbf{R}$: under $\mathbf{R}$ diagonal, the Kronecker-product matrix that appears in the vectorised expression becomes block-diagonal, with each block corresponding to one row of $\mathbf{\Lambda}$ and inverted independently. The row-by-row formulation is therefore not an approximation but the natural computational realization of the same estimator. It offers a more direct interpretation as a sequence of M independent weighted regressions, where each row is optimized over its specific information set $\mathcal{T}_i$. This results in a formulation that is more computationally efficient (each $\Phi_i = \sum_{t\in\mathcal{T}_i} \hat{w}_t^{\varepsilon}\mathbb{E}[f_t f_t' \mid \mathbf{Y}]$ is an $r \times r$ inversion, in contrast to the global $Mr \times Mr$ inversion of the vectorised form) and more transparent in its treatment of ragged-edge data through the observation sets.

The weights are also adjusted for missing data. The posterior weights $\hat{w}_t^{\varepsilon}$ that enter (103) are not those of the fully-observed case: they are the missing-data-aware quantities (102) computed with shape $(\nu_\varepsilon + m_t)/2$ and the observed-components Mahalanobis residual $d_t^{\varepsilon,\mathrm{obs}}$. In a period t with sparse observations, $\hat{w}_t^{\varepsilon}$ is therefore informed by only the m_t observed series — coherently with the row-by-row restriction to $\mathcal{T}_i$.

Two observations:

- **Each row is a separate regression.** Equation (103) is the weighted least-squares estimator from regressing $\{y_{i,t}\}_{t\in\mathcal{T}_i}$ on the factor moments $\{\mathbb{E}[f_t' \mid \mathbf{Y}]\}_{t\in\mathcal{T}_i}$, using weights $\hat{w}_t^{\varepsilon}$. No information about series $j \neq i$ enters the calculation.
- **Different rows use different data.** Each row of $\mathbf{\Lambda}$ is estimated on its own observation set $\mathcal{T}_i$, which generally differs across i in ragged-edge data. This is the key advantage of the row-by-row formulation: a series available through time T and a series available only through time $T-3$ are handled coherently, each with its own observation set.

Update for R, Entry by Entry

Since $\mathbf{R}$ is diagonal, its i-th diagonal entry r_i is estimated from the residuals of series i alone. By the same logic as the $\mathbf{\Lambda}$ update:

$$\boxed{r_i^{(j+1)} \;=\; \frac{1}{|\mathcal{T}_i|} \sum_{t\in\mathcal{T}_i} \hat{w}_t^{\varepsilon}\, \mathbb{E}\Big[\, (y_{i,t} - \Lambda_{i\cdot}^{(j+1)}\, f_t)^2 \,\Big|\, \mathbf{Y} \Big],} \tag{104}$$

where $|\mathcal{T}_i|$ is the number of observations for series i. Expanding the inner expectation:

$$\mathbb{E}[(y_{i,t} - \Lambda_{i\cdot}^{(j+1)} f_t)^2 \mid \mathbf{Y}] \;=\; (y_{i,t} - \Lambda_{i\cdot}^{(j+1)}\, \hat{f}_{t|T})^2 \;+\; \Lambda_{i\cdot}^{(j+1)}\, P_{t|T}\, \Lambda_{i\cdot}^{(j+1)\prime}.$$

Note that all the formulas in this section use the *updated* loading matrix $\mathbf{\Lambda}^{(j+1)}$ from (103), not the previous-iteration $\mathbf{\Lambda}^{(j)}$. This is the same sequential conditional maximisation scheme adopted in the full-data case: we first update $\mathbf{\Lambda}$, then use the updated value when computing the residuals that feed into the $\mathbf{R}$ update.

Note the division by $|\mathcal{T}_i|$, not by T: each series is normalised by the effective number of observations available for it. This is the correct generalisation of the $1/T$ factor in the full-data formula (81).

Why R normalises by $|\mathcal{T}_i|$ but Q keeps the full T. The division by $|\mathcal{T}_i|$ in (104) invites a natural question: if the idiosyncratic-variance update restricts itself to the observed periods, why do the $\mathbf{A}$ and $\mathbf{Q}$ updates — declared "unchanged" at the start of this section — continue to sum over all T periods, even though many of those periods have missing observations?

The distinction is between the *observation* equation and the *state* equation, and it hinges on what the Kalman smoother delivers. The variance r_i is a property of the measurement error $\varepsilon_{i,t} = y_{i,t} - \mathbf{\Lambda}_{i\cdot} f_t$ in the observation equation. A residual for series i can be formed only when $y_{i,t}$ is actually recorded; at a missing entry there is no measurement, no residual, and therefore no contribution to r_i. The sum runs over $\mathcal{T}_i$, and the normalisation $|\mathcal{T}_i|$ counts exactly those contributions.

The innovation covariance $\mathbf{Q}$, by contrast, is a property of the state equation $f_t = \mathbf{A} f_{t-1} + u_t$, which involves only the latent factors and never the observations directly. The key fact is that the factors exist at *every* time period — including months in which some, most, or even all series are missing. This is precisely what the Kalman filter and smoother provide: by propagating the state through the transition dynamics and updating it with whatever data are available at each date (through the selection matrix $\mathbf{W}_t$), they return a smoothed estimate $\hat{f}_{t|T}$ and its posterior covariance $P_{t|T}$ for all $t = 1, \ldots, T$. The gaps in the observation panel are filled, within the state space, by the model's own dynamics conditioned on the available information; the latent factor path itself has no gaps.

Consequently the factor innovation $f_t - \mathbf{A} f_{t-1}$ and its posterior second moment are defined at every transition, regardless of how many series happened to be observed on those dates. There is one innovation per transition and T of them, so the determinant-penalty count in the first-order condition for $\mathbf{Q}$ is the full sample length. Missingness does not remove transitions from the sum; it only makes the corresponding $\hat{f}_{t|T}$ less precisely estimated, which enters through a larger posterior covariance $P_{t|T}$ in the smoothed moment — a change in precision, not in the count. In short, $\mathbf{R}$ counts *observations* and must skip the missing ones, whereas $\mathbf{Q}$ counts *state transitions*, which the smoother makes available at every date. The two normalisations differ because they sit on opposite sides of the state-space representation, not because of any inconsistency.

Our entry-by-entry formulation (104) is algebraically equivalent to the global matrix update in Bańbura and Modugno (2014). While BM use a diagonal masking matrix W_t and a $1/T$ normalization by imputing the variance for missing observations, we achieve the same result by restricting the summation to the effective observation set $\mathcal{T}_i$ and normalizing by its cardinality $|\mathcal{T}_i|$. This approach is numerically more direct and highlights the robust weighting mechanism $\hat{w}_t^{\varepsilon}$.

Update for ν_ε with Missing Data

The first-order condition for ν_ε derived in the full-data case (89) acquires a single modification under missing data: the constant dimension M that appeared in the conjugacy argument is replaced by the time-varying effective dimension m_t. Repeating the derivation done earlier starting from the modified Block A (92), the FOC becomes

$$\boxed{\log\tfrac{\nu_\varepsilon}{2} \;-\; \psi\!\left(\tfrac{\nu_\varepsilon}{2}\right) \;+\; 1 \;+\; \overline{\log w}^{\varepsilon} \;-\; \bar{w}^{\varepsilon} \;=\; 0,} \tag{105}$$

where the posterior summaries are computed using the missing-data weight update (102) with shape $(\nu_\varepsilon + m_t)/2$:

$$\bar{w}^{\varepsilon} \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\hat{w}_t^{\varepsilon}, \qquad \overline{\log w}^{\varepsilon} \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\widehat{\log w}_t^{\varepsilon},$$

with the log-mean derived from the Gamma posterior:

$$\widehat{\log w}_t^{\varepsilon} \;=\; \psi\!\left(\tfrac{\nu_\varepsilon+m_t}{2}\right) \;-\; \log\!\left(\frac{\nu_\varepsilon+d_t^{\varepsilon,\mathrm{obs}}}{2}\right).$$

The functional form of (105) is identical to the full-data case (89), so the same properties (monotonicity, uniqueness of the root) and the same numerical solver (Brent's method on a bracket of the form [2.001, 1000]) carry over without modification. The only difference lies in the inputs $(\bar{w}^{\varepsilon}, \overline{\log w}^{\varepsilon})$, which now reflect the time-varying effective dimensions.

Note that *the FOC for ν_u is unaffected by missing data*: the factor-side residual d_t^u depends only on the latent factors, which are always fully present, and the dimension r in the conjugacy argument does not change.

Implementation Notes

The selection-matrix formulation is mathematically elegant but computationally wasteful: forming and multiplying by $\mathbf{W}_t$ is essentially equivalent to *indexing* the observed components of y_t, $\mathbf{\Lambda}$, and $\mathbf{R}$. In practice, one never constructs $\mathbf{W}_t$ explicitly. Instead, the algorithm stores (once, before the EM loop starts) two sets of index arrays:

- For each time t: the list $\mathcal{I}_t = \{i : y_{i,t} \text{ observed}\}$ of observed series at time t, so that $y_t^{\mathrm{obs}} = y_t[\mathcal{I}_t]$ in indexing notation.
- For each series i: the observation set $\mathcal{T}_i = \{t : y_{i,t} \text{ observed}\}$, used for the row-by-row $\mathbf{\Lambda}$ update.

Every $\mathbf{W}_t\mathbf{\Lambda}$ in the filter then becomes `Lambda[I_t, :]` in Python, and every $\mathbf{W}_t\mathbf{R}\mathbf{W}_t'$ becomes `R[I_t][:, I_t]`. The computational overhead of the missing-data adjustment is therefore essentially zero — a great practical advantage of the BM formulation.

Summary. The ragged-edge extension requires four modifications to the full-data algorithm:

1. In the Kalman filter update, use $\mathbf{W}_t\mathbf{\Lambda}$ and $\mathbf{W}_t\mathbf{R}\mathbf{W}_t'$ in place of $\mathbf{\Lambda}$ and $\mathbf{R}$ (equivalently: slice the full matrices by the indices of observed series at each t).
2. In the idiosyncratic weight update, compute the Mahalanobis residual using only observed components, and use shape parameter $(\nu_\varepsilon + m_t)/2$.
3. In the M-step, update each row of $\mathbf{\Lambda}$ and each diagonal entry of $\mathbf{R}$ using only the observation set $\mathcal{T}_i$ of that series.
4. In the ν_ε update, compute the posterior summaries $(\bar{w}^{\varepsilon}, \overline{\log w}^{\varepsilon})$ using the missing-data weight posteriors with time-varying shape $(\nu_\varepsilon + m_t)/2$.

Everything else — state transition dynamics, factor smoothing, weight posteriors on the factor side, updates of $\mathbf{A}, \mathbf{Q}, \nu_u$ — is unchanged.

Mixed-Frequency Data: the Mariano–Murasawa Approach

In nowcasting applications, the data set typically contains series at *different frequencies*. The target variable — in our case the quarterly real GDP — is observed only at the end of each calendar quarter. Many of its predictors (payrolls, industrial production, retail sales, financial indicators) are observed monthly. A smaller number of series may be observed weekly or daily, but we restrict attention to the most relevant case: a mix of monthly and quarterly data. As noted by Cascaldi-Garcia (2023) and Bańbura et al. (2013), including daily or weekly variables does not improve the quality of the nowcast, because these indicators are detached from the real economy and often noisy.

The previous section handled the case where some observations are missing because they have not yet been released (ragged edge). The present section handles a structurally different problem: some observations are missing because they *do not exist* at that frequency — a quarterly series has no meaning in a non-terminal month of a quarter. The two problems often combine: GDP is both ragged (with publication lag) and intrinsically quarterly.

The standard solution in the nowcasting literature is the approach of Mariano and Murasawa (2003), which postulates a *latent monthly version* of each quarterly observable and specifies an exact linear relation between the observed quarterly value and the latent monthly values from the current and adjacent quarters. This allows quarterly and monthly series to coexist in the same state-space model, with the Kalman filter handling the aggregation automatically.

Two roles for MM in our framework. The Mariano–Murasawa framework plays two distinct roles in our estimation pipeline, and it is useful to distinguish them up front because they involve different objects and different stages of the algorithm.

First, MM is used during initialisation as a panel-completion device. Recall that the construction of $\theta^{(0)}$ relies on Principal Component Analysis applied to the data matrix $\widetilde{\mathbf{Y}}$, which requires a balanced dataset. Quarterly series — which are structurally missing in non-quarter-end months — cannot be handled by a generic Gaussian fill, because they are not really "missing" in the statistical sense: they obey an aggregation constraint linking them to a latent monthly process. We therefore use the MM identity to recover the implied monthly values consistent with the quarterly observation, fill those into the panel, and run PCA on the resulting balanced version. These imputed monthly values are used *only* to compute $\theta^{(0)}$, and are discarded as soon as the EM iterations begin.

Second, MM enters as a structural ingredient of the state-space model used by the EM iterations. Once the EM loop is running, the quarterly observation is no longer "filled in" — it is treated rigorously as a non-trivial linear combination of latent monthly factors. To represent this, the state vector is augmented to include not only the contemporaneous monthly factors f_t but also their lags $f_{t-1}, \ldots, f_{t-k}$, and the row of $\mathbf{\Lambda}$ corresponding to a quarterly series is replaced by a block that combines these lagged factors using the MM aggregation weights. The quarterly observation enters the likelihood only at the end of each quarter (treated as missing in the other months via the selection matrix machinery developed before). The filter and smoother then handle the cross-frequency aggregation automatically.

The remainder of this section develops both ingredients. We first recall the MM aggregation identity, then derive the monthly-fill formula used at initialisation, then describe how the model's state vector and loading matrix are augmented to incorporate the aggregation as a structural feature of the EM iterations.

Motivation: The Log-Differences Aggregation Problem

Before stating the Mariano–Murasawa restriction formally, it is worth understanding where the specific coefficients $\{1, 2, 3, 2, 1\}$ come from. This is instructive because the result is not obvious at first sight.

Setting. Consider a real variable whose level is observed at monthly frequency: X_t is its level at the end of month t, for $t = 1, 2, \ldots$. The corresponding *quarterly level* is the average of the three monthly levels that constitute the quarter: for a quarter ending at month $3m$,

$$X^Q_{3m} \;=\; \frac{1}{3}\big(X_{3m} + X_{3m-1} + X_{3m-2}\big).$$

This is the standard *flow convention*: the quarterly GDP is approximately the sum of monthly GDP over the three months of the quarter, divided by 3 (or equivalently, the average).

The problem. In macroeconomic applications we rarely work with levels. Data are typically transformed to *log-differences* (month-over-month or quarter-over-quarter growth rates), because log-differences are approximately stationary and directly interpretable as growth rates.

Define:

- $x_t \equiv \log X_t - \log X_{t-1}$: the monthly log-difference;
- $x^Q_{3m} \equiv \log X^Q_{3m} - \log X^Q_{3m-3}$: the quarterly log-difference.

We would like to express x^Q_{3m} as a linear function of monthly log-differences $\{x_\tau\}$. The obvious guess — that x^Q_{3m} is the sum or average of the x_τ in the quarter — turns out to be *wrong*. The quarterly log-difference is *not* a simple average of the monthly log-differences within the quarter, because taking logs does not commute with averaging.

Deriving the correct aggregation. We now show, with a direct calculation, that the quarterly log-difference is a specific *weighted average* of the five most recent monthly log-differences, with weights $\{1, 2, 3, 2, 1\}/3$. The argument relies on the standard approximation $\log(1 + z) \approx z$ for small z, which we apply to monthly and quarterly growth rates alike (typical magnitudes are in the order of 10^{-2} or 10^{-3}).

Start from

$$\log X^Q_{3m} - \log X^Q_{3m-3} \;=\; \log \frac{X^Q_{3m}}{X^Q_{3m-3}} \;=\; \log \frac{\frac{1}{3}(X_{3m} + X_{3m-1} + X_{3m-2})}{\frac{1}{3}(X_{3m-3} + X_{3m-4} + X_{3m-5})}.$$

Write each monthly level as $X_\tau = X_{3m-5} \cdot \exp(S_\tau)$, where $S_\tau = \sum_{k=3m-4}^{\tau} x_k$ is the cumulative sum of monthly log-differences from some fixed anchor. Substituting and using the approximation $\exp(S_\tau) \approx 1 + S_\tau$ to linear order:

$$\begin{aligned}
\log \frac{X^Q_{3m}}{X^Q_{3m-3}} \;&\approx\; \log \frac{3 + S_{3m} + S_{3m-1} + S_{3m-2}}{3 + S_{3m-3} + S_{3m-4} + S_{3m-5}} \\
&\approx\; \tfrac{1}{3}\big[(S_{3m} + S_{3m-1} + S_{3m-2}) - (S_{3m-3} + S_{3m-4} + S_{3m-5})\big],
\end{aligned}$$

where the last step uses $\log((3 + A)/(3 + B)) \approx (A - B)/3$ for small A, B. Substituting $S_\tau = \sum_{k=3m-4}^{\tau} x_k$ and grouping terms by x_k:

- x_{3m} appears only in S_{3m}: coefficient $+\frac{1}{3}$;
- x_{3m-1} appears in S_{3m}, S_{3m-1}: coefficient $+\frac{2}{3}$;
- x_{3m-2} appears in $S_{3m}, S_{3m-1}, S_{3m-2}$: coefficient $+\frac{3}{3} = 1$;
- x_{3m-3} appears in $S_{3m}, S_{3m-1}, S_{3m-2}$ with coefficient $+1$, and in S_{3m-3} with coefficient $-\frac{1}{3}$. Net: $+\frac{2}{3}$;
- x_{3m-4} appears in all three positive S's with coefficient $+1$ and in S_{3m-3}, S_{3m-4} with coefficient $-\frac{2}{3}$. Net: $+\frac{1}{3}$.

Collecting:

$$\boxed{x^Q_{3m} \;\approx\; \tfrac{1}{3}\,x_{3m} \;+\; \tfrac{2}{3}\,x_{3m-1} \;+\; x_{3m-2} \;+\; \tfrac{2}{3}\,x_{3m-3} \;+\; \tfrac{1}{3}\,x_{3m-4}.} \tag{106}$$

This is the *Mariano–Murasawa restriction*. The weights $\{1, 2, 3, 2, 1\}/3$ form a triangular pattern centred on x_{3m-2} (the middle month of the quarter in question), sum to 3 (reflecting that a quarter contains three months of growth), and decay symmetrically. They have a clean interpretation: the middle month's log-difference enters with full weight because it contributes to both the numerator quarter (in full) and differs from the denominator quarter (in full); the adjacent months contribute with weight $2/3$ because they "overlap" between the two quarterly averages; the extreme months contribute with weight $1/3$.

Key implication. Equation (106) relates the observed quarterly log-difference x^Q_{3m} to the *five* most recent monthly log-differences. Crucially, these monthly values are *latent* — if the original quarterly series is, say, GDP, we do not observe its monthly counterpart directly. But if we have a DFM with monthly latent factors, and if the quarterly GDP loads on a latent "monthly GDP factor component" via loadings λ_i, then the quarterly observation can be connected to the current and past values of the monthly latent factor via the same triangular pattern.

Using the MM Identity for Initialisation

Before discussing how the MM aggregation enters the state-space model used during the EM iterations, we describe its first role: as a *panel-completion device* during the construction of $\theta^{(0)}$. The PCA-based initialisation requires a balanced version of $\widetilde{\mathbf{Y}}$ at monthly frequency, including for those series — most notably GDP — that are observed only quarterly. The MM identity (106) provides exactly the tool needed to produce this monthly version.

The fill problem. For a quarterly series i (e.g. real GDP), we observe $x^Q_{i,3m}$ at quarter-end months $t = 3, 6, 9, \ldots$ but not at non-quarter-end months. Equation (106) relates each observed $x^Q_{i,3m}$ to the five latent monthly values $x_{i,3m}, x_{i,3m-1}, x_{i,3m-2}, x_{i,3m-3}, x_{i,3m-4}$. To construct an initial monthly series consistent with the MM restriction, we must solve a system of linear constraints — one per quarter — for the unknown monthly values.

Locally constant assumption: the simplest fill. The standard solution is to assume that, *within each quarter*, the three monthly log-differences are equal. That is, for each quarter m separately, we postulate

$$x_{i,3m} \;=\; x_{i,3m-1} \;=\; x_{i,3m-2} \;\equiv\; \xi_{i,m}.$$

The interpretation is straightforward: given a quarterly observation of, say, 0.6% growth, in the absence of additional information we cannot tell whether the economy grew faster at the beginning or at the end of the quarter, so we attribute the same monthly growth to each of the three months. The label *locally constant* reflects this: the monthly log-differences are constant *locally inside each quarter*, but *not* across quarters — the value $\xi_{i,m}$ generally differs from $\xi_{i,m-1}$, $\xi_{i,m+1}$, and so on. Each quarter has its own constant.

This is a minimal-information assumption: it imposes only what the data already tells us (the quarterly growth) and refrains from fabricating intra-quarter variation that we cannot identify.

Deriving the recursion. Substituting the locally constant assumption into (106), the three terms with weights $\frac{1}{3}, \frac{2}{3}, 1$ are all multiplied by $\xi_{i,m}$ (the constant of the current quarter), and the remaining two terms with weights $\frac{2}{3}, \frac{1}{3}$ are multiplied by $\xi_{i,m-1}$ (the constant of the previous quarter):

$$\begin{aligned} x^Q_{i,3m} &\approx \left(\tfrac{1}{3} + \tfrac{2}{3} + 1\right) \xi_{i,m} \; + \; \left(\tfrac{2}{3} + \tfrac{1}{3}\right) \xi_{i,m-1} \\ &= 2\,\xi_{i,m} \; + \; \xi_{i,m-1}. \end{aligned}$$

Solving for $\xi_{i,m}$ gives the recursion that produces the monthly fill quarter by quarter:

$$\xi_{i,m} \; = \; \tfrac{1}{2}\left(x^Q_{i,3m} \; - \; \xi_{i,m-1}\right). \tag{107}$$

At each quarter m, given the previous quarter's monthly value $\xi_{i,m-1}$, the formula reads off the current quarter's monthly value $\xi_{i,m}$ from the observed quarterly value $x^Q_{i,3m}$.

Initialising the recursion at the first quarter. The recursion (107) requires a starting value for $\xi_{i,0}$, which corresponds to a quarter *before* the sample begins and is therefore not observed. This is purely a boundary issue: only the first quarter of the sample has this problem. A simple and standard choice is to assign the first observed quarterly growth uniformly across its three months, i.e. to set the boundary condition at

$$\xi_{i,0} \; = \; \frac{x^Q_{i,3}}{3},$$

which corresponds to "treat the first observed quarter as if all three of its months had grown at the same rate, equal to one-third of the quarter's total log-growth." This boundary choice influences only the first few months of the filled series and is washed out quickly, since the EM iteration discards the fill anyway.

It is important to stress that this special treatment applies *only* to the very first quarter of the sample, as a mathematical artefact of starting the recursion. The locally constant assumption itself is applied to every quarter in the sample, not just the first.

Why this is more principled than linear interpolation. A naive alternative would be to interpolate linearly between observed quarterly values, or to repeat the quarterly value across the three months of each quarter. Both procedures violate the MM aggregation (106): the implied monthly series, when re-aggregated using the MM weights $\{1, 2, 3, 2, 1\}/3$, would not match the observed quarterly value. The locally-constant fill described above, by contrast, satisfies (106) exactly under its own assumption — every observed quarterly value is reproduced by re-aggregating its three filled monthly values.

Discarded after initialisation. We emphasise once more that this monthly fill is used *only* to construct $\theta^{(0)}$ via PCA. Once the EM iterations begin, the quarterly observation is no longer represented as three filled-in monthly values: it re-enters the state-space machinery as a single quarterly data point, and the missing months are formally treated as missing via the selection-matrix formalism developed earlier. The aggregation identity (106) is then incorporated structurally into the model through the augmented state vector, which we describe in the next subsection.

Adapting the State-Space Model: the Structural Role of MM

We now turn to the second role of the Mariano–Murasawa identity in our framework: as a structural ingredient of the state-space model used during the EM iterations. While the MM-based monthly fill described in the previous subsection was a one-shot device used *only* to construct $\theta^{(0)}$ via PCA, the construction we describe now is permanent: it modifies the state-space model so that the EM algorithm treats quarterly observations rigorously, without ever filling them in. The aggregation identity (106) is now imposed as an exact restriction on how quarterly series load on the latent monthly factors.

The key insight is that the quarterly observation depends on the *contemporaneous and four lagged* values of the monthly factor. In a standard first-order state-space representation, the state vector contains only f_t — not its lags. We therefore need to *augment the state vector* to include the lagged factors.

Augmented state vector. Define

$$\tilde{f}_t \equiv \begin{pmatrix} f_t \\ f_{t-1} \\ f_{t-2} \\ f_{t-3} \\ f_{t-4} \end{pmatrix} \in \mathbb{R}^{5r}, \tag{108}$$

where each block f_{t-k} is the r-dimensional factor vector at time $t-k$, and $5r$ is the total dimension of the augmented state. The "5" corresponds to the five monthly values that enter the aggregation (106).

Augmented transition equation. The monthly dynamics $f_t = \mathbf{A} f_{t-1} + u_t$ extend to the augmented state via the *companion form*:

$$\tilde{f}_t = \tilde{\mathbf{A}} \tilde{f}_{t-1} + \tilde{u}_t, \tag{109}$$

with

$$\tilde{\mathbf{A}} = \begin{pmatrix} \mathbf{A} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{5r \times 5r}, \qquad \tilde{u}_t = \begin{pmatrix} u_t \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix} \in \mathbb{R}^{5r}.$$

The first block row encodes the genuine dynamic equation: reading only the first block of $\tilde{f}_t$ gives $f_t = \mathbf{A} f_{t-1} + u_t$, the original VAR(1) specification. The remaining four block rows are *identity shifts* that encode the tautologies $f_{t-k} = f_{t-k}$ for $k = 1, 2, 3, 4$: they propagate the lagged factors forward through time by carrying them unchanged into the next period. In state-space language, these are "definitional" rather than "dynamic" equations. They are needed because the Kalman filter requires a first-order representation, and to keep the lagged factors available at time t we must include them explicitly in the state.

Augmented innovation covariance. The innovation $\tilde{u}_t$ has all blocks except the first equal to zero. Its conditional covariance is therefore

$$\tilde{\mathbf{Q}}_t = \operatorname{Cov}(\tilde{u}_t \mid w_t^u) = \begin{pmatrix} \mathbf{Q}/w_t^u & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \end{pmatrix}, \tag{110}$$

a $5r \times 5r$ matrix with $\mathbf{Q}/w_t^u$ in the top-left block and zeros elsewhere. This is a *singular* covariance matrix (rank r, not $5r$), because the lagged-factor blocks are deterministic functions of past shocks rather than new innovations. The Kalman filter handles singular covariance matrices without issue, provided one uses the Moore–Penrose pseudo-inverse where needed — or, more practically, avoids inverting $\tilde{\mathbf{Q}}_t$ explicitly, which is what most numerical implementations do by default.

Initial state. The augmented initial state $\tilde{f}_0 \in \mathbb{R}^{5r}$ needs a distribution. A standard choice is

$$\tilde{f}_0 \sim \mathcal{N}(\mathbf{0}, \tilde{\mathbf{\Sigma}}_0),$$

with $\tilde{\mathbf{\Sigma}}_0$ obtained either by solving the discrete Lyapunov equation $\tilde{\mathbf{\Sigma}}_0 = \tilde{\mathbf{A}}\,\tilde{\mathbf{\Sigma}}_0\,\tilde{\mathbf{A}}' + \tilde{\mathbf{Q}}_0$ (for a reference value of w_0^u, typically 1), or by setting $\tilde{\mathbf{\Sigma}}_0 = I_{5r}$ for a flat prior.

Adapting the Observation Equation

We now adapt the observation equation to handle both monthly and quarterly series simultaneously. We partition the full vector of observed series into two groups:

- *Monthly series*, collected in $y_t^M \in \mathbb{R}^{M_M}$ — observed every month;
- *Quarterly series*, collected in $y_t^Q \in \mathbb{R}^{M_Q}$ — observed only when t is a quarter-end month ($t = 3, 6, 9, \ldots$), missing otherwise.

The full observation dimension is $M = M_M + M_Q$, and the observed vector at time t is $y_t = (y_t^{M\prime}, y_t^{Q\prime})'$, with the quarterly components being missing at non-quarter-end months.

Monthly block: loadings on the contemporaneous factor. Monthly series load on the contemporaneous factor f_t only. In the augmented representation, each monthly loading row $\mathbf{\Lambda}_{i\cdot}^M \in \mathbb{R}^{1\times r}$ acts on the *first block* of $\tilde{f}_t$:

$$y_{i,t}^M = \mathbf{\Lambda}_{i\cdot}^M f_t + \varepsilon_{i,t}^M = \left(\mathbf{\Lambda}_{i\cdot}^M, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}\right) \tilde{f}_t + \varepsilon_{i,t}^M.$$

The $1 \times 5r$ row vector $(\mathbf{\Lambda}_{i\cdot}^M, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0})$ is the *effective loading* for monthly series i.

Quarterly block: loadings via the Mariano–Murasawa pattern. Quarterly series load on all five blocks of $\tilde{f}_t$ with the aggregation weights from (106). Let $\mathbf{\Lambda}_{j\cdot}^Q \in \mathbb{R}^{1\times r}$ denote the quarterly loadings — these are the coefficients that the *latent monthly version* of quarterly series j would carry, if we could observe it monthly. The observed quarterly series, via the Mariano–Murasawa restriction, satisfies

$$y_{j,3m}^Q = \tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^Q f_{3m} + \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^Q f_{3m-1} + \mathbf{\Lambda}_{j\cdot}^Q f_{3m-2} + \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^Q f_{3m-3} + \tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^Q f_{3m-4} + \varepsilon_{j,3m}^Q, \tag{111}$$

at quarter-end months $t = 3m$, and is missing otherwise. In augmented notation this becomes

$$y_{j,t}^Q = \left(\tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^Q, \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^Q, \mathbf{\Lambda}_{j\cdot}^Q, \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^Q, \tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^Q\right) \tilde{f}_t + \varepsilon_{j,t}^Q, \qquad t = 3m.$$

The effective loading row for quarterly series j is

$$\tilde{\mathbf{\Lambda}}^{Q}_{j\cdot} \;\equiv\; \left(\tfrac{1}{3}\mathbf{\Lambda}^{Q}_{j\cdot},\; \tfrac{2}{3}\mathbf{\Lambda}^{Q}_{j\cdot},\; \mathbf{\Lambda}^{Q}_{j\cdot},\; \tfrac{2}{3}\mathbf{\Lambda}^{Q}_{j\cdot},\; \tfrac{1}{3}\mathbf{\Lambda}^{Q}_{j\cdot}\right) \;\in\; \mathbb{R}^{1\times 5r},$$

which is a *linear function of the unconstrained* $\mathbf{\Lambda}^{Q}_{j\cdot}$: the r-dimensional loading that we would estimate if the quarterly series were observed at monthly frequency. The extra structure — the specific coefficients $\{1/3, 2/3, 1, 2/3, 1/3\}$ — is *known and fixed*, not estimated. It is a *restriction on the loading matrix*, dictated by the log-difference aggregation derived previously.

Assembling the full effective loading matrix. Stacking all M series, the effective loading matrix in the augmented representation takes the form

$$\tilde{\mathbf{\Lambda}} \;\equiv\; \begin{pmatrix} \mathbf{\Lambda}^{M} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \tfrac{1}{3}\mathbf{\Lambda}^{Q} & \tfrac{2}{3}\mathbf{\Lambda}^{Q} & \mathbf{\Lambda}^{Q} & \tfrac{2}{3}\mathbf{\Lambda}^{Q} & \tfrac{1}{3}\mathbf{\Lambda}^{Q} \end{pmatrix} \;\in\; \mathbb{R}^{M\times 5r}, \tag{112}$$

where $\mathbf{\Lambda}^{M} \in \mathbb{R}^{M_M\times r}$ is the monthly loading matrix (unconstrained), and $\mathbf{\Lambda}^{Q} \in \mathbb{R}^{M_Q\times r}$ is the underlying monthly loading matrix for the quarterly series (also unconstrained — these are the loadings we actually estimate). The zeros in the first block row reflect that monthly series do not load on lagged factors.

The observation equation in augmented form is then

$$y_t \;=\; \tilde{\mathbf{\Lambda}}\,\tilde{f}_t \;+\; \varepsilon_t,$$

with the understanding that the quarterly components of y_t are *missing* at non-quarter-end months. The missing pattern is handled by the selection-matrix formalism, which applies uniformly to both publication-lag and frequency-induced missingness.

Ragged edge of quarterly data. It is worth noting that quarterly data are also subject to a ragged edge: GDP for the quarter ending in month $3m$ is published with a delay (typically 45 days or more after quarter-end in the US, longer in Europe). In a real-time vintage dataset, GDP is therefore missing both for the current quarter (which has not yet ended) and for the most recent completed quarter (published too late for the current vintage). The selection matrix $\mathbf{W}_t$ handles both cases without any special treatment: for non-quarter-end months all quarterly series are missing because they do not exist at monthly frequency; for quarter-end months with an unresolved release lag, the corresponding quarterly series remain missing because they have not yet been released. The formalism of the selection matrix applies uniformly, which is one of the practical advantages of embedding the Mariano–Murasawa aggregation inside the selection-matrix framework.

The M-Step in the Mixed-Frequency Model

The M-step for the mixed-frequency model follows the same logic, with two important modifications: the factor moments are extracted from the first block of the augmented smoothed covariance (which is the $r \times r$ top-left sub-matrix of $\tilde{P}_{t|T}$), and the quarterly loadings require a new derivation based on the composite regressor.

(i) Updates of $\mathbf{A}$, $\mathbf{Q}$, ν_u: unchanged formulas. These updates only involve the *monthly* factor dynamics — the state equation $f_t = \mathbf{A} f_{t-1} + u_t$ is a purely monthly relationship. In the augmented representation, the factor moments needed for these updates — $\mathbb{E}[f_t f_t' \mid \mathbf{Y}]$, $\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}]$, and $\mathbb{E}[f_{t-1} f_{t-1}' \mid \mathbf{Y}]$ — are extracted from the appropriate sub-blocks of the augmented smoothed moments. Specifically, $\mathbb{E}[f_t f_t' \mid \mathbf{Y}]$ is the top-left $r \times r$ block of $\tilde{P}_{t|T} + \hat{\tilde{f}}_{t|T} \hat{\tilde{f}}_{t|T}'$, and the cross-moment $\mathbb{E}[f_t f_{t-1}' \mid \mathbf{Y}]$ is the top-left-to-top-second-block cross-piece of the same matrix. With these substitutions, the update formulas for $\mathbf{A}$, $\mathbf{Q}$, and ν_u are identical to those derived before.

(ii) Update of $\mathbf{\Lambda}^M$: the monthly part. Each row of the monthly loading matrix is updated exactly using the missing data formalism (row-by-row weighted OLS on observed time periods), using only the contemporaneous factor f_t as the regressor:

$$\mathbf{\Lambda}_{i\cdot}^{M,(k+1)} \;=\; \Big(\sum_{t \in \mathcal{T}_i^M} \hat{w}_t^{\varepsilon}\, y_{i,t}^M \, \mathbb{E}[f_t' \mid \mathbf{Y}] \Big) \Big(\sum_{t \in \mathcal{T}_i^M} \hat{w}_t^{\varepsilon}\, \mathbb{E}[f_t f_t' \mid \mathbf{Y}] \Big)^{-1}, \tag{113}$$

where $\mathcal{T}_i^M$ is the observation set of monthly series i and (k) denotes the current EM iteration. The moment $\mathbb{E}[f_t f_t' \mid \mathbf{Y}]$ is read off from the top-left $r \times r$ block of $\tilde{P}_{t|T}$ plus the outer product of the first-block mean; and similarly for $\mathbb{E}[f_t \mid \mathbf{Y}]$.

(iii) Update of $\mathbf{\Lambda}^Q$: the composite-regressor form. The quarterly loading update is the new piece. Each row $\mathbf{\Lambda}_{j\cdot}^Q$ multiplies all five blocks of $\tilde{f}_t$ via the Mariano–Murasawa weights. We adapt the single-equation derivation to this case.

For quarterly series j, the observation equation at quarter-end month $t = 3m$ reads

$$y_{j,3m}^Q \;=\; \mathbf{\Lambda}_{j\cdot}^Q \; \underbrace{\left[\tfrac{1}{3} f_{3m} + \tfrac{2}{3} f_{3m-1} + f_{3m-2} + \tfrac{2}{3} f_{3m-3} + \tfrac{1}{3} f_{3m-4}\right]}_{\equiv\, \phi_{3m}} \;+\; \varepsilon_{j,3m}^Q.$$

We define the *composite regressor* $\phi_t \in \mathbb{R}^r$ — a weighted average of the current and past four monthly factors, with the Mariano–Murasawa weights:

$$\phi_t \;\equiv\; \tfrac{1}{3} f_t + \tfrac{2}{3} f_{t-1} + f_{t-2} + \tfrac{2}{3} f_{t-3} + \tfrac{1}{3} f_{t-4}. \tag{114}$$

With this definition, quarterly series j at quarter-end month t satisfies

$$y_{j,t}^Q \;=\; \mathbf{\Lambda}_{j\cdot}^Q\, \phi_t \;+\; \varepsilon_{j,t}^Q,$$

which is formally the *same* observation equation as the monthly case — except that the regressor is the composite ϕ_t (a linear combination of the current and four lagged factors) rather than the single factor f_t.

The M-step update for $\mathbf{\Lambda}^Q$ is therefore a weighted OLS of $y_{j,t}^Q$ on ϕ_t, taken over the observation set $\mathcal{T}_j^Q$ of quarterly series j:

$$\boxed{\mathbf{\Lambda}_{j\cdot}^{Q,(k+1)} \;=\; \Big(\sum_{t \in \mathcal{T}_j^Q} \hat{w}_t^{\varepsilon}\, y_{j,t}^Q \, \mathbb{E}[\phi_t' \mid \mathbf{Y}] \Big) \Big(\sum_{t \in \mathcal{T}_j^Q} \hat{w}_t^{\varepsilon}\, \mathbb{E}[\phi_t \phi_t' \mid \mathbf{Y}] \Big)^{-1}.} \tag{115}$$

The quantities $\mathbb{E}[\phi_t \mid \mathbf{Y}]$ and $\mathbb{E}[\phi_t \phi_t' \mid \mathbf{Y}]$ are computable from the smoothed moments of the augmented state. Using the linearity of expectation and the definition (114):

$$\mathbb{E}[\phi_t \mid \mathbf{Y}] \;=\; \tfrac{1}{3}\,\hat{f}_{t|T} + \tfrac{2}{3}\,\hat{f}_{t-1|T} + \hat{f}_{t-2|T} + \tfrac{2}{3}\,\hat{f}_{t-3|T} + \tfrac{1}{3}\,\hat{f}_{t-4|T}, \tag{116}$$

$$\mathbb{E}[\phi_t \phi_t' \mid \mathbf{Y}] \;=\; \sum_{s=0}^{4}\sum_{\ell=0}^{4} c_s\, c_\ell\, \mathbb{E}[f_{t-s} f_{t-\ell}' \mid \mathbf{Y}], \tag{117}$$

where the coefficients $(c_0, c_1, c_2, c_3, c_4) = (\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3})$ are the Mariano–Murasawa weights. The cross-moments $\mathbb{E}[f_{t-s} f_{t-\ell}' \mid \mathbf{Y}]$ are read off from the augmented smoothed covariance $\tilde{P}_{t|T}$: the (s, ℓ)-block of $\tilde{P}_{t|T}$ is $\mathrm{Cov}(f_{t-s}, f_{t-\ell} \mid \mathbf{Y})$, and adding $\hat{f}_{t-s|T}\,\hat{f}_{t-\ell|T}'$ gives the full second-order moment.

(iv) Update of R: monthly and quarterly parts. The idiosyncratic variances are updated entry by entry, but the residual computation for quarterly series uses ϕ_t instead of f_t.

Monthly part. For monthly series i:

$$r_i^{M,(k+1)} \;=\; \frac{1}{|\mathcal{T}_i^M|} \sum_{t \in \mathcal{T}_i^M} \hat{w}_t^{\varepsilon}\, \mathbb{E}\Big[\big(y_{i,t}^M - \mathbf{\Lambda}_{i\cdot}^{M,(k+1)}\, f_t\big)^2 \,\Big|\, \mathbf{Y}\Big]\,. \tag{118}$$

Quarterly part. For quarterly series j:

$$r_j^{Q,(k+1)} \;=\; \frac{1}{|\mathcal{T}_j^Q|} \sum_{t \in \mathcal{T}_j^Q} \hat{w}_t^{\varepsilon}\, \mathbb{E}\Big[\big(y_{j,t}^Q - \mathbf{\Lambda}_{j\cdot}^{Q,(k+1)}\, \phi_t\big)^2 \,\Big|\, \mathbf{Y}\Big]\,. \tag{119}$$

Each conditional expectation expands using the posterior first and second moments derived above — in particular, the quarterly-case expansion uses $\mathbb{E}[\phi_t \phi_t' \mid \mathbf{Y}]$ from (117).

(v) Update of ν_ε: pooled across monthly and quarterly. The update for ν_ε follows the same Brent's-method procedure, but the first-order condition now sums over *all* observed entries — both monthly and quarterly — weighted by their posterior weights $\hat{w}_t^{\varepsilon}$ and using the appropriate residuals ($y_{i,t}^M - \mathbf{\Lambda}_{i\cdot}^M f_t$ for monthly series, $y_{j,t}^Q - \mathbf{\Lambda}_{j\cdot}^Q \phi_t$ for quarterly series). The formula for the update is conceptually identical to the pure-monthly case; the only change is the residual construction, which uses ϕ_t for the quarterly rows.

A note on $\hat{w}_t^{\varepsilon}$ in the mixed-frequency case. The posterior weight $\hat{w}_t^{\varepsilon}$ used in all of the formulas above is the missing-data-aware quantity with shape $(\nu_\varepsilon + m_t)/2$, where m_t now varies substantially across periods: in non-quarter-end months only the M_M monthly series contribute, while at quarter-end months (when GDP is released) both monthly and quarterly series contribute, increasing m_t to $M_M + M_Q$. The Mahalanobis residual $d_t^{\varepsilon,\mathrm{obs}}$ likewise pools both monthly and quarterly residuals at quarter-end months — using f_t as regressor for monthly series and ϕ_t for quarterly series.

Boundary at the start of the sample. A natural concern when looking at the composite regressor (114) is what happens at the beginning of the sample: the first quarter-end month is $t = 3$, and at that point the formula

$$\phi_3 \;=\; \tfrac{1}{3} f_3 + \tfrac{2}{3} f_2 + f_1 + \tfrac{2}{3} f_{-1} + \tfrac{1}{3} f_{-2}$$

seems to require pre-sample factors f_{-1} and f_{-2} that we never observe. The augmented state-space formulation handles this automatically, and no special boundary treatment is needed.

The reason is that the augmented initial state already encodes the pre-sample lags. Recall that $\tilde{f}_0 = (f_0, f_{-1}, f_{-2}, f_{-3}, f_{-4})'$ is given a prior distribution $\mathcal{N}(\mathbf{0}, \tilde{\mathbf{\Sigma}}_0)$ when the filter is initialised. The pre-sample factors are therefore *latent variables* of the model — not missing data — and the Kalman smoother delivers their posterior moments together with the in-sample ones. Concretely, the augmented smoothed mean $\hat{\tilde{f}}_{3|T} \in \mathbb{R}^{5r}$ contains

$$\hat{\tilde{f}}_{3|T} = \left(\hat{f}_{3|T},\ \hat{f}_{2|T},\ \hat{f}_{1|T},\ \hat{f}_{-1|T},\ \hat{f}_{-2|T}\right)',$$

and the composite regressor at $t = 3$ is computed by combining the five blocks with the Mariano–Murasawa weights — exactly as for any other quarter-end month.

The only quantitative consequence of this construction is that the posterior moments $\hat{f}_{-1|T}, \hat{f}_{-2|T}$ are estimated with substantial uncertainty: they are constrained only by the initial-state prior and indirectly through the dynamics, not by any observation. As a result, the first quarterly observation contributes less information to the M-step than later ones, where ϕ_t aggregates only in-sample factors that are tightly constrained by data on both sides. This is the appropriate finite-sample behaviour: in the absence of pre-sample information about the factor realisations, the model defers to the prior, and its uncertainty about the implied quarterly observation is correctly reflected in $\mathbb{E}[\phi_3 \phi_3' \mid \mathbf{Y}]$ via the smoothed covariance $\tilde{P}_{3|T}$.

Summary. Despite the visual complexity of the augmented representation, the M-step for the mixed-frequency model is conceptually clean:

- $\mathbf{A}$, $\mathbf{Q}$, and ν_u use the same update formulas as in the normal M-step, drawing factor moments from the top-left $r \times r$ sub-blocks of the augmented smoother output.
- Monthly loadings $\mathbf{\Lambda}^M$ are updated using the missing data formalism, row by row, using f_t as the regressor.
- Quarterly loadings $\mathbf{\Lambda}^Q$ are updated row by row with a weighted OLS formula of the same structure, but with the composite regressor ϕ_t in place of f_t.
- Idiosyncratic variances $\mathbf{R}$ are updated entry by entry: using residuals $y^M_{i,t} - \mathbf{\Lambda}^M_{i\cdot} f_t$ for monthly series and $y^Q_{j,t} - \mathbf{\Lambda}^Q_{j\cdot} \phi_t$ for quarterly series.
- The degrees of freedom ν_ε are updated by pooling monthly and quarterly residuals together, with the same functional form as in the pure-monthly case.

The entire mixed-frequency machinery thus reduces to the substitution $f_t \to \phi_t$ for the observation equation of quarterly series, plus the augmented Kalman filter developed before. The conceptual integrity of the algorithm is preserved throughout.

A practical note on identification. In the mixed-frequency setting, $\mathbf{\Lambda}^Q_{j\cdot}$ are the loadings of the *latent monthly version* of quarterly series j. They are not directly observable, and their interpretation is always as “what the loading would be if we could measure this variable monthly”. For GDP, this gives the monthly latent factor behaviour — a natural input to Growth-at-Risk analysis, where monthly tail risk is economically more informative than quarterly tail risk.

A note on the normalisation of R updates. The normalisation $1/|\mathcal{T}_i^M|$ we adopt in (118) (and analogously for the quarterly series) differs from the $1/T$ normalisation used in the original Bańbura-Modugno (2014) implementation. The difference arises because BM include the full period count T in the denominator even for series that are missing in some periods, yielding an estimator

$$\hat{r}_i^{\mathrm{BM}} = \frac{|\mathcal{T}_i^M|}{T}\,\hat{r}_i,$$

where $\hat{r}_i$ is our estimator. The two coincide when the series is fully observed ($|\mathcal{T}_i^M| = T$), but for series with substantial missing data — in particular for quarterly series, which are missing in two-thirds of the months — the BM normalisation systematically underestimates the idiosyncratic variance by a factor $|\mathcal{T}_i^M|/T$. The normalisation $1/|\mathcal{T}_i^M|$ we adopt corresponds to the standard maximum-likelihood estimator of the idiosyncratic variance given the observed sample, and is consistent with the refinement of the BM approach found in the more recent literature such as Banbura, Giannone, Modugno, and Reichlin (2013) and Jungbacker and Koopman (2014).

Block Factor Structure: Economic Identification

We now introduce the restrictions that give the model its economic meaning. Up to this point, the r factors have been anonymous mathematical objects, identified only up to a rotation: any linear transformation of the factors, compensated by the inverse transformation applied to the loadings, yields an observationally equivalent model. This *rotational indeterminacy* is well known in factor analysis — we will discuss it formally later. For now, we note simply that without further restrictions, we cannot interpret the estimated factors as "real activity", "financial conditions", or anything else economically meaningful — they are arbitrary linear combinations of whatever the data-generating process contains.

The standard way out is to impose *exclusion restrictions* on the loading matrix: specific entries of $\boldsymbol{\Lambda}$ are constrained to be zero, so that each factor can only load on a pre-specified subset of the observed series. This makes the factor identification *economic* rather than statistical.

Motivation and Setup

Consider a dataset containing three economically distinct groups of series:

- *Real-activity indicators* (group R): industrial production, retail sales, employment, etc.;
- *Financial indicators* (group F): credit spreads, stock market indices, term spreads, etc.;
- *Other indicators* (group X): a third set that we will leave generic for now (sentiment surveys, commodities, labour-market soft data, and so on — the choice will be made based on the specific application).

Let M_R, M_F, M_X denote the number of series in each group, with $M = M_R + M_F + M_X$. Order the observation vector y_t so that the first M_R entries are real, the next M_F are financial, and the last M_X are other:

$$y_t \;=\; \begin{pmatrix} y_t^R \\ y_t^F \\ y_t^X \end{pmatrix}, \quad y_t^R \in \mathbb{R}^{M_R},\;\; y_t^F \in \mathbb{R}^{M_F},\;\; y_t^X \in \mathbb{R}^{M_X}.$$

The block-diagonal loading matrix. We correspondingly use $r = 3$ latent factors, ordered as $f_t = (f_t^R, f_t^F, f_t^X)' \in \mathbb{R}^3$, and impose the *block-diagonal loading structure*:

$$\boldsymbol{\Lambda} \;=\; \begin{pmatrix} \boldsymbol{\Lambda}^R & \mathbf{0}_{M_R\times 1} & \mathbf{0}_{M_R\times 1} \\ \mathbf{0}_{M_F\times 1} & \boldsymbol{\Lambda}^F & \mathbf{0}_{M_F\times 1} \\ \mathbf{0}_{M_X\times 1} & \mathbf{0}_{M_X\times 1} & \boldsymbol{\Lambda}^X \end{pmatrix}, \tag{120}$$

where $\boldsymbol{\Lambda}^R \in \mathbb{R}^{M_R\times 1}$, $\boldsymbol{\Lambda}^F \in \mathbb{R}^{M_F\times 1}$, and $\boldsymbol{\Lambda}^X \in \mathbb{R}^{M_X\times 1}$ are the three within-group loading vectors. The off-diagonal blocks are *zero*, encoding the economic restriction that real series respond only to the real factor, financial series only to the financial factor, and other series only to the third factor.

The block-diagonal loading matrix in expanded form. Writing out the entries explicitly, with $\lambda_i^R, \lambda_i^F, \lambda_i^X$ denoting the loadings of the i-th series of each group:

$$\boldsymbol{\Lambda} = \left(\begin{array}{ccc} \lambda_1^R & 0 & 0 \\ \lambda_2^R & 0 & 0 \\ \vdots & \vdots & \vdots \\ \lambda_{M_R}^R & 0 & 0 \\ \hline 0 & \lambda_1^F & 0 \\ 0 & \lambda_2^F & 0 \\ \vdots & \vdots & \vdots \\ 0 & \lambda_{M_F}^F & 0 \\ \hline 0 & 0 & \lambda_1^X \\ 0 & 0 & \lambda_2^X \\ \vdots & \vdots & \vdots \\ 0 & 0 & \lambda_{M_X}^X \end{array}\right).$$

The pattern is clear: each row has exactly one non-zero entry, in the column corresponding to the factor of its block. There are $M_R + M_F + M_X = M$ rows and 3 columns, with $M_R + M_F + M_X$ non-zero entries — *three times fewer* parameters than an unrestricted $M \times 3$ loading matrix.

Compatibility with the mixed-frequency formulation. The block-diagonal structure is fully compatible with the Mariano–Murasawa augmentation introduced earlier. If GDP belongs to the real-activity group R — as is natural — then its row in the augmented loading matrix carries the MM weights only on the *real-factor* blocks of the augmented state vector. Concretely, for the GDP row j_{GDP}:

$$\tilde{\boldsymbol{\Lambda}}_{j_{\text{GDP}}\cdot} = \Big(\underbrace{\tfrac{1}{3}\lambda_{j_{\text{GDP}}}^R,\ 0,\ 0,}_{t}\ \underbrace{\tfrac{2}{3}\lambda_{j_{\text{GDP}}}^R,\ 0,\ 0,}_{t-1}\ \underbrace{\lambda_{j_{\text{GDP}}}^R,\ 0,\ 0,}_{t-2}\ \underbrace{\tfrac{2}{3}\lambda_{j_{\text{GDP}}}^R,\ 0,\ 0,}_{t-3}\ \underbrace{\tfrac{1}{3}\lambda_{j_{\text{GDP}}}^R,\ 0,\ 0}_{t-4} \Big),$$

a 1×15 row (with $r = 3$, $5r = 15$) where the only non-zero entries are in the columns corresponding to the *real-factor blocks* of $\tilde{f}_t$. The financial and other factor blocks contribute zero, encoding the economic restriction that GDP is identified as a real-side variable. The composite regressor ϕ_t for GDP therefore reduces to a function of f^R only:

$$\phi_{j_{\text{GDP}},t}^R = \tfrac{1}{3}f_t^R + \tfrac{2}{3}f_{t-1}^R + f_{t-2}^R + \tfrac{2}{3}f_{t-3}^R + \tfrac{1}{3}f_{t-4}^R.$$

The MM aggregation is a constraint on *which lags of the real factor* contribute to the quarterly GDP observation; the block-diagonal structure is a constraint on *which factor* contributes. The two constraints are independent and combine multiplicatively in the augmented loading matrix.

The full factor dynamics. While the loadings are restricted to be block-diagonal, the factor dynamics $\mathbf{A}$ remain *unrestricted* — a full 3×3 matrix:

$$\mathbf{A} = \begin{pmatrix} a^{RR} & a^{RF} & a^{RX} \\ a^{FR} & a^{FF} & a^{FX} \\ a^{XR} & a^{XF} & a^{XX} \end{pmatrix},$$

where the diagonal entries a^{kk} capture the persistence of each factor (how strongly f_t^k depends on its own past), and the off-diagonal entries $a^{k\ell}$ ($k \neq \ell$) capture the *dynamic spillovers* between factors. Economically, a^{RF} measures how shocks to the financial factor propagate to the real factor with a one-period lag — the transmission of financial conditions to the real economy. Symmetrically, a^{FR} measures real-to-financial transmission. The state-transition equation

$$f_t \;=\; \mathbf{A}\, f_{t-1} + u_t, \qquad \mathbf{A} \in \mathbb{R}^{3\times 3} \text{ unrestricted},$$

therefore preserves the economic richness of the model: the contemporaneous separation is encoded in $\mathbf{\Lambda}$ (who loads on what), the dynamic spillovers in $\mathbf{A}$ (how factors evolve jointly).

This structure — block-diagonal $\mathbf{\Lambda}$, full $\mathbf{A}$ — is the standard specification of Doz, Giannone, and Reichlin (2012) and Cascaldi-Garcia (2023), and underlies many modern nowcasting applications. Its combination of economic identification (via $\mathbf{\Lambda}$) and dynamic flexibility (via $\mathbf{A}$) makes it especially suited for our purpose: Growth-at-Risk applications that track how financial stress propagates to real activity.

Note on generality. The notation above assumes that each block contains a single factor (one-dimensional $\mathbf{\Lambda}^R, \mathbf{\Lambda}^F, \mathbf{\Lambda}^X$). The construction extends straightforwardly to cases where a block contains multiple factors — for example, two "real" factors (production-side and consumption-side) and one financial factor. In that case, $\mathbf{\Lambda}^R$ would be $M_R \times r_R$ for some $r_R \geq 1$, with $r = r_R + r_F + r_X$. Everything that follows applies with minor notational adjustments.

M-Step with Block Restrictions

The block-diagonal structure affects only one update: that of the loading matrix $\mathbf{\Lambda}$. The other M-step updates ($\mathbf{A}, \mathbf{Q}, \mathbf{R}, \nu_u, \nu_\varepsilon$) are untouched, because they do not involve exclusion restrictions on $\mathbf{\Lambda}$.

The key simplification is that, with exclusion restrictions, the row-by-row formulation used for missing data applies *naturally*: each row of $\mathbf{\Lambda}$ is estimated using only the factors on which that row is allowed to load.

Consider series i in the real-activity group ($i \in \{1, \dots, M_R\}$). Its observation equation is

$$y^R_{i,t} \;=\; \mathbf{\Lambda}^R_i\, f^R_t \;+\; \varepsilon^R_{i,t},$$

where $\mathbf{\Lambda}^R_i$ is the scalar loading on the real factor (non-zero), and the zero restrictions ensure that f^F_t and f^X_t do not enter. The M-step update for $\mathbf{\Lambda}^R_i$ is the weighted OLS of $y^R_{i,t}$ on f^R_t alone:

$$\mathbf{\Lambda}^{R,(j+1)}_i \;=\; \frac{\sum_{t\in\mathcal{T}^R_i} \hat{w}^\varepsilon_t\, y^R_{i,t}\, \mathbb{E}[f^R_t \mid \mathbf{Y}]}{\sum_{t\in\mathcal{T}^R_i} \hat{w}^\varepsilon_t\, \mathbb{E}[(f^R_t)^2 \mid \mathbf{Y}]}, \tag{121}$$

where $\mathcal{T}^R_i$ is the observation set of real series i (may incorporate both ragged edge and, if applicable, quarterly missingness through the MM framework). Analogously for the financial and other groups:

$$\mathbf{\Lambda}^{F,(j+1)}_i \;=\; \frac{\sum_{t\in\mathcal{T}^F_i} \hat{w}^\varepsilon_t\, y^F_{i,t}\, \mathbb{E}[f^F_t \mid \mathbf{Y}]}{\sum_{t\in\mathcal{T}^F_i} \hat{w}^\varepsilon_t\, \mathbb{E}[(f^F_t)^2 \mid \mathbf{Y}]}, \qquad \mathbf{\Lambda}^{X,(j+1)}_i \;=\; \frac{\sum_{t\in\mathcal{T}^X_i} \hat{w}^\varepsilon_t\, y^X_{i,t}\, \mathbb{E}[f^X_t \mid \mathbf{Y}]}{\sum_{t\in\mathcal{T}^X_i} \hat{w}^\varepsilon_t\, \mathbb{E}[(f^X_t)^2 \mid \mathbf{Y}]}. \tag{122}$$

Each of the three updates is a *scalar weighted OLS*: a one-variable regression of each observation series on the single factor it loads on, weighted by $\hat{w}^\varepsilon_t$. The first- and second-order moments of the individual factors f^R_t, f^F_t, f^X_t are readily extracted from the joint smoothed moments of f_t: the mean of f^R_t is the first component of $\hat{f}_{t|T}$, the variance of f^R_t is the $(1,1)$ entry of $P_{t|T}$, and so on.

Extracting block-specific moments from the joint Kalman output. A subtle but important point: the Kalman filter and smoother do *not* run separately for each block. They operate on the full joint state $f_t = (f_t^R, f_t^F, f_t^X)' \in \mathbb{R}^3$, which evolves as $f_t = \mathbf{A} f_{t-1} + u_t$ with $\mathbf{A}$ unrestricted (full 3×3). The output of the smoother is therefore the full vector mean $\hat{f}_{t|T} \in \mathbb{R}^3$ and the full covariance $P_{t|T} \in \mathbb{R}^{3\times 3}$, including all cross-block covariances $\mathrm{Cov}(f_t^k, f_t^\ell \mid \mathbf{Y})$ for $k \neq \ell$.

The block-specific moments needed for the loading updates are extracted by indexing this joint output: for the real-factor block ($k = R$), the first component of $\hat{f}_{t|T}$ gives $\mathbb{E}[f_t^R \mid \mathbf{Y}]$, and the (R, R) diagonal entry of $P_{t|T}$ — combined with the outer product of the first component — gives $\mathbb{E}[(f_t^R)^2 \mid \mathbf{Y}]$. Symmetrically for the financial and other blocks. The cross-block covariances are *not used* in the loading updates because the block-diagonal structure decouples the row-by-row regressions.

The cross-block covariances are, however, essential for the update of $\mathbf{A}$. The off-diagonal entry a^{RF}, for instance, depends on the cross-moment $\mathbb{E}[f_t^R f_{t-1}^F \mid \mathbf{Y}]$, which in turn requires the (R, F) entry of the lag-one cross-covariance $P_{t,t-1|T}$. The full Kalman output is therefore needed for the joint dynamics, even though it provides more than what each individual loading update consumes.

Comparison with the unrestricted update. Without the block-diagonal restrictions, the M-step update for each row of $\mathbf{\Lambda}$ would be an r-dimensional weighted OLS (as in equation (103)), requiring the inversion of a 3×3 matrix of weighted factor moments. With the block-diagonal restrictions, each row reduces to a scalar regression on a single factor — numerically cheaper and with a direct economic interpretation.

Interaction with Mariano–Murasawa. For quarterly series in the block-structured model, the same logic applies with ϕ_t replacing f_t. If the quarterly GDP is a real-activity variable, its observation equation becomes

$$y_{j,t}^Q \;=\; \mathbf{\Lambda}_j^{Q,R}\, \phi_t^R \;+\; \varepsilon_{j,t}^Q, \qquad \phi_t^R \;=\; \tfrac{1}{3} f_t^R + \tfrac{2}{3} f_{t-1}^R + f_{t-2}^R + \tfrac{2}{3} f_{t-3}^R + \tfrac{1}{3} f_{t-4}^R,$$

where ϕ_t^R is the *real-factor composite regressor* — the MM aggregation applied only to the real factor, since the quarterly GDP is a real variable and cannot load on financial or other factors by exclusion. The M-step update for $\mathbf{\Lambda}_j^{Q,R}$ is a scalar weighted OLS of $y_{j,t}^Q$ on ϕ_t^R over the observation set $\mathcal{T}_j^Q$.

Identification Status

Block-diagonal restrictions on $\mathbf{\Lambda}$ resolve much but *not all* of the rotational indeterminacy present in the unconstrained model. Specifically:

- The *assignment* of each factor to an economic block is fixed — the real factor is "the factor that drives real series, not financial series". This rules out the arbitrary relabelling of factors.
- The *scale* of each factor remains free: multiplying f_t^R by a constant c and dividing $\mathbf{\Lambda}^R$ by the same c yields an observationally equivalent model.
- The *sign* of each factor remains free: flipping f_t^R to $-f_t^R$ and $\mathbf{\Lambda}^R$ to $-\mathbf{\Lambda}^R$ is observationally equivalent.

The scale and sign ambiguities are innocuous for nowcasting purposes — the fitted values of y_t are invariant to them — and are resolved by standard normalisations applied as a post-processing step once the EM algorithm has converged. The two conventions commonly adopted in the literature are: (i) fixing the first non-zero entry of each $\mathbf{\Lambda}^R, \mathbf{\Lambda}^F, \mathbf{\Lambda}^X$ to a positive value, leaving $\mathbf{Q}$ free; and (ii) normalising the total (marginal) variance of each factor to one, with the corresponding rescaling of $\mathbf{\Lambda}^k$. The two yield observationally equivalent estimates of all economic quantities of interest (fitted values, predictive distributions, factor moments after appropriate re-scaling). A formal discussion of the indeterminacy and the corresponding normalisations is deferred to later section.

If each block contains a single factor ($r_R = r_F = r_X = 1$, our main case of interest), no *within-block* rotational indeterminacy remains: there is nothing to rotate when a block is one-dimensional. If a block contains multiple factors, within-block rotations remain unidentified — but this is again innocuous for the purposes of fitting and forecasting.

Operational implementation. In our implementation, we adopt the post-hoc normalisation approach. The EM algorithm is run with no constraints on the entries of $\mathbf{\Lambda}^k$, $\mathbf{A}$, or $\mathbf{Q}$, and the resulting estimates are free up to the residual sign-and-scale ambiguity discussed above. Once the EM iterations have converged, two distinct normalising transformations are applied to the final output.

The first, the *sign normalisation*, fixes the orientation of each block factor and is universal: it is applied identically regardless of any further choice. The second, the *scale normalisation*, fixes the unit of measurement of each block factor and admits two equivalent conventions in the literature; we describe both for completeness and indicate which one we adopt.

Sign normalisation. For each block $k \in \{R, F, X\}$, if the first non-zero entry of $\mathbf{\Lambda}^k$ is negative — equivalently, if the loading on the reference series of the block is negative — we flip the sign of both $\mathbf{\Lambda}^k$ and $\hat{f}_t^k$ (and the corresponding row and column of $\mathbf{A}$, $\mathbf{Q}$, $P_{t|T}$). This guarantees that the factor is interpretable as "the higher, the more of the reference quantity" — for instance, the real factor is high when real activity is strong, the financial factor is high when financial conditions are tight (or relaxed, depending on the choice of reference series). The sign normalisation is applied unconditionally and independently of which scale convention is adopted.

Two conventions for the scale normalisation. The chapter on rotational indeterminacy (later in this document) shows that two alternative scale conventions are observationally equivalent: both pin down a unique representation of the parameters within the equivalence class of rotational transformations, and both yield identical fitted values, identical forecasts, and identical conditional quantiles. The two conventions differ only in the units used to express the factors and the corresponding loadings.

- *Convention 1 — factor variance normalisation.* Rescale each block factor so that its total (marginal) variance over the sample equals one:

$$\widehat{\mathrm{Var}}(f^k) \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(\hat{f}^k_{t|T} - \bar{f}^k\big)^2 \;+\; \frac{1}{T}\sum_{t=1}^{T}\big(P_{t|T}\big)_{kk} \;=\; 1, \qquad k \in \{R, F, X\}.$$

  The corresponding $\mathbf{\Lambda}^k$ is rescaled by the same standard deviation, and $\mathbf{Q}_{kk}$ adjusts implicitly. This is the total (marginal) variance of the factor, dominated by the sample-variance term; normalising it standardises the factor in the sample sense, making it dimensionless and comparable across blocks.

- *Convention 2 — loading anchor normalisation.* Alternatively, fix one entry of each $\mathbf{\Lambda}^k$ to a chosen value (typically the first non-zero entry equal to one). This pins down the unit of measurement of f^k to that of the anchoring series, leaving $\mathbf{Q}_{kk}$ free to be estimated. It is the convention adopted in the standard DFM literature (Bańbura and Modugno, 2014).

The convention adopted in this thesis. We adopt *Convention 1* (factor variance normalisation) throughout the empirical analysis, in line with the Growth-at-Risk literature on which the second-stage quantile regression is based. The alternative would yield identical conclusions modulo a re-scaling. A more detailed discussion of the equivalence between the two conventions is provided in the chapter on rotational indeterminacy. Note that the normalisations described above only affect the parameters that pertain to the latent factor space: $\mathbf{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$, and the smoothed covariances $P_{t|T}$. The idiosyncratic scale matrix $\mathbf{R}$ and the degrees-of-freedom parameters ν_u, ν_ε are invariant under rotation of the factors and are therefore not subject to either the sign or the scale normalisation.

Rotational Indeterminacy

We have postponed a formal discussion of the identification status of the model until now, after all the structural restrictions (block-diagonal $\mathbf{\Lambda}$, Mariano–Murasawa aggregation) have been introduced. This section provides that discussion. The aim is twofold: first, to *characterise* precisely which aspects of the parameter vector θ can be identified from the observed data $\mathbf{Y}$, and which cannot; second, to *argue* — carefully — that the residual indeterminacy does not compromise the practical usefulness of the model for the purposes of nowcasting and Growth-at-Risk analysis.

The Source of the Indeterminacy

Consider the observation equation without the structural restrictions:

$$y_t \;=\; \mathbf{\Lambda} f_t \;+\; \varepsilon_t.$$

The factors f_t and the loadings $\mathbf{\Lambda}$ appear only through their *product* $\mathbf{\Lambda} f_t$. This product is invariant under a simultaneous transformation of the two:

$$(\mathbf{\Lambda}, f_t) \;\longmapsto\; (\mathbf{\Lambda}\mathbf{H}^{-1},\; \mathbf{H} f_t), \tag{123}$$

for any invertible $r \times r$ matrix $\mathbf{H}$. The transformed product satisfies $(\mathbf{\Lambda}\mathbf{H}^{-1})\,(\mathbf{H} f_t) = \mathbf{\Lambda} f_t$, so the observed data y_t is *the same* for the original and the transformed parameters. This is the fundamental source of indeterminacy in factor models: the latent factors are identified only up to a linear transformation.

The transformation must be consistent across equations. For the transformation (123) to yield a valid model, the other parameters must transform compatibly. Specifically, if f_t is replaced by $\mathbf{H} f_t$, then the state equation $f_t = \mathbf{A} f_{t-1} + u_t$ must become $\mathbf{H} f_t = \mathbf{H}\mathbf{A}\mathbf{H}^{-1}\,\mathbf{H} f_{t-1} + \mathbf{H} u_t$, so that

$$\mathbf{A} \;\longmapsto\; \mathbf{H}\mathbf{A}\mathbf{H}^{-1}, \qquad \mathbf{Q} \;\longmapsto\; \mathbf{H}\mathbf{Q}\mathbf{H}', \tag{124}$$

where the latter transformation of the covariance $\mathbf{Q}$ follows from $\mathrm{Cov}(\mathbf{H} u_t) = \mathbf{H}\,\mathrm{Cov}(u_t)\,\mathbf{H}'$.

The idiosyncratic parameters $\mathbf{R}$, ν_u, ν_ε are unaffected: they do not depend on the factor basis.

The equivalence class. Putting it together, the parameter vectors

$$\theta \;=\; \{\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}, \nu_u, \nu_\varepsilon\}$$

and

$$\theta^{\mathbf{H}} \;=\; \{\mathbf{\Lambda}\mathbf{H}^{-1},\; \mathbf{H}\mathbf{A}\mathbf{H}^{-1},\; \mathbf{H}\mathbf{Q}\mathbf{H}',\; \mathbf{R},\; \nu_u,\; \nu_\varepsilon\}$$

produce *identical* distributions of the observed data $\mathbf{Y}$. They have the same likelihood, the same fitted values, the same forecasts. The observed data cannot distinguish them.

This is the rotational indeterminacy: the parameter space is partitioned into *equivalence classes* indexed by invertible $r \times r$ matrices $\mathbf{H}$, and the likelihood is constant within each class.

What Block Restrictions Identify

The block-diagonal structure on $\mathbf{\Lambda}$ discussed before restricts the set of admissible transformations $\mathbf{H}$. Specifically: for $\mathbf{\Lambda}\mathbf{H}^{-1}$ to retain the block-diagonal structure imposed on $\mathbf{\Lambda}$, the matrix $\mathbf{H}^{-1}$ (and hence $\mathbf{H}$) must itself be block-diagonal. Any off-diagonal entry in $\mathbf{H}^{-1}$ would mix factors across blocks and destroy the exclusion restrictions.

In our specific case, with three one-dimensional blocks (real, financial, other), the admissible transformations reduce to

$$\mathbf{H} \;=\; \mathrm{diag}(h_R, h_F, h_X), \qquad h_R, h_F, h_X \in \mathbb{R} \setminus \{0\}. \tag{125}$$

That is, each factor can be rescaled by an arbitrary non-zero scalar, independently of the others, but the factors cannot be *mixed*. The three scalars h_R, h_F, h_X represent the residual indeterminacy.

Two observations:

(a) Across-block identification is achieved. The *assignment* of factors to economic blocks is now unambiguous: the real factor is "the factor that drives real series" — no other linear combination of the three latent quantities can satisfy this restriction. The block-diagonal structure identifies factors *up to a rescaling within each block*, which is the strongest identification one can hope for without further normalisations.

(b) Within-block scale and sign remain free. Each scalar h_k encodes two sub-ambiguities:

- *Scale*: multiplying f_t^R by $|h_R|$ and dividing $\mathbf{\Lambda}^R$ by $|h_R|$ is observationally equivalent. The factor could be measured in any unit — standard deviations, percentage points, raw units — and the model cannot tell which.
- *Sign*: taking $h_R = -1$ flips the sign of the real factor. Whether a positive f_t^R corresponds to economic expansion or contraction is a matter of convention, not data.

These are the residual indeterminacies.

Why This Is Not a Problem in Practice

At first sight, the residual scale and sign indeterminacies may seem worrying: the EM algorithm, applied from different initial points, could converge to different rescalings and sign conventions. Does this mean the estimates are meaningless?

The answer is no, and the reason is that *all quantities of substantive interest are invariant* to the transformation (125). Specifically:

Fitted values are invariant. The fitted conditional mean of the observation y_t under the model is $\mathbf{\Lambda}\hat{f}_{t|T}$. Under the transformation $(\mathbf{\Lambda}, f_t) \mapsto (\mathbf{\Lambda}\mathbf{H}^{-1}, \mathbf{H}f_t)$, this becomes $(\mathbf{\Lambda}\mathbf{H}^{-1})\,(\mathbf{H}\hat{f}_{t|T}) = \mathbf{\Lambda}\hat{f}_{t|T}$ — unchanged. The algorithm produces the same fitted values regardless of which member of the equivalence class it converges to.

Likelihood and marginal density of y_t are invariant. Both the log-likelihood and the marginal distribution of y_t depend on θ only through the product $\mathbf{\Lambda}f_t$ and through the covariance $\mathbf{\Lambda}\mathrm{Var}(f_t)\mathbf{\Lambda}' + \mathbf{R}$. The first is invariant by construction; the second is invariant because

$$(\mathbf{\Lambda}\mathbf{H}^{-1})\,\mathrm{Var}(\mathbf{H}f_t)\,(\mathbf{\Lambda}\mathbf{H}^{-1})' = \mathbf{\Lambda}\mathbf{H}^{-1}\,\mathbf{H}\,\mathrm{Var}(f_t)\,\mathbf{H}'\,(\mathbf{H}^{-1})'\,\mathbf{\Lambda}' = \mathbf{\Lambda}\,\mathrm{Var}(f_t)\,\mathbf{\Lambda}'.$$

Consequently, any function of the likelihood — AIC, BIC, LR tests, model comparison — is also invariant.

Forecasts of y_t are invariant. The forecast $\mathbb{E}[y_{T+h} \mid \mathbf{Y}]$ involves the same product $\mathbf{\Lambda}\mathbb{E}[f_{T+h} \mid \mathbf{Y}]$, which is invariant. Forecast *confidence intervals*, which depend on the forecast variance $\mathbf{\Lambda}\text{Var}(f_{T+h})\mathbf{\Lambda}' + \mathbf{R}$, are similarly invariant.

Quantile forecasts — the Growth-at-Risk object — are invariant. This is the quantity we ultimately care about. In the second stage of our framework, quantile regressions are run of a target variable (in our case, quarterly GDP growth) on the estimated factors $\hat{f}_{t|T}$. Under a within-block rescaling $(f_t, \mathcal{L}_F) \mapsto (\mathbf{H}f_t, \mathcal{L}_F\mathbf{H}^{-1})$ where $\mathcal{L}_F$ collects the quantile-regression coefficients, the regressors $\hat{f}_{t|T}$ are scaled by $\mathbf{H}$ and the coefficients $\mathcal{L}_F$ are scaled by $\mathbf{H}^{-1}$, in such a way that the conditional quantile

$$Q_\tau(y_{T+h} \mid \mathbf{Y}) \;=\; \mathcal{L}_F\, \hat{f}_{T+h|T}$$

is unchanged. The individual coefficients $\mathcal{L}_F$ are *not* identified — they depend on the chosen scale of the factors — but the conditional quantile itself, the Growth-at-Risk tail estimate, is identified and unambiguous.

Summary. Every object of interest — likelihood, fitted values, forecasts, forecast intervals, conditional quantiles — is invariant to the residual rotational indeterminacy. The estimator is identified *up to an equivalence class*, but the *quantities we care about* are identified without ambiguity.

The interpretation of the individual quantile-regression coefficients — how a standardised increase in financial conditions translates into percentage-point movements of the conditional quantile of GDP growth — requires fixing the scale and sign of the factors via the post-hoc normalisation discussed earlier. We return to this point in the treatment of the second-stage quantile regression.

Resolving the Indeterminacy for Interpretation

While the indeterminacy is innocuous for forecasting, it does matter for *interpretation*. If we want to discuss "by how many standard deviations the real factor rose in month t" or "whether the financial factor is above its historical average", we need to fix the scale and sign.

The standard approach is to impose two normalisations *ex post*, after the EM has converged. There are two equivalent conventions commonly adopted in the literature, and we briefly describe both.

Convention 1: factor variance normalisation. Rescale each factor so that its total (marginal) variance over the sample equals one. By the law of total variance, the marginal variance of the smoothed factor decomposes into the dispersion of its point estimate plus the average filtering uncertainty:

$$\widehat{\text{Var}}(f^k) \;=\; \underbrace{\frac{1}{T}\sum_{t=1}^{T}\left(\hat{f}^k_{t|T} - \bar{f}^k\right)^2}_{\text{sample variance of the estimates}} \;+\; \underbrace{\frac{1}{T}\sum_{t=1}^{T}\text{Var}\big(f^k_t \mid Y\big)}_{\text{mean posterior variance}} \;=\; 1, \qquad k \in \{R, F, X\},$$

where $\bar{f}^k = \frac{1}{T}\sum_t \hat{f}^k_{t|T}$ and $\mathrm{Var}(f^k_t \mid Y) = \left(P_{t|T}\right)_{kk}$. The corresponding $\mathbf{\Lambda}^k$ is rescaled by the same standard deviation $\widehat{\mathrm{Var}}(f^k)^{1/2}$ (so that the fitted values $\mathbf{\Lambda}^k f^k$ are unchanged), and $\mathbf{Q}_{kk}$ adjusts implicitly.

It is important that this is the *total* variance and not the posterior variance alone: the posterior variance $(P_{t|T})_{kk}$ measures only the smoother's residual *uncertainty* about the factor, not the amplitude with which the factor actually moves over the sample. In our estimates the posterior term accounts for only a few percent of the total (the sample-variance term dominates), so normalising the total variance standardises the factor in the *sample* sense. This convention makes each factor dimensionless and directly comparable across blocks: a movement of "one unit" in the real factor corresponds to one sample standard deviation. It is the convention adopted in the Growth-at-Risk literature (Adrian, Boyarchenko, and Giannone, 2019), where factors enter the second-stage quantile regression as standardised quantities and coefficients are interpretable as "effects of a one-standard-deviation movement".

Convention 2: loading anchor normalisation. Alternatively, fix one entry of each $\mathbf{\Lambda}^k$ to a chosen value (typically 1). This pins down the unit of measurement of f^k to that of the anchoring series, leaving $\mathbf{Q}$ free to be estimated. This is the convention adopted in the DFM literature (Bańbura and Modugno, 2014). It has the advantage of being more "transparent" algebraically — no sample-based rescaling is needed — but the resulting factor is harder to compare across different samples or specifications.

Sign normalisation. Both conventions still leave the *sign* of each factor free. In both cases, the sign is fixed by choosing a reference series in each block — typically an index that uncontroversially represents "favourable conditions" (non-farm payrolls for the real block, a broad stock market index for the financial block, an aggregate sentiment measure for the third block) — and requiring its loading to be positive. If the loading is negative at convergence, we multiply both the factor and its loadings by -1.

The two conventions are observationally equivalent. Either convention pins down a unique representation of the estimated parameters within the equivalence class of rotational transformations. They yield identical fitted values, identical forecasts, and identical conditional quantiles — they differ only in the units used to express the factors and the corresponding loadings. Throughout the empirical analysis we adopt *Convention 1* (factor variance normalisation), consistent with the operational implementation discussed earlier, as it aligns with the standardised-factor convention of the Growth-at-Risk literature; results are reported using this normalisation. The alternative would yield identical conclusions modulo a re-scaling.

Note on timing. These normalisations are applied only *after* the EM has converged. Applying them during the EM iterations would not affect convergence (since the likelihood is invariant) but would add unnecessary computational cost. It is simpler and cleaner to let the EM find any rescaling it prefers, and then impose the economic normalisation as a final post-processing step.

Implications for Initialisation and Convergence

One might worry that the indeterminacy implies that the EM, starting from different initial points, converges to different parameter values — that the algorithm is "unstable". This is technically true, but it does not reflect instability: the different convergence points are all within the *same equivalence class*, and produce identical observations of y_t, identical likelihoods, and identical forecasts.

If we initialise the EM from two different perturbations of the PCA starting point and run it to convergence, we typically obtain:

- Different numerical values of $\mathbf{\Lambda}^R, \mathbf{\Lambda}^F, \mathbf{\Lambda}^X$;
- Correspondingly different numerical values of the smoothed factors $\hat{f}^R_{t|T}, \hat{f}^F_{t|T}, \hat{f}^X_{t|T}$;
- But *identical* fitted values $\mathbf{\Lambda}\hat{f}_{t|T}$, *identical* likelihoods, and *identical* conditional quantile estimates.

The two runs differ by a block-diagonal rescaling, which is eliminated by the ex post normalisations of the previous subsection. A standard robustness diagnostic is to verify that the two runs, *after normalisation*, produce essentially identical factor paths — any residual numerical difference reflects only the finite-precision arithmetic of the optimisation.

Closing remark. Rotational indeterminacy is an inherent feature of factor models, not a bug. The block-diagonal restrictions we have imposed eliminate most of it, leaving only a scale-and-sign ambiguity that is innocuous for all practical purposes. The estimated quantities relevant for nowcasting and Growth-at-Risk — fitted values, likelihoods, forecasts, conditional quantiles — are all identified in the strong sense.

Putting It All Together: The Complete Model

We have built the EM algorithm for the Student-t Dynamic Factor Model in layers. We derived the basic E-step under full-data assumptions; then we derived the M-step updates. The subsequent sections extended the model to increasingly realistic conditions: ragged-edge missing data, mixed-frequency observations with Mariano-Murasawa aggregation, block-diagonal factor restrictions for economic identification, and the management of residual rotational indeterminacy.

This section synthesises all these pieces into a single, unified model specification. The aim is to provide a consolidated reference for implementation and for the reader who wants to see how the ingredients fit together without re-reading the earlier sections.

The Full State-Space Model

Let the observed data matrix be $\mathbf{Y} \in \mathbb{R}^{M\times T}$, with M total series — partitioned into three economic blocks (real, financial, other) — observed at monthly frequency, some subject to quarterly frequency and ragged-edge publication delays. Let $r = r_R + r_F + r_X$ be the total number of factors, with r_R, r_F, r_X allocated to each block.

The augmented state. To accommodate Mariano-Murasawa mixed-frequency aggregation, the state vector is augmented to include the current and four lagged factors:

$$\tilde{f}_t \;=\; \big(f_t',\, f_{t-1}',\, f_{t-2}',\, f_{t-3}',\, f_{t-4}'\big)' \;\in\; \mathbb{R}^{5r},$$

with the underlying monthly factor vector partitioned by block: $f_t = (f_t^R,\, f_t^F,\, f_t^X)'$.

The state transition equation. The monthly factor dynamics follow a VAR(1) with Student-t innovations, lifted to the augmented state via the companion form:

$$\tilde{f}_t \;=\; \tilde{\mathbf{A}}\,\tilde{f}_{t-1} \;+\; \tilde{u}_t, \qquad \tilde{u}_t \mid w_t^u \;\sim\; \mathcal{N}\big(\mathbf{0},\, \tilde{\mathbf{Q}}/w_t^u\big), \tag{126}$$

where

$$\tilde{\mathbf{A}} \;=\; \begin{pmatrix} \mathbf{A} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} \end{pmatrix}, \qquad \tilde{\mathbf{Q}} \;=\; \mathrm{diag}(\mathbf{Q}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}),$$

and the weight $w_t^u \sim \mathrm{Gamma}(\nu_u/2, \nu_u/2)$ governs the heavy-tailed distribution of the innovation u_t via the scale-mixture representation. The matrix $\mathbf{A} \in \mathbb{R}^{r\times r}$ is the unrestricted monthly VAR(1) coefficient matrix — it captures dynamic spillovers *across* blocks.

The observation equation. Order the series as $y_t = (y_t^{M\prime},\, y_t^{Q\prime})'$, with monthly series stacked first and quarterly series last. The observation equation in the augmented representation is:

$$y_t^{\mathrm{obs}} \;=\; \big(\mathbf{W}_t\,\tilde{\mathbf{\Lambda}}\big)\,\tilde{f}_t \;+\; \mathbf{W}_t\,\varepsilon_t, \qquad \varepsilon_t \mid w_t^{\varepsilon} \;\sim\; \mathcal{N}\big(\mathbf{0},\, \mathbf{R}/w_t^{\varepsilon}\big), \tag{127}$$

where:

- $\mathbf{W}_t \in \{0,1\}^{m_t\times M}$ is the selection matrix for time t, handling both ragged-edge missingness and the absence of quarterly observations at non-quarter-end months;

- $\tilde{\mathbf{\Lambda}} \in \mathbb{R}^{M\times 5r}$ is the effective loading matrix, combining the monthly loading block $\mathbf{\Lambda}^M$ (with block-diagonal structure) and the Mariano-Murasawa aggregation pattern for the quarterly loadings $\mathbf{\Lambda}^Q$:

$$\tilde{\mathbf{\Lambda}} \;=\; \begin{pmatrix} \mathbf{\Lambda}^M & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \frac{1}{3}\mathbf{\Lambda}^Q & \frac{2}{3}\mathbf{\Lambda}^Q & \mathbf{\Lambda}^Q & \frac{2}{3}\mathbf{\Lambda}^Q & \frac{1}{3}\mathbf{\Lambda}^Q \end{pmatrix}. \tag{128}$$

Block-diagonal loading structure. Both $\mathbf{\Lambda}^M$ and $\mathbf{\Lambda}^Q$ exhibit the block-diagonal structure: each series loads only on the factor of its economic group. In particular, the (monthly) loading matrix is

$$\mathbf{\Lambda}^M \;=\; \begin{pmatrix} \mathbf{\Lambda}^{M,R} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{\Lambda}^{M,F} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Lambda}^{M,X} \end{pmatrix},$$

and similarly for $\mathbf{\Lambda}^Q$. The scalar elements of each block are the free parameters to be estimated. The zero off-diagonal blocks are the economic exclusion restrictions.

Idiosyncratic covariance. The idiosyncratic noise covariance $\mathbf{R}$ is diagonal:

$$\mathbf{R} \;=\; \mathrm{diag}(r_1, r_2, \ldots, r_M),$$

with heavy-tailed scaling governed by the idiosyncratic weight $w_t^\varepsilon \sim \mathrm{Gamma}(\nu_\varepsilon/2, \nu_\varepsilon/2)$.

The full parameter vector. The complete parameter vector is:

$$\theta \;=\; \big\{\, \mathbf{A},\;\; \mathbf{Q},\;\; \mathbf{\Lambda}^M,\;\; \mathbf{\Lambda}^Q,\;\; \mathbf{R},\;\; \nu_u,\;\; \nu_\varepsilon \,\big\}, \tag{129}$$

where $\mathbf{A} \in \mathbb{R}^{r\times r}$ is unrestricted, $\mathbf{Q} \in \mathbb{R}^{r\times r}$ is symmetric positive definite, $\mathbf{\Lambda}^M$ and $\mathbf{\Lambda}^Q$ are block-diagonal, $\mathbf{R}$ is diagonal, and $\nu_u, \nu_\varepsilon > 2$ for the existence of finite second moments.

The complete observation system in matrix form. For concreteness, we display the full observation system in expanded matrix form, with all blocks visible and dimensions explicit. The construction supports any number of quarterly series in each economic block: $\mathbf{\Lambda}^{Q,R}, \mathbf{\Lambda}^{Q,F}, \mathbf{\Lambda}^{Q,X}$ may each contain zero, one, or several rows depending on which series in the dataset are observed at quarterly frequency. In our applied setting GDP is the only quarterly series and belongs to the real block, so $\mathbf{\Lambda}^{Q,R}$ has one row and $\mathbf{\Lambda}^{Q,F}, \mathbf{\Lambda}^{Q,X}$ are empty; we nonetheless display the general expression below to make the structure visible for any future extension.

Stack the data column by column: $\mathbf{Y} = (y_1, y_2, \ldots, y_T) \in \mathbb{R}^{M\times T}$, and partition the rows of $\mathbf{Y}$ into six blocks — three for monthly series (R, F, X) and three for quarterly series (R, F, X):

$$\mathbf{Y} \;=\; \left(\begin{array}{c} \mathbf{Y}^{M,R} \\ \mathbf{Y}^{M,F} \\ \mathbf{Y}^{M,X} \\ \hline \mathbf{Y}^{Q,R} \\ \mathbf{Y}^{Q,F} \\ \mathbf{Y}^{Q,X} \end{array}\right), \qquad \mathbf{Y}^{M,k} \in \mathbb{R}^{M_M^k \times T}, \quad \mathbf{Y}^{Q,k} \in \mathbb{R}^{M_Q^k \times T}.$$

The total number of series is $M = \sum_k (M_M^k + M_Q^k)$.

The observation equation in companion form. The observation system $\mathbf{Y} = \tilde{\mathbf{\Lambda}}\,\tilde{\mathbf{F}} + \mathbf{E}$ takes the following expanded form (without missing data; the selection matrix $\mathbf{W}_t$ is applied column by column at runtime):

$$
\underbrace{\left(\begin{array}{c} \mathbf{Y}^{M,R} \\ \mathbf{Y}^{M,F} \\ \mathbf{Y}^{M,X} \\ \hline \mathbf{Y}^{Q,R} \\ \mathbf{Y}^{Q,F} \\ \mathbf{Y}^{Q,X} \end{array}\right)}_{M\times T}
=
\underbrace{\left(\begin{array}{ccccc}
\mathbf{\Lambda}^{M,R} & \mathbf{0} & \mathbf{0} & \mathbf{0}_{M_M^R\times 4r} & \\
\mathbf{0} & \mathbf{\Lambda}^{M,F} & \mathbf{0} & \mathbf{0}_{M_M^F\times 4r} & \\
\mathbf{0} & \mathbf{0} & \mathbf{\Lambda}^{M,X} & \mathbf{0}_{M_M^X\times 4r} & \\
\hline
c_1\mathbf{\Lambda}^{Q,R}\,\mathbf{0}\,\mathbf{0} & c_2\mathbf{\Lambda}^{Q,R}\,\mathbf{0}\,\mathbf{0} & c_3\mathbf{\Lambda}^{Q,R}\,\mathbf{0}\,\mathbf{0} & c_4\mathbf{\Lambda}^{Q,R}\,\mathbf{0}\,\mathbf{0} & c_5\mathbf{\Lambda}^{Q,R}\,\mathbf{0}\,\mathbf{0} \\
\mathbf{0}\,c_1\mathbf{\Lambda}^{Q,F}\,\mathbf{0} & \mathbf{0}\,c_2\mathbf{\Lambda}^{Q,F}\,\mathbf{0} & \mathbf{0}\,c_3\mathbf{\Lambda}^{Q,F}\,\mathbf{0} & \mathbf{0}\,c_4\mathbf{\Lambda}^{Q,F}\,\mathbf{0} & \mathbf{0}\,c_5\mathbf{\Lambda}^{Q,F}\,\mathbf{0} \\
\mathbf{0}\,\mathbf{0}\,c_1\mathbf{\Lambda}^{Q,X} & \mathbf{0}\,\mathbf{0}\,c_2\mathbf{\Lambda}^{Q,X} & \mathbf{0}\,\mathbf{0}\,c_3\mathbf{\Lambda}^{Q,X} & \mathbf{0}\,\mathbf{0}\,c_4\mathbf{\Lambda}^{Q,X} & \mathbf{0}\,\mathbf{0}\,c_5\mathbf{\Lambda}^{Q,X}
\end{array}\right)}_{\tilde{\mathbf{\Lambda}}\,:\,M\times 5r}
\underbrace{\left(\begin{array}{c} \mathbf{F}^{R} \\ \mathbf{F}^{F} \\ \mathbf{F}^{X} \\ \hline \mathbf{F}^{R}_{-1} \\ \mathbf{F}^{F}_{-1} \\ \mathbf{F}^{X}_{-1} \\ \hline \vdots \\ \hline \mathbf{F}^{R}_{-4} \\ \mathbf{F}^{F}_{-4} \\ \mathbf{F}^{X}_{-4} \end{array}\right)}_{\tilde{\mathbf{F}}\,:\,5r\times T}
+
\underbrace{\left(\begin{array}{c} \mathbf{E}^{M,R} \\ \mathbf{E}^{M,F} \\ \mathbf{E}^{M,X} \\ \hline \mathbf{E}^{Q,R} \\ \mathbf{E}^{Q,F} \\ \mathbf{E}^{Q,X} \end{array}\right)}_{M\times T}.
$$

Reading the loading matrix $\tilde{\mathbf{\Lambda}}$. Two restrictions are visible in the structure of $\tilde{\mathbf{\Lambda}}$:

Block-diagonal restriction (vertical direction). Within each of the five lag-blocks of width r (columns labelled by lag $\ell = 0, 1, 2, 3, 4$), the columns are partitioned as $(\mathbf{F}^{R}_{t-\ell}, \mathbf{F}^{F}_{t-\ell}, \mathbf{F}^{X}_{t-\ell})$. Each row of $\tilde{\mathbf{\Lambda}}$ has non-zero entries only in the columns corresponding to the factor of its own block. The notation "$c_k\mathbf{\Lambda}^{Q,F}\,\mathbf{0}\,\mathbf{0}$" inside a lag-block means that, within that lag-block, only the financial-factor columns contain $c_k\mathbf{\Lambda}^{Q,F}$, while the real-factor and other-factor columns are zero.

Mariano-Murasawa restriction (horizontal direction). Quarterly rows load on *all five* lag-blocks with weights $c = (\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3})$, encoding the quarterly aggregation of monthly factors. Monthly rows load only on the contemporaneous lag-block ($\ell = 0$), with zero on the four lagged copies of the factor — they have no MM aggregation because they are observed at the same frequency as the factor.

The state equation in companion form. Symmetrically, the augmented state evolves as $\tilde{f}_t = \tilde{\mathbf{A}}\,\tilde{f}_{t-1} + \tilde{u}_t$. With the same partition of the factor block $f_t = (f_t^R, f_t^F, f_t^X)$ and the same lag-block ordering of $\tilde{f}_t$, the system reads:

$$
\underbrace{\left(\begin{array}{c} f_t^R \\ f_t^F \\ f_t^X \\ \hline f_{t-1}^R \\ f_{t-1}^F \\ f_{t-1}^X \\ \hline f_{t-2}^R \\ f_{t-2}^F \\ f_{t-2}^X \\ \hline \vdots \\ \hline f_{t-4}^R \\ f_{t-4}^F \\ f_{t-4}^X \end{array}\right)}_{\tilde{f}_t\,:\,5r\times 1}
=
\underbrace{\left(\begin{array}{cccc}
\mathbf{A}^{RR} & \mathbf{A}^{RF} & \mathbf{A}^{RX} & \mathbf{0}_{r\times 4r} \\
\mathbf{A}^{FR} & \mathbf{A}^{FF} & \mathbf{A}^{FX} & \mathbf{0}_{r\times 4r} \\
\mathbf{A}^{XR} & \mathbf{A}^{XF} & \mathbf{A}^{XX} & \mathbf{0}_{r\times 4r} \\
\hline
I_{r_R} & \mathbf{0} & \mathbf{0} & \mathbf{0}_{r\times 4r} \\
\mathbf{0} & I_{r_F} & \mathbf{0} & \mathbf{0}_{r\times 4r} \\
\mathbf{0} & \mathbf{0} & I_{r_X} & \mathbf{0}_{r\times 4r} \\
\hline
\mathbf{0} & \cdots & \cdots & \cdots \\
\hline
\vdots & \vdots & \vdots & \vdots
\end{array}\right)}_{\tilde{\mathbf{A}}\,:\,5r\times 5r}
\underbrace{\left(\begin{array}{c} f_{t-1}^R \\ f_{t-1}^F \\ f_{t-1}^X \\ \hline f_{t-2}^R \\ \vdots \\ f_{t-5}^X \end{array}\right)}_{\tilde{f}_{t-1}\,:\,5r\times 1}
+
\underbrace{\left(\begin{array}{c} u_t^R \\ u_t^F \\ u_t^X \\ \hline \mathbf{0} \\ \hline \mathbf{0} \\ \hline \vdots \\ \hline \mathbf{0} \end{array}\right)}_{\tilde{u}_t\,:\,5r\times 1}.
$$

The first block-row encodes the genuine VAR(1) dynamics $f_t = \mathbf{A}f_{t-1} + u_t$ with $\mathbf{A}$ a full $r \times r$ matrix (real, financial, and other factors interacting dynamically across blocks). The remaining four block-rows are identity shifts: they propagate the lagged factors forward in time without any dynamics, because they are deterministic copies of past factor values. The innovation $\tilde{u}_t$ has only its first block non-zero (distributed as $\mathcal{N}(\mathbf{0}, \mathbf{Q}/w_t^u)$), reflecting that genuine shocks enter only at the contemporaneous step.

The two equations side by side. Together, the augmented observation equation $\mathbf{Y} = \tilde{\mathbf{\Lambda}}\,\tilde{\mathbf{F}} + \mathbf{E}$ and the augmented state equation $\tilde{f}_t = \tilde{\mathbf{A}}\,\tilde{f}_{t-1} + \tilde{u}_t$ form a linear Gaussian state-space model (conditional on the weights w_t^u, w_t^ε) of dimension $5r$, ready for the Kalman filter and smoother. The block-diagonal restriction enters *horizontally* in $\tilde{\mathbf{\Lambda}}$ (which factor of which block); the MM aggregation enters *vertically* via the weights c on the 5 lag-blocks (which lag of the factor); the VAR dynamics enter in $\tilde{\mathbf{A}}$ via the unrestricted $\mathbf{A}$ in the top block-row; and the heavy-tailed structure enters via w_t^u, w_t^ε governing the conditional covariances $\mathbf{Q}/w_t^u$ and $\mathbf{R}/w_t^\varepsilon$.

A small concrete example. To make the structure of $\tilde{\mathbf{\Lambda}}$ tangible, consider a minimal dataset with three series:

- Series 1: *payrolls*, monthly, real block;
- Series 2: *stock market index*, monthly, financial block;
- Series 3: *GDP*, quarterly, real block.

With one factor per economic block ($r_R = r_F = r_X = 1$, hence $r = 3$), the augmented state vector $\tilde{f}_t \in \mathbb{R}^{15}$ stacks the three factors at five consecutive lags:

$$\tilde{f}_t \;=\; \big(\,\underbrace{f_t^R,\, f_t^F,\, f_t^X}_{\text{lag 0}},\, \underbrace{f_{t-1}^R,\, f_{t-1}^F,\, f_{t-1}^X}_{\text{lag 1}},\, \underbrace{f_{t-2}^R,\, f_{t-2}^F,\, f_{t-2}^X}_{\text{lag 2}},\, \underbrace{f_{t-3}^R,\, f_{t-3}^F,\, f_{t-3}^X}_{\text{lag 3}},\, \underbrace{f_{t-4}^R,\, f_{t-4}^F,\, f_{t-4}^X}_{\text{lag 4}}\,\big)'.$$

The corresponding loading matrix $\tilde{\mathbf{\Lambda}} \in \mathbb{R}^{3\times 15}$ takes the fully-expanded form:

$$\tilde{\mathbf{\Lambda}} \;=\; \begin{pmatrix} \lambda_1^{M,R} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1^{M,F} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3}\lambda_1^{Q,R} & 0 & 0 & \frac{2}{3}\lambda_1^{Q,R} & 0 & 0 & \lambda_1^{Q,R} & 0 & 0 & \frac{2}{3}\lambda_1^{Q,R} & 0 & 0 & \frac{1}{3}\lambda_1^{Q,R} & 0 & 0 \end{pmatrix}.$$

Reading the rows row by row:

- *Row 1 (payrolls, monthly real):* a single non-zero entry in column 1, corresponding to f_t^R at lag 0. All other 14 entries are zero — payrolls do not load on financial or other factors (block-diagonal restriction), and as a monthly series they do not load on lagged factors (no MM aggregation needed).

- *Row 2 (stock index, monthly financial):* a single non-zero entry in column 2, corresponding to f_t^F at lag 0. The reasoning is symmetric to row 1.

- *Row 3 (GDP, quarterly real):* five non-zero entries, one in each lag-block, all in the column corresponding to f^R (the real factor) — columns 1, 4, 7, 10, 13. The values are $\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}$ times $\lambda_1^{Q,R}$, the loading of GDP on the underlying monthly real factor. The financial and other columns are zero in every lag (block-diagonal restriction applied to all five lags).

The two patterns visible in $\tilde{\mathbf{\Lambda}}$ — block-diagonal in *which factor* a row loads on, and Mariano–Murasawa in *which lags* a quarterly row loads on — are now fully concrete: $\lambda_1^{M,R}$ enters once, $\lambda_1^{M,F}$ enters once, and $\lambda_1^{Q,R}$ enters five times with the MM weights, all in the real-factor columns.

The E-Step in the Unified Model

The E-step integrates all the extensions developed in the previous sections. At each outer iteration j, it proceeds as follows.

Inner ECM loop. Starting from $\hat{w}_t^{u,[0]} = \hat{w}_t^{\varepsilon,[0]} = 1$ for all t, iterate for $k = 1, 2, \ldots$ until convergence:

1. *Kalman filter with* $\mathbf{W}_t$ *selection.* Run the filter on the augmented state $\tilde{f}_t$ using time-varying covariances $\tilde{\mathbf{Q}}_t = \tilde{\mathbf{Q}}/\hat{w}_t^{u,[k-1]}$ and $\tilde{\mathbf{R}}_t = \mathbf{R}/\hat{w}_t^{\varepsilon,[k-1]}$. Apply $\mathbf{W}_t$ in the update step to handle missing data.

2. *RTS smoother on the augmented state.* Obtain $\{\hat{\tilde{f}}_{t|T}, \tilde{P}_{t|T}\}$, from which the monthly factors and their cross-lag covariances are extracted as sub-blocks.

3. *Compute Mahalanobis residuals.* For $t = 1, \ldots, T$:
$$\begin{aligned} d_t^{u,[k]} &= \mathbb{E}[(f_t - \mathbf{A}f_{t-1})' \, \mathbf{Q}^{-1} \, (f_t - \mathbf{A}f_{t-1}) \mid \mathbf{Y}], \\ d_t^{\varepsilon,\text{obs},[k]} &= \mathbb{E}[(y_t^{\text{obs}} - \mathbf{W}_t\tilde{\mathbf{\Lambda}}\tilde{f}_t)' \, (\mathbf{W}_t\mathbf{R}\mathbf{W}_t')^{-1} \, (y_t^{\text{obs}} - \mathbf{W}_t\tilde{\mathbf{\Lambda}}\tilde{f}_t) \mid \mathbf{Y}]. \end{aligned}$$
Both include the correction for the posterior uncertainty of the factor via the trace terms introduced previously.

4. *Update the weights*:
$$\hat{w}_t^{u,[k]} = \frac{\nu_u + r}{\nu_u + d_t^{u,[k]}}, \qquad \hat{w}_t^{\varepsilon,[k]} = \frac{\nu_\varepsilon + m_t}{\nu_\varepsilon + d_t^{\varepsilon,\text{obs},[k]}}.$$

5. *Check convergence.* If $\max_t |\hat{w}_t^{[k]} - \hat{w}_t^{[k-1]}| < \text{tol}_{\text{inner}}$ for both weight sequences, exit the inner loop. Otherwise set $k \leftarrow k + 1$ and repeat.

Log-weight expectations. After the inner loop converges, compute the posterior log-weights needed for the ν-update:
$$\widehat{\log w_t}^{u} = \psi\big(\tfrac{\nu_u + r}{2}\big) - \log\big(\tfrac{\nu_u + d_t^u}{2}\big), \qquad \widehat{\log w_t}^{\varepsilon} = \psi\big(\tfrac{\nu_\varepsilon + m_t}{2}\big) - \log\big(\tfrac{\nu_\varepsilon + d_t^{\varepsilon,\text{obs}}}{2}\big).$$

The M-Step in the Unified Model

With the posterior moments in hand, update the parameters sequentially. The block-by-block order is:

(a) Update of A and Q. Using only the first block of the augmented smoothed moments (the contemporaneous monthly factor):
$$\mathbf{A}^{(j+1)} = \mathcal{P}_{10}^u \, (\mathcal{P}_{00}^u)^{-1}, \qquad \mathbf{Q}^{(j+1)} = \frac{1}{T}\Big(\mathcal{P}_{11}^u - \mathbf{A}^{(j+1)} \, (\mathcal{P}_{10}^u)'\Big),$$
where $\mathcal{P}_{ab}^u = \sum_t \hat{w}_t^u \, \mathbb{E}[f_{t-a}f_{t-b}' \mid \mathbf{Y}]$.

(b) Update of $\mathbf{\Lambda}^M$ — monthly loadings, row by row. For each monthly series i in block $k \in \{R, F, X\}$, update the scalar loading on factor f^k:
$$\mathbf{\Lambda}_i^{M,k,(j+1)} = \frac{\sum_{t \in \mathcal{T}_i^M} \hat{w}_t^\varepsilon \, y_{i,t}^M \, \mathbb{E}[f_t^k \mid \mathbf{Y}]}{\sum_{t \in \mathcal{T}_i^M} \hat{w}_t^\varepsilon \, \mathbb{E}[(f_t^k)^2 \mid \mathbf{Y}]}.$$

(c) Update of $\mathbf{\Lambda}^Q$ — quarterly loadings, row by row. For each quarterly series j in block k, using the composite regressor ϕ_t^k:

$$\mathbf{\Lambda}_j^{Q,k,(j+1)} = \frac{\sum_{t\in\mathcal{T}_j^Q} \hat{w}_t^\varepsilon \, y_{j,t}^Q \, \mathbb{E}[\phi_t^k \mid \mathbf{Y}]}{\sum_{t\in\mathcal{T}_j^Q} \hat{w}_t^\varepsilon \, \mathbb{E}[(\phi_t^k)^2 \mid \mathbf{Y}]},$$

with $\phi_t^k = \frac{1}{3}f_t^k + \frac{2}{3}f_{t-1}^k + f_{t-2}^k + \frac{2}{3}f_{t-3}^k + \frac{1}{3}f_{t-4}^k$.

(d) Update of R — entry by entry. For each series i:

$$r_i^{(j+1)} = \frac{1}{|\mathcal{T}_i|} \sum_{t\in\mathcal{T}_i} \hat{w}_t^\varepsilon \, \mathbb{E}\big[\, e_{i,t}^2 \,\big|\, \mathbf{Y} \,\big], \qquad e_{i,t} = \begin{cases} y_{i,t}^M - \mathbf{\Lambda}_{i\cdot}^{M,(j+1)} f_t^k & \text{(monthly)}, \\ y_{i,t}^Q - \mathbf{\Lambda}_{i\cdot}^{Q,(j+1)} \phi_t^k & \text{(quarterly)}. \end{cases}$$

where the residual uses f_t^k for monthly series and ϕ_t^k for quarterly series.

(e) Update of ν_u and ν_ε via Brent. Solve the root-finding problems:

$$g_u(\nu_u^{(j+1)}) = 0, \qquad g_\varepsilon(\nu_\varepsilon^{(j+1)}) = 0,$$

with $g(\cdot)$ defined by the first-order conditions (86) and (87).

Interactions Between the Extensions

It is worth pausing on how the various extensions interact cleanly:

(i) Ragged edge and mixed-frequency share the same formalism. Both are handled by the selection matrix $\mathbf{W}_t$. For monthly series, $\mathbf{W}_t$ drops those not yet released. For quarterly series, $\mathbf{W}_t$ drops them at non-quarter-end months, and additionally drops them at quarter-end months where the publication lag has not elapsed. A single, uniform mechanism handles both.

(ii) Mixed-frequency interacts with block restrictions through ϕ_t^k. For a quarterly series in block k, the composite regressor ϕ_t^k is the MM aggregation applied to the *block-specific* factor. Quarterly GDP, being a real-activity variable, loads only on ϕ_t^R — never on ϕ_t^F or ϕ_t^X. The two restrictions (MM aggregation + economic exclusion) compose cleanly: they live on different dimensions of $\tilde{f}_t$ and do not interfere.

(iii) Student-t extensions affect only the *weights* in the sums. All updates for $\mathbf{A}, \mathbf{\Lambda}^M, \mathbf{\Lambda}^Q, \mathbf{R}$ have the *same structural form* as in the Gaussian DFM, differing only by the presence of $\hat{w}_t^u$ or $\hat{w}_t^\varepsilon$ inside the sums. The heavy-tailed framework is therefore a strict generalisation: setting $\hat{w}_t = 1$ for all t (equivalent to $\nu \to \infty$) recovers exactly the Gaussian DFM of Bańbura and Modugno (2014).

(iv) Block structure reduces the $\mathbf{\Lambda}$-update to scalar regressions. Without block restrictions, each row of $\mathbf{\Lambda}$ requires a 3×3 matrix inversion. With block restrictions — given that each block is one-dimensional in our specification — the update reduces to scalar divisions. The algorithm is both cheaper and more numerically stable.

(v) Rotational indeterminacy leaves all quantities of interest invariant. As established before, the fitted values, likelihood, forecasts, and conditional quantiles — the quantities used in the Growth-at-Risk stage of the analysis — are all invariant to the residual sign-and-scale indeterminacy. The model is identified in the strong sense for all practical purposes.

Summary: A Single Unified Algorithm

The entire framework can now be described compactly:

1. *Data structure.* Observe $\mathbf{Y}$ at monthly frequency, with M series partitioned into three economic blocks; quarterly series are flagged as missing outside quarter-end months; ragged edge handled by marking unreleased observations as missing.

2. *Model specification.* State-space model with augmented state $\tilde{f}_t \in \mathbb{R}^{5r}$, companion-form transition $\tilde{\mathbf{A}}$, effective loading $\tilde{\mathbf{\Lambda}}$ with block-diagonal structure and MM aggregation pattern, time-varying $\mathbf{W}_t$ selection for missing data, and Student-t scale-mixture representation for heavy-tailed shocks.

3. *Algorithm.* EM with inner ECM iteration for the coupling between factors and weights, sequential block-wise M-step updates, Brent root-finding for the degrees of freedom, PCA initialisation, and ex post scale-and-sign normalisation.

Everything fits together in a single algorithmic specification, which we state formally as pseudocode in the next section.

The Complete Algorithm: Pseudocode

This section consolidates the entire EM algorithm into a single, unified pseudocode. The intent is both pedagogical (showing how the pieces fit together in one procedural description) and practical (serving as a specification for implementation).

The pseudocode makes explicit the *outer EM loop* (which updates the parameter vector θ), the *inner ECM loop* (which resolves the coupling between factor and weight posterior moments within the E-step), and the *sequential M-step* (which updates $\mathbf{A}, \mathbf{Q}, \mathbf{\Lambda}, \mathbf{R}, \nu_u, \nu_\varepsilon$ one at a time, each conditional on the previously updated parameters).

Inputs and Outputs

Inputs.

- $\mathbf{Y} \in \mathbb{R}^{M \times T}$: observed data matrix, with `NaN` entries for missing observations (ragged edge and quarterly non-quarter-end months).
- Block-partitioning of the M series into three groups $\{R, F, X\}$, with sizes $M_R + M_F + M_X = M$.
- Number of factors per block: r_R, r_F, r_X (typically each equal to 1), with $r = r_R + r_F + r_X$.
- Frequency flag for each series: monthly or quarterly.
- Maximum number of outer iterations: $J_{\max}$ (typically 200–500).
- Outer tolerance: $\mathrm{tol}_{\mathrm{outer}}$ (typically 10^{-5}).
- Inner tolerance: $\mathrm{tol}_{\mathrm{inner}}$ (typically 10^{-4}).
- Bracket for the ν root-finding: $[\nu_{\min}, \nu_{\max}] = [2.001, 1000]$.

Outputs.

- Estimated parameters $\hat{\theta} = \{\hat{\mathbf{A}}, \hat{\mathbf{Q}}, \hat{\mathbf{\Lambda}}^M, \hat{\mathbf{\Lambda}}^Q, \hat{\mathbf{R}}, \hat{\nu}_u, \hat{\nu}_\varepsilon\}$.
- Smoothed factor moments $\{\hat{f}_{t|T}, P_{t|T}\}_{t=1}^T$.
- Posterior weight moments $\{\hat{w}_t^u, \hat{w}_t^\varepsilon\}_{t=1}^T$.
- Convergence trace $\{L^{(j)}\}$ for diagnostic purposes.

The Complete Pseudocode

Algorithms 1 and 2 together constitute the complete procedure. Algorithm 1 handles preprocessing, initialisation, and the E-step (the iterative factor–weight coupling resolution); Algorithm 2 handles the sequential M-step, convergence check, and post-processing.

Algorithm 1 EM algorithm for the Student-t DFM (Part 1 of 2): Preprocessing, Initialisation, and E-step

Require: $\mathbf{Y}$, block partitioning, $\{r_R, r_F, r_X\}$, $J_{\max}$, $\text{tol}_{\text{outer}}$, $\text{tol}_{\text{inner}}$, $[\nu_{\min}, \nu_{\max}]$

Ensure: $\hat{\theta}$, $\{\hat{f}_{t|T}, P_{t|T}\}$, $\{\hat{w}_t^u, \hat{w}_t^\varepsilon, \widehat{\log w_t^u}, \widehat{\log w_t^\varepsilon}\}$

// 1. Preprocessing

Build selection matrices $\{\mathbf{W}_t\}_{t=1}^T$ from the missing-data pattern of $\mathbf{Y}$

Build observation sets $\mathcal{T}_i^M$, $\mathcal{T}_j^Q$ for each series

Standardise $\mathbf{Y}$ column-wise

// 2. Initialisation

Fill missing entries: Mariano–Murasawa identity for quarterly series (locally constant fill via recursion (107)); $\mathcal{N}(0,1)$ draws for ragged-edge missings

Compute block-by-block PCA: for each block $b \in \{R, F, X\}$, eigendecompose $\widehat{\mathbf{\Sigma}}_Y^b$ and form $\mathbf{F}^{b,(0)} = (\mathbf{V}^b)'\tilde{\mathbf{Y}}^b$

$\mathbf{\Lambda}^{(0)} \leftarrow$ block-by-block scalar OLS: each row of $\tilde{\mathbf{Y}}$ regressed on the factor of its economic block (zeros enforced on off-block columns)

$\mathbf{A}^{(0)}, \mathbf{Q}^{(0)} \leftarrow$ VAR(1) OLS on $\mathbf{F}^{(0)}$

$\mathbf{R}^{(0)} \leftarrow$ diagonal residual variance

$\nu_u^{(0)}, \nu_\varepsilon^{(0)} \leftarrow 10$

$\tilde{\mathbf{\Sigma}}_0^{(0)} \leftarrow I_{5r}$ or solution to the discrete Lyapunov equation $\tilde{\mathbf{\Sigma}}_0 = \tilde{\mathbf{A}}^{(0)}\,\tilde{\mathbf{\Sigma}}_0\,\tilde{\mathbf{A}}^{(0)\prime} + \tilde{\mathbf{Q}}^{(0)}$

$\tilde{\mathbf{\Sigma}}_0$ is held fixed throughout the EM iterations (not updated)

// 3. Outer EM loop (continues in Algorithm 2)

for $j = 0, 1, \ldots, J_{\max} - 1$ **do**

    // 3a. E-step: inner ECM loop

    Set $k \leftarrow 0$, $\hat{w}_t^{u,[0]} \leftarrow 1$, $\hat{w}_t^{\varepsilon,[0]} \leftarrow 1$ for all t

    repeat

        $k \leftarrow k + 1$

        (F-update) Run Kalman filter on $\tilde{f}_t \in \mathbb{R}^{5r}$ with
        $\tilde{\mathbf{Q}}_t = \tilde{\mathbf{Q}}/\hat{w}_t^{u,[k-1]}$, $\tilde{\mathbf{R}}_t = \mathbf{R}/\hat{w}_t^{\varepsilon,[k-1]}$, and $\mathbf{W}_t$

        Run RTS smoother on augmented state

        Track incremental log-likelihood:
        $L^{[k]} \leftarrow -\frac{1}{2}\sum_t \left[\log|S_t^{\text{obs}}| + (\eta_t^{\text{obs}})'(S_t^{\text{obs}})^{-1}\eta_t^{\text{obs}} + m_t \log(2\pi)\right]$

        Extract monthly factor moments from augmented sub-blocks:
        $\hat{f}_{t|T} \leftarrow \hat{\tilde{f}}_{t|T}[1{:}r]$
        $P_{t|T} \leftarrow \tilde{P}_{t|T}[1{:}r,\ 1{:}r]$
        $P_{t,t-1|T} \leftarrow \tilde{P}_{t|T}[1{:}r,\ r{+}1{:}2r]$

        (W-update) Compute Mahalanobis residuals $d_t^{u,[k]}$ and $d_t^{\varepsilon,\text{obs},[k]}$

        Update weights:
        $\hat{w}_t^{u,[k]} \leftarrow (\nu_u + r)/(\nu_u + d_t^{u,[k]})$
        $\hat{w}_t^{\varepsilon,[k]} \leftarrow (\nu_\varepsilon + m_t)/(\nu_\varepsilon + d_t^{\varepsilon,\text{obs},[k]})$

    until $\max_t |\hat{w}_t^{[k]} - \hat{w}_t^{[k-1]}| < \text{tol}_{\text{inner}}$

    Store $\{\hat{f}_{t|T}, P_{t|T}, P_{t,t-1|T}, \hat{w}_t^u, \hat{w}_t^\varepsilon\}$

    Compute $\widehat{\log w_t^u}, \widehat{\log w_t^\varepsilon}$ via digamma formula

(Continued in Algorithm 2)

end for

Algorithm 2 EM algorithm for the Student-t DFM (Part 2 of 2): M-step, Convergence Check, and Post-processing

(Continuing from Algorithm 1; the line numbering restarts from 1, as the M-step constitutes a self-contained module.)

// 3b. M-step
Update A and Q:
$\mathcal{P}^u_{ab} \leftarrow \sum_t \hat{w}^u_t \mathbb{E}[f_{t-a} f'_{t-b} \mid \mathbf{Y}]$ for $(a,b) \in \{(1,1),(1,0),(0,0)\}$
$\mathbf{A}^{(j+1)} \leftarrow \mathcal{P}^u_{10}(\mathcal{P}^u_{00})^{-1}$
$\mathbf{Q}^{(j+1)} \leftarrow \frac{1}{T}(\mathcal{P}^u_{11} - \mathbf{A}^{(j+1)}(\mathcal{P}^u_{10})')$
Update $\mathbf{\Lambda}^M$ (monthly loadings), row by row:
for all blocks $k \in \{R, F, X\}$, monthly series i in block k **do**
$$\mathbf{\Lambda}^{M,k,(j+1)}_i \leftarrow \frac{\sum_{t \in \mathcal{T}^M_i} \hat{w}^\varepsilon_t \, y^M_{i,t} \, \mathbb{E}[f^k_t \mid \mathbf{Y}]}{\sum_{t \in \mathcal{T}^M_i} \hat{w}^\varepsilon_t \, \mathbb{E}[(f^k_t)^2 \mid \mathbf{Y}]}$$
end for
Update $\mathbf{\Lambda}^Q$ (quarterly loadings), row by row:
for all quarterly series j in block k **do**
Compute $\mathbb{E}[\phi^k_t \mid \mathbf{Y}]$ and $\mathbb{E}[(\phi^k_t)^2 \mid \mathbf{Y}]$ with MM weights
$$\mathbf{\Lambda}^{Q,k,(j+1)}_j \leftarrow \frac{\sum_{t \in \mathcal{T}^Q_j} \hat{w}^\varepsilon_t \, y^Q_{j,t} \, \mathbb{E}[\phi^k_t \mid \mathbf{Y}]}{\sum_{t \in \mathcal{T}^Q_j} \hat{w}^\varepsilon_t \, \mathbb{E}[(\phi^k_t)^2 \mid \mathbf{Y}]}$$
end for
Update R, entry by entry:
for all series $i = 1, \ldots, M$ **do**
$r^{(j+1)}_i \leftarrow \frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} \hat{w}^\varepsilon_t \, \mathbb{E}[(\text{residual}_{i,t})^2 \mid \mathbf{Y}]$
end for
Update ν_u and ν_ε via Brent's method:
$\bar{w}^u \leftarrow \frac{1}{T}\sum_t \hat{w}^u_t$, $\overline{\log w}^u \leftarrow \frac{1}{T}\sum_t \widehat{\log w}^u_t$
$g_u(\nu) \leftarrow \log(\nu/2) - \psi(\nu/2) + 1 + \overline{\log w}^u - \bar{w}^u$
$\nu^{(j+1)}_u \leftarrow \text{Brent}(g_u; [\nu_{\min}, \nu_{\max}])$
(Analogous computation for $\nu^{(j+1)}_\varepsilon$)
// 3c. Convergence check
$L^{(j+1)} \leftarrow$ full ELBO $=$ Kalman marginal log-likelihood $+ \mathbb{E}_q[\log p(W \mid \nu)] + H[q(W)]$ at converged inner loop
if $|L^{(j+1)} - L^{(j)}|/|L^{(j)}| < \text{tol}_{\text{outer}}$ **then**
break
end if
// 4. Post-processing - Rotational indeterminacies
Rescale each factor to unit empirical variance; adjust loadings correspondingly
Fix signs via reference series
return $\hat{\theta}$, $\{\hat{f}_{t|T}, P_{t|T}\}$, $\{\hat{w}^u_t, \hat{w}^\varepsilon_t, \widehat{\log w}^u_t, \widehat{\log w}^\varepsilon_t\}$

Remarks on the Pseudocode

Modular structure. The pseudocode is organised into four modular blocks: preprocessing, initialisation, the outer EM loop, and post-processing. Each of these modules can be implemented and tested independently, which is a significant advantage during development.

Nested loops. The algorithm contains *three levels of loops*:

- The outer EM loop (up to $J_{\max}$ iterations);

- The inner ECM loop (typically 3–10 iterations per outer step);
- The block-by-block and series-by-series loops in the M-step (finite, deterministic length).

The inner ECM loop is the only one with a data-dependent number of iterations; the others are bounded in advance. The overall computational cost is dominated by the Kalman filter/smoother pass, which has complexity $O(T \cdot r^3)$ per iteration (due to the matrix inversions in the update step).

Numerical stability considerations. Several implementation details require care:

- *Matrix inversions.* The Kalman gain and the M-step updates involve matrix inversions of potentially ill-conditioned matrices, especially in the early iterations where estimates are imprecise. In practice one uses the pseudoinverse or Cholesky-based solves (`numpy.linalg.solve`, `scipy.linalg.cho_solve`) rather than explicit matrix inversion.
- *Singular* $\tilde{\mathbf{Q}}$. As noted before, the augmented innovation covariance $\tilde{\mathbf{Q}}$ has rank r in a $5r \times 5r$ matrix. The Kalman filter remains numerically stable provided one uses the formulation above (propagating $\tilde{P}_{t|t}$ rather than attempting to invert $\tilde{\mathbf{Q}}$ directly).
- *Brent's method for* ν. Verify at each iteration that the bracket $[\nu_{\min}, \nu_{\max}]$ still contains the root by checking $g_u(\nu_{\min}) > 0$ and $g_u(\nu_{\max}) < 0$. If the test fails, extend the bracket or switch to a safeguarded Newton step.
- *Posterior weights.* The weights $\hat{w}_t^u, \hat{w}_t^\varepsilon$ are guaranteed to be positive by construction (they are posterior means of Gamma-distributed random variables). In implementation, it is nevertheless prudent to floor them at a small positive value (e.g., 10^{-6}) to avoid division-by-zero in the Kalman update when dividing by the weight.

Structure of the inner ECM loop. The inner ECM loop resolves the coupling. Typical behaviour: starting from all weights equal to 1, the first iteration produces a Gaussian-like Kalman pass; the weights then update based on the resulting residuals, down-weighting outlier time periods. Subsequent iterations refine the weights until convergence. Empirically, 3–5 inner iterations are sufficient in most datasets; we have never observed a case requiring more than 15.

Extracting outputs. The pseudocode returns the smoothed monthly factors $\hat{f}_{t|T}$ and their covariances $P_{t|T}$. These are the inputs to the second stage of the framework — the quantile regression for Growth-at-Risk — which we will describe in the next chapter. In implementation, one also typically returns the final $\tilde{\mathbf{\Lambda}}$ (the effective augmented loading), the fitted values $\mathbf{\Lambda}\hat{f}_{t|T}$ for diagnostic purposes, and the history $\{L^{(j)}\}$ of ELBO values for convergence monitoring.

Convergence and Stopping Criteria

The EM algorithm is guaranteed to converge to a *stationary point* of the observed-data log-likelihood — either a local maximum, a saddle point, or (in pathological cases) a boundary point. This guarantee is asymptotic: it does not say *how fast* convergence occurs, nor does it tell us *when to stop* in a finite-time implementation. In this section we discuss the practical question of how to monitor and terminate the algorithm.

Monotonicity of the EM Algorithm

A foundational property of EM is the *monotonicity of the observed-data log-likelihood* across outer iterations:

$$\ell(\theta^{(j+1)} \mid \mathbf{Y}) \;\geq\; \ell(\theta^{(j)} \mid \mathbf{Y}) \qquad \text{for every } j. \tag{130}$$

That is, the log-likelihood is guaranteed to increase (weakly) at every step. This is not a consequence of the specific parameterisation — it is a general property that follows from the ELBO decomposition and from the fact that the M-step maximises the expected complete-data log-likelihood while the E-step tightens the ELBO.

Implication for monitoring. Equation (130) provides a strong diagnostic: if at any iteration the log-likelihood *decreases*, the algorithm has a bug — most likely in the M-step derivatives or in the sign of one of the updates. In practice, monitoring the log-likelihood path across iterations is an invaluable debugging tool: a bug that corrupts a single derivative will almost always produce a visibly non-monotone trajectory.

Caveat under variational EM. Strictly speaking, the monotonicity (130) applies to *exact* EM, where the E-step returns the true posterior $p(\mathcal{Z} \mid \mathbf{Y}, \theta^{(j)})$. In our implementation we use *variational* EM, where the E-step returns a mean-field approximation. Under the variational relaxation, the monotonic quantity is the *ELBO* $\mathcal{L}(q^{(j+1)}, \theta^{(j+1)})$, not the log-likelihood itself. However, the ELBO differs from the log-likelihood by a KL divergence term — typically small when the mean-field factorisation is a good approximation — so in practice the log-likelihood remains nearly monotone.

In our setting, the mean-field factorisation between factors and weights is an excellent approximation (the true posterior is well-approximated by the product of its marginals). Empirically, we observe clean monotone increases in both the log-likelihood and the ELBO; deviations from monotonicity, if any, are numerical artifacts at the scale of floating-point precision.

The ELBO as a Monitoring Quantity

The principled quantity to monitor across iterations is the ELBO

$$\mathcal{L}(q, \theta) \;=\; \mathbb{E}_q[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)] \;-\; \mathbb{E}_q[\log q(\mathcal{Z})], \tag{131}$$

evaluated at the output of each outer iteration. Under the mean-field factorisation, $\mathcal{L}$ admits a closed-form expression in terms of the moments computed in the E-step, so no additional computational burden is required to track it.

Computing the ELBO in practice. The ELBO decomposes into five blocks corresponding to the five blocks of the complete-data log-likelihood:

- *Observation block*: $-\frac{1}{2}\log|\mathbf{R}|$ terms, log-likelihood of the observed y_t^{obs} given factors and weights;
- *State transition block*: $-\frac{T}{2}\log|\mathbf{Q}|$ terms, plus the weighted state residuals;
- *Weight priors block D*: $-T\log\Gamma(\nu_u/2)$, plus log-linear terms in $\hat{w}_t^u$ and $\widehat{\log w}_t^u$;
- *Weight priors block E*: analogous for ν_ε;
- *Entropy of the variational posterior*: $-\mathbb{E}_q[\log q]$, available in closed form under the mean-field factorisation. Since each weight posterior is a Gamma density, $q(W_{s,t}) = \text{Gamma}(\alpha_{s,t}, \beta_{s,t})$, and the weights are conditionally independent across t and across the two families $s \in \{u, \varepsilon\}$, the entropy is the sum of the per-component Gamma entropies,

$$H[q(W_s)] = \sum_t \Big[\alpha_{s,t} - \log\beta_{s,t} + \log\Gamma(\alpha_{s,t}) + (1-\alpha_{s,t})\,\psi(\alpha_{s,t})\Big], \tag{132}$$

where ψ is the digamma function. Using the posterior identity $\log\beta_{s,t} = \psi(\alpha_{s,t}) - \mathbb{E}_q[\log W_{s,t}]$, this reduces to the form actually evaluated in the implementation,

$$H[q(W_s)] = \sum_t \Big[\alpha_{s,t}\big(1-\psi(\alpha_{s,t})\big) + \log\Gamma(\alpha_{s,t})\Big] + T_s\,\overline{\log w}_s, \tag{133}$$

with shape $\alpha_{u,t} = (\nu_u + r)/2$ (constant in t) on the factor side and $\alpha_{\varepsilon,t} = (\nu_\varepsilon + m_t)/2$ on the idiosyncratic side, m_t being the number of series observed at date t.

Each of these blocks is computable from the posterior moments already produced by the E-step. Implementing the ELBO calculation is then straightforward.

The weight correction vanishes in the Gaussian limit. The weight-prior blocks (D, E) and the entropy block above are the *only* parts of the ELBO that depend on the latent weights; together they form the correction $\Delta_W = \mathbb{E}_q[\log p(W \mid \nu)] + H[q(W)] = -\text{KL}\big(q(W)\,\|\,p(W \mid \nu)\big)$ that turns the Kalman conditional log-likelihood into the full ELBO. As $\nu_s \to \infty$ the prior $p(W_s \mid \nu_s) = \text{Gamma}(\nu_s/2, \nu_s/2)$ concentrates on a point mass at $w = 1$, and so does the posterior, since its shape $\alpha_{s,t} = (\nu_s + \cdot)/2 \to \infty$ with the same scale. Both the entropy (132) and the prior term individually diverge (the prior grows as $\frac{T_s}{2}\log\nu_s + O(1)$), but they diverge by offsetting amounts: their sum is the negative KL divergence between two laws that collapse onto the *same* degenerate limit, so $\Delta_W \to 0$. In the Gaussian case the weights are therefore identically one, no posterior uncertainty remains, and the full ELBO reduces term by term to the Kalman conditional log-likelihood. The implementation reflects this exactly: when $\nu_s = \infty$ the corresponding contributions to Δ_W are skipped (set to zero) rather than evaluated, avoiding the spurious cancellation of two infinities.

A practical alternative: the Kalman marginal log-likelihood. Computing the full ELBO at each iteration is feasible but adds a non-trivial implementation effort. A common practical alternative is to monitor the *marginal log-likelihood* of $\mathbf{Y}$ produced as a by-product of the Kalman filter:

$$\log p(\mathbf{Y} \mid \theta^{(j)}) \;=\; -\tfrac{1}{2}\sum_t \big[\log|S_t^{\text{obs}}| + (\eta_t^{\text{obs}})'(S_t^{\text{obs}})^{-1}\eta_t^{\text{obs}} + m_t\log(2\pi)\big].$$

This quantity differs from the ELBO by the KL divergence between the mean-field q and the true posterior, which is small when the factorisation is accurate. Empirically, the two trajectories are nearly indistinguishable in our applications. While this proxy is available and is adequate whenever ν is finite and well-identified, we nonetheless adopt the *full* ELBO as the monitored quantity in our implementation, accepting the modest additional computational cost. The reason is that the proxy ceases to be faithful in the Gaussian limit $\nu \to \infty$: there the Kalman marginal log-likelihood and the ELBO diverge, and only the full ELBO — the quantity the EM algorithm actually maximises — behaves correctly as a convergence target. This case arises in the nesting experiment of the simulation study, where it is analysed in detail. The two additional terms of the ELBO, $\mathbb{E}_q[\log p(W \mid \nu)]$ and $H[q(W)]$, are available in closed form from the weight sufficient statistics already returned by the E-step, so the extra cost is a handful of scalar evaluations per iteration.

A necessary caveat: the proxy is faithful only for finite ν. The justification just given — that the Kalman marginal log-likelihood and the ELBO are numerically indistinguishable — requires an important qualification, and the standard statement of it is slightly imprecise. The gap between the two is *not* merely the Kullback–Leibler divergence between the mean-field posterior and the true posterior. Writing the ELBO as

$$\mathrm{ELBO}(\theta) = \underbrace{\mathbb{E}_q[\log p(Y \mid W, \theta)]}_{\text{conditional (Kalman) log-likelihood}} + \mathbb{E}_q[\log p(W \mid \nu)] + H[q(W)],$$

the difference between the monitored conditional log-likelihood and the full ELBO is the sum of the weight-prior term $\mathbb{E}_q[\log p(W \mid \nu)]$ and the posterior-entropy term $H[q(W)]$ — the part of the lower bound that concerns the latent weights. These two terms are stable, and the conditional log-likelihood is therefore a reliable proxy, *only when the degrees of freedom ν converge to a finite value*. Once $\nu^{(j)}$ settles near a finite optimum ν^*, the weight terms become constant across iterations and the two trajectories differ by a fixed additive constant that leaves the relative change —and hence the stopping decision— unaffected. This is the situation in every finite-tail application considered here, with our data we got the calibrated $\nu^* \approx 4$, and it is why the proxy works in practice.

The qualification matters because the proxy *does* break down in one well-defined limit: when the data are genuinely Gaussian, the maximiser drives $\nu \to \infty$, the weight-prior term grows without bound as $\frac{T}{2} \log \nu + O(1)$, and the conditional log-likelihood drifts slightly downward while the full ELBO continues to increase. The conditional log-likelihood then ceases to be a faithful surrogate, and convergence must be assessed on the full ELBO itself. This situation arises in the Gaussian-DGP nesting test of the simulation study (Experiment B), where it is analysed in detail. For this reason we adopt the *full* ELBO as the default monitored quantity throughout, for both the Student-t and the Gaussian estimator: it is the quantity the EM algorithm actually maximises and it remains a faithful convergence target in every regime, including the Gaussian limit $\nu \to \infty$. The conditional-log-likelihood proxy is retained only as an inexpensive cross-check, exact up to an additive constant whenever ν converges to a finite value, but it is never relied upon as the sole criterion. The two cases are distinct. For the Gaussian estimator the degrees of freedom are fixed at $\nu = \infty$, the weights are identically one, the weight-prior and entropy corrections vanish exactly, and the full ELBO coincides term by term with the Kalman conditional log-likelihood — so the same default criterion is trivially correct. For the Student-t estimator confronted with Gaussian data, by contrast, $\nu^{(j)}$ is estimated and drifts towards large finite values: there the weight-prior term grows and the conditional log-likelihood separates from the full ELBO, which is precisely why the full ELBO must be the monitored quantity. In both cases the default criterion is the full ELBO, with no runtime switch.

A caveat on monotonicity under variational EM. Strict monotonicity of the ELBO holds exactly under classical EM, where the E-step computes the true posterior. Our E-step uses a *mean-field* factorisation between the dynamic factors and the Student-t weights, so the guarantee holds only up to the Kullback–Leibler gap of that factorisation. In practice this gap is negligible: the monitored ELBO is monotone to within a relative residual of order 10^{-6}, well below the convergence tolerance $\mathrm{tol}_{\mathrm{outer}} = 10^{-5}$, and visible only at the plateau when the per-iteration change is itself near zero. The monotonicity diagnostic is therefore applied as a *relative* test, flagging a genuine decrease only when it exceeds $\mathrm{tol}_{\mathrm{outer}}\,|L^{(j)}|$; smaller fluctuations are the expected signature of the mean-field approximation, not a defect of the algorithm.

Interpretation of the ELBO trajectory. A well-behaved run typically exhibits the following pattern:

- *Early iterations* (rapid growth): the ELBO increases rapidly in the first 10–30 iterations, as the algorithm moves away from the PCA-based initial point.
- *Middle iterations* (slow growth): the ELBO continues to increase, but more slowly — typically a geometric decay of the per-iteration improvement.
- *Late iterations* (plateau): the ELBO approaches an asymptote, with improvements smaller than $\mathrm{tol}_{\mathrm{outer}}$ per step.

Empirically, typical convergence is achieved in 50–200 outer iterations for datasets of size $M \sim 20$–50, $T \sim 200$–400. Runs that require substantially more iterations often indicate poor initialisation or weak identification.

Stopping Criteria

Three stopping criteria are standard:

(i) Relative change in the ELBO. Terminate when

$$\frac{|\mathcal{L}^{(j+1)} - \mathcal{L}^{(j)}|}{|\mathcal{L}^{(j)}| + \epsilon} < \mathrm{tol}_{\mathrm{outer}},$$

where $\epsilon = 10^{-10}$ is a small constant to prevent division by zero for small ELBO values. Typical tolerance: $\mathrm{tol}_{\mathrm{outer}} = 10^{-5}$ for standard applications, 10^{-6} for high-precision requirements.

(ii) Relative change in parameters. Terminate when

$$\frac{\|\theta^{(j+1)} - \theta^{(j)}\|}{\|\theta^{(j)}\| + \epsilon} < \mathrm{tol}_{\mathrm{outer}}.$$

This is an alternative or complementary criterion to (i). Parameter stability is often a more demanding criterion than ELBO stability — it is possible for parameters to continue drifting on a flat ELBO surface, indicating weak identification of certain directions.

(iii) Maximum iteration limit. Terminate when $j = J_{\max}$, independently of any tolerance check. Typical value: $J_{\max} = 500$. This is a safety mechanism: if the algorithm has not converged in 500 iterations, it is almost certainly stuck in a local maximum or encountering numerical issues, and further iterations are unlikely to help.

Recommended practice. In our implementation we apply criteria (i) and (iii) jointly: stop at the first j for which either condition is satisfied. We also *record* the parameter-change trajectory (criterion ii) for diagnostic purposes, but do not use it as a stopping criterion in the main loop.

Local Maxima and Multiple Restarts

The EM algorithm converges to a stationary point, but this stationary point may not be the global maximum of the likelihood. In practice, for DFM models of the complexity we consider, the likelihood surface typically has:

- *One global maximum*, generically attainable from a wide range of initialisations;
- *A small number of poor local maxima*, arising from pathological configurations (for example, solutions where one block of factors is near-singular, or where ν collapses to its bracket boundary);
- *Flat regions* corresponding to the rotational indeterminacy, which are not true local maxima but equivalence classes of equivalent solutions.

Multiple random restarts. The standard robustness check is to run the algorithm from several initialisations and compare the resulting log-likelihoods. We recommend:

1. *PCA-based initialisation* (the default);
2. *PCA perturbations*: add small Gaussian noise to the PCA factors before running;
3. *Random block-structured starts*: draw $\mathbf{\Lambda}^M$ entries from $\mathcal{N}(0, 1)$ respecting the block restrictions.

If all runs converge to the same log-likelihood (up to numerical precision) and to the same fitted values (after rotational normalisation), the global maximum is likely to have been found. If runs converge to different log-likelihoods, select the run with the highest log-likelihood as the final estimate.

Diagnostic heuristic. A rough heuristic: if 3 out of 3 random restarts converge to log-likelihoods within 0.1% of each other, the global maximum is likely unique (or very robust). If there is substantial spread across restarts, the likelihood surface is multi-modal and more careful analysis is warranted — possibly including a finer initialisation strategy or a reduced factor count.

Early-iteration dynamics. Another diagnostic is the behaviour of $\hat{\nu}_u$ and $\hat{\nu}_\varepsilon$ in the first few iterations. Starting from $\nu = 10$, we typically observe one of three behaviours:

- *Heavy-tailed data*: $\hat{\nu}$ decreases toward 4–8, indicating significant departures from Gaussianity;
- *Near-Gaussian data*: $\hat{\nu}$ increases toward the upper bracket $\nu_{\max} = 1000$, indicating that the Student-t flexibility is not exploited;
- *Pathological collapse*: $\hat{\nu}$ drops toward $\nu_{\min} = 2.001$, indicating that the algorithm is fitting extreme outliers at the cost of overall fit — often a sign of data-quality issues or model misspecification.

The first two are expected and benign; the third should prompt inspection of the data and of the initialisation.

Practical Diagnostics

We close with a checklist of diagnostics that we apply to every run:

1. *ELBO monotonicity*: plot $\mathcal{L}^{(j)}$ against j. The curve should be monotonically non-decreasing. Any decrease indicates a bug.

2. *Convergence rate*: the log-change $\log |\mathcal{L}^{(j+1)} - \mathcal{L}^{(j)}|$ should decrease approximately linearly in j after the initial phase, indicating geometric convergence.

3. *Inner ECM convergence*: the number of inner iterations per outer step should stabilise to a small number (3–10) after the initial phase. Persistently high inner-iteration counts (say, > 20) suggest numerical issues in the weight-factor coupling.

4. *Parameter stability*: ν_u, ν_ε should stabilise after 50–100 iterations; erratic behaviour in the degrees of freedom typically reflects weak identification of the heavy-tailed component.

5. *Posterior weights*: the cross-sectional distribution of $\{\hat{w}_t^\varepsilon\}$ should have most mass near 1 with a few periods of small values (outlier down-weighting). A distribution uniformly below 1 suggests model misspecification.

6. *Fitted-value plots*: overlay $\hat{y}_{i,t} = \mathbf{\Lambda}_{i\cdot}^{(j+1)} \hat{f}_{t|T}$ against $y_{i,t}$ for a selection of series. Large persistent residuals suggest omitted variables or insufficient factor dimension.

These diagnostics, applied systematically, provide confidence that the algorithm has converged to an economically meaningful and numerically stable solution. They are the final check before using the estimates in the subsequent Growth-at-Risk analysis.

This concludes our derivation of the EM algorithm for the Student-t Dynamic Factor Model with block structure, ragged-edge missing data, and mixed-frequency observations. The next chapter develops the second stage of the framework: quantile regression on the estimated factors, which produces the conditional quantile forecasts underlying the Growth-at-Risk analysis.

Using the Estimated Model: Forecasts, Backdating, and Conditional Expectations

The previous sections have derived a complete EM algorithm for the Student-t Dynamic Factor Model with mixed-frequency observations, ragged-edge missingness, and block-diagonal economic identification. Once the algorithm has converged, the output consists of the estimated parameters $\hat{\theta} = (\hat{\mathbf{\Lambda}}, \hat{\mathbf{A}}, \hat{\mathbf{Q}}, \hat{\mathbf{R}}, \hat{\nu}_u, \hat{\nu}_\varepsilon)$ and the smoothed factor moments $\{\hat{\tilde{f}}_{t|T}, \tilde{P}_{t|T}\}$.

This section explains how these outputs are used in practice. Following the structure of Bańbura and Modugno (2014), three applications stand out: *forecasting* (filling in observations beyond the end of the sample), *backdating* (filling in observations before a series begins), and *conditional expectation of any missing entry* (filling in missing observations within the sample, including the latent monthly values implied by quarterly series). All three rely on the same underlying machinery — the Kalman filter and smoother applied to the augmented state-space representation — and differ only in which observations are conditioned on.

A natural question arises at this point. During initialisation, we already filled in missing entries: Mariano–Murasawa identity for quarterly series, and $\mathcal{N}(0,1)$ draws for the ragged edge. Why are we revisiting the question of conditional expectations now, after estimation?

The answer is that the two filling operations have entirely different purposes and entirely different statistical properties.

The initialisation fill is a *numerical scaffold*. It exists only to provide a complete data matrix on which PCA can be computed, so as to obtain a starting value $\theta^{(0)}$. The values produced by this fill are crude approximations: the MM identity uses a locally-constant extrapolation of quarterly observations; the Gaussian draw assigns a common-mean value to the ragged-edge missings without any reference to the cross-sectional or dynamic structure of the panel. These values are *discarded immediately* after PCA. From the very first E-step onward, the EM algorithm operates exclusively on the genuinely observed entries, encoded via the selection matrix $\mathbf{W}_t$; the initial fill plays no further role in the estimation.

The conditional expectations introduced in this section are conceptually different. They are produced *after* convergence of the EM algorithm, using the estimated parameters $\hat{\theta}$ and the smoothed factor estimates $\hat{f}_{t|T}$. They are the *best linear predictors* of the unobserved entries under the estimated model: they exploit the entire panel of observed series, the estimated cross-sectional correlations encoded in $\hat{\mathbf{\Lambda}}$, and the estimated dynamic relationships encoded in $\hat{\mathbf{A}}$. In statistical terms, they are optimal — not heuristic.

The distinction can be summarised as follows. The initialisation fill is a tool for *starting* the algorithm; the conditional expectations are an *output* of the algorithm. The first is discarded; the second is reported. The first ignores the model structure (because the model has not yet been estimated); the second uses the full model structure. The first is rough and serves only PCA; the second is principled and serves forecasting, backdating, and high-frequency interpolation as we shall discuss below.

The Master Formula

Once the EM has converged, the model implies a precise relationship between the latent factors and the observed data. For any series i and any time t, the conditional expectation given the full observed panel $\mathbf{Y}^{\text{obs}}$ is:

Two clarifications before stating the formula. First, the conditional expectation $\hat{\mathbb{E}}[\cdot \mid \mathbf{Y}^{\text{obs}}]$ is computed as a *best linear predictor* of $y_{i,t}$ given the observed panel — the same projection structure that underlies the Kalman filter and smoother throughout this thesis. Under the Student-t scale-mixture representation, conditional on the estimated weights $\hat{w}_t^u, \hat{w}_t^\varepsilon$ the model is Gaussian and the linear projection is exact; the conditional expectation operator $\hat{\mathbb{E}}[\cdot]$ should be read in this sense throughout the section.

Second, the row $\hat{\mathbf{\Lambda}}_{i\cdot}$ refers to the i-th row of the estimated loading matrix as it exits the EM algorithm — that is, the row that already incorporates all the extensions of the model: the Mariano–Murasawa quarterly aggregation for quarterly series, the appropriate block of the block-diagonal structure for the block to which series i belongs, and the constraints imposed by ragged-edge missingness through the selection matrices $\mathbf{W}_t$ during estimation. Equivalently, in the augmented representation, $\hat{\mathbf{\Lambda}}_{i\cdot}$ is the appropriate row of $\hat{\hat{\mathbf{\Lambda}}}$ that pairs the observation $y_{i,t}$ with the corresponding sub-block of $\hat{\hat{f}}_{t|T}$.

$$\hat{\mathbb{E}}[y_{i,t} \mid \mathbf{Y}^{\text{obs}}] \;=\; \hat{\mathbf{\Lambda}}_{i\cdot}\, \hat{f}_{t|T} \;+\; \hat{\mathbb{E}}[\varepsilon_{i,t} \mid \mathbf{Y}^{\text{obs}}], \tag{134}$$

where $\hat{\mathbf{\Lambda}}_{i\cdot}$ is the i-th row of the estimated loading matrix and $\hat{f}_{t|T}$ is the smoothed factor estimate (extracted from the relevant block of the augmented smoothed state $\hat{\hat{f}}_{t|T}$). The conditional expectation of the idiosyncratic component depends on whether $y_{i,t}$ has been observed:

- If $y_{i,t}$ was observed, $\hat{\mathbb{E}}[\varepsilon_{i,t} \mid \mathbf{Y}^{\text{obs}}] = y_{i,t} - \hat{\mathbf{\Lambda}}_{i\cdot} \hat{f}_{t|T}$ (the observed residual);
- If $y_{i,t}$ was missing, $\hat{\mathbb{E}}[\varepsilon_{i,t} \mid \mathbf{Y}^{\text{obs}}] = 0$, because the idiosyncratic component is uncorrelated with all observed series (by the diagonal structure of $\mathbf{R}$) and therefore the best linear predictor given the panel is its unconditional mean of zero.

Hence for missing entries the formula simplifies to:

$$\hat{\mathbb{E}}[y_{i,t} \mid \mathbf{Y}^{\text{obs}}] \;=\; \hat{\mathbf{\Lambda}}_{i\cdot}\, \hat{f}_{t|T}, \qquad y_{i,t} \text{ missing.} \tag{135}$$

This is the master formula from which all the applications below follow. Different patterns of "missingness" produce different applications.

Forecasts

The simplest application is forecasting beyond the end of the sample. We derive the forecast formula step by step, starting from the state equation of the underlying VAR(1):

$$f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t.$$

Setting $t = T+1, T+2, \ldots, T+h$ and iterating forward, we obtain the following recursive expression for f_{T+h}:

$$\begin{aligned} f_{T+1} &= \mathbf{A} f_T + u_{T+1}, \\ f_{T+2} &= \mathbf{A} f_{T+1} + u_{T+2} = \mathbf{A}^2 f_T + \mathbf{A} u_{T+1} + u_{T+2}, \\ f_{T+3} &= \mathbf{A} f_{T+2} + u_{T+3} = \mathbf{A}^3 f_T + \mathbf{A}^2 u_{T+1} + \mathbf{A} u_{T+2} + u_{T+3}, \\ &\vdots \end{aligned}$$

A simple induction yields the general formula for $h \geq 1$:

$$f_{T+h} = \mathbf{A}^h f_T + \sum_{k=0}^{h-1} \mathbf{A}^k u_{T+h-k}. \tag{136}$$

The first term carries forward the state at time T through h applications of the transition matrix. The second term accumulates the future shocks $u_{T+1}, u_{T+2}, \ldots, u_{T+h}$, each propagated by the appropriate power of $\mathbf{A}$ to reflect how long it has had to spread through the dynamics.

From the state equation to the forecast. We now take the conditional expectation of (136) given the observed panel up to time T:

$$\hat{f}_{T+h|T} \equiv \mathbb{E}[f_{T+h} \mid \mathbf{Y}^{\text{obs}}_{1:T}].$$

The future shocks are, by construction, unpredictable given current information: $\mathbb{E}[u_{T+h-k} \mid \mathbf{Y}^{\text{obs}}_{1:T}] = \mathbf{0}$ for every $k = 0, 1, \ldots, h-1$. Substituting into (136):

$$\hat{f}_{T+h|T} = \mathbf{A}^h \, \mathbb{E}[f_T \mid \mathbf{Y}^{\text{obs}}_{1:T}] + \sum_{k=0}^{h-1} \mathbf{A}^k \underbrace{\mathbb{E}[u_{T+h-k} \mid \mathbf{Y}^{\text{obs}}_{1:T}]}_{=\mathbf{0}}.$$

The shock term vanishes, leaving the deterministic propagation of the current smoothed state through h applications of $\mathbf{A}$:

$$\boxed{\hat{f}_{T+h|T} = \hat{\mathbf{A}}^h \hat{f}_{T|T}, \qquad h = 1, 2, \ldots} \tag{137}$$

where $\hat{f}_{T|T}$ is the smoothed state at the end of the sample and $\hat{\mathbf{A}}^h$ is the h-th matrix power of the estimated transition matrix.

In the augmented state-space. In our setting the relevant state is the augmented vector $\tilde{f}_t = (f'_t, f'_{t-1}, f'_{t-2}, f'_{t-3}, f'_{t-4})' \in \mathbb{R}^{5r}$, which evolves under the companion-form matrix $\tilde{\mathbf{A}}$. The same derivation applies block-by-block:

$$\hat{\tilde{f}}_{T+h|T} = \hat{\tilde{\mathbf{A}}}^h \hat{\tilde{f}}_{T|T}, \tag{138}$$

and the monthly factor forecast is the first r-block of the augmented forecast:

$$\hat{f}_{T+h|T} = \big[\hat{\tilde{f}}_{T+h|T}\big]_{1:r}.$$

From the state forecast to the observation forecast. The observation forecast at horizon h is obtained by applying the loading matrix to the state forecast. From the observation equation $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$ and the fact that the future idiosyncratic shocks are also unpredictable ($\mathbb{E}[\varepsilon_{T+h} \mid \mathbf{Y}_{1:T}^{\text{obs}}] = \mathbf{0}$), the forecast of the observation vector simplifies to:

$$\hat{y}_{T+h|T} \;=\; \hat{\mathbf{\Lambda}}\, \hat{f}_{T+h|T} \;=\; \hat{\mathbf{\Lambda}}\, \hat{\mathbf{A}}^h\, \hat{f}_{T|T}. \tag{139}$$

For quarterly series in the mixed-frequency formulation, the corresponding aggregation via the composite regressor $\phi^k_{T+h|T}$ — itself a weighted sum of the five lagged augmented factors at the forecast horizon — is used in place of $\hat{f}_{T+h|T}$.

Forecast uncertainty. The forecast variance is obtained by propagating the state covariance forward via the state equation:

$$\tilde{P}_{T+h|T} \;=\; \tilde{\mathbf{A}}\, \tilde{P}_{T+h-1|T}\, \tilde{\mathbf{A}}' \;+\; \tilde{\mathbf{Q}}, \qquad h = 1, 2, \ldots \tag{140}$$

starting from $\tilde{P}_{T|T}$ at the end of the sample, and the corresponding forecast variance for the observation is

$$\operatorname{Var}(y_{T+h} \mid \mathbf{Y}_{1:T}^{\text{obs}}) \;=\; \hat{\mathbf{\Lambda}}\, \tilde{P}_{T+h|T}\, \hat{\mathbf{\Lambda}}' \;+\; \hat{\mathbf{R}}. \tag{141}$$

The forecast variance decays geometrically with horizon h if and only if all eigenvalues of $\hat{\mathbf{A}}$ lie strictly inside the unit circle; otherwise it either remains bounded (unit-root case) or diverges (explosive case).

The role of the Student-t tails. The Student-t structure of the innovations u_t and ε_t implies that the forecast uncertainty is captured not only through the second moment $\operatorname{Var}(y_{T+h} \mid \mathbf{Y}_{1:T}^{\text{obs}})$ but also through the conditional density of y_{T+h}. Even at short horizons, the predictive density inherits heavier tails than the Gaussian case: extreme realisations are explicitly accommodated by the model. This is the methodological advantage of the proposed framework over its Gaussian counterpart for tail-risk analyses such as Growth-at-Risk, where the focus is on the lower quantiles of the predictive distribution rather than on its mean. Operationally, the forecast variance reported above captures the second moment of this predictive density, while the full Student-t structure governs the tails and is therefore directly relevant for quantile-based summaries of the forecast.

Nowcasting as a special case. In real-time applications, the final months of the sample exhibit *ragged-edge missingness*: some series have been released, others have not. The selection matrix $\mathbf{W}_t$ encodes this pattern. The Kalman filter handles missing entries automatically by reducing the dimension of the observation equation at each t, so the forecasts produced by the filter at the end of the sample naturally account for the incomplete cross-section. This is the basic machinery of *nowcasting*: at the current date T, the model produces $\hat{y}_{T|T}$ for series that have not yet been released (such as GDP), using the dynamic relationships estimated on the historical sample. Nowcasting is therefore not a separate technique but a specific application of the master formula (135) to the entries at the right edge of the panel.

Backdating

The complementary application is the estimation of historical values for series that begin only in the middle of the sample. Let series i have first observation at time $t_i > 1$. For all $t < t_i$, the value $y_{i,t}$ is treated as missing, and the conditional expectation follows from the master formula (135):

$$\hat{y}_{i,t|T} = \hat{\mathbf{\Lambda}}_{i\cdot} \hat{f}_{t|T}, \qquad t < t_i. \tag{142}$$

The expectation exploits the dynamic correlations between series i and all other series in the panel: even though $y_{i,t}$ itself is not observed at t, the other series at t together with the dynamics imply a posterior expectation for the latent factor $\hat{f}_{t|T}$, which in turn implies a fitted value for $y_{i,t}$ through the loading vector $\hat{\mathbf{\Lambda}}_{i\cdot}$.

Why backdating matters. A naive reaction would be: *if a series begins only at t_i, why should we care about its values before t_i — surely the analysis can simply start from t_i?* The answer is that backdating addresses two distinct concerns that arise routinely in applied macroeconomic work.

The first is *consistency of historical comparisons*. Many modern macroeconomic indicators — financial conditions indices, sentiment measures, attention-based variables drawn from text data, high-frequency credit metrics — have been introduced only in recent decades. Comparing recent realisations of such indices against historical episodes (the 1990 recession, the dot-com collapse, the global financial crisis) requires a measure of where the index *would have stood* during those earlier episodes. Backdating provides exactly this counterfactual, derived from the estimated factor model rather than from ad-hoc extrapolation.

The second is *forecast-stage homogeneity*. A two-stage forecasting framework such as the one developed in this thesis estimates the first-stage factor model on the full panel and then uses the smoothed factor history as a regressor in the second-stage quantile regression for GDP. If some series in the panel were unavailable in the early sample, the corresponding cross-sectional information for those years would be lost; backdating recovers it by reconstructing the implied historical values consistent with the model's estimated structure. This produces a longer effective sample for the second-stage regression, which translates into more informative estimates of conditional quantiles, particularly in the tails — the case most relevant for Growth-at-Risk.

A practical example: in our application, the financial-conditions indices used in the financial block are available only from the mid-1990s onward, while many real-activity indicators in the real block extend back to the late 1950s. Backdating allows the financial-block contribution to the historical factor estimate to be reconstructed for the earlier decades, on the basis of the estimated cross-block relationships. The result is a continuous financial-factor history that can be used to characterise the behaviour of conditional quantiles over the full sample.

What backdating is, and what it is not. The procedure should be read with care. Equation (142) provides the model-consistent expectation of $y_{i,t}$ for $t < t_i$, conditional on the observed panel and on the estimated parameters. This is the best linear predictor produced by the estimated DFM. It is *not* a reconstruction of the unobserved truth: the actual value of series i at $t < t_i$ may have differed from the backdated value, and the discrepancy is contained in the smoothed forecast variance.

A caveat on backdating quality. The reliability of backdated values depends crucially on the *time-invariance* of the loading vector $\mathbf{\Lambda}_{i\cdot}$. The model assumes that the relationship between series i and the common factors is the same throughout the sample, including the pre-t_i period that we are reconstructing. If this relationship has in fact changed structurally over time — for example, because of methodological revisions in how the series is collected, or because of deeper economic shifts that alter its factor exposure — the backdated values inherit the loading estimated on the post-t_i data and may therefore misrepresent the true historical realisations.

The backdating formula is a model-consistent expectation, not a guaranteed recovery of the unobserved truth; it is most reliable when the series is structurally stable.

Quarterly Interpolation

A third application — which has been treated extensively in the mixed-frequency section — is the construction of *monthly counterparts* of quarterly observations. A quarterly series is treated as a monthly series observed only at quarter-end months, with the intervening monthly values latent. The Kalman smoother applied to the augmented state automatically delivers the monthly counterparts:

$$\hat{y}^{Q}_{j,t|T} \;=\; \hat{\mathbf{\Lambda}}^{Q}_{j\cdot}\,\hat{\phi}^{k}_{t|T}, \qquad t \notin \{\text{quarter-end months}\}, \tag{143}$$

where $\hat{\phi}^{k}_{t|T}$ is the composite regressor at time t involving the smoothed factor of the relevant block at five consecutive lags. Compared with traditional univariate interpolation methods such as Chow and Lin (1971), this approach uses *cross-sectional information* from the entire panel of monthly indicators in addition to the temporal structure, producing interpolations that are coherent with the broader macroeconomic dynamics.

From the initial fill to the final interpolation. The same logic that distinguishes the initialisation fill from the final conditional expectations — discussed at the start of this section — applies in a particularly transparent way to quarterly interpolation. Recall that during initialisation we filled the non-quarter-end monthly entries of each quarterly series using the Mariano–Murasawa identity: a locally-constant assumption that distributed the quarterly observation across the three months of the quarter with the appropriate fixed weights. This fill served only to enable the PCA initialisation; it ignored the cross-sectional correlation structure of the panel and the dynamic relationships across months.

Equation (143) produces something entirely different. After convergence of the EM, the smoothed factor history $\{\hat{f}_{t|T}\}$ incorporates information from the entire monthly panel and from the estimated VAR(1) dynamics. The composite regressor $\hat{\phi}^{k}_{t|T}$ — itself a weighted sum of smoothed factor values at five consecutive lags — captures the dynamic propagation of factor movements through time, and the loading $\hat{\mathbf{\Lambda}}^{Q}_{j\cdot}$ ties this composite to the specific quarterly series j. The resulting monthly value $\hat{y}^{Q}_{j,t|T}$ is therefore an *estimate* of the latent monthly counterpart that is consistent with the full panel and the estimated model — not a mechanical disaggregation of the quarterly observation.

The two procedures use the same identity, but with profoundly different meanings: the initial fill applies the Mariano–Murasawa identity as a *heuristic* that ignores the cross-section; the final interpolation applies the same algebraic structure within a properly estimated state-space model that exploits the cross-section in full.

Why this matters for the empirical application. The latent monthly path of a quarterly target variable is the object of central interest in real-time nowcasting. Most macroeconomic targets — and in particular GDP — are observed only at quarterly frequency, but policy decisions, financial markets, and forecasters operate at monthly (or higher) frequency. The monthly counterpart $\hat{y}^{Q}_{j,t|T}$ provides a continuous-time read on the underlying activity of the target variable, and is the quantity against which monthly news releases of related indicators are interpreted. Without this interpolation, the model would deliver coarse quarterly updates; with it, the quarterly target is given a coherent monthly trajectory that follows the high-frequency information flow.

A Unified Reading

All three applications — forecasts, backdating, and interpolation — are instances of the same underlying operation: computing the conditional expectation of an unobserved entry given the observed panel under the estimated model. The unified machinery is the Kalman filter and smoother applied to the augmented state-space representation. The estimated parameters $\hat{\theta}$ encode the dynamic and cross-sectional structure of the data; the filter and smoother propagate this structure through time to deliver conditional expectations for any unobserved entry. The three labels ("forecast", "backdating", "interpolation") refer to different patterns of missingness — at the end of the sample, at the beginning of the history of a specific series, or in the middle of the sample at high-frequency intervals — but the algorithmic treatment is identical. Ragged-edge missingness, where some monthly series have not yet been released at the end of the sample, is naturally subsumed in the forecast case (it is a "partial forecast" where some entries of y_T are observed and others must be conditioned out).

This unity is one of the methodological appeals of the state-space approach: a single estimation framework yields a coherent prediction object for any combination of observed and missing data.

Connection to the second stage. The forecasts $\hat{f}_{T+h|T}$ and the historical factor estimates $\hat{f}_{t|T}$ produced in this section are the direct inputs to the second-stage quantile regression for Growth-at-Risk. The factor histories serve as regressors in the quantile regression of the target variable (GDP growth) on lagged factors; the factor forecasts serve as out-of-sample predictors for constructing real-time predictive quantiles. The connection between the two stages is therefore mediated by the smoothed factor estimates produced by the Kalman machinery of this section.

News and Forecast Revisions in Real Time

The previous section developed the conditional expectation machinery of the estimated model — forecasts, backdating, interpolation. We now consider a richer real-time setting in which information arrives *sequentially* and the forecast is updated as new data are released. The central object of this section is the notion of *news*: the difference between the value of a newly released observation and the value the model would have predicted for it before the release. We derive how news translate into forecast revisions, and we discuss the role of the heavy-tailed Student-t structure in this propagation.

The treatment follows Bańbura and Modugno (2014) and Cascaldi-Garcia (2023), adapted to the notation of our framework.

Data Vintages and the Real-Time Information Set

In a real-time application the data are not given as a fixed panel $\mathbf{Y} \in \mathbb{R}^{M \times T}$. Different observations are released at different calendar dates, and the model must be updated as each release arrives. We formalise this through the notion of *vintage*.

Vintage data. A *vintage* v is a particular calendar date — typically the date of a statistical release. We denote by $\mathbf{Y}^{(v)}$ the data matrix as it was known at vintage v, treating as missing all observations that had not yet been released by that date. Different vintages $v_1 < v_2 < \cdots$ correspond to progressively richer information sets:

$$\mathbf{Y}^{(v_1)} \subset \mathbf{Y}^{(v_2)} \subset \cdots$$

where the inclusion is in the sense that each entry observed at vintage v_1 is also observed at all later vintages (we abstract from data revisions throughout this section).

Equivalently, at each vintage v there is an associated selection matrix $\mathbf{W}_t^{(v)}$ for every t, which selects the entries of y_t that have been released by vintage v. Earlier vintages have "smaller" $\mathbf{W}_t^{(v)}$ (fewer observed entries) at the right edge of the sample; later vintages have "larger" $\mathbf{W}_t^{(v)}$ as more series are released. The mechanism is the natural generalisation of the ragged-edge framework developed previously to a dynamically expanding panel.

Two vintages and the new information. Consider two consecutive vintages v and $v+1$. The information set expands from $\mathbf{Y}^{(v)}$ to $\mathbf{Y}^{(v+1)}$ through the release of a set of *new observations*:

$$\Delta\mathbf{Y}^{(v+1)} \equiv \mathbf{Y}^{(v+1)} \setminus \mathbf{Y}^{(v)} = \{\, y_{i_j,t_j} \,:\, j = 1, \ldots, J_{v+1} \,\}, \tag{144}$$

where J_{v+1} is the number of new entries released between v and $v+1$. Each new entry is indexed by a series i_j and a time period t_j. In general the new releases at vintage $v+1$ pertain to different time periods (the publication delay differs across series): $t_j \neq t_\ell$ for $j \neq \ell$ is the generic case.

The symbol "$\setminus$" denotes *set difference*: $\mathbf{Y}^{(v+1)} \setminus \mathbf{Y}^{(v)}$ is the set of entries that belong to $\mathbf{Y}^{(v+1)}$ but not to $\mathbf{Y}^{(v)}$ — i.e. exactly the observations released between vintages v and $v+1$.

The Definition of a News

The new releases $\Delta\mathbf{Y}^{(v+1)}$ contain two kinds of information: a *predictable* component, namely the value that the model conditional on $\mathbf{Y}^{(v)}$ would have assigned to each y_{i_j,t_j}, and an *unpredictable* component, namely the difference between the actual release and the model expectation. Only the unpredictable component carries new information for the model. We formalise this distinction as follows.

News. For each new release y_{i_j,t_j} between vintages v and $v+1$, the corresponding *news* is

$$\mathcal{I}_{v+1,j} \;\equiv\; y_{i_j,t_j} \;-\; \mathbb{E}\big[y_{i_j,t_j} \,\big|\, \mathbf{Y}^{(v)}\big], \tag{145}$$

the difference between the observed value and the model expectation given the previous vintage. Collecting the news for the J_{v+1} releases into a single vector,

$$\mathcal{I}_{v+1} \;=\; \big(\mathcal{I}_{v+1,1},\, \mathcal{I}_{v+1,2},\, \ldots,\, \mathcal{I}_{v+1,J_{v+1}}\big)' \;\in\; \mathbb{R}^{J_{v+1}}. \tag{146}$$

The vector $\mathcal{I}_{v+1}$ collects all the unpredictable information that arrived between v and $v+1$.

News versus releases. A crucial distinction: *news is not the same as a release*. If the actual release y_{i_j,t_j} happens to coincide with the model expectation $\mathbb{E}[y_{i_j,t_j} \mid \mathbf{Y}^{(v)}]$, then $\mathcal{I}_{v+1,j} = 0$: the release contributes *no news*, and the forecast of the target variable will not be revised. It is the surprise component, not the release itself, that drives forecast revisions.

This is the methodological point of the news framework: forecast revisions are not driven by "data arriving", but by "data arriving differently from what the model predicted". Two statistically equivalent releases — one expected and one unexpected — produce very different forecast revisions even though both consist of one new data point.

Orthogonality of news. By construction, the news vector $\mathcal{I}_{v+1}$ is orthogonal to the information set $\mathbf{Y}^{(v)}$:

$$\mathbb{E}\big[\mathcal{I}_{v+1} \,\big|\, \mathbf{Y}^{(v)}\big] \;=\; \mathbf{0}.$$

This is immediate from the definition: subtracting the conditional expectation orthogonalises by construction. It is the analogue of the innovation property in the Kalman filter: news are the "innovations" of the real-time recursion.

Notation in the mixed-frequency setting. A clarification is in order before computing the cross-moments explicitly. In our framework, the state vector is the augmented $\tilde{f}_t = (f_t', f_{t-1}', f_{t-2}', f_{t-3}', f_{t-4}')' \in \mathbb{R}^{5r}$, and the loading matrix $\tilde{\mathbf{\Lambda}}$ has rows that reflect the frequency of each series:

- For a *monthly* series i, the row $\tilde{\mathbf{\Lambda}}_{i\cdot}$ has the form $[\mathbf{\Lambda}^M_{i\cdot}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}]$, so that $y^M_{i,t} = \mathbf{\Lambda}^M_{i\cdot} f_t + \varepsilon_{i,t}$ only loads on the current factor;

- For a *quarterly* series j, the row $\tilde{\mathbf{\Lambda}}_{j\cdot}$ has the Mariano–Murasawa form

$$\big[\tfrac{1}{3}\mathbf{\Lambda}^Q_{j\cdot},\; \tfrac{2}{3}\mathbf{\Lambda}^Q_{j\cdot},\; \mathbf{\Lambda}^Q_{j\cdot},\; \tfrac{2}{3}\mathbf{\Lambda}^Q_{j\cdot},\; \tfrac{1}{3}\mathbf{\Lambda}^Q_{j\cdot}\big],$$

  so that the quarterly observation aggregates the composite regressor

$$\phi^k_t \;=\; \tfrac{1}{3} f_t \;+\; \tfrac{2}{3} f_{t-1} \;+\; f_{t-2} \;+\; \tfrac{2}{3} f_{t-3} \;+\; \tfrac{1}{3} f_{t-4}.$$

In what follows, $\boldsymbol{\Lambda}_{i_j\cdot}$ and f_{t_j} should be read as the appropriate sub-blocks of the augmented $\tilde{\boldsymbol{\Lambda}}$ and $\tilde{f}_{t_j}$, depending on whether the released series i_j is monthly or quarterly. The cross-moments and news weights derived below automatically reflect this distinction, since they are computed from the augmented smoothed covariances $\tilde{P}_{t,s|v}$.

Forecast Revision as Function of the News

We now derive the central result: the revision of a forecast between two consecutive vintages can be written as a linear function of the news vector. Let y_{k,t_k} be a target variable of interest (in our application, the latent monthly GDP or quarterly GDP at some horizon). Denote by:

$$\begin{aligned}
\hat{y}^{(v)}_{k,t_k} &\equiv \mathbb{E}\big[y_{k,t_k} \,\big|\, \mathbf{Y}^{(v)}\big] \qquad \text{(forecast at vintage } v\text{)},\\
\hat{y}^{(v+1)}_{k,t_k} &\equiv \mathbb{E}\big[y_{k,t_k} \,\big|\, \mathbf{Y}^{(v+1)}\big] \qquad \text{(forecast at vintage } v+1\text{)}.
\end{aligned}$$

The forecast revision is the difference between the two:

$$\text{revision}_{v+1} \equiv \hat{y}^{(v+1)}_{k,t_k} - \hat{y}^{(v)}_{k,t_k}.$$

We seek an expression of this revision as a function of the news vector $\mathcal{I}_{v+1}$.

Decomposition via orthogonal projection. By the orthogonal decomposition of conditional expectations (also known as the law of total expectation in projective form),

$$\underbrace{\mathbb{E}\big[y_{k,t_k} \,\big|\, \mathbf{Y}^{(v+1)}\big]}_{\text{new forecast}} = \underbrace{\mathbb{E}\big[y_{k,t_k} \,\big|\, \mathbf{Y}^{(v)}\big]}_{\text{old forecast}} + \underbrace{\mathbb{E}\big[y_{k,t_k} \,\big|\, \mathcal{I}_{v+1}\big]}_{\text{revision}}. \tag{147}$$

The new forecast splits into the old forecast plus the projection of y_{k,t_k} on the news vector. The decomposition is exact because $\mathcal{I}_{v+1}$ is, by construction, orthogonal to $\mathbf{Y}^{(v)}$: had we projected on $\mathbf{Y}^{(v+1)}$ as a whole, the result would have collapsed onto $\hat{y}^{(v)}_{k,t_k}$ plus the projection on the new information beyond it, which is precisely $\mathcal{I}_{v+1}$.

The projection formula. Under joint Gaussianity (conditional on the weight processes w^u_t, w^ε_t in our Student-t scale-mixture formulation), the projection on $\mathcal{I}_{v+1}$ has a closed form. Standard Gaussian projection algebra gives

$$\mathbb{E}\big[y_{k,t_k} \,\big|\, \mathcal{I}_{v+1}\big] = \mathbb{E}\big[y_{k,t_k}\,\mathcal{I}'_{v+1}\big]\,\mathbb{E}\big[\mathcal{I}_{v+1}\,\mathcal{I}'_{v+1}\big]^{-1}\mathcal{I}_{v+1}, \tag{148}$$

since the news vector has zero mean by construction ($\mathbb{E}[\mathcal{I}_{v+1} \mid \mathbf{Y}^{(v)}] = \mathbf{0}$). The projection involves two quantities: the cross-moment $\mathbb{E}[y_{k,t_k}\,\mathcal{I}'_{v+1}]$ between the target and the news, and the second moment matrix $\mathbb{E}[\mathcal{I}_{v+1}\,\mathcal{I}'_{v+1}]$ of the news itself.

A remark on the Student-t case. The projection formula (148) delivers the *best linear predictor* of y_{k,t_k} given the news vector — that is, the linear combination of $\mathcal{I}_{v+1}$ that minimises the mean-squared prediction error. Under joint Gaussianity, the best linear predictor coincides exactly with the conditional expectation, and the formula recovers $\mathbb{E}[y_{k,t_k} \mid \mathcal{I}_{v+1}]$ in full.

Under the Student-t scale-mixture representation of our framework, the situation is more nuanced. The *marginal* conditional expectation $\mathbb{E}[y_{k,t_k} \mid \mathcal{I}_{v+1}]$ — integrating over the latent weights w^u_t, w^ε_t — is in general a non-linear function of the news vector, and the formula (148) delivers only its best linear approximation. However, *conditional* on the scale-mixture weights, the model is Gaussian and the projection recovers the true conditional expectation exactly.

The Kalman filter and smoother developed in earlier sections operate precisely under this conditional-Gaussian interpretation: at each iteration, the smoothed estimates use the current weight path $\hat{w}_t^u, \hat{w}_t^\varepsilon$, and the resulting moments are best linear predictors that become exact conditional expectations when conditioned on those weights.

The news framework inherits the same logic. Within each EM iteration — at fixed weights — the linear projection is exact, and the news decomposition recovers the true revision of the conditional expectation. After marginalising over the weights, the linear decomposition becomes a best linear approximation rather than an exact conditional expectation, but with negligible practical consequences: it is the same projection structure that underlies all the predictive output of the model, from the smoothed states to the forecasts. The Student-t tail structure enters through the time-varying covariances $\mathbf{Q}/\hat{w}_t^u$ and $\mathbf{R}/\hat{w}_t^\varepsilon$ used in the Kalman recursion, which downweight outlier periods — a point we develop further later in this section.

Expressing news as a function of factors and idiosyncratic shocks. To make the cross-moments computable within the DFM, we re-express each news $\mathcal{I}_{v+1,j}$ in terms of the structural components of the model. From the observation equation

$$y_{i_j,t_j} \;=\; \mathbf{\Lambda}_{i_j\cdot}\, f_{t_j} \;+\; \varepsilon_{i_j,t_j},$$

we have

$$\begin{aligned}
\mathcal{I}_{v+1,j} \;&=\; y_{i_j,t_j} \;-\; \mathbb{E}\big[y_{i_j,t_j} \,\big|\, \mathbf{Y}^{(v)}\big] \\
&=\; \mathbf{\Lambda}_{i_j\cdot}\, f_{t_j} \;+\; \varepsilon_{i_j,t_j} \;-\; \mathbf{\Lambda}_{i_j\cdot}\, \mathbb{E}\big[f_{t_j} \,\big|\, \mathbf{Y}^{(v)}\big] \\
&=\; \mathbf{\Lambda}_{i_j\cdot}\, \Big(f_{t_j} \;-\; \mathbb{E}\big[f_{t_j} \,\big|\, \mathbf{Y}^{(v)}\big]\Big) \;+\; \varepsilon_{i_j,t_j},
\end{aligned} \tag{149}$$

where we have used $\mathbb{E}[\varepsilon_{i_j,t_j} \mid \mathbf{Y}^{(v)}] = 0$ (since the idiosyncratic shock is mean-zero and uncorrelated with the factors). Equation (149) decomposes the news into two components: the *factor surprise* $f_{t_j} - \mathbb{E}[f_{t_j} \mid \mathbf{Y}^{(v)}]$, projected onto the loading vector of the released series, plus the *idiosyncratic shock* ε_{i_j,t_j} that pertains specifically to that observation.

This representation makes clear that a news reflects two kinds of surprise: (i) the latent factor has moved differently from what the panel up to vintage v implied; (ii) the released series has a purely idiosyncratic component that no factor model could have predicted. Both contribute to the news, and both will enter the revision formula.

Computing the cross-moments. Using (149), the j-th entry of the cross-moment vector $\mathbb{E}[y_{k,t_k}\, \mathcal{I}_{v+1}']$ is

$$\begin{aligned}
\mathbb{E}\big[y_{k,t_k}\, \mathcal{I}_{v+1,j}\big] \;&=\; \mathbb{E}\Big[\, \mathbf{\Lambda}_{k\cdot} f_{t_k} \Big(\mathbf{\Lambda}_{i_j\cdot}\, \big(f_{t_j} - \mathbb{E}[f_{t_j} \mid \mathbf{Y}^{(v)}]\big) + \varepsilon_{i_j,t_j}\Big) \,\Big] \\
&=\; \mathbf{\Lambda}_{k\cdot}\, \mathbb{E}\Big[\big(f_{t_k} - \mathbb{E}[f_{t_k} \mid \mathbf{Y}^{(v)}]\big)\, \big(f_{t_j} - \mathbb{E}[f_{t_j} \mid \mathbf{Y}^{(v)}]\big)'\,\Big]\, \mathbf{\Lambda}_{i_j\cdot}' \\
&=\; \mathbf{\Lambda}_{k\cdot}\, \mathrm{Cov}\big(f_{t_k}, f_{t_j} \,\big|\, \mathbf{Y}^{(v)}\big)\, \mathbf{\Lambda}_{i_j\cdot}',
\end{aligned} \tag{150}$$

where the ε_{i_j,t_j} term drops out because the idiosyncratic shock of series i_j is uncorrelated with the target y_{k,t_k} (and with the factors). The remaining expectation is the conditional covariance of the latent factors at the two relevant time points, taken under the vintage-v information set — a quantity directly produced by the Kalman smoother run on $\mathbf{Y}^{(v)}$.

The news second-moment matrix. We now compute the second moment of the news vector itself, which will appear inverted in the revision formula. Letting j and ℓ index two arbitrary news within the vector $\mathcal{I}_{v+1}$ (so that $j, \ell \in \{1, \ldots, J_{v+1}\}$), we expand the (j, ℓ) entry using (149) applied twice:

$$\mathbb{E}\big[\mathcal{I}_{v+1,j}\,\mathcal{I}_{v+1,\ell}\big] \;=\; \mathbb{E}\Big[\,\big(\boldsymbol{\Lambda}_{i_j\cdot}\widetilde{f}_{t_j} + \varepsilon_{i_j,t_j}\big)\,\big(\boldsymbol{\Lambda}_{i_\ell\cdot}\widetilde{f}_{t_\ell} + \varepsilon_{i_\ell,t_\ell}\big)'\,\Big],$$

where we have written $\widetilde{f}_t \equiv f_t - \mathbb{E}[f_t \mid \mathbf{Y}^{(v)}]$ for compactness (the "factor surprise" at time t).

Expanding the product yields four terms, three of which vanish: the factor surprise is uncorrelated with the idiosyncratic shocks, so the two cross terms $\mathbb{E}[\boldsymbol{\Lambda}_{i_j\cdot}\widetilde{f}_{t_j}\varepsilon_{i_\ell,t_\ell}]$ and $\mathbb{E}[\varepsilon_{i_j,t_j}\widetilde{f}'_{t_\ell}\boldsymbol{\Lambda}'_{i_\ell\cdot}]$ are zero. The only two surviving terms are

$$\mathbb{E}\big[\mathcal{I}_{v+1,j}\,\mathcal{I}_{v+1,\ell}\big] \;=\; \underbrace{\boldsymbol{\Lambda}_{i_j\cdot}\,\mathrm{Cov}\big(f_{t_j}, f_{t_\ell}\,\big|\,\mathbf{Y}^{(v)}\big)\,\boldsymbol{\Lambda}'_{i_\ell\cdot}}_{\text{factor contribution}} \;+\; \underbrace{\mathbb{E}\big[\varepsilon_{i_j,t_j}\,\varepsilon_{i_\ell,t_\ell}\big]}_{\text{idiosyncratic contribution}}\,. \tag{151}$$

The first term is the factor contribution, expressed via the conditional covariance of the factor at the two relevant time points. The second term is the idiosyncratic contribution, which needs careful interpretation.

The idiosyncratic term and the indicator function. The idiosyncratic contribution $\mathbb{E}[\varepsilon_{i_j,t_j}\,\varepsilon_{i_\ell,t_\ell}]$ is determined by the structure of $\mathbf{R}$, which in our framework is diagonal: idiosyncratic shocks are uncorrelated both across series and across time. This means that the contribution is non-zero *only* when the two news $\mathcal{I}_{v+1,j}$ and $\mathcal{I}_{v+1,\ell}$ pertain to the exact same observation — that is, when both the series index and the time index coincide ($i_j = i_\ell$ and $t_j = t_\ell$). In that case, the contribution reduces to the idiosyncratic variance of that series, R_{i_j,i_j}, where R_{i_j,i_j} denotes the (i_j, i_j) entry of $\mathbf{R}$ (i.e. the diagonal entry corresponding to series i_j).

We can write this compactly using an *indicator function*:

$$\mathbb{E}\big[\varepsilon_{i_j,t_j}\,\varepsilon_{i_\ell,t_\ell}\big] \;=\; \mathbf{1}_{\{i_j=i_\ell,\,t_j=t_\ell\}} \cdot R_{i_j,i_j},$$

where $\mathbf{1}_{\{\cdot\}}$ equals 1 when its argument is true and 0 otherwise. The indicator equals 1 precisely when the two news indices j, ℓ refer to exactly the same observation (typically only when $j = \ell$, i.e. on the diagonal of the second-moment matrix), and 0 in all other cases.

Substituting into (151), the full (j, ℓ) entry of the second-moment matrix becomes:

$$\begin{aligned}\mathbb{E}\big[\mathcal{I}_{v+1,j}\,\mathcal{I}_{v+1,\ell}\big] \;=\;& \boldsymbol{\Lambda}_{i_j\cdot}\,\mathrm{Cov}\big(f_{t_j}, f_{t_\ell}\,\big|\,\mathbf{Y}^{(v)}\big)\,\boldsymbol{\Lambda}'_{i_\ell\cdot}\\ &+\; \mathbf{1}_{\{i_j=i_\ell,\,t_j=t_\ell\}}\,R_{i_j,i_j}.\end{aligned} \tag{152}$$

The factor contribution applies to every pair (j, ℓ); the idiosyncratic contribution applies only on the diagonal entries corresponding to the same observation.

The revision in closed form. Combining (148), (150), and (152):

$$\boxed{\;\mathrm{revision}_{v+1} \;=\; \mathbf{B}^{(v+1)}_{k,t_k}\,\mathcal{I}_{v+1}, \qquad \mathbf{B}^{(v+1)}_{k,t_k} \;=\; \mathbb{E}\big[y_{k,t_k}\,\mathcal{I}'_{v+1}\big]\,\mathbb{E}\big[\mathcal{I}_{v+1}\,\mathcal{I}'_{v+1}\big]^{-1}.\;} \tag{153}$$

The row vector $\mathbf{B}^{(v+1)}_{k,t_k} \in \mathbb{R}^{1\times J_{v+1}}$ contains the *news weights*: the j-th entry $b^{(v+1)}_{k,t_k,j}$ is the contribution of the j-th news $\mathcal{I}_{v+1,j}$ to the revision of the target. The structural breakdown of the cross-moments in (150) and (152) shows that the news weights are entirely determined by three ingredients of the DFM: the loadings $\boldsymbol{\Lambda}$, the smoothed conditional covariances of the factors under vintage v, and the idiosyncratic scale matrix $\mathbf{R}$.

Connection to the Kalman filter update. Formula (153) is, in essence, a generalisation of the standard Kalman filter update equation (e.g. Harvey, 1989, Eq. 3.2.3a) to the case in which the new observations arrive *non-synchronously* across series and time periods. In the synchronous case — a single full cross-section y_{t+1} released at once — the formula reduces exactly to the familiar Kalman gain multiplied by the innovation. The vintage-based formulation extends this to a setting where the new information is a sparse subset of the panel, identified by the index pairs (i_j, t_j).

Computing the News Weights in Practice

The formulas (150) and (152) derived in the previous subsection express the cross-moments needed to compute the news weights $\mathbf{B}_{k,t_k}^{(v+1)}$ as combinations of three structural ingredients: the estimated loading matrix $\hat{\mathbf{\Lambda}}$, the smoothed conditional covariances of the factors under vintage v, $\mathrm{Cov}(f_t, f_s \mid \mathbf{Y}^{(v)})$, and the estimated idiosyncratic scale matrix $\hat{\mathbf{R}}$. We now describe how these ingredients are obtained operationally and combined into the weight matrix.

The role of the Kalman smoother. The conditional covariances $\mathrm{Cov}(f_t, f_s \mid \mathbf{Y}^{(v)})$ that appear in both cross-moment expressions are exactly the smoothed cross-covariances produced by the Kalman smoother applied to the vintage panel $\mathbf{Y}^{(v)}$. The augmented state-space representation developed in earlier sections already produces these covariances as a by-product of the smoother pass — no additional computation is required beyond what is already performed during estimation.

Specifically, two distinct cases arise depending on the time gap $|t-s|$ between the two relevant time points:

- *Short lags* ($|t-s| \leq 4$). Because the augmented state vector $\tilde{f}_t = (f_t', f_{t-1}', f_{t-2}', f_{t-3}', f_{t-4}')'$ contains the current factor together with its first four lags, the smoothed within-time covariance matrix $\tilde{P}_{t|v}$ — produced directly by the smoother at every t — already contains $\mathrm{Cov}(f_t, f_{t-k} \mid \mathbf{Y}^{(v)})$ for $k = 0, 1, 2, 3, 4$ as off-diagonal sub-blocks. No additional smoothing operation is needed: the relevant $r \times r$ sub-block of $\tilde{P}_{t|v}$ delivers the answer.

- *Long lags* ($|t-s| > 4$). For time gaps beyond four months, the within-time augmented covariance $\tilde{P}_{t|v}$ is insufficient: it does not contain the factor lagged more than four periods. The required quantity is the augmented *cross-time* covariance $\tilde{P}_{t,s|v}$, which the Kalman smoother delivers via a standard recursion (De Jong–Shephard or the equivalent backward pass). The appropriate sub-block of $\tilde{P}_{t,s|v}$ then yields $\mathrm{Cov}(f_t, f_s \mid \mathbf{Y}^{(v)})$ for any (t, s).

In practice, when news arrive within a few months of each other — which is the typical scenario in real-time data flow — the short-lag case applies and the smoother output is directly usable. The long-lag case is rarer and requires the additional cross-time recursion.

Operational algorithm. The full procedure for computing the news weights at vintage $v+1$ is therefore:

1. Run the Kalman filter and smoother on the vintage panel $\mathbf{Y}^{(v)}$, using the parameter estimate $\hat{\theta}$ obtained from the most recent EM run. This produces the smoothed augmented states and covariances $\{\hat{\tilde{f}}_{t|v}, \tilde{P}_{t,s|v}\}$.

2. Identify the new releases between vintages v and $v+1$: the index pairs (i_j, t_j) for $j = 1, \ldots, J_{v+1}$.

3. Extract the monthly-factor cross-covariances $\mathrm{Cov}(f_t, f_s \mid \mathbf{Y}^{(v)})$ from the appropriate sub-blocks of $\tilde{P}_{t,s|v}$.

4. Assemble the cross-moment vector $\mathbb{E}[y_{k,t_k}\, \mathcal{I}'_{v+1}]$ using (150) for each $j = 1, \ldots, J_{v+1}$.

5. Assemble the second-moment matrix $\mathbb{E}[\mathcal{I}_{v+1}\, \mathcal{I}'_{v+1}]$ using (152) for each pair (j, ℓ).

6. Solve the linear system

$$\mathbf{B}^{(v+1)}_{k,t_k} \;=\; \mathbb{E}[y_{k,t_k}\, \mathcal{I}'_{v+1}]\, \mathbb{E}[\mathcal{I}_{v+1}\, \mathcal{I}'_{v+1}]^{-1},$$

obtaining the row of news weights.

The same Kalman machinery developed for likelihood estimation thus yields, with minimal additional work, the decomposition of forecast revisions into news contributions. In implementation, the inversion of $\mathbb{E}[\mathcal{I}_{v+1}\, \mathcal{I}'_{v+1}]$ is performed via Cholesky decomposition rather than explicit matrix inversion to preserve numerical stability when the news vector is high-dimensional.

Interpretation of the weights. The weight $b^{(v+1)}_{k,t_k,j}$ has a precise interpretation: *the marginal effect on the target forecast of a unit news at release j*. Large weights identify releases that are informative for the target — series that move the forecast significantly when they surprise. Small weights identify releases that are largely redundant — series whose information content is already captured by other observed series through the cross-sectional correlation structure.

This interpretation has both ex-post and ex-ante operational value: *ex post*, the weights allow attribution of a forecast revision to the specific releases that drove it; *ex ante*, they allow the analyst to identify which upcoming releases are likely to move the forecast the most, and therefore deserve the closest attention. In the context of nowcasting, this is the kind of attribution statement that traditional point-forecast frameworks cannot deliver.

Decomposition of Forecast Revisions Across Multiple News

Equation (153) extends naturally to the case in which several releases occur simultaneously between vintages — for instance, a single Friday morning may bring the release of payrolls, manufacturing PMI, and consumer confidence. The total revision then decomposes additively into contributions from each individual news:

$$\underbrace{\hat{y}^{(v+1)}_{k,t_k} - \hat{y}^{(v)}_{k,t_k}}_{\text{total revision}} = \sum_{j=1}^{J_{v+1}} \underbrace{b^{(v+1)}_{k,t_k,j} \cdot \mathcal{I}_{v+1,j}}_{\text{contribution of release } j} . \tag{154}$$

This is the central result of the news framework: *the total revision is a weighted sum of the individual news, with weights given by the row vector* $\mathbf{B}^{(v+1)}_{k,t_k}$. Each contribution is the product of two quantities: the size of the news (*how surprising* the release was) and the weight (*how relevant* that particular release is for the target). A small but very surprising news may contribute more to the revision than a large news with weak structural relevance, and vice versa.

Equation (154) is simply the component-wise expansion of the matrix product $\mathbf{B}^{(v+1)}_{k,t_k} \mathcal{I}_{v+1}$ in (153); what makes it operationally useful is that each term in the sum can now be interpreted as the contribution of a specific release to the total revision.

An operational reading. Each term $b^{(v+1)}_{k,t_k,j} \cdot \mathcal{I}_{v+1,j}$ quantifies the contribution of the j-th release to the revision. We can therefore say, after a given vintage update:

> *"The forecast of GDP at quarter t_k has been revised upward by $\Delta = +0.12$ percentage points between vintage v and $v+1$. Of this revision, $+0.08$ comes from a positive surprise in industrial production, $+0.05$ from a positive surprise in consumer confidence, and -0.01 from a slightly negative surprise in retail sales."*

This is the kind of attribution statement that the news framework makes possible, and that is missing from any single-pass forecasting exercise.

Aggregation by economic block. The decomposition (154) can be summed across subsets of releases to identify contributions of *groups* of indicators. In our block-structured DFM, the natural aggregation is by economic block:

$$\hat{y}^{(v+1)}_{k,t_k} - \hat{y}^{(v)}_{k,t_k} = \Delta^R_{v+1} + \Delta^F_{v+1} + \Delta^X_{v+1},$$

where

$$\Delta^b_{v+1} = \sum_{j : i_j \in \text{block } b} b^{(v+1)}_{k,t_k,j} \cdot \mathcal{I}_{v+1,j}, \qquad b \in \{R, F, X\}.$$

This aggregation makes the decomposition directly interpretable in terms of which area of the economy is driving the forecast revision — real activity, financial conditions, or the residual "other" block. In a Growth-at-Risk context, the financial-block contribution Δ^F_{v+1} is of particular interest, as it quantifies how the latest financial-conditions data flow has updated the central GDP forecast. The extension of this decomposition to the conditional quantiles of GDP — rather than the central forecast — is the natural next step.

The Role of the Heavy-Tailed Structure

The derivation above proceeded under the standard Gaussian projection machinery, which is exact in our framework *conditional* on the weight processes w^u_t, w^ε_t. The Student-t structure of our model introduces additional features that do not appear in the Gaussian DFM of Bańbura and Modugno (2014), and that are worth making explicit. Three distinct points emerge: the conditional Gaussianity of the projection, the outlier-aware structure of the news weights, and a genuinely non-linear feedback between news magnitude and weight estimation.

The conditional Gaussianity. Under the scale-mixture representation of the Student-t, the model is conditionally Gaussian given the latent weights w^u_t and w^ε_t: at each t, the covariance structure of the innovations is $\mathbf{Q}/w^u_t$ and $\mathbf{R}/w^\varepsilon_t$. The Kalman smoother applied within the EM iteration uses time-varying versions of these covariances: $\tilde{\mathbf{Q}}_t = \tilde{\mathbf{Q}}/\hat{w}^u_t$ and $\tilde{\mathbf{R}}_t = \mathbf{R}/\hat{w}^\varepsilon_t$. The covariance moments produced by the smoother — and therefore the news weights computed from them via (150)–(152) — are *conditional* on this estimated weight path. The orthogonal projection on the news vector remains the best linear predictor, and the news decomposition retains its closed-form structure within each EM iteration.

Outlier-aware news weights. A direct consequence is that the news weights in the Student-t DFM become *outlier-aware* in a way that those of the Gaussian DFM are not. The covariance contributions involving any time period t are inflated by $1/\hat{w}^\varepsilon_t$ (for idiosyncratic contributions) and $1/\hat{w}^u_t$ (for factor contributions). Concretely:

- Releases pertaining to time periods that the model has identified as "well-behaved" ($\hat{w}_t \approx 1$) receive standard news weights, essentially identical to those produced by the Gaussian DFM;
- Releases pertaining to periods that the model has identified as outlier-prone ($\hat{w}_t < 1$) receive *attenuated* news weights, reflecting the model's reduced trust in observations from those periods.

The attenuation factor is essentially $\hat{w}_t^\varepsilon$ for idiosyncratic contributions: a period with $\hat{w}_t^\varepsilon = 0.3$ contributes to the news weights with one-third the strength of a Gaussian-equivalent period. This is the mechanism through which the Student-t extension protects the forecast revision from being unduly driven by outlier-contaminated periods.

Quantitative magnitude of the effect. The size of the outlier-aware adjustment depends on the estimated degrees of freedom $\hat{\nu}_\varepsilon$ and on the empirical distribution of the Mahalanobis residuals. In well-behaved macroeconomic samples with $\hat{\nu}_\varepsilon$ in the range 8–15, most periods will have $\hat{w}_t^\varepsilon$ close to one, and the news weights will be effectively indistinguishable from their Gaussian counterparts. The deviation becomes substantive only in periods that the model identifies as genuinely outlier-prone — typically crisis episodes such as 2008–2009 or 2020. Empirically, the practical relevance of the outlier-aware adjustment is concentrated precisely in the periods where real-time nowcasting is most challenging and where robustness to outliers is most valuable.

The Gaussian limit. In the limit $\nu_u, \nu_\varepsilon \to \infty$ (the Gaussian case), all $\hat{w}_t^u, \hat{w}_t^\varepsilon \to 1$, and the news weights reduce exactly to the Bańbura–Modugno (2014) formulation. The Student-t extension therefore preserves the linearity of the news decomposition within each EM iteration, while introducing an outlier-driven modulation of the weights across periods. The two frameworks are *nested*: the Student-t news decomposition is a strict generalisation of the Gaussian case, recovering it exactly when the data are well-described by a Gaussian model.

A genuinely non-linear feedback: news magnitude affects weight estimation. The most distinctive feature of the Student-t news framework is a *second-order non-linearity* that is entirely absent from the Gaussian case. To make this concrete, recall that in a real-time implementation the EM is re-estimated at every vintage: when a new release arrives at vintage $v+1$, the algorithm is re-run on the expanded panel $\mathbf{Y}^{(v+1)}$ and produces an updated parameter estimate $\hat{\theta}^{(v+1)}$. The Student-t-specific non-linearity emerges from how the new observation interacts with the re-estimation of the weights, in two distinct steps.

First, the news $\mathcal{I}_{v+1,j}$ is computed using the parameter estimates $\hat{\theta}^{(v)}$ obtained at vintage v (before the new release is incorporated). The forecast revision is then given by $b_{k,t_k,j}^{(v+1)} \cdot \mathcal{I}_{v+1,j}$ as derived above.

Second, when the EM is re-run at vintage $v+1$ incorporating the new observation y_{i_j,t_j}, the Mahalanobis residual at t_j will reflect this new information. If y_{i_j,t_j} turned out to be far from the model prediction — that is, if the news was large in magnitude — then the new Mahalanobis residual at t_j is large, and the re-estimated weight $\hat{w}_{t_j}^{\varepsilon,(v+1)}$ is *revised downward* relative to its vintage-v value $\hat{w}_{t_j}^{\varepsilon,(v)}$. The model interprets the large news as partial evidence that the period t_j is outlier-contaminated, and attenuates its contribution to subsequent estimation.

The consequence is twofold:

- *Within-vintage*: the news weight $b_{k,t_k,j}^{(v+1)}$ used to compute the revision at vintage $v+1$ does *not* fully reflect the magnitude of the surprise — it was computed under the previous weight estimate $\hat{w}_{t_j}^{\varepsilon,(v)}$, which has not yet been updated;

- *Across-vintage*: at all subsequent vintages $v+2, v+3, \ldots$, the weight $\hat{w}_{t_j}^{\varepsilon}$ that enters the covariance computations will be lower than at vintage v, and all future news weights involving t_j will be correspondingly attenuated.

A fully time-consistent treatment would iterate the news computation and the weight re-estimation until convergence within each vintage. We do not develop this extension here — the standard one-pass implementation is sufficient for practical purposes, and the within-vintage non-linearity is empirically small except for genuinely extreme news. The point is to flag this property as a distinctive feature of the heavy-tailed news framework that has no analogue in the Gaussian case, and that opens an avenue for future methodological refinement.

Implications for real-time nowcasting. The outlier-aware structure of the news weights has a clear practical interpretation in real-time nowcasting. During a crisis episode, the standard Gaussian news framework would assign exaggerated weights to any release pertaining to the crisis periods, because it treats the high-variance shocks of those periods as representative of the underlying covariance structure. The Student-t news framework, in contrast, recognises the crisis periods as outlier-prone, attenuates their weight, and produces forecast revisions that are more robust to the contaminating effects of extreme realisations. This is the operational counterpart of the identification of ν_u, ν_ε at the estimation stage: the heavy-tailed structure that protects the parameter estimates from outlier contamination also protects the real-time forecast updates.

Looking Ahead

The news framework developed in this section operates at the level of *point forecasts*: it decomposes the revision of $\mathbb{E}[y_{k,t_k} \mid \mathbf{Y}^{(v)}]$ into contributions from individual news. Two natural extensions are deferred to subsequent work.

Extension to the quantile framework. The most economically interesting extension is to the second-stage quantile regression: how does a new release update not the expected GDP, but the conditional quantiles $Q_\tau(y_{T+h} \mid \mathbf{Y}^{(v)})$? In a heavy-tailed setting such as Growth-at-Risk, news weights are likely to be *asymmetric across quantiles*: a positive surprise in credit spreads should affect the lower tail of the GDP distribution disproportionately, with a larger weight at $\tau = 0.05$ than at $\tau = 0.95$. This asymmetry is a property that emerges naturally from the combination of a heavy-tailed DFM (Stage 1) and a quantile regression (Stage 2), and represents the natural integration of the news framework into the Growth-at-Risk literature. We develop this extension in the chapter dedicated to the second stage.

Vintage data in the Monte Carlo design. The validity of the news framework in the proposed Student-t DFM can be assessed through the Monte Carlo design discussed later in this work: by simulating realistic vintage data and tracking forecast revisions across releases, one can quantify how well the algorithm reproduces the theoretical news weights, and how the heavy-tailed structure affects the precision of the decomposition. This validation is the natural counterpart for the news framework of the parameter recovery experiments described in the calibration Monte Carlo, and we develop it in conjunction with the broader second-stage simulation.

Extension: Persistent Idiosyncratic Errors via AR(1) Dynamics

The model developed so far attributes *all* temporal persistence in the panel to the common factors, through the VAR(1) dynamics on f_t, and treats the idiosyncratic errors ε_t as serially uncorrelated draws from a heavy-tailed Student-t. This section describes — as a methodological extension, not as part of the estimated baseline — how the model *would* be enriched to allow each idiosyncratic error to carry its own persistence, modelled as a first-order autoregression with heavy-tailed innovations. The development is entirely at the level of the model: the data-generating process, the augmented state-space representation, and the resulting modifications to the E- and M-step equations. We close with the identification questions the extension raises and an explicit statement of its status as future work.

Motivation: Transitory Outliers versus Persistent Anomalies

It is essential to separate two distinct phenomena that are easily conflated, because the model already handles one of them and the present extension is concerned exclusively with the other.

Transitory outliers (already handled). The Student-t structure on ε_t already provides robustness to *isolated* idiosyncratic outliers. Through the scale-mixture representation, an anomalous observation in a single period receives a small posterior weight w_t^{ε}, which down-weights its contribution to the parameter updates. This mechanism, however, is inherently *memoryless*: each weight w_t^{ε} is drawn independently across time, so an extreme realisation at period t tells the model nothing about period $t+1$. The heavy tails absorb spikes that are large but *transitory*.

Persistent anomalies (the novelty). A different empirical regularity is the *persistence* of series-specific deviations: an idiosyncratic shock to one variable that *lasts several periods* — a sectoral disruption, a measurement or data-collection irregularity that propagates across consecutive vintages, a slow-moving series-specific trend not shared by the common factors. Such behaviour produces autocorrelation in $\varepsilon_{i,t}$ that the i.i.d. assumption cannot represent and that the heavy tails, being memoryless, do not capture either. Persistence and tail-heaviness are therefore *orthogonal* features: the former concerns *how long* a series-specific deviation lasts, the latter *how extreme* the fresh shock in any single period can be.

This distinction was anticipated in the discussion of the idiosyncratic errors, where serial uncorrelation was adopted deliberately and the AR(1) idiosyncratic specification — in the manner of Bańbura and Modugno (2014) and Cascaldi-Garcia (2023) — was flagged as a natural direction for enrichment. The present section makes that direction explicit.

What the Baseline Captures, and What It Does Not

In the baseline DGP, the only channel for serial dependence in the observed data $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$ is the factor block. Because ε_t is i.i.d. over time, every autocorrelation in any series is forced to load onto the common factors via $\mathbf{A}$. This is a deliberate modelling choice with a clear identification rationale: it keeps the factors as the sole carriers of dynamics and so makes the common component the unique source of co-movement *and* persistence.

The cost of this choice is that genuinely series-specific persistence has nowhere to go. If a single variable exhibits autocorrelated idiosyncratic behaviour that is *not* shared by the rest of the panel, the baseline must either ignore it (treating successive deviations as independent draws) or distort the common factors in the attempt to track it. The extension below adds a dedicated, series-specific channel for this persistence, leaving the common factors free to represent only genuinely shared dynamics.

The Extended Data-Generating Process

The factor block is left untouched. As before,

$$f_t \;=\; \mathbf{A}\, f_{t-1} \;+\; u_t, \qquad u_t \mid w_t^u \sim \mathcal{N}\big(0,\, \mathbf{Q}/w_t^u\big), \qquad w_t^u \overset{iid}{\sim} \mathrm{Gamma}\big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\big). \tag{155}$$

The idiosyncratic block is the locus of the extension. Each series $i = 1, \ldots, M$ is endowed with its own autoregressive dynamics,

$$\varepsilon_{i,t} \;=\; \rho_i\, \varepsilon_{i,t-1} \;+\; u_{i,t}^{\varepsilon}, \qquad u_{i,t}^{\varepsilon} \mid w_{i,t}^{\varepsilon} \sim \mathcal{N}\big(0,\, \sigma_i^2/w_{i,t}^{\varepsilon}\big), \qquad w_{i,t}^{\varepsilon} \overset{iid}{\sim} \mathrm{Gamma}\big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\big), \tag{156}$$

where $\rho_i \in (-1, 1)$ is the series-specific persistence coefficient and $u_{i,t}^{\varepsilon}$ is the *fresh idiosyncratic innovation*. The observation equation is unchanged,

$$y_t \;=\; \mathbf{\Lambda}\, f_t \;+\; \varepsilon_t. \tag{157}$$

Two objects that must not be confused: the scale R and the weight w^{ε}. The extension keeps intact the role of the diagonal scale matrix and adds a clearly separate role for the weights. It is worth stating the two explicitly, because they are often conflated.

- The diagonal matrix $\mathbf{R} = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_M^2)$ is the *baseline scale* of the idiosyncratic innovations. It carries one variance parameter σ_i^2 per series, it is *constant over time*, and it remains diagonal and structurally unchanged from the baseline model. It encodes the typical, ordinary-times magnitude of the series-specific shock $u_{i,t}^{\varepsilon}$.

- The weight $w_{i,t}^{\varepsilon}$ is a *random, time-varying multiplier* that generates the heavy tails: conditional on the weight, the innovation has variance $\sigma_i^2/w_{i,t}^{\varepsilon}$. A small weight in a given period inflates the effective variance for that period only — this is precisely the transitory-outlier mechanism. The weight does not alter the baseline scale σ_i^2; it modulates it stochastically over time.

In words: $\mathbf{R}$ fixes *how large the shock is on average*; $w_{i,t}^{\varepsilon}$ governs *how much larger than average it may be in any one period*; and the new coefficient ρ_i governs *how long a deviation, once it occurs, persists*. The three are distinct, and the extension keeps them so.

Per-series weights versus a shared weight. There are two coherent ways to attach the heavy-tail weights to the idiosyncratic innovations, and the distinction concerns the *weight* w^{ε} alone, never the scale $\mathbf{R}$.

- **Per-series weights (the natural formulation).** Each series receives its own weight $w_{i,t}^{\varepsilon}$, as written in (156). The model then becomes coherently series-specific along *every* idiosyncratic dimension: series i has its own scale σ_i^2, its own persistence ρ_i, and its own time-varying down-weighting $w_{i,t}^{\varepsilon}$, giving the effective innovation variance $\sigma_i^2/w_{i,t}^{\varepsilon}$. An anomalous period for one series no longer mechanically down-weights the others.

- **A shared weight (a simplification).** One may instead retain a single weight w_t^{ε} common to all series at time t, as in the baseline's multivariate Student-t. This is lighter, but introduces a structural asymmetry once the AR(1) dynamics are present: persistence is modelled *series by series* (each ρ_i distinct), whereas the tail down-weighting remains *global* (one w_t^{ε} for the whole cross-section). Having a series-specific level of dynamics but a panel-wide level of robustness is internally inconsistent, and this asymmetry is the main argument in favour of the per-series formulation.

In what follows we develop the per-series formulation as the conceptually correct one, and note the shared-weight case as the available simplification where it changes the equations.

Exact nesting of the baseline. The extension is a strict generalisation. Setting $\rho_i = 0$ for all i collapses (156) to $\varepsilon_{i,t} = u^{\varepsilon}_{i,t}$, with $\varepsilon_{i,t} \mid w^{\varepsilon}_{i,t} \sim \mathcal{N}(0, \sigma_i^2 / w^{\varepsilon}_{i,t})$, which is exactly the i.i.d. heavy-tailed idiosyncratic error of the baseline. The scale matrix $\mathbf{R}$ recovers its original meaning (the scale of ε_t itself), and the only formal difference is the per-series rather than shared weight. The baseline is therefore the $\rho = 0$ special case of the extended model, which makes the extension a continuous enrichment rather than a distinct model.

Scale-Mixture Coherence: A Mirror of the Factor Side

The single most important structural property of the specification in (156) is that it preserves the conditionally-Gaussian scale-mixture architecture on which the entire EM apparatus rests. The heavy tails are placed on the *innovation* $u^{\varepsilon}_{i,t}$, exactly as the factor-side heavy tails are placed on the innovation u_t. Conditional on the weights $w^{\varepsilon}_{i,t}$, the idiosyncratic recursion (156) is a *linear Gaussian* autoregression, and the Gamma–Gaussian conjugacy that delivers the closed-form weight posteriors carries over unchanged.

This is the decisive point: the extension is *not a new architecture*. It is the mirror image, on the idiosyncratic side, of a mechanism the model already possesses on the factor side. The factor block is already a persistent process (the VAR) whose heavy tails sit on its fresh innovation u_t and whose anomalies are governed by an innovation-level weight w^u_t. The AR(1) idiosyncratic block reproduces the same triad — persistence in a recursion, heavy tails on the innovation, conjugate weight on the innovation — one series at a time. Every derivation that works for (f_t, u_t, w^u_t) has a scalar counterpart for $(\varepsilon_{i,t}, u^{\varepsilon}_{i,t}, w^{\varepsilon}_{i,t})$. Had the heavy tails instead been placed on the *accumulated* error $\varepsilon_{i,t}$ rather than on its innovation, the conditionally-Gaussian linear state-space structure would be lost and the conjugate weight posterior would no longer hold; this is the reason the innovation-level placement is the coherent one.

The Augmented State-Space Representation

Because $\varepsilon_{i,t}$ now depends on $\varepsilon_{i,t-1}$, the idiosyncratic errors acquire *memory* and can no longer be relegated to the observation noise. They must be carried inside the state vector. This is the structural cost of the extension, and it raises the state dimension from $5r$ to $5r + M$.

The further-augmented state. We append the contemporaneous idiosyncratic vector to the Mariano–Murasawa augmented state $\tilde{f}_t \in \mathbb{R}^{5r}$ already in use:

$$\check{f}_t \;=\; \big(\tilde{f}_t',\; \varepsilon_t'\big)' \;=\; \big(f_t',\, f_{t-1}',\, f_{t-2}',\, f_{t-3}',\, f_{t-4}',\; \varepsilon_t'\big)' \;\in\; \mathbb{R}^{5r+M}. \tag{158}$$

Only the contemporaneous ε_t is appended: the AR(1) has a single lag, and ε_{t-1} is supplied by the state at the previous period through the transition.

The augmented transition. The state transition is block-diagonal between the factor sub-state and the idiosyncratic sub-state, since the two evolve independently:

$$\check{f}_t \;=\; \check{\mathbf{A}}\,\check{f}_{t-1} \;+\; \check{u}_t, \qquad \check{\mathbf{A}} \;=\; \begin{pmatrix} \tilde{\mathbf{A}} & \mathbf{0}_{5r\times M} \\ \mathbf{0}_{M\times 5r} & \mathbf{P} \end{pmatrix}, \qquad \mathbf{P} \;=\; \operatorname{diag}(\rho_1, \rho_2, \ldots, \rho_M), \tag{159}$$

where the factor companion block is exactly the one already derived,

$$\tilde{\mathbf{A}} = \begin{pmatrix} \mathbf{A} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & I_r & \mathbf{0} \end{pmatrix},$$

and the new diagonal block $\mathbf{P}$ collects the M idiosyncratic persistence coefficients. The block-diagonal form makes explicit that the factor dynamics and the idiosyncratic dynamics do not interact within the transition — their only meeting point is the observation equation.

The augmented innovation covariance. The augmented innovation $\check{u}_t = (\tilde{u}_t', u_t^{\varepsilon\prime})'$ stacks the factor innovation (lifted to companion form) and the idiosyncratic innovation vector $u_t^{\varepsilon} = (u_{1,t}^{\varepsilon}, \ldots, u_{M,t}^{\varepsilon})'$. Its conditional covariance, given the weights, is block-diagonal and time-varying through the weights:

$$\operatorname{Cov}\big(\check{u}_t \mid w_t^u, w_{1,t}^{\varepsilon}, \ldots, w_{M,t}^{\varepsilon}\big) = \begin{pmatrix} \tilde{\mathbf{Q}}/w_t^u & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_t^{\varepsilon} \end{pmatrix}, \qquad \mathbf{D}_t^{\varepsilon} = \operatorname{diag}\left(\frac{\sigma_1^2}{w_{1,t}^{\varepsilon}}, \ldots, \frac{\sigma_M^2}{w_{M,t}^{\varepsilon}}\right), \tag{160}$$

with $\tilde{\mathbf{Q}} = \operatorname{diag}(\mathbf{Q}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0})$ the singular companion noise of the factor block. The idiosyncratic block of (160) is where the separation of $\mathbf{R}$ and the weights becomes operational: the numerators σ_i^2 are the fixed entries of $\mathbf{R}$, while the denominators $w_{i,t}^{\varepsilon}$ are the time-varying tail multipliers. Under the shared-weight simplification, $\mathbf{D}_t^{\varepsilon}$ would reduce to $\mathbf{R}/w_t^{\varepsilon}$ with a single scalar denominator common to all series; the per-series formulation replaces that one scalar with the M distinct weights.

The noise-free observation equation. With the idiosyncratic errors absorbed into the state, the observation equation no longer has a separate noise term: all stochastic disturbance now lives in the state innovation. The observed data are an exact linear function of the augmented state,

$$y_t^{\text{obs}} = \mathbf{W}_t \check{\boldsymbol{\Lambda}} \check{f}_t, \qquad \check{\boldsymbol{\Lambda}} = \big[\ \tilde{\boldsymbol{\Lambda}} \ \big|\ I_M\ \big] \in \mathbb{R}^{M \times (5r+M)}, \tag{161}$$

where $\tilde{\boldsymbol{\Lambda}} \in \mathbb{R}^{M \times 5r}$ is the effective loading matrix of the mixed-frequency model — carrying the block-diagonal monthly loadings $\boldsymbol{\Lambda}^M$ and the Mariano–Murasawa aggregation pattern for the quarterly loadings — and the appended identity block I_M maps each idiosyncratic state component directly onto its own series. The selection matrix $\mathbf{W}_t$ continues to pick out the rows observed at time t, now selecting jointly the corresponding factor-loading rows of $\tilde{\boldsymbol{\Lambda}}$ and the corresponding columns of I_M. The effective measurement noise is *zero*: in scale-matrix terms, the idiosyncratic scale that previously sat in the observation equation has migrated into the state through (160). This "noiseless observation" form is the standard device for embedding autocorrelated measurement errors into a state-space model.

Consequences for the E-Step and the M-Step

The estimation algorithm retains its overall shape — a smoother conditional on the weights, a conjugate weight update, and closed-form parameter updates — but each ingredient acquires an idiosyncratic counterpart. We describe the changes at the level of the equations.

Smoothed idiosyncratic moments. Because ε_t is now part of $\check{f}_t$, the smoother that runs on the augmented state returns, alongside the factor moments, the posterior moments of the idiosyncratic errors:

$$\mathbb{E}[\varepsilon_{i,t} \mid \mathbf{Y}], \qquad \mathbb{E}[\varepsilon_{i,t}^2 \mid \mathbf{Y}], \qquad \mathbb{E}[\varepsilon_{i,t}\, \varepsilon_{i,t-1} \mid \mathbf{Y}].$$

These play, for the idiosyncratic AR(1), exactly the role that the weighted factor second moments $\mathcal{P}^u_{00}, \mathcal{P}^u_{10}, \mathcal{P}^u_{11}$ play for the VAR: they are the sufficient statistics for the updates of ρ_i and σ_i^2.

The idiosyncratic Mahalanobis residual is now on the innovation. In the baseline, the weight posterior is driven by the residual $d^\varepsilon_t = (y_t - \boldsymbol{\Lambda} f_t)'\mathbf{R}^{-1}(y_t - \boldsymbol{\Lambda} f_t)$, the squared Mahalanobis distance of the *contemporaneous* idiosyncratic error. In the extension, the heavy-tailed quantity is the *innovation* $u^\varepsilon_{i,t} = \varepsilon_{i,t} - \rho_i \varepsilon_{i,t-1}$, so the relevant residual is, per series,

$$d^\varepsilon_{i,t} \;=\; \frac{\mathbb{E}\big[\, (\varepsilon_{i,t} - \rho_i \varepsilon_{i,t-1})^2 \,\big|\, \mathbf{Y} \,\big]}{\sigma_i^2}, \tag{162}$$

where the expectation is taken under the smoother and therefore includes the posterior-variance correction (the smoothed second moments of $\varepsilon_{i,t}$ and $\varepsilon_{i,t-1}$ and their cross moment), not merely the squared point estimate. The conjugate posterior of the weight keeps its Gamma form, but with the dimension count appropriate to a *scalar* innovation:

$$w^\varepsilon_{i,t} \mid \mathbf{Y}, \theta \;\sim\; \text{Gamma}\!\left(\frac{\nu_\varepsilon + 1}{2},\; \frac{\nu_\varepsilon + d^\varepsilon_{i,t}}{2}\right), \qquad \hat{w}^\varepsilon_{i,t} \;=\; \frac{\nu_\varepsilon + 1}{\nu_\varepsilon + d^\varepsilon_{i,t}}. \tag{163}$$

The shape parameter is $(\nu_\varepsilon + 1)/2$ rather than the baseline's $(\nu_\varepsilon + M)/2$ precisely because each per-series weight now governs a single one-dimensional innovation rather than the M-dimensional vector ε_t. Under the shared-weight simplification, by contrast, one would aggregate the innovation residuals across series into a single $d^\varepsilon_t = \sum_i d^\varepsilon_{i,t}$ and retain the shape $(\nu_\varepsilon + m_t)/2$ of the baseline, where m_t is the number of series observed at time t.

The new update of the persistence coefficients ρ_i. The persistence coefficients are new free parameters, and their M-step update is the scalar analogue of the VAR transition update $\mathbf{A}^{(j+1)} = \mathcal{P}^u_{10}\,(\mathcal{P}^u_{00})^{-1}$. Maximising the expected complete-data log-likelihood over ρ_i yields a weighted least-squares ratio of smoothed cross-moments,

$$\rho_i^{(j+1)} \;=\; \frac{\displaystyle\sum_t \hat{w}^\varepsilon_{i,t}\; \mathbb{E}[\varepsilon_{i,t}\, \varepsilon_{i,t-1} \mid \mathbf{Y}]}{\displaystyle\sum_t \hat{w}^\varepsilon_{i,t}\; \mathbb{E}[\varepsilon_{i,t-1}^2 \mid \mathbf{Y}]}, \tag{164}$$

with each contribution down-weighted by the posterior weight $\hat{w}^\varepsilon_{i,t}$ — the same robust weighting that protects the factor dynamics from innovation outliers, now protecting the estimation of idiosyncratic persistence.

The update of the baseline scale σ_i^2. Evaluated at the just-updated $\rho_i^{(j+1)}$, the update of the idiosyncratic scale is the scalar analogue of the innovation-covariance update for $\mathbf{Q}$. The cross-terms cancel at the optimum in ρ_i, leaving

$$\sigma_i^{2,(j+1)} \;=\; \frac{1}{T}\sum_t \hat{w}^\varepsilon_{i,t}\, \Big(\mathbb{E}[\varepsilon_{i,t}^2 \mid \mathbf{Y}] \;-\; \rho_i^{(j+1)}\, \mathbb{E}[\varepsilon_{i,t}\, \varepsilon_{i,t-1} \mid \mathbf{Y}]\Big). \tag{165}$$

The matrix $\mathbf{R} = \text{diag}(\sigma_1^2, \ldots, \sigma_M^2)$ remains diagonal throughout, and these per-series updates simply refresh its entries; the off-diagonal structure of the model is untouched. The degrees-of-freedom update for ν_ε retains the digamma-based first-order condition of the baseline, now formed from the per-series weights $\hat{w}^\varepsilon_{i,t}$ and their logarithms aggregated over the full panel.

Summary of the algorithmic change. At the level of equations, the extension adds one new parameter vector ($\{\rho_i\}$) with a closed-form update (164), re-expresses the idiosyncratic residual on the innovation (162), attaches per-series weights through (163), and enlarges the smoother to the state (158). Every step has a direct precedent in the factor-side machinery; nothing in the conjugacy or the closed-form M-step is sacrificed.

Identification: Common versus Idiosyncratic Persistence

The extension introduces a genuine identification tension that the baseline avoided by construction. With both the factors (through $\mathbf{A}$) and the idiosyncratic errors (through the ρ_i) now able to generate autocorrelation in the observed series, the data must *allocate* persistence between a common source and a series-specific one. The baseline resolves this by fiat: with $\rho_i \equiv 0$, all persistence is common. Once the ρ_i are free, the allocation becomes an estimation problem.

What identifies the split. The identifying feature is the *cross-sectional* dimension. A common factor drives *many* series simultaneously: its persistence shows up as *co-movement* across the panel. An idiosyncratic AR(1) drives *one* series only: its persistence is, by construction, uncorrelated across series. The block-diagonal loading structure reinforces this separation, since it ties each series to a single factor and leaves the ρ_i as purely series-specific nuisance dynamics. In principle, therefore, shared persistence is absorbed by the factors and series-specific persistence by the ρ_i. In practice the two channels can still compete, particularly for series weakly related to their block factor, where the data may be nearly indifferent between a small common loading and a moderate idiosyncratic persistence.

Cautions. Several safeguards are natural in the extended model. Stationarity of each idiosyncratic process requires $|\rho_i| < 1$, which should be imposed (or the update (164) otherwise constrained) to keep the state transition $\check{\mathbf{A}}$ stable. Where the common/idiosyncratic split is weakly identified, some form of regularisation — a shared ρ across series, a prior shrinking the ρ_i toward zero, or a restriction to a subset of series believed to exhibit idiosyncratic persistence — would stabilise the estimates and preserve the parsimony that motivated the baseline's serial-uncorrelation assumption in the first place. These are modelling decisions to be settled when the extension is taken from specification to estimation.

Interaction with the Mixed-Frequency and Block Structure

The AR(1) idiosyncratic block interacts cleanly, but not trivially, with the two other extensions of the model.

Block structure. The persistence coefficients ρ_i are entirely series-specific and live in the diagonal block $\mathbf{P}$ of $\check{\mathbf{A}}$, orthogonal to the block-diagonal loading structure of $\boldsymbol{\Lambda}$. They do not enter the economic identification of the factors and do not disturb the exclusion restrictions that define the real, financial, and other blocks. They add one nuisance dynamic per series, no more.

Mixed frequency. The interaction with the Mariano–Murasawa aggregation requires care for the quarterly series. For a monthly series the appended idiosyncratic component enters the observation through the identity block of $\check{\boldsymbol{\Lambda}}$ in the obvious way. For a quarterly series, the observed quarterly value aggregates the underlying monthly quantities through the MM weights; a *persistent* idiosyncratic error for such a series would, strictly, call for the same temporal-aggregation treatment applied to the factor — that is, the idiosyncratic state of a quarterly series would itself need to be aggregated across the relevant months rather than entering only contemporaneously. In the applied setting this concerns the single quarterly series (GDP), and the cleanest specification would either confine idiosyncratic persistence to the monthly series or extend the MM aggregation to the idiosyncratic state of the quarterly series as well. We flag this as a modelling choice internal to the extension rather than resolving it here.

Status: A Methodological Extension for Future Validation

The development in this section is a *methodological extension*, not a component of the estimated and validated baseline. The baseline Student-t DFM — with i.i.d. heavy-tailed idiosyncratic errors — is the model whose parameter-recovery properties are assessed in the calibrated Monte Carlo experiment that follows. The AR(1) idiosyncratic specification described here has *not* been subjected to that validation: no Monte Carlo recovery experiment for the extended data-generating process is reported, and no empirical results from estimating the extended model are presented.

What this section establishes is the *coherent path* for the extension: the extended DGP (155)–(157), its augmented state-space representation (158)–(161), and the corresponding modifications to the E- and M-step equations (162)–(165). The construction shows that persistent idiosyncratic errors with heavy-tailed innovations fit within the existing scale-mixture apparatus as a mirror of the factor-side mechanism, and that the baseline is recovered exactly as the $\rho \to 0$ limit. A full assessment — a dedicated Monte Carlo design that calibrates the ρ_i, simulates the extended DGP, and verifies recovery of the persistence coefficients jointly with the loadings, the factor dynamics, and the two tail indices — is the natural next step, and is left to future work.

Monte Carlo Validation: A Calibrated-DGP Experiment

The previous sections derived the EM algorithm for the Student-t Dynamic Factor Model in full detail, with all the extensions (mixed-frequency, ragged-edge, block-diagonal identification) that make it applicable to real macroeconomic panels. We now turn to the question of *validation*: how do we assess whether the proposed estimator actually recovers the parameters of the data-generating process? This question cannot be answered using real data alone, and the answer requires a controlled simulation experiment of the kind described in this section.

Motivation: Why a Calibrated Monte Carlo Experiment

The fundamental limitation of real-data estimation. When the EM algorithm is applied to the real macroeconomic panel, it returns a vector of parameter estimates $\hat{\boldsymbol{\theta}}$. The estimates themselves are observable: we can inspect their numerical values, their relative magnitudes, their stability under perturbations of the data. What we *cannot* observe is the true parameter vector $\boldsymbol{\theta}^*$ that generated the data. Without ground truth, there is no benchmark against which to compare $\hat{\boldsymbol{\theta}}$, and no way to distinguish between two equally plausible hypotheses:

- the algorithm is correctly identifying the underlying structure, or
- the algorithm is producing systematically biased estimates that happen to look reasonable.

Real data alone provide an *application* of the method, not a *validation* of it.

The role of controlled simulation. A Monte Carlo experiment supplies the missing ground truth. We specify a true parameter vector $\boldsymbol{\theta}^*$, generate synthetic data from the assumed data-generating process, and apply the algorithm to this synthetic data without using any knowledge of $\boldsymbol{\theta}^*$. Comparing the resulting estimates $\hat{\boldsymbol{\theta}}^{(s)}$ across many independent replications $s = 1, \dots, S$ with the known truth $\boldsymbol{\theta}^*$ produces an empirical distribution of the estimator that allows us to answer four distinct questions:

1. *Is the estimator unbiased, or does it systematically over- or under-estimate certain parameters?*
2. *Does the estimator concentrate around the truth as the sample size grows* (a finite-sample analogue of consistency)?
3. *Does the algorithm converge reliably from a given initialisation, or is it prone to local maxima or numerical pathologies?*
4. *How robust is the estimator to model misspecification* — for instance, what happens if the data are truly Gaussian and we apply a Student-t EM, or vice versa?

None of these questions can be answered from real data alone. The Monte Carlo is therefore not optional or supplementary: it is the only methodological route through which the proposed estimator can be subjected to a meaningful test.

Why *calibrated* on real data. A second design decision concerns the choice of $\boldsymbol{\theta}^*$. The simplest option would be to pick textbook parameter values — round numbers, simple persistence coefficients, generic loadings. This is the approach taken in many methodological papers, but it has a serious drawback: the resulting synthetic data may bear little resemblance to the empirical setting in which the algorithm will ultimately be used. An estimator that performs flawlessly on stylised data may still fail on the real macroeconomic panel, because the relevant features of the data — persistence levels, correlation structures, tail behaviour, missingness patterns — are quantitatively different from those of the simulation.

The calibration approach avoids this gap. We set

$$\boldsymbol{\theta}^* \;:=\; \hat{\boldsymbol{\theta}} \text{ (estimated on real macro data),}$$

so that the synthetic data inhabit the same statistical neighbourhood as the real series: the same persistence, the same loadings, the same heaviness of tails, the same idiosyncratic structure. The macroeconomic panel becomes the *empirical reference point* for the Monte Carlo experiment.

This is the standard approach in the DFM Monte Carlo literature (Doz, Giannone, Reichlin, 2012; Bańbura and Modugno, 2014) and is explicitly endorsed by Stock and Watson (2016, Chapter 8) as the methodological reference point for evaluating dynamic factor model estimators. It avoids both the artificiality of textbook parameter choices and the trivial circularity of validating on the same data used for estimation.

The economic-econometric goal. The Monte Carlo experiment described in this section is not merely a sanity check on the implementation. Its goal is to characterise the finite-sample properties of the proposed estimator under realistic empirical conditions, and to document its relative performance against the Gaussian benchmark of Bańbura and Modugno (2014). In particular, we aim to show that the Student-t extension is a *strict generalisation* of the Gaussian case: it matches the benchmark when the data are well-described by a Gaussian model, and outperforms it when the data exhibit heavy tails or contamination by outliers. This is the central econometric claim of the proposed methodology, and the Monte Carlo experiment is the vehicle through which it is empirically substantiated.

The Three-Step Design

The Monte Carlo design proceeds in three sequential steps, each addressing a distinct aspect of the validation problem. The high-level logic is summarised in Figure 3

Step 1 — Calibration on real data. The EM algorithm developed in this thesis is applied once to the real macroeconomic panel $\mathbf{Y}^{\text{real}} \in \mathbb{R}^{M\times T}$, with all extensions in operation: mixed-frequency aggregation, ragged-edge missingness via the selection matrices $\mathbf{W}_t$, and block-diagonal identification of the real, financial, and "other" factor blocks. The algorithm is run to convergence, producing the estimated parameter vector

$$\hat{\boldsymbol{\theta}} \;=\; \big(\hat{\boldsymbol{\Lambda}},\; \hat{\mathbf{A}},\; \hat{\mathbf{Q}},\; \hat{\mathbf{R}},\; \hat{\nu}_u,\; \hat{\nu}_\varepsilon\big).$$

This vector is then *re-labelled* as the true parameter vector of the Monte Carlo experiment:

$$\boldsymbol{\theta}^* \;:=\; \hat{\boldsymbol{\theta}}.$$

The change of notation is conceptual rather than numerical: the same numbers are now treated as the data-generating parameters of a hypothetical world, rather than as estimates of an unknown truth. From Step 2 onward, $\boldsymbol{\theta}^*$ plays the role of ground truth.

In addition to the parameter values, Step 1 also records two features of the real panel that will be reused in Step 2: the *missingness pattern* $\{\mathbf{W}_t^{\text{real}}\}_{t=1}^T$ encoding the ragged-edge structure, and the *mixed-frequency calendar* indicating which months are quarter-end for the quarterly series. These two ingredients are not parameters in the usual sense, but they are part of the empirical setting that we want to reproduce.

Step 2 — Generation of synthetic samples. A collection of S independent synthetic samples $\{\mathbf{Y}^{(s)}\}_{s=1}^{S}$ is generated by drawing from the Student-t DFM with parameters $\boldsymbol{\theta}^*$. Each sample has length T^* — typically equal to T_{real} for the headline results, but also varied for the finite-sample diagnostics described later. The detailed simulation procedure — how scale-mixture innovations are drawn, how the ragged-edge pattern is imposed, how the mixed-frequency aggregation is applied, and how contamination is injected for Experiment C — is the subject of later sections.

For each replication s, the simulation produces not only the observable synthetic panel $\mathbf{Y}^{(s)}$ but also the latent objects that drive the data: the factor path $\{f_t^{(s)}\}$, the scale-mixture weight paths $\{w_t^{u,(s)}, w_t^{\varepsilon,(s)}\}$, and the underlying idiosyncratic shocks $\{\varepsilon_t^{(s)}\}$. These latent objects are stored alongside $\mathbf{Y}^{(s)}$: they are the ground truth against which the smoothed estimates produced in Step 3 will be compared. The combination of known parameters $\boldsymbol{\theta}^*$ and known latent paths is what makes the controlled experiment possible.

From a balanced panel to an empirically realistic one. A clarification on what Step 2 produces. The data-generating process in the previous paragraph generates, for each replication s, a panel $\{y_t^{(s)}\}_{t=1}^{T^*} = \{\boldsymbol{\Lambda}^* f_t^{(s)} + \varepsilon_t^{(s)}\}_{t=1}^{T^*}$ that is, by construction, *balanced and at monthly frequency*: every entry is observed, all series have a value for every time period, and no calendar structure distinguishes monthly from quarterly observations. This is the natural output of the Student-t DFM, which has no inherent concept of mixed-frequency or ragged-edge: these features are properties of the empirical release calendar, not of the underlying generative model.

To make the synthetic data comparable to the real panel — and to ensure that the re-estimation in Step 3 confronts the same difficulty that the EM faces empirically — we then *impose* the missing-data structure of the real panel on the generated data:

- Apply the recorded missingness pattern $\{\mathbf{W}_t^{\text{real}}\}_{t=1}^{T_{\text{real}}}$ to $\{y_t^{(s)}\}$, masking entries that were not yet released at each calendar date in the real data;

- For quarterly series, mask all non-quarter-end months, keeping only the entries at quarter-end months consistent with the empirical release schedule.

This step is purely deterministic — no random sampling is involved — and turns the balanced synthetic panel into a ragged-edge, mixed-frequency panel whose missing-data structure replicates the real one. The resulting $\mathbf{Y}^{(s)}$ that the EM in Step 3 will see has the same difficulty as the empirical estimation problem.

The details of the simulation pipeline are developed in a later subsection.

Step 3 — Re-estimation and comparison. On each synthetic sample $\mathbf{Y}^{(s)}$, the EM algorithm is run from scratch — that is, with a fresh PCA initialisation that uses no knowledge of either $\boldsymbol{\theta}^*$ or of the latent paths. The algorithm converges to its own estimate $\hat{\boldsymbol{\theta}}^{(s)}$ and produces its own smoothed factor path $\{\hat{f}_t^{(s)}\}$. The collection $\{\hat{\boldsymbol{\theta}}^{(s)}, \hat{f}_t^{(s)}\}_{s=1}^{S}$ is the empirical distribution of the estimator under the assumed DGP.

The comparison with ground truth proceeds along two axes:

- *Parameter recovery*: comparing $\hat{\boldsymbol{\theta}}^{(s)}$ against $\boldsymbol{\theta}^*$ via Bias, Variance, and RMSE for each component;

- *Factor recovery*: comparing the smoothed estimates $\{\hat{f}_t^{(s)}\}$ against the true factor paths $\{f_t^{(s)}\}$ via correlation and trajectory error.

The detailed list of validation metrics — and how they translate across the contamination grid and the sample-size grid — is the subject of later sections.

What this design tests. By the end of Step 3, the Monte Carlo has produced enough information to characterise the proposed estimator under the conditions of the real panel: how biased it is, how variable, how well it recovers the latent factors, how reliably it converges. The same design, applied in parallel to the Gaussian benchmark of Bańbura and Modugno (2014), then enables the comparative analysis that is the central econometric claim of this thesis.

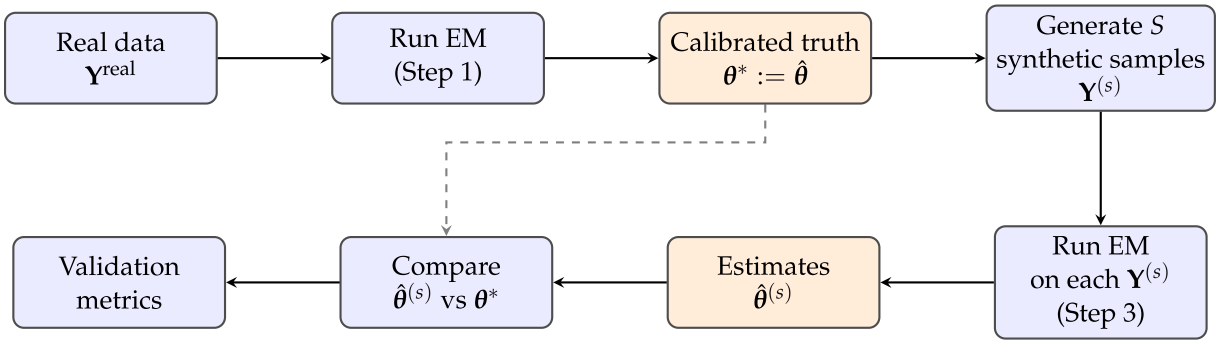


Figure 3: The three-step Monte Carlo design. The real data are used to calibrate the true parameter vector $\boldsymbol{\theta}^*$ (orange boxes mark ground truth). Synthetic samples are then generated from this $\boldsymbol{\theta}^*$ and re-estimated; the resulting estimates $\hat{\boldsymbol{\theta}}^{(s)}$ are compared against the calibrated truth to assess parameter recovery.

Simulating Realistic Empirical Conditions

A Monte Carlo experiment is only as informative as its similarity to the empirical setting it aims to validate. If the synthetic data generated in Step 2 are a stylised idealisation of the real panel — balanced, single-frequency, free of outliers — then the algorithm's performance on this synthetic data will overstate its performance on the real macroeconomic series, where none of these idealisations hold. The credibility of the validation therefore depends on reproducing, in the simulated data, the empirical features that make the real estimation problem hard.

This subsection describes how each such feature is reproduced operationally. Six aspects of the empirical setting are considered; we briefly introduce each of them here, before developing them in dedicated subsections.

(i) Student-t scale-mixture innovations. The data-generating process at the heart of the experiment is the Student-t DFM developed in this thesis. The innovations to the factor process u_t and to the observation equation ε_t are generated via the scale-mixture representation: Gaussian shocks whose covariance is scaled by an independent Gamma weight, with degrees of freedom equal to the calibrated $\nu_u^*, \nu_\varepsilon^*$. This construction produces the heavy tails that the Gaussian benchmark cannot reproduce, and it stores the latent weight paths $\{w_t^{u,(s)}, w_t^{\varepsilon,(s)}\}$ as ground truth for the validation of the algorithm's weight-recovery step.

(ii) Ragged-edge missingness. Real macroeconomic data are released with publication delays that differ across series: at any sample date T, some series are already available for the current period while others are still to be released. The cross-section at the right edge of the panel is therefore *ragged*, with progressively sparser observations as one approaches T. The Monte Carlo reproduces this pattern by recording the missingness structure of the real data — encoded by the selection matrices $\{\mathbf{W}_t^{\text{real}}\}$ — and imposing it on every synthetic sample before re-estimation. This ensures that the synthetic estimation problem confronts the same information structure as the empirical one, including the asymmetric concentration of missing data at the most recent periods.

(iii) Mixed-frequency calendar. A fraction of the series in the panel are observed at quarterly rather than monthly frequency — including the target variable GDP. The model handles this through the Mariano–Murasawa augmentation of the state vector and a corresponding loading pattern. To reproduce the same difficulty in the synthetic setting, we generate the data at monthly frequency for all series, including those that are observed only quarterly in the real panel; we then mask all non-quarter-end months for the quarterly series, so that the re-estimation step sees the same pattern of quarterly observations that the real data exhibits. The latent monthly values of the quarterly series remain stored as ground truth, allowing us to evaluate whether the algorithm correctly interpolates them.

(iv) Block-diagonal economic identification. The DFM developed in this thesis imposes a block-diagonal structure on the loading matrix: real-activity series load only on the real factor, financial series only on the financial factor, and so on. This identification restriction distinguishes our framework from the unrestricted DFM of Doz, Giannone, and Reichlin (2012) and gives the factors an economic interpretation. The same restriction is preserved both in the data generation and in the re-estimation of the synthetic samples, so that the Monte Carlo evaluates the algorithm *within* the identified model, not in an unrestricted setting. This in turn allows us to test whether the block identification is itself correctly recovered — that is, whether the estimated real factor tracks the true real factor rather than some linear combination involving the other blocks.

(v) Contamination by extreme observations. The Student-t framework is motivated by the empirical observation that macroeconomic series occasionally exhibit extreme realisations — crisis episodes, pandemics, abrupt policy interventions — that the Gaussian model would dismiss as overwhelmingly unlikely. To quantify the robustness margin of the proposed estimator under this empirical reality, a third experiment generates data from a Gaussian DGP contaminated by a small fraction π of outlier periods drawn from a heavy-tailed distribution. The contamination intensity π is varied across the grid $\{0.01, 0.05, 0.10\}$, tracing a continuous path from near-Gaussian conditions to substantially contaminated ones. This experiment is the most economically meaningful of the three, as it reproduces the realistic case in which the baseline data are well-behaved but a small minority of periods is governed by a different mechanism that produces extreme observations.

(vi) Vintages and the real-time data flow. The news framework developed in the previous section operates on *vintages* — snapshots of the panel at different calendar dates, reflecting the sequential arrival of new releases. To validate the news framework in the Monte Carlo, the simulation must reproduce the vintage structure of the real data: the release calendar of each series, the sequence of vintages it generates, and the corresponding pattern of forecast revisions. Although the systematic empirical analysis of the news framework is deferred to future work, the vintage-generation infrastructure is built into the simulation pipeline described below, so that the validation exercise can be extended to the news framework without further infrastructure development.

Why all six matter. The six features cannot be sacrificed individually without compromising the credibility of the validation. Generating data without scale-mixture innovations would amount to using a Gaussian DGP in all experiments, which would prevent any test of the Student-t machinery (a problem we address by varying the DGP across the three experiments below). Synthetic data without ragged-edge would make the estimation problem easier than the empirical one. Synthetic data without mixed-frequency would not test the Mariano–Murasawa machinery that is central to the model. Synthetic data without block identification would not test the economic interpretation of the factors. A Monte Carlo without contamination experiments would not reveal the robustness margin that motivates the heavy-tailed extension in the first place. Synthetic data without vintages would not test the news framework. Taken together, the six features turn the calibrated Monte Carlo from a stylised exercise into a genuine reproduction of the empirical estimation problem. The remaining subsections describe how each feature is implemented in practice.

(i) The Student-t Scale-Mixture

Where we are in the design. Recall the structure of the Monte Carlo design: Step 1 estimated the Student-t DFM on the real data, producing the calibrated parameter vector

$$\boldsymbol{\theta}^* \;=\; (\boldsymbol{\Lambda}^*,\, \mathbf{A}^*,\, \mathbf{Q}^*,\, \mathbf{R}^*,\, \nu_u^*,\, \nu_\varepsilon^*),$$

which is now treated as the ground truth of the Monte Carlo. Step 2 — which is what this subsection describes — generates the S synthetic samples $\{\mathbf{Y}^{(s)}\}_{s=1}^{S}$ that the EM in Step 3 will see. The current question is therefore: *given the calibrated parameters $\boldsymbol{\theta}^*$, how do we generate a synthetic sample from the Student-t DFM?*

Why a scale-mixture, and not direct sampling. The Student-t distribution does not have a standard primitive random-number generator in most numerical libraries, and even where one exists, sampling directly from a multivariate Student-t would not produce the latent objects (the weights) that the algorithm recovers in the E-step. We therefore exploit the *scale-mixture representation* of the multivariate Student-t:

$$y \sim t_\nu(\mathbf{0}, \boldsymbol{\Sigma}) \iff y \,|\, w \sim \mathcal{N}\big(\mathbf{0}, \boldsymbol{\Sigma}/w\big), \quad w \sim \text{Gamma}\big(\nu/2,\, \nu/2\big).$$

This identity says that a Student-t draw is equivalent to a two-step procedure: first draw a Gamma weight w, then draw a Gaussian conditional on w with covariance scaled by $1/w$. Marginalising over w, the resulting y has exactly a Student-t distribution with ν degrees of freedom.

This representation is both *numerically efficient* — Gaussian draws and Gamma draws are both standard primitives — and *methodologically transparent*: it makes the latent weights w_t^u, w_t^ε explicit at every period, so that we can store them as ground truth against which the EM-estimated weights $\hat{w}_t^u, \hat{w}_t^\varepsilon$ will later be compared.

A clarification on the weights. A subtle but important point: the weights drawn in this subsection are entirely *new*, independent draws from the marginal Gamma distribution. They are not the smoothed weights $\hat{w}_t^{u,\text{real}}, \hat{w}_t^{\varepsilon,\text{real}}$ that the EM in Step 1 produced on the real data — those were specific to the real realisation and would have no role here. The key insight is that in the scale-mixture representation, the weights are not parameters of the model; they are *latent random variables* whose distribution is fully determined by the parameter ν^*. Once ν_u^* and ν_ε^* are known — as they are, from the calibration step — generating new weights for the synthetic samples is a direct probabilistic operation: draw from Gamma$(\nu^*/2, \nu^*/2)$. No estimation or smoothing is involved.

This is the central conceptual point of the Monte Carlo construction: each synthetic replication s is a fresh realisation of the data-generating process, with its own independent latent weight paths and its own independent factor and idiosyncratic shocks. The EM that re-estimates each synthetic sample in Step 3 will, in turn, recover smoothed weight paths $\hat{w}_t^{u,(s)}, \hat{w}_t^{\varepsilon,(s)}$ specific to that replication, and these can be compared against the ground-truth $w_t^{u,(s)}, w_t^{\varepsilon,(s)}$ drawn here to assess the accuracy of the weight-recovery step.

The three-step procedure for each replication. With this setup in mind, the procedure for each replication s proceeds in three sub-steps, described in order below: draw the two latent weight paths (Step i.a), draw the factor path conditional on the factor weights (Step i.b), and draw the observations conditional on both factors and idiosyncratic weights (Step i.c).

Step (i.a) — Draw the latent weight paths. We begin by drawing two independent weight paths for the entire synthetic sample $t = 1, \dots, T^*$, one for the factor process and one for the observation process:

$$w_t^{u,(s)} \sim \text{Gamma}\big(\nu_u^*/2,\ \nu_u^*/2\big), \tag{166}$$

$$w_t^{\varepsilon,(s)} \sim \text{Gamma}\big(\nu_\varepsilon^*/2,\ \nu_\varepsilon^*/2\big), \tag{167}$$

with the draws independent across time and across the two processes. The Gamma distribution is parametrised with shape and rate both equal to $\nu^*/2$, a convention that gives the weights two convenient properties: their unconditional mean is one ($\mathbb{E}[w_t] = 1$) and their variance is $2/\nu^*$.

The role of ν^* is therefore entirely captured by this single quantity: smaller ν^* produces larger variance in the weights, which in turn produces a wider spread of the conditional covariances of the Gaussian shocks across t, which finally translates into heavier tails of the marginal Student-t distribution. The calibrated values $\nu_u^*, \nu_\varepsilon^*$ that we extracted from the real data therefore directly determine the empirical realism of the heavy tails in the synthetic samples. A typical macroeconomic calibration yields $\nu^* \in [8, 15]$, which produces weights tightly distributed around one for most periods and substantially below one for a small fraction of outlier-prone periods.

Step (i.b) — Draw the factor path conditional on weights. With the factor weights $\{w_t^{u,(s)}\}$ drawn, the factor innovations $u_t^{(s)}$ are conditionally Gaussian with time-varying covariance:

$$u_t^{(s)} \mid w_t^{u,(s)} \sim \mathcal{N}\Big(\mathbf{0},\ \mathbf{Q}^*/w_t^{u,(s)}\Big), \qquad t = 1, \dots, T^*.$$

A period with $w_t^{u,(s)} \approx 1$ produces a near-typical innovation; a period with $w_t^{u,(s)}$ substantially below one produces an innovation drawn from a distribution with inflated covariance — exactly the mechanism through which Student-t heavy tails are operationalised.

The factor path is then constructed recursively from the state equation, after initialising at the stationary distribution of the VAR(1):

$$f_0^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_0), \qquad \text{vec}(\mathbf{\Sigma}_0) = \big(I_{r^2} - \mathbf{A}^* \otimes \mathbf{A}^*\big)^{-1}\text{vec}(\mathbf{Q}^*), \tag{168}$$

$$f_t^{(s)} = \mathbf{A}^* f_{t-1}^{(s)} + u_t^{(s)}, \qquad t = 1, \dots, T^*. \tag{169}$$

The choice to initialise at the stationary distribution rather than at the origin is deliberate: it avoids the transient effect that would arise if $f_0 = \mathbf{0}$, in which the first several periods of the simulated path would be artificially close to zero while the dynamics propagate the recent shocks. By drawing f_0 directly from the stationary distribution, we ensure that the entire simulated path is statistically homogeneous across time.

Step (i.c) — Draw the observations conditional on factors and weights. With the factor path $\{f_t^{(s)}\}$ constructed and the idiosyncratic weights $\{w_t^{\varepsilon,(s)}\}$ already drawn in Step (i.a), the idiosyncratic shocks $\varepsilon_t^{(s)}$ are conditionally Gaussian with time-varying diagonal covariance:

$$\varepsilon_t^{(s)} \,\big|\, w_t^{\varepsilon,(s)} \;\sim\; \mathcal{N}\Big(\mathbf{0},\, \mathbf{R}^*/w_t^{\varepsilon,(s)}\Big),$$

and the synthetic observations are assembled from the observation equation:

$$y_t^{(s)} \;=\; \mathbf{\Lambda}^* f_t^{(s)} \;+\; \varepsilon_t^{(s)}, \qquad t = 1, \dots, T^*.$$

The product $\mathbf{\Lambda}^* f_t^{(s)}$ is the common-factor component of $y_t^{(s)}$, capturing the cross-sectional comovement driven by the latent factors; $\varepsilon_t^{(s)}$ is the series-specific idiosyncratic component, with its own heavy-tailed weight process operating independently from the factor weights.

What is stored for ground truth. At the end of the three steps above, the simulation produces — for each replication s — two distinct categories of objects:

- *Observable panel*: the synthetic observations $\{y_t^{(s)}\}_{t=1}^{T^*}$. This is what the EM algorithm in Step 3 of the Monte Carlo design has access to. The ragged-edge missingness pattern and the mixed-frequency calendar are then imposed on this panel as described in the subsections that follow, before the re-estimation step.

- *Latent ground truth*: the factor path $\{f_t^{(s)}\}$, the weight paths $\{w_t^{u,(s)}, w_t^{\varepsilon,(s)}\}$, and the idiosyncratic shocks $\{\varepsilon_t^{(s)}\}$. These are the objects that the EM algorithm *would* need to know but does not — and that the Monte Carlo uses to assess how well the EM-recovered smoothed counterparts $\hat{f}_t^{(s)}, \hat{w}_t^{u,(s)}, \hat{w}_t^{\varepsilon,(s)}$ approximate them.

The strict separation between observable panel and latent ground truth is what makes the Monte Carlo a meaningful validation exercise: the algorithm operates on what it would see in the real world, while the validation has access to what really drove the data.

(ii) Ragged-Edge Missingness

The empirical feature to reproduce. Real macroeconomic data are released with publication delays that vary substantially across series: GDP is published roughly one month after the end of the reference quarter, employment data are released with a one-month delay, industrial production with about six weeks, while financial conditions indices and stock-market indicators are available essentially in real time. The practical consequence is that, at any given sample date T, the most recent months of the panel exhibit a characteristic *ragged* pattern: the last column of the data matrix is sparse, populated only by the few series with no delay; the second-to-last column is denser; and so on, until past months in which all series have eventually been released.

This pattern is not a curiosity — it is the defining empirical challenge of *nowcasting*, the activity of producing real-time estimates of variables (such as GDP) that have not yet been officially released. The mixed-frequency framework, the Kalman filter and smoother with selection matrices $\mathbf{W}_t$, and the news framework developed earlier in this thesis are specifically designed to handle this ragged structure. The Monte Carlo must therefore confront the same difficulty: a synthetic panel that is fully balanced at all t would not test the nowcasting machinery at all.

Why the DGP alone does not generate ragged-edge. A clarification is in order. The Student-t DFM described in Step (i) generates, for each $t = 1, \ldots, T^*$, a complete cross-section of M observations $y_t^{(s)}$. There is no mechanism within the model itself that would make any entry of $y_t^{(s)}$ missing: missingness in the real data is a property of the *release calendar*, not of the data-generating process. We therefore impose ragged-edge missingness on the synthetic data *ex post*, after the balanced panel has been generated, by applying the same selection matrices that the EM uses on the real data.

Imposing the real-data pattern. During the calibration of Step 1, we record the selection matrix $\mathbf{W}_t^{\text{real}}$ for every $t = 1, \ldots, T_{\text{real}}$ in the real panel. Each $\mathbf{W}_t^{\text{real}}$ is a diagonal $M \times M$ matrix whose (i, i) entry equals 1 if series i has been observed at time t in the real data, and 0 otherwise. The full sequence $\{\mathbf{W}_t^{\text{real}}\}_{t=1}^{T_{\text{real}}}$ encodes the entire publication-delay structure of the empirical sample.

For each synthetic sample $\mathbf{Y}^{(s)}$ of length $T^* = T_{\text{real}}$, we apply this same selection matrix sequence to construct the version of the synthetic data that the EM algorithm will actually see:

$$\mathbf{Y}_t^{(s),\text{ragged}} \;=\; \mathbf{W}_t^{\text{real}}\, y_t^{(s)}, \qquad t = 1, \ldots, T_{\text{real}}.$$

The operation is deterministic: it sets to zero (or, equivalently, to "missing") exactly those entries of $y_t^{(s)}$ that would have been missing in the corresponding period of the real data. Two properties of this construction are worth emphasising:

- The fully observed synthetic vector $y_t^{(s)}$ remains *stored as ground truth* alongside the ragged-edge version. The masked entries are not lost — they are simply hidden from the EM. This allows us to evaluate, ex post, how well the algorithm imputes the missing entries using only the observed ones.

- The EM in Step 3 sees only $\mathbf{Y}^{(s),\text{ragged}}$, never $y_t^{(s)}$. Its task is exactly the empirical task: recover the underlying parameters and the latent factor path from a panel whose right edge is sparse, using the selection matrices to handle the missing entries.

The resulting synthetic estimation problem is therefore informationally equivalent to the empirical one, by construction.

Synthetic samples of different length. A subtle issue arises when the Monte Carlo varies the sample size $T^* \neq T_{\text{real}}$, as required by the finite-sample analysis described earlier. The recorded missingness pattern $\{\mathbf{W}_t^{\text{real}}\}$ has length exactly T_{real}, so it cannot be applied verbatim to a synthetic sample of different length.

We adopt a simple convention: the ragged-edge structure is concentrated at the right edge of the panel, and its depth in real data is typically only a few months (the so-called *ragged region*, of length $T_{\text{real,ragged}}$, which is generally between 3 and 6 months depending on the publication delays of the series included in the panel). For a synthetic sample of length T^*, we apply the real ragged-edge pattern to the last $T_{\text{real,ragged}}$ periods of the sample (mapping the relative position of each release to its real-data counterpart), and treat the remaining $T^* - T_{\text{real,ragged}}$ periods as a fully observed balanced panel.

This convention preserves the qualitative difficulty of ragged-edge estimation — the EM still faces a sparse right edge in each replication — while accommodating synthetic samples of arbitrary length. It is consistent with the assumption that the publication-delay structure is a recent-period phenomenon, not a sample-wide one: in historical data, sufficiently old periods are fully observed.

(iii) Mixed-Frequency Observations

The empirical feature to reproduce. A fraction of the series in the panel — and in particular the target variable GDP — are observed at *quarterly* rather than monthly frequency. This is not a feature of the data-generating process: at the underlying level, factors and shocks evolve at monthly frequency, and the macroeconomic activity they describe is itself a continuous flow. What is quarterly is the *measurement*: the statistical agency aggregates three months of activity into a single quarterly observation, releases the result roughly one month after the end of the quarter, and that single number is all that the analyst sees.

Throughout this thesis we have handled this mismatch through the Mariano–Murasawa augmented state-space representation, in which each quarterly series is treated as a monthly series observed only at quarter-end months, loaded on the composite regressor

$$\phi_t^k \;=\; \tfrac{1}{3}f_t \;+\; \tfrac{2}{3}f_{t-1} \;+\; f_{t-2} \;+\; \tfrac{2}{3}f_{t-3} \;+\; \tfrac{1}{3}f_{t-4}.$$

The five lagged factors and the specific MM weights reflect the log-difference structure of the quarter-on-quarter growth rate of the underlying monthly process. The Monte Carlo must therefore generate synthetic data that confront the EM with this same mismatch: a panel in which some series are observed at every month and others only at quarter-end months, with the quarterly entries representing the MM-aggregated monthly path.

Why the DGP generates everything at monthly frequency. A clarification analogous to the one made for ragged-edge missingness applies here. The Student-t DFM specified in Step (i) generates a complete monthly path $f_t^{(s)}$ for the latent factors, and from this path a complete monthly observation $y_{i,t}^{(s)} = \mathbf{\Lambda}_{i\cdot}^{*} f_t^{(s)} + \varepsilon_{i,t}^{(s)}$ is produced for *every* series in the panel, regardless of whether that series is observed monthly or quarterly in the real data. There is no separate "quarterly" DGP: the model assumes that the underlying economic activity of every series exists at monthly frequency, and that quarterly observability is purely a measurement convention imposed by the release calendar.

The synthetic monthly path of a quarterly series is therefore the *latent* monthly counterpart of the quarterly observations that the real data would release. As with ragged-edge missingness, the calendar-induced unobservability is imposed on this full monthly path in a separate, deterministic step.

Imposing the quarterly aggregation. Let j denote a quarterly series — i.e. a series that in the real data is only observed at quarter-end months. Step (i) has generated a full *latent monthly path* for this series:

$$y_{j,t}^{(s)} \;=\; \mathbf{\Lambda}_{j\cdot}^{Q,*}\,\tilde{f}_t^{(s)} \;+\; \varepsilon_{j,t}^{(s)}, \qquad t = 1, \ldots, T^*,$$

where $\mathbf{\Lambda}_{j\cdot}^{Q,*}$ is the Mariano–Murasawa row of the loading matrix for series j — with the five weighted entries $[\tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^{Q},\, \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^{Q},\, \mathbf{\Lambda}_{j\cdot}^{Q},\, \tfrac{2}{3}\mathbf{\Lambda}_{j\cdot}^{Q},\, \tfrac{1}{3}\mathbf{\Lambda}_{j\cdot}^{Q}]$ Note that $y_{j,t}^{(s)}$ is a *monthly* value, even though the series j is intrinsically quarterly: it is the value that the series would have if it were observed monthly.

The quarterly aggregation that the real data imposes is then applied by masking this latent monthly path at all months other than quarter-end:

$$\tilde{y}_{j,t}^{(s)} = \begin{cases} y_{j,t}^{(s)} & \text{if } t \text{ is a quarter-end month,} \\ \text{missing} & \text{otherwise.} \end{cases}$$

The EM in Step 3 sees only the masked panel $\tilde{y}_{j,t}^{(s)}$: one observation every three months at quarter-end calendar dates, exactly as in the real data. The latent monthly values at non-quarter-end months are hidden from the algorithm but stored alongside the masked panel as ground truth.

What this enables. This design provides two distinct ground-truth objects against which the EM in Step 3 can be evaluated.

First, the algorithm produces smoothed estimates of the latent monthly values $\hat{y}_{j,t}^{(s)}$ at the non-quarter-end months through the Kalman smoother applied to the augmented state-space. These estimates are the model's reconstruction of the latent monthly path of the quarterly series j — that is, the very task that the Mariano–Murasawa framework was designed to perform. The Monte Carlo allows us to compare them directly against the true generated values $y_{j,t}^{(s)}$, quantifying how well the algorithm interpolates between consecutive quarter-end observations.

Second, the algorithm produces estimates of the quarterly loading matrix $\hat{\mathbf{\Lambda}}^Q$, including the specific Mariano–Murasawa pattern of five weighted entries $[\frac{1}{3}\mathbf{\Lambda}^Q, \frac{2}{3}\mathbf{\Lambda}^Q, \mathbf{\Lambda}^Q, \frac{2}{3}\mathbf{\Lambda}^Q, \frac{1}{3}\mathbf{\Lambda}^Q]$. These estimates can be compared against the true $\mathbf{\Lambda}^{Q,*}$ that generated the synthetic data, testing whether the EM correctly identifies the magnitude of the loading and the specific MM structure imposed by the augmented representation.

Together, these two validation outputs test the mixed-frequency machinery of the model from two complementary angles: the recovery of the latent monthly path (a smoother quality test) and the recovery of the parameter that maps factors to quarterly series (a maximum-likelihood quality test).

(iv) Block-Diagonal Economic Identification

The empirical feature to reproduce. The DFM developed in this thesis imposes a *block-diagonal* structure on the loading matrix: real-activity series load only on the real factor, financial series only on the financial factor, and the residual "other" series load only on the third block. This is an *economic identification restriction* that distinguishes our setting from the unrestricted DFM of Doz, Giannone, and Reichlin (2012), where the factors are extracted purely statistically and have no inherent economic interpretation.

The motivation for the restriction is interpretability: by forcing each factor to capture variation in a specific economic domain, we obtain factors that can be unambiguously labelled as "real activity", "financial conditions", and "other". This labelling is essential for the Growth-at-Risk analysis of the second stage, where the financial-conditions factor plays a distinct role in shaping the lower tail of the GDP distribution. A factor model that mixes real and financial influences across the same factor would not support this kind of structural interpretation.

The Monte Carlo must therefore preserve the block-diagonal structure throughout: in the data generation, in the re-estimation, and in the comparison between estimates and ground truth.

Why the structure must be imposed everywhere. Block-diagonality is not a feature of the data that emerges spontaneously from the DGP — it is a *constraint* on the parameter space. Without it, the EM has no reason to allocate each series to a specific factor: the rotational indeterminacy discussed earlier in the thesis means that an unrestricted DFM can always rotate the factors to redistribute the loading mass across them, yielding statistically equivalent but economically meaningless representations. Imposing the block structure removes this freedom and pins down a specific economic identification.

For the Monte Carlo to evaluate the proposed estimator meaningfully, the same restriction must be applied at every step:

- In the *calibration* of Step 1, the EM applied to $\mathbf{Y}^{\text{real}}$ imposes the block-diagonal structure during estimation, producing $\boldsymbol{\Lambda}^*$ with the appropriate zero entries off the diagonal blocks.
- In the *data generation* of Step 2, the same $\boldsymbol{\Lambda}^*$ is used verbatim: there are no off-block loadings, by construction. The synthetic factor paths $\{f^{R,(s)}, f^{F,(s)}, f^{X,(s)}\}$ each drive only their respective block of series.
- In the *re-estimation* of Step 3, the EM applied to each synthetic sample imposes the same block-diagonal constraint on its loading estimates $\hat{\boldsymbol{\Lambda}}^{(s)}$. The algorithm therefore operates within the same identified model that we want to validate.

Without this consistency, the Monte Carlo would be testing two different models — and any disagreement between estimates and truth could equally well be attributed to the mismatch in identifying restrictions rather than to genuine estimation error.

What this lets us test. With the block-diagonal structure preserved across all three steps, the Monte Carlo enables two distinct validation tests specific to this identification.

The first test concerns the *within-block loading magnitudes*. For each block $b \in \{R, F, X\}$, the EM produces estimates $\hat{\boldsymbol{\Lambda}}^{b,(s)}$ of the rows of the loading matrix corresponding to series in that block. These can be compared against the true $\boldsymbol{\Lambda}^{b,*}$ via the bias, variance, and RMSE metrics introduced later. A successful recovery indicates that the EM correctly identifies how strongly each series in a given block loads on its block-specific factor.

The second test concerns the *recovery of the block-specific factors themselves*. The smoothed factor estimates $\hat{f}^{R,(s)}, \hat{f}^{F,(s)}, \hat{f}^{X,(s)}$ produced by the EM can be compared against the true factor paths $f^{R,(s)}, f^{F,(s)}, f^{X,(s)}$ via per-block correlations and trajectory errors. The key question this test answers is whether each estimated block factor tracks the corresponding true factor specifically — for example, whether $\hat{f}^R$ correlates with $f^{R,*}$ but *not* with $f^{F,*}$. A high within-block correlation combined with a near-zero cross-block correlation $\text{Corr}(\hat{f}^R, f^{F,*})$ provides strong evidence that the block-diagonal identification is empirically respected by the algorithm.

These two tests together address whether the economic identification of the model is statistically robust: the loadings are correctly assigned within blocks, and the factors are correctly assigned across blocks.

(v) Contamination by Extreme Observations

The empirical feature to reproduce. The Student-t framework developed in this thesis is motivated by a specific empirical observation about macroeconomic data: *most of the time, the data behave well* — series move together, shocks are moderate in magnitude, and the standard Gaussian DFM provides a reasonable approximation. But a small minority of periods are governed by entirely different mechanisms: crisis episodes (the global financial crisis of 2008–2009), abrupt policy interventions, pandemics (the Covid-19 shock of 2020), or geopolitical events that produce extreme realisations the Gaussian model would treat as overwhelmingly unlikely.

These extreme periods are not just statistical curiosities: they are precisely the periods where a Growth-at-Risk analysis is most needed — when the lower tail of the GDP distribution is at its most informative, and when a forecasting model must remain reliable. A Gaussian DFM, applied to such data, absorbs the extreme realisations into its covariance estimates, inflating them and distorting the factor recovery. The Student-*t* extension is designed precisely to recognise these periods as outliers, down-weight their contribution, and preserve the integrity of the estimates from the bulk of the sample.

What this experiment is designed to quantify. Experiments A and B describe two limiting cases: a fully Student-*t* DGP (Experiment A, where the heavy-tailed extension should outperform the Gaussian benchmark) and a fully Gaussian DGP (Experiment B, where the two estimators should agree by the nesting property). Real macroeconomic data lie between these two extremes: predominantly Gaussian, with sporadic outlier-prone periods. Experiment C — the contamination experiment described in this subsubsection — fills this intermediate region by varying the fraction π of contaminated periods across a grid, tracing a continuous path from the Gaussian endpoint to substantially contaminated regimes.

The experiment provides a quantitative measure of the *robustness margin* of the Student-*t* extension: how much worse the Gaussian benchmark performs as π grows, and how stably the Student-*t* estimator maintains its accuracy. This is the most economically meaningful of the three experiments, because it most closely reproduces the empirical reality of the data we actually want to analyse.

The contamination mechanism. Let $\pi \in \{0, 0.01, 0.05, 0.10\}$ denote the *contamination intensity* — the fraction of time periods governed by the outlier mechanism. The three positive values correspond roughly to one in a hundred, one in twenty, and one in ten time periods being affected; the value $\pi = 0$ is included deliberately as an anchor, since at $\pi = 0$ the data-generating process is purely Gaussian and the experiment reduces *exactly* to Experiment B (this provides an internal consistency check: the two estimators must coincide there by the nesting property).

For each t in the synthetic sample, we draw an independent Bernoulli indicator

$$z_t \sim \text{Bernoulli}(\pi),$$

which determines whether period t is contaminated ($z_t = 1$) or not ($z_t = 0$). The indicator is indexed by *time*, not by the series–time pair: when a period is flagged as contaminated, the outlier mechanism applies to the *entire* idiosyncratic vector at that period, affecting all M series simultaneously. This reflects the empirical observation that extreme periods — a crisis month, a policy shock — are system-wide events felt across the whole cross-section, not isolated glitches in a single series. Conditional on $z_t = 1$, the idiosyncratic shock at t is *replaced* (not augmented) by a draw from a heavy-tailed Student-*t* distribution with low degrees of freedom and inflated scale:

$$\varepsilon_t^{(s)} \,\big|\, z_t = 1 \sim t_{\nu_{\text{contam}}}(\mathbf{0}, \kappa^2 \mathbf{R}^*),$$

with $\nu_{\text{contam}} = 3$ — low enough to produce visible, genuinely heavy-tailed outlier realisations — and $\kappa = 5$, so that the conditional idiosyncratic variance is $\kappa^2 = 25$ times the Gaussian baseline (roughly an order of magnitude larger, making the contamination clearly distinguishable from ordinary fluctuations). The substitutive mechanism — the contaminated shock replaces the Gaussian one rather than adding to it — ensures that $\pi = 0$ recovers the Gaussian baseline exactly, period by period. For t with $z_t = 0$, the standard Gaussian shock applies. In implementation, the Student-t draw is generated through the same scale-mixture representation used throughout this thesis: a single mixing weight $w_t^{\text{contam}} \sim \text{Gamma}(\nu_{\text{contam}}/2, \nu_{\text{contam}}/2)$, shared across the M series of the contaminated period, scales the inflated covariance $\kappa^2 \mathbf{R}^*$.

The factor innovations $u_t^{(s)}$ are not contaminated in this experiment: only the idiosyncratic component is affected. This is a deliberate choice that mirrors the empirical regularity that extreme periods typically manifest as deviations *from* the underlying factor structure rather than as movements of the factor itself — a stock-market crash, for instance, is felt as a spike in the residual of financial series rather than as a shift in the latent financial factor. (The case in which the factor is also contaminated is a natural extension that we do not pursue here.)

What this experiment isolates. The contamination mechanism produces a data-generating process that is *predominantly Gaussian* but with periodic excursions into heavy-tailed territory. Three concrete predictions follow from the comparative structure of the two estimators:

- As π grows, the Gaussian EM should exhibit *increasing bias and variance* in $\hat{\mathbf{\Lambda}}, \hat{\mathbf{Q}}, \hat{\mathbf{R}}$. The mechanism is direct: the Gaussian model has no way to distinguish contaminated from clean periods, so the extreme observations enter the covariance estimates as if they were typical realisations, inflating them and distorting the factor recovery.

- The Student-t EM should remain *comparatively stable* as π grows. Its estimated $\hat{\nu}_\varepsilon$ adjusts to reflect the empirical tail thickness, decreasing as π increases, and the posterior weights $\hat{w}_t^\varepsilon$ concentrate the down-weighting on the periods that were in fact contaminated.

- The Student-t EM should *identify* the truly contaminated periods through its posterior weights $\hat{w}_t^\varepsilon$: the contaminated periods, having generated extreme residuals, should receive the smallest weights. The effectiveness of this identification is quantified by a detection metric defined precisely below.

At $\pi = 0$ the experiment reduces exactly to Experiment B (Gaussian DGP, both estimators should agree). As π grows the DGP approaches a fully heavy-tailed regime, and the comparison approaches Experiment A. Experiment C therefore traces a continuous interpolation between the two endpoints, with the fraction of contamination π as the natural parameter.

(vi) Vintages and the Real-Time Data Flow

The empirical feature to reproduce. The news framework developed earlier in this thesis does not operate on a single static panel: it operates on a *sequence* of panels, each corresponding to a different calendar date and reflecting the cumulative information available up to that date. Each such snapshot is a *vintage*. As new data releases occur, the vintage expands: industrial production for January is released around mid-February; payrolls for January is released on the first Friday of February; the advance estimate of GDP for the quarter ending January is released about one month after the quarter ends. Each release moves one or more entries of the data matrix from "not yet available" to "available", producing a new vintage that is strictly richer than its predecessor.

This sequential arrival of information is the empirical reality of real-time nowcasting, and the news framework was specifically constructed to quantify how each new release updates the forecast of the target variable. For the Monte Carlo to validate the news framework, the synthetic data must reproduce the same vintage structure: a release calendar that determines when each (i, t) observation becomes available, and a corresponding sequence of expanding panels.

Why the DGP alone does not generate vintages. The data-generating process described in Step (i) produces a full synthetic panel $\{y_t^{(s)}\}_{t=1}^{T^*}$ in which all entries are simultaneously available — there is no notion of "release date" in the DGP. As with ragged-edge missingness and quarterly masking, the vintage structure is not a feature of the underlying model but a feature of the empirical *release calendar*, which the Monte Carlo must impose ex post on the generated data.

Recording and applying the release calendar. The real data come with a documented release calendar: for each series i and each reference period t, we know the calendar date $v_{i,t}$ at which the corresponding observation became available in real time. We record this calendar during the calibration of Step 1 and apply it verbatim to each synthetic sample $\mathbf{Y}^{(s)}$.

The calendar defines, implicitly, a partition of time into a sequence of distinct *release dates*: between two consecutive release dates the information set is unchanged, and at each release date one or more new observations are added. Let $v_1 < v_2 < \cdots < v_V$ denote the ordered sequence of distinct release dates encountered in the synthetic sample. For each vintage v_k, we construct the data panel that would have been observable at that date:

$$\mathbf{Y}^{(s),v_k} \;=\; \big\{\, y_{i,t}^{(s)} \;:\; v_{i,t} \leq v_k \,\big\},$$

with all not-yet-released entries treated as missing. By construction, the sequence

$$\mathbf{Y}^{(s),v_1} \;\subset\; \mathbf{Y}^{(s),v_2} \;\subset\; \cdots \;\subset\; \mathbf{Y}^{(s),v_V}$$

forms a filtration that mirrors the real-time information flow: each vintage extends its predecessor by exactly the observations released between the two corresponding calendar dates.

Sample-size invariance of the release structure. A subtle point arises when the Monte Carlo varies the sample size $T^* \neq T_{\text{real}}$. The release calendar of the real data is encoded not by specific calendar dates, but by publication *lags* that are intrinsic to each series: GDP is released approximately one month after the end of the reference quarter, industrial production with a six-week delay, payrolls with a one-week delay, financial conditions essentially in real time. Let d_i denote the publication lag of series i (in months); then the release date of the observation $y_{i,t}^{(s)}$ in the synthetic sample is

$$v_{i,t} \;=\; t \;+\; d_i.$$

This formulation depends only on t and on the series-specific lag d_i, not on the total length of the sample. The vintage sequence and the filtration construction described above therefore apply identically to samples of any length T^*: the structure of the release calendar is sample-size invariant. Only the lags $\{d_i\}$ must be recorded during the calibration of Step 1, not the specific calendar dates.

What this enables for the news framework. With the vintage sequence available for each synthetic replication, the Monte Carlo can directly test the news framework developed in the previous section. For each pair of consecutive vintages (v_k, v_{k+1}) and each replication s, two distinct objects can be computed and compared:

- The *theoretical news weights* $\mathbf{B}_{k,t_k}^{(v_{k+1})}$, obtained from the analytical expressions derived earlier, evaluated at the estimated parameters and smoothed covariances of the model at vintage v_k;
- The *realised forecast revisions*, obtained by re-running the EM on the two vintage panels $\mathbf{Y}^{(s),v_k}$ and $\mathbf{Y}^{(s),v_{k+1}}$ and recording the change in the implied target forecast.

The product of the theoretical weight matrix with the realised news vector, $\mathbf{B}_{k,t_k}^{(v_{k+1})}\,\mathcal{I}_{v_{k+1}}$, should equal the realised revision up to numerical precision. Agreement between the two constitutes evidence that the news framework, as implemented within the Student-t DFM, is internally consistent and quantitatively accurate.

The Monte Carlo can also assess two more nuanced properties of the framework. First, the distribution of news weights across vintages reveals how the model attributes forecast revisions to specific releases — a diagnostic that can be compared against prior economic intuition. Second, the empirical role of the Student-t outlier-aware mechanism (the modulation of the news weights by the estimated weights $\hat{w}_t^{\varepsilon}$ in high-stress periods) can be quantified by comparing the weights produced by the Student-t EM against those produced by the Gaussian benchmark applied to the same synthetic panels.

A note on scope. The vintage-based validation described here is a natural extension of the parameter-recovery Monte Carlo developed in the rest of this section, and one that we plan to develop fully in future work. In the present thesis, we focus on the parameter recovery experiments (Experiments A, B, and C as described above), and we report the vintage-based validation as a planned extension rather than as a fully implemented result. The infrastructure for generating vintage data — the release calendar, the sequence of expanding panels, the framework for comparing theoretical and realised revisions — is already built into the simulation pipeline, so that the vintage-based validation can be carried out without additional infrastructure development. The systematic empirical analysis of news weights and their associated forecast revisions is the natural complement to the empirical application of the news framework on real data, and the two will be developed together in subsequent work.

Summary of the Simulation Pipeline

The six elements together. The six elements described in this subsection — Student-t scale-mixture innovations, ragged-edge missingness, mixed-frequency aggregation, block-diagonal economic identification, outlier contamination, and the vintage structure of real-time releases — together produce synthetic samples whose statistical features mirror those of the real macroeconomic panel. Each element targets a specific empirical challenge that the proposed estimator must confront: heavy-tailed innovations, ragged data, quarterly target variables, structured factor loadings, periodic outlier regimes, and a sequentially expanding information set. Together, they turn the calibrated Monte Carlo from a stylised exercise into a genuine reproduction of the empirical estimation problem.

The order of operations. The order in which these elements are applied to construct each synthetic sample $\mathbf{Y}^{(s)}$ deserves to be made explicit, because it reflects the conceptual structure of the data-generating mechanism. For each replication s, the operations proceed in the following sequence:

1. *Draw the latent weight paths* $\{w_t^{u,(s)}, w_t^{\varepsilon,(s)}\}_{t=1}^{T^*}$ from independent Gamma distributions with parameters derived from ν_u^* and ν_ε^*.

2. *Draw the latent factor path* $\{f_t^{(s)}\}_{t=1}^{T^*}$ recursively, conditional on the factor weights, starting from the stationary distribution of the VAR(1).

3. *Draw the latent idiosyncratic shocks* $\{\varepsilon_t^{(s)}\}_{t=1}^{T^*}$ conditional on the idiosyncratic weights, with the block-diagonal covariance structure preserved.

4. *Inject contamination (Experiment C only)*: for each t, draw a Bernoulli indicator z_t and, if $z_t = 1$, replace the idiosyncratic shock with a draw from the heavy-tailed contamination distribution.

5. *Compose the balanced observations* $y_t^{(s)} = \mathbf{\Lambda}^* f_t^{(s)} + \varepsilon_t^{(s)}$ for all $t = 1, \ldots, T^*$, producing a complete monthly panel.

6. *Apply the empirical observability structure*: mask the entries of the balanced panel according to the ragged-edge selection matrices $\{\mathbf{W}_t^{\text{real}}\}$ and to the quarter-end mask for the quarterly series.

7. *Construct the vintage sequence*: organise the masked observations into the filtration $\mathbf{Y}^{(s),v_1} \subset \mathbf{Y}^{(s),v_2} \subset \cdots$ implied by the publication lags $\{d_i\}$ of the individual series.

Latent variables vs. observed panel. The order of operations reflects a clear conceptual separation between the *data-generating mechanism* and the *empirical observability structure*. Steps 1–5 produce the latent and observable quantities that the Student-t DFM would generate in a fully transparent world, where every entry of the monthly panel is visible at every t. Steps 6–7 then impose the empirical reality on this idealised panel: missingness from publication delays, quarterly aggregation, and the sequential arrival of releases. In particular:

- The latent weights, factors, and idiosyncratic shocks exist for *every* $t = 1, \ldots, T^*$ — they are the random variables of the DGP and are not affected by the calendar of releases.

- The fully observed monthly panel $\{y_t^{(s)}\}$ exists for every t in the synthetic mechanism, but only the masked version $\mathbf{Y}^{(s)}$ is seen by the EM in Step 3 of the Monte Carlo design.

- All the entries hidden by Steps 6–7 are *stored as ground truth* alongside the masked panel: the EM does not see them, but the Monte Carlo uses them to evaluate the algorithm's ability to recover the latent monthly path of quarterly series and to forecast the not-yet-released entries at the ragged edge.

This asymmetric design — generate everything, then hide some of it — is what makes the controlled experiment meaningful. The data-generating mechanism is fully known by construction, so the Monte Carlo has access to the ground truth of every latent and masked quantity, while the algorithm in Step 3 confronts exactly the same difficulty (missing data, mixed frequency, sequential releases) that it would face on the real panel.

Putting it all together. The full simulation pipeline therefore combines the six elements in a coherent generation procedure, run independently for each replication s and for each combination of sample size T^* and contamination level π. The pipeline is formalised in the algorithmic description that follows.

Handling Rotational Indeterminacy

The DFM is identified only up to a rotation of the latent factors, as discussed in detail in earlier sections of this thesis: for any invertible matrix $\mathbf{H} \in \mathbb{R}^{r\times r}$, the pair $(\boldsymbol{\Lambda}^*, f_t^*)$ and the pair $(\boldsymbol{\Lambda}^*\mathbf{H}^{-1}, \mathbf{H}f_t^*)$ produce statistically indistinguishable observed data. The block-diagonal economic identification imposed in our framework restricts $\mathbf{H}$ to be block-diagonal itself, but does not eliminate it altogether: within each block, the factors can still be rescaled and sign-flipped freely. For the Monte Carlo to produce meaningful comparisons across replications — and between estimates and truth — this residual ambiguity must be resolved.

The convention adopted. The Monte Carlo applies the same post-hoc normalisation that the empirical analysis adopts elsewhere in this thesis. After each EM run produces its estimates $\hat{\boldsymbol{\theta}}^{(s)}$, two distinct normalising transformations are applied to the output, in sequence.

Sign normalisation. For each block $b \in \{R, F, X\}$, a *reference series* is chosen — a series that uncontroversially represents the economic content of the block (e.g. industrial production for the real block, a credit-spread index for the financial block, an aggregate sentiment measure for the third block). If the loading on this reference series is negative at convergence, the entire block factor is multiplied by -1, and the corresponding row and column of $\hat{\mathbf{A}}^{(s)}, \hat{\mathbf{Q}}^{(s)}, \hat{\boldsymbol{\Lambda}}^{(s)}$ and the smoothed covariances are sign-flipped accordingly. This guarantees that the orientation of each block factor is consistent across replications.

Scale normalisation (Convention 1). Each block factor $\hat{f}^{b,(s)}$ is rescaled so that its total (marginal) variance over the sample equals one:

$$\frac{1}{T^*}\sum_{t=1}^{T^*}\big(\hat{f}^{b,(s)}_{t|T^*} - \bar{f}^{b,(s)}\big)^2 \;+\; \frac{1}{T^*}\sum_{t=1}^{T^*}\big(P^{(s)}_{t|T^*}\big)_{bb} \;=\; 1, \qquad b \in \{R, F, X\}.$$

The corresponding rows of $\hat{\boldsymbol{\Lambda}}^{(s)}$ are rescaled by the inverse standard deviation, and the block-diagonal entries of $\hat{\mathbf{Q}}^{(s)}$ adjust implicitly. This is the same convention adopted in the empirical analysis: it makes the factors dimensionless and comparable across blocks and across replications.

The same two transformations are applied to $\boldsymbol{\theta}^*$ when it is constructed from the real-data calibration in Step 1. The comparison between $\hat{\boldsymbol{\theta}}^{(s)}$ and $\boldsymbol{\theta}^*$ is therefore made on a common scale and sign convention, which is what makes the metrics introduced below interpretable across replications.

Procrustes alignment as a robustness check. Even after sign and scale normalisations, a residual rotational ambiguity may remain in finite samples. Two replications that are rotationally equivalent in the population may differ in finite samples by an unidentified within-block rotation, particularly if the reference series used for sign normalisation happen to have weak estimated loadings in some replications. For robustness, we recommend reporting — alongside the standard normalised-estimate metrics — a *Procrustes-aligned* version of the comparison.

For each replication s, the orthogonal rotation $\mathbf{H}^{(s)}$ that minimises the Frobenius distance between the estimated loading and the calibrated truth

$$\mathbf{H}^{(s)} \;=\; \underset{\mathbf{H}^\top\mathbf{H}=I}{\arg\min}\; \big\|\, \hat{\boldsymbol{\Lambda}}^{(s)}\mathbf{H} \;-\; \boldsymbol{\Lambda}^* \,\big\|_F$$

is computed via the singular value decomposition of $\hat{\boldsymbol{\Lambda}}^{(s)\top}\boldsymbol{\Lambda}^*$ (the classical Procrustes solution). The aligned estimate $\hat{\boldsymbol{\Lambda}}^{(s)}\mathbf{H}^{(s)}$ is then compared with $\boldsymbol{\Lambda}^*$ in the usual way, with the rotation applied also to the smoothed factor path and the relevant parameters of $\hat{\mathbf{A}}^{(s)}$ and $\hat{\mathbf{Q}}^{(s)}$.

This procedure isolates the genuine estimation error from any residual rotational ambiguity that the sign-and-scale normalisation may have failed to fully resolve. In practice, the two sets of metrics — the normalised-only and the Procrustes-aligned — should agree closely; large discrepancies between them would indicate that the rotational normalisation is insufficient and that the experiment should be reported with the Procrustes alignment as the primary measure.

Validation Metrics

What the Monte Carlo produces. For each combination of sample size T^*, contamination intensity π, and estimator EM $\in \{\mathrm{EM}_t, \mathrm{EM}_{\mathrm{Gauss}}\}$, the Monte Carlo described in the previous subsections produces an empirical distribution of estimates:

$$\left\{\hat{\boldsymbol{\theta}}^{(s)},\ \hat{f}_t^{(s)},\ \hat{w}_t^{u,(s)},\ \hat{w}_t^{\varepsilon,(s)}\right\}_{s=1}^{S},$$

together with the corresponding ground truth $\boldsymbol{\theta}^*$ and $\{f_t^{(s)}, w_t^{u,(s)}, w_t^{\varepsilon,(s)}\}_{s=1}^{S}$ stored at the generation step. From the joint comparison of estimates and ground truth, we extract the validation metrics that support the empirical conclusions of the simulation study.

The four families of metrics. The metrics fall into four conceptually distinct families, each addressing a different dimension of estimator quality:

1. *Parameter recovery*: how well does $\hat{\boldsymbol{\theta}}^{(s)}$ recover the calibrated truth $\boldsymbol{\theta}^*$?
2. *Factor recovery*: how well does the smoothed factor path $\hat{f}_t^{(s)}$ track the true factor path $f_t^{(s)}$?
3. *Outlier identification*: does the estimated weight sequence $\hat{w}_t^{(s)}$ correctly flag the periods that are truly outliers under the data-generating process — i.e. does the Student-t mechanism down-weight the right observations?
4. *Algorithmic reliability*: does the EM converge consistently, in a finite number of iterations, with a well-behaved likelihood path?
5. *Behaviour across sample sizes*: how do the metrics above change as T^* varies, and what does this tell us about the asymptotic properties of the estimator?

The first two families address *statistical accuracy* — the extent to which the algorithm recovers the latent and parametric content of the model. The third family addresses *algorithmic correctness* — the extent to which the implementation behaves as the theory predicts. The fourth family addresses *finite-sample behaviour* — the extent to which the estimator approximates its asymptotic properties at the sample sizes used in the empirical application.

Parameter Recovery: Bias, Variance, and RMSE

For each scalar parameter θ_k — a single entry of $\mathbf{\Lambda}^*$, $\mathbf{A}^*$, $\mathbf{Q}^*$, $\mathbf{R}^*$, or one of the heavy-tail parameters $\nu_u^*, \nu_\varepsilon^*$ — we compute three standard summary statistics across the S replications:

$$\mathrm{Bias}(\theta_k) \;=\; \frac{1}{S}\sum_{s=1}^{S}\hat{\theta}_k^{(s)} \;-\; \theta_k^*, \tag{170}$$

$$\mathrm{Var}(\theta_k) \;=\; \frac{1}{S}\sum_{s=1}^{S}\big(\hat{\theta}_k^{(s)} - \bar{\theta}_k\big)^2, \tag{171}$$

$$\mathrm{RMSE}(\theta_k) \;=\; \sqrt{\frac{1}{S}\sum_{s=1}^{S}\big(\hat{\theta}_k^{(s)} - \theta_k^*\big)^2}, \tag{172}$$

where $\bar{\theta}_k = (1/S)\sum_s \hat{\theta}_k^{(s)}$ is the Monte Carlo mean. The three statistics are linked by the standard decomposition

$$\mathrm{RMSE}(\theta_k)^2 \;=\; \mathrm{Bias}(\theta_k)^2 \;+\; \mathrm{Var}(\theta_k),$$

which makes the RMSE a comprehensive summary of estimator quality: a low RMSE requires both small bias (systematic accuracy) and small variance (precision). The bias and variance can therefore be read together to diagnose the dominant source of estimation error: a large bias relative to the variance indicates a systematic under- or over-estimation that more replications would not correct; a large variance relative to the bias indicates an unbiased estimator with high noise that more replications would smooth out.

Matrix-valued summaries. For matrix-valued parameters such as $\mathbf{\Lambda}^*$ or $\mathbf{A}^*$, the entry-wise metrics above are complemented by a global summary based on the Frobenius norm:

$$\mathrm{RMSE}_F(\mathbf{\Lambda}) \;=\; \sqrt{\frac{1}{S}\sum_{s=1}^{S}\|\,\hat{\mathbf{\Lambda}}^{(s)} - \mathbf{\Lambda}^*\,\|_F^2}.$$

The Frobenius RMSE provides a single scalar summary of how well the full matrix has been recovered, complementing the entry-wise diagnostics that allow one to identify which specific entries contribute most to the global error.

Special attention to the degrees-of-freedom parameters. The heavy-tail parameters ν_u, ν_ε deserve particular focus, because they are the parameters that distinguish the Student-t DFM from the Gaussian benchmark. Their recovery under the Monte Carlo provides a direct test of whether the EM correctly identifies the tail thickness from the data. Two qualitative properties are expected:

- Under a Student-t DGP (Experiment A) with $\nu^* \in [5, 30]$ — the empirically relevant range — the estimated $\hat{\nu}$ should concentrate around the truth, with bias that vanishes as T^* grows.
- Under a Gaussian DGP (Experiment B), the estimated $\hat{\nu}$ should drift toward the upper bracket of the Brent search interval, which is the algorithm's empirical signal that the tail thickness is indistinguishable from the Gaussian limit.

The second property is the empirical analogue of the *nesting relation* between the Student-t and Gaussian DFMs: in the limit $\nu \to \infty$ the Student-t collapses to the Gaussian, and the Monte Carlo should display this collapse directly.

Factor Recovery

In addition to the parameter estimates, the EM produces smoothed factor paths $\hat{f}_t^{(s)} = \mathbb{E}[f_t \mid \mathbf{Y}^{(s)}, \hat{\boldsymbol{\theta}}^{(s)}]$. Because the synthetic data are generated from a known true factor path $\{f_t^{(s)}\}$, the smoothed estimate and the truth can be compared directly. Three metrics summarise the quality of factor recovery.

Correlation. The Pearson correlation between the smoothed estimate and the true path,

$$\mathrm{Corr}\big(\hat{f}^{(s)}, f^{(s)}\big),$$

is computed separately for each block factor (real, financial, "other"). Under the post-hoc normalisation discussed in the previous subsection — sign flip plus factor-variance rescaling — both quantities are on a common scale and the correlation has its standard interpretation: a value close to one indicates that the smoother has identified the correct latent factor, and a value close to zero indicates that the estimated factor has no relation to the truth.

Trajectory error. The root mean-squared trajectory error,

$$\mathrm{RMSE}_{\mathrm{traj}}^{(s)} = \sqrt{\frac{1}{T^*} \sum_{t=1}^{T^*} \| \hat{f}_t^{(s)} - f_t^{(s)} \|^2},$$

measures the average distance between the two factor paths. Trajectory error is more sensitive than correlation to systematic distortions that preserve the timing of the factor but bias its amplitude — for example, a smoother that systematically under-shoots peaks would show high correlation but elevated trajectory error. The two metrics are therefore complementary: correlation captures the alignment of the time-series patterns, trajectory error captures the absolute fidelity of the reconstruction.

Per-block recovery and cross-block consistency. For the block-structured model, the recovery of each block factor is reported separately:

$$\mathrm{Corr}(\hat{f}^R, f^R), \quad \mathrm{Corr}(\hat{f}^F, f^F), \quad \mathrm{Corr}(\hat{f}^X, f^X).$$

A high correlation within each block confirms that the EM has not mixed the block factors. In parallel, the cross-block correlations

$$\mathrm{Corr}(\hat{f}^R, f^F), \quad \mathrm{Corr}(\hat{f}^R, f^X), \quad \mathrm{Corr}(\hat{f}^F, f^X),$$

should be near zero, providing a consistency check that the block-diagonal identification has been correctly recovered by the algorithm. A pattern in which some cross-block correlation is systematically far from zero would indicate that the identification has not been pinned down despite the imposed block-diagonal constraint.

Weight Recovery: Outlier Identification by Rank Overlap

The Student-t weights $\hat{w}_t^u$ and $\hat{w}_t^\varepsilon$ are the mechanism through which the model achieves robustness: outlying periods receive small weights and therefore contribute little to the parameter updates. A natural question for the recovery study is whether the estimated weights identify the *same* periods as outliers that the data-generating process actually made outlying. It is tempting to answer this with the linear correlation between the estimated and the true weight sequences, but that statistic is misleading here, and understanding why motivates a more appropriate diagnostic.

The difficulty is that the linear correlation is dominated by the *normal* periods, which form the overwhelming majority of the sample and whose weights cluster near one. Small, idiosyncratic discrepancies among these near-unit weights inject noise into the correlation, even though those periods are irrelevant for robustness: a small discrepancy between two near-unit weights changes nothing about the estimator's behaviour. What matters for robustness is whether the estimator correctly identifies the *tail* of the weight distribution — the genuinely outlying periods that, if not down-weighted, would distort the M-step. A moderate full-sample correlation is therefore fully compatible with excellent robustness, provided the outliers are correctly flagged.

We accordingly assess weight recovery by a *rank-overlap* diagnostic targeted at the outliers. Let $\mathcal{O}_k^{\text{true}}$ be the set of the k periods with the *smallest* true weights (the most strongly down-weighted under the DGP), and $\mathcal{O}_k^{\text{hat}}$ the analogous set under the estimated weights, with $k = \lceil \alpha\, T \rceil$ for a small fraction α (we report $\alpha \in \{5\%, 10\%\}$). The primary statistic is the *precision at k*,

$$P_k \;=\; \frac{|\, \mathcal{O}_k^{\text{true}} \cap \mathcal{O}_k^{\text{hat}} \,|}{k},$$

the fraction of the truly most-outlying periods that the estimator also places among its most-outlying (the quantity known as *precision@k* in the ranking literature). To make this scale-free we compare it against the overlap a random ranking would achieve, $\text{rand} = k/T$, and report the *lift* P_k/rand: a lift far above one means the estimator identifies real outliers far better than chance. As a finer check we also compute the Spearman rank correlation between $\hat{w}_t$ and w_t^{true} *restricted to the true outlier set*, which asks not merely whether the outlier set is recovered but whether the estimator also orders the outliers by severity.

This diagnostic targets exactly the property that underpins the robustness claim of the model. An estimator with a modest full-sample weight correlation but a high outlier lift is doing precisely what the Student-t specification is designed to do: it may be imprecise about the uninformative bulk of near-unit weights, while being sharp about the handful of periods whose down-weighting actually protects the parameter estimates. The idiosyncratic weights, estimated from many series per period, are expected to be recovered more sharply than the factor-side weights, which depend on only r factor-innovation dimensions; the diagnostic should reflect this asymmetry.

Algorithmic Reliability

A complementary set of metrics tracks the algorithm's behaviour rather than its statistical output. These metrics are diagnostics of the implementation itself — they detect numerical pathologies or convergence failures that would invalidate the statistical conclusions drawn from the other metric families.

Convergence rate. The fraction of replications in which the outer EM loop attains its stopping criterion within the maximum allowed number of iterations $J_{\max}$. A convergence rate substantially below one indicates that some replications failed to converge within budget; the corresponding estimates are flagged separately in the analysis, and the convergence rate itself becomes a metric of practical reliability.

Average iteration count. The mean number of outer EM iterations required to reach convergence, averaged over the converged replications. This is informative about the practical computational cost of the algorithm under the considered (T^*, π) configuration, and about how this cost scales as the parameter configuration varies.

Inner ECM convergence. The mean number of inner fixed-point iterations per outer step, required by the conditional-maximisation structure of the M-step. Pathological behaviour at this level — such as a sudden jump in inner-iteration counts on a subset of replications — indicates that the inner fixed-point is becoming difficult to solve, possibly due to a near-singular configuration of the parameters in the neighbourhood of the EM trajectory.

Likelihood-path monotonicity. The fraction of replications in which the marginal log-likelihood path is monotone non-decreasing across outer iterations. The EM algorithm is theoretically guaranteed to produce a monotone likelihood path under correct implementation; any violation is therefore a direct diagnostic of a coding error or a numerical instability. A monotonicity rate of one across replications is essentially a correctness guarantee for the implementation; values below one signal that the implementation requires inspection before the statistical metrics can be trusted.

Behaviour Across Sample Sizes

To approximate a finite-sample analogue of consistency, the entire Monte Carlo is repeated for several values of T^*:

$$T^* \in \{100, 200, 400, 800, T_{\text{real}}\}.$$

For each T^*, all of the metrics above are recomputed. Two qualitative properties are expected of a well-behaved estimator:

- *Variance and RMSE should decrease* as T^* grows, reflecting the increasing precision of the estimator with more information.
- *Bias should decrease or remain stable*; sharp increases in bias for small T^* would indicate finite-sample distortions that need to be documented and, if possible, characterised theoretically.

The convergence curve and its empirical interpretation. Plotting RMSE versus T^* on log–log axes typically reveals a near-linear trajectory, whose slope provides an empirical estimate of the finite-sample rate of convergence. A slope close to $-1/2$ is consistent with the standard parametric rate $T^{-1/2}$ that the literature has established for related estimators in the large-T, fixed-M regime. A slope below $-1/2$ in absolute value (i.e. a flatter curve) would signal that the estimator converges at a rate slower than the parametric benchmark, with quantifiable consequences for the precision attainable at finite sample sizes.

Why this matters for the proposed estimator. The Student-t EM developed in this thesis has not been studied asymptotically: no formal proof of consistency or asymptotic normality is available for the joint estimation of $(\mathbf{\Lambda}, \mathbf{A}, \mathbf{Q}, \mathbf{R}, \nu_u, \nu_\varepsilon)$ in the mixed-frequency, ragged-edge, block-diagonal setting. The Monte Carlo analysis across sample sizes is therefore the primary route through which the finite-sample properties of the proposed estimator are documented. The qualitative shapes of the bias and RMSE curves — particularly for the heavy-tail parameters ν_u, ν_ε specific to the extension — provide the empirical evidence that substantiates the claim that the proposed estimator is a viable inference tool for the kind of macroeconomic data used in the empirical application.

Three Monte Carlo Experiments

The validation strategy is articulated through three Monte Carlo experiments, each designed to assess a distinct property of the proposed estimator. Together they characterise the relative performance of the Student-t DFM developed in this thesis against the Gaussian benchmark of Bańbura and Modugno (2014), under three data-generating conditions: heavy-tailed data, Gaussian data, and intermediate cases with outlier contamination.

The three experiments share the same calibration step (Step 1) and the same simulation pipeline introduced earlier. They differ only in the data-generating process used to produce the synthetic samples and in the estimator applied to recover the parameters. The expected outcomes, derived from the structure of the two models, constitute a precise set of falsifiable predictions.

Experiment A: Student-t DGP, Both Estimators

Setup. Data are generated from the Student-t DGP described earlier, with parameters $\boldsymbol{\theta}^*$ calibrated on the real macroeconomic panel. Both estimators are applied to each synthetic sample: the proposed Student-t EM and the Gaussian EM of Bańbura and Modugno (2014). Each replication therefore produces two estimates, $\hat{\boldsymbol{\theta}}_t^{(s)}$ and $\hat{\boldsymbol{\theta}}_{\text{Gauss}}^{(s)}$, which are compared against the common ground truth $\boldsymbol{\theta}^*$.

Metrics compared. For each parameter common to both models — $\boldsymbol{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$, $\mathbf{R}$ — we report Bias, RMSE, and Frobenius RMSE under both estimators. Note that the interpretation of $\mathbf{Q}$ and $\mathbf{R}$ differs across the two models: in the Student-t DFM they are *scale* matrices of the conditional Gaussian innovations, while in the Gaussian DFM they are the unconditional *covariance* matrices. For comparison purposes, we report both estimators on the same scale by reinterpreting the Gaussian $\hat{\mathbf{Q}}$ and $\hat{\mathbf{R}}$ as scale matrices: the implied marginal covariance of the Student-t at $\nu^* < \infty$ is $\nu^*/(\nu^* - 2)$ times the scale matrix, and we apply the corresponding rescaling when necessary.

In addition, factor recovery is compared on a common scale via the post-hoc factor variance normalisation. The full likelihood at convergence is also reported, with the appropriate adjustment for the different parameter dimensionality.

Expected outcome. The proposed Student-t EM is expected to outperform the Gaussian benchmark by quantifiable margins. The mechanism is the following:

- The Student-t EM correctly identifies the heavy-tailed structure through the estimated weights $\hat{w}_t^u, \hat{w}_t^\varepsilon$, which down-weight observations from periods with extreme innovations;
- The Gaussian EM, lacking this mechanism, treats every observation as symmetrically informative for the covariance estimates, and therefore over-estimates $\hat{\mathbf{Q}}, \hat{\mathbf{R}}$ in the presence of the heavy-tailed shocks generated by the DGP;
- As a consequence, the Gaussian factor estimates are biased toward extreme realisations, while the Student-t factor estimates concentrate around the truth.

Quantitatively, the RMSE on $\mathbf{Q}$ and $\mathbf{R}$ is expected to be substantially smaller for the Student-t EM, with the gap widening as ν^* decreases (heavier tails imply larger improvements). The factor recovery correlations are expected to favour the Student-t estimator across all block factors.

What this experiment establishes. Experiment A establishes the *baseline advantage* of the Student-t DFM in its native setting: when the data are genuinely heavy-tailed, the proposed estimator outperforms the Gaussian benchmark. This is the conventional case for the Student-t DFM and the result is consistent with the heavy-tailed extensions of the DFM proposed by Antolín-Díaz, Drechsel, and Petrella (2024) and the broader robust-estimation literature.

Experiment B: Gaussian DGP, Both Estimators

Setup. Data are generated from a Gaussian DGP — the limit $\nu_u^* \to \infty$, $\nu_\varepsilon^* \to \infty$ of the Student-t DGP — with all other parameters held at their calibrated values. Both estimators are applied to each synthetic sample. The same set of metrics as in Experiment A is reported.

Construction of the Gaussian DGP: removing the tails, keeping everything else. A methodological choice underlies this construction, and it is worth making explicit. The Gaussian parameter vector $\boldsymbol{\theta}_B^*$ is obtained from the *same* calibrated $\boldsymbol{\theta}^*$ used in Experiment A — the one estimated on real data in the first-stage fit — by setting only the degrees of freedom to their Gaussian limit, $\nu_u^* = \nu_\varepsilon^* = \infty$, and leaving the loadings $\boldsymbol{\Lambda}^*$, the transition matrix $\mathbf{A}^*$, the innovation covariance $\mathbf{Q}^*$ and the idiosyncratic variances $\mathbf{R}^*$ exactly as calibrated. Operationally, the scale-mixture weights are no longer drawn from a Gamma distribution but fixed at $w_t^u = w_t^\varepsilon = 1$ for all t, so the factor innovations and measurement errors become exactly Gaussian.

We deliberately do *not* re-estimate a separate Gaussian DFM on the real data to obtain $\boldsymbol{\theta}_B^*$. Doing so would produce a parameter vector that differs from the Experiment A calibration along *every* dimension — loadings, dynamics and variances would all shift to absorb, in a Gaussian likelihood, the heavy-tailed features that the Student-t fit attributes to the weights. The comparison between Experiments A and B would then confound two distinct changes: the removal of the tails (the object of interest) and the wholesale re-optimisation of all remaining parameters. By instead changing *only* ν and freezing $(\boldsymbol{\Lambda}^*, \mathbf{A}^*, \mathbf{Q}^*, \mathbf{R}^*)$, the two experiments share an identical structure and differ in exactly one respect: the presence or absence of heavy tails. This isolation is what makes the A-versus-B contrast clean — any difference in the estimators' relative performance across the two experiments is attributable to the tails alone, and not to an incidental change in the underlying loadings or dynamics. It also sharpens the nesting test itself: the Student-t estimator is confronted with data drawn from precisely the structure it fits in Experiment A, with only the tails switched off, so that a high estimated $\hat{\nu}$ is unambiguous evidence that the estimator detects their absence.

Expected outcome. The expectation here is the converse of Experiment A. Because the data-generating process is now correctly specified for the Gaussian benchmark, the Gaussian EM should perform optimally. The Student-t EM, applied to data that have no heavy tails, should:

- Recover $\hat{\nu}_u, \hat{\nu}_\varepsilon$ at values near the upper bracket of the Brent search interval — the algorithm's empirical signal that the tail thickness is indistinguishable from Gaussian;
- Produce parameter estimates $\hat{\boldsymbol{\Lambda}}, \hat{\mathbf{A}}, \hat{\mathbf{Q}}, \hat{\mathbf{R}}$ that are *statistically indistinguishable* from the Gaussian EM estimates — same RMSE, same factor recovery, same convergence behaviour.

The two estimators should therefore agree, on average, when applied to Gaussian data. Any systematic difference would indicate that the Student-t EM incurs an efficiency loss for fitting an over-parameterised model in the absence of heavy tails — a small but measurable cost.

A subtlety in the Gaussian limit: monitoring the ELBO, not its conditional component. Experiment B exposes a delicate interaction between the data-generating process and the convergence criterion of the EM algorithm that does not arise in Experiment A, and that deserves a careful treatment because it is generic to any Student-t model estimated by EM and tested for Gaussian nesting.

Recall the structure of the estimator. The Student-t DFM is written as a Gaussian scale mixture: the factor innovations and the idiosyncratic errors are conditionally Gaussian given latent weights,

$$u_t \mid w_t^u \sim \mathcal{N}\big(0,\, (w_t^u)^{-1}\mathbf{Q}\big), \qquad \varepsilon_t \mid w_t^\varepsilon \sim \mathcal{N}\big(0,\, (w_t^\varepsilon)^{-1}\mathbf{R}\big),$$

with the weights drawn a priori from $w_t^u \sim \text{Gamma}(\nu_u/2, \nu_u/2)$ and $w_t^\varepsilon \sim \text{Gamma}(\nu_\varepsilon/2, \nu_\varepsilon/2)$.

The variational EM algorithm maximises the evidence lower bound (ELBO), introduced previously in its general form

$$\mathcal{L}(q, \theta) = \mathbb{E}_q\big[\log p(\mathbf{Y}, \mathcal{Z} \mid \theta)\big] - \mathbb{E}_q\big[\log q(\mathcal{Z})\big],$$

where $\mathcal{Z}$ collects *all* latent variables. The step that specialises this to our model, and that isolates the role of the degrees of freedom, is worth making fully explicit. In the Student-t DFM there are two distinct kinds of latent object, and the latent set factorises as

$$\mathcal{Z} = (F,\, W), \qquad W = (W_u,\, W_\varepsilon),$$

where $F = (f_1, \ldots, f_T)$ are the *dynamic factors* — the state-space latent states — and W are the *scale-mixture weights* that generate the heavy tails, one per period for the factor innovations (W_u) and one per period for the idiosyncratic errors (W_ε). Under the mean-field factorisation adopted throughout,

$$q(\mathcal{Z}) = q(F \mid W)\, q(W), \qquad q(W) = \prod_t q(w_t^u) \prod_t q(w_t^\varepsilon),$$

the expectation defining the ELBO can be taken in two stages: first over the factors F given the weights, then over the weights. The inner stage is the key simplification. *Conditional on the weights*, the model is exactly Gaussian — the innovations and errors are $\mathcal{N}(0, (w_t^u)^{-1}\mathbf{Q})$ and $\mathcal{N}(0, (w_t^\varepsilon)^{-1}\mathbf{R})$ — so integrating out the factors is precisely what the Kalman filter does. It returns the weight-conditional marginal likelihood

$$\log p(\mathbf{Y} \mid W, \theta) = \int p(\mathbf{Y}, F \mid W, \theta)\, dF,$$

in which the factors no longer appear: they have been marginalised analytically. Carrying this through, the ELBO reorganises into three interpretable pieces,

$$\mathcal{L}(q, \theta) = \underbrace{\mathbb{E}_{q(W)}\big[\log p(\mathbf{Y} \mid W, \theta)\big]}_{\text{(1) conditional (Kalman) log-likelihood}} + \underbrace{\mathbb{E}_{q(W)}\big[\log p(W \mid \nu)\big]}_{\text{(2) weight prior term}} + \underbrace{H\big[q(W)\big]}_{\text{(3) posterior entropy}},$$

where $q(W)$ is the mean-field variational posterior over the weights. The factors F have disappeared into term (1), where the Kalman filter handles them exactly; what remains explicit are the weights, through their prior term (2) and their posterior entropy (3). This is the precise sense in which the decomposition "conditions on the weights": the state-space latent variables are integrated out analytically, and the only latent quantities left to monitor are the ones that control the tail thickness.

Term (1) is the marginal log-likelihood of the observations *conditional on the imputed weights*, and it is a free by-product of the Kalman filter run inside the E-step. For computational convenience, the implementation a natural and computationally cheap choice is to monitor term (1) alone, treating it as a proxy for the full ELBO, — a choice justified, as discussed previously, by the fact that the two quantities differ only by the weight-related terms (2) and (3), which stabilise quickly once the degrees of freedom ν settle.

This proxy is faithful whenever ν converges to a finite, well-identified value, which is precisely the situation in Experiment A and in the empirical fit on real data, where the calibrated $\nu^* \approx 4$. There, once $\nu^{(j)} \approx \nu^*$, the terms (2) and (3) become essentially constant across iterations, so tracking term (1) is equivalent — up to a stable additive constant that does not affect the relative change — to tracking the full ELBO. The algorithm converges cleanly, as documented.

The Gaussian DGP of Experiment B breaks this equivalence. When the data carry no heavy tails, the true degrees of freedom are infinite, and the estimator can only reproduce the Gaussian limit by driving $\hat{\nu} \to \infty$. But $\nu = \infty$ lies on the *boundary* of the parameter space and is never reached in finite steps: the M-step keeps increasing ν a little at every iteration (in our diagnostic run, $\hat{\nu}_u$ climbs monotonically from 10 toward 142 and beyond, while $\hat{\nu}_\varepsilon$ reaches the upper search bound of 1000). This unbounded drift has a direct consequence on the three terms of the ELBO:

- Term (2), the weight-prior contribution, *grows* without bound as ν increases. To see this concretely, take the factor side (the idiosyncratic side is identical). The weight prior is $w_t^u \sim \text{Gamma}(\nu_u/2, \nu_u/2)$, so

  $$\mathbb{E}_q[\log p(W_u \mid \nu_u)] = T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma\big(\tfrac{\nu_u}{2}\big)\big] + T\big(\tfrac{\nu_u}{2} - 1\big)\,\overline{\log w}_u - T\,\tfrac{\nu_u}{2}\,\bar{w}_u.$$

  For Gaussian data the posterior weights concentrate sharply around one, so $\bar{w}_u \to 1$ and $\overline{\log w}_u \to 0$. Substituting these limiting values, the two weight-average terms collapse to $T(\frac{\nu_u}{2} - 1)\cdot 0 - T\frac{\nu_u}{2}\cdot 1 = -T\frac{\nu_u}{2}$, leaving

  $$\mathbb{E}_q[\log p(W_u \mid \nu_u)] \;\approx\; T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma\big(\tfrac{\nu_u}{2}\big) - \tfrac{\nu_u}{2}\big]\,.$$

  Now apply Stirling's approximation to the log-gamma term. For large argument x, $\log\Gamma(x) = x\log x - x - \frac{1}{2}\log x + \frac{1}{2}\log(2\pi) + O(1/x)$. With $x = \nu_u/2$,

  $$\log\Gamma\big(\tfrac{\nu_u}{2}\big) \;=\; \tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \tfrac{\nu_u}{2} - \tfrac{1}{2}\log\tfrac{\nu_u}{2} + \tfrac{1}{2}\log(2\pi) + O(1/\nu_u).$$

  Substituting this into the bracket, the two leading pieces $\frac{\nu_u}{2}\log\frac{\nu_u}{2}$ cancel, the two $-\frac{\nu_u}{2}$ cancel, and what survives is

  $$\mathbb{E}_q[\log p(W_u \mid \nu_u)] \;\approx\; T\big[\tfrac{1}{2}\log\tfrac{\nu_u}{2} - \tfrac{1}{2}\log(2\pi)\big] + O(1) \;=\; \tfrac{T}{2}\log\nu_u + O(1).$$

  This term is strictly increasing in ν_u and unbounded: every increase in ν raises the ELBO through this channel. It is precisely the force that drives the M-step to push ν upward on Gaussian data — the algorithm climbs the ELBO by sending the degrees of freedom toward infinity, as it should when there are no heavy tails to detect.

- Term (1), the conditional Kalman log-likelihood, *decreases* slightly along the path. The reason is that the M-step recomputes $\mathbf{\Lambda}, \mathbf{Q}, \mathbf{R}$ as a weighted (GLS) fit using the *current* weights $\hat{w}^{(j)}$, which tend to one but shift at every step as ν grows. These updates are optimal for the current weights, not for the pure unit-weight Gaussian fit that would maximise term (1) in the limit, so term (1) drifts down as the algorithm approaches the Gaussian solution from a slightly oblique direction.

- The full ELBO, the sum (1)+(2)+(3), nonetheless *increases monotonically*. This is guaranteed by construction: the M-step maximises the variational objective, so $\text{ELBO}(\theta^{(j+1)}) \geq \text{ELBO}(\theta^{(j)})$ at every iteration. The increase of term (2) more than compensates the small decrease of term (1).

The practical symptom is that the convergence test, which watches the relative change of term (1), never triggers within the iteration cap: term (1) keeps drifting down by a vanishing but non-zero amount at every step, so the relative change settles just above the tolerance rather than below it. (In our run at $T = 800$ the relative change of term (1) reaches 1.011×10^{-5} at iteration 199, i.e. only one percent above the tolerance 10^{-5}: the algorithm *is* converging, merely more slowly than the cap allows.) It is essential to stress that this is *not* a failure of the EM algorithm, nor a violation of its monotonicity: the quantity the algorithm actually maximises, the ELBO, is increasing throughout. What fails is the *proxy* — monitoring term (1) in isolation — in the one regime where (2) and (3) do not stabilise.

Resolution: monitoring the full ELBO. The diagnosis points to an unambiguous fix. Since the failure is not in the algorithm but in the quantity used to declare convergence, the remedy is to monitor the object the EM algorithm actually maximises — the full ELBO, $\mathcal{L} = (1) + (2) + (3)$ — rather than its weight-conditional component (1) alone. We discuss why this is the right choice, how it is computed, and why the cheaper alternatives we considered are inferior.

Why the full ELBO is the correct monitor. By the variational argument, the M-step maximises the ELBO at each iteration, so $\mathcal{L}(\theta^{(j+1)}) \geq \mathcal{L}(\theta^{(j)})$ *by construction*: the full ELBO is monotone non-decreasing along the entire EM path, in every regime, with no exception. Monitoring it therefore restores exact agreement between the convergence criterion and the algorithm's guarantee. The apparent "monotonicity violations" seen when tracking term (1) in the Gaussian limit are revealed for what they are — an artefact of watching a component that is free to decrease while the total increases — and they disappear once the correct quantity is monitored.

How it is computed, at no extra cost. The two missing pieces are available in closed form from quantities the E-step already returns. With $\bar{w}$ and $\overline{\log w}$ the posterior weight averages, the prior term is

$$\mathbb{E}_q[\log p(W_s \mid \nu_s)] = T_s\Big[\tfrac{\nu_s}{2}\log\tfrac{\nu_s}{2} - \log\Gamma\big(\tfrac{\nu_s}{2}\big) + \big(\tfrac{\nu_s}{2} - 1\big)\overline{\log w}_s - \tfrac{\nu_s}{2}\,\bar{w}_s\Big], \qquad s \in \{u, \varepsilon\},$$

and the posterior entropy, using the closed-form entropy of a Gamma(α, β) density together with the identity $\log\beta = \psi(\alpha) - \mathbb{E}[\log w]$, reduces to

$$H[q(W_s)] = \sum_t \Big[\alpha_{s,t}\big(1 - \psi(\alpha_{s,t})\big) + \log\Gamma(\alpha_{s,t})\Big] + T_s\,\overline{\log w}_s,$$

with shape $\alpha_{u,t} = (\nu_u + r)/2$ on the factor side (constant in t, since all r factors are always present) and $\alpha_{\varepsilon,t} = (\nu_\varepsilon + m_t)/2$ on the idiosyncratic side, where m_t is the number of series observed at date t. The dependence on m_t is the only place where the ragged edge enters the correction: periods with fewer observed series contribute a different posterior shape, exactly as they do in the M-step update of $\mathbf{R}$ and $\mathbf{\Lambda}$. The full ELBO is then $\mathcal{L} = L_{\text{Kalman}} + \Delta_W$, where Δ_W sums the prior and entropy terms over both weight families; it adds a handful of scalar operations per iteration and no new passes through the data.

Why the full ELBO flattens correctly. Once ν is large enough that the data are statistically indistinguishable from Gaussian, the ELBO stops rising appreciably. Concretely, the ν-component of the objective behaves as $\frac{T}{2}\log\nu + O(1)$, whose derivative $\frac{T}{2\nu}$ tends to zero. Each Brent step still finds the stationary point of the first-order condition at a slightly larger ν, but the resulting ELBO gain per step shrinks to zero, so the relative change of the *full* ELBO falls below tolerance at a finite, large $\hat{\nu}$ — the algorithm's honest report that the tails are absent. Convergence is declared; the estimate $\hat{\nu}$ is large but finite, which is the correct finite-sample signature of a Gaussian DGP.

Alternatives considered and rejected. Three cheaper fixes were available, and we record why each is inferior. (i) *Freezing* ν after a fixed number of iterations stops the drift, but the number is arbitrary and the device suppresses precisely the information the nesting test is meant to reveal — how large ν becomes. (ii) *Switching the criterion to the relative change of the parameters* rather than the likelihood is robust to the drift but departs from the likelihood-based stopping rule used everywhere else in the thesis, and the parameters can still move while the objective is effectively flat. (iii) *A hard threshold* (declare convergence once ν exceeds, say, 500) is pragmatic but a pure heuristic, with no link to the optimised objective. Monitoring the full ELBO is preferable to all three because it is not a patch: it is the principled criterion that the variational formulation prescribes in the first place, exact in every regime, and reducing to the term (1) proxy whenever ν is finite and stable — which is why the simpler proxy was, and remains, adequate for Experiments A and C and for the empirical fit on real data.

What this experiment establishes. Experiment B is the empirical analogue of the *nesting* property that relates the Student-t and Gaussian DFMs. The Student-t extension reduces to the Gaussian model in the limit $\nu \to \infty$, and this nesting should be visible in finite samples: when the data are Gaussian, the Student-t EM should detect this and collapse to the Gaussian solution. Documenting this collapse is one of the central contributions of the simulation study. It shows that the proposed estimator is a *strict generalisation* of the Gaussian benchmark:

- it matches the benchmark when the benchmark is correctly specified (Experiment B), and
- it outperforms the benchmark when the benchmark is misspecified (Experiment A and, as we shall see, Experiment C).

The proposed methodology is therefore never worse than the standard approach. It is the kind of *Pareto-improving* property that justifies the additional methodological investment required by the Student-t extension.

Experiment C: Contamination Experiments

The empirical motivation. Experiments A and B describe two limiting cases that are methodologically illuminating but empirically extreme. Real macroeconomic data are neither fully Gaussian (the Experiment B limit) nor fully heavy-tailed with a low calibrated ν^* (the Experiment A limit). They sit between the two: most periods are well-behaved and well-approximated by a Gaussian model, but a small minority of periods is governed by a different mechanism that produces extreme observations — crisis episodes, abrupt policy interventions, pandemics, geopolitical shocks. This is the empirical regime in which the proposed methodology will ultimately be applied, and the robustness of the estimator to this kind of contamination is the central applied question that the Monte Carlo must answer.

Experiment C is designed precisely to fill this intermediate region between A and B. It varies the fraction of contaminated periods π along a grid, allowing us to trace a continuous path from near-Gaussian conditions (small π) to substantially contaminated regimes (larger π). The relative performance of the two estimators along this grid is the central empirical output of the experiment.

Setup. Data are generated from a baseline Gaussian DGP — the same DGP as in Experiment B, with $\nu_u^*, \nu_\varepsilon^* \to \infty$ — but a fraction π of time periods is contaminated by extreme outliers drawn from a heavy-tailed contamination distribution. The contamination mechanism, detailed earlier in this section, is a Bernoulli switch on each period t: with probability π, the period is contaminated and the idiosyncratic shock is drawn from a $t_{\nu_{\text{contam}}}(\mathbf{0}, \kappa^2\mathbf{R}^*)$ distribution with $\nu_{\text{contam}} = 3$; otherwise, the period follows the standard Gaussian draw.

We consider the contamination grid

$$\pi \in \{0, 0.01, 0.05, 0.10\},$$

where $\pi = 0$ is included as an anchor (no contamination: the DGP is exactly that of Experiment B) and the three positive levels correspond respectively to a contamination of approximately one period in a hundred, one in twenty, and one in ten. The inflation factor is fixed at $\kappa = 5$ throughout the experiment (so that the contaminated idiosyncratic variance is $\kappa^2 = 25$ times the Gaussian baseline), with the contamination affecting all M series of a flagged period simultaneously. The three positive levels are chosen to span empirically plausible contamination intensities: $\pi = 0.01$ corresponds roughly to the proportion of truly extreme months in a typical post-war macroeconomic sample (e.g. a few months around the early 1980s recession, the 2008 crisis, and the 2020 pandemic); $\pi = 0.05$ represents a more contaminated regime; $\pi = 0.10$ is an extreme stress test, in which one in ten periods is governed by the outlier mechanism.

Both estimators are applied to each synthetic sample. The same set of metrics as in Experiments A and B is reported, with all results broken down by π so that the dependence on the contamination intensity is fully visible.

The expected pattern. Three concrete predictions follow from the comparative structure of the two estimators, each addressing a distinct dimension of the contamination effect.

First — the degradation of the Gaussian benchmark. As π grows, the Gaussian EM should exhibit increasing bias and variance in $\hat{\mathbf{\Lambda}}, \hat{\mathbf{Q}}, \hat{\mathbf{R}}$. The mechanism is direct: the Gaussian model has no way to distinguish contaminated from clean periods, so the extreme observations are absorbed into the covariance estimates as if they were typical realisations, inflating them and distorting the factor recovery. The cost should grow approximately linearly with π — each additional contaminated period adds a comparable amount of distortion to the parameter estimates.

Second — the stability of the Student-t extension. The Student-t EM, by contrast, should remain comparatively stable as π grows. Two adaptive mechanisms are at work. The estimated degree of freedom $\hat{\nu}_\varepsilon$ adjusts to reflect the empirical tail thickness, decreasing as π increases; this widens the conditional distribution of the weights and allows them to absorb more contamination. The posterior weights $\hat{w}_t^\varepsilon$ then concentrate the down-weighting on the periods that were in fact contaminated, isolating them from the covariance estimates and protecting the parameter recovery from their influence.

Third — the detection of contaminated periods. A direct measure of how effectively the Student-t EM identifies the outliers is provided by the posterior weights $\hat{w}_t^\varepsilon$, which should be smallest exactly at the periods that were in fact contaminated. To quantify this without introducing an arbitrary cutoff on the weights, we adopt a *precision-at-k* criterion. Let $n_c = \sum_t z_t$ be the number of truly contaminated periods in a replication. We rank all periods by their estimated weight and flag the $k = n_c$ periods with the lowest weights — those the estimator most strongly down-weights. The *detection rate* is the fraction of these flagged periods that are genuinely contaminated. Because exactly n_c periods are flagged, precision and recall coincide and the metric cannot be inflated by over-flagging: the estimator commits to precisely n_c suspects, no more.

To interpret the detection rate we compare it against a purely random ranking, which would in expectation capture a fraction $n_c/T \approx \pi$ of the contaminations — simply the prevalence of contamination in the sample. The *lift*,

$$\text{lift} \;=\; \frac{\text{detection rate}}{n_c/T} \;\approx\; \frac{\text{detection rate}}{\pi},$$

measures how many times better than chance the estimator locates the outliers. The lift is largest precisely when contamination is *rare*: at small π a random guess almost never lands on a contaminated period (with, say, only five contaminated months in a sample of five hundred, blind selection captures essentially none), so a high detection rate yields a very large lift, of order $1/\pi$; as π grows the random baseline itself improves, so the same near-perfect detection rate produces a smaller lift. A lift that *declines* with π therefore does not indicate worsening detection — it reflects the mechanical improvement of the random baseline — and for this reason the detection rate, the direct measure of identification accuracy, is the quantity to read alongside it. The Gaussian estimator, whose weights are identically one by construction, performs no down-weighting and admits no informative ranking: its detection metric is undefined, which is itself the point — a model that cannot tell outliers apart cannot locate them.

The two endpoints and the continuum. Experiment C is constructed so that the limiting cases $\pi = 0$ and $\pi \to 1$ recover Experiments B and A respectively. At $\pi = 0$ no contamination is injected and the DGP is exactly the Gaussian DGP of Experiment B; the two estimators should agree by the nesting property. As π grows, the DGP departs gradually from the Gaussian and approaches a fully Student-t regime — in the limit $\pi \to 1$ with appropriately chosen κ^2 and ν_{contam}, the DGP would coincide with a Student-t DGP equivalent to that of Experiment A. The contamination grid therefore parametrises a continuous interpolation between the two endpoints, and the empirical curves of bias, variance, and RMSE versus π document the smooth transition.

What this experiment establishes. Three substantive conclusions follow.

The robustness margin of the proposed estimator. The gap in RMSE between the Gaussian and Student-t estimators at each π measures the practical cost of using the Gaussian benchmark when the data contain even a small fraction of contaminated periods, and the corresponding gain from adopting the heavy-tailed extension. This is the most directly applied result of the entire Monte Carlo: it tells the practitioner what they stand to gain (or lose) by choosing one estimator over the other under the empirical conditions they actually face.

The relevance to macroeconomic practice. Real macroeconomic data are very rarely fully Gaussian and very rarely fully described by a low-ν Student-t. They are almost always somewhere in the intermediate contamination range, with π implicitly determined by the historical episodes covered by the sample. Experiment C is therefore the experiment most relevant to the actual choice between estimators in applied work.

The mechanism through which robustness operates. Beyond the headline RMSE comparison, the detection rate and the estimated $\hat{\nu}_\varepsilon$ trace the channel through which the Student-t extension achieves its robustness: outlier identification via the latent weights, and adaptive adjustment of the tail thickness. Documenting these channels separately — rather than only their aggregate effect on RMSE — turns the Monte Carlo from a performance comparison into a mechanism-revealing exercise.

Summary of the Three Experiments

Together, the three experiments characterise the proposed estimator:

- *Experiment A* (Student-*t* DGP): the baseline advantage in the heavy-tailed setting that motivates the extension;
- *Experiment B* (Gaussian DGP): the nesting property — the Student-*t* EM collapses to the Gaussian benchmark when the data are Gaussian;
- *Experiment C* (contamination grid): the robustness margin under the empirically realistic intermediate regime.

Taken together, they document that the proposed estimator is a strict generalisation of the Gaussian DFM: never worse, and substantially better whenever the data exhibit heavy tails or outlier contamination.

The Full Monte Carlo Algorithm

The complete validation procedure is split into two algorithms: Algorithm 3 formalises the calibration and the generation of the synthetic samples; Algorithm 4 formalises the re-estimation, the aggregation across replications, and the production of validation outputs.

Algorithm 3 Monte Carlo design — calibration and synthetic data generation

Require: Real-data panel $\mathbf{Y}^{\text{real}} \in \mathbb{R}^{M \times T}$, block partitioning $\{r_R, r_F, r_X\}$, number of replications S, sample-size grid $\mathcal{T}^* = \{T_1^*, \ldots, T_K^*\}$, contamination grid $\Pi = \{0, 0.01, 0.05, 0.10\}$.

Ensure: Calibrated truth $\boldsymbol{\theta}^*$ and a collection of synthetic panels $\{\mathbf{Y}^{(s)}\}$ indexed by (s, T^*, π), together with their ground-truth latent paths.

// Step 1 — Calibration on real data
Run the Student-t EM on $\mathbf{Y}^{\text{real}}$ to convergence, with all extensions in operation (mixed-frequency, ragged-edge, block-diagonal).
Apply post-hoc normalisation: rescale each block factor to unit total (marginal) variance; fix signs via reference series.
Set $\boldsymbol{\theta}^* \leftarrow \hat{\boldsymbol{\theta}}_{\text{real}}$ (the calibrated truth).
Record the missingness pattern $\{\mathbf{W}_t^{\text{real}}\}_{t=1}^{T_{\text{real}}}$ and the mixed-frequency calendar of the real panel.

// Step 2 — Synthetic data generation
for $T^* \in \mathcal{T}^*$ **do**
  for $\pi \in \Pi$ **do**
    for $s = 1, \ldots, S$ **do**
      // 2a. Latent paths
      Set the random seed as a deterministic function of (s, T^*, π) for reproducibility.
      Draw factor weights $\{w_t^{u,(s)}\}_{t=1}^{T^*}$ from Gamma$(\nu_u^*/2,\ \nu_u^*/2)$.
      Draw idiosyncratic weights $\{w_t^{\varepsilon,(s)}\}_{t=1}^{T^*}$ from Gamma$(\nu_\varepsilon^*/2,\ \nu_\varepsilon^*/2)$.
      Draw factor innovations $\{u_t^{(s)}\}$, conditionally Gaussian with covariance $\mathbf{Q}^*/w_t^{u,(s)}$.
      Draw idiosyncratic shocks $\{\varepsilon_t^{(s)}\}$, conditionally Gaussian with covariance $\mathbf{R}^*/w_t^{\varepsilon,(s)}$.
      Draw the initial state $f_0^{(s)}$ from the stationary distribution of the VAR(1).
      Construct the factor path $f_t^{(s)} = \mathbf{A}^* f_{t-1}^{(s)} + u_t^{(s)}$ for $t = 1, \ldots, T^*$.
      // 2b. Observations
      Generate $y_t^{(s)} = \boldsymbol{\Lambda}^* f_t^{(s)} + \varepsilon_t^{(s)}$ for $t = 1, \ldots, T^*$.
      // 2c. Contamination (only if $\pi > 0$)
      if $\pi > 0$ **then**
        for $t = 1, \ldots, T^*$ **do**
          Draw $z_t \sim$ Bernoulli(π).
          if $z_t = 1$ **then**
            Replace $\varepsilon_t^{(s)}$ with a draw from $t_{\nu_{\text{contam}}}(\mathbf{0},\ \kappa^2 \mathbf{R}^*)$, with $\nu_{\text{contam}} = 3$.
            Update $y_t^{(s)}$ accordingly.
          end if
        end for
      end if
      // 2d. Apply empirical structure
      Apply the missingness pattern $\{\mathbf{W}_t^{\text{real}}\}$ (truncated or extended to length T^*) to $\{y_t^{(s)}\}$ to obtain the ragged-edge version $\mathbf{Y}^{(s)}$.
      For quarterly series, mask all non-quarter-end months as missing.
      // 2e. Construct the vintage sequence
      From the publication lags $\{d_i\}$ recorded in Step 1, construct the release dates $v_{i,t} = t + d_i$ for each (i, t).
      Assemble the filtration $\mathbf{Y}^{(s),v_1} \subset \mathbf{Y}^{(s),v_2} \subset \cdots$ by ordering the distinct release dates and including, at each vintage v_k, all entries (i, t) with $v_{i,t} \leq v_k$.
      // 2f. Store the replication
      Store $\mathbf{Y}^{(s)}$ and its vintage sequence together with the latent ground truth $\{f_t^{(s)}, w_t^{u,(s)}, w_t^{\varepsilon,(s)}, \varepsilon_t^{(s)}\}$.
    end for
  end for
end for
return The calibrated truth $\boldsymbol{\theta}^*$ and the collection of synthetic panels with their latent paths.

From data construction to inference. The output of Algorithm 3 is the raw material of the validation experiment: a calibrated truth and a collection of synthetic panels that span the four dimensions of variation (sample size, contamination intensity, replication index, and —

implicit in the latent path storage — the choice of DGP-specification given by the contamination parameter π). The next stage, formalised in Algorithm 4, applies each estimator (EM_t and $\mathrm{EM}_{\mathrm{Gauss}}$) to each synthetic panel, computes the replication-level metrics, and aggregates them into the validation outputs that support the empirical conclusions of the simulation study.

Algorithm 4 Monte Carlo design — re-estimation and aggregation

Require: Calibrated truth $\boldsymbol{\theta}^*$ from Algorithm 3, collection of synthetic panels $\{\mathbf{Y}^{(s)}\}$ indexed by (s, T^*, π) with latent ground-truth paths, estimator set $\mathcal{E} = \{\mathrm{EM}_t,\ \mathrm{EM}_{\mathrm{Gauss}}\}$, maximum outer iterations $J_{\max}$, convergence tolerance η.

Ensure: For each (T^*, π, EM) triple: empirical distribution of $\{\hat{\boldsymbol{\theta}}^{(s)}, \hat{f}^{(s)}_{t|T^*}\}_{s=1}^{S}$, validation metrics, and convergence diagnostics.

```
// Step 1 — Re-estimation loop
for T* ∈ 𝒯* do
    for π ∈ Π do
        for EM ∈ ℰ do
            for s = 1, ..., S do
                // 1a. Re-estimate on the synthetic data
                Initialise via PCA on the balanced pre-filled version of Y^(s).
                Run estimator EM on Y^(s), with the block-diagonal constraint on Λ imposed throughout, until
                convergence or until J_max iterations are reached.
                Apply post-hoc normalisation to θ̂^(s) (sign flip via reference series, then factor-variance rescaling —
                Convention 1).
                // 1b. Record replication-level metrics
                Store θ̂^(s), smoothed factors {f̂_t^(s)}, smoothed weights {ŵ_t^{u,(s)}, ŵ_t^{ε,(s)}}, and convergence diagnostics
                (number of outer iterations, monotonicity flag, terminal likelihood).
                Compute parameter recovery errors θ̂^(s) − θ* (entry-wise and Frobenius).
                Compute factor recovery metrics Corr(f̂^(s), f^(s)) and trajectory error, per block.
            end for
        end for
    end for
end for
// Step 2 — Aggregation across replications
for each (T*, π, EM) combination do
    Compute Bias, Variance, and RMSE for each parameter component across the S replications.
    Compute Frobenius RMSE for matrix-valued parameters.
    Compute factor recovery distributions (mean and interquartile range of correlations and trajectory errors).
    Compute convergence statistics (convergence rate, mean iteration count, monotonicity rate).
    Optionally: compute Procrustes-aligned versions of the parameter recovery metrics for robustness.
end for
// Step 3 — Cross-cutting analyses
Plot RMSE versus T* on log–log axes (finite-sample rate of convergence).
Plot RMSE versus π for fixed T* (robustness curves).
Plot the distribution of ν̂_u, ν̂_ε across replications under each DGP (recovery of the heavy-tail parameters).
Compute the detection rate: fraction of true contaminated periods (those with z_t = 1) for which ŵ_t^ε falls below a
chosen threshold, and the corresponding false-positive rate on non-contaminated periods (Experiment C).
Plot the histogram of estimated posterior weights ŵ_t^ε separately for contaminated and non-contaminated periods,
as a graphical complement to the detection rate.
return Aggregated validation tables and figures for all (T*, π, EM) combinations.
```

Reading the algorithms. Algorithm 3 produces the data; Algorithm 4 produces the inference. The conceptual separation is useful both for implementation — the two stages can be coded, tested, and parallelised independently — and for reasoning about computational cost: the data generation is relatively cheap, dominated by random number draws and matrix multiplications, while the re-estimation is the computationally intensive stage where the EM iterations are run on each synthetic panel. Splitting the algorithm makes this asymmetry explicit.

Parallelisation. The Monte Carlo design is *embarrassingly parallel*: each (s, T^*, π, EM) combination in Algorithm 4 is an independent re-estimation that requires no communication with the others. Implementation distributes the workload across cores or compute nodes by assigning ranges of replications, and aggregates the results at the end. The deterministic random seed in Algorithm 3 ensures reproducibility regardless of the order in which the replications are executed.

The role of $J_{\max}$. A maximum outer iteration count $J_{\max}$ is needed because the EM algorithm, while guaranteed to monotonically increase the marginal likelihood, is not guaranteed to converge within a bounded number of iterations. In practice, the vast majority of replications converge within 50–200 outer iterations. We adopt $J_{\max} = 500$ as a generous cap that should accommodate even the slow-converging cases, and report the convergence rate as part of the diagnostic output. Replications that fail to converge within $J_{\max}$ are flagged but not excluded from the metrics — their inclusion provides a conservative measure of algorithm reliability.

Practical Considerations

The Monte Carlo design described in the preceding subsections is flexible across a number of dimensions: the number of replications, the grid of sample sizes, the choice of computational architecture, and the seed management strategy. This subsection collects the practical decisions that translate the design into an executable experiment.

Choice of S (Number of Replications)

The number of replications S governs the precision of the Monte Carlo estimates of bias, variance, and RMSE. The Monte Carlo standard error of an RMSE estimate scales as $1/\sqrt{S}$ — the familiar rate for averaging S independent draws.

Concretely, with $S = 1000$ and an RMSE of order 0.10, the Monte Carlo standard error of the estimated RMSE is approximately $\text{RMSE}/\sqrt{2S} \approx 0.002$ — the factor $\sqrt{2}$ arises from applying the delta method to the variance of the squared deviations under approximate Gaussianity. This is small enough that comparisons between estimators are statistically reliable. With $S = 100$ the precision drops by a factor of $\sqrt{10} \approx 3$, which is acceptable for exploratory work but insufficient for the final reported results.

We adopt the following two-tier strategy:

- For exploratory work — tuning convergence tolerances, inspecting raw outputs, debugging the simulation pipeline — we use $S = 100$. This is fast enough for iteration but gives a rough sense of estimator behaviour.
- For the final reported results, we use $S = 1000$. This is the standard value in the DFM Monte Carlo literature (Doz, Giannone, and Reichlin, 2012; Bańbura and Modugno, 2014) and produces estimates with Monte Carlo standard errors that are smaller than the substantive differences between estimators.

Larger values of S (e.g. $S = 5000$) would further refine the estimates but at proportional computational cost. The marginal gain is typically small once S exceeds a thousand, and the diminishing returns argue for the conventional choice.

Choice of T^* (Synthetic Sample Length)

The Monte Carlo is run not only at a single sample size, but across a grid of values T^*:

$$T^* \in \{100, 200, 400, 800\} \text{ months,}$$

in addition to the headline case $T^* = T_{\text{real}}$. Each value in the grid plays a distinct role in characterising the finite-sample behaviour of the estimator.

The headline case $T^* = T_{\text{real}}$. For our application, T_{real} corresponds to approximately 33–35 years of monthly observations (roughly 400 months in the real panel). The headline validation sets $T^* = T_{\text{real}}$ and asks the most directly relevant question: how well does the estimator perform under the same sample length as the empirical analysis to follow? The bias, variance, and RMSE reported at this sample size are quantitative analogues of the finite-sample uncertainty one should expect when applying the method to the real panel.

Smaller sample sizes: stress-testing. The values $T^* = 100$ and $T^* = 200$ are below the empirical sample length and serve as a stress test: they document how the estimator performs under conditions of limited data. This is operationally relevant for several realistic scenarios — applying the method to a subset of the historical record (e.g. post-2008 only), or to a country with shorter data availability than the United States or the Euro Area. If the estimator deteriorates sharply at small sample sizes, this is a practical limitation worth documenting; if it remains stable (low bias, manageable RMSE), this is evidence of robustness to sample size that strengthens the applicability of the method.

Larger sample sizes: approaching the asymptotic regime. The value $T^* = 800$ extends beyond the empirical length and approximates the asymptotic behaviour that the literature predicts for related estimators but does not formally establish for our specific setting. By doubling the synthetic sample length, we allow the estimator to operate under conditions closer to the "large-T" regime in which the parametric rate $T^{-1/2}$ is expected to apply. The progression from $T^* = 100$ to $T^* = 800$ traces a continuous trajectory toward the asymptotic limit, and the empirical curve of RMSE against T^* provides an estimate of the convergence rate.

Reading the convergence curve. Plotting RMSE against T^* on log–log axes typically reveals a near-linear trajectory, whose slope provides an empirical estimate of the finite-sample rate of convergence. A slope close to $-1/2$ is consistent with the standard parametric rate $T^{-1/2}$. A slower slope would signal that the estimator converges at a rate below the parametric benchmark, and quantitatively characterises the finite-sample efficiency loss relative to the asymptotic ideal.

Units of sample size. For our mixed-frequency model the natural unit of T^* is the *monthly* observation count: the latent factor process evolves at monthly frequency, and the Kalman recursion operates on monthly time steps. A sample of $T^* = 200$ months corresponds to roughly $T^*/3 \approx 67$ quarters of quarterly observations, which is the relevant unit when interpreting the recovery of the quarterly target variable (e.g. GDP). We therefore interpret the grid as values of monthly T^* throughout, and report quarterly equivalents whenever the discussion turns to quarterly aggregates.

Computational Cost

The full design covers four dimensions of variation, which together determine the total computational effort. Before discussing the wall-clock implications, we briefly recall what each dimension contains and why.

The four dimensions of variation.

- *Replications* ($S = 1000$): the number of independent synthetic samples generated for each parameter configuration. As discussed in the previous subsection, $S = 1000$ delivers a Monte Carlo standard error small enough to support reliable comparisons between estimators.

- *Sample sizes* ($|\mathcal{T}^*| = 5$): the grid of synthetic sample lengths $\mathcal{T}^* = \{100, 200, 400, 800, T_{\text{real}}\}$, ranging from a fourth of the empirical sample (stress test) to twice the empirical sample (asymptotic approximation), with the headline case $T^* = T_{\text{real}}$ at the centre of the grid. Each value plays a distinct role in characterising the finite-sample behaviour of the estimator.

- *Contamination levels* ($|\Pi| = 4$): the grid of contamination intensities $\Pi = \{0, 0.01, 0.05, 0.10\}$, ranging from no contamination (the case in which the data are generated from the assumed DGP without any extreme periods) to ten-percent contamination (a non-trivial fraction of time periods governed by an outlier-generating mechanism). This grid traces the continuous path from the Gaussian benchmark of Experiment B (at $\pi = 0$) to the progressively more contaminated regimes of Experiment C.

- *Estimators* ($|\mathcal{E}| = 2$): the two estimators applied to each synthetic sample — the proposed Student-t EM EM_t and the Gaussian EM of Bańbura and Modugno (2014), EM_{Gauss}. Each estimator is run on each combination of the other three dimensions, allowing the direct head-to-head comparison that is the methodological core of the validation.

Total number of EM runs. The total number of EM runs is the product of the four dimensions:

$$S \cdot |\mathcal{T}^*| \cdot |\Pi| \cdot |\mathcal{E}| \;=\; 1000 \cdot 5 \cdot 4 \cdot 2 \;=\; 40{,}000.$$

Each EM run has cost of order $O(T^* \cdot r^3)$ per outer iteration, multiplied by the number of outer iterations to convergence — typically 50–200 in our setting. On a single modern workstation, the wall-clock time of a single EM run on a panel of the size of our real data is of the order of seconds; the full 40,000 runs therefore translate to wall-clock times in the range of tens of hours on a single core.

Parallelisation strategy. As discussed in the previous subsection, the Monte Carlo loop is *embarrassingly parallel*: each (s, T^*, π, EM) combination is an independent re-estimation that requires no communication with the others. We exploit this by distributing the workload across the available cores, with each core handling a slice of the replications. This reduces the wall-clock time to a few hours on a modern multi-core workstation, and to a fraction of an hour on a small compute cluster. The deterministic random seeds (function of s, T^*, and π) guarantee that the results are reproducible regardless of the parallelisation pattern adopted.

Practical scaling considerations. The 40,000 EM runs are budgeted under the design described above. Adjustments are straightforward:

- Reducing S to 100 (for exploratory work or for time-constrained settings) divides the cost by 10, bringing the experiment within a few hours on a single machine.

- Pruning the grid $\mathcal{T}^*$ (e.g. keeping only $T^* \in \{T_{\text{real}}, 2T_{\text{real}}\}$) divides the cost by another factor of 2 or 3, at the price of losing the finite-sample convergence analysis.
- Reducing $|\Pi|$ to a single intermediate value ($\pi = 0.05$) reduces the contamination grid to a single comparison point, at the price of losing the robustness curve.

We adopt the full design for the main reported results, and selectively prune it for sensitivity analyses where appropriate.

Reproducibility

The Monte Carlo design is built to be fully reproducible. Each synthetic dataset $\mathbf{Y}^{(s)}$ is generated from a deterministic random seed that is a function of the triple (s, T^*, π). The seeds are recorded together with the estimation results, so that any individual replication can be re-generated from the same initial conditions and the same synthetic data. The post-hoc normalisation and the Procrustes alignment are themselves deterministic given the inputs.

The codebase is structured so that the calibration (Algorithm 3), the data generation, and the re-estimation (Algorithm 4) can each be executed independently. The aggregated outputs — tables of bias, variance, RMSE, factor correlations, and convergence diagnostics — are stored alongside the raw replication-level results, so that any downstream analysis can be re-run without re-executing the underlying simulation.

Looking Ahead: The Second Stage

The Monte Carlo experiment described in this section validates the first stage of our two-stage methodology: the EM-based estimation of the Student-t DFM. A separate question is whether the estimated factors $\{\hat{f}_t^{(s)}\}$ are sufficiently informative *regressors* for the second-stage quantile regression used in the Growth-at-Risk component of the methodology — that is, whether the propagation of first-stage estimation error into the second-stage quantile estimates preserves the validity of the conditional quantile forecasts.

The methodological challenge. The second-stage Monte Carlo design is more delicate than the first. The quantile regression of Koenker and Bassett (1978) is *distribution-free*: it does not assume any specific form for the conditional distribution of GDP given the factors. The decision of how to simulate a ground-truth conditional distribution for GDP — and how to assess the second-stage performance against it — involves methodological choices that go beyond the calibrated-DGP strategy adopted for the first stage. Two natural options have to be evaluated:

- Specifying conditional quantile coefficients $\beta_0(\tau), \beta_1(\tau)$ that vary with τ, so that the implicit conditional distribution emerges from the variation in coefficients (the "implicit" approach, most faithful to the Koenker–Bassett framework);
- Specifying a parametric ground-truth conditional density for GDP — for instance, the skewed Student-t of Adrian, Boyarchenko, and Giannone (2019), with parameters depending on the factors — and deriving the quantiles analytically (the "explicit" approach, more tractable for tail-focused metrics).

Each option has its trade-offs. The implicit approach is more econometrically conservative; the explicit approach allows for sharper analytical comparisons.

Beyond conditional quantiles: generated regressors. Independently of the choice of ground-truth distribution, the second stage involves a *generated regressors problem* (Pagan, 1984): the regressors in the second-stage quantile regression are themselves estimates produced by the first stage, and carry their own estimation error. The Monte Carlo can quantify how this error propagates into the second-stage quantile estimates, and whether the resulting bias or coverage distortion is substantively important for the Growth-at-Risk application. This question is genuinely interesting from an econometric standpoint and has no clear analogue in the existing Growth-at-Risk literature, which typically treats the first-stage regressors as observable.

Scope of the present work. We defer the second-stage Monte Carlo design to a separate methodological chapter. The infrastructure developed in this section — the calibrated DGP, the simulation pipeline, the replication architecture — is built to accommodate the second-stage extension once the methodological choices have been resolved. In particular, the synthetic factor paths $\{f_t^{(s)}\}$ and their EM-recovered counterparts $\{\hat{f}_t^{(s)}\}$ are already generated and stored as part of the first-stage Monte Carlo: they serve as the natural input to the second-stage simulation as soon as the corresponding ground-truth conditional distribution for GDP has been specified.

The first-stage Monte Carlo described in this section is therefore a self-contained validation exercise, but also the foundation on which the second-stage validation will be built. The two-stage character of the methodology is reflected in the two-stage character of its validation strategy.

A Limitation of Idiosyncratic Robustness: Common Extreme Movements

The robustness of the Student-t specification, which constitutes the central methodological advantage of the First Stage, rests on the scale-mixture mechanism: an observation that produces a large idiosyncratic residual receives a small posterior weight w_t^ε and is therefore discounted in the M-step. This is precisely the desired behaviour when the large residual reflects *idiosyncratic* contamination — a measurement error, an anomalous data revision, a one-off disturbance confined to a single series — as deliberately constructed in the contamination experiment. In that setting, down-weighting protects the estimates of $\mathbf{\Lambda}$, $\mathbf{A}$, $\mathbf{Q}$ and $\mathbf{R}$ from being distorted by noise that carries no information about the common factors.

A more delicate situation arises, however, when the extreme observation is not idiosyncratic noise but the manifestation of a *common* extreme movement — a systemic event such as the 2008 financial crisis or the 2020 pandemic contraction, in which all series move sharply and *in a correlated fashion*. Here the large residuals are not noise to be discarded: they are the signal itself. The scale-mixture mechanism, which weights each period solely on the magnitude of its idiosyncratic residual, cannot by construction distinguish the two cases. Confronted with a period in which every series records an extreme reading, the Student-t estimator interprets the resulting large residuals as idiosyncratic outliers and assigns a vanishingly small weight w_t^ε, thereby discarding precisely the information that identifies the crisis. The phenomenon is self-reinforcing: a factor path calibrated to ordinary, moderate fluctuations fails to track the abrupt common movement, producing a large Mahalanobis residual, which in turn drives w_t^ε towards zero, which further suppresses the model's response to the event.

This limitation is not a numerical artefact but a structural feature of any estimator that achieves robustness through residual magnitude alone. It is therefore worth stating the underlying distinction precisely. Write the idiosyncratic residual at time t as $\boldsymbol{\varepsilon}_t = \mathbf{y}_t - \mathbf{\Lambda}\mathbf{f}_t$. A large $\|\boldsymbol{\varepsilon}_t\|$ admits two qualitatively different explanations:

1. an *idiosyncratic* outlier, in which one (or few) coordinates of $\boldsymbol{\varepsilon}_t$ are large while the others remain ordinary and the cross-sectional pattern is *uncorrelated* — the case in which down-weighting is correct;
2. a *common* extreme, in which the coordinates of $\boldsymbol{\varepsilon}_t$ are large *and mutually correlated*, reflecting a factor movement that the fitted dynamics failed to anticipate — the case in which down-weighting is harmful.

The current specification reacts identically to both, because the posterior weight depends on $\boldsymbol{\varepsilon}_t$ only through a scalar measure of its size, ignoring its cross-sectional structure. Distinguishing the two regimes — discounting case (1) while preserving case (2) — would require enriching the model so that the weighting mechanism becomes sensitive to whether an extreme is shared across the cross-section or confined to it. We outline four routes, in increasing order of ambition, and indicate why a definitive treatment lies beyond the scope of the present work.

(i) Heavy tails on the common factor. The most parsimonious remedy works within the existing architecture. The model already endows the factor innovation with heavy tails, $\mathbf{u}_t \sim t_{\nu_u}(\mathbf{0}, \mathbf{Q})$, through its own mixing weight w_t^u. In principle a common extreme should be absorbed by a large factor innovation — the factor itself moving sharply — rather than discarded as idiosyncratic noise. In practice, however, the EM algorithm tends to attribute correlated extreme residuals to the idiosyncratic component, because lowering w_t^ε offers a cheaper local improvement of the objective than accommodating a large factor shock. A targeted reform would bias this attribution towards the common channel: for instance by allowing a heavier tail on the factor (a smaller ν_u) or by re-parametrising the relative scales of w_t^u and w_t^ε so that, when residuals are correlated, the factor weight rather than the idiosyncratic weight adjusts. This route remains close to the estimated model and to the heavy-tailed dynamic-factor tradition (Antolín-Díaz et al.), but it does not by itself *identify* the common-versus-idiosyncratic nature of an extreme; it only shifts the default attribution.

(ii) Cross-sectional weighting. A more direct attack replaces the magnitude-based weight with one that is explicitly sensitive to the cross-sectional structure of the residuals. The intuition is that an extreme which is *shared* across many series in a correlated way is common (and should be retained), whereas one *confined* to a single series is idiosyncratic (and should be discounted). Concretely, in place of a weight driven by the scalar Mahalanobis distance $d_t = \boldsymbol{\varepsilon}_t^\top \mathbf{R}^{-1} \boldsymbol{\varepsilon}_t$, one could construct a statistic that decomposes the residual vector into a component aligned with the loading structure (the common part) and a residual orthogonal to it (the genuinely idiosyncratic part), and down-weight only the latter. A period in which the large residuals project strongly onto $\boldsymbol{\Lambda}$ — i.e. are explained by a coordinated factor movement — would retain a high weight, whereas a period whose large residuals are orthogonal to the common structure would be discounted. This is no longer a standard scale-mixture: it requires defining and justifying a new weighting statistic, establishing the properties of the resulting (non-standard) EM updates, and verifying that the modified algorithm retains the monotonicity and convergence guarantees that the scale-mixture representation confers automatically. It is, in effect, an original robust estimator, and its development would constitute a research contribution in its own right.

(iii) Separate weights for factor and idiosyncratic channels. The present model already maintains two distinct weight sequences, w_t^u for the factor innovation and w_t^ε for the idiosyncratic shock, but the mechanism that determines their relative adjustment is driven by the likelihood and not by any explicit notion of commonality. A refinement would make the split deliberate: when the cross-section of residuals is coherent (suggesting a common event), the adjustment is loaded onto w_t^u, allowing the factor to move and the common signal to be preserved; when the residual is isolated (suggesting idiosyncratic noise), the adjustment is loaded onto w_t^ε, discounting the single aberrant series. This is conceptually related to route (ii) — both hinge on reading the cross-sectional structure — but operates through the existing two-weight decomposition rather than through a redefined statistic. It shares the same difficulty: the coupling between the two weights would have to be specified and its estimation-theoretic consequences worked out.

(iv) Regime-switching robustness. The most ambitious route treats the distinction between regimes as a latent state to be inferred. One would posit two regimes governing the weighting mechanism: a "tranquil" regime, in which the Student-t down-weighting operates as in the present model and protects the estimates from idiosyncratic noise; and a "crisis" regime, in which the mechanism is switched off (or its tails are made effectively Gaussian and reactive) so that common extremes are absorbed rather than discarded. A Markov-switching process would govern the transition between the two, and the regime probabilities would be inferred from the data — in particular from the cross-sectional coherence and persistence of the extremes. Such a *Markov-switching dynamic factor model with state-dependent robustness* is conceptually appealing precisely because it formalises the idea that an estimator should be robust in normal times and reactive in crises. It is, however, considerably more complex than the present framework: it would require a substantially expanded state space, an inference scheme combining the Kalman filter with a Hamilton-type regime filter, and a careful treatment of the identification and convergence of the resulting algorithm. We regard it as a natural but distinct line of research rather than an extension to be undertaken here.

Relation to the Second Stage. Crucially, the architecture of this thesis already addresses the limitation at the system level, through the division of labour between its two stages, so that no single estimator is asked to be simultaneously robust and reactive. The First Stage is charged with extracting clean, robustly estimated factors and parameters, shielded from idiosyncratic contamination; for that task, discounting large idiosyncratic residuals is exactly right. The capture of *tail risk* — the asymmetry and the heavy lower tail of the GDP distribution that materialise precisely during systemic events — is instead the responsibility of the Second Stage, in which quantile regression (in the spirit of Adrian, Boyarchenko and Giannone) models how the *conditional distribution* of GDP, including its extreme quantiles, depends on the factors delivered by the First Stage. The smoothing of common extremes by the robust First Stage is therefore not an unaddressed defect but a motivated design choice: the First Stage supplies conditioning information purged of idiosyncratic noise, and the Second Stage reads the risk of extreme common outcomes off that information. Viewed in this light, the empirical evidence that the robust First Stage attenuates the 2008 and 2020 contractions does not weaken the framework; it provides a concrete rationale for the two-stage construction, by demonstrating why a single robust filter cannot, on its own, serve both the estimation of common dynamics and the assessment of tail risk.

The Second Stage: From Point Nowcasts to a Predictive Density

The limit of the point nowcast. The first stage delivers, for each reference quarter, a single number: the conditional mean of GDP growth implied by the dynamic factor model. As a summary of the central tendency of the economy this object is informative, and it is the natural output of a model estimated by least-squares-type criteria. Yet for the two uses that motivate a nowcasting exercise in the first place — the conduct of policy and the assessment of risk — a point is not enough. A policymaker does not only want to know where growth is expected to land; she wants to know how *uncertain* that expectation is and, above all, how *large the danger of an adverse outcome* is: the probability that the economy slips into recession, the depth of a plausible bad scenario, the magnitude of the loss that would be exceeded only with small probability. None of these quantities is a statement about the centre of the distribution; each is a statement about its *left tail*. What is required, therefore, is not a point but a *predictive density* for GDP growth, and in particular a faithful description of its lower quantiles. This is the object that the *Growth-at-Risk* literature places at the centre of macroeconomic monitoring, and it is the object that the second stage is built to recover.

Every probabilistic model already implies a density — but a rigid one. It would be a mistake to think that moving from a point to a distribution requires abandoning the model and starting afresh. Any probabilistic model already *implies* a predictive distribution: around the conditional mean it places a whole density, whose dispersion quantifies forecast uncertainty. The difficulty is not the existence of this implied density but its *shape*. In a model with a *fixed* (homoskedastic) variance, the predictive density is symmetric about the point forecast and of *constant width*: the spread it assigns to the next observation is the same in every period, because it is governed by a single, time-invariant variance parameter. Such a density cannot, by construction, represent the central stylised fact of macroeconomic risk — that uncertainty is *not* constant. In tranquil times the distribution of growth is narrow and roughly symmetric; in turbulent times it ought to *widen* and develop a long, heavy *lower* tail, as probability mass shifts towards severely negative outcomes. A constant-width, symmetric density is frozen in its tranquil-times shape: it neither widens in a crisis nor turns asymmetric, and so it systematically understates downside risk exactly when that risk is greatest. This is the conceptual ceiling of *classical density nowcasting*: it does deliver a density, but a rigid one, blind to the regime.

Quantile regression as a remedy — and its limitations. One way to obtain a density whose shape responds to the regime, without first solving the harder problem of making the model's own variance time-varying, is to model the quantiles *directly*. This is the route of the Growth-at-Risk literature: instead of a single conditional mean, one estimates a family of conditional quantiles of GDP growth as functions of a set of predictors, allowing each quantile its own sensitivity. The lower quantiles are then free to react more strongly than the upper ones, so that the left tail widens precisely when the predictors signal danger, reproducing the asymmetric, regime-dependent shape that the fixed-variance density cannot. Operationally this is quantile regression: for a quantile level τ the coefficients minimise the asymmetric check (pinball) loss, and the fitted value estimates the conditional τ-quantile rather than the mean. As a remedy this is effective, but it carries two limitations that are worth separating.

(a) Quantile regression on the factors of a block model. Suppose the predictors are the factors of a factor model in which each factor is constructed to explain the series of *its own* block. Then the factor of the block to which the target variable belongs will, by construction, absorb a large share of the target's own dynamics, while the factors of the other blocks — built to summarise

variables of a different nature — carry comparatively little information about it. In a quantile regression of the target on all the factors, the own-block factor tends to *dominate* the tail: it is the one whose movements line up with the target's, so it is the one whose coefficient the check-loss makes large, while the remaining factors contribute only marginally. The estimated tail is then essentially a one-factor object, and the multivariate conditioning set is, in effect, collapsed onto the single block to which the target was assigned. This is a structural feature of block factor models, stated here only as a conceptual point and without reference to any particular empirical configuration.

(b) Quantile regression on the own-block factor alone. One may respond by regressing the target's quantiles on its own-block factor only. This works: the regression maps the extreme realisations of that factor into a steeply sloped lower tail, and the resulting quantiles do widen in the right episodes. But notice what kind of object this is. It is a *separate* regression, estimated on top of an already estimated factor model — a second, free-standing equation bolted onto the first stage. The predictive density it produces is not *generated* by the model that produced the factors; it is reconstructed, after the fact, by fitting a curve through estimated quantiles. The two stages are coherent only by stipulation, not by construction. What is missing is *generative coherence*: a single probabilistic model that, on its own, implies the regime-dependent density we are after.

Solving the problem inside the model: stochastic volatility and Markov switching. The deeper remedy is therefore to attack the problem *inside* the model rather than alongside it. The reason the classical density was rigid is that its variance was fixed; the reason quantile regression feels like an add-on is that it leaves the model's variance fixed and repairs the symptom externally. If, instead, we endow the model's *volatility* with a *dynamic* of its own, the predictive density inherits that dynamic directly: it *widens by itself* in turbulent regimes and contracts in calm ones, and the heavy, asymmetric lower tail emerges as a property of a *single, generative* model rather than of an auxiliary regression. There are two canonical ways to give volatility a dynamic, and we present both on an equal footing before committing to one.

Stochastic volatility (SV). Here the (log-)volatility is itself a latent stochastic process that evolves *continuously* and *persistently* over time, typically as a first-order autoregression in logs,

$$\log \sigma_t^2 \;=\; c + \phi \, \log \sigma_{t-1}^2 + \eta_t.$$

Volatility drifts smoothly, with shocks that fade gradually rather than switching abruptly; the persistence parameter ϕ controls how long a high-volatility phase lingers. The familiar fixed-volatility model is nested as the special case in which this process has *no* persistence and no innovation variance, so that σ_t^2 collapses to a constant. Stochastic volatility thus generalises the homoskedastic model along the single dimension that matters for tail risk.

Markov switching. Here volatility is governed not by a continuous process but by a small number of *discrete states* — for instance a "calm" regime and a "crisis" regime — with the economy moving between them according to a Markov chain whose transition matrix encodes how likely a switch is and how persistent each regime tends to be. The predictive density is then a mixture across regimes, and its tail thickens whenever the crisis state carries appreciable probability.

Both routes deliver what quantile regression cannot: a predictive density that is regime-dependent *by construction*, produced by one internally coherent model. This is the cardinal point of the whole second stage. Quantile regression repairs the density from the *outside*, as a separate equation appended to the factor model; stochastic volatility and Markov switching repair it from the *inside*, by making time-varying risk a generative feature of the model itself. It is precisely this difference — an external patch versus internal, coherent generation — that justifies moving beyond the quantile-regression remedy. The detailed development that follows takes the *stochastic-volatility* route as its main line, while retaining Markov switching as the natural discrete-state alternative; the next section sets out the specific variants in full.

Formal Specification of the Stochastic-Volatility Variants

We now make the variants explicit. To avoid carrying out a separate derivation for each of them, we first restate the current model in our own notation, then embed it — together with all the proposed variants — inside a single *parent specification* controlled by a small set of switches. Each variant is then a particular setting of those switches, and the nesting of the current model is a matter of reading off which switches to disable. Throughout we specify *only the data-generating process*; the estimation of the additional latent volatility states is deferred to the sections that follow.

The baseline DGP, recalled. In the notation of (15)–(16), the model estimated in the first stage is the heavy-tailed dynamic factor model written as a Gaussian scale mixture,

$$f_t = \mathbf{A}\, f_{t-1} + u_t, \qquad u_t \mid w_t^u \sim \mathcal{N}\big(0,\, \mathbf{Q}/w_t^u\big), \qquad w_t^u \overset{iid}{\sim} \text{Gamma}\big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\big), \tag{173}$$

$$y_t = \mathbf{\Lambda}\, f_t + \varepsilon_t, \qquad \varepsilon_t \mid w_t^\varepsilon \sim \mathcal{N}\big(0,\, \mathbf{R}/w_t^\varepsilon\big), \qquad w_t^\varepsilon \overset{iid}{\sim} \text{Gamma}\big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\big), \tag{174}$$

with $\mathbf{R} = \text{diag}(r_1, \ldots, r_M)$ and the weights mutually independent across time and across the two families. The single scalar weight w_t^u scales the entire factor-innovation vector, and the single scalar weight w_t^ε scales the entire idiosyncratic vector; the conditional variance of each shock is therefore *constant up to its iid tail weight*. There is no notion of a *persistent* swing in volatility: a calm month and the month at the centre of a crisis carry the same scale matrices $\mathbf{Q}, \mathbf{R}$, modulated only by independent draws of w_t^u, w_t^ε that carry no memory from one period to the next. This is exactly the rigidity diagnosed above: the density implied by (173)–(174) cannot widen of its own accord in a turbulent regime.

A parent specification with switches. The single idea behind all the variants is to let the *conditional variance* of each shock vary over time, and to do so through *two* multiplicative channels that play distinct roles. We factor that variance into a fixed base scale — the matrices $\mathbf{Q}$ and $\mathbf{R}$ of the baseline — times two strictly positive multipliers: a persistent stochastic-volatility term h_t and a tail term s_t. For the factor innovation this reads

$$u_t \mid w_t^u, h_t^u \;\sim\; \mathcal{N}\Big(0,\; \underbrace{\mathbf{Q}}_{\substack{\text{base}\\ \text{scale}}} \cdot \underbrace{h_t^u}_{\substack{\text{persistent}\\ \text{volatility}}} \underbrace{\tfrac{1}{w_t^u}}_{\substack{\text{tail}\\ s_t^u}}\Big),$$

and for the idiosyncratic vector, writing $\mathbf{H}_t^\varepsilon = \text{diag}(h_{1,t}^\varepsilon, \ldots, h_{M,t}^\varepsilon)$,

$$\varepsilon_t \mid w_t^\varepsilon, \mathbf{H}_t^\varepsilon \;\sim\; \mathcal{N}\big(0,\; \mathbf{H}_t^\varepsilon\, \mathbf{R} \,/\, w_t^\varepsilon\big),$$

so that series i alone carries the conditional variance

$$\text{Var}\big(\varepsilon_{i,t} \mid \cdot\big) \;=\; \underbrace{r_i}_{\substack{\text{base}\\ \text{scale}}} \cdot \underbrace{h_{i,t}^\varepsilon}_{\substack{\text{persistent}\\ \text{volatility}}} \cdot \underbrace{\tfrac{1}{w_t^\varepsilon}}_{\substack{\text{tail}\\ s_t^\varepsilon}}$$

The two multipliers are deliberately different objects, and it is worth saying plainly what each one is and what it does.

- h_t **— the persistent volatility (slow, with memory).** A strictly positive process that rescales the *entire* conditional variance smoothly over time. It sits near one in normal times, drifts up as a turbulent regime sets in and — crucially — *stays* up for a while once it has risen, because it evolves with persistence. This is the channel that lets the predictive density "breathe": widen during a crisis and contract again as calm returns. Its signature in the data is *volatility clustering* (turbulent months bunch together), and that persistence is exactly what identifies it.
- s_t **— the tail multiplier (abrupt, memoryless).** A strictly positive but *serially independent* rescaling, drawn afresh each period. Having no memory it cannot describe a regime, but it does what a smooth process cannot: it absorbs a single, isolated observation lying many standard deviations from its neighbours. It is therefore the channel responsible for the *heavy-tailed shape* of the distribution at a given instant, and its functional form is the object of switch D1 below. Its signature in the data is the *isolated spike*.

The persistent component follows, for each active state, an independent log-volatility AR(1) process,

$$\log h_t^u = \mu_u + \phi_u\big(\log h_{t-1}^u - \mu_u\big) + \eta_t^u, \qquad \eta_t^u \sim \mathcal{N}(0, \sigma_u^2), \tag{175}$$

$$\log h_{i,t}^\varepsilon = \mu_{\varepsilon,i} + \phi_{\varepsilon,i}\big(\log h_{i,t-1}^\varepsilon - \mu_{\varepsilon,i}\big) + \eta_{i,t}^\varepsilon, \qquad \eta_{i,t}^\varepsilon \sim \mathcal{N}(0, \sigma_{\varepsilon,i}^2), \tag{176}$$

whose three parameters read cleanly: μ is the long-run level of log-volatility (normalised so that $\mu = 0$ lets the process fluctuate around $h_t = 1$; once *frozen*, $\sigma^2 = 0$ and $\mu = 0$, it sits at $h_t \equiv 1$ exactly); $\phi \in (-1, 1)$ is the *persistence* — how long a high-volatility phase lingers, with ϕ close to one meaning slow, drawn-out regimes; and σ^2 is the variance of the log-volatility innovation, i.e. how far volatility can move from one period to the next. All η innovations are mutually independent and independent of the tail weights and of the base shocks; this keeps the two channels statistically distinct, so that h_t is identified by the *persistence* of volatility and s_t by *isolated spikes*.

Within this parent specification, a variant is fixed by *two independent switches, one acting on each multiplier*: switch D1 chooses the *form of the tail multiplier s_t*, while switch D2 chooses *which volatility multipliers h_t* are allowed to move. The first reshapes the tails of the per-period distribution; the second decides where time-varying volatility is active. Because s_t and h_t enter the variance as separate factors, the two switches operate independently and combine freely.

Switch D1 — the tail multiplier s_t (the *shape* of the per-period tails): it sets what multiplies the base scale period by period.

D1-a $s_t \equiv 1$ — the tail channel is off (degenerate weights $w_t \equiv 1$), so the shock is Gaussian given its volatility: *thin tails*;

D1-b $s_t = 1/w_t$, $w_t \sim \text{Gamma}(\nu/2, \nu/2)$ — the Student-t tail of the current model: *heavy tails* from a continuous, every-period mixing weight;

D1-c s_t an *outlier* multiplier (defined below) — a Gaussian base inflated at rare, large events, whose timing is either fixed in advance (*dated* version) or inferred by the model (*stochastic* version): *heavy tails from a few identified events*.

Switch D2 — the volatility scope (which h_t are allowed to *move*): a state is *active* when its innovation variance $\sigma^2 > 0$ and *frozen* when $\sigma^2 = 0$, $\mu = 0$ (forcing $h_t \equiv 1$ for all t — the persistence ϕ then irrelevant — which switches the channel off). The two scopes are:

D2-a only the common factor volatility h_t^u is active;

D2-b the common volatility *and* every idiosyncratic volatility $h^{\varepsilon}_{i,t}$, $i = 1, \dots, M$, are active.

The tail-mechanism axis (D1). Holding the volatility scope (the active D2 set) fixed, the three D1 variants change only the tail multiplier s_t. For each we write out the *complete* DGP and then the model it reduces to. They do *not* all nest the same baseline: only the Student-t choice (D1-b) recovers the heavy-tailed model estimated in the first stage, whereas D1-a and D1-c reduce to a thin-tailed Gaussian DFM.

D1-a *(pure stochastic volatility).* The tail channel is switched off, $s_t \equiv 1$ (i.e. $w^u_t \equiv w^{\varepsilon}_t \equiv 1$), so the shocks are Gaussian conditional on their volatility states. The complete DGP is

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad && u_t \mid h^u_t \sim \mathcal{N}\big(0,\, h^u_t\, \mathbf{Q}\big), \\ y_t &= \boldsymbol{\Lambda}\, f_t + \varepsilon_t, && \varepsilon_t \mid \mathbf{H}^{\varepsilon}_t \sim \mathcal{N}\big(0,\, \mathbf{H}^{\varepsilon}_t\, \mathbf{R}\big), \end{aligned}$$

with $\mathbf{H}^{\varepsilon}_t = \operatorname{diag}(h^{\varepsilon}_{1,t}, \dots, h^{\varepsilon}_{M,t})$ and the active log-volatilities following (175)–(176). Heavy tails come *only* from movements in h_t; a smooth, persistent h_t widens the density gradually and cannot reproduce an isolated one-period jump.

Nesting. Freezing every volatility state ($\sigma^2 = 0, \mu = 0 \Rightarrow h^u_t \equiv 1$, $\mathbf{H}^{\varepsilon}_t \equiv \mathbf{I}$) removes the SV channel and leaves the thin-tailed **Gaussian DFM**,

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad && u_t \sim \mathcal{N}(0,\, \mathbf{Q}), \\ y_t &= \boldsymbol{\Lambda}\, f_t + \varepsilon_t, && \varepsilon_t \sim \mathcal{N}(0,\, \mathbf{R}). \end{aligned}$$

Because $s_t \equiv 1$, no setting of the parameters recovers our Student-t model: D1-a nests the Gaussian DFM *only*.

D1-b *(stochastic volatility with Student-t tails).* Keep the Gamma weights, $s_t = 1/w_t$, on top of the volatility states. The complete DGP is

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad && u_t \mid w^u_t, h^u_t \sim \mathcal{N}\big(0,\, h^u_t\, \mathbf{Q}/w^u_t\big), \\ y_t &= \boldsymbol{\Lambda}\, f_t + \varepsilon_t, && \varepsilon_t \mid w^{\varepsilon}_t, \mathbf{H}^{\varepsilon}_t \sim \mathcal{N}\big(0,\, \mathbf{H}^{\varepsilon}_t\, \mathbf{R}/w^{\varepsilon}_t\big), \end{aligned}$$

with $w^u_t \sim \text{Gamma}(\nu_u/2, \nu_u/2)$, $w^{\varepsilon}_t \sim \text{Gamma}(\nu_{\varepsilon}/2, \nu_{\varepsilon}/2)$, and the active log-volatilities as in (175)–(176). Two sources of tail behaviour act together: persistent volatility (h_t) for slow regime shifts and the Student-t weight ($1/w_t$) for abrupt within-regime outliers.

Nesting. Freezing every volatility state ($h^u_t \equiv 1$, $\mathbf{H}^{\varepsilon}_t \equiv \mathbf{I}$) returns **our Student-$t$ DFM** (173)–(174),

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad && u_t \mid w^u_t \sim \mathcal{N}\big(0,\, \mathbf{Q}/w^u_t\big), \\ y_t &= \boldsymbol{\Lambda}\, f_t + \varepsilon_t, && \varepsilon_t \mid w^{\varepsilon}_t \sim \mathcal{N}\big(0,\, \mathbf{R}/w^{\varepsilon}_t\big), \end{aligned}$$

the first-stage model — and *this is the only D1 variant that does so.* Letting in addition $\nu_u, \nu_{\varepsilon} \to \infty$ collapses $1/w_t \to 1$ and yields the Gaussian DFM, so this column alone realises the chain Gaussian $\subset$ our Student-t $\subset$ Student-t-with-SV.

D1-c *(stochastic volatility with explicit outliers).* Replace the continuous tail weight by a Gaussian base scaled by a discrete outlier multiplier $s_t = 1 + (\kappa^2 - 1)\, o_t$ ($\kappa > 1$, $o_t \in \{0, 1\}$). The complete DGP is

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad && u_t \mid h^u_t, o^u_t \sim \mathcal{N}\big(0,\, (1 + (\kappa_u^2 - 1) o^u_t)\, h^u_t\, \mathbf{Q}\big), \\ y_t &= \boldsymbol{\Lambda}\, f_t + \varepsilon_t, && \varepsilon_t \mid \mathbf{H}^{\varepsilon}_t, o^{\varepsilon}_t \sim \mathcal{N}\big(0,\, (1 + (\kappa_{\varepsilon}^2 - 1) o^{\varepsilon}_t)\, \mathbf{H}^{\varepsilon}_t\, \mathbf{R}\big), \end{aligned}$$

where the indicators $o^u_t, o^{\varepsilon}_t \in \{0, 1\}$ flag the rare large events, $\kappa_u, \kappa_{\varepsilon} > 1$ set their size, and the active log-volatilities follow (175)–(176). Tail mass is now attributed to rare, identified events rather than to a heavy-tailed law operating in every period. The mechanism comes in *two versions*, differing in *who decides when an outlier occurs.*

Stochastic version (preferred). The indicators are *latent and estimated*: $o_t^u, o_t^\varepsilon \sim \text{Bernoulli}(p)$ with a small occurrence probability p to be inferred, so that the model itself discovers which periods are outliers from the data. This is the honest, automatic option — it needs no manual intervention and no look-ahead in identifying the events, and it stays within the spirit of a single generative model. Its limit, which we state plainly, is that identification is *hard*: the model must separate a "rare outlier" from "persistent high volatility" — exactly the distinction D1-c is meant to draw — and with very few extreme episodes in the sample (the 2020 collapse is essentially a *single* event) the data carry little information with which to estimate p and the size parameters κ. Learning what outliers "look like" from one or two observations is statistically fragile.

Dated version (robust fallback). The indicators are instead *fixed by the researcher* at known dates, $o_t = \mathbb{1}\{t \in \mathcal{O}\}$ for a pre-specified calendar set $\mathcal{O}$ of recognised disruptions (the Covid months of 2020 being the leading case); the model is told *when* the outliers are and estimates only their size κ. This buys very clean identification — there is no volatility-versus-outlier confusion, because the timing is given — and it is exactly the device of Carriero, Clark, Marcellino and Mertens (2022) for keeping the 2020 observations from distorting the estimated volatility process. The price is that it is interventionist and *ex post*: it uses hindsight about which dates were anomalous (it is not a predictive mechanism), and the choice of $\mathcal{O}$ is to some degree ad hoc. It is best read as a robustness device for *estimation* rather than as a generative model of tail events.

We adopt the stochastic version as the default, on grounds of honesty and coherence with the generative framework, and fall back on the dated version when the stochastic one fails to identify the events cleanly — which, in a sample containing essentially one pandemic, is a genuine possibility, and is why the dated treatment remains the literature standard for the Covid period.

Nesting. Switching the outliers off — either $\kappa_u = \kappa_\varepsilon = 1$, or $o_t^u = o_t^\varepsilon = 0$ for all t (in the dated version this means $\mathcal{O} = \varnothing$, in the stochastic version $p = 0$) — and freezing every volatility state leaves the **Gaussian DFM**,

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad & u_t &\sim \mathcal{N}(0, \mathbf{Q}), \\ y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \mathbf{R}). \end{aligned}$$

The base law is Gaussian, so — exactly like D1-a — D1-c nests the Gaussian DFM but *not* our Student-t model.

The volatility-scope axis (D2). The second switch fixes which volatility states are active:

D2-a *common only* — only h_t^u is active, every idiosyncratic state frozen ($\mathbf{H}_t^\varepsilon \equiv \mathbf{I}$); systemic, economy-wide swings in volatility are captured, series-specific ones are not.

D2-b *common plus all idiosyncratic* — h_t^u and every $h_{i,t}^\varepsilon$ are active, so each series carries its own persistent volatility on top of the common one. This is the most general scope and our preferred one.

We deliberately omit a scope that activates idiosyncratic volatility on the *target series alone*: it would privilege GDP for no structural reason — only because GDP happens to be this thesis's contingent target — and would make no sense were the target another, similar series (industrial production, say). A target-specific volatility is a property of the application, not of the model, so we do not consider it.

The six variants as instances of the parent. The two switches are *orthogonal*, so combining D1 $\in \{a, b, c\}$ with D2 $\in \{a, b\}$ yields *six* models. They share everything except the shock variances — the measurement equation $y_t = \mathbf{\Lambda} f_t + \varepsilon_t$, the factor transition $f_t = \mathbf{A} f_{t-1} + u_t$, the Gamma weights where present, and the log-volatility processes (175)–(176) for the active states — and differ *only* in the tail multiplier s_t and in which h are active. Writing h_t^u for the (always-active) common volatility and $\mathbf{H}_t^\varepsilon = \operatorname{diag}(h_{1,t}^\varepsilon, \ldots, h_{M,t}^\varepsilon)$ for the idiosyncratic ones (with $\mathbf{H}_t^\varepsilon \equiv \mathbf{I}$ under D2-a), the six conditional laws of the shocks are:

D1a-D2a $u_t \mid \cdot \sim \mathcal{N}(0, h_t^u \mathbf{Q}),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, \mathbf{R})$. *Nests* the Gaussian DFM.

D1a-D2b $u_t \mid \cdot \sim \mathcal{N}(0, h_t^u \mathbf{Q}),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, \mathbf{H}_t^\varepsilon \mathbf{R})$. *Nests* the Gaussian DFM.

D1b-D2a $u_t \mid \cdot \sim \mathcal{N}(0, h_t^u \mathbf{Q}/w_t^u),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, \mathbf{R}/w_t^\varepsilon)$. *Nests* our Student-*t* DFM.

D1b-D2b *(preferred)* $u_t \mid \cdot \sim \mathcal{N}(0, h_t^u \mathbf{Q}/w_t^u),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, \mathbf{H}_t^\varepsilon \mathbf{R}/w_t^\varepsilon)$. *Nests* our Student-*t* DFM.

D1c-D2a $u_t \mid \cdot \sim \mathcal{N}(0, s_t^u h_t^u \mathbf{Q}),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, s_t^\varepsilon \mathbf{R})$. *Nests* the Gaussian DFM.

D1c-D2b $u_t \mid \cdot \sim \mathcal{N}(0, s_t^u h_t^u \mathbf{Q}),\ \varepsilon_t \mid \cdot \sim \mathcal{N}(0, s_t^\varepsilon \mathbf{H}_t^\varepsilon \mathbf{R})$. *Nests* the Gaussian DFM.

where $s_t^u = 1 + (\kappa_u^2 - 1) o_t^u$ and $s_t^\varepsilon = 1 + (\kappa_\varepsilon^2 - 1) o_t^\varepsilon$ are the outlier multipliers of D1-c, and (in D1-a, D1-b) the Gamma weights are $w_t^u \sim \operatorname{Gamma}(\nu_u/2, \nu_u/2)$, $w_t^\varepsilon \sim \operatorname{Gamma}(\nu_\varepsilon/2, \nu_\varepsilon/2)$.

For concreteness we write two cells out in full. The preferred cell **D1b-D2b** is

$$
\begin{aligned}
f_t &= \mathbf{A}\, f_{t-1} + u_t, & u_t \mid w_t^u, h_t^u &\sim \mathcal{N}\big(0,\ h_t^u\, \mathbf{Q}/w_t^u\big),\\
y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, & \varepsilon_t \mid w_t^\varepsilon, \mathbf{H}_t^\varepsilon &\sim \mathcal{N}\big(0,\ \mathbf{H}_t^\varepsilon\, \mathbf{R}/w_t^\varepsilon\big),
\end{aligned}
$$

with w_t^u, w_t^ε Gamma as above, $\mathbf{H}_t^\varepsilon = \operatorname{diag}(h_{1,t}^\varepsilon, \ldots, h_{M,t}^\varepsilon)$, and $\log h_t^u$ and every $\log h_{i,t}^\varepsilon$ following (175)–(176). The most distant **contrast** cell **D1a-D2a** (Gaussian tails, common volatility only) is

$$
\begin{aligned}
f_t &= \mathbf{A}\, f_{t-1} + u_t, & u_t \mid h_t^u &\sim \mathcal{N}\big(0,\ h_t^u\, \mathbf{Q}\big),\\
y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}\big(0,\ \mathbf{R}\big),
\end{aligned}
$$

with only $\log h_t^u$ active ((175)) and no tail weight on either shock.

Nesting, in one picture. The tail switch D1 alone decides what a cell reduces to when its volatility states are frozen: the two D1-b cells collapse to **our Student-*t* DFM** (173)–(174), whereas every D1-a and D1-c cell collapses to the thin-tailed **Gaussian DFM** (for D1-c after also switching the outliers off, $\kappa_u = \kappa_\varepsilon = 1$ or $o_t^u = o_t^\varepsilon = 0$). The scope D2 changes the model but not its floor. Hence the chain

$$
\text{Gaussian DFM} \ \subset\ \text{our Student-}t\text{ DFM} \ \subset\ \text{Student-}t\text{ DFM with stochastic volatility}
$$

runs only through the D1-b column ($\nu_u, \nu_\varepsilon \to \infty$ giving the first inclusion, activating the h states the second); no D1-a or D1-c cell ever passes through our Student-*t* model. Our preferred specification is **D1b-D2b**, and the development that follows builds the stochastic-volatility block along this line, with the outlier mechanism (D1-c) and the common-only scope (D2-a) as the principal robustness variants.

Remark (mixed frequency). The stochastic-volatility multipliers act on the *innovation* scale matrices $\mathbf{Q}$ and $\mathbf{R}$ of the monthly model. In the mixed-frequency companion form introduced earlier, where the state is augmented to $\tilde{f}_t = (f_t', f_{t-1}', \ldots, f_{t-4}')'$ and the loadings to $\tilde{\mathbf{\Lambda}}$, the volatility enters *only* through the covariance of the monthly innovation u_t that feeds the leading block of $\tilde{f}_t$, leaving the companion matrix $\tilde{\mathbf{A}}$ and the Mariano–Murasawa aggregation weights embedded in $\tilde{\mathbf{\Lambda}}$ unchanged. The variants above can therefore be read directly off the compact monthly equations, exactly as the baseline DGP is.

Asymmetry: From Symmetric Heavy Tails to a Skewed Predictive Density

Symmetric heavy tails are not enough. The variants of the previous subsection share one feature: although they make the tails heavy and regime-dependent, those tails are *symmetric*. It is enough to see this on the preferred specification D1b-D2b (Student-t tails, full stochastic volatility); the argument is identical for the others. Both shocks are conditionally Gaussian:

$$u_t \mid h_t^u, w_t^u \sim \mathcal{N}\big(0,\, h_t^u\, \mathbf{Q}/w_t^u\big), \qquad \varepsilon_t \mid \mathbf{H}_t^\varepsilon, w_t^\varepsilon \sim \mathcal{N}\big(0,\, \mathbf{H}_t^\varepsilon\, \mathbf{R}/w_t^\varepsilon\big),$$

or, exposing the Gaussian base shocks $z_t^u \sim \mathcal{N}(0, \mathbf{I}_r)$ and $z_t^\varepsilon \sim \mathcal{N}(0, \mathbf{I}_M)$ in *standardized* form,

$$u_t \;=\; \sqrt{h_t^u}\, \mathbf{Q}^{1/2}\, \frac{z_t^u}{\sqrt{w_t^u}}, \qquad \varepsilon_{i,t} \;=\; \sqrt{h_{i,t}^\varepsilon\, r_i}\, \frac{z_{i,t}^\varepsilon}{\sqrt{w_t^\varepsilon}}.$$

Now the symmetry is visible *by construction*. In each case the base shock (z_t^u or $z_{i,t}^\varepsilon$) is Gaussian, hence symmetric about zero, while the volatility multiplier (h_t^u or $h_{i,t}^\varepsilon$) and the tail weight ($1/w_t^u$ or $1/w_t^\varepsilon$) are *nonnegative scale factors*. Multiplying a symmetric shock by a positive scale leaves it symmetric: it stretches the distribution equally on both sides. A large h (turbulent regime) or a small w (tail event) therefore widens the left and the right tail by the *same* amount. Stochastic volatility and the Student-t weight thus deliver excess *kurtosis* — fat but symmetric tails — and regime-dependent *scale*, but essentially zero *skewness*: a scale mixture of mean-zero Gaussians, $\int \mathcal{N}(0, \sigma^2)\, dG(\sigma^2)$, is symmetric for *any* mixing law G. The predictive density can grow wider or narrower with the regime, but it cannot *lean*.

Why symmetry is a limitation for Growth-at-Risk. The signature finding of the Growth-at-Risk literature is precisely an *asymmetry*: as conditions deteriorate, the *left* tail of the growth distribution lengthens and thickens — downside risk rises — while the *right* tail stays comparatively stable. The lower quantiles move with the regime; the upper quantiles barely move. A symmetric mechanism cannot reproduce this by construction: when it widens, it widens both tails together, predicting that booms become as likely and as extreme as crashes in turbulent times — contrary to the evidence. Symmetric stochastic volatility captures the change in *scale* (the density breathes in and out with the regime) but not the change in *shape* (the density leaning to the left in bad times). To turn regime-dependent scale into regime-dependent *downside* risk, the model needs one further ingredient. We adopt a *single* mechanism — the *leverage effect* — and we choose it deliberately: it makes the asymmetry *emerge* from the structure of the model rather than imposing it by assumption.

The leverage effect: economic intuition. The mechanism originates in finance, with Black (1976). When the price of a firm's equity falls, its debt-to-equity ratio rises: the firm becomes more leveraged, hence riskier, and the volatility of its returns increases. A *fall* in the level is therefore followed by *more* volatility than a rise of the same size — the asymmetric "leverage effect," one of the most robust stylised facts of asset returns. The macroeconomic analogue, for a target such as GDP growth, is equally familiar: recessions are intrinsically more turbulent and uncertain than expansions — activity, financial conditions and forecasts all become more volatile precisely when growth turns down. The mechanism is thus not a modelling trick but a documented phenomenon, and this is what gives the resulting asymmetry an economic, rather than merely statistical, justification.

The formal mechanism. We implement leverage by *correlating*, for each shock that carries a stochastic volatility, the innovation that moves the *level* with the innovation that moves its *(log-)volatility*. The coupling applies *uniformly* to every volatility process in the model — the common factor and, under scope D2-b, each idiosyncratic component — so that no series is singled out.

Common factor. In the standardized form $u_t = \sqrt{h_t^u}\,\mathbf{Q}^{1/2} z_t^u$ of the previous subsection, the level shock $z_t^u \in \mathbb{R}^r$ and the common log-volatility innovation η_t^u of (175) — previously independent — are given the joint Gaussian law

$$\begin{pmatrix} z_t^u \\ \eta_t^u \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \mathbf{I}_r & \sigma_u \boldsymbol{\rho} \\ \sigma_u \boldsymbol{\rho}' & \sigma_u^2 \end{pmatrix} \right), \qquad \boldsymbol{\rho} \in \mathbb{R}^r, \quad \boldsymbol{\rho}'\boldsymbol{\rho} < 1,$$

so that $\operatorname{Corr}(z_{j,t}^u, \eta_t^u) = \rho_j$ for each factor j. The diagonal block $\mathbf{I}_r$ is the (standardized) covariance of the level shock, σ_u^2 that of the log-volatility innovation, and the off-diagonal $\sigma_u \boldsymbol{\rho}$ carries the leverage — the only new object relative to the symmetric model.

Idiosyncratic components. The same coupling attaches to *every* idiosyncratic component with an active stochastic volatility (all M of them under scope D2-b). For series i, writing $\varepsilon_{i,t} = \sqrt{h_{i,t}^\varepsilon\, r_i}\, z_{i,t}^\varepsilon$ and using its log-volatility innovation $\eta_{i,t}^\varepsilon$ of (176), the scalar analogue is

$$\begin{pmatrix} z_{i,t}^\varepsilon \\ \eta_{i,t}^\varepsilon \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{\varepsilon,i}\, \rho_{\varepsilon,i} \\ \sigma_{\varepsilon,i}\, \rho_{\varepsilon,i} & \sigma_{\varepsilon,i}^2 \end{pmatrix} \right), \qquad \rho_{\varepsilon,i} \in (-1, 1),$$

with $\operatorname{Corr}(z_{i,t}^\varepsilon, \eta_{i,t}^\varepsilon) = \rho_{\varepsilon,i}$ — one scalar leverage correlation per series. The tail weights w_t^u, w_t^ε remain independent of everything else.

In each block the constraint ($\boldsymbol{\rho}'\boldsymbol{\rho} < 1$ for the factor, $|\rho_{\varepsilon,i}| < 1$ for series i) is exactly positive-definiteness of the stacked covariance: the Schur complement is $\sigma_u^2(1 - \boldsymbol{\rho}'\boldsymbol{\rho}) > 0$, respectively $\sigma_{\varepsilon,i}^2(1 - \rho_{\varepsilon,i}^2) > 0$. This is the correlated-error, or "leverage," stochastic-volatility specification studied by Jacquier, Polson and Rossi (2004) and Yu (2005). The leverage sign is *negative*, $\rho_j, \rho_{\varepsilon,i} < 0$: a negative level shock ($z < 0$, the economy or a series contracting) is accompanied by a positive log-volatility innovation ($\eta > 0$, volatility rising), so downturns are more turbulent than upturns. *Nesting:* setting all correlations to zero ($\boldsymbol{\rho} = 0$ and $\rho_{\varepsilon,i} = 0$ for all i) makes the level and volatility shocks independent again and returns the symmetric stochastic-volatility model of the previous subsection; freezing the volatilities on top of that returns the baseline.

A note on notation. The standardized forms used above, $u_t = \sqrt{h_t^u}\,\mathbf{Q}^{1/2} z_t^u$ and $\varepsilon_{i,t} = \sqrt{h_{i,t}^\varepsilon\, r_i}\, z_{i,t}^\varepsilon$, suppress the Student-$t$ tail weights $1/\sqrt{w_t^u}$ and $1/\sqrt{w_t^\varepsilon}$ that appeared in the previous subsection. This is purely for readability: the weights are conditionally independent of both the level shocks and the log-volatility innovations, so they take no part in the leverage coupling and simply ride along as positive scale factors. They are reinstated wherever the full conditional law is needed.

The timing of the leverage: a conscious choice. The joint laws above pair the level shock z_t^u with the log-volatility innovation η_t^u that drives the *same* period's volatility h_t^u through (175) — and, identically, $z_{i,t}^\varepsilon$ with $\eta_{i,t}^\varepsilon$. This is only one of two admissible timings for the coupling, and which one we adopt is a genuine modelling decision rather than a notational detail; we make it explicit here.

Two admissible timings.

- **Contemporaneous.** The period-t level shock is correlated with the log-volatility innovation of the *same* period, η_t^u, which determines h_t^u: $\operatorname{Corr}(z_{j,t}^u, \eta_t^u) = \rho_j$. The drop in the level and the rise in volatility are one and the same event — this is exactly the joint law written above.

- **Lagged.** The period-t level shock is correlated instead with the log-volatility innovation of the *next* period, η^u_{t+1}, which determines h^u_{t+1}: $\mathrm{Corr}(z^u_{j,t}, \eta^u_{t+1}) = \rho_j$. Today's adverse shock *anticipates* tomorrow's turbulence.

The structure of the stacked covariance — the correlation entries, and the positive-definiteness constraints $\boldsymbol{\rho}'\boldsymbol{\rho} < 1$ and $|\rho_{\varepsilon,i}| < 1$ with their Schur complements — is *identical* in the two cases. The only thing that changes is *which* log-volatility innovation, the contemporaneous η_t or the one-step-ahead η_{t+1}, is stacked together with the level shock z_t.

Applied coherently across blocks. Whichever timing we choose, we apply it *uniformly* to every leveraged volatility process: the common shock (z^u_t coupled with its η^u) and each idiosyncratic component ($z^\varepsilon_{i,t}$ coupled with its η^ε_i) obey the *same* temporal scheme. Mixing a contemporaneous coupling in one block with a lagged one in another would have no economic rationale and is excluded by construction.

Economic reading. The two timings encode genuinely different stories. Under the contemporaneous form the shock and the turbulence are the same episode: the bad draw *is* the volatile period. Under the lagged form the negative shock is the *trigger* of a subsequent volatile phase — the downturn arrives first and the turbulence follows. The contemporaneous coupling is the one used by Jacquier, Polson and Rossi (2004); Yu (2005) draws the distinction explicitly and argues for the lagged form as the discrete-time analogue of the continuous-time leverage diffusion.

Our choice. We adopt the **contemporaneous** timing as the principal specification — it is the standard correlated-error form, it keeps the level and volatility innovations within a single period (so that the joint law, and hence the sampler, factorises one period at a time), and it is already the law written in the equations above — and we retain the **lagged** timing as an explicit variant.[1] The substantive point is that this timing is a *deliberate* choice, not an artefact of notation.

Which channels matter for Growth-at-Risk. Although leverage is specified uniformly across series, the block-diagonal loading structure implies that GDP's predictive density depends only on the common factor (through its loading) and on GDP's *own* idiosyncratic component — the idiosyncratic shocks of the other series never enter the GDP equation. For the Growth-at-Risk object, therefore, the leverage that actually shapes the target's skewness flows through exactly two channels, the common factor ($\boldsymbol{\rho}$) and the GDP idiosyncratic component ($\rho_{\varepsilon,g}$), while every other $\rho_{\varepsilon,i}$ shapes the predictive density of its own series. GDP is thus not privileged by assumption — the specification is symmetric across series — it is simply the series whose predictive density we read off.

Why this generates asymmetry. The correlation turns a change in *scale* into a change in *shape*. Because downward moves of the level systematically bring higher volatility, the adverse scenarios are drawn from a more dispersed distribution than the favourable ones: very negative outcomes become both more likely and more extreme, while very positive ones are damped by the lower volatility that accompanies upward moves. Integrating the volatility out, the predictive density acquires a *longer, heavier left tail* and a comparatively contained right tail — a

[1] Provisional: the final choice between the contemporaneous and the lagged timing is to be confirmed with the supervisor. The two specifications share the same parameters and the same covariance structure, so the model code and the estimator carry over to the lagged form by re-indexing the volatility innovation that is coupled to the level shock ($\eta_t \to \eta_{t+1}$).

negatively skewed density, exactly the Growth-at-Risk shape. In short, the negative correlation *stretches the left tail*: the more negative ρ_j, the stronger the downside skew, and $\rho_j = 0$ recovers symmetry.

Leverage as the natural completion of stochastic volatility. It is worth being explicit about why leverage is the *right* way to add skewness here, rather than an external device. Pure stochastic volatility is *conditionally* symmetric: at any fixed instant t, given h_t^u, the shock is a symmetric Gaussian (or symmetric Student-t). Yet its *unconditional* distribution — averaging over the joint law of level and volatility — can be skewed, but *only if* high volatility and low level systematically coincide. Under independence (the previous subsection) they do not, and the unconditional law stays symmetric. The leverage effect is precisely what makes them coincide: by tying a negative level shock to a positive volatility shock, it *activates* the asymmetry already latent in the stochastic-volatility structure, instead of adding a new one from outside. Leverage is therefore the natural *completion* of stochastic volatility — the minimal modification that lets a per-instant-symmetric model generate an asymmetric predictive density. This is also what distinguishes our approach from standard Growth-at-Risk practice: Adrian, Boyarchenko and Giannone (2019) obtain the asymmetry by *imposing* a skew-t (Azzalini) distribution when they smooth their estimated quantiles, whereas we prefer a mechanism in which the asymmetry *emerges* endogenously, rather than being built in through an assumed distributional form with no structural motivation of its own.

Cost, and what leverage does not solve. Two honest qualifications. First, estimation: the level and volatility shocks, previously separable, must now be handled *jointly* in the sampler, and the model carries one leverage correlation per volatility process (the factor vector $\boldsymbol{\rho}$ and a scalar $\rho_{\varepsilon,i}$ for each idiosyncratic series). This is a genuine — if mild — complication relative to the independent case, and we defer its treatment to the estimation chapter. Second, and more importantly, what leverage does *not* do: it supplies the *structural* asymmetry of the predictive density — recessions more turbulent than expansions — but the volatility–level link operates only *once a shock arrives*, not before it. Leverage therefore does not, and cannot, anticipate an exogenous, non-forecastable shock such as the 2020 pandemic: it shapes *how* risk is distributed around the forecast, not *whether* an unforeseeable event can be predicted. It is the right mechanism for skewness, but it leaves the non-forecastability of events like Covid exactly where it was — a property of the world, discussed earlier, not a defect the model can repair.

The Complete D1b–D2b Data-Generating Process with Leverage

The previous subsections introduced each ingredient of the richest specification one at a time: the stochastic-volatility channel, the Student-t tail weight, and the leverage coupling, in both its contemporaneous and its lagged timing. We now collect them in a single place and write the **D1b–D2b** cell — Student-t tails (D1-b), full stochastic volatility on the common *and* every idiosyncratic block (D2-b), with leverage — as one complete law of motion. This is the most general model of the family: every other variant studied above is one of its restrictions, recovered by switching a single component off. After stating the full DGP we show, in three explicit steps, how it collapses back to the symmetric Student-t-with-SV model, then to the first-stage Student-t DFM, and finally to the baseline Gaussian DFM.

The full law of motion

Levels: state and observation. The factors follow a VAR(1) and the observables load on them contemporaneously,

$$f_t = \mathbf{A}\, f_{t-1} + u_t, \tag{177}$$

$$y_t = \mathbf{\Lambda}\, f_t + \varepsilon_t, \tag{178}$$

with $f_t \in \mathbb{R}^r$, $y_t \in \mathbb{R}^M$, autoregressive matrix $\mathbf{A} \in \mathbb{R}^{r\times r}$, and loadings $\mathbf{\Lambda} \in \mathbb{R}^{M\times r}$. The factor-innovation base scale $\mathbf{Q} \in \mathbb{R}^{r\times r}$ is a *full* (dense), symmetric positive-definite matrix — the common shocks are correlated across factors — whereas the idiosyncratic base scale $\mathbf{R} = \operatorname{diag}(r_1, \ldots, r_M)$ is diagonal, so that the series-specific shocks are mutually uncorrelated and all cross-series co-movement is channelled through the common factors.

Shocks: three multiplicative channels. Each innovation is a Gaussian base shock rescaled by a persistent volatility h_t and a memoryless tail weight $1/w_t$. In conditional form,

$$u_t \mid w_t^u, h_t^u \sim \mathcal{N}\Big(0,\ h_t^u\, \mathbf{Q} \,/\, w_t^u\Big), \tag{179}$$

$$\varepsilon_t \mid w_t^\varepsilon, \mathbf{H}_t^\varepsilon \sim \mathcal{N}\Big(0,\ \mathbf{H}_t^\varepsilon\, \mathbf{R} \,/\, w_t^\varepsilon\Big), \qquad \mathbf{H}_t^\varepsilon = \operatorname{diag}\big(h_{1,t}^\varepsilon, \ldots, h_{M,t}^\varepsilon\big), \tag{180}$$

or, exposing the standardized Gaussian base shocks $z_t^u \sim \mathcal{N}(0, \mathbf{I}_r)$ and $z_{i,t}^\varepsilon \sim \mathcal{N}(0,1)$,

$$u_t \;=\; \sqrt{h_t^u}\; \mathbf{Q}^{1/2}\, \frac{z_t^u}{\sqrt{w_t^u}}, \qquad \varepsilon_{i,t} \;=\; \sqrt{h_{i,t}^\varepsilon\, r_i}\; \frac{z_{i,t}^\varepsilon}{\sqrt{w_t^\varepsilon}}. \tag{181}$$

Channel 1 — persistent volatility (D2-b: all states active). Every log-volatility, common and idiosyncratic, evolves as an independent AR(1),

$$\log h_t^u = \mu_u + \phi_u\big(\log h_{t-1}^u - \mu_u\big) + \eta_t^u, \tag{182}$$

$$\log h_{i,t}^\varepsilon = \mu_{\varepsilon,i} + \phi_{\varepsilon,i}\big(\log h_{i,t-1}^\varepsilon - \mu_{\varepsilon,i}\big) + \eta_{i,t}^\varepsilon, \qquad i = 1, \ldots, M, \tag{183}$$

with persistences $\phi_u, \phi_{\varepsilon,i} \in (-1,1)$ and log-volatility innovation variances $\sigma_u^2, \sigma_{\varepsilon,i}^2 > 0$.

Channel 2 — Student-t tail weights (D1-b). The memoryless tail multipliers are inverse Gamma weights, drawn afresh each period and shared within a block,

$$w_t^u \overset{iid}{\sim} \operatorname{Gamma}\big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\big), \qquad w_t^\varepsilon \overset{iid}{\sim} \operatorname{Gamma}\big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\big), \tag{184}$$

mutually independent across time, across the two families, and independent of the base shocks and of the log-volatility innovations. Each block carries a *single* scalar weight: w_t^u rescales the whole factor-innovation vector and w_t^ε the whole idiosyncratic vector. Integrating that weight out — while keeping the volatility h_t fixed — turns the conditional Gaussian into a Student-t, the usual Gaussian scale-mixture representation of the t:

$$u_t \mid h_t^u \;\sim\; t_{\nu_u}(0,\, h_t^u\,\mathbf{Q}), \qquad \varepsilon_t \mid \mathbf{H}_t^\varepsilon \;\sim\; t_{\nu_\varepsilon}(0,\, \mathbf{H}_t^\varepsilon\,\mathbf{R}),$$

a multivariate t in each case, because the one shared weight couples the components of the vector through their common mixing scale.

Channel 3 — leverage (contemporaneous timing). The base shock that moves the *level* is correlated with the innovation that moves the *(log-)volatility* of the same period. For the common factor, the level shock $z_t^u \in \mathbb{R}^r$ and the common log-volatility innovation η_t^u of (182) obey the joint Gaussian law

$$\begin{pmatrix} z_t^u \\ \eta_t^u \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \mathbf{I}_r & \sigma_u\,\boldsymbol{\rho} \\ \sigma_u\,\boldsymbol{\rho}' & \sigma_u^2 \end{pmatrix}\right), \qquad \boldsymbol{\rho} \in \mathbb{R}^r, \quad \boldsymbol{\rho}'\boldsymbol{\rho} < 1, \tag{185}$$

so that $\mathrm{Corr}(z_{j,t}^u, \eta_t^u) = \rho_j$. For each idiosyncratic component the scalar analogue couples $z_{i,t}^\varepsilon$ with $\eta_{i,t}^\varepsilon$ of (183),

$$\begin{pmatrix} z_{i,t}^\varepsilon \\ \eta_{i,t}^\varepsilon \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{\varepsilon,i}\,\rho_{\varepsilon,i} \\ \sigma_{\varepsilon,i}\,\rho_{\varepsilon,i} & \sigma_{\varepsilon,i}^2 \end{pmatrix}\right), \qquad \rho_{\varepsilon,i} \in (-1, 1), \tag{186}$$

with $\mathrm{Corr}(z_{i,t}^\varepsilon, \eta_{i,t}^\varepsilon) = \rho_{\varepsilon,i}$. The positive-definiteness constraints $\boldsymbol{\rho}'\boldsymbol{\rho} < 1$ and $|\rho_{\varepsilon,i}| < 1$ are exactly the Schur complements $\sigma_u^2(1 - \boldsymbol{\rho}'\boldsymbol{\rho}) > 0$ and $\sigma_{\varepsilon,i}^2(1 - \rho_{\varepsilon,i}^2) > 0$. The leverage signs are negative, $\rho_j, \rho_{\varepsilon,i} < 0$: a contraction of the level ($z < 0$) comes with rising volatility ($\eta > 0$), so downturns are more turbulent than upturns and the predictive density leans to the left. The tail weights w_t^u, w_t^ε stay independent of the (z, η) pairs and play no part in the coupling.

Channel 3′ — leverage (lagged timing, variant). The lagged form is obtained by re-indexing a *single* object: the level shock of period t is coupled with the log-volatility innovation of period $t{+}1$ rather than t. The covariance structure of (185)–(186) is unchanged; only the time index of the volatility innovation moves,

$$\begin{pmatrix} z_t^u \\ \eta_{t+1}^u \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \mathbf{I}_r & \sigma_u\,\boldsymbol{\rho} \\ \sigma_u\,\boldsymbol{\rho}' & \sigma_u^2 \end{pmatrix}\right), \qquad \begin{pmatrix} z_{i,t}^\varepsilon \\ \eta_{i,t+1}^\varepsilon \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{\varepsilon,i}\,\rho_{\varepsilon,i} \\ \sigma_{\varepsilon,i}\,\rho_{\varepsilon,i} & \sigma_{\varepsilon,i}^2 \end{pmatrix}\right), \tag{187}$$

so that $\mathrm{Corr}(z_{j,t}^u, \eta_{t+1}^u) = \rho_j$ and $\mathrm{Corr}(z_{i,t}^\varepsilon, \eta_{i,t+1}^\varepsilon) = \rho_{\varepsilon,i}$: today's adverse shock anticipates tomorrow's turbulence. The timing is applied uniformly across all blocks; the contemporaneous form (185)–(186) is our principal specification and the lagged form (187) the retained variant.

The complete generative recipe. Collecting (177)–(184) with the leverage law (185)–(186) (contemporaneous) or (187) (lagged), one period of the DGP unfolds as follows: (i) draw the tail weights w_t^u, w_t^ε from (184); (ii) draw the coupled pairs (z_t^u, η_t^u) and $(z_{i,t}^\varepsilon, \eta_{i,t}^\varepsilon)$ jointly from the leverage law; (iii) advance the log-volatilities through (182)–(183) using the η's; (iv) form the innovations u_t, ε_t via the standardized map (181); and (v) update the level through the state and observation equations (177)–(178). The latent objects carried across periods are the factors f_t, the log-volatilities $\log h_t^u, \log h_{i,t}^\varepsilon$, and (for the lagged timing) the one-period link between z_t and η_{t+1}.

Collapsing back: leverage $\to$ Student-t $\to$ Gaussian

The general model contains the simpler ones as nested limits. Three restrictions, applied in sequence, walk back the entire construction.

Step 1 — switch off leverage ($\rho = 0$, $\rho_{\varepsilon,i} = 0$). Setting every leverage correlation to zero makes the off-diagonal blocks in (185)–(186) vanish, so the level shocks and the log-volatility innovations become independent again,

$$\begin{pmatrix} z_t^u \\ \eta_t^u \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & \sigma_u^2 \end{pmatrix}\right), \qquad \begin{pmatrix} z_{i,t}^{\varepsilon} \\ \eta_{i,t}^{\varepsilon} \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & \sigma_{\varepsilon,i}^2 \end{pmatrix}\right).$$

The level law is untouched — the shocks keep their volatility and tail channels — but the density can no longer lean: this is the *symmetric* Student-t-with-SV model,

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad u_t \mid w_t^u, h_t^u \sim \mathcal{N}(0,\, h_t^u\, \mathbf{Q}/w_t^u), \\ y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, \qquad \varepsilon_t \mid w_t^{\varepsilon}, \mathbf{H}_t^{\varepsilon} \sim \mathcal{N}(0,\, \mathbf{H}_t^{\varepsilon}\, \mathbf{R}/w_t^{\varepsilon}), \end{aligned}$$

with (182)–(184) still in force. Skewness disappears; excess kurtosis and regime-dependent scale remain. (The lagged variant (187) collapses to the same point, since it too has zero off-diagonal once $\rho = 0$.)

Step 2 — freeze the volatilities ($\sigma^2 = 0$, $\mu = 0$). Freezing every log-volatility state ($\sigma_u^2 = 0, \mu_u = 0$ and $\sigma_{\varepsilon,i}^2 = 0, \mu_{\varepsilon,i} = 0$ for all i) forces $h_t^u \equiv 1$ and $\mathbf{H}_t^{\varepsilon} \equiv \mathbf{I}_M$ for all t, so (182)–(183) become degenerate and drop out. The persistent channel is gone and only the Student-t tail weight survives, returning the **first-stage Student-t DFM** (173)–(174),

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad u_t \mid w_t^u \sim \mathcal{N}(0,\, \mathbf{Q}/w_t^u), \\ y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, \qquad \varepsilon_t \mid w_t^{\varepsilon} \sim \mathcal{N}(0,\, \mathbf{R}/w_t^{\varepsilon}), \end{aligned}$$

with $w_t^u \sim \text{Gamma}(\nu_u/2, \nu_u/2)$, $w_t^{\varepsilon} \sim \text{Gamma}(\nu_{\varepsilon}/2, \nu_{\varepsilon}/2)$ — the heavy-tailed model estimated by EM in the first stage. Equivalently, marginalising the weights, $u_t \sim t_{\nu_u}(0, \mathbf{Q})$ and $\varepsilon_t \sim t_{\nu_{\varepsilon}}(0, \mathbf{R})$.

Step 3 — thin the tails ($\nu \to \infty$). Sending the degrees of freedom to infinity, $\nu_u, \nu_{\varepsilon} \to \infty$, concentrates the Gamma weights at one, $w_t^u, w_t^{\varepsilon} \to 1$ in probability, so $1/w_t \to 1$ and the scale mixture collapses to its Gaussian core. What remains is the thin-tailed **baseline Gaussian DFM**,

$$\begin{aligned} f_t &= \mathbf{A}\, f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0,\, \mathbf{Q}), \\ y_t &= \mathbf{\Lambda}\, f_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\, \mathbf{R}). \end{aligned}$$

The nesting chain. The three steps trace the inclusion

$$\underbrace{\text{Gaussian DFM}}_{\nu\to\infty,\ \sigma^2=0,\ \rho=0} \subset \underbrace{\text{Student-}t\text{ DFM}}_{\sigma^2=0,\ \rho=0} \subset \underbrace{\text{Student-}t\text{ with SV}}_{\rho=0} \subset \underbrace{\text{D1b–D2b with leverage}}_{\text{full model}},$$

each arrow switching exactly one channel back on: the Student-t tail (ν finite), the persistent volatility ($\sigma^2 > 0$), and the leverage asymmetry ($\rho \neq 0$). Because D1b–D2b sits at the top of this chain, deriving the estimator once for the full model — the task of the next section — automatically covers every restriction below it.

Estimating the Model: From EM to MCMC

We now turn from the data-generating process to its *estimation*: given the observed data $\mathbf{Y}$, how do we recover the unknown parameters and the unobserved paths — factors, volatilities, tail weights — of the model just written? We carry out the derivation for the preferred cell **D1b–D2b with leverage**: Student-t tails, full stochastic volatility on both the common and the idiosyncratic blocks, and the correlated-error leverage coupling of the previous subsection. This is a deliberate economy of effort. D1b–D2b is the *most general* specification of the family, and, as the nesting chain of the previous section showed, every other cell is one of its restrictions, reached by switching a single component off: freezing a volatility path ($\sigma^2 = 0, \mu = 0$), setting a leverage correlation to zero ($\boldsymbol{\rho} = 0$), or sending a degrees-of-freedom parameter to infinity ($\nu \to \infty$). An estimator that handles the richest model therefore handles all the others for free — one simply *disables* the corresponding sampling block — so we derive it once, here, and make the restrictions explicit later.

The thesis of this section is that the move from the first stage to the second is *less* of a rupture than it first appears. We change the *estimation engine* — from the EM algorithm to Markov chain Monte Carlo (MCMC) — but we keep almost all of the apparatus built for EM, because the hard, model-specific half of it (the linear-Gaussian state-space and its Kalman recursions) survives untouched *inside* the new engine. The plan is accordingly as follows. We first explain *why* EM can no longer be used once stochastic volatility and leverage enter (§); we then restate the inferential target in Bayesian terms — the whole posterior rather than a single point (§); we recall why such a posterior must be *simulated* rather than computed (§) and why the simulation is organised as a *Gibbs* sweep over blocks (§); and we finally make explicit the *bridge* — the precise list of what the first stage already gives us for free (§). The block-by-block samplers — for the states, the volatility paths, the tail weights and the parameters — are then developed one at a time in the sections that follow.

Why the estimator must change

What made EM work: conjugacy and closed forms. It is worth recalling, in plain terms, *why* the EM algorithm was the right tool for the first-stage model, because the reason it worked is exactly the reason it now fails. EM is an *optimisation* method: it seeks the single parameter vector $\hat{\boldsymbol{\theta}}$ that maximises the marginal likelihood $p(\mathbf{Y} \mid \boldsymbol{\theta})$ — the data density obtained *after* integrating the latent variables out. Maximising that likelihood directly is hard, because the latent factors and tail weights sit *inside* the integral. EM sidesteps the difficulty by alternating two steps. In the **E-step** it forms the expected complete-data log-likelihood — it "fills in" the latent quantities by their conditional expectations given the data and the current parameters, which here meant running the Kalman smoother for the factors and computing the Gamma posterior mean $\hat{w}_t$ for each tail weight. In the **M-step** it maximises that expected complete-data log-likelihood over $\boldsymbol{\theta}$, which delivered the closed-form weighted-OLS and weighted-moment updates for $\mathbf{A}, \boldsymbol{\Lambda}, \mathbf{Q}, \mathbf{R}$.

The linchpin of this construction is *conjugacy*, and it is worth saying precisely what that means here. Conditional on the latent weights and the latent states, the complete-data likelihood factors into *Gaussian* pieces. A Gaussian likelihood in the regression coefficients, paired with its moment sufficient statistics, has a maximiser available *in closed form* — it is ordinary (here, weighted) least squares — and the conditional expectations the E-step needs are themselves available in closed form, because the smoother of a linear-Gaussian model returns exact Gaussian moments and the weight posterior is an exact Gamma. Every step is therefore an explicit formula: no numerical integration, no iterative inner optimisation. This is what made the first stage fast and reliable — and these are the updates (173)'s estimator derived for $\mathbf{A}$, $\mathbf{\Lambda}$, $\mathbf{Q}$ and $\mathbf{R}$ in the M-step section.

What stochastic volatility breaks. Stochastic volatility severs this conjugacy at its root. The conditional variance of each shock is now h_t/w_t times its base scale, and the volatility enters through $h_t = \exp(\log h_t)$, with $\log h_t$ following the AR(1) recursions (175)–(176). The offending feature is the *exponential*: the unknown log-volatility sits inside an $\exp(\cdot)$ that multiplies the variance, and there is no prior on $\log h_t$ conjugate to a likelihood of this shape. Two distinct closed forms collapse as a consequence.

- *The E-step expectation is no longer analytic.* It would now have to average the complete-data log-likelihood over the joint conditional law of the factors *and* the entire volatility paths $h^u_{1:T}, h^\varepsilon_{1:T}$. But those paths are not Gaussian conditional on the data — the exp nonlinearity destroys the very Gaussianity the Kalman smoother relies on — so the expectation has no closed form. The smoother that delivered the factor moments simply does not apply to the volatility states.
- *The M-step has no explicit maximiser.* Even granting the expectation, the parameters of the volatility processes — the long-run level μ, the persistence ϕ and the innovation variance σ^2 of each AR(1) — enter the log-likelihood nonlinearly and admit no weighted-least-squares solution. There is simply no formula to write down for their update.

Leverage tightens the knot. The leverage coupling makes things worse. By correlating each level shock z_t with its volatility innovation η_t — the joint laws (185)–(186) — it renders the level block and the volatility block *statistically inseparable*: one can no longer integrate out the states while holding the volatilities fixed, nor profile the volatilities while holding the states fixed, because the two now share a single joint Gaussian innovation. Whatever hope one might have had of treating the volatility approximately and the rest in closed form is removed by this coupling.

Why not simply patch EM numerically. One might ask why we do not keep EM and merely compute the awkward pieces by numerical optimisation or quadrature. The obstacle is dimensionality. The intractable E-step expectation is an integral over the *entire* volatility path — a space whose dimension grows with T (one $\log h_t$ per period for the common factor, and M more per period under D2-b) — and numerical integration in thousands of dimensions is hopeless. A numerical inner maximisation in the M-step, nested inside an already iterative outer loop and re-run every sweep, would be both fragile and astronomically expensive, and it would still leave the E-step integral unsolved. EM is simply the wrong *shape* of algorithm for a model whose latent structure can no longer be integrated out analytically.

The change of paradigm. We therefore stop trying to *optimise*. Rather than search for the single point that maximises the likelihood (or, equivalently, the ELBO of the variational/EM view), we move to *Bayesian inference by simulation*: we treat the parameters as random, place the latent paths on the same footing, and *draw* from the joint distribution of all the unknowns instead of maximising over them. As the next subsections make clear, this is not merely a way around the broken closed forms — it is also the more natural target for a Growth-at-Risk exercise, whose object is a whole predictive *density*, not a point.

The Bayesian object: the whole posterior, not a point

From a point to a distribution. The first change is conceptual. A maximum-likelihood estimator returns a single "best" parameter vector and treats the parameters as fixed, unknown constants. The Bayesian paradigm instead treats the parameters as *random variables* and describes our knowledge of them by a whole probability distribution. Before seeing the data we encode what we know — or, deliberately, how little we know — in a *prior* $p(\boldsymbol{\theta})$; the data then update that prior into a *posterior*. The output of estimation is no longer a point but a distribution, and uncertainty is quantified from the inside, not bolted on afterwards through asymptotic standard errors.

Bayes' theorem, term by term. Collecting the parameters in $\boldsymbol{\theta}$ — here $\boldsymbol{\theta}$ gathers $(\mathbf{A}, \mathbf{Q}, \boldsymbol{\Lambda}, \mathbf{R})$, the log-volatility parameters (μ, ϕ, σ^2) of every active AR(1), the leverage correlations $(\boldsymbol{\rho}, \rho_{\varepsilon,i})$ and the degrees of freedom (ν_u, ν_ε) — Bayes' theorem reads

$$p(\boldsymbol{\theta} \mid \mathbf{Y}) \;=\; \frac{p(\mathbf{Y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathbf{Y})}, \qquad p(\mathbf{Y}) \;=\; \int p(\mathbf{Y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta}.$$

Each term has a clean reading. The *likelihood* $p(\mathbf{Y} \mid \boldsymbol{\theta})$ measures how well a given parameter value explains the observed data — it is the same object EM maximised. The *prior* $p(\boldsymbol{\theta})$ carries our pre-data beliefs. Their product, renormalised, is the *posterior* $p(\boldsymbol{\theta} \mid \mathbf{Y})$, our post-data beliefs. The denominator $p(\mathbf{Y})$ — the *evidence*, or marginal likelihood — is just the integral of the numerator over $\boldsymbol{\theta}$; it does not depend on $\boldsymbol{\theta}$ and serves only to make the posterior integrate to one. This is the single most important structural fact for what follows: *the posterior is known only up to a constant*,

$$p(\boldsymbol{\theta} \mid \mathbf{Y}) \;\propto\; \underbrace{p(\mathbf{Y} \mid \boldsymbol{\theta})}_{\text{likelihood}}\; \underbrace{p(\boldsymbol{\theta})}_{\text{prior}},$$

and that constant — the integral $p(\mathbf{Y})$ — is what we cannot compute. The symbol "$\propto$" ("proportional to") is doing real work here: it says we can write down the *shape* of the posterior everywhere, as a function of $\boldsymbol{\theta}$, while remaining ignorant of its overall scale. Knowing the shape is in principle enough — the missing constant is fixed, once and for all, by the requirement that the density integrate to one — but pinning it down means evaluating that high-dimensional integral, and *that* is the obstacle. Every device in the simulation strategy below is, at bottom, a way of working with the shape alone and never having to compute the constant.

A trivial worked example: one unknown mean. It helps to see all four terms in a case small enough to solve by hand. Strip the model down to a single unknown scalar θ — think of it as an unknown average growth rate — about which, before seeing data, we hold a Gaussian belief centred at m_0 with variance s_0^2:

$$\text{prior} \qquad \theta \sim \mathcal{N}\big(m_0,\, s_0^2\big), \qquad p(\theta) \;\propto\; \exp\Big(-\tfrac{(\theta - m_0)^2}{2 s_0^2}\Big).$$

A large s_0^2 means a vague, non-committal prior; a small one, a sharp pre-data conviction. We then observe n noisy measurements $y_1, \ldots, y_n$, each a draw of θ blurred by Gaussian noise of *known* variance σ^2,

$$\text{likelihood} \qquad y_i = \theta + \text{noise}, \quad \text{noise} \sim \mathcal{N}(0, \sigma^2), \qquad p(\mathbf{y} \mid \theta) \;\propto\; \exp\Big(-\tfrac{1}{2\sigma^2} \textstyle\sum_{i=1}^{n} (y_i - \theta)^2\Big).$$

By Bayes' theorem the posterior is the product of these two Gaussians (up to the constant we are entitled to ignore),

$$p(\theta \mid \mathbf{y}) \;\propto\; \exp\Big(-\tfrac{(\theta - m_0)^2}{2 s_0^2}\Big)\, \exp\Big(-\tfrac{1}{2\sigma^2} \textstyle\sum_i (y_i - \theta)^2\Big).$$

Both exponents are quadratic in θ, so their sum is again a quadratic, and *a quadratic exponent is a Gaussian*: completing the square gives, with sample mean $\bar{y} = \frac{1}{n}\sum_i y_i$,

$$\theta \mid \mathbf{y} \;\sim\; \mathcal{N}(m_1,\, s_1^2), \qquad \underbrace{\frac{1}{s_1^2}}_{\text{posterior precision}} = \underbrace{\frac{1}{s_0^2}}_{\text{prior}} + \underbrace{\frac{n}{\sigma^2}}_{\text{data}}, \qquad m_1 = s_1^2\Big(\frac{m_0}{s_0^2} + \frac{n\,\bar{y}}{\sigma^2}\Big).$$

Read off the three lessons this little formula carries, because each one recurs, unchanged in spirit, in the full model.

- *Information adds up.* The *precision* (inverse variance) of the posterior is the prior precision *plus* the data precision n/σ^2. Every observation makes the posterior a little sharper; uncertainty only ever shrinks as data accrue.
- *The posterior is a compromise.* The posterior mean m_1 is a *precision-weighted average* of the prior mean m_0 and the sample mean $\bar{y}$: whichever source is more precise pulls m_1 towards itself. The data do not erase the prior — they are blended with it, in proportion to how much each one knows.
- *The limits are the ones intuition demands.* With no data ($n = 0$) the posterior *is* the prior. As data pile up ($n \to \infty$) the data precision dominates, $m_1 \to \bar{y}$ and $s_1^2 \to 0$: the prior is washed out and the posterior collapses onto the value the data alone would suggest — the point a maximum-likelihood estimator would have returned. The Bayesian answer thus *contains* the classical one as its large-sample limit, while saying more (a whole distribution) when data are scarce.

Two further points make this toy the right rehearsal for what follows. First, we never needed the evidence $p(\mathbf{y})$ to find the posterior: recognising the quadratic exponent as a Gaussian told us the *shape*, and normalisation was automatic — exactly the "work with the shape, ignore the constant" move written above. Second, this example is easy only because prior and likelihood are *conjugate* (Gaussian times Gaussian is Gaussian), so the posterior stays in a known family and the constant happens to be computable. In our model that good fortune fails — stochastic volatility and leverage destroy conjugacy, as § explained — the posterior is no recognisable density, and the normalising integral is genuinely intractable. That is precisely why, beyond this toy, we must *simulate* the posterior rather than write it down.

Latent variables on the same footing: data augmentation. In a latent-variable model the unobserved quantities are not a nuisance to be eliminated before inference begins; they are unknowns like any other, and the cleanest course is to give them distributions and infer them *jointly* with the parameters. This is the idea of *data augmentation*: we deliberately *enlarge* the list of unknowns to include the latent paths, because doing so makes each conditional distribution simple even though the marginal is hard. Here the augmented unknowns are the factor path $f_{1:T}$, the volatility paths $h_{1:T} = (h^u_{1:T}, h^\varepsilon_{1:T})$ and the tail weights $w = (w^u_{1:T}, w^\varepsilon_{1:T})$, so the proper target of inference is the *joint* posterior

$$p\big(f_{1:T},\, h_{1:T},\, w,\, \boldsymbol{\theta} \,\big|\, \mathbf{Y}\big).$$

The object of inference is thus no longer a single maximiser but this entire distribution. Whenever we later want the parameters alone, or the volatility alone, we simply ignore the draws of the other blocks: with samples, marginalisation is free — one keeps the coordinates of interest and discards the rest.

Why this fits Growth-at-Risk especially well. The payoff is immediate for the second stage. Every quantity of interest is a *functional* of the posterior, obtained by integration: a posterior mean, a credible interval, and — the object we actually care about — the *predictive density* of GDP and its lower quantiles for Growth-at-Risk. Because the posterior is a distribution from the outset, the predictive density falls out of it directly: one propagates each posterior draw through the model's forward dynamics and collects the resulting GDP draws, whose spread *is* the predictive density, already inclusive of both shock uncertainty and parameter uncertainty. This is a genuine improvement over the first stage, where EM returned a point and we had to *append* a density to it after the fact and by hand. Here the density is not an add-on; it is what the method produces.

Why simulate: Markov chain Monte Carlo

The obstacle: an intractable normalising constant. We have just seen that the posterior is known only *up to* the constant $p(\mathbf{Y})$. That constant is an integral over a space of enormous dimension — the states, the two families of volatility paths, the tail weights and the parameter blocks, all stacked across the T time periods, together running to many thousands of unknowns. The integral has no closed form, and quadrature is again hopeless at this dimension. So we can write the posterior's *shape* (the numerator) everywhere, but we can neither normalise it nor, it would seem, sample from it: we know the relative heights of the density at any two points, but not its absolute scale.

The Monte Carlo idea: represent a distribution by samples. The way out begins with a simple observation. To answer essentially any question about a distribution — its mean, a quantile, the probability of a region, the shape of a tail — it suffices to possess a large *sample* of draws from it; one then reads the answer off the empirical distribution of the draws (a sample average, a sample quantile). We need neither the density's formula nor its normalising constant; we need representative points. This is the Monte Carlo principle: *trade analysis for sampling*. The problem is therefore not to compute the posterior but to *draw* from it — and direct sampling is blocked precisely because we lack the normalising constant.

The Markov-chain device. Markov chain Monte Carlo resolves exactly this. Instead of drawing independent points from the posterior — which we cannot do — one constructs a *Markov chain*: a sequence in which each new state is drawn from a transition rule depending only on the current state. The rule is engineered so that the chain's *stationary* (invariant) distribution is precisely the target posterior. Run the chain long enough and it "forgets" where it started and settles into exploring the posterior, visiting each region in proportion to its posterior probability. The crucial point is that building such a transition rule requires only the *unnormalised* posterior — ratios of the density at different points, in which the unknown constant $p(\mathbf{Y})$ cancels — which is exactly what we have. The intractable normaliser is never evaluated.

Reading the output: burn-in and ergodic averages. Two practical notions follow. First, *burn-in*: the early portion of the chain still reflects the arbitrary starting value, before stationarity is reached, and is discarded. Second, *ergodic averaging*: once stationary, the draws — although *serially correlated*, since each depends on the previous one — can still be averaged to approximate posterior expectations, with the approximation improving as the chain lengthens. The serial correlation does not bias these averages; it only makes them less informative per draw than independent samples would be, so a correlated chain must be run longer to reach a given precision (its *effective* sample size is smaller than its length). In practice one therefore also monitors *convergence* — that stationarity has been reached — and *mixing* — how quickly the chain traverses the posterior — but the principle is the one just stated: a long stationary run *is* a sample from the posterior, and the uncomputable constant $p(\mathbf{Y})$ is never touched.

Why Gibbs: divide the posterior into blocks

The difficulty of moving in all directions at once. It remains to build a transition rule with the right stationary distribution. The unknowns form several *heterogeneous* blocks — the states $f_{1:T}$, the volatility paths $h_{1:T}$, the tail weights w, and the parameters $\boldsymbol{\theta}$ — living on very different scales and geometries. Proposing a sensible joint move over all of them at once is intractable: a blind joint update would almost never land in a region of appreciable posterior probability, and the chain would crawl.

The key fact: hard joint, easy conditionals. The model has a feature that rescues the situation. While the *joint* posterior is intractable, each block's *full conditional* — its distribution given the data and the current values of *all* the other blocks — is tractable, and indeed usually a standard, recognisable density. This is no accident: it is exactly what the data-augmentation choice of § bought us. Conditioning on the volatility paths and weights makes the state block linear-Gaussian; conditioning on the states makes each weight an exact Gamma and each parameter block a conjugate draw; and so on down the list. The dependence among the blocks is real, but *once we fix all but one block, what remains is simple*.

The Gibbs sampler. This is precisely the setting the Gibbs sampler is designed for. One cycles through the blocks in turn, redrawing each from its full conditional given the current values of the others,

$$\text{draw } f_{1:T} \mid \text{rest}, \quad \text{then } h_{1:T} \mid \text{rest}, \quad \text{then } w \mid \text{rest}, \quad \text{then } \boldsymbol{\theta} \mid \text{rest},$$

and repeats the sweep. The intuition is that of solving a hard problem one piece at a time: each conditional draw refreshes one block in the light of the others, and the coupling between blocks is respected because every draw conditions on the latest values of all the rest. Under mild conditions the joint posterior is the invariant distribution of the resulting chain — cycling through the full conditionals leaves the joint law unchanged — so iterating the sweep produces draws from exactly the joint posterior we want. The Gibbs sampler is due to Geman and Geman (1984); Casella and George (1992) give the standard intuition for why cycling through full conditionals targets the joint law, and Robert and Casella (2004) provide the general Monte Carlo treatment.

Why blocking, and what it costs. A last point of design. We update each block *jointly* — the whole factor path in one move, the whole volatility path in another — rather than one scalar at a time. The reason is that variables *within* a block are strongly dependent: consecutive factors f_t and f_{t+1}, or consecutive log-volatilities, are highly correlated through the dynamics, and a sampler that updated them individually would move in tiny, correlated steps and mix painfully slowly. Drawing a whole path at once breaks that bottleneck. Where a block's full conditional is itself non-standard — as it will be for the volatility paths, on account of the very exp nonlinearity that broke EM — the Gibbs step for that block is replaced by a Metropolis–Hastings step (or by an auxiliary-variable device that restores a tractable conditional), with the chain still targeting the same posterior. The skeleton, however, is the Gibbs cycle above; the next subsection records how much of each block we already possess from the first stage.

The bridge: what the first stage already gives us

It would be discouraging if changing the estimation engine meant rebuilding the model from scratch. It does not. The decisive — and reassuring — point is that almost all of the machinery built for EM survives intact, because it lives *inside* the Gibbs blocks rather than in the optimisation that wrapped them. Three pieces carry over, and it is worth being explicit about each.

(i) Conditional on volatility, the model is again linear-Gaussian, and Kalman is reused verbatim. This is the heart of the bridge. The thing that broke EM was the $\exp(\log h_t)$ nonlinearity — but inside the Gibbs sweep the volatility block is *conditioned on* when we update the states. Once the paths $h_{1:T}$ and the weights w are held fixed, the multipliers h_t/w_t are just *known numbers*: the nonlinearity is frozen out. The conditional innovation covariances become the *deterministic* time-varying matrices

$$\mathbf{Q}_t \;=\; \frac{h_t^u}{w_t^u}\,\mathbf{Q}, \qquad \mathbf{R}_{i,t} \;=\; \frac{h_{i,t}^{\varepsilon}}{w_t^{\varepsilon}}\,r_i.$$

A linear state-space model with Gaussian shocks and *known* time-varying covariances is precisely what the Kalman filter and smoother handle: the recursions derived for the first stage apply unchanged, now fed the time-varying $\mathbf{Q}_t, \mathbf{R}_{i,t}$ in place of the constant scales. The one genuine difference is the *purpose* we put the smoother to. EM needed the conditional *mean* of the states (a point, for the E-step); the Gibbs states block instead needs a *draw* of the whole state path from its conditional distribution. This is supplied by a *simulation smoother* — forward-filtering, backward-sampling — which runs the very same Kalman forward pass and then samples backwards instead of merely averaging. Same filter underneath; a draw rather than a mean on the way out.

(ii) The mixed-frequency construction is untouched. The apparatus that reconciles monthly factors with quarterly GDP carries over without a single change. As noted in the remark on mixed frequency in the previous chapter, the stochastic-volatility multipliers act *only* on the covariance of the monthly innovation feeding the leading block of the augmented state; they do not touch the deterministic bookkeeping that maps states to observations. The companion form $\tilde{f}_t = (f_t', f_{t-1}', \ldots, f_{t-4}')'$, the companion matrix $\tilde{\mathbf{A}}$ and the Mariano–Murasawa aggregation weights embedded in $\tilde{\mathbf{\Lambda}}$ are exactly as derived there. Volatility changes *what variance* the leading innovation carries, not the structure that aggregates months into quarters, so everything in that chapter transfers verbatim.

(iii) The weighted M-step updates survive as conditional draws. The closed-form M-step updates for $\mathbf{A}$, $\mathbf{\Lambda}$ and the scales $\mathbf{Q}$, r_i were weighted-OLS and weighted-moment expressions built from the weighted cross-moments of the E-step. These do not go to waste. Under conjugate priors — a (matrix-)Normal prior on the regression coefficients $\mathbf{A}$ and $\mathbf{\Lambda}$, an inverse-Wishart prior on $\mathbf{Q}$, and inverse-Gamma priors on the r_i — the *same* weighted cross-moments are the sufficient statistics that parameterise the full conditional posteriors, and those posteriors belong to the *same* (Normal, inverse-Wishart, inverse-Gamma) families. The algebra of the weighted moments is thus reused line for line; the only thing that changes is the final act. Where EM read off the *mode* of that conditional object and reported it as a point update, Gibbs takes a *random draw* from it. A point becomes a sample from the same family — the distinction between optimising and simulating, in microcosm.

The central message. We keep "model + Kalman" — the difficult, model-specific half of the problem, already constructed and validated in the first stage — and we replace only "the estimation engine," EM by MCMC. The conditionally linear-Gaussian state-space, the mixed-frequency aggregation and the conjugate parameter algebra all migrate from the EM derivation into the Gibbs sweep, essentially unchanged. The genuinely *new* work is confined to what EM could never do at all: the sampler for the volatility paths and their parameters, the leverage coupling that ties level and volatility innovations together, and the step that updates the degrees of freedom. Those are the blocks the following sections build; the state-space we already have.

The skeleton of the sampler

With the paradigm and the bridge in place, the estimator takes the shape of a single Gibbs sweep that cycles, in turn, through the states, the volatility paths, the tail weights and the parameters, drawing each from its full conditional given the others; iterating the sweep and, after a burn-in, collecting the draws yields a sample from the joint posterior of all the unknowns — and hence, by simulating forward, the predictive density and the lower quantiles that the second stage requires. The next section makes this sweep precise: it inventories the quantities to be sampled, fixes the order of the blocks and states the algorithm, while the derivation of each individual full conditional is left to the sections that follow.

The Gibbs Sampler: Sweep and Algorithm

We now make the sweep anticipated above precise, still for the master specification D1b-D2b with leverage. This section is the *map* of the sampler, not its derivations: it says *what* is sampled and in *what order*, organising the unknowns into blocks and giving the algorithm that cycles through them. The individual full conditional distributions — *how* each block is drawn — are derived one at a time in the sections that follow, to which we point as we go.

The quantities to be sampled

The sampler must draw two qualitatively different kinds of unknown.

Latent variables (time-varying). These grow with the sample length T and are the reason the posterior is high-dimensional:

- the **states** $f_{1:T} = (f_1, \ldots, f_T)$ — the latent factor path (in the mixed-frequency model, the augmented companion state $\tilde{f}_{1:T}$);
- the **volatility paths** $h_{1:T} = \big(h^u_{1:T}, \{h^\varepsilon_{i,1:T}\}_{i=1}^M\big)$ — the common log-volatility path and, under scope D2-b, one idiosyncratic path per series;
- the **tail weights** $w = \big(w^u_{1:T}, w^\varepsilon_{1:T}\big)$ — the Student-t scale-mixture weights of switch D1-b, one common and one idiosyncratic weight per period.

Parameters (time-invariant). Fixed in number, collected in $\boldsymbol{\theta}$:

- the state-space matrices $\mathbf{A}, \mathbf{Q}, \boldsymbol{\Lambda}, \mathbf{R}$;
- the **log-volatility parameters** $(\mu, \phi, \sigma^2_\eta)$ for *each* active AR(1) process — $(\mu_u, \phi_u, \sigma^2_u)$ for the common volatility and $(\mu_{\varepsilon,i}, \phi_{\varepsilon,i}, \sigma^2_{\varepsilon,i})$ for each idiosyncratic volatility;
- the **leverage correlations** — the common vector $\boldsymbol{\rho}$ and the scalar $\rho_{\varepsilon,i}$ of each idiosyncratic block;
- the **degrees of freedom** (ν_u, ν_ε) governing the heaviness of the two tail families.

Sampling the joint posterior of all of these at once is intractable; the Gibbs sampler is the device that breaks it into manageable pieces.

The conditional decomposition into blocks

Rather than the joint posterior $p\big(f_{1:T}, h_{1:T}, w, \boldsymbol{\theta} \mid \mathbf{Y}\big)$, the Gibbs sampler draws each block in turn from its *full conditional* — its distribution given the data and the current values of all the other blocks. One full sweep visits, in order:

- **(a) States**, $f_{1:T} \sim p\big(f_{1:T} \mid h_{1:T}, w, \boldsymbol{\theta}, \mathbf{Y}\big)$. Given the volatilities and weights the model is linear-Gaussian, so the entire factor path is drawn in one block. *Derived in the section on the states block.*
- **(b) Volatility paths**, $h_{1:T} \sim p\big(h_{1:T} \mid f_{1:T}, w, \boldsymbol{\theta}, \mathbf{Y}\big)$. The genuinely new, non-conjugate block, where the AR(1) log-volatility structure and the leverage coupling are handled. *Derived in the section on the volatility block.*
- **(c) Tail weights**, $w \sim p\big(w \mid f_{1:T}, h_{1:T}, \boldsymbol{\theta}, \mathbf{Y}\big)$. A conjugate draw, the sampling counterpart of the closed-form weight update already used in the EM E-step. *Derived in the section on the weights block.*

(d) Parameters, $\boldsymbol{\theta} \sim p\big(\boldsymbol{\theta} \mid f_{1:T}, h_{1:T}, w, \mathbf{Y}\big)$. Itself a short ordered sub-sweep over $\{\mathbf{A}, \mathbf{Q}\}$, $\{\boldsymbol{\Lambda}, \mathbf{R}\}$, the log-volatility parameters, the leverage correlations and the degrees of freedom, each drawn from its own conditional. *Derived in the section on the parameter block.*

Each block conditions on the *most recently drawn* value of every other block (a systematic-scan Gibbs sampler), so that within a sweep the output of (a) feeds (b), and so on.

The algorithm

Algorithm 5 states the sweep. It is deliberately schematic: every **draw** step stands for a full conditional whose form is the subject of a later section.

Algorithm 5 Gibbs sampler for the Student-t DFM with stochastic volatility and leverage (master cell D1b-D2b)

Require: data $\mathbf{Y}$; priors on the parameters $\boldsymbol{\theta}$; number of iterations N; burn-in length N_b
Ensure: posterior draws $\big\{f_{1:T}^{(n)}, h_{1:T}^{(n)}, w^{(n)}, \boldsymbol{\theta}^{(n)}\big\}_{n=N_b+1}^{N}$

// 1. Initialise the chain
Set $h_{1:T}^{(0)}$, $w^{(0)}$, $\boldsymbol{\theta}^{(0)}$ to starting values (e.g. from the first-stage EM fit)
// 2. Gibbs sweeps
for $n = 1, \ldots, N$ **do**
    (a) States: draw $f_{1:T}^{(n)} \sim p\big(f_{1:T} \mid h_{1:T}^{(n-1)}, w^{(n-1)}, \boldsymbol{\theta}^{(n-1)}, \mathbf{Y}\big)$
    (b) Volatility: draw $h_{1:T}^{(n)} \sim p\big(h_{1:T} \mid f_{1:T}^{(n)}, w^{(n-1)}, \boldsymbol{\theta}^{(n-1)}, \mathbf{Y}\big)$
    (c) Tail weights: draw $w^{(n)} \sim p\big(w \mid f_{1:T}^{(n)}, h_{1:T}^{(n)}, \boldsymbol{\theta}^{(n-1)}, \mathbf{Y}\big)$
    (d) Parameters: draw $\boldsymbol{\theta}^{(n)} \sim p\big(\boldsymbol{\theta} \mid f_{1:T}^{(n)}, h_{1:T}^{(n)}, w^{(n)}, \mathbf{Y}\big)$
end for
// 3. Discard burn-in
return $\big\{f_{1:T}^{(n)}, h_{1:T}^{(n)}, w^{(n)}, \boldsymbol{\theta}^{(n)}\big\}_{n=N_b+1}^{N}$ as draws from the joint posterior

After the burn-in N_b — the initial transient during which the chain forgets its starting values — the retained draws are (correlated) samples from the joint posterior, and any posterior summary is computed as an average over them.

Blocking and order

A point of principle underlies steps (a) and (b): we sample the *whole* path at once, not one period at a time. The factor path $f_{1:T}$ is drawn as a single block — by a simulation smoother, the forward-filtering backward-sampling recursion built on the same Kalman filter inherited from the first stage, whose details are the subject of the states section — and the volatility paths $h_{1:T}$ are blocked in the same spirit.

The reason is the mixing of the chain. The successive states of a latent path are strongly dependent through their dynamics, so a sampler that updated f_t one period at a time, each conditional on its neighbours f_{t-1} and f_{t+1}, could move each coordinate only slightly per sweep: successive draws of the chain would be highly autocorrelated, the chain would explore the posterior slowly, and the effective sample size for a given number of iterations would be small. Drawing the entire correlated path jointly removes that within-path dependence from the chain's bottleneck: it produces far less autocorrelated draws and hence faster, more reliable convergence. This is precisely the gain delivered by the forward-filtering backward-sampling algorithm of Carter and Kohn (1994) and Frühwirth-Schnatter (1994), which we use for the states.

As for the *order* of the four blocks, any fixed, valid scan leaves the joint posterior invariant, so the order is not a matter of correctness; it can, however, affect the mixing speed. We adopt the natural sequence states $\rightarrow$ volatility $\rightarrow$ weights $\rightarrow$ parameters, which lets each conditioning set use the freshest available draws.

Practical ingredients

Four practical elements complete the skeleton; each is taken up in detail in its own place and is only named here.

- **Priors.** The Bayesian treatment requires a prior on every parameter in $\boldsymbol{\theta}$. We use proper, weakly informative, conjugate-where-possible priors; their exact form is specified together with the parameter block, since it is there that they combine with the likelihood to give the conditionals of step (d).
- **Initialisation.** The chain must be started somewhere; we initialise it at the first-stage EM estimates, which place it in a region of high posterior mass and shorten the burn-in.
- **Burn-in.** The initial draws, still influenced by the starting values, are discarded; only the post-burn-in draws are treated as posterior samples.
- **Convergence diagnostics.** Whether the chain has reached and adequately explored its stationary distribution is assessed *ex post* — by trace plots, autocorrelation and effective sample size, and the potential-scale-reduction statistic of Gelman and Rubin (1992) across independent chains.

With the skeleton fixed, the sections that follow derive the four blocks in turn, beginning with the states.

Sampling the States $f_{1:T}$ (Gibbs Step (a))

We begin the sweep with step (a), the states. This is the friendliest block of the four, for a reason already announced in the bridge of the introductory section: *conditional on the volatility paths $h_{1:T}$, the tail weights w and the parameters $\boldsymbol{\theta}$*, the model collapses back to a linear-Gaussian state-space model — an ordinary one, save that its innovation covariances are time-varying. Drawing the whole factor path from such a model is a standard and fully solved problem, the *simulation smoother*. Almost all the machinery this requires — the companion construction and the Kalman filter — is already in hand from the first stage; the only genuinely new ingredient is a backward *sampling* pass in place of the backward *averaging* pass of the smoother.

The conditional model is linear-Gaussian

Fix the volatility paths, the weights and the parameters. The two multipliers attached to each shock are then known numbers, and the conditional innovation covariances are the *deterministic* time-varying matrices

$$\mathbf{Q}_t \;=\; \frac{h_t^u}{w_t^u}\,\mathbf{Q}, \qquad \mathbf{R}_{i,t} \;=\; \frac{h_{i,t}^{\varepsilon}}{w_t^{\varepsilon}}\,r_i, \qquad \mathbf{R}_t \;=\; \operatorname{diag}\bigl(\mathbf{R}_{1,t},\ldots,\mathbf{R}_{M,t}\bigr), \tag{188}$$

where h_t^u and the $h_{i,t}^{\varepsilon}$ carry the persistent stochastic volatility and $1/w_t^u$, $1/w_t^{\varepsilon}$ the Student-t tail rescaling. With these in place the data-generating process is exactly a linear-Gaussian state-space model with time-varying covariances,

$$f_t = \mathbf{A}\,f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0,\mathbf{Q}_t), \qquad y_t = \boldsymbol{\Lambda}\,f_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,\mathbf{R}_t),$$

identical in form to the weighted system the EM E-step already filtered, with the single change that the factor-innovation covariance now also carries the volatility multiplier h_t^u and the idiosyncratic covariance the multipliers $h_{i,t}^{\varepsilon}$.

In the mixed-frequency model this is run, exactly as before, on the *augmented companion state* $\tilde{f}_t = (f_t', f_{t-1}', \ldots, f_{t-4}')' \in \mathbb{R}^{5r}$. The companion matrix $\tilde{\mathbf{A}}$ and the observation matrix $\tilde{\boldsymbol{\Lambda}}$ — which carries the Mariano–Murasawa aggregation weights — are *unchanged* from the baseline: the volatility enters *only* through the covariance of the monthly innovation feeding the leading block, so that the augmented innovation covariance is the singular matrix

$$\tilde{\mathbf{Q}}_t \;=\; \begin{pmatrix} \mathbf{Q}_t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{5r\times 5r}, \qquad \mathbf{Q}_t = \frac{h_t^u}{w_t^u}\,\mathbf{Q}, \tag{189}$$

with $\mathbf{Q}_t$ in the top-left $r \times r$ block and zeros elsewhere, exactly as in the baseline companion form but with $\mathbf{Q}_t$ in place of $\mathbf{Q}/w_t^u$. Everything structural in the mixed-frequency construction therefore transfers verbatim; volatility changes only *what variance* the leading innovation has.

Draw, not mean: what changes relative to EM

In the EM algorithm the Kalman smoother returned, for each t, the *smoothed mean* $\hat{f}_{t|T} = \mathbb{E}[f_t \mid \mathbf{Y}, \cdot]$ and its covariance $P_{t|T}$ — a single best (minimum mean-square-error) point estimate of the factor path and its uncertainty. The Gibbs sampler needs something different: not the mean of the conditional distribution of the states, but a *draw* from it — an entire sampled path $f_{1:T}$ distributed according to $p\bigl(f_{1:T} \mid h_{1:T}, w, \boldsymbol{\theta}, \mathbf{Y}\bigr)$. This is what a *simulation smoother* produces. The drawn path is random; averaged over many sweeps its mean reproduces $\hat{f}_{t|T}$ and its sampling covariance reproduces $P_{t|T}$, but each individual draw is a full trajectory, as the joint posterior over states requires.

The simulation smoother: forward filtering, backward sampling

The classical construction is forward-filtering backward-sampling (FFBS), due to Carter and Kohn (1994) and Frühwirth-Schnatter (1994). It rests on the backward factorisation of the joint conditional density, which follows from the Markov structure of the state-space model:

$$p(\tilde{f}_{1:T} \mid \mathbf{Y}, \cdot) \;=\; p(\tilde{f}_T \mid \mathbf{Y}, \cdot) \prod_{t=1}^{T-1} p(\tilde{f}_t \mid \tilde{f}_{t+1}, y_{1:t}, \cdot), \tag{190}$$

where the conditioning on $(h_{1:T}, w, \boldsymbol{\theta})$ is suppressed as a dot. The key simplification is in the last factor: given $\tilde{f}_{t+1}$, the state $\tilde{f}_t$ is independent of all future observations $y_{t+1:T}$ — the future depends on $\tilde{f}_t$ only through $\tilde{f}_{t+1}$ — so $p(\tilde{f}_t \mid \tilde{f}_{t+1}, y_{1:T}, \cdot) = p(\tilde{f}_t \mid \tilde{f}_{t+1}, y_{1:t}, \cdot)$, and only the *filtered* information up to t is needed at each backward step. Equation (190) prescribes the algorithm: sample $\tilde{f}_T$, then descend, sampling each $\tilde{f}_t$ given the already-drawn $\tilde{f}_{t+1}$.

Forward pass. Run the weighted Kalman filter of the E-step unchanged — now fed the time-varying covariances $\tilde{\mathbf{Q}}_t$ of (189) and $\tilde{\mathbf{R}}_t = \operatorname{diag}(\mathbf{R}_{i,t})$ — storing, for $t = 1, \ldots, T$, the filtered moments $(\hat{f}_{t|t}, P_{t|t})$ and the one-step predicted moments $(\hat{f}_{t+1|t}, P_{t+1|t})$, with $\hat{f}_{t+1|t} = \tilde{\mathbf{A}}\hat{f}_{t|t}$ and $P_{t+1|t} = \tilde{\mathbf{A}}P_{t|t}\tilde{\mathbf{A}}' + \tilde{\mathbf{Q}}_{t+1}$.

Backward sampling pass. Initialise at the end of the sample by drawing from the filter's final distribution,

$$\tilde{f}_T \;\sim\; \mathcal{N}(\hat{f}_{T|T}, P_{T|T}). \tag{191}$$

Then, for $t = T-1, T-2, \ldots, 1$, draw $\tilde{f}_t$ from its conditional given the already-sampled $\tilde{f}_{t+1}$ and the data up to t. That conditional is Gaussian and is obtained by standard conditioning on the joint law of $(\tilde{f}_t, \tilde{f}_{t+1})$ given $y_{1:t}$, in which

$$\tilde{f}_t \mid y_{1:t} \sim \mathcal{N}(\hat{f}_{t|t}, P_{t|t}), \qquad \operatorname{Cov}(\tilde{f}_t, \tilde{f}_{t+1} \mid y_{1:t}) = P_{t|t}\tilde{\mathbf{A}}', \qquad \tilde{f}_{t+1} \mid y_{1:t} \sim \mathcal{N}(\hat{f}_{t+1|t}, P_{t+1|t}).$$

Writing the conditioning gain as

$$J_t \;=\; P_{t|t}\,\tilde{\mathbf{A}}'\,P_{t+1|t}^{-1}, \tag{192}$$

which is *exactly* the Rauch–Tung–Striebel smoother gain already derived for the E-step, the backward draw is

$$\boxed{\tilde{f}_t \mid \tilde{f}_{t+1}, y_{1:t}, \cdot \;\sim\; \mathcal{N}(m_t, V_t), \qquad \begin{aligned} m_t &= \hat{f}_{t|t} + J_t(\tilde{f}_{t+1} - \hat{f}_{t+1|t}), \\ V_t &= P_{t|t} - J_t\,P_{t+1|t}\,J_t' \;=\; P_{t|t} - J_t\,\tilde{\mathbf{A}}\,P_{t|t}. \end{aligned}} \tag{193}$$

The conditional mean m_t has the same form as the RTS smoothed mean (56), except that the *drawn* value $\tilde{f}_{t+1}$ replaces the smoothed mean $\hat{f}_{t+1|T}$; the conditional covariance V_t is the filtered covariance reduced by the information that fixing $\tilde{f}_{t+1}$ contributes. Carrying this recursion down to $t = 1$ produces a single trajectory $\tilde{f}_{1:T}$ drawn jointly from $p(\tilde{f}_{1:T} \mid \mathbf{Y}, \cdot)$ — the entire path in one block, exactly the blocking advocated in the previous section.

The singular companion covariance. A practical caveat specific to the mixed-frequency form: because $\tilde{\mathbf{Q}}_t$ in (189) has rank r, not $5r$, the conditional covariance V_t in (193) is singular. This is not a defect but the deterministic identity-shift structure of the companion state: the lagged blocks of $\tilde{f}_t$ already appear inside $\tilde{f}_{t+1}$, so they are fixed once $\tilde{f}_{t+1}$ is drawn, and only the leading r-dimensional factor block carries a genuine random draw. In practice one therefore samples only the non-degenerate component — equivalently, draws the monthly factor f_t and reads the lagged blocks off $\tilde{f}_{t+1}$ — rather than attempting to sample the full $5r$-vector from a rank-deficient Gaussian.

An efficient alternative: the Durbin–Koopman smoother

When the state dimension is large — here $5r$ in the companion form — storing every filtered covariance $P_{t|t}$ and performing an explicit Gaussian draw at each t becomes costly. Durbin and Koopman (2002) give a simulation smoother that avoids both. The idea is disarmingly simple: draw an *unconditional* path and pseudo-observations $(\tilde{f}^{+}_{1:T}, y^{+}_{1:T})$ by simulating forward from the model; run the ordinary (mean) Kalman smoother once on the artificial data $y^{+}_{1:T}$ and once on the real data $y_{1:T}$ (or, more cheaply, on their difference); and form

$$\tilde{f}^{\text{draw}}_{1:T} \;=\; \tilde{f}^{+}_{1:T} \;+\; (\hat{f}_{1:T|T} - \hat{f}^{+}_{1:T|T}),$$

the simulated path corrected by the smoothed discrepancy. The result is *distributionally identical* to the FFBS draw of (193), but it reuses only the mean smoother — no backward sampling and no storage of the full covariance sequence — which makes it the preferred route for higher-dimensional states. Either sampler may be used interchangeably for this block.

What is reused

It is worth stating plainly how little of this block is new. The augmented companion construction ($\tilde{\mathbf{A}}$, $\tilde{\mathbf{\Lambda}}$, the Mariano–Murasawa weights, the singular $\tilde{\mathbf{Q}}_t$), the forward Kalman filter and even the smoother gain J_t of (192) are all inherited verbatim from the EM machinery; the only modifications are that the time-varying covariances now also carry the volatility multipliers $h^u_t, h^\varepsilon_{i,t}$ through (188)–(189), and that the backward pass *samples* from (193) instead of *averaging* as in (56). In code terms it is a thin layer over functions that already exist — the genuinely new work of the sampler lies in the next block, the volatility paths.

Sampling the Volatility Paths $h_{1:T}$: Base Case (Gibbs Step (b))

We now reach step (b), and with it the genuinely new part of the estimator — the block that justifies the move to MCMC in the first place. We derive it here in its *base* form, *without* the leverage coupling; the correlation between level and volatility shocks is added in the next section, as a modification of the joint law sampled here. Throughout we condition, as the Gibbs sweep prescribes, on the states $f_{1:T}$ drawn in step (a), on the tail weights w and on the parameters $\boldsymbol{\theta}$.

The problem: volatility enters nonlinearly

Conditional on the states, the weights and the parameters, the residuals that drive each volatility process are *known*. For the common factor, the level shock is reconstructed from the sampled path,

$$u_t \;=\; f_t - \mathbf{A}\, f_{t-1} \;\in \mathbb{R}^r, \tag{194}$$

and for each idiosyncratic series the residual is

$$\varepsilon_{i,t} \;=\; y_{i,t} - (\boldsymbol{\Lambda}\, f_t)_i \;\in \mathbb{R}. \tag{195}$$

In the base specification these residuals obey, recalling the data-generating process,

$$u_t \mid h_t^u, w_t^u \sim \mathcal{N}\big(0,\, h_t^u\, \mathbf{Q}/w_t^u\big), \qquad \varepsilon_{i,t} \mid h_{i,t}^\varepsilon, w_t^\varepsilon \sim \mathcal{N}\big(0,\, h_{i,t}^\varepsilon\, r_i/w_t^\varepsilon\big).$$

Sampling the volatility path is hard because h_t enters *nonlinearly*: it is the conditional *variance* of the residual, not a location, so a residual is of the form $\text{residual}_t = \sqrt{h_t} \times (\text{Gaussian})$ and the latent state we wish to recover, $\log h_t$, is related to the observable residual through a squared, non-Gaussian map. This is precisely the structure that broke the closed-form M-step and forced the change of paradigm. The standard resolution is that of Kim, Shephard and Chib (1998), which we now follow.

The linearising transformation

Treat first a generic *scalar* residual e_t with conditional variance h_t, i.e. $e_t \mid h_t \sim \mathcal{N}(0, h_t)$; we map the two processes onto this template below. Write $e_t = \sqrt{h_t}\, z_t$ with $z_t \sim \mathcal{N}(0,1)$, square, and take logs:

$$\underbrace{\log\left(e_t^2\right)}_{y_t^*} \;=\; \log h_t \;+\; \underbrace{\log\left(z_t^2\right)}_{\xi_t}. \tag{196}$$

This is a remarkable simplification: (196) is *linear* in the state $\log h_t$. The price is the error term $\xi_t = \log(z_t^2)$, which is the log of a χ_1^2 variate — a *fixed, non-Gaussian* distribution, left-skewed, with mean $\mathbb{E}[\xi_t] = -1.2704$ and variance $\mathrm{Var}(\xi_t) = \pi^2/2$. In practice one guards against $e_t^2 \approx 0$ by the usual "offset," $y_t^* = \log(e_t^2 + c)$ with a small constant c, whence the name *offset mixture*. Paired with the AR(1) dynamics of the log-volatility already specified in the data-generating process,

$$\log h_t \;=\; \mu + \phi\big(\log h_{t-1} - \mu\big) + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \tag{197}$$

the pair (196)–(197) is a *linear* state-space model with state $\log h_t$ — but with a non-Gaussian measurement error. That is the only obstacle that remains.

The Kim–Shephard–Chib mixture approximation

The device of Kim, Shephard and Chib (1998) is to approximate the $\log \chi_1^2$ density of ξ_t by a finite *mixture of seven Gaussian components*,

$$p(\xi_t) \;\approx\; \sum_{j=1}^{7} q_j \, \mathcal{N}\big(\xi_t;\, m_j,\, v_j^2\big), \tag{198}$$

where the weights, means and variances $\{q_j, m_j, v_j^2\}_{j=1}^{7}$ are *fixed, known constants* — tabulated once and for all in Kim, Shephard and Chib (1998, Table 4) by matching the moments of the $\log \chi_1^2$ distribution — and are not estimated. The mixture is made tractable by data augmentation: introduce, for each t, a *component indicator* $s_t \in \{1, \ldots, 7\}$ with prior $\Pr(s_t = j) = q_j$, such that

$$\xi_t \mid s_t \;\sim\; \mathcal{N}\big(m_{s_t},\, v_{s_t}^2\big). \tag{199}$$

Conditional on the indicator s_t, the error is Gaussian, and so the observation equation (196) becomes

$$y_t^* \mid \log h_t,\, s_t \;\sim\; \mathcal{N}\big(\log h_t + m_{s_t},\, v_{s_t}^2\big) : \tag{200}$$

a *linear-Gaussian* observation, with a component-dependent mean offset m_{s_t} and variance $v_{s_t}^2$. Conditional on $s_{1:T}$, then, the model (200)–(197) is exactly a linear-Gaussian state-space system for the scalar state $\log h_t$.

The seven constants are reported in Table 3.

j	q_j (weight)	m_j (mean)	v_j^2 (variance)
1	0.04395	1.50746	0.16735
2	0.24566	0.52478	0.34023
3	0.34001	−0.65098	0.64009
4	0.25750	−2.35859	1.26261
5	0.10556	−5.24321	2.61369
6	0.00002	−9.83726	5.17950
7	0.00730	−11.40039	5.79596

Table 3: Seven-component Gaussian mixture approximating the $\log \chi_1^2$ density of ξ_t, used in the base volatility sampler. Original mixture of Kim, Shephard and Chib (1998); values as reproduced in Omori, Chib, Shephard and Nakajima (2007, Table 1), checked digit by digit against the original PDF. Internal consistency checks also pass ($\sum_j q_j = 1$ and $\sum_j q_j m_j \approx -1.2704$).

The volatility sub-sweep

The augmentation turns step (b) into a two-move Gibbs sub-sweep that alternates between the indicators and the log-volatility path.

(i) Draw the indicators $s_{1:T} \mid \log h_{1:T}$. Given the current log-volatility path and the residuals, the indicators are *conditionally independent across* t, each drawn from a discrete distribution on $\{1, \ldots, 7\}$ obtained by Bayes' rule from (198):

$$\Pr\big(s_t = j \mid \log h_t,\, y_t^*\big) \;\propto\; q_j \, \mathcal{N}\big(y_t^*;\, \log h_t + m_j,\, v_j^2\big), \qquad j = 1, \ldots, 7, \tag{201}$$

normalised over the seven components — a trivial multinomial draw per period.

(ii) Draw the path $\log h_{1:T} \mid s_{1:T}$. Given the indicators, (200)–(197) is a linear-Gaussian state-space model in the scalar state $\log h_t$. We therefore sample the whole path by *exactly the same* *forward-filtering backward-sampling simulation smoother derived in the previous section* — this is the reuse that makes the volatility block manageable. The only adaptation is bookkeeping: the "observation" is y_t^*, the measurement equation has the known offset m_{s_t} and the known variance $v_{s_t}^2$, the state is the scalar $\log h_t$ with transition ϕ and innovation variance σ_η^2, and the stationary initialisation is $\log h_0 \sim \mathcal{N}(\mu, \sigma_\eta^2/(1-\phi^2))$. Running the scalar Kalman filter forward and the backward-sampling pass (193) of the previous section yields a draw of the entire log-volatility path in one block.

Iterating (i) and (ii) within the sweep — or simply performing each once per Gibbs iteration, since the outer sweep already cycles — draws $(s_{1:T}, \log h_{1:T})$ from their joint conditional.

Repetition across all volatility processes

Under scope D2-b the model carries $M+1$ independent log-volatility processes — the common h^u and the M idiosyncratic h_i^ε — and the sub-sweep above is applied to *each*, with its own residuals and its own AR(1) parameters $(\mu, \phi, \sigma_\eta^2)$.

- **Idiosyncratic processes.** Each $\varepsilon_{i,t}$ of (195) is already scalar; the template applies directly with the whitened residual $e_{i,t} = \sqrt{w_t^\varepsilon/r_i}\,\varepsilon_{i,t}$, so that $e_{i,t} \mid h_{i,t}^\varepsilon \sim \mathcal{N}(0, h_{i,t}^\varepsilon)$ and $y_{i,t}^* = \log(e_{i,t}^2 + c)$.
- **Common factor.** Here h_t^u is a single scalar that scales the whole r-vector u_t. Whitening by the base scale and the tail weight, $\tilde{u}_t = \sqrt{w_t^u}\,\mathbf{Q}^{-1/2}u_t$, gives $\tilde{u}_t \mid h_t^u \sim \mathcal{N}(0, h_t^u I_r)$, i.e. r conditionally independent scalar measurements $\tilde{u}_{t,k} \mid h_t^u \sim \mathcal{N}(0, h_t^u)$ of the *same* state. The template applies with r log-square observations $y_{t,k}^* = \log(\tilde{u}_{t,k}^2 + c)$ per period, each with its own indicator $s_{t,k}^u$; the backward-sampling pass simply runs on a scalar state fed r measurements per time point.

The cost is explicit: one volatility sub-sweep per process, hence $M+1$ simulation-smoother passes per Gibbs iteration. This is the principal computational burden the stochastic-volatility extension adds.

What is reused, what is new

The decomposition is clean. *Reused:* the forward-filtering backward-sampling simulation smoother of the previous section is invoked verbatim, now on a scalar log-volatility state, to perform move (ii) for every process. *New to this block:* the log-square transformation (196) that linearises the measurement; the Kim–Shephard–Chib mixture (198) that restores conditional Gaussianity; and the indicator draw (201) that the mixture requires. This — the transformation, the offset-mixture approximation, and the auxiliary indicators — is the genuinely new estimation machinery introduced by the stochastic-volatility model, and the reason the whole estimator had to become a sampler. The leverage coupling, taken up next, modifies the *joint* law of the level and volatility innovations sampled here, but leaves this scaffolding in place.

Adding Leverage to the Volatility Sampler (Gibbs Step (b), Full Version)

The previous section built the volatility sampler in its base form, under the assumption that the level shock and the log-volatility innovation are independent. We now restore the leverage coupling of the asymmetry section. As anticipated, leverage modifies only the *joint law* of the level and volatility innovations; the scaffolding of the base sampler — the log-square transformation, the Kim–Shephard–Chib mixture, the forward-filtering backward-sampling pass — remains in place. What changes is one thing: the conditional distribution of the log-volatility innovation acquires a *drift* that depends on the observed level residual.

The problem leverage introduces

In the base case the level shock z_t and the log-volatility innovation η_t were independent, so the volatility path could be sampled *ignoring* the level residuals beyond their squared magnitude: the information $\{y_t^*\}$ of (196) was all that mattered. Leverage breaks this. Under the correlated-error specification of the asymmetry section, the standardized level shock z_t and the innovation η_t that moves the log-volatility are jointly Gaussian with a non-zero correlation ρ. Consequently the distribution of η_t — and hence of the next log-volatility — *depends on the observed level shock*: a draw of the volatility path can no longer treat the level residual as uninformative about η_t. The sampler must condition the volatility innovation on the level residual.

The conditional law of the volatility innovation

The correction follows from the joint Gaussian law posited in the data-generating process, by elementary conditioning. Take first a generic *scalar* process with log-volatility innovation standard deviation σ_η and leverage correlation ρ. The data-generating process couples the standardized level shock $z_t \sim \mathcal{N}(0,1)$ and the innovation η_t through

$$\begin{pmatrix} z_t \\ \eta_t \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho\,\sigma_\eta \\ \rho\,\sigma_\eta & \sigma_\eta^2 \end{pmatrix} \right),$$

where z_t is recovered from the known residual by standardising out the base scale, the tail weight and the current volatility — for an idiosyncratic series, $z_{i,t} = \varepsilon_{i,t} \sqrt{w_t^\varepsilon / r_i} / \sqrt{h_{i,t}^\varepsilon}$, i.e. $z_{i,t} = e_{i,t} / \sqrt{h_{i,t}^\varepsilon}$ in the notation of the base case. Conditioning the bivariate normal on z_t gives the Gaussian

$$\boxed{\eta_t \mid z_t \;\sim\; \mathcal{N}\Big(\underbrace{\rho\,\sigma_\eta\,z_t}_{\text{leverage drift}} \;,\; \underbrace{\sigma_\eta^2(1-\rho^2)}_{\text{reduced variance}} \Big).} \tag{202}$$

The independent case of the previous section is recovered exactly at $\rho = 0$, where the mean collapses to zero and the variance to σ_η^2. For the common factor the same conditioning, applied to the joint law $(z_t^u, \eta_t^u) \sim \mathcal{N}(0, [\,I_r,\ \sigma_u \boldsymbol{\rho};\ \sigma_u \boldsymbol{\rho}',\ \sigma_u^2\,])$ of the asymmetry section, yields

$$\eta_t^u \mid z_t^u \;\sim\; \mathcal{N}\Big(\sigma_u\,\boldsymbol{\rho}'\,z_t^u,\ \sigma_u^2(1-\boldsymbol{\rho}'\boldsymbol{\rho})\Big), \qquad z_t^u = \tilde{u}_t / \sqrt{h_t^u}, \tag{203}$$

with $\tilde{u}_t = \sqrt{w_t^u}\,\mathbf{Q}^{-1/2} u_t$ the whitened factor innovation of the base case. Because the leverage sign is negative, a contractionary level shock ($z_t < 0$) raises the conditional mean of η_t and so pushes the next log-volatility up — the asymmetry, now entering the *estimator* exactly as it entered the model.

Timing: which residual corrects which transition

The conditional law (202) fixes the *shape* of the correction; the *timing* convention of the asymmetry section fixes *which* log-volatility transition it modifies, since it determines which η is coupled to z_t.

- **Contemporaneous timing** (our principal specification): z_t is correlated with η_t, the innovation that drives $\log h_t$ in (197). The leverage drift then enters the *same-period* transition,

$$\log h_t \mid \log h_{t-1}, z_t \;\sim\; \mathcal{N}\Big(\mu + \phi\big(\log h_{t-1} - \mu\big) + \rho\,\sigma_\eta\, z_t,\;\; \sigma_\eta^2\big(1-\rho^2\big)\Big). \tag{204}$$

- **Lagged timing** (the retained variant): z_t is correlated with η_{t+1}, which drives $\log h_{t+1}$, so the very same drift $\rho\sigma_\eta z_t$ corrects the *next-period* transition $\log h_t \to \log h_{t+1}$ instead.

In both cases the modification is identical in form — the transition mean of the log-volatility AR(1) is shifted by $\rho\sigma_\eta z_t$ and its innovation variance shrunk to $\sigma_\eta^2(1-\rho^2)$ — and only the index of the corrected transition differs. We adopt the contemporaneous form (204) to remain coherent with the principal specification declared in the asymmetry section, flagging the timing as a choice with consequences for the sampler, to which we return below.

The modified sampler

The leverage thus enters the volatility sub-sweep of the base case at a single point. Move (i), the indicator draw (201), is unchanged — it concerns only the measurement side (200), which leverage does not touch. Move (ii), the forward-filtering backward-sampling draw of $\log h_{1:T}$, runs on the *same* linear-Gaussian state-space, with the *only* change that the state transition is now (204) in place of the symmetric (197): a leverage-dependent intercept $\rho\sigma_\eta z_t$ is added to the AR(1) mean and the state-innovation variance is reduced from σ_η^2 to $\sigma_\eta^2(1-\rho^2)$. Everything else — the log-square transform (196), the Kim–Shephard–Chib observation density (200), the indicators, and the backward-sampling recursion (193) itself — is identical to the base case. In one line: *leverage re-drifts the state equation of the volatility filter and leaves the measurement equation alone.*

Idiosyncratic leverage: the same coupling on all M series

So far the leverage has been written for the generic scalar template and for the common factor. Under scope D2-b it applies with equal force to the idiosyncratic block: each of the M series carries its own stochastic volatility and, with it, its own leverage coupling. These are M processes on the same footing as the common one, not a detail, and we make them explicit.

For series i the asymmetry section couples the standardized idiosyncratic shock $z^{\varepsilon}_{i,t}$ with its log-volatility innovation $\eta^{\varepsilon}_{i,t}$ through the *scalar* joint law

$$\begin{pmatrix} z^{\varepsilon}_{i,t} \\ \eta^{\varepsilon}_{i,t} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{\varepsilon,i}\,\sigma_{\varepsilon,i} \\ \rho_{\varepsilon,i}\,\sigma_{\varepsilon,i} & \sigma^2_{\varepsilon,i} \end{pmatrix}\right), \qquad z^{\varepsilon}_{i,t} = \frac{e_{i,t}}{\sqrt{h^{\varepsilon}_{i,t}}},$$

with $e_{i,t} = \sqrt{w^{\varepsilon}_t/r_i}\,\varepsilon_{i,t}$ the whitened idiosyncratic residual of the base case and $\rho_{\varepsilon,i}, \sigma_{\varepsilon,i}$ the leverage correlation and the log-volatility innovation standard deviation *specific to series i*. Conditioning on the observed standardized shock gives, exactly as in (202) but now once per series,

$$\eta^{\varepsilon}_{i,t} \mid z^{\varepsilon}_{i,t} \;\sim\; \mathcal{N}\Big(\rho_{\varepsilon,i}\,\sigma_{\varepsilon,i}\, z^{\varepsilon}_{i,t},\; \sigma^2_{\varepsilon,i}\big(1-\rho^2_{\varepsilon,i}\big)\Big), \qquad i = 1,\ldots,M. \tag{205}$$

This is the scalar conditional (202) replicated M times, with a single scalar leverage correlation $\rho_{\varepsilon,i}$ per series — the idiosyncratic counterpart of the vector $\boldsymbol{\rho}$ that governs the common block in (203), here scalar because each idiosyncratic shock is univariate.

Entry into each idiosyncratic sampler. The drift enters the volatility sub-sweep of series i exactly as it does for the common factor and the generic scalar. Under the contemporaneous timing, the AR(1) transition of $\log h^{\varepsilon}_{i,t}$ — the per-series analogue of (204) — acquires the leverage intercept and the reduced innovation variance,

$$\log h^{\varepsilon}_{i,t} \mid \log h^{\varepsilon}_{i,t-1}, z^{\varepsilon}_{i,t} \;\sim\; \mathcal{N}\Big(\mu_{\varepsilon,i} + \phi_{\varepsilon,i}\big(\log h^{\varepsilon}_{i,t-1} - \mu_{\varepsilon,i}\big) + \rho_{\varepsilon,i}\,\sigma_{\varepsilon,i}\,z^{\varepsilon}_{i,t},\;\; \sigma^2_{\varepsilon,i}\big(1-\rho^2_{\varepsilon,i}\big)\Big), \tag{206}$$

while the measurement side of series i — its log-square transform and its Kim–Shephard–Chib indicators — is left untouched, just as in the modified sampler above. The lagged variant again shifts the same drift $\rho_{\varepsilon,i}\sigma_{\varepsilon,i}z^{\varepsilon}_{i,t}$ onto the $t \to t+1$ transition. In short, the idiosyncratic block is the common construction repeated M times, one scalar leverage correlation at a time.

An honest caveat: the mixture and the correlation interact

The presentation above is exact at the level of the population conditional (202), but it understates a real difficulty in turning that conditional into a sampler, and we flag it plainly. Two complications arise.

First, the standardized shock $z_t = e_t/\sqrt{h_t}$ that enters the drift *itself depends on the volatility* h_t being sampled. The leverage term $\rho\sigma_\eta z_t$ is therefore not a constant intercept but a function of the state, so the transition (204) is not, strictly, linear-Gaussian in $\log h_t$ — the very nonlinearity the base transformation removed creeps back through the leverage channel. Second, the log-square map (196) discards the *sign* of the level residual, $d_t = \operatorname{sign}(e_t)$, yet the leverage correlation is with the *signed* shock z_t, so the sign must be reinstated as additional conditioning information.

These two obstacles — a state-dependent drift and a discarded sign — attend equally each of the $M+1$ volatility processes, and they admit two distinct resolutions, each paired with one of the two timing conventions of the asymmetry section. Rather than settle the matter here, we treat it in full once the weights block is in place: the section "Two Routes for the Leverage" below derives *both* estimation strategies — the contemporaneous timing handled by a Metropolis update, and the lagged timing handled by the sign-augmented mixture of Omori, Chib, Shephard and Nakajima (2007) — as two branches of a common trunk, and weighs the trade-off between them.

Sampling the Tail Weights w (Gibbs Step (c))

After the volatility block — the most technical of the four — step (c) is a relief. The Student-t tail weights are *conjugate*: their full conditional is again a Gamma, drawn analytically in closed form, with no mixture, no Metropolis step and no simulation smoother. The block reuses, almost verbatim, the weight algebra already present in the EM treatment of the heavy-tailed model; the single new ingredient is that the residual is now standardised by the volatility before it enters the update.

The scale-mixture structure, recalled

The Student-t shocks of the model are, by construction, Gaussian *conditional on a latent weight*, the weight carrying a Gamma prior — the scale-mixture-of-normals representation of the t distribution. Two weights appear: w_t^u, which scales the entire factor innovation, and w_t^ε, which scales the entire idiosyncratic vector, with priors

$$w_t^u \overset{iid}{\sim} \mathrm{Gamma}\Big(\tfrac{\nu_u}{2}, \tfrac{\nu_u}{2}\Big), \qquad w_t^\varepsilon \overset{iid}{\sim} \mathrm{Gamma}\Big(\tfrac{\nu_\varepsilon}{2}, \tfrac{\nu_\varepsilon}{2}\Big), \tag{207}$$

independently across t. In the full stochastic-volatility model the conditional law of each shock now carries *both* multipliers,

$$u_t \mid h_t^u, w_t^u \sim \mathcal{N}\big(0,\, h_t^u\,\mathbf{Q}/w_t^u\big), \qquad \varepsilon_t \mid \mathbf{H}_t^\varepsilon, w_t^\varepsilon \sim \mathcal{N}\big(0,\, \mathbf{H}_t^\varepsilon\,\mathbf{R}/w_t^\varepsilon\big),$$

with $\mathbf{H}_t^\varepsilon = \mathrm{diag}(h_{1,t}^\varepsilon, \ldots, h_{M,t}^\varepsilon)$: the persistent volatility h_t and the tail weight $1/w_t$ act *together* on the variance, the first slow and serially dependent, the second iid and abrupt.

Gaussian–Gamma conjugacy and the deflated residual

The key property is the one already exploited in the EM E-step: a Gamma prior on a precision multiplier, times a Gaussian likelihood in which that multiplier scales the precision, yields a Gamma posterior. Conditional on the states, the volatility path and the parameters, the residual driving each shock is known, and the only twist relative to the EM derivation is that the variance now contains the sampled volatility as well as the weight. The residual must therefore enter the update *already deflated by the volatility*.

Concretely, for the idiosyncratic vector the likelihood contribution of w_t^ε, as a function of w_t^ε, is

$$p\big(y_t \mid f_t, \mathbf{H}_t^\varepsilon, w_t^\varepsilon, \theta\big) \;\propto\; (w_t^\varepsilon)^{M/2}\,\exp\Big(-\tfrac{w_t^\varepsilon}{2}\,\check{d}_t^\varepsilon\Big),$$

where the *volatility-deflated* squared Mahalanobis residual is

$$\check{d}_t^\varepsilon \;\equiv\; (y_t - \mathbf{\Lambda} f_t)'\,\big(\mathbf{H}_t^\varepsilon \mathbf{R}\big)^{-1}(y_t - \mathbf{\Lambda} f_t) \;=\; \sum_{i=1}^{M} \frac{\varepsilon_{i,t}^2}{h_{i,t}^\varepsilon\, r_i}, \qquad \varepsilon_{i,t} = y_{i,t} - (\mathbf{\Lambda} f_t)_i. \tag{208}$$

This is exactly the EM quantity d_t^ε of (43), except that each squared residual is divided by its own sampled volatility $h_{i,t}^\varepsilon$: the residual is purged of the volatility that the previous block already explained, so that what remains for the tail weight to absorb is the *excess* dispersion beyond the volatility regime. Multiplying by the Gamma prior (207) and collecting powers and exponentials gives the conjugate posterior

$$\boxed{w_t^\varepsilon \mid y_t, f_t, \mathbf{H}_t^\varepsilon, \theta \;\sim\; \mathrm{Gamma}\left(\frac{\nu_\varepsilon + M}{2},\; \frac{\nu_\varepsilon + \check{d}_t^\varepsilon}{2}\right).} \tag{209}$$

The factor-side weight is identical in form, with the state residual in place of the observation residual and its deflation by the scalar common volatility h_t^u:

$$\check{d}_t^u \;\equiv\; \frac{1}{h_t^u}\,(f_t - \mathbf{A} f_{t-1})'\,\mathbf{Q}^{-1}(f_t - \mathbf{A} f_{t-1}) \;=\; \frac{d_t^u}{h_t^u},$$

$$\boxed{w_t^u \mid f_t, f_{t-1}, h_t^u, \theta \;\sim\; \text{Gamma}\left(\frac{\nu_u + r}{2},\; \frac{\nu_u + \check{d}_t^u}{2}\right),} \tag{210}$$

with d_t^u the EM residual (45). Setting all volatilities to one ($h_t^u = h_{i,t}^\varepsilon = 1$) collapses (209)–(210) back to the EM posteriors (44)–(46), as it must: the stochastic-volatility model adds the deflation and nothing else.

From expectation to draw

The relationship to the existing machinery is the same "mode-to-draw" correspondence seen for the states. In the EM E-step the weight algebra served to compute the *posterior mean* $\hat{w}_t = \mathbb{E}[w_t \mid \cdot] = \alpha/\beta$ — the plug-in precision the Kalman filter then used. In the Gibbs sampler the *same* algebra produces the *same* Gamma parameters (α, β), but instead of taking the mean we *draw*

$$w_t^\varepsilon \sim \text{Gamma}\Big(\tfrac{\nu_\varepsilon + M}{2}, \tfrac{\nu_\varepsilon + \check{d}_t^\varepsilon}{2}\Big), \qquad w_t^u \sim \text{Gamma}\Big(\tfrac{\nu_u + r}{2}, \tfrac{\nu_u + \check{d}_t^u}{2}\Big).$$

A single Gamma random draw per weight per period replaces the expectation; the existing Student-t weight routine is reused unchanged save for that final substitution and the volatility deflation.

Both weights, independent across time

The conditional-independence structure that made the EM update time-separable carries over intact: given the states, the volatilities and the parameters, each w_t^u depends only on the period-t factor residual and each w_t^ε only on the period-t idiosyncratic residual. The $2T$ weight draws are therefore mutually independent and can be performed in a single vectorised pass over $t = 1, \ldots, T$ — two Gamma variates per period, one for each family.

The link to the degrees of freedom

The degrees of freedom ν_u, ν_ε enter (209)–(210) as fixed quantities, because in the sweep they are *parameters*, sampled in the parameter block of step (d). They are nonetheless tied to this block: ν governs the Gamma prior of the weights, so the weights are precisely the latent data through which ν is identified. The sufficient statistics the ν-update will need are the posterior log-weights $\mathbb{E}[\log w_t] = \psi(\alpha) - \log\beta$ (with ψ the digamma function), available in closed form from the same Gamma parameters. We record the link here and defer the sampling of ν itself to the next section.

What is reused, what is new

Reused: the Gaussian–Gamma conjugacy and the entire weight-update algebra of the EM E-step, reinterpreted — exactly as for the states — as a *draw* from the Gamma rather than its mean. *New:* only the conditioning on the sampled volatility, i.e. the deflation of the Mahalanobis residual by h_t in (208) and (210). This is, with the states, one of the two friendliest blocks of the sweep: a closed-form draw and almost no new code.

An honest qualification. The clean conjugacy of (209)–(210) is exact under the specification of the asymmetry section, in which the tail weights are independent of the level–volatility *leverage* channel: once the volatility path is fixed, the weight then sees only the deflated residual, and the Gamma draw is exact. Whether, and how, the leverage might instead be allowed to touch the weight's conditional is one facet of a broader question — how leverage is estimated at all — that we take up systematically in the next section, where it reappears as one entry in the comparison of the two estimation routes.

Two Routes for the Leverage: Common Trunk, Two Branches

The leverage has now surfaced twice in the estimation: in the volatility block, where it re-drifts the log-volatility transition, and in the weights block, where it threatens the conjugacy of the Gamma draw. Both appearances trace to a single root, and both can be handled in two distinct ways. Rather than pick one and suppress the other, we derive *both* estimation routes here, as two branches growing from one common trunk. This keeps the exposition free of repetition — the shared structure is established once — and it leaves the final choice a genuine methodological decision, to be settled empirically and with the supervisor, with both instruments already in hand.

The common trunk: one coupling, three manifestations

Everything specific to leverage in the sampler descends from one fact: the leverage couples the *level* innovation and the *log-volatility* innovation through a single off-diagonal correlation ρ. From that one coupling, three manifestations have appeared.

(i) **Volatility transition (block b).** The log-volatility transition ceases to be linear-Gaussian, for two reasons: the leverage drift $\rho\sigma_\eta z_t$ involves the standardized shock $z_t = e_t/\sqrt{h_t}$, which *depends on the very state h_t being sampled*; and the linearising map $y_t^* = \log e_t^2$ *discards the sign $d_t = \operatorname{sign}(e_t)$*, while the correlation is with the *signed* shock.

(ii) **Tail weights (block c).** If the leverage is allowed to propagate into the weight's conditional, conditioning on the sampled volatility re-centres the underlying Gaussian base shock, and the w-kernel is no longer exactly Gamma.

(iii) **Common root.** Both are the same phenomenon: a correlation between level and volatility that the conditionally independent blocks of the base sampler did not have to face.

The object common to every route is the conditional law of the volatility innovation given the level shock, derived earlier,

$$\eta_t \mid z_t \;\sim\; \mathcal{N}\big(\rho\,\sigma_\eta\,z_t,\; \sigma_\eta^2(1-\rho^2)\big), \qquad \text{(202 recalled)}$$

together with its common-factor and idiosyncratic versions (203) and (205). The two obstacles of manifestation (i) — the state-dependence of z_t and the lost sign — are shared by both routes; what differs is only *how* they are resolved. And the choice of resolution is dictated by the *timing* convention fixed in the asymmetry section: the contemporaneous timing leads to a Metropolis update (Branch A), the lagged timing to a sign-augmented mixture (Branch B).

Branch A: contemporaneous timing with a Metropolis update

Under the contemporaneous timing the level shock z_t is correlated with η_t, the innovation driving $\log h_t$, so the leverage drift $\rho\sigma_\eta z_t = \rho\sigma_\eta\, e_t/\sqrt{h_t}$ enters the same-period transition and is genuinely a function of the state. The route of Jacquier, Polson and Rossi (2004) accepts this and abandons the mixture altogether: a Metropolis–Hastings step needs only to *evaluate* an unnormalised conditional density, never to recognise a conjugate form, so the state-dependent drift is no obstacle. Working with the *signed* residual e_t directly, the second obstacle — the lost sign — never arises, since no log-square transform is taken.

Concretely, a single-move sampler updates $\log h_t$ one period at a time. Its full conditional collects the three terms in which $\log h_t$ appears — the level likelihood, the transition into t (carrying the state-dependent leverage drift), and the transition out of t —

$$\pi(\log h_t) \;\propto\; \underbrace{h_t^{-1/2}\exp\Big(-\frac{e_t^2}{2h_t}\Big)}_{\text{level likelihood}}\;\underbrace{\exp\Big(-\frac{\big(\log h_t-\mu-\phi(\log h_{t-1}-\mu)-\rho\sigma_\eta e_t/\sqrt{h_t}\big)^2}{2\sigma_\eta^2(1-\rho^2)}\Big)}_{\text{transition into } t}$$
$$\times\;\underbrace{\exp\Big(-\frac{\big(\log h_{t+1}-\mu-\phi(\log h_t-\mu)-\rho\sigma_\eta z_{t+1}\big)^2}{2\sigma_\eta^2(1-\rho^2)}\Big)}_{\text{transition out of } t}. \tag{211}$$

A proposal $\log h_t^\star \sim q(\cdot \mid \log h_t)$ — for instance a random walk, or a Gaussian fitted to a Taylor expansion of $\log \pi$ — is accepted with the usual probability

$$\alpha \;=\; \min\Big\{1,\;\frac{\pi(\log h_t^\star)\,q(\log h_t \mid \log h_t^\star)}{\pi(\log h_t)\,q(\log h_t^\star \mid \log h_t)}\Big\}. \tag{212}$$

Pro: the route is faithful to the contemporaneous timing adopted in the initial presentation of the asymmetry, so no part of the model specification has to be revised. *Con:* it discards the mixture-and-FFBS scaffolding of the base volatility block; a single-move Metropolis sampler mixes more slowly than a blocked draw and requires proposal tuning to achieve a workable acceptance rate.

Branch B: lagged timing with the sign-augmented mixture

Under the lagged timing the level shock z_t is correlated with η_{t+1}, the innovation driving $\log h_{t+1}$. This is the specification Yu (2005) singles out as the proper discrete-time leverage, and Omori, Chib, Shephard and Nakajima (2007) show it is exactly the form that lets the Kim–Shephard–Chib machinery survive. The two obstacles are removed by *augmenting the sampler with the sign* $d_t = \operatorname{sign}(e_t)$ and refining the mixture.

The key identity recovers the signed shock from the discarded magnitude: since $e_t = \sqrt{h_t}\,z_t$ and $y_t^* = \log e_t^2 = \log h_t + \xi_t$ with $\xi_t = \log z_t^2$,

$$z_t \;=\; d_t\,|z_t| \;=\; d_t\,\exp(\xi_t/2), \qquad \xi_t = y_t^* - \log h_t, \tag{213}$$

so the leverage drift of (202), written for η_{t+1}, becomes $\rho\sigma_\eta z_t = \rho\sigma_\eta\, d_t \exp(\xi_t/2)$ — the sign carried explicitly by d_t, the magnitude by $\exp(\xi_t/2)$. Omori et al. approximate the *joint* law of (ξ_t, η_{t+1}) by a *ten-component* mixture (richer than the seven of the symmetric case, to capture the joint accurately), introducing the indicator $s_t \in \{1,\dots,10\}$ such that, conditional on the indicator *and* the sign,

$$\begin{aligned}\xi_t \mid s_t = j \;&\sim\; \mathcal{N}(m_j, v_j^2),\\ \eta_{t+1} \mid (s_t = j,\, d_t,\, \xi_t) \;&\sim\; \mathcal{N}\Big(d_t\,\rho\sigma_\eta\, e^{m_j/2}\big(a_j + b_j(\xi_t - m_j)\big),\; \sigma_\eta^2(1-\rho^2)\Big),\end{aligned} \tag{214}$$

where $\{q_j, m_j, v_j^2, a_j, b_j\}_{j=1}^{10}$ are the fixed constants tabulated in Omori et al. (2007); the constants a_j, b_j come from a linear approximation of $\exp(\xi_t/2)$ within component j. The payoff is that, since $\xi_t = y_t^* - \log h_t$, the conditional mean in (214) is *linear* in $\log h_t$, so that — conditional on the indicators $s_{1:T}$ and the signs $d_{1:T}$ — the measurement equation $y_t^* = \log h_t + \xi_t$ and the leverage-bearing state transition together form a genuinely linear-Gaussian state-space. Both obstacles dissolve at once: the state-dependent magnitude $\exp(\xi_t/2)$ is absorbed into the component constants and linearised in $\log h_t$, and the sign is supplied by conditioning on d_t.

The path $\log h_{1:T}$ is then drawn by exactly the forward-filtering backward-sampling pass of the base block. *Pro:* the entire Kim–Shephard–Chib-plus-FFBS scaffolding is retained, and the route rests on Yu's (2005) theoretical case for the lagged form. *Con:* it requires adopting the lagged timing, a change from the contemporaneous convention used when the asymmetry was first presented.

The ten constants per component are reported in Table 4.

j	q_j (weight)	m_j (mean)	v_j^2 (variance)	a_j	b_j
1	0.00609	1.92677	0.11265	1.01418	0.50710
2	0.04775	1.34744	0.17788	1.02248	0.51124
3	0.13057	0.73504	0.26768	1.03403	0.51701
4	0.20674	0.02266	0.40611	1.05207	0.52604
5	0.22715	−0.85173	0.62699	1.08153	0.54076
6	0.18842	−1.97278	0.98583	1.13114	0.56557
7	0.12047	−3.46788	1.57469	1.21754	0.60877
8	0.05591	−5.55246	2.54498	1.37454	0.68728
9	0.01575	−8.68384	4.16591	1.68327	0.84163
10	0.00115	−14.65000	7.33342	2.50097	1.25049

Table 4: Ten-component mixture for the lagged leverage sampler of Branch B, approximating the joint law of (ξ_t, η_{t+1}): $\{q_j, m_j, v_j^2\}$ approximate $\log \chi_1^2$ as in the base block, while $\{a_j, b_j\}$ are the per-component coefficients of the linear approximation of $\exp(\xi_t/2)$ entering (214). Constants from Omori, Chib, Shephard and Nakajima (2007, Table 1), checked digit by digit against the original PDF. Internal consistency checks also pass ($\sum_j q_j = 1$, $\sum_j q_j m_j \approx -1.2704$, $b_j = a_j/2$).

Both branches across all $M + 1$ processes

Either branch applies, process by process, to the common factor and to each of the M idiosyncratic series, exactly as the base volatility block did. For an idiosyncratic series the leverage correlation is the scalar $\rho_{\varepsilon,i}$ and both branches operate on the univariate residual $e_{i,t}$ without change. For the common factor the correlation is the vector $\boldsymbol{\rho}$: Branch A handles this directly, since the Metropolis target (211) only needs the multivariate conditional evaluated; Branch B, whose sign-and-mixture construction is natively univariate, requires the leverage to act through the relevant scalar combination $\boldsymbol{\rho}' z_t^u$, a mild extra step. In both cases the cost is the one already noted — the leverage treatment is repeated $M + 1$ times per sweep.

The trade-off and the choice

Table 5 sets the two branches side by side.

	Branch A	Branch B
Timing	contemporaneous, $\mathrm{Corr}(z_t, \eta_t)$	lagged, $\mathrm{Corr}(z_t, \eta_{t+1})$
Volatility sampler	single-move Metropolis–Hastings	sign-augmented 10-component mixture + FFBS
Reuse of base block	abandons KSC mixture and FFBS	retains KSC mixture (refined) and FFBS
State-dependent drift	handled by direct density evaluation	linearised into the component constants
Lost sign	no issue (signed residual used)	restored via the sign indicator d_t
Tail-weight conjugacy	leverage folded into the MH step	weights kept conjugate (separable channel)
Theoretical basis	Jacquier, Polson & Rossi (2004)	Yu (2005); Omori et al. (2007)
Match to initial presentation	yes (contemporaneous adopted)	requires switching the timing
Mixing / tuning	slower, needs proposal tuning	fast, blocked, no tuning
Immediate implementability	yes: self-contained, no external table	no: needs the 10-component table (Table 4) first

Table 5: The two estimation routes for the leverage, as branches of the common trunk established above.

The trade-off is clean and there is no dominant choice. Branch A preserves the timing of the model as first presented at the cost of the efficient blocked sampler; Branch B preserves the efficient sampler and a stronger theoretical pedigree at the cost of switching the timing convention. Which to prefer depends on considerations the derivation alone cannot settle — the empirical sensitivity of the Growth-at-Risk results to the timing, the mixing actually achieved on our data, and the supervisor's reading of the modelling priorities. We therefore leave the final choice open as a deliberate methodological decision. What matters for the derivation is that *both* routes are now complete: whichever timing is ultimately adopted, the corresponding sampler is already in hand. Carrying both to completion, rather than committing prematurely, demonstrates command of the full estimation problem and keeps every door open.

A practical note on the work sequence. The two routes differ in one respect that is invisible in the mathematics but decisive for the order of implementation. Branch A is implementable *immediately* from first principles: its Metropolis target (211) is self-contained, built only from the model parameters $(\mu, \phi, \sigma_\eta, \rho)$ and the data, and the proposal is a design choice, so no external constant is required. Branch B, by contrast, cannot be coded until the ten-component mixture table (Table 4) — and, for the base block it relies on, the seven-component table (Table 3) — has been retrieved digit by digit from the source. The consequence for the work plan is concrete: if one wishes to have a working leverage sampler without first depending on tabulated constants from the literature, Branch A can be built and tested now, while Branch B is gated on recovering and validating those tables.

Sampling the Parameters (Gibbs Step (d))

The final sampling block draws the parameters $\boldsymbol{\theta}$ given the states, the volatility paths and the tail weights. It is a *mixed* block: some families are a direct reuse of the EM M-step, reinterpreted as draws (the friendly part); others — the log-volatility parameters and the leverage correlations — are genuinely new. We organise the block by *family*. Throughout, we exploit a fact peculiar to the Gibbs sampler: because the states are now *sampled* rather than averaged, the cross-moments that the M-step formed as posterior expectations $\mathbb{E}[f_t f_t' \mid \mathbf{Y}]$ become *realised* outer products $f_t f_t'$ of the drawn path — the same algebra, with the expectation dropped.

What the volatility scalars multiply. Before proceeding it is worth fixing the *dimensions* of the stochastic-volatility terms, since they reappear throughout the block. Under the master cell (scope D2-b) the model carries $M+1$ *scalar* log-volatility processes, and each multiplies a *different* object. The common volatility h_t^u is a single positive number that scales the *entire* factor-innovation covariance: the conditional variance of the VAR(1) innovation is

$$\operatorname{Var}(u_t \mid \cdot) = \frac{h_t^u}{w_t^u}\,\mathbf{Q}, \tag{215}$$

i.e. the one scalar h_t^u premultiplies the whole matrix $\mathbf{Q}$. This is the coherent way to attach a stochastic volatility to the factor block precisely because the factors do *not* evolve as separate, independent AR(1) lines but as a single *vector* VAR(1) whose innovation covariance $\mathbf{Q}$ is non-diagonal: the off-diagonal entries of $\mathbf{Q}$ encode the contemporaneous co-movement of the factor shocks. A single scalar h_t^u multiplying the whole matrix dilates the overall magnitude of those shocks over time while leaving their correlation structure — the *shape* of $\mathbf{Q}$ — exactly intact (the scalar cancels in the implied correlation matrix); giving each factor its own volatility would instead be a heavier, full multivariate-SV specification, distorting the cross-factor correlations, which we do not adopt. The tail weight w_t^u enters in the same matrix-wide, scalar fashion and for the same reason.

The idiosyncratic side is different precisely because $\mathbf{R}$ is *diagonal*: conditional on the factors the M series are independent, so there is no cross-series covariance to preserve, and the volatility is attached *per series*. Each $h_{i,t}^\varepsilon$ is one number scaling the i-th diagonal entry r_i of $\mathbf{R}$,

$$\operatorname{Var}(\varepsilon_{i,t} \mid \cdot) = \frac{h_{i,t}^\varepsilon}{w_t^\varepsilon}\,r_i, \tag{216}$$

so that the M idiosyncratic volatilities $h_{1,t}^\varepsilon, \dots, h_{M,t}^\varepsilon$ act element-by-element along the diagonal of $\mathbf{R}$, just as the single h_t^u acts matrix-wide on $\mathbf{Q}$. These are exactly the conditional variances that produce the combined precision weights $g_t^u = w_t^u / h_t^u$ and $g_{i,t}^\varepsilon = w_t^\varepsilon / h_{i,t}^\varepsilon$ used in Family A below.

This block is also where the *priors*, deferred until now, must be specified, since it is they that combine with the likelihood to give the conditionals below.

Priors

We use proper, weakly informative, conjugate-where-possible priors:

- **Loadings and transition $\boldsymbol{\Lambda}, \mathbf{A}$**: matrix-normal (Gaussian) priors, conjugate to the linear-Gaussian regressions below.
- **Innovation scales $\mathbf{Q}, \mathbf{R}$**: an inverse-Wishart prior on the factor scale $\mathbf{Q}$ and independent inverse-Gamma priors on the diagonal idiosyncratic scales $r_1, \dots, r_M$ (the conjugate choices for Gaussian covariances).

- **Log-volatility parameters** $(\mu, \phi, \sigma_\eta^2)$ of each AR(1): a Normal–inverse-Gamma prior on the regression coefficients and the innovation variance, with the persistence ϕ restricted to the stationary region $|\phi| < 1$ (e.g. a normal truncated to $(-1, 1)$, or a Beta prior on $(\phi + 1)/2$).
- **Leverage correlations** ρ (common, vector) and $\rho_{\varepsilon,i}$ (scalar): priors supported on the admissible region — $|\rho_{\varepsilon,i}| < 1$ for each series and $\boldsymbol{\rho}'\boldsymbol{\rho} < 1$ for the common block — for instance a uniform or a transformed-normal law on that set.
- **Degrees of freedom** ν_u, ν_ε: a proper prior on $(2, \infty)$ (so the variance exists), e.g. an exponential or a uniform on a bounded range.

Family A — the linear-Gaussian parameters $\mathbf{A}, \mathbf{Q}, \boldsymbol{\Lambda}, \mathbf{R}$

These are the friendliest. Conditional on the sampled states, volatilities and weights, the model is the linear-Gaussian regression of the M-step, and the weighted-least-squares updates survive verbatim as the *means* of conjugate posteriors — we now *draw* from those posteriors instead of taking the mode. The only change relative to the EM moments is that each observation now carries a *combined* precision weight that folds in the volatility alongside the tail weight,

$$g_t^u \;\equiv\; \frac{w_t^u}{h_t^u}, \qquad g_{i,t}^\varepsilon \;\equiv\; \frac{w_t^\varepsilon}{h_{i,t}^\varepsilon}, \tag{217}$$

since the conditional innovation variances are $h_t^u \mathbf{Q}/w_t^u$ and $h_{i,t}^\varepsilon r_i / w_t^\varepsilon$.

The deflated regression. Both updates rest on the same device. Conditional on the sampled states, volatilities and weights, every innovation is Gaussian with a *known*, observation-specific variance; dividing each equation by the square root of its own scale restores homoskedasticity and returns an ordinary Gaussian regression to which the conjugate algebra applies verbatim. For the factor VAR $f_t = \mathbf{A} f_{t-1} + u_t$ with $u_t \mid \cdot \sim \mathcal{N}(0, h_t^u \mathbf{Q}/w_t^u)$, multiply through by $\sqrt{g_t^u}$:

$$\tilde{f}_t \;=\; \mathbf{A}\,\tilde{f}_{t-1} \;+\; \tilde{u}_t, \qquad \tilde{f}_t \equiv \sqrt{g_t^u}\, f_t, \quad \tilde{f}_{t-1} \equiv \sqrt{g_t^u}\, f_{t-1}, \quad \tilde{u}_t \equiv \sqrt{g_t^u}\, u_t \sim \mathcal{N}(0, \mathbf{Q}), \tag{218}$$

a homoskedastic Gaussian regression with innovation covariance $\mathbf{Q}$. The realised, combined-weight moments

$$\mathcal{P}_{11}^u = \sum_t g_t^u\, f_t f_t', \quad \mathcal{P}_{10}^u = \sum_t g_t^u\, f_t f_{t-1}', \quad \mathcal{P}_{00}^u = \sum_t g_t^u\, f_{t-1} f_{t-1}',$$

are then nothing but the cross-products $\sum_t \tilde{f}_t \tilde{f}_t'$, $\sum_t \tilde{f}_t \tilde{f}_{t-1}'$, $\sum_t \tilde{f}_{t-1} \tilde{f}_{t-1}'$ of the deflated data — the exact analogues of the M-step moments (64) with $\hat{w}_t^u$ replaced by g_t^u and the expectation dropped. The same deflation by $\sqrt{g_{i,t}^\varepsilon}$ turns each observation row into a scalar-variance regression below.

Transition A and factor scale Q: the derivation. Work with $\mathbf{B} \equiv \mathbf{A}'$, so that (218) reads $\tilde{f}_t' = \tilde{f}_{t-1}'\mathbf{B} + \tilde{u}_t'$ in the conventional "response-on-regressor" layout. The conditional likelihood is

$$p(f_{1:T} \mid \mathbf{A}, \mathbf{Q}, \cdot) \;\propto\; |\mathbf{Q}|^{-T/2} \exp\Big\{ -\tfrac{1}{2}\,\mathrm{tr}\Big[\mathbf{Q}^{-1} \sum_t (\tilde{f}_t - \mathbf{B}'\tilde{f}_{t-1})(\tilde{f}_t - \mathbf{B}'\tilde{f}_{t-1})'\Big]\Big\}. \tag{219}$$

Expanding the inner sum and completing the square in $\mathbf{B}$ gives the exact quadratic identity

$$\sum_t (\tilde{f}_t - \mathbf{B}'\tilde{f}_{t-1})(\tilde{f}_t - \mathbf{B}'\tilde{f}_{t-1})' \;=\; \mathcal{S}^u \;+\; (\mathbf{B} - \widehat{\mathbf{B}})'\,\mathcal{P}_{00}^u\,(\mathbf{B} - \widehat{\mathbf{B}}), \tag{220}$$

where $\widehat{\mathbf{B}} = (\mathcal{P}_{00}^u)^{-1}(\mathcal{P}_{10}^u)'$ is the weighted-LS coefficient — equivalently $\widehat{\mathbf{A}} = \widehat{\mathbf{B}}' = \mathcal{P}_{10}^u(\mathcal{P}_{00}^u)^{-1}$, the M-step value (67) — and $\mathcal{S}^u = \mathcal{P}_{11}^u - \mathcal{P}_{10}^u(\mathcal{P}_{00}^u)^{-1}(\mathcal{P}_{10}^u)'$ is the residual scatter. Substituted into (219), the first term of (220) depends on $\mathbf{Q}$ alone and the second is a Gaussian kernel in $\mathbf{B}$ with precision $\mathbf{Q}^{-1} \otimes \mathcal{P}_{00}^u$: the likelihood is thus matrix-normal in $\mathbf{B}$ times inverse-Wishart in $\mathbf{Q}$. This is the conjugate *matrix-normal–inverse-Wishart* (MNIW) family, which we adopt as prior:

$$\mathbf{Q} \sim \mathcal{IW}(\Psi_0, \nu_0), \qquad \mathbf{B} \mid \mathbf{Q} \sim \mathcal{MN}\big(\mathbf{B}_0,\ \Lambda_0^{-1},\ \mathbf{Q}\big) \iff \operatorname{vec}(\mathbf{B}) \mid \mathbf{Q} \sim \mathcal{N}\big(\operatorname{vec}(\mathbf{B}_0), \mathbf{Q} \otimes \Lambda_0^{-1}\big), \tag{221}$$

with Λ_0 a prior precision in regressor space. Multiplying (221) by the likelihood and completing the square once more — now pooling the prior precision Λ_0 with the data precision $\mathcal{P}_{00}^u$ — returns the posterior in the same family,

$$\mathbf{Q} \mid \cdot \sim \mathcal{IW}(\Psi_n, \nu_0 + T), \qquad \mathbf{B} \mid \mathbf{Q}, \cdot \sim \mathcal{MN}\big(\mathbf{B}_n,\ \Lambda_n^{-1},\ \mathbf{Q}\big), \tag{222}$$

$$\Lambda_n = \Lambda_0 + \mathcal{P}_{00}^u, \quad \mathbf{B}_n = \Lambda_n^{-1}\big(\Lambda_0 \mathbf{B}_0 + (\mathcal{P}_{10}^u)'\big), \quad \Psi_n = \Psi_0 + \mathcal{P}_{11}^u + \mathbf{B}_0' \Lambda_0 \mathbf{B}_0 - \mathbf{B}_n' \Lambda_n \mathbf{B}_n, \tag{223}$$

the standard conjugate updates (note $\mathcal{P}_{00}^u \widehat{\mathbf{B}} = (\mathcal{P}_{10}^u)'$). In the flat-prior limit $\Lambda_0 \to 0$, $\Psi_0 \to 0$, $\nu_0 \to 0$ these collapse to $\Lambda_n = \mathcal{P}_{00}^u$, $\mathbf{B}_n = \widehat{\mathbf{B}}$ and $\Psi_n = \mathcal{P}_{11}^u - \widehat{\mathbf{B}}' \mathcal{P}_{00}^u \widehat{\mathbf{B}} = \mathcal{S}^u$, so the sequential draw of $\mathbf{Q}$ then $\mathbf{A} = \mathbf{B}'$ becomes

$$\boxed{\mathbf{Q} \mid \cdot \ \sim\ \mathcal{IW}\big(\Psi_0 + \mathcal{S}^u,\ \nu_0 + T\big), \qquad \operatorname{vec}(\mathbf{A}) \mid \mathbf{Q}, \cdot\ \sim\ \mathcal{N}\big(\operatorname{vec}(\widehat{\mathbf{A}}),\ \mathbf{Q} \otimes (\mathcal{P}_{00}^u)^{-1}\big),} \tag{224}$$

the among-row covariance being $\mathbf{Q}$ and the among-column (regressor) covariance $(\mathcal{P}_{00}^u)^{-1}$. The posterior *mean* of $\mathbf{A}$ is exactly the EM update (67); the sampler merely adds the dispersion the M-step discarded.

Loadings Λ and idiosyncratic scales r_i: the derivation. Because $\mathbf{R}$ is diagonal the observation block factorises over series, and each row is the scalar-variance regression $\tilde{y}_{i,t} = \mathbf{\Lambda}_i' \tilde{f}_t + \tilde{\varepsilon}_{i,t}$, with $\tilde{\varepsilon}_{i,t} \sim \mathcal{N}(0, r_i)$ after deflation by $\sqrt{g_{i,t}^{\varepsilon}}$ (the block restriction zeroes the loadings on factors series i does not load on, so $\mathbf{\Lambda}_i$ is the restricted i-th row of $\mathbf{\Lambda}$). Its conditional likelihood is

$$p(y_{i,1:T} \mid \mathbf{\Lambda}_i, r_i, \cdot) \ \propto\ r_i^{-T/2} \exp\Big\{ -\tfrac{1}{2r_i} \sum_t g_{i,t}^{\varepsilon} \, (y_{i,t} - \mathbf{\Lambda}_i' f_t)^2 \Big\}. \tag{225}$$

With $\mathcal{F}_i = \sum_t g_{i,t}^{\varepsilon} f_t f_t'$ and $\rfloor_i = \sum_t g_{i,t}^{\varepsilon} f_t \, y_{i,t}$, the completion of the square in $\mathbf{\Lambda}_i$ is the exact identity

$$\sum_t g_{i,t}^{\varepsilon} (y_{i,t} - \mathbf{\Lambda}_i' f_t)^2 = (\mathbf{\Lambda}_i - \mathcal{F}_i^{-1} \rfloor_i)' \mathcal{F}_i (\mathbf{\Lambda}_i - \mathcal{F}_i^{-1} \rfloor_i) + \Big(\textstyle\sum_t g_{i,t}^{\varepsilon} y_{i,t}^2 - \rfloor_i' \mathcal{F}_i^{-1} \rfloor_i \Big), \tag{226}$$

which exhibits (225) as Gaussian in $\mathbf{\Lambda}_i$ (first term) times inverse-gamma in r_i (second term). The conjugate *normal–inverse-gamma* prior $r_i \sim \mathcal{IG}(a_0, b_0)$, $\mathbf{\Lambda}_i \mid r_i \sim \mathcal{N}(m_0, r_i M_0)$ combines with it to give the posterior

$$r_i \mid \cdot \sim \mathcal{IG}\big(a_0 + \tfrac{T}{2},\ b_n\big), \qquad \mathbf{\Lambda}_i \mid r_i, \cdot \sim \mathcal{N}\big(\bar{\mathcal{F}}_i^{-1} \bar{\rfloor}_i,\ r_i \bar{\mathcal{F}}_i^{-1}\big), \tag{227}$$

$$\bar{\mathcal{F}}_i = \mathcal{F}_i + M_0^{-1}, \quad \bar{\rfloor}_i = \rfloor_i + M_0^{-1} m_0, \quad b_n = b_0 + \tfrac{1}{2}\Big(\textstyle\sum_t g_{i,t}^{\varepsilon} y_{i,t}^2 + m_0' M_0^{-1} m_0 - \bar{\rfloor}_i' \bar{\mathcal{F}}_i^{-1} \bar{\rfloor}_i \Big). \tag{228}$$

In the flat-prior limit $M_0^{-1} \to 0$, $a_0, b_0 \to 0$ the precision $\bar{\mathcal{F}}_i \to \mathcal{F}_i$, the mean $\bar{\mathcal{F}}_i^{-1} \bar{\rfloor}_i \to \mathcal{F}_i^{-1} \rfloor_i$ (the weighted-LS loading), and the inverse-gamma scale collapses, by (226), to $b_n \to \frac{1}{2} \sum_t g_{i,t}^{\varepsilon} \varepsilon_{i,t}^2$ with $\varepsilon_{i,t} = y_{i,t} - \widehat{\mathbf{\Lambda}}_i' f_t$ and $\widehat{\mathbf{\Lambda}}_i = \mathcal{F}_i^{-1} \rfloor_i$, recovering the draw

$$\boxed{r_i \mid \cdot \ \sim\ \mathcal{IG}\Big(a_0 + \tfrac{T}{2},\ b_0 + \tfrac{1}{2} \sum_t g_{i,t}^{\varepsilon} \, \varepsilon_{i,t}^2 \Big), \qquad \mathbf{\Lambda}_i \mid r_i, \cdot\ \sim\ \mathcal{N}\big(\mathcal{F}_i^{-1} \rfloor_i,\ r_i \, \mathcal{F}_i^{-1}\big).} \tag{229}$$

Again the posterior mean $\mathcal{F}_i^{-1} \rfloor_i$ is the weighted-LS loading update; setting all $h = 1$ and replacing realised by expected moments returns the EM M-step exactly. This family is therefore pure reuse.

Family B — the log-volatility parameters $(\mu, \phi, \sigma_\eta^2)$

These are new, but easy: conditional on the sampled log-volatility path $\log h_{1:T}$ (from step (b)), each AR(1) (197) is an ordinary linear Gaussian regression in its own parameters, so the same normal–inverse-gamma machinery as Family A applies — only now the "data" is the sampled volatility path, not the states.

The regression. Write $x_t = \log h_t$ and put the AR(1) in regression form

$$x_t = \alpha + \phi\, x_{t-1} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \qquad t = 2, \dots, T, \tag{230}$$

with intercept $\alpha = \mu(1-\phi)$. Collecting the $T-1$ equations, stack the response $\mathbf{x} = (x_2, \dots, x_T)'$ and the design X whose t-th row is $(1, x_{t-1})$, and write $\beta = (\alpha, \phi)'$, so that $\mathbf{x} = X\beta + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_\eta^2 I_{T-1})$. The conditional likelihood is

$$p(\mathbf{x} \mid \beta, \sigma_\eta^2) \;\propto\; (\sigma_\eta^2)^{-(T-1)/2} \exp\Big\{ -\tfrac{1}{2\sigma_\eta^2} (\mathbf{x} - X\beta)'(\mathbf{x} - X\beta) \Big\}. \tag{231}$$

Completing the square in β around the OLS estimate $\widehat{\beta}_{\text{ols}} = (X'X)^{-1}X'\mathbf{x}$ gives the exact identity

$$(\mathbf{x} - X\beta)'(\mathbf{x} - X\beta) = \text{RSS}_{\text{ols}} + (\beta - \widehat{\beta}_{\text{ols}})'X'X(\beta - \widehat{\beta}_{\text{ols}}), \qquad \text{RSS}_{\text{ols}} = \mathbf{x}'\mathbf{x} - \widehat{\beta}_{\text{ols}}'X'X\widehat{\beta}_{\text{ols}}, \tag{232}$$

which exhibits (231) as Gaussian in β times inverse-gamma in σ_η^2 — the normal–inverse-gamma form.

Prior and posterior. The conjugate prior is $\sigma_\eta^2 \sim \mathcal{IG}(a_\eta, b_\eta)$ and $\beta \mid \sigma_\eta^2 \sim \mathcal{N}(\beta_0, \sigma_\eta^2 V_\eta)$ (so V_η^{-1} is the prior precision in the same units as $X'X$). Multiplying the prior by (231) pools the two quadratics in β via $(\beta - \widehat{\beta}_{\text{ols}})'X'X(\beta - \widehat{\beta}_{\text{ols}}) + (\beta - \beta_0)'V_\eta^{-1}(\beta - \beta_0) = (\beta - \widehat{\beta})'V_n^{-1}(\beta - \widehat{\beta}) +$ const, with posterior precision and mean

$$V_n^{-1} = X'X + V_\eta^{-1}, \qquad \widehat{\beta} = V_n\big(X'\mathbf{x} + V_\eta^{-1}\beta_0\big) = (X'X + V_\eta^{-1})^{-1}(X'\mathbf{x} + V_\eta^{-1}\beta_0), \tag{233}$$

and the residual constant lifts the inverse-gamma scale, leaving the standard conjugate draws

$$\boxed{\sigma_\eta^2 \mid \cdot \sim \mathcal{IG}\Big(a_\eta + \tfrac{T-1}{2},\; b_\eta + \tfrac{1}{2}\,\text{SSR}\Big), \qquad \begin{pmatrix} \alpha \\ \phi \end{pmatrix} \Big|\, \sigma_\eta^2, \cdot \sim \mathcal{N}\big(\widehat{\beta},\, \sigma_\eta^2 (X'X + V_\eta^{-1})^{-1}\big)} \tag{234}$$

where the exact inverse-gamma scale is $b_\eta + \frac{1}{2}\big(\mathbf{x}'\mathbf{x} + \beta_0'V_\eta^{-1}\beta_0 - \widehat{\beta}'V_n^{-1}\widehat{\beta}\big)$, which in the flat-prior limit $V_\eta^{-1} \to 0$ reduces to $\widehat{\beta} \to \widehat{\beta}_{\text{ols}}$ and to $b_\eta + \frac{1}{2}\,\text{SSR}$ with $\text{SSR} = (\mathbf{x} - X\widehat{\beta})'(\mathbf{x} - X\widehat{\beta})$ the residual sum of squares at the posterior mean — the shorthand in the box. As in Family A, the posterior mean $\widehat{\beta}$ is exactly the (prior-shrunk) least-squares fit and the sampler adds only the draw.

Constraints and repetition. The draw of (α, ϕ) is restricted to $|\phi| < 1$ (reject, or use the truncated normal), and the long-run mean is recovered as $\mu = \alpha/(1-\phi)$. The stationary initial condition $x_1 \sim \mathcal{N}(\mu, \sigma_\eta^2/(1-\phi^2))$ contributes one extra factor to the likelihood of $(\mu, \phi, \sigma_\eta^2)$; it is either included (a small Metropolis correction) or, as is common, conditioned upon by starting the regression at $t = 2$. This draw is performed independently for each of the $M+1$ processes — the common h^u and each idiosyncratic h_i^ε — with its own residuals and its own $(\mu, \phi, \sigma_\eta^2)$.

Family C — the leverage correlations $\rho, \rho_{\varepsilon,i}$

The leverage correlation has no standard closed-form conditional in *either* route, because it enters the volatility dynamics both through the drift ($\propto \rho$) and through the reduced innovation variance ($\propto 1-\rho^2$); it is therefore sampled by a Metropolis step, and the form of that step is dictated by which branch of the "Two Routes" section is adopted.

- **Branch A (contemporaneous, Metropolis).** The correlation is sampled by the *same* Metropolis machinery that draws the volatility path in this branch: ρ enters the single-move target (211) through the drift $\rho\sigma_\eta z_t$ and the variance $\sigma_\eta^2(1-\rho^2)$, and a proposal in ρ is accepted with the corresponding ratio, restricted to $|\rho| < 1$.
- **Branch B (lagged, Omori).** In the sign-augmented linear-Gaussian representation (214), conditional on the signs and indicators the leverage enters the state-transition mean *linearly* in ρ, so ρ (jointly with σ_η) is drawn from a regression-type conditional — in practice by a Metropolis or tailored proposal that handles the residual $1-\rho^2$ variance term — again restricted to $|\rho| < 1$.

The explicit conditional. In both branches the correlation enters through the *same* algebraic skeleton, which makes the target concrete and verifiable. Let

$$\eta_t \;\equiv\; (\log h_t - \mu) - \phi(\log h_{t-1} - \mu) \tag{235}$$

denote the realised AR(1) innovation of the sampled log-volatility path, and let k_t be the period-t leverage regressor — the quantity that ρ multiplies in the conditional mean of η_t. Collecting the $T-1$ leverage-bearing transitions, each Gaussian with mean $\rho\, k_t$ and variance $\sigma_\eta^2(1-\rho^2)$, the full conditional of ρ is

$$p(\rho \mid \cdot) \;\propto\; p(\rho)\,(1-\rho^2)^{-(T-1)/2} \exp\Big\{-\frac{1}{2\sigma_\eta^2(1-\rho^2)} \sum_t \big(\eta_t - \rho\, k_t\big)^2\Big\}, \qquad |\rho| < 1, \tag{236}$$

with the regressor specialised to the branch:

$$k_t = \sigma_\eta z_t \;\text{(Branch A, contemporaneous)}, \qquad k_t = \sigma_\eta\, d_{t-1}\, e^{m_j/2}\big(a_j + b_j(\xi_{t-1} - m_j)\big),\; j = s_{t-1} \;\text{(Branch B, Omori} \tag{237}$$

the latter read off the Omori conditional mean (214) (Branch B additionally conditions on the mixture indicators $s_{1:T}$ and the signs $d_{1:T}$). The form of (236) shows why no conjugate draw exists in *either* route: although the exponent is quadratic in ρ, the prefactor $(1-\rho^2)^{-(T-1)/2}$ and the $1-\rho^2$ inside the variance make the density non-Gaussian. It is therefore sampled by a Metropolis step — a proposal $\rho^\star \sim q(\cdot \mid \rho)$ on $(-1, 1)$ accepted with

$$\alpha = \min\Big\{1,\; \frac{p(\rho^\star \mid \cdot)\, q(\rho \mid \rho^\star)}{p(\rho \mid \cdot)\, q(\rho^\star \mid \rho)}\Big\}. \tag{238}$$

In Branch A this is the very step (212) that already drives the volatility path; in Branch B, where conditioning on the signs and indicators makes the conditional mean *linear* in ρ, (236) is close to Gaussian and a tailored (Gaussian-proposal or transformed) Metropolis move attains a high acceptance rate.

For the common factor the same logic applies to the vector $\boldsymbol{\rho}$ — with ρ^2 replaced by $\boldsymbol{\rho}'\boldsymbol{\rho}$ in (236) — the draw being restricted to the admissible region $\boldsymbol{\rho}'\boldsymbol{\rho} < 1$; for each idiosyncratic series it is the scalar $\rho_{\varepsilon,i}$. This family is new, and it inherits the branch dependence of the volatility block rather than admitting a single universal sampler.

Family D — the degrees of freedom ν_u, ν_ε

Deferred from the weights block, the degrees of freedom are the least friendly parameters: their conditional is *not* a standard conjugate form. *Derivation.* In the scale mixture the tail weights are conditionally i.i.d. $w_t^u \mid \nu_u \sim \text{Gamma}(\nu_u/2, \nu_u/2)$ (shape and rate equal, so $\mathbb{E}[w_t^u] = 1$), hence

$$p(w_{1:T}^u \mid \nu_u) = \prod_{t=1}^{T} \frac{(\nu_u/2)^{\nu_u/2}}{\Gamma(\nu_u/2)} (w_t^u)^{\nu_u/2-1} e^{-(\nu_u/2)w_t^u}. \tag{239}$$

Multiplying by the prior $p(\nu_u)$ and discarding the ν_u-free factor $\prod_t (w_t^u)^{-1}$, the realised weights enter only through $\sum_t \log w_t^u$ and $\sum_t w_t^u$, giving the conditional

$$p(\nu_u \mid w_{1:T}^u) \;\propto\; p(\nu_u) \left[\frac{(\nu_u/2)^{\nu_u/2}}{\Gamma(\nu_u/2)}\right]^T \exp\Big(\tfrac{\nu_u}{2}\sum_t \log w_t^u - \tfrac{\nu_u}{2}\sum_t w_t^u\Big), \tag{240}$$

and likewise for ν_ε, with the two sufficient statistics $\sum_t \log w_t$ and $\sum_t w_t$. This is the very quantity the EM treatment optimised — there the digamma identity supplied $\mathbb{E}[\log w_t] = \psi(\alpha) - \log\beta$ as the relevant statistic; here, the weights being *sampled*, we use the realised $\sum_t \log w_t^u$ directly.

Log-target, log-concavity, and the draw. Writing $S \equiv \sum_t \log w_t^u - \sum_t w_t^u$, the log of (240) is

$$\log p(\nu_u \mid \cdot) = \text{const} + \log p(\nu_u) + T\Big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma(\tfrac{\nu_u}{2})\Big] + \tfrac{\nu_u}{2} S. \tag{241}$$

Differentiating twice, the data-and-mixture part has

$$\frac{\partial^2}{\partial\nu_u^2}\Big\{T\big[\tfrac{\nu_u}{2}\log\tfrac{\nu_u}{2} - \log\Gamma(\tfrac{\nu_u}{2})\big]\Big\} = \frac{T}{2}\Big[\frac{1}{\nu_u} - \tfrac{1}{2}\,\psi'(\tfrac{\nu_u}{2})\Big] < 0, \tag{242}$$

negative for all $\nu_u > 0$ because the trigamma function obeys $\psi'(\nu_u/2) > 2/\nu_u$; with a log-concave prior $p(\nu_u)$ the whole target (241) is therefore log-concave. It is not a standard density, so it is sampled either by a Metropolis step — a proposal $\nu_u^\star \sim q(\cdot \mid \nu_u)$ on $(2, \infty)$ accepted with $\alpha = \min\{1, [p(\nu_u^\star \mid \cdot)q(\nu_u \mid \nu_u^\star)]/[p(\nu_u \mid \cdot)q(\nu_u^\star \mid \nu_u)]\}$ — or, exploiting log-concavity, by a griddy-Gibbs draw: evaluate (240) on a grid $\{\nu^{(1)}, \dots, \nu^{(G)}\}$, normalise to probabilities $\propto p(\nu^{(g)} \mid \cdot)$, and sample one grid point (optionally with interpolation). Either way this reuses the same one-dimensional ν-update logic of the M-step, now as a draw rather than a maximisation: the M-step solved $\partial_\nu \log p = 0$ at the *expected* $\mathbb{E}[\log w_t]$, whereas the sampler explores (240) around that mode at the *realised* $\sum_t \log w_t$.

What is reused, what is new

Reused: the weighted-least-squares M-step updates reappear intact as the conjugate draws of Family A — the posterior means are precisely the EM updates, with $g_t = w_t/h_t$ in place of $\hat{w}_t$ and realised moments in place of expectations — and the one-dimensional ν-update logic returns as the Metropolis draw of Family D. *New:* the log-volatility AR(1) regressions of Family B and the leverage-correlation draws of Family C, the latter inheriting the branch structure of the volatility sampler. With this block the sweep is complete: states, volatilities, weights and parameters have each been given their full conditional, and iterating the four steps delivers the joint posterior from which the predictive density and its Growth-at-Risk quantiles are read off.

Variants by Switch: Every Grid Cell as a Restriction of the Master Sampler

This closes the estimation derivation, and it introduces *no* new mathematics. It is the bookkeeping that shows how every cell of the D1 $\times$ D2 grid of Section — and the symmetric, no-leverage cases — is obtained from the single sampler already derived, by switching blocks of the sweep on or off. It is the estimation-side mirror of the mother-form DGP: there one data-generating process specialised to each cell by setting switches; here one Gibbs sampler specialises to each cell by omitting or modifying the corresponding blocks.

The principle

We derived the sampler for the *master* cell — D1-b $\times$ D2-b with leverage, the most general specification — in Sections –, summarised in Algorithm 5. Every other cell is a *restriction* of it. The skeleton of the Gibbs sweep never changes — states (a), volatilities (b), tails (c), parameters (d), each drawn from its full conditional; what changes from cell to cell is only *which* steps are executed and, in one case, *what* a step samples. The generality of the master therefore costs no extra derivation: one derivation covers the entire grid.

The restrictions, one per variant

D2-a (common volatility only). Drop the idiosyncratic volatility. In step (b) the idiosyncratic part of the volatility sampler is omitted and the $h^{\varepsilon}_{i,t}$ are frozen at 1; the sampler runs only for the common path h^u. Correspondingly, Family B of step (d) is drawn only for the single process h^u rather than for all $M + 1$. Everything else — states, tails, Families A, C, D — is unchanged.

D1-a (Gaussian, pure stochastic volatility). Switch the tail multiplier off, $w \equiv 1$. The whole weights block, step (c) of Section , is omitted, and with it the degrees-of-freedom draw, Family D of step (d): there is no ν to sample. The shocks are Gaussian conditional on their volatility, so the combined precision weight of Family A collapses to $g_t = 1/h_t$. Steps (a), (b) and Families A, B — and C, if leverage is retained — run unchanged.

D1-c (explicit outliers). *Replace*, rather than omit, the weights block. The Gamma scale-mixture draw of step (c) is substituted by the sampling of the outlier indicators (and their associated parameters; in the *stochastic* version also the occurrence probability p). The structure is identical — a latent block that modulates the per-period variance — only the latent object changes: a continuous Gamma weight becomes a discrete outlier indicator. Consistently, Family D (ν) is replaced by the draw of the outlier parameters; the rest of the sweep is untouched.

No leverage ($\rho = 0$). Set every leverage correlation to zero. In step (b) the leverage correction of Section drops out and the volatility sampler reverts to the symmetric base case of Section — the plain Kim–Shephard–Chib sub-sweep (198); in step (d) the leverage draw, Family C, is omitted. Being symmetric, this case does not even need the ten-component Omori constants of Table 4: the seven-component KSC mixture suffices.

The limiting case: the current model

The first-stage model of this thesis — the Student-t DFM *without* stochastic volatility — is the restriction that freezes *all* volatility (every $\sigma^2_\eta = 0$ and $\mu = 0$, so $h \equiv 1$ and the persistence ϕ

is inoperative) while keeping the Student-t weights (D1-b) and no leverage. On the sampler side this omits the *entire* volatility step (b) — there is nothing to sample — together with the leverage block (Family C); what remains is step (a) states, step (c) tails, and step (d) parameters in Families A and D. That is exactly the MCMC counterpart of the EM algorithm estimated in the first stage: the same weighted-least-squares updates and the same one-dimensional ν-update, now realised as draws rather than maximisations. The general sampler, restricted, reproduces the estimation of the starting model — closing the circle with the DGP, where the current model was already the degenerate case of the mother form.

Two levels of leverage choice

The leverage deserves a clarification, because it involves *two distinct* decisions that must not be conflated.

First level — whether to include leverage ($\rho \neq 0$ vs. $\rho = 0$). This is the symmetric/asymmetric axis, and it is a genuine *restriction*: setting $\rho = 0$ omits the leverage correction in step (b) and Family C in step (d), exactly as in the no-leverage variant above. This is the on/off switch for leverage.

Second level — how to specify the leverage when it is present. Even in the richest cell (the master, $\rho \neq 0$) a further choice remains: the *timing* of the leverage — contemporaneous or lagged. This is *not* a restriction; it switches nothing off. It is an *internal bifurcation* of the full case: the master "D1-b $\times$ D2-b + leverage" exists in *two* versions, equal in modelling richness but differing in timing and therefore in sampler, as derived in the Two Routes section (Section):

- contemporaneous timing $\rightarrow$ Branch A, the Metropolis update of Section ;
- lagged timing $\rightarrow$ Branch B, the sign-augmented Omori mixture of Section .

This timing choice *persists* into the master cell: it is not resolved by restricting any other axis, and it remains an open methodological decision — to be settled empirically, and with the advisor — with both samplers already derived and ready. Consequently every leveraged cell of the grid splits further into a contemporaneous (Branch A) and a lagged (Branch B) variant.

The map at a glance

Table 6 maps each grid cell to the active and omitted blocks of the sweep.

Cell	(a) states	step (b) vol		step (c)	step (d) parameters			
		h^u	h^ε	tails	A	B	C (ρ)	D (ν)
D1a×D2a	✓	✓	—	—	✓	h^u	†	—
D1a×D2b	✓	✓	✓	—	✓	all	†	—
D1b×D2a	✓	✓	—	Gamma w	✓	h^u	†	✓
D1b×D2b*	✓	✓	✓	Gamma w	✓	all	†	✓
D1c×D2a	✓	✓	—	outlier ind.	✓	h^u	†	outlier par.
D1c×D2b	✓	✓	✓	outlier ind.	✓	all	†	outlier par.
current model	✓	—	—	Gamma w	✓	—	—	✓

Table 6: Each D1 × D2 cell as a restriction of the master sweep. ✓ = active; = — omitted (the state frozen at $h \equiv 1$, or the block not executed); "h^u"/"all" = Family B drawn for the common process only / for all $M+1$ processes; "outlier ind./par." = the weights and ν blocks replaced by the outlier indicators and their parameters. †: Family C and the step-(b) leverage correction are active *only* if $\rho \neq 0$; when they are, the cell splits further into Branch A (contemporaneous) or Branch B (lagged) of Section . *: the master cell, the fully general one from which every other row is obtained by restriction. The last row, "current model", is the no-SV limiting case of Section .

With this map the estimation derivation is complete: a single sampler, derived once for the master cell, delivers — by switching blocks on, off, or sideways — the estimator of every specification in the grid, exactly as the mother-form DGP delivered every data-generating process.

Vectors, Matrices, Determinants, Volumes and Eigenvalues

Vectors and the Geometry of $\mathbb{R}^n$

Before turning to matrices and eigenvalues, we briefly review the geometric concept of a vector. A vector $v \in \mathbb{R}^n$ is an ordered tuple of n real numbers,

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix},$$

where the integer n is the *dimension* of the space in which v lives. The components v_i can be thought of as coordinates of v in a fixed reference system, and equivalently as the lengths of the projections of v on each coordinate axis.

Geometric interpretation: vectors as arrows. Geometrically, a vector is an arrow with a specified direction and length, drawn from the origin to the point with the given coordinates. Two vectors are equal if and only if they have the same direction and the same length — their geometric position in space is determined entirely by the components $(v_1, \ldots, v_n)$. In two dimensions, a vector $v = (v_1, v_2)' \in \mathbb{R}^2$ corresponds to an arrow from the origin to the point with horizontal coordinate v_1 and vertical coordinate v_2.

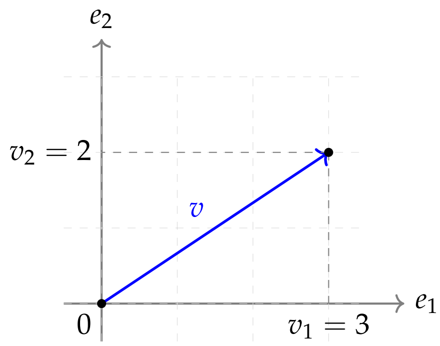


Figure 4: A vector $v = (3, 2)' \in \mathbb{R}^2$ visualised as an arrow from the origin to the point with coordinates $v_1 = 3$ and $v_2 = 2$.

Length (norm) and inner product. Two operations on vectors are central to all linear algebra. The *Euclidean norm* of v is the length of the corresponding arrow:

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} = \sqrt{v'v}.$$

The *inner product* (or dot product) of two vectors $u, v \in \mathbb{R}^n$ is the scalar

$$u'v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

The geometric meaning of the inner product is captured by the identity $u'v = \|u\| \, \|v\| \cos\theta$, where θ is the angle between the two vectors. Two consequences are particularly useful: $u'v = 0$ if and only if u and v are *orthogonal* (perpendicular), and $u'v$ is positive (resp. negative) if the angle between them is acute (resp. obtuse).

Vector addition and scalar multiplication. Two more operations complete the basic algebra. *Addition* of two vectors is component-wise: $u + v = (u_1 + v_1, \ldots, u_n + v_n)'$. Geometrically, $u + v$ is the diagonal of the parallelogram formed by u and v. *Scalar multiplication* of a vector by a real number c is also component-wise: $c\,v = (cv_1, \ldots, cv_n)'$. Geometrically, multiplication by c stretches or compresses v by the factor $|c|$, and reverses its direction if $c < 0$.

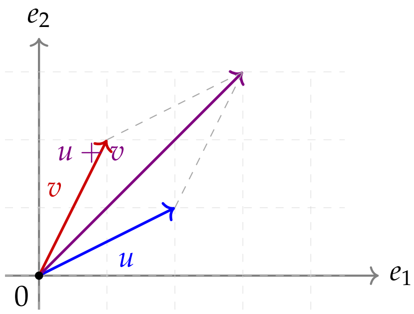


Figure 5: Addition of two vectors $u, v \in \mathbb{R}^2$. The sum $u + v$ is the diagonal of the parallelogram spanned by u and v.

Standard basis and coordinates. The *standard basis* of $\mathbb{R}^n$ is the set of n unit vectors $\{e_1, e_2, \ldots, e_n\}$, where each e_i has a 1 in position i and zeros elsewhere. Any vector $v \in \mathbb{R}^n$ admits the unique decomposition

$$v \;=\; v_1\, e_1 \;+\; v_2\, e_2 \;+\; \cdots \;+\; v_n\, e_n,$$

which expresses v as a linear combination of the basis vectors with coefficients equal to its coordinates. This decomposition is so natural that we usually identify v with its column of coordinates $(v_1, \ldots, v_n)'$. However, it is important to keep in mind that the coordinates depend on the chosen basis: in a different basis (for instance, the basis of eigenvectors of a matrix, used extensively in the spectral decomposition), the same vector v has different coordinates. The vector itself is a geometric object; its coordinates are merely a representation in a specific reference system.

Why vectors matter for the DFM. In the dynamic factor model, several objects are vectors of various dimensions:

- the observation vector $y_t \in \mathbb{R}^M$, with one component per observed series at time t;
- the latent factor vector $f_t \in \mathbb{R}^r$, with one component per common factor;
- the augmented state vector $\tilde{f}_t = (f_t', f_{t-1}', \ldots, f_{t-4}')' \in \mathbb{R}^{5r}$, stacking the current and four lagged factors;
- the idiosyncratic shock $\varepsilon_t \in \mathbb{R}^M$ and the factor innovation $u_t \in \mathbb{R}^r$, both vectors of suitable dimension.

Each of these vectors lives in its own Euclidean space, and the geometric operations introduced in this section — norm, inner product, addition — apply to all of them. The role of matrices, to which we turn next, is to map vectors from one such space to another and to encode the linear relationships among them.

Matrices as Linear Transformations

Having introduced vectors as the basic geometric objects of $\mathbb{R}^n$, we now turn to matrices. A matrix $A \in \mathbb{R}^{m\times n}$ is a rectangular array of mn real numbers, organised in m rows and n columns:

$$A \;=\; \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Matrices admit two complementary readings, both of which will be useful in what follows.

Matrices as collections of vectors. The first reading sees a matrix as a *collection of vectors*. Specifically, A can be viewed either as a horizontal stack of n column vectors $a_{\cdot 1}, a_{\cdot 2}, \dots, a_{\cdot n} \in \mathbb{R}^m$,

$$A \;=\; \big(a_{\cdot 1} \mid a_{\cdot 2} \mid \cdots \mid a_{\cdot n} \big),$$

or as a vertical stack of m row vectors $a_{1\cdot}, a_{2\cdot}, \dots, a_{m\cdot} \in \mathbb{R}^{1\times n}$. This reading is particularly natural when the columns of A have a direct interpretation: in our DFM, for instance, the columns of the loading matrix $\mathbf{\Lambda}$ are the loadings of all M observed series on each individual factor, and the rows of $\mathbf{\Lambda}$ are the loadings of each individual series on all r factors.

Matrices as linear transformations. The second, and more important, reading sees a matrix as a *linear transformation* from one Euclidean space to another. The matrix $A \in \mathbb{R}^{m\times n}$ defines a function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ via

$$T_A(v) \;=\; Av, \qquad v \in \mathbb{R}^n.$$

The transformation T_A is *linear* in the sense that it preserves vector addition and scalar multiplication:

$$T_A(v+w) \;=\; T_A(v) + T_A(w), \qquad T_A(c\,v) \;=\; c\,T_A(v).$$

This linearity has a striking geometric consequence: T_A maps the origin to the origin, straight lines to straight lines, and parallel lines to parallel lines. In particular, it maps the unit square (in $\mathbb{R}^2$) to a parallelogram, and the unit cube (in $\mathbb{R}^3$) to a parallelepiped — the shape is determined entirely by the columns of A, which are the images of the standard basis vectors $e_1, e_2, \dots, e_n$. Indeed,

$$A\,e_i \;=\; a_{\cdot i}, \qquad i = 1, 2, \dots, n,$$

so the i-th column of A is precisely the image of the i-th basis vector under T_A. This is the geometric meaning of the columns of a matrix: *they tell us where the basis vectors are sent*, and the transformation of any other vector follows by linearity.

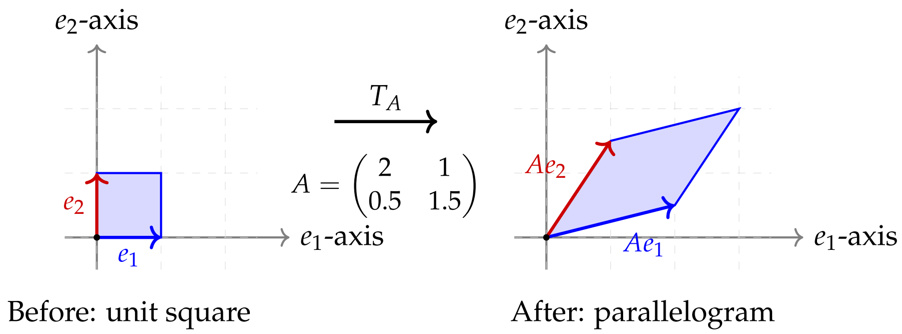


Figure 6: A linear transformation T_A defined by a matrix $A \in \mathbb{R}^{2\times 2}$ maps the unit square (left) to a parallelogram (right). The columns of A are the images of the standard basis vectors: Ae_1 is the first column and Ae_2 is the second column. The transformation preserves the origin, parallel lines, and ratios of lengths along any single direction.

Three special cases. A few elementary linear transformations are worth visualising explicitly, because more general matrices can be understood by combining them.

The identity matrix I_n maps every vector to itself: $T_{I_n}(v) = v$. The unit square is sent to itself, unchanged.

Diagonal matrices apply independent stretching along each coordinate axis. For $D = \operatorname{diag}(d_1, d_2)$, the transformation T_D sends the unit square to a rectangle of width d_1 and height d_2 (or to a reflection, if some $d_i < 0$). There is no rotation; only axis-aligned stretching.

Rotation matrices preserve lengths and angles, rotating every vector around the origin by a fixed angle θ. In two dimensions, the rotation matrix is

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The unit square is sent to another unit square, rotated by θ. No stretching, only rotation.

A general matrix $A \in \mathbb{R}^{2\times 2}$ produces a combination of these elementary operations: rotation, stretching, and possibly reflection. The spectral decomposition of a symmetric positive-definite matrix, derived in a later section, will make this decomposition explicit.

Composition of transformations. If $A : \mathbb{R}^n \to \mathbb{R}^m$ and $B : \mathbb{R}^m \to \mathbb{R}^p$ are two linear transformations, their composition is itself linear and is represented by the *product matrix* $BA \in \mathbb{R}^{p\times n}$:

$$T_B \circ T_A(v) = B(Av) = (BA)\,v.$$

This is the source of the matrix-multiplication formula, which encodes the algebraic rule for composing two linear maps. The order matters: in general $AB \neq BA$, reflecting the fact that the order in which two transformations are applied affects the final result.

Matrices in the DFM. The matrices appearing in the dynamic factor model can now be interpreted as linear transformations on the relevant spaces.

- The loading matrix $\boldsymbol{\Lambda} \in \mathbb{R}^{M\times r}$ is a map from the factor space $\mathbb{R}^r$ to the observation space $\mathbb{R}^M$, sending each factor configuration f_t to its observation-space image $\boldsymbol{\Lambda} f_t$.

- The transition matrix $\mathbf{A} \in \mathbb{R}^{r \times r}$ is a map from the factor space to itself, propagating factors one step forward in time: $f_t \mapsto \mathbf{A} f_t$.
- The selection matrix $\mathbf{W}_t \in \{0,1\}^{m_t \times M}$ is a projection from the full observation space onto the subspace of currently observed components.

The geometric perspective on matrices as transformations is the key to making sense of these objects; the algebraic perspective on matrices as collections of numbers gives us the operational tools to compute with them.

The Determinant as a Stretching Factor

The determinant of a square matrix has both an algebraic definition and a geometric one. The algebraic definition — as a polynomial in the matrix entries with specific sign properties — is computationally useful but offers little intuition. The geometric definition — as a *volume scaling factor* — captures the meaning that we use throughout the thesis.

Geometric definition. Let $A \in \mathbb{R}^{n \times n}$ be a square matrix, viewed as a linear transformation $T_A : \mathbb{R}^n \to \mathbb{R}^n$. The *determinant* of A, denoted $\det(A)$ or $|A|$, is the factor by which T_A scales the volume of any region in $\mathbb{R}^n$. Specifically, if $S \subset \mathbb{R}^n$ is a region of volume $\mathrm{Vol}(S)$, then its image under T_A has volume

$$\mathrm{Vol}(T_A(S)) \;=\; |\det(A)| \cdot \mathrm{Vol}(S).$$

The volume scaling factor is independent of the chosen region S — it depends only on the matrix. The sign of $\det(A)$ encodes *orientation*: $\det(A) > 0$ means T_A preserves orientation (does not perform a reflection), while $\det(A) < 0$ means T_A reverses orientation (it includes a reflection). The absolute value $|\det(A)|$ is the pure volume scaling.

The two-dimensional case. In two dimensions, the determinant has an especially clean interpretation. Recall from the previous section that T_A maps the unit square (with vertices $0, e_1, e_1 + e_2, e_2$) to a parallelogram with vertices $0, Ae_1, Ae_1 + Ae_2, Ae_2$. The unit square has area 1, so the parallelogram has area $|\det(A)|$.

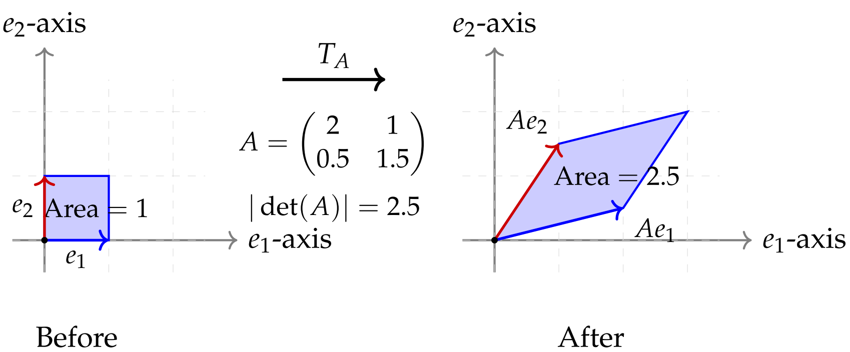


Figure 7: Geometric meaning of the determinant. The matrix $A = \begin{pmatrix} 2 & 1 \\ 0.5 & 1.5 \end{pmatrix}$ has $\det(A) = 2 \cdot 1.5 - 1 \cdot 0.5 = 2.5$. The unit square (left, area 1) is mapped to a parallelogram (right, area 2.5) — the determinant is the area-scaling factor.

Three special cases worth visualising. *Case 1:* $\det(A) > 1$ — *expansion.* The transformation enlarges volumes. Long vectors get longer, regions become bigger. This is the case of the figure above, where a unit area is stretched into a 2.5-area parallelogram.

Case 2: $0 < \det(A) < 1$ — *contraction.* The transformation shrinks volumes. The unit square is mapped to a parallelogram of area less than 1. Notably, repeated application of A produces geometrically vanishing regions: $\det(A^k) = (\det A)^k \to 0$ as $k \to \infty$.

Case 3: $\det(A) = 0$ — *collapse.* The transformation collapses the entire space to a lower-dimensional subspace. The unit square is mapped to a line segment, or even to a single point. Geometrically, the parallelogram *degenerates*: two of its sides become parallel (in 2D) and the area shrinks to zero. Algebraically, this corresponds to the columns of A being linearly dependent — they no longer span the full space. A matrix with $\det(A) = 0$ is called *singular*, and its action is irreversible: distinct vectors can be mapped to the same image, so T_A cannot be inverted.

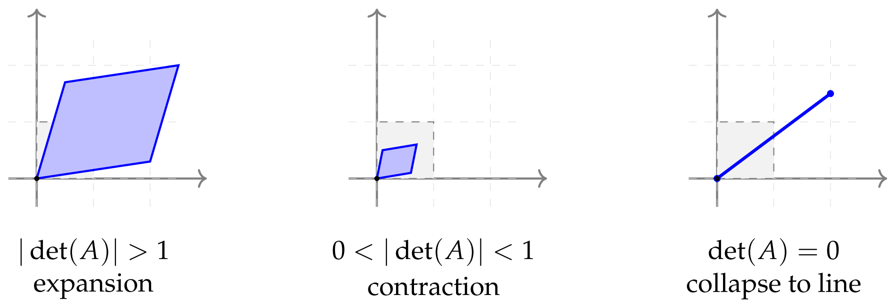


Figure 8: Three regimes of the determinant. From left to right: expansion ($|\det(A)| > 1$), contraction ($|\det(A)| < 1$), and collapse to a lower-dimensional subspace ($\det(A) = 0$, the singular case). The original unit square is shown in dashed grey for reference.

Algebraic computation. For practical computations, the determinant of a 2×2 matrix is

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For larger matrices, the determinant can be computed via cofactor expansion (recursive but expensive for $n > 4$) or via LU-decomposition (much faster, $O(n^3)$). Three properties are used repeatedly throughout the thesis:

- *Multiplicativity*: $\det(AB) = \det(A)\det(B)$ for any two square matrices of the same size.
- *Inverse*: $\det(A^{-1}) = 1/\det(A)$, valid whenever $\det(A) \neq 0$.
- *Diagonal and triangular matrices*: the determinant is the product of the diagonal entries. In particular, for $D = \mathrm{diag}(d_1, \ldots, d_n)$, $\det(D) = d_1 d_2 \cdots d_n$.

Why the determinant matters in the DFM. The determinant appears repeatedly in the thesis through the multivariate Gaussian and Student-t densities. The normalising constant of $\mathcal{N}(\mu, \Sigma)$ in m dimensions involves $|\Sigma|^{1/2}$, which — by the volume interpretation derived in the Geometric Interlude — is proportional to the volume of the Mahalanobis ellipsoid $\{x : (x-\mu)'\Sigma^{-1}(x-\mu) \leq 1\}$. A large $|\Sigma|$ means a large ellipsoid, hence a flat density; a small $|\Sigma|$ means a concentrated ellipsoid, hence a peaked density. In the M-step derivations, the term $-\frac{T}{2}\log|\Sigma|$ that appears in the expected complete-data log-likelihood is precisely a penalty against distributions whose ellipsoid is too large — the algorithm balances the fit quality (driven by Mahalanobis distances of the residuals) against this volumetric penalty.

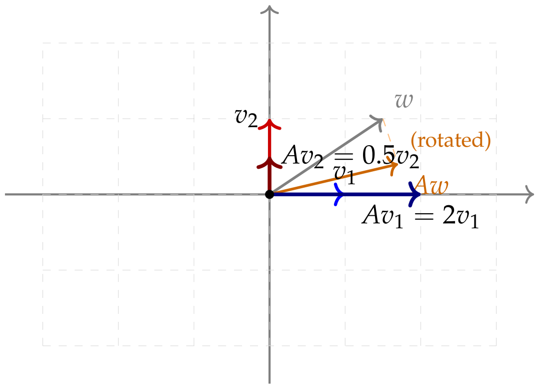
Eigenvalues and Eigenvectors: A Visual Introduction

We now arrive at the central concept of this appendix: eigenvalues and eigenvectors. Their formal treatment will be developed in detail in the next section; here we focus on the geometric intuition, visualising what eigenvectors and eigenvalues look like in two dimensions.

The defining property. A non-zero vector $v \in \mathbb{R}^n$ is called an *eigenvector* of a square matrix $A \in \mathbb{R}^{n \times n}$ if its image Av lies on the same line as v. Equivalently, there exists a scalar λ such that

$$Av \;=\; \lambda\, v.$$

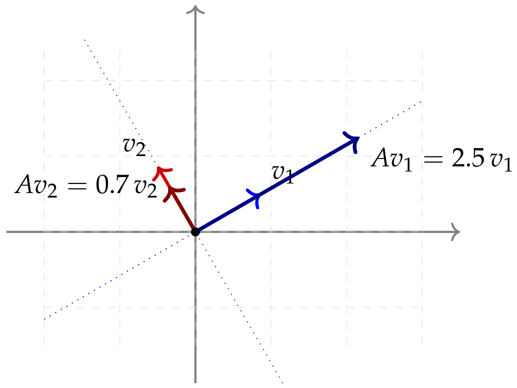
The scalar λ is the *eigenvalue* corresponding to v. The defining condition has a vivid geometric interpretation: an eigenvector is a direction in which A acts by *pure stretching*, with no rotation. Most directions in space are *not* eigendirections — they get both rotated and stretched by A. Eigendirections are the special, privileged directions in which the action of A simplifies to a scalar multiplication.



Eigenvectors of $A = \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}$:
v_1 stretched by $\lambda_1 = 2$, v_2 shrunk by $\lambda_2 = 0.5$.
Generic w is both rotated and stretched.

Figure 9: Eigenvectors as "privileged directions" under a linear transformation. The matrix $A = \mathrm{diag}(2, 0.5)$ has the coordinate axes as eigendirections: $v_1 = e_1$ is stretched by $\lambda_1 = 2$, and $v_2 = e_2$ is shrunk by $\lambda_2 = 0.5$. A generic vector w that is not aligned with the eigenvectors is both rotated and stretched by A.

Why eigendirections are special. The previous figure displays a particularly simple case: A is diagonal, so the eigenvectors are the coordinate axes. For a generic matrix A, the eigenvectors will *not* coincide with the coordinate axes — they will be rotated. The key insight is that *some* pair of perpendicular directions (in the symmetric case) or *some* pair of directions (in the general case) does play the role of the coordinate axes: those are the eigendirections.



Eigenvectors of a generic symmetric A are perpendicular, but not aligned with the coordinate axes.

Figure 10: For a generic symmetric matrix A, the eigenvectors v_1, v_2 are perpendicular to each other (a property guaranteed by the spectral theorem) but rotated relative to the coordinate axes. The matrix A stretches space along v_1 by λ_1 and along v_2 by λ_2. The dotted lines indicate the eigendirections, which form a rotated coordinate system in which A acts diagonally.

Eigenvectors and the spectral decomposition. The fact that perpendicular eigenvectors exist for any symmetric matrix is the content of the *spectral theorem*, which we state formally in the next section. For now, the geometric picture is sufficient: the spectral decomposition $\Sigma = V\Lambda V'$ that we introduced in the Geometric Interlude is exactly the algebraic expression of this geometric fact — V is the matrix whose columns are the eigenvectors, Λ is the diagonal matrix of eigenvalues, and the product $V\Lambda V'$ describes the transformation as "rotate to the eigenvector basis, stretch by the eigenvalues, rotate back".

Three regimes for two-dimensional eigenvalues. The qualitative behaviour of A as a transformation depends on the sign and magnitude of its eigenvalues. Three regimes are worth visualising:

- *Both* $\lambda_1, \lambda_2 > 1$: pure expansion in both eigendirections. The unit square in the eigenvector basis is stretched into a larger rectangle.
- *Both* $0 < \lambda_1, \lambda_2 < 1$: pure contraction. Repeated application of A^k shrinks every vector to zero.
- $\lambda_1 > 1 > \lambda_2 > 0$: stretching in one direction, compression in the other. This is the typical regime of empirical covariance matrices: most variance concentrates in a few principal directions, with the remaining directions carrying very little variance.

The third regime is particularly important for our setting: when we apply PCA to macroeconomic data, the empirical covariance matrix $\widehat{\mathbf{\Sigma}}_Y$ typically has a few large eigenvalues (corresponding to common factors) and many small eigenvalues (corresponding to idiosyncratic noise), exactly the configuration of one stretched direction and many compressed directions.

Towards the formal treatment. The visual intuition developed in this section is the geometric core of the appendix that follows. The next section formalises the spectral decomposition, derives its algebraic consequences (powers, inverses, square roots of matrices), and connects eigenvalues to positive definiteness, ellipsoids, and the volumetric meaning of the determinant

— all of which we have already used informally in the Geometric Interlude. The remaining sections of the appendix then apply these tools to specific problems: stability of linear dynamical systems (the unit-circle condition for the VAR transition matrix **A**), Principal Component Analysis (eigendecomposition of the sample covariance), and noise floors of random matrices (the Circular and Marchenko–Pastur laws).

Symmetric Positive-Definite Matrices and Ellipsoids

A particularly important class of symmetric matrices is that of *positive-definite* (PD) matrices. A symmetric matrix $\Sigma \in \mathbb{R}^{m\times m}$ is positive definite if

$$v'\Sigma\, v \;>\; 0 \qquad \text{for every non-zero } v \in \mathbb{R}^m.$$

Geometrically, this condition says that the quadratic form $v \mapsto v'\Sigma v$ takes positive values everywhere except at the origin. The level sets of this quadratic form are *ellipsoids* centred at the origin, and their geometry is entirely controlled by the eigenstructure of Σ.

Eigenvalues of a PD matrix are positive. Suppose v is an eigenvector of Σ with eigenvalue λ. Then $\Sigma v = \lambda v$, and

$$v'\Sigma\, v \;=\; \lambda\, \|v\|^2.$$

Since $v'\Sigma v > 0$ by positive definiteness and $\|v\|^2 > 0$ for any non-zero vector, we must have $\lambda > 0$. The same argument in reverse shows the converse: a symmetric matrix with all positive eigenvalues is PD. So, for symmetric matrices, *positive definiteness is equivalent to all eigenvalues being positive*.

The ellipsoid associated with Σ. Recall from the Geometric Interlude that the level set of the Mahalanobis distance,

$$E_c \;\equiv\; \{x \in \mathbb{R}^m : x'\Sigma^{-1}x \le c\},$$

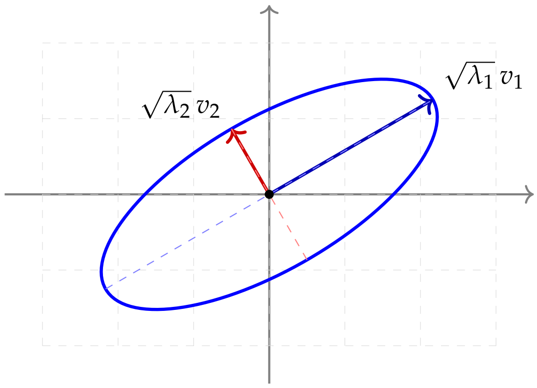
is an ellipsoid centred at the origin. Using the spectral decomposition $\Sigma = V\Lambda V'$ and the change of variable $y = V'x$, this ellipsoid takes the equivalent form

$$\sum_{i=1}^{m} \frac{y_i^2}{\lambda_i} \;\le\; c.$$

This is the standard form of an ellipsoid with axes along the coordinate directions $y_1, \ldots, y_m$ and semi-axis lengths $\sqrt{c\,\lambda_1}, \ldots, \sqrt{c\,\lambda_m}$. Going back to the original coordinates via $x = Vy$, the axes of E_c are aligned with the eigenvectors of Σ, and their semi-axis lengths are $\sqrt{c\,\lambda_i}$.

Visualising the ellipsoid in two dimensions. In two dimensions, the ellipsoid is an ordinary ellipse. Its geometric properties can be read directly off the spectral decomposition of Σ:

- The *orientation* of the ellipse is given by the eigenvectors v_1, v_2;
- The *semi-axis lengths* are $\sqrt{\lambda_1}$, $\sqrt{\lambda_2}$ (taking $c = 1$);
- The *eccentricity* (how elongated the ellipse is) is determined by the ratio λ_1/λ_2.



Mahalanobis ellipse $\{x : x'\Sigma^{-1}x = 1\}$:
axes along v_1, v_2, semi-axis lengths $\sqrt{\lambda_1}, \sqrt{\lambda_2}$.

Figure 11: The Mahalanobis ellipse associated with a symmetric positive-definite matrix Σ. The principal axes of the ellipse are aligned with the eigenvectors v_1, v_2 of Σ, and the semi-axis lengths are the square roots of the corresponding eigenvalues. A large eigenvalue produces a long axis (high variance in that direction); a small eigenvalue produces a short axis (low variance in that direction).

Why $\sqrt{\lambda_i}$ and not λ_i? The square root in the semi-axis length is initially counterintuitive but becomes clear with the following calculation. The eigenvalue equation $\Sigma v_i = \lambda_i v_i$ shows that Σ stretches v_i by a factor λ_i. The quadratic form $v'\Sigma v$, when evaluated at the eigenvector, gives $v_i'\Sigma v_i = \lambda_i \|v_i\|^2 = \lambda_i$ (assuming $\|v_i\| = 1$). So a unit displacement along v_i contributes λ_i to the quadratic form. To reach the level set $x'\Sigma^{-1}x = 1$, we need a displacement of length $\sqrt{\lambda_i}$ along v_i — because the inverse Σ^{-1} contracts by $1/\lambda_i$ in that direction. The square root is therefore the natural "half-length" associated with the eigenvalue, and it is the right measure of dispersion in the corresponding principal direction.

Special cases worth visualising. *Case 1: $\Sigma = \sigma^2 I_2$ (isotropic).* Both eigenvalues equal σ^2, eigenvectors are arbitrary (any orthonormal basis works), and the ellipse degenerates to a *circle* of radius σ. The distribution is equally spread in all directions.

Case 2: Σ diagonal but unequal entries. Eigenvectors are the coordinate axes; semi-axis lengths are $\sqrt{\Sigma_{11}}$ and $\sqrt{\Sigma_{22}}$. The ellipse is axis-aligned but elongated.

Case 3: Σ with non-zero off-diagonals. The eigenvectors are rotated relative to the coordinate axes (the angle is determined by the off-diagonal entries), and the ellipse is correspondingly tilted. This is the case shown in Figure 11.

Connection with the DFM. The scale matrix $\mathbf{Q}$ of the factor innovations and the scale matrix $\mathbf{R}$ of the idiosyncratic shocks are both symmetric and positive definite. Their ellipsoids describe the geometry of the corresponding distributions: a large eigenvalue of $\mathbf{Q}$ in some direction means that the factor innovation is volatile in that direction; a large eigenvalue of $\mathbf{R}$ for some series means that the idiosyncratic shock for that series is large. The smoothed posterior covariance $P_{t|T}$ produced by the Kalman filter is also a PD ellipsoid, this time describing the posterior uncertainty about the factor at time t — small eigenvalues mean tight identification, large eigenvalues mean weak identification. The distinction between scale and covariance is relevant when $\nu < \infty$: in the Student-t scale-mixture representation, $\mathbf{Q}$ and $\mathbf{R}$ are scale matrices, and the actual covariance matrices are $\frac{\nu_u}{\nu_u - 2}\mathbf{Q}$ and $\frac{\nu_\varepsilon}{\nu_\varepsilon - 2}\mathbf{R}$ respectively (when $\nu > 2$). The geometric interpretation in terms of ellipsoids applies equally to both: covariance and scale ellipsoids differ only by a uniform inflation factor in each dimension.

The Mahalanobis Distance: A Visual Perspective

The Mahalanobis distance was introduced algebraically in the Geometric Interlude and used extensively throughout the EM derivations. We now give it a complementary visual treatment.

The Euclidean distance: a benchmark. Recall the ordinary Euclidean distance between x and μ:

$$d_E(x, \mu) \;=\; \|x - \mu\| \;=\; \sqrt{(x-\mu)'(x-\mu)}.$$

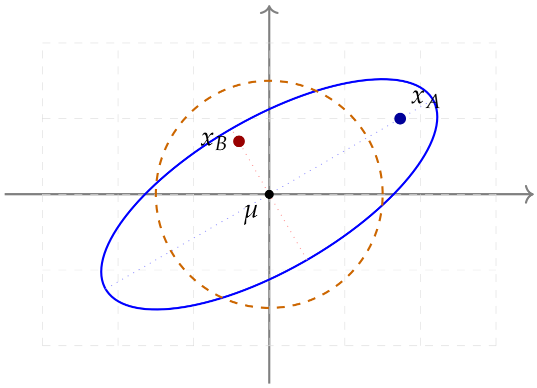
The level sets of d_E are *circles* (or spheres in higher dimensions) centred at μ. All points equidistant from μ lie on the same circle; the circles get larger as d_E increases. This is the geometry of the standard isotropic Gaussian $\mathcal{N}(\mu, I_m)$.

The Mahalanobis distance: a stretched and rotated alternative. The Mahalanobis distance generalises the Euclidean distance by incorporating the scale matrix Σ:

$$d_M(x, \mu; \Sigma) \;=\; \sqrt{(x-\mu)'\Sigma^{-1}(x-\mu)}.$$

When $\Sigma = I_m$, the Mahalanobis distance reduces to the Euclidean distance. For general Σ, the level sets of d_M are no longer circles but *ellipsoids* — exactly the ones discussed in the previous section. Two points equidistant from μ in the Mahalanobis sense lie on the same ellipsoid; points along the long axis of the ellipsoid are "Mahalanobis-far" even if they are Euclidean-near.

Why this difference matters. The intuition is that the Mahalanobis distance *accounts for the shape of the distribution*. If Σ has a large eigenvalue along some direction (say, the v_1 direction), then movements along v_1 are "cheap" — they cover little Mahalanobis ground because the distribution is spread out in that direction, so a large displacement is unsurprising. Conversely, movements perpendicular to v_1 — along the short axis of the ellipsoid — are "expensive": they cover a lot of Mahalanobis ground per unit of Euclidean distance, because the distribution is tightly concentrated in that direction.



Blue ellipse: $d_M(x, \mu) = 1$

Orange circle: $d_E(x, \mu) = 1.5$

x_A: far in Euclidean, on ellipse (Mahalanobis = 1)

x_B: near in Euclidean, also Mahalanobis ≈ 1

Figure 12: Comparison of Euclidean and Mahalanobis distances. The orange dashed circle is a level set of the Euclidean distance from μ. The blue ellipse is a level set of the Mahalanobis distance under a non-isotropic Σ. Two points x_A and x_B at similar Mahalanobis distance from μ can be at very different Euclidean distances: x_A is far Euclidean-wise but "cheap" Mahalanobis-wise (because it lies along the long axis of the ellipse), whereas x_B is close Euclidean-wise but "expensive" Mahalanobis-wise (because it lies along the short axis).

Whitening transformation: making Mahalanobis Euclidean. The Mahalanobis distance can be reduced to a Euclidean distance via a coordinate change. Define the *whitening transformation* $z = \Sigma^{-1/2}(x - \mu)$, where $\Sigma^{-1/2}$ is the symmetric inverse square root of Σ. Then

$$\|z\|^2 = (x - \mu)' \, \Sigma^{-1/2} \, \Sigma^{-1/2} \, (x - \mu) = (x - \mu)' \Sigma^{-1} (x - \mu),$$

so the Euclidean norm of z equals the squared Mahalanobis distance of x from μ. Geometrically, $\Sigma^{-1/2}$ is the linear transformation that rotates the eigenvectors of Σ to the coordinate axes *and* contracts each axis by $1/\sqrt{\lambda_i}$, turning the ellipsoid into a unit circle. In this transformed "whitened" space, the Mahalanobis distance is just the ordinary Euclidean distance, and the distribution is just the standard isotropic Gaussian.

Why this matters in the EM derivations. The Mahalanobis distance appears throughout the M-step in two distinct forms.

First, as the *quadratic kernel* of the Gaussian/Student-t density: $-\frac{1}{2}(x - \mu)'\Sigma^{-1}(x - \mu)$ in the Gaussian log-density, polynomial in the Student-t case. The expected value of this quantity, taken over the posterior of the latent variables, drives the M-step updates of the loading matrix $\mathbf{\Lambda}$ and the transition matrix $\mathbf{A}$.

Second, as the *weight assignment rule* in the Student-t scale-mixture representation. The posterior weight $\hat{w}_t = (\nu + p)/(\nu + d_t)$ for an observation depends on its Mahalanobis distance d_t from the centre: observations with small d_t (near the centre of the ellipsoid) receive weights near 1; observations with large d_t (in the tails of the ellipsoid) are down-weighted with $\hat{w}_t \to 0$. The Mahalanobis distance is therefore the geometric quantity that the algorithm uses to decide which observations to "trust" and which to discount as outliers.

Volumes, Determinants, and the Geometry of Concentration

We have seen that the determinant of a matrix is a volume-scaling factor, and that the Mahalanobis ellipsoid associated with a covariance matrix Σ has axes determined by the eigenvectors of Σ and semi-axis lengths $\sqrt{\lambda_i}$. We now connect these two facts: the volume of the Mahalanobis ellipsoid is determined by $|\Sigma|^{1/2}$, and the determinant $|\Sigma|$ is therefore a measure of *how spread out* the distribution is.

The volume of a 2D ellipse. In two dimensions, the area of an ellipse with semi-axis lengths a and b is

$$\text{Area} \;=\; \pi\, a\, b.$$

For the Mahalanobis ellipse $\{x : x'\Sigma^{-1}x \le 1\}$, the semi-axis lengths are $a = \sqrt{\lambda_1}$ and $b = \sqrt{\lambda_2}$, so

$$\text{Area}(E_1) \;=\; \pi\sqrt{\lambda_1\lambda_2} \;=\; \pi\sqrt{|\Sigma|}.$$

The area of the unit Mahalanobis ellipse is therefore proportional to $\sqrt{|\Sigma|}$, with π as the proportionality constant.

Why $|\Sigma| = \prod_i \lambda_i$. The identity $|\Sigma| = \lambda_1\lambda_2\cdots\lambda_m$ is a general property of any square matrix: the determinant equals the product of the eigenvalues, counted with multiplicity. For a symmetric positive-definite Σ, this identity has two particularly clean consequences. First, all $\lambda_i > 0$, so $|\Sigma| > 0$ — the determinant is strictly positive (and not just non-zero), and $|\Sigma|^{1/2} = \sqrt{\lambda_1\cdots\lambda_m}$ is well-defined as a real number without any sign or branch ambiguity. Second, the level set $\{x : x'\Sigma^{-1}x \le 1\}$ is a bounded, closed ellipsoid (not a hyperboloid or other unbounded surface) — a property guaranteed exactly by the positivity of the eigenvalues. The volume of this ellipsoid is then simply the product of its semi-axis lengths $\sqrt{\lambda_i}$, multiplied by a dimension-dependent constant.

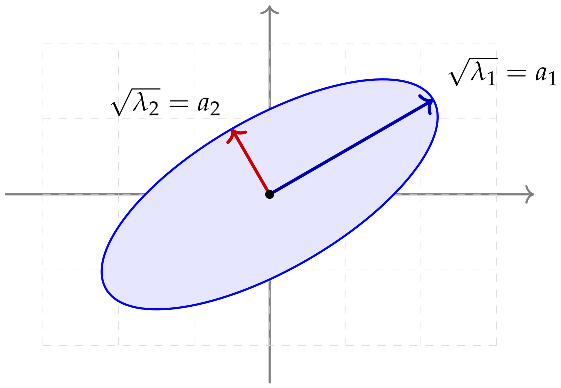
The volume in m dimensions. In m dimensions, the volume of an ellipsoid with semi-axes $a_1, \ldots, a_m$ is

$$\text{Volume} \;=\; \frac{\pi^{m/2}}{\Gamma(m/2+1)}\, a_1\, a_2\cdots a_m,$$

where Γ is the gamma function (a generalisation of the factorial: $\Gamma(n+1) = n!$ for integer n). For the unit Mahalanobis ellipsoid, $a_i = \sqrt{\lambda_i}$, so

$$\text{Volume}(E_1) \;=\; \frac{\pi^{m/2}}{\Gamma(m/2+1)}\sqrt{\lambda_1\lambda_2\cdots\lambda_m} \;=\; \frac{\pi^{m/2}}{\Gamma(m/2+1)}\,|\Sigma|^{1/2}.$$

The volume scales with $|\Sigma|^{1/2}$, with a dimension-dependent constant prefactor. For practical purposes, this is the unifying identity: the determinant of Σ, raised to the power $1/2$, *is* the volume of the Mahalanobis ellipsoid (up to a dimensional constant).



Area of ellipse $= \pi\, a_1\, a_2 = \pi\sqrt{\lambda_1\lambda_2} = \pi|\Sigma|^{1/2}$
Large $|\Sigma|$: large ellipse, spread-out distribution.
Small $|\Sigma|$: small ellipse, concentrated distribution.

Figure 13: The area (volume in 2D) of the Mahalanobis ellipse is $\pi\sqrt{|\Sigma|}$. The semi-axes are $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$, the square roots of the eigenvalues of Σ. The determinant therefore captures the volumetric "size" of the ellipse: a large determinant means a large ellipse, and hence a distribution that is spread out over a large region of space; a small determinant means a concentrated distribution.

The role of $|\Sigma|^{1/2}$ in the density. This volumetric interpretation explains why the normalising constant of the multivariate Gaussian density contains the factor $|\Sigma|^{1/2}$ in the denominator:

$$p(x) \;=\; \frac{1}{(2\pi)^{m/2}\,|\Sigma|^{1/2}}\,\exp\!\Big(-\tfrac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\Big)\,.$$

The factor $|\Sigma|^{1/2}$ in the denominator is exactly the volume of the unit Mahalanobis ellipsoid (up to the constant $(2\pi)^{m/2}/\pi^{m/2}\cdot\Gamma(m/2+1)$), which ensures the density integrates to one. A distribution with a large ellipsoid (large $|\Sigma|^{1/2}$) must have a low density value at every point — the probability mass is spread thin. A distribution with a small ellipsoid (small $|\Sigma|^{1/2}$) is concentrated, and the density value is correspondingly high.

Why $-\frac{1}{2}\log|\Sigma|$ acts as a regulariser. Taking the logarithm of the density,

$$\log p(x) \;=\; -\tfrac{m}{2}\log(2\pi) \;-\; \tfrac{1}{2}\log|\Sigma| \;-\; \tfrac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu).$$

The term $-\frac{1}{2}\log|\Sigma|$ is a *penalty* that grows with the volume of the ellipsoid. In the M-step derivations of the DFM, when we maximise the expected log-likelihood with respect to Σ (whether it is $\mathbf{Q}$ or $\mathbf{R}$), this term *prevents* the algorithm from making the ellipsoid arbitrarily large to fit any single observation perfectly. Without the log-determinant penalty, a degenerate solution with $|\Sigma| \to 0$ in some direction (concentrating all probability mass on a lower-dimensional subspace) would maximise the kernel term but is prevented by the divergence of $-\log|\Sigma| \to +\infty$ — an infinite penalty. The interplay between the kernel (Mahalanobis-distance) term and the log-determinant term is what makes the maximum-likelihood estimator of Σ well-defined and non-degenerate.

Random Matrices and the Circular Law: A Visual Introduction

The eigenvalues of a matrix that is constructed from *random* entries — rather than from data with structure — exhibit a remarkable regularity. Understanding this regularity provides the "noise floor" against which the genuine signal eigenvalues of empirical covariance matrices must be compared. We give a brief visual treatment of the Circular Law here; a more formal discussion can be found in the next section of the appendix.

The Circular Law. Let $A_n \in \mathbb{R}^{n\times n}$ be a random matrix whose entries $A_{n,ij}$ are independent and identically distributed with mean zero and unit variance. The Circular Law (Ginibre, 1965; Tao and Vu, 2010) states that, as $n \to \infty$, the eigenvalues of the *rescaled* matrix $A_n/\sqrt{n}$ converge in distribution to the uniform distribution on the unit disk in the complex plane:

$$\{ z \in \mathbb{C} : |z| \le 1 \}.$$

The scaling by $\sqrt{n}$ is essential. Without it, the eigenvalues would diverge as n grows, scattered ever more wildly across the complex plane. With the $1/\sqrt{n}$ scaling, the eigenvalues populate the unit disk uniformly, regardless of the specific distribution of the entries (only the first two moments matter).

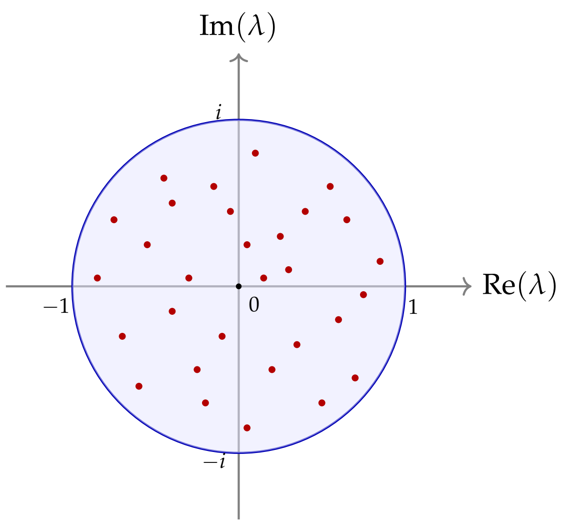


Figure 14: Eigenvalues of a random matrix $A_n/\sqrt{n}$ in the complex plane, for n moderately large. The blue circle is the boundary of the unit disk. As $n \to \infty$, the empirical distribution of the eigenvalues fills the unit disk uniformly. The scaling by $\sqrt{n}$ is what places the spectrum on the natural scale of the unit disk; without this scaling, the eigenvalues would diverge.

Why this matters: the noise floor. The Circular Law is the eigenvalue distribution of a matrix with *no structure* — pure noise. When we apply factor analysis to a finite sample of macroeconomic data, the empirical covariance matrix $\widehat{\mathbf{\Sigma}}_Y$ contains both signal (from genuine common factors) and noise (from finite-sample variation in the idiosyncratic component). The Circular Law characterises the noise-only contribution: if there were no factors, the eigenvalues of $\widehat{\mathbf{\Sigma}}_Y/\sqrt{n}$ would, in the limit, be uniformly distributed in the unit disk. Genuine factor eigenvalues stand out above this noise floor: they appear as "outliers" to the bulk of the random-matrix spectrum.

The Marchenko–Pastur companion. For the specific case of *symmetric positive semi-definite* matrices — such as sample covariance matrices computed from n observations of p random variables — the analogue of the Circular Law is the *Marchenko–Pastur Law* (Marchenko and Pastur, 1967). This result describes the asymptotic distribution of the eigenvalues of $\widehat{\boldsymbol{\Sigma}}$, when the underlying population covariance is the identity matrix. The eigenvalues are no longer uniform on the disk but are distributed on a real interval $[\lambda_-, \lambda_+]$ with a specific density. The boundary value λ_+ provides a sharper estimate of the noise floor for sample covariance matrices, and is the basis for many modern factor-selection criteria (such as the Onatski criterion or the Bai-Ng information criterion).

Bridge to the unit-circle stability condition. Beyond providing a noise floor for empirical covariances, the unit circle in the complex plane plays a second crucial role in the stability of dynamical systems. We turn to this in the next section.

Stability of Linear Dynamical Systems: A Visual Perspective

The unit circle in the complex plane plays two distinct but related roles in our analysis. First, as we have just seen, it bounds the "noise floor" of empirical covariance matrices. Second, it is the geometric criterion that separates stable from unstable VAR processes. We now visualise this second role.

Recap: the stability condition. Consider a linear dynamical system

$$x_{t+1} \;=\; \mathbf{A}\, x_t \;+\; u_{t+1}, \qquad t = 0, 1, 2, \ldots$$

where $\mathbf{A} \in \mathbb{R}^{n\times n}$ is the transition matrix. As discussed extensively earlier, the process $\{x_t\}$ is covariance-stationary if and only if all eigenvalues of $\mathbf{A}$ satisfy

$$|\lambda_i| \;<\; 1, \qquad i = 1, 2, \ldots, n.$$

Geometrically, all eigenvalues must lie strictly *inside* the unit circle in the complex plane. Equivalently: the eigenvalues live in the same set that bounds the noise floor of the Circular Law, but for a completely different reason.

Three geometric scenarios. The position of the eigenvalues in the complex plane immediately classifies the dynamic behaviour of the process. Three scenarios are worth visualising side by side.

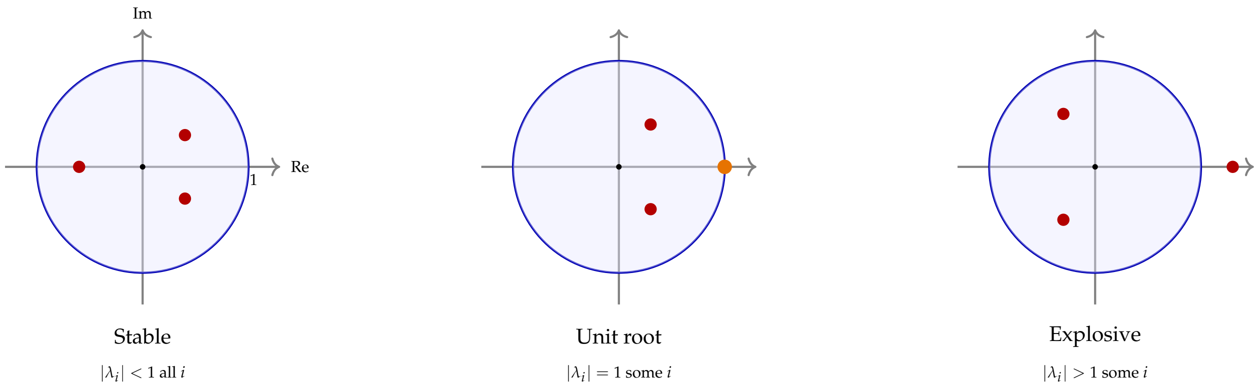


Figure 15: The three regimes of a linear dynamical system, classified by the position of the eigenvalues of $\mathbf{A}$ in the complex plane. *Left*: all eigenvalues strictly inside the unit circle; the process is covariance-stationary, shocks decay exponentially, the system has a well-defined long-run distribution. *Middle*: at least one eigenvalue exactly on the unit circle; the process has a unit root, shocks have permanent effects, and the variance grows linearly in t. *Right*: at least one eigenvalue outside the unit circle; the process is explosive, shocks amplify exponentially, the system diverges to infinity.

Why the eigenvalues control everything. Recall that the iterated state at time t depends on $\mathbf{A}^t$ applied to the initial state. Using the spectral decomposition (when $\mathbf{A}$ is diagonalisable),

$$\mathbf{A}^t \;=\; V\Lambda^t V^{-1}, \qquad \Lambda^t = \operatorname{diag}(\lambda_1^t, \ldots, \lambda_n^t).$$

The behaviour of $\mathbf{A}^t$ is governed by what happens to each λ_i^t as t grows. If $|\lambda_i| < 1$, then $\lambda_i^t \to 0$ exponentially fast — the corresponding mode of the system decays into oblivion. If $|\lambda_i| = 1$, then λ_i^t either remains constant (real $\lambda_i = 1$) or oscillates (complex λ_i on the unit circle); the mode persists indefinitely. If $|\lambda_i| > 1$, then $\lambda_i^t \to \infty$ exponentially fast — the mode amplifies without bound. The eigenvalue *moduli* control whether each mode dies out, persists, or explodes; the *arguments* (when complex) control whether each mode oscillates and at what frequency.

Complex eigenvalues and macroeconomic cycles. A pair of complex-conjugate eigenvalues $\lambda = r\,e^{\pm i\theta}$ inside the unit circle ($r < 1$) produces a damped oscillation. The modulus r controls the rate of damping (closer to 1 means slower decay), and the argument θ controls the frequency (closer to zero means longer cycles). The corresponding mode of the system traces an inward spiral in two-dimensional projections, with period $2\pi/\theta$ and decay rate r^t.

This is the algebraic content of business cycles: the cyclical behaviour observed in macroeconomic data is the geometric expression of complex eigenvalues of $\mathbf{A}$ that lie inside but not too close to the origin of the unit disk. After estimation of the DFM, plotting the eigenvalues of $\widehat{\mathbf{A}}$ in the complex plane provides a compact diagnostic of both stability (do they lie inside the unit circle?) and cyclicality (are there complex eigenvalues close to the boundary, indicating long, slowly damped cycles?).

Practical implication for the EM algorithm. The EM algorithm we use to estimate $\mathbf{A}$ does not impose the unit-circle constraint during optimisation: each M-step solves the unconstrained least-squares problem for $\mathbf{A}$. In practice, when the data have been pre-processed to be stationary (log-differences, first-differences as appropriate), the estimated $\widehat{\mathbf{A}}$ at convergence has eigenvalues comfortably inside the unit circle. Should an estimated eigenvalue approach the boundary, this is a signal of either insufficient pre-processing or of a misspecified factor structure that fails to capture all the non-stationary content of the data.

An equivalent formulation: roots of the characteristic polynomial. The reader with a background in classical time-series econometrics will be familiar with an alternative formulation of the stability condition. Writing the VAR(1) in lag-operator notation as

$$(I - \mathbf{A}L)\,x_t \;=\; u_t,$$

where L is the lag operator, the characteristic polynomial is $\det(I - \mathbf{A}z) = 0$ for $z \in \mathbb{C}$. The standard econometric stability condition (Hamilton, 1994; Lütkepohl, 2005) requires that all roots z_i of this polynomial lie strictly *outside* the unit circle:

$$|z_i| \;>\; 1, \qquad i = 1, 2, \ldots, n.$$

The two conditions are equivalent. The roots of the characteristic polynomial are the reciprocals of the eigenvalues of $\mathbf{A}$: $z_i = 1/\lambda_i$. Therefore $|\lambda_i| < 1$ if and only if $|z_i| > 1$. Both conditions describe the same geometric fact — that the system's modes decay rather than amplify over time — but expressed via different objects: eigenvalues of the transition matrix (the state-space convention used here) versus roots of the characteristic polynomial (the classical econometric convention). We adopt the eigenvalue formulation throughout, in keeping with the state-space framework that underlies the DFM.

Matrix Rank: Independent Directions and the Selection Matrix

The final concept we visualise in this appendix is the *rank* of a matrix. Rank is a notion of "effective dimensionality" that plays a central role in the missing-data formalism we developed earlier.

Definition. The *rank* of a matrix $A \in \mathbb{R}^{m\times n}$ is the maximum number of linearly independent columns of A — equivalently, the maximum number of linearly independent rows. Formally,

$$\operatorname{rank}(A) \;=\; \dim\left(\text{columnspaceof} A\right) \;=\; \dim\left(\text{rowspaceof} A\right).$$

The rank is bounded above by both m and n: $\operatorname{rank}(A) \leq \min(m, n)$. A matrix is said to have *full rank* if its rank equals this upper bound, and *deficient rank* otherwise.

Geometric interpretation. The rank of A is the dimension of the image of T_A — the "effective dimensionality" of the linear transformation. If $A \in \mathbb{R}^{m\times n}$ has rank k, then the transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$ maps every vector in $\mathbb{R}^n$ into a k-dimensional subspace of $\mathbb{R}^m$ — the column space of A. Vectors in $\mathbb{R}^n$ that are mapped to zero by T_A form the *null space* of A, which has dimension $n - k$ (the rank-nullity theorem).

For a square matrix $A \in \mathbb{R}^{n\times n}$:

- $\operatorname{rank}(A) = n$ (full rank): A is invertible, $\det(A) \neq 0$, and T_A is a bijection of $\mathbb{R}^n$;
- $\operatorname{rank}(A) < n$ (rank-deficient): A is singular, $\det(A) = 0$, and T_A collapses $\mathbb{R}^n$ to a lower-dimensional subspace.

This is why the determinant being zero corresponds geometrically to the volume collapse we visualised previously — a singular matrix has deficient rank, and its image is a flat, lower-dimensional region.

The selection matrix $\mathbf{W}_t$ as a rank operator. The most concrete appearance of rank in our framework is in the selection matrix $\mathbf{W}_t \in \{0, 1\}^{m_t\times M}$ that handles missing data. By construction, $\mathbf{W}_t$ has exactly m_t rows, each of which is a standard basis vector of $\mathbb{R}^M$ (a row with a single 1 and the rest zeros). These rows are linearly independent because they correspond to distinct basis vectors. Therefore

$$\operatorname{rank}(\mathbf{W}_t) \;=\; m_t.$$

The action of $\mathbf{W}_t$ on the full observation vector y_t is to extract its m_t observed components:

$$y_t^{\text{obs}} \;=\; \mathbf{W}_t\, y_t \;\in\; \mathbb{R}^{m_t}.$$

The selection matrix is therefore a *projection* from the ambient M-dimensional observation space to the m_t-dimensional subspace spanned by the currently observed series. As t varies, the subspace varies — its dimension depends on how many series are released by time t — but at every t the projection is well-defined and has rank exactly equal to m_t.

Why the rank of $\mathbf{W}_t$ matters. Several quantities in the EM derivations depend explicitly on $m_t = \operatorname{rank}(\mathbf{W}_t)$. Three are especially worth recalling:

- In the weight update for the idiosyncratic shocks,

$$\hat{w}_t^{\varepsilon} \;=\; \frac{\nu_\varepsilon + m_t}{\nu_\varepsilon + d_t^{\varepsilon,\text{obs}}},$$

the numerator includes m_t, the dimension of the observed-data space at time t. This dimension changes across t because of ragged-edge missingness, and the weight update correctly reflects the variable amount of information available at each t.

- In the log-likelihood from the Kalman filter,

$$\log p(y_t^{\text{obs}} \mid \theta) \;=\; -\tfrac{1}{2}\big[\log|S_t^{\text{obs}}| + (\eta_t^{\text{obs}})'(S_t^{\text{obs}})^{-1}\eta_t^{\text{obs}} + m_t \log(2\pi)\big],$$

  the constant term scales with m_t, so periods with fewer observations contribute proportionally less to the log-likelihood.

- In the Mahalanobis distance computed at time t, only the $m_t \times m_t$ sub-block of $\mathbf{R}$ enters the calculation. This sub-block has rank m_t (it is positive definite), and the inversion $(\mathbf{W}_t\mathbf{R}\mathbf{W}_t')^{-1}$ is well-defined precisely because $\mathbf{W}_t$ has full row rank.

The rank concept is therefore not a mathematical abstraction but a direct counterpart of the practical question "how much information is observable at time t?". The answer is: m_t pieces of information, encoded by an m_t-rank projection from the full observation space.

Linear dependence and collinearity. The most common way for a matrix to fail to have full rank is *linear dependence* among its columns (or rows). Two vectors $v_1, v_2 \in \mathbb{R}^n$ are said to be linearly dependent if one is a scalar multiple of the other, $v_2 = c\, v_1$ for some $c \in \mathbb{R}$. More generally, a set of vectors $\{v_1, v_2, \ldots, v_k\}$ is linearly dependent if at least one of them can be written as a linear combination of the others,

$$v_k \;=\; c_1 v_1 \;+\; c_2 v_2 \;+\; \cdots \;+\; c_{k-1} v_{k-1},$$

for some real coefficients $c_1, \ldots, c_{k-1}$. When this happens, the column space spanned by $\{v_1, \ldots, v_k\}$ has dimension strictly less than k — the redundant vector v_k adds no new direction.

In a data matrix $\mathbf{Y} \in \mathbb{R}^{M\times T}$ whose columns are T observations of M time series, the practical expression of this phenomenon is *collinearity*: two or more series move together (up to a scalar) so that one of them is redundant. If, for instance, two interest rate series are identical at every t, then the corresponding two rows of $\mathbf{Y}$ are scalar multiples of each other, and $\operatorname{rank}(\mathbf{Y})$ drops by one. Since for any real matrix the identity $\operatorname{rank}(\mathbf{Y}\mathbf{Y}') = \operatorname{rank}(\mathbf{Y})$ holds, the sample covariance $\widehat{\boldsymbol{\Sigma}}_Y = \frac{1}{T}\mathbf{Y}\mathbf{Y}'$ inherits the same rank deficiency. By the spectral decomposition, the number of non-zero eigenvalues of a symmetric matrix equals its rank — so at least one eigenvalue of $\widehat{\boldsymbol{\Sigma}}_Y$ is exactly zero. Consequently, the determinant $|\widehat{\boldsymbol{\Sigma}}_Y| = \prod_i \lambda_i$ is zero, and the matrix is singular: it cannot be inverted, and the Mahalanobis distance based on it is undefined.

Perfect collinearity rarely occurs in real macroeconomic datasets, but *near-collinearity* — situations where two or more series are very highly correlated, though not exactly proportional — is common. In near-collinearity, the smallest eigenvalues of $\widehat{\boldsymbol{\Sigma}}_Y$ are not exactly zero but very close to it. Numerically, this produces an *ill-conditioned* covariance matrix: $|\widehat{\boldsymbol{\Sigma}}_Y|$ is tiny, $\widehat{\boldsymbol{\Sigma}}_Y^{-1}$ has very large entries, and operations involving the inverse (such as the Mahalanobis distance or the Kalman gain) are numerically unstable. This is one of several reasons why empirical applications often pre-process data by selecting representative series within each group, by projecting onto a smaller number of principal components, or by introducing explicit regularisation (such as a small ridge δI added to the covariance) to ensure invertibility.

The factor model itself is a structural response to collinearity: by postulating that M observed series are driven by $r \ll M$ common factors, the model formalises the empirical observation that macroeconomic series carry redundant information and admit a low-dimensional summary.

The factor structure replaces the high-dimensional, near-collinear **Y** with a lower-dimensional, well-conditioned **F** that captures the genuinely independent directions of variation.

Bridge to the formal appendix. The visual and intuitive treatment provided in the preceding sections is intended as preparation for the formal appendix that follows. The next sections develop the algebraic content: the spectral theorem proper, the connection between PCA and the eigendecomposition of the sample covariance, and the explicit formulae used in the initialisation of the DFM. The geometric intuitions developed here — vectors as arrows, matrices as transformations, determinants as volume scaling, eigenvectors as privileged directions, ellipsoids as level sets, ranks as effective dimensionalities — are the conceptual scaffolding on which the algebraic machinery rests.

Appendix: Eigenvalues, Eigenvectors, and Principal Components

This appendix collects the linear algebra used throughout the thesis: eigenvalues, eigenvectors, the spectral decomposition, their geometric meaning, their role in characterising stability of linear dynamical systems, and their role in Principal Component Analysis (PCA). The material is intended as a reference — the reader who is already comfortable with these concepts can skip to the summary at the end. No new results are proved; all statements are classical and can be found in any graduate-level linear algebra text, for example Horn and Johnson (2013) or Meyer (2000).

Eigenvalues and Eigenvectors: Definition and Intuition

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. A non-zero vector $v \in \mathbb{R}^n$ is called an *eigenvector* of A, with corresponding *eigenvalue* $\lambda \in \mathbb{C}$, if

$$Av \;=\; \lambda\, v. \tag{243}$$

The defining condition says that A acts on v by simple *scalar multiplication*: the vector direction is preserved, and only its magnitude changes, by the factor λ. Any non-zero scalar multiple of an eigenvector is itself an eigenvector with the same eigenvalue, so eigenvectors are naturally thought of as *directions* rather than as specific vectors; one usually normalises them to unit length.

The characteristic polynomial. Finding eigenvalues reduces to a polynomial-root problem. Rewriting (243) as $(A - \lambda I)v = 0$, we see that a non-zero v can satisfy this equation only if the matrix $A - \lambda I$ is *singular*. Equivalently,

$$\det(A - \lambda I) \;=\; 0. \tag{244}$$

The left-hand side, viewed as a function of λ, is a polynomial of degree n called the *characteristic polynomial* of A. By the fundamental theorem of algebra, it has exactly n roots in $\mathbb{C}$ counting multiplicity — so every $n \times n$ matrix has n eigenvalues (possibly repeated, possibly complex).

Geometric intuition in two dimensions. The meaning of the eigenvalue equation is clearest in low dimension. Consider $A \in \mathbb{R}^{2\times 2}$ acting on vectors in the plane. A generic vector v is mapped by A to some new vector Av that is in general *neither* aligned with v nor of the same length — A both *stretches* and *rotates*. An eigenvector is a special direction in which the rotation part vanishes: A acts on an eigenvector by pure stretching, by the factor λ.

Three illustrative cases in two dimensions sharpen the picture.

Case 1 — Real positive eigenvalues: pure stretching. When $A = \mathrm{diag}(2, 1/2)$, the two eigenvectors are the coordinate axes, with eigenvalues 2 and $1/2$. The matrix stretches vectors by a factor of 2 along the first axis and compresses them by a factor of $1/2$ along the second. Applied repeatedly, A^k stretches the first component geometrically and shrinks the second — the typical behaviour of a stable system with one dominant direction.

Case 2 — Real negative eigenvalues: stretching with reflection. If $\lambda < 0$, the eigenvector direction is preserved but its *orientation* is reversed. Applied repeatedly, A^k alternates sign: oscillation with possible amplification or damping depending on $|\lambda|$.

Case 3 — Complex eigenvalues: rotation. When A is a rotation matrix or has complex conjugate eigenvalues $\lambda = re^{\pm i\theta}$, the eigenvectors are also complex, and A acts on the real plane as a composition of a rotation by angle θ and a scaling by r. Applied repeatedly, A^k traces a spiral — inward if $r < 1$, outward if $r > 1$, and a circle if $r = 1$.

This third case is important: even when A has only real entries, its eigenvalues can be complex. Complex eigenvalues correspond to *rotational* or *oscillatory* modes in the dynamics of repeated multiplication. When we study the stability of a VAR process below, the complex eigenvalues of the transition matrix capture the cyclical behaviour of the system.

Spectral Decomposition and Symmetric Matrices

The most important case for our purposes is that of *symmetric* matrices, which are sufficiently well-behaved to admit a complete geometric description through eigenvalues and eigenvectors.

The spectral theorem. If $A \in \mathbb{R}^{n\times n}$ is symmetric, that is $A = A^\top$, then the spectral theorem guarantees:

1. all n eigenvalues of A are real;
2. there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A;
3. A admits the factorisation
$$A = V\Lambda V^\top, \tag{245}$$
where $V = [v_1, v_2, \ldots, v_n]$ is an orthogonal matrix whose columns are the eigenvectors (so $V^\top V = I$), and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ is the diagonal matrix of the eigenvalues.

Three consequences of the spectral theorem are used repeatedly.

(i) Geometric reading. The factorisation $A = V\Lambda V^\top$ reads as a sequence of three operations on any vector x: first, $V^\top x$ rotates x into the basis of the eigenvectors; second, $\Lambda V^\top x$ stretches each rotated component by the corresponding eigenvalue; third, $V(\Lambda V^\top x)$ rotates back to the original basis. In other words, A is a *rotation + diagonal stretching + inverse rotation*. The eigenvectors mark the directions along which A's action is diagonal; the eigenvalues measure the stretching in those directions.

(ii) Powers and functions of matrices. Once $A = V\Lambda V^\top$, powers and general functions of A are trivial to compute:

$$A^k = V\Lambda^k V^\top, \qquad f(A) = V\, f(\Lambda)\, V^\top,$$

where $\Lambda^k = \mathrm{diag}(\lambda_1^k, \ldots, \lambda_n^k)$ and $f(\Lambda) = \mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_n))$. The same logic extends to non-symmetric diagonalisable matrices via $A^k = V\Lambda^k V^{-1}$ (with V^{-1} in place of $V^\top$): the stability of the iteration $x_{t+1} = Ax_t$ is therefore governed by the eigenvalues of A, since the iterated map $A^k x_0$ amounts to raising each λ_i to the k-th power. The unit-circle stability condition $|\lambda_i| < 1$ for the VAR transition matrix is the direct application of this principle.

(iii) Determinant and trace. Since the determinant and trace are invariant under change of basis, we have

$$\det(A) = \prod_{i=1}^{n} \lambda_i, \qquad \mathrm{tr}(A) = \sum_{i=1}^{n} \lambda_i.$$

These identities let us read off two global summaries of A's action directly from the eigenvalues: the determinant measures the volume change induced by A, and the trace measures the total stretching along the eigenvector directions.

Positive Definite Matrices

A symmetric matrix A is *positive definite* (PD) if $x^\top Ax > 0$ for all $x \neq 0$, and *positive semi-definite* (PSD) if the inequality is relaxed to $\geq$. The two notions correspond exactly to sign conditions on the eigenvalues:

- A is PD $\iff$ all eigenvalues $\lambda_i > 0$;
- A is PSD $\iff$ all eigenvalues $\lambda_i \geq 0$.

This follows immediately from the spectral decomposition: writing $y = V^\top x$, the quadratic form $x^\top Ax = y^\top \Lambda y = \sum_i \lambda_i y_i^2$, which is positive for all $x \neq 0$ precisely when all $\lambda_i > 0$.

Geometric meaning. Positive definite symmetric matrices arise naturally as covariance matrices and as scale matrices of multivariate densities (such as those of the Gaussian and multivariate Student-t). Their level sets

$$\{x \in \mathbb{R}^n : x^\top A^{-1} x \leq c\}$$

are *ellipsoids* centred at the origin, with axes aligned with the eigenvectors of A and semi-axis lengths proportional to $\sqrt{\lambda_i}$. A degenerate case — with one or more eigenvalues equal to zero — corresponds to an ellipsoid that has collapsed to a lower-dimensional subspace, geometrically representing a degenerate distribution. Throughout the DFM estimation, ensuring that the estimated scale matrices $\hat{\mathbf{Q}}$ and $\hat{\mathbf{R}}$ remain strictly positive definite is a practical concern precisely because their eigenvalues must not be allowed to approach zero.

The square root of a PD matrix. Given a PD matrix $A = V\Lambda V^\top$, its *symmetric square root* is

$$A^{1/2} \;=\; V\Lambda^{1/2}V^\top, \qquad \Lambda^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}).$$

This satisfies $A^{1/2}A^{1/2} = A$ and is used, for example, in the whitening transformation $z = A^{-1/2}(x - \mu)$ that maps a multivariate Gaussian $\mathcal{N}(\mu, A)$ to the standard Gaussian $\mathcal{N}(0, I)$. Only PD matrices admit a real symmetric square root; this is one of several ways in which positive definiteness is the algebraic property that makes covariance matrices analytically tractable.

Eigenvalues and the Geometry of Random Matrices

A final topic worth mentioning is the distribution of eigenvalues of *random* matrices, which provides useful intuition for finite-sample behaviour in PCA and factor analysis. The relevant result here is the Circular Law.

The Circular Law. Consider an $n \times n$ matrix A whose entries A_{ij} are independent and identically distributed random variables with mean zero and unit variance. The Circular Law (Ginibre, 1965; Tao and Vu, 2010) states that, as $n \to \infty$, the eigenvalues of $A/\sqrt{n}$ converge in distribution to the uniform distribution on the unit disk in the complex plane: $\{z \in \mathbb{C} : |z| \leq 1\}$.

The scaling by $\sqrt{n}$ is essential — it places the spectrum on a natural scale. Without it, the eigenvalues of A itself would diverge as n grows; with it, they populate the complex plane uniformly inside the unit circle.

Why this matters. The Circular Law has two practical consequences relevant to our setting. First, it describes the "noise floor" of empirical covariance matrices: when we compute the sample covariance of n i.i.d. random variables from T observations, the resulting matrix has eigenvalues that, for large n relative to T, populate a bounded region of the complex plane. Signal eigenvalues (reflecting true common factors) stand out above this noise floor only if they are sufficiently large. This intuition is formalised by the Marchenko-Pastur law (Marchenko and Pastur, 1967), which is the symmetric-matrix version of the Circular Law — and it underlies many criteria for selecting the number of factors in large-dimensional factor models.

Second, the unit circle in the complex plane plays a central role in the stability analysis of linear dynamical systems: a VAR process is stationary if and only if the eigenvalues of its transition matrix lie strictly *inside* the unit circle. This connection is the subject of the next subsection.

Eigenvalues and the Stability of Linear Dynamical Systems

Consider a linear dynamical system driven by a transition matrix $A \in \mathbb{R}^{n\times n}$:

$$x_{t+1} \;=\; A\,x_t \;+\; u_{t+1}, \qquad t = 0, 1, 2, \ldots$$

where u_t is a zero-mean shock with covariance $\mathbf{Q}$. This is a VAR(1) process — and it is exactly the law of motion of the factors f_t in our DFM, with $A = \mathbf{A}$ and $u_t \sim t_{\nu_u}(0, \mathbf{Q})$. A natural question is: under what conditions does the process $\{x_t\}$ remain bounded over time, and does it admit a well-defined long-run distribution?

The answer is given by the eigenvalues of A and involves one of the most important geometric objects in the theory of dynamical systems — the *unit circle* in the complex plane.

Propagation of shocks through A^k. Iterating the VAR equation, the state at time t is expressed as

$$x_t \;=\; A^t x_0 \;+\; \sum_{k=0}^{t-1} A^k u_{t-k}.$$

The behaviour of the system as $t \to \infty$ is therefore dictated by the powers A^k. Using the spectral decomposition (when A is diagonalisable),

$$A^k \;=\; V\Lambda^k V^{-1}, \qquad \Lambda^k = \mathrm{diag}(\lambda_1^k, \ldots, \lambda_n^k).$$

The matrix A^k grows, shrinks, or oscillates depending entirely on whether each $|\lambda_i|^k$ grows, shrinks, or stays bounded. The modulus of each eigenvalue is the critical quantity.

The stability condition: eigenvalues strictly inside the unit circle. The VAR(1) process is *stable* — and hence *covariance-stationary* — if and only if all eigenvalues of A satisfy

$$|\lambda_i| \;<\; 1, \qquad i = 1, 2, \ldots, n. \tag{246}$$

Geometrically, all eigenvalues must lie *strictly inside* the unit circle in the complex plane: $\{z \in \mathbb{C} : |z| < 1\}$. When this condition holds, $|\lambda_i|^k \to 0$ as $k \to \infty$ for every i, so the shock propagated through A^k decays exponentially, the infinite sum $\sum_{k=0}^{\infty} A^k \mathbf{Q}(A^\top)^k$ converges to a well-defined long-run covariance matrix $\mathbf{\Sigma}_\infty$, and the process has a stationary distribution.

Three regimes emerge from this condition:

- *All $|\lambda_i| < 1$: stable.* The process is stationary; shocks dissipate exponentially; the factors fluctuate around a well-defined mean level (in our case, zero, since the factors are centred by construction).

- *Some* $|\lambda_i| = 1$*: unit root*. The process has one or more unit roots; shocks have permanent effects, and the variance of x_t grows linearly in t. This is the regime of the integrated processes I(1) — random walks and their multivariate generalisations.

- *Some* $|\lambda_i| > 1$*: explosive*. Shocks amplify exponentially, and the process diverges to infinity. This regime is not economically realistic for macroeconomic time series and corresponds to a misspecified model.

Why "strict" inequality? The condition $|\lambda_i| < 1$ must be strict: an eigenvalue *on* the unit circle gives a unit root and a non-stationary process. For our DFM, we require all eigenvalues of $\mathbf{A}$ to lie strictly inside the unit circle so that the estimated factor process has a stationary distribution and the associated Kalman filter produces well-defined long-run covariances. In practice, the EM algorithm we use does not *impose* this constraint during estimation, but in all applications we verify that the estimated $\widehat{\mathbf{A}}$ satisfies it. Should an eigenvalue approach unity during the iteration, this would signal either an integrated process mistakenly modelled as stationary, or a numerical problem requiring regularisation.

The connection with pre-processing. This is why the stationarity pre-processing discussed at the beginning is essential: we transform each series (log-differences for real and price variables, first differences for rates) so that the input to the DFM is stationary. Without this pre-processing, the common factors extracted from the panel would inherit non-stationary behaviour, causing one or more eigenvalues of $\mathbf{A}$ to sit on or near the unit circle, and breaking the state-space machinery. The unit-circle condition on $\widehat{\mathbf{A}}$ is, in this sense, the analytical counterpart of the empirical choice to work with growth rates rather than levels.

Complex eigenvalues and macroeconomic cycles. Even in the stable regime, real macroeconomic transition matrices typically have *complex* eigenvalues. A complex eigenvalue $\lambda = re^{i\theta}$ with $r < 1$ produces a damped oscillation: the impulse response to a shock traces a spiralling path in the factor space, decaying at rate r^k and cycling with period $2\pi/\theta$. This is the algebraic expression of the business cycle: the cyclical pattern in the factors corresponds to complex eigenvalues of $\mathbf{A}$, with the period determined by their argument and the damping by their modulus. Plotting the eigenvalues of $\widehat{\mathbf{A}}$ in the complex plane, after estimation, provides a compact visualisation of the cyclical content of the estimated dynamics.

Principal Component Analysis

Having established the centrality of eigendecomposition in both the geometry of covariance matrices and the stability of linear dynamics, we turn to *Principal Component Analysis* (PCA). PCA is, at its core, an eigenvalue problem applied to the empirical covariance matrix. We derive it from first principles and then connect it to our DFM initialisation.

The problem. Let $\mathbf{Y} \in \mathbb{R}^{M \times T}$ be a data matrix whose columns are T observations of an M-dimensional random vector, centred to have zero sample mean. Define the $M \times M$ sample covariance matrix

$$\widehat{\mathbf{\Sigma}}_Y = \frac{1}{T}\mathbf{Y}\mathbf{Y}^\top.$$

PCA seeks a small number $r < M$ of linear combinations of the M series that capture as much of the variability in $\mathbf{Y}$ as possible. Formally, we seek directions $v_1, v_2, \ldots, v_r \in \mathbb{R}^M$ with $\|v_k\| = 1$ such that the scalar time series

$$z_{k,t} = v_k^\top y_t$$

have maximal variance, subject to v_k being orthogonal to all previous directions.

The first principal component. The first principal direction v_1 is the solution of

$$v_1 \;=\; \underset{\|v\|=1}{\arg\max}\; \mathrm{Var}(v^\top y_t) \;=\; \underset{\|v\|=1}{\arg\max}\; v^\top \widehat{\mathbf{\Sigma}}_Y\, v. \tag{247}$$

This is a constrained quadratic optimisation: maximise the quadratic form $v^\top \widehat{\mathbf{\Sigma}}_Y v$ on the unit sphere $\{v : \|v\| = 1\}$. The Lagrangian is

$$\mathcal{L}(v, \mu) \;=\; v^\top \widehat{\mathbf{\Sigma}}_Y\, v \;-\; \mu\,(v^\top v - 1),$$

with first-order condition $\widehat{\mathbf{\Sigma}}_Y\, v \;=\; \mu\, v$. This is *exactly* the eigenvalue equation for $\widehat{\mathbf{\Sigma}}_Y$, with Lagrange multiplier μ playing the role of the eigenvalue. The solution is therefore an eigenvector of $\widehat{\mathbf{\Sigma}}_Y$, and at the optimum, $v_1^\top \widehat{\mathbf{\Sigma}}_Y\, v_1 = \mu$: the variance explained by v_1 equals its eigenvalue. To maximise the explained variance, we choose the eigenvector *corresponding to the largest eigenvalue* λ_1.

Subsequent principal components. The second principal direction v_2 is defined by the same optimisation, with the additional constraint $v_2 \perp v_1$. By the spectral theorem, the eigenvectors of the symmetric $\widehat{\mathbf{\Sigma}}_Y$ are mutually orthogonal, so the solution is simply the eigenvector corresponding to the second-largest eigenvalue λ_2. More generally, the k-th principal direction v_k is the eigenvector corresponding to the k-th largest eigenvalue λ_k.

The PCA identity. Collecting the eigenvalues in decreasing order $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_M \geq 0$ (the inequality ≥ 0 holds because $\widehat{\mathbf{\Sigma}}_Y$ is positive semi-definite, being a scaled sum of outer products), and the corresponding eigenvectors $v_1, \ldots, v_M$ as the columns of an orthogonal matrix V, the spectral decomposition of $\widehat{\mathbf{\Sigma}}_Y$ reads

$$\widehat{\mathbf{\Sigma}}_Y \;=\; V\,\Lambda\,V^\top, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_M).$$

This is the formal content of PCA: it *is* the eigendecomposition of the sample covariance matrix. The principal components themselves are obtained by projecting the data onto the eigenvector basis:

$$Z \;=\; V^\top\,\mathbf{Y} \;\in\; \mathbb{R}^{M\times T},$$

where the k-th row of Z is the k-th principal component time series.

Variance explained and dimensionality reduction. The total variance of the data is $\mathrm{tr}(\widehat{\mathbf{\Sigma}}_Y) = \sum_k \lambda_k$ (the trace equals the sum of eigenvalues). The fraction of total variance explained by the first r principal components is therefore

$$\text{variance explained by } z_1, \ldots, z_r \;=\; \frac{\sum_{k=1}^{r} \lambda_k}{\sum_{k=1}^{M} \lambda_k}.$$

If the spectrum of $\widehat{\mathbf{\Sigma}}_Y$ is *concentrated* — that is, a few large eigenvalues followed by many small ones — then a small number r of principal components captures most of the variance, and the data admit a *low-dimensional summary* in terms of those r components. This is the empirical foundation of factor modelling: high-dimensional macroeconomic data typically have a few dominant eigenvalues corresponding to common factors, and many small eigenvalues corresponding to idiosyncratic noise. The "scree plot" of eigenvalues in decreasing order — and the identification of the "elbow" separating large from small eigenvalues — is the traditional graphical tool for selecting r.

PCA as Initialisation for the DFM

We now return to the DFM of this thesis and explain why PCA provides a natural starting point for the EM algorithm — the role of PCA in the initialisation step.

The intuition. The DFM postulates that the M observed series are driven by r common factors through a linear loading matrix $\mathbf{\Lambda}$, plus idiosyncratic noise:

$$y_t = \mathbf{\Lambda} f_t + \varepsilon_t.$$

If the idiosyncratic component ε_t is "small" relative to the common component $\mathbf{\Lambda} f_t$, then the covariance of y_t is dominated by $\mathbf{\Lambda}\,\mathrm{Var}(f_t)\,\mathbf{\Lambda}^\top$ — a rank-r matrix. Its top r eigenvectors span the same column space as $\mathbf{\Lambda}$, and the corresponding eigenvalues reflect the variance contributions of the factors. PCA — which extracts precisely these top eigenvectors from $\widehat{\mathbf{\Sigma}}_Y$ — therefore recovers a basis for the factor space, up to an orthogonal rotation.

This observation is classical and underlies the equivalence between PCA and static factor analysis. For our initialisation purposes, PCA gives us three quantities "for free": a basis for the factors (the top r eigenvectors of $\widehat{\mathbf{\Sigma}}_Y$), starting values for the factor loadings $\mathbf{\Lambda}^{(0)}$ (read off from the same eigenvectors), and an initial sense of which directions in data space carry the most variance.

The standard initialisation procedure. Concretely, the initialisation of the DFM proceeds as follows. With the data balanced and pre-processed to be centred and unit-variance:

1. Form the sample covariance $\widehat{\mathbf{\Sigma}}_Y = \frac{1}{T}\mathbf{Y}\mathbf{Y}^\top$.
2. Compute its spectral decomposition $\widehat{\mathbf{\Sigma}}_Y = V\Lambda V^\top$, with eigenvalues ordered decreasingly $\lambda_1 \geq \ldots \geq \lambda_M$.
3. Retain the top r eigenvectors $V_r = [v_1, v_2, \ldots, v_r] \in \mathbb{R}^{M\times r}$.
4. Set the initial loading matrix to $\mathbf{\Lambda}^{(0)} = V_r \Lambda_r^{1/2}$, where $\Lambda_r = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$.
5. Set the initial factors to $\mathbf{F}^{(0)} = \Lambda_r^{-1/2}\, V_r^\top\, \mathbf{Y}$, a $r \times T$ matrix whose rows have unit variance and are mutually uncorrelated in sample.

With these initialisations in hand, the remaining parameters $\mathbf{A}^{(0)}, \mathbf{Q}^{(0)}, \mathbf{R}^{(0)}$ are obtained from ordinary least squares applied to the PCA factors, and the degrees of freedom are initialised at $\nu_u^{(0)} = \nu_\varepsilon^{(0)} = 10$ as discussed previously. The EM iteration then takes over.

Why PCA initialisation works in practice. PCA is an *approximation* to the correct initialisation for the DFM, not the exact solution. Three approximations are made:

- *Static rather than dynamic.* PCA extracts factors from a *static* covariance, ignoring the VAR dynamics of f_t specified in the model. The true factor estimator should account for $\mathbf{A}$ and the time structure.
- *Gaussian rather than heavy-tailed.* PCA assumes Gaussian errors implicitly (least-squares projection); the Student-t structure of our model is ignored at this stage.
- *Equal-weight over time.* PCA weights every observation equally; the correct weighting in the Student-t framework would use the posterior weights $\hat{w}_t^u, \hat{w}_t^\varepsilon$.

Despite these approximations, PCA produces a starting point that is *close enough* to the true optimum that the EM algorithm converges within 50–200 iterations in typical applications. The EM iteration itself corrects for each of the three approximations: the Kalman filter introduces the dynamic structure, the posterior weights introduce the correct Student-t down-weighting, and the M-step updates realign the parameters with all of these refinements. The role of PCA is therefore not to provide a final answer, but to provide a *good enough starting point* from which the EM iteration can descend — or, more precisely, ascend — to the maximum-likelihood estimate efficiently.

Eigenvalue inspection during estimation. Beyond initialisation, the eigenvalues of $\widehat{\mathbf{\Sigma}}_Y$ are a useful diagnostic throughout. The scree plot of the eigenvalues — or, equivalently, the "variance explained" curve — helps confirm that the chosen number of factors r is appropriate: ideally, $\lambda_1, \ldots, \lambda_r$ should stand clearly above $\lambda_{r+1}, \lambda_{r+2}, \ldots$. If there is no clear separation, this is a signal that either r is mis-specified, or that the signal-to-noise ratio in the data is too low for reliable factor extraction. Such diagnostics are standard in applied factor modelling and inform our choice of r in the empirical work.

Summary: Eigenvalues in the DFM Pipeline

We close this appendix by collecting, in a single place, the distinct roles played by eigenvalues and eigenvectors throughout the DFM estimation.

Covariance and scale matrices: positive definite structure. The innovation covariance $\mathbf{Q}$, the idiosyncratic scale $\mathbf{R}$, and the posterior covariances $\tilde{P}_{t|T}$ produced by the Kalman smoother are all symmetric positive definite. Their positive eigenvalues ensure that the Mahalanobis distances, quadratic forms, and log-determinants appearing in the EM derivations are well-defined. A numerical collapse of any eigenvalue toward zero signals that the estimation is producing a degenerate distribution and requires regularisation.

Transition matrix: stability via the unit circle. The VAR transition matrix $\mathbf{A}$ governs the dynamic stability of the factor process. Its eigenvalues must lie strictly inside the unit circle for the process to be stationary and for the Kalman filter to produce bounded long-run covariances. The modulus of each eigenvalue measures persistence; complex eigenvalues correspond to cyclical modes. After estimation, the spectrum of $\widehat{\mathbf{A}}$ is inspected to confirm stability and to interpret the cyclical content of the estimated dynamics.

Data covariance: PCA initialisation. The sample covariance $\widehat{\mathbf{\Sigma}}_Y$ admits a spectral decomposition whose top r eigenvectors provide initial values for the factor loadings, and whose top eigenvalues measure the variance captured by each initial factor. PCA is the algebraic expression of "find the directions of maximal variance", and these directions are, in turn, the eigenvectors of the covariance. The EM algorithm corrects the approximations inherent in this starting point through iterated Kalman filtering and M-step updates.

Random matrices: noise-floor intuition. The Circular Law and its relatives (the Marchenko-Pastur law in particular) characterise the distribution of eigenvalues of empirical covariance matrices in the absence of a true signal. They provide a "noise floor" against which genuine factor eigenvalues must stand out for the DFM to extract meaningful structure. While we do not use these results formally in estimation, they provide the theoretical backdrop against which choices such as the number of factors r must be made.

In sum: eigenvalues quantify stability, dispersion, and information content; eigenvectors identify principal directions, factor loadings, and modes of dynamics. The repeated appearance of eigendecomposition in the DFM pipeline is not a coincidence but the algebraic expression of the fact that multivariate linear models are best understood in the basis that diagonalises their fundamental operators.